25 July - Morning Shift

Physics

Q. 1 If momentum $[P]$, area $[A]$ and time $[t]$ are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is :

Option 1:
$[PA^1\ T^0]$

Option 2:
$[PAT^{-1}]$

Option 3:
$[PA^{-1}\ T]$

Option 4:
$[PA^{-1}\ T^{-1}]$

Correct Answer:
$[PA^1\ T^0]$

Solution:
Let,

$\eta\ \alpha\ P^x\ A^y\ T^z$

η $\rightarrow$Coefficient of viscosity

$\therefore [\eta] = [P]^x\ [A]^y\ [T]^z$

$[M'L^{-1}T^{-1}] = [M'L'T^{-1}]^x\ [L^2]^y\ [T]^z$

$[M'L^{-1}T^{-1}] = [M^x\ L^{x+2y}\ \ T^{-x+z}]$

Comparing both the sides ,we get,

$x = 9 \rightarrow (1)$

$x + 2y = -1$

$y = -1 \rightarrow (2)$

$-x + z = -1$

$z = 0 \rightarrow (3)$

$\therefore [\eta] = \left[P^1 A^{-1} T^0\right]$

Hence (1) is correct option.

Q. 2 Which of the following physical quantities have the same dimensions?

Option 1:
Electric displacement $(\vec{D})$ and surface charge density

Option 2:
Displacement current and electric field

Option 3:
Current density and surface charge density

Option 4:
Electric potential and energy

Correct Answer:
Electric displacement $(\vec{D})$ and surface charge density

Solution:

$$[\text{Electric displacement}] = [\text{D}] = \left[\frac{\text{charge}}{\text{area}}\right]$$

$$= [\text{A}'\text{T}'\text{L}^{-2}]$$

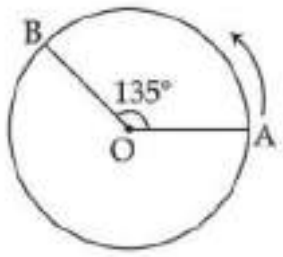
$$\left[\begin{array}{c}\text{Surface charge} \\ \text{density}\end{array}\right] = \left[\frac{\text{Q}}{\text{A}}\right] = [\text{A}'\text{T}'\text{L}^{-2}]$$

$$[\text{Current density}] = [\text{J}] = \left[\frac{\text{I}}{\text{A}}\right] = [\text{A}'\text{L}^{-2}]$$

$$[\text{Electric field}] = [\text{M}^{1}\text{L}^{1}\text{T}^{-3}\text{A}^{-1}]$$

Hence (1) is correct option.

Q. 3 A person moved from A to B on a circular path as shown in figure. If the distance travelled by him is $60\ \text{m}$, then the magnitude of displacement would be :
(Given $\cos 135^{\circ} = -0.7$)



Option 1:
$42\ \text{m}$

Option 2:
$47\ \text{m}$

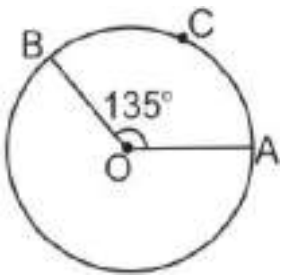
Option 3:
$19\ \text{m}$

Option 4:
$40\ \text{m}$

Correct Answer:
$47\ \text{m}$

Solution:
Distance

$$\text{ACB} = \text{r}\left(135^{\circ} \times \frac{\pi}{180}\right)$$

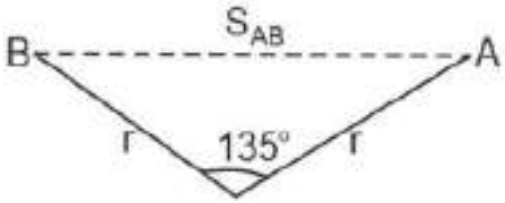


$$60 = \frac{3}{4}\pi r$$

$$r = \frac{80}{\pi} \rightarrow (1)$$

Magnitude of displacement AB is $S_{AB} = \sqrt{r^2 + r^2 - 2r^2 \cos 135^0}$

(from cosine laws)



$$S_{AB} = \sqrt{2r} \times \sqrt{1.7}$$

$$S_{AB} = \sqrt{3.4}\ r$$

From eq.1

$$S_{AB} = \sqrt{3.4} \times \frac{80}{\pi}$$

$$= 47\ \text{m}$$

Hence (2) is correct option.

Q. 4 A body of mass $0.5\ \text{kg}$ travels on straight line path with velocity $v = (3x^2 + 4)\ \text{m/s}$. The net workdone by the force during its displacement from $x = 0$ to $x = 2\ \text{m}$ is :

Option 1:
$64\ \text{J}$

Option 2:
$60\ \text{J}$

Option 3:
$120\ \text{J}$

Option 4:
$128\ \text{J}$

Correct Answer:

$60\ \mathrm{J}$

Solution:

By work-energy theorem ,

$$\mathrm{W} = \Delta \mathrm{kE} = \frac{1}{2}\mathrm{mv_f^2} - \frac{1}{2}\mathrm{mv_i^2}$$

It is given that,

$$v = 3x^2 + 4$$

$$\text{at } \mathrm{x} = 0, \mathrm{v_i} = 4\ \mathrm{m/s}$$

$$\text{at } \mathrm{x} = 2, \mathrm{v_f} = 16\ \mathrm{m/s}$$

$$\therefore \mathrm{W} = \frac{1}{2} \times 0.5(16)^2 - \frac{1}{2}(0.5)(4)^2$$

$$\mathrm{W} = \frac{1}{4}\left[240\right] = 60\ \mathrm{J}$$

Hence (2) is correct

Q. 5 A solid cylinder and a solid sphere, having same mass m and radius R, roll down the same inclined plane from top without slipping. They start from rest. The ratio of velocity of the solid cylinder to that of the solid sphere, with which they reach the ground, will be :

Option 1:

$\sqrt{\frac{5}{3}}$

Option 2:

$\sqrt{\frac{4}{5}}$

Option 3:

$\sqrt{\frac{3}{5}}$

Option 4:

$\sqrt{\frac{14}{15}}$

Correct Answer:

$\sqrt{\frac{14}{15}}$

Solution:

For a body undergoing rolling motion on inclined plane,

$$V = \sqrt{\frac{2gh}{\left(1+\frac{k^2}{R^2}\right)}}$$

For soild cylinder ,$\frac{k^2}{R^2} = \frac{1}{2}$

For solid sphere,$\frac{k^2}{R^2} = \frac{2}{5}$

$$\frac{V_{cylinder}}{V_{sphere}} = \sqrt{\frac{1+2/5}{1+\frac{1}{2}}}$$

$$= \sqrt{\frac{7/5}{3/2}}$$

$$= \sqrt{\frac{14}{15}}$$

Hence (4) is corect option.

Q. 6 Three identical particles A, B and C of mass $100\ kg$ each are placed in a straight line with $AB = BC = 13\ m$. The gravitational force on a fourth particle P of the same mass is f, when placed at a distance $13\ m$ from the particle B on the perpendicular bisector of the line AC. The value of F will be approximately :

Option 1:

$21\ G$

Option 2:

$100\ G$

Option 3:

$59\ G$

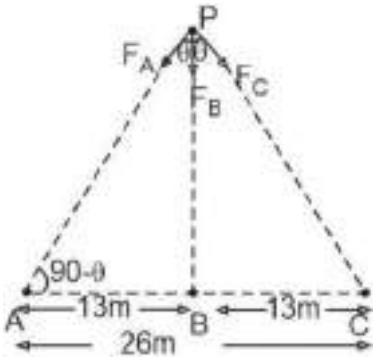
Option 4:

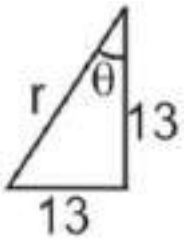
$42\ \mathrm{G}$

Correct Answer:

$100\ \mathrm{G}$

Solution:





Let, $\mathrm{F_A}, \mathrm{F_B}$ and $\mathrm{F_c}$ be the gravitational force on particle P by particle A, B and C respectively .

$$\bar{F} = \bar{F}_A + \bar{F}_B + \bar{F}_C \quad \text{(Given)}$$

$$\mathrm{F} = \mathrm{F_A}\cos\theta + \mathrm{F_B} + \mathrm{F_C}.\cos\theta$$

$$= \frac{2\mathrm{G}(100)(100)}{\mathrm{r}^2} \times \frac{13}{\mathrm{r}} + \frac{\mathrm{G}(100)^2}{13^2}$$

$$\mathrm{r} = \sqrt{(13)^2 + (13)^2}$$

$$\mathrm{F} = \frac{2\mathrm{G}\,(100)^2 \times 13}{(13)^2 \times 2 \times 13\sqrt{2}} + \frac{\mathrm{G}\,(100)^2}{13^2}$$

$$= \frac{\mathrm{G}\,(10)^4}{(13)^2}\left[\frac{1}{\sqrt{2}} + 1\right]$$

$$\mathrm{F} = 100\ \mathrm{G}$$

Hence (2) is correct option

Q. 7 A certain amount of gas of volume V at $27^{\circ}\mathrm{C}$ temperature and pressure $2 \times 10^{7}\mathrm{Nm}^{-2}$ expands isothermally until its volume gets doubled. Later it expands adiabatically until its volume gets redoubled. The final pressure of the gas will be (Use $\gamma = 1.5$) :

Option 1:
$3.536 \times 10^{5}\ \mathrm{Pa}$

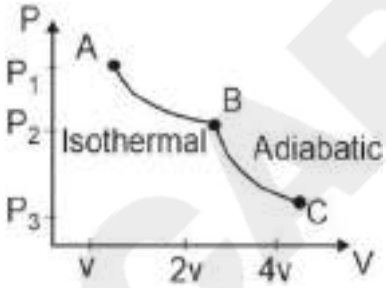
Option 2:
$3.536 \times 10^{6}\ \mathrm{Pa}$

Option 3:
$1.25 \times 10^{6}\ \mathrm{Pa}$

Option 4:
$1.25 \times 10^{5}\ \mathrm{Pa}$

Correct Answer:
$3.536 \times 10^{6}\ \mathrm{Pa}$

Solution:



Process $\mathrm{A} \rightarrow \mathrm{B}$ is isothermal process

$$\therefore \mathrm{P_1 V} = \mathrm{P_2}\,(2\mathrm{v})$$

$$\mathrm{P_2} = \frac{\mathrm{P_1}}{2} \rightarrow (1)$$

Process $\mathrm{B} \rightarrow \mathrm{C}$ is adiabatic process

$$\mathrm{P_2}(2\mathrm{v})^{\mathrm{v}} = \mathrm{P_3}(4\mathrm{v})^{\mathrm{v}}$$

$$\frac{P_2}{P_3} = (2)^v = (2)^3/2$$

$$\frac{P_2}{P_3} = 2\sqrt{2} \rightarrow 2$$

From eq.(1)&(2)

$$\frac{P_1}{2} = 2\sqrt{2}\ P_3$$

$$P_3 = \frac{P_1}{4\sqrt{2}}$$

$$= \frac{2 \times 10^7}{4\sqrt{2}}$$

$$P_3 = 0.3536 \times 10^7$$

The final pressure of the gas will be $= 3.536 \times 10^6$ pa

Hence (2) is correct option.

Q. 8 Following statements are given :
(A) The average kinetic energy of a gas molecule decreases when the temperature is reduced.
(B) The average kinetic energy of a gas molecule increases with increase in pressure at constant temperature.
(C) The average kinetic energy of a gas molecule decreases with increase in volume.
(D) Pressure of a gas increases with increase in temperature at constant pressure.
(E) The volume of gas decreases with increase in temperature.
Choose the correct answer from the options given below :

Option 1:
(A) and (D) only

Option 2:
(A), (B) and (D) only

Option 3:
(B) and (D) only

Option 4:
(A), (B) and (E) only

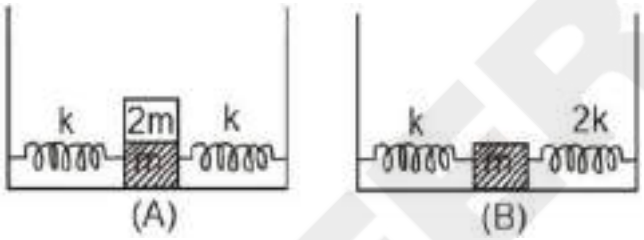
Correct Answer:
(A) and (D) only

Solution:
The average Kinetic energy of a gas molecule is say KE

$$\mathrm{KE}=\frac{3}{2}\mathrm{K_B T}=\frac{3}{2}\mathrm{PV}$$

$\rightarrow$ with reduction in temperature, the KE of a gas molecule decreases
$\rightarrow$ Pressure of a gas increases with increase in temperature at constant volume
Hence (1) is correct option.

Q. 9



In figure (A), mass $'2\ \mathrm{m}'$ is fixed on mass $'\mathrm{m}'$ which is attached to two springs of spring constant K. In figure (B), mass ' m ' is attached to two springs of spring constant ' K ' and ' $2\mathrm{k}$ '. If mass $'\mathrm{m}$ ' in (A) and in (B) are displaced by distance $'\mathrm{x}$ 'horizontally and then released, then time period $\mathrm{T_1}$ and $\mathrm{T_2}$ corresponding to $(\mathrm{A})\mathrm{and}(\mathrm{B})$ respectively follow the relation.

Option 1:
$\frac{\mathrm{T_1}}{\mathrm{T_2}}=\frac{3}{\sqrt{2}}$

Option 2:
$\frac{\mathrm{T_1}}{\mathrm{T_2}}=\sqrt{\frac{3}{2}}$

Option 3:

$$\frac{T_1}{T_2}=\sqrt{\frac{2}{3}}$$

Option 4:

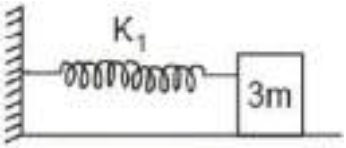
$$\frac{T_1}{T_2}=\frac{\sqrt{2}}{3}$$

Correct Answer:

$$\frac{T_1}{T_2}=\frac{3}{\sqrt{2}}$$

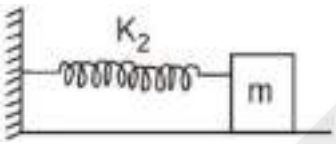
Solution:

Equivalent of figure (A) is



$$k_1=k+k=2k$$

Equivalent of figure (B) is



$$k_2=k+2k=3k$$

$$T_1=2\pi\sqrt{\frac{3m}{k_1}}=2\pi\sqrt{\frac{3m}{2k}}$$

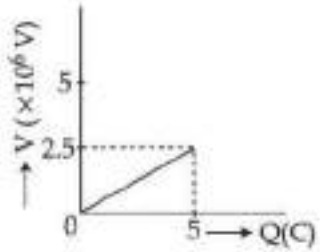
$$T_2=2\pi\sqrt{\frac{m}{k_2}}=2\pi\sqrt{\frac{m}{3k}}$$

$$\frac{T_1}{T_2}=\frac{3}{\sqrt{2}}$$

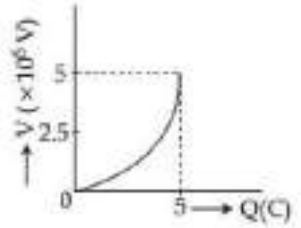
Hence (1) is correct option.

Q. 10 A condenser of $2\mu\mathrm{F}$ capacitance is charged steadily from $0\ to\ 5\mathrm{C}$. Which of the following graph represents correctly the variation of potential difference (V) across it's plates with respect to the charge (Q) on the condenser ?

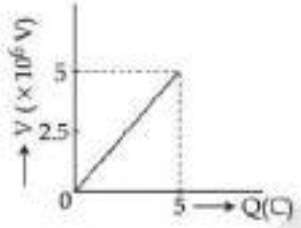
Option 1:



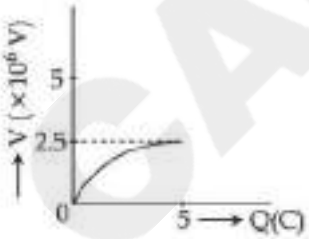
Option 2:



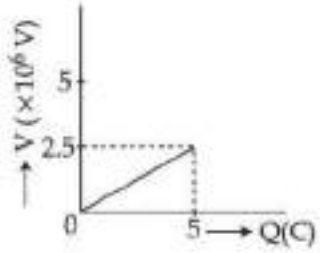
Option 3:



Option 4:



Correct Answer:



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Solution:

$$C = \frac{Q}{V} = \text{ constant}$$

The ratio $\left(\frac{Q}{v}\right)$ is constant for a given material in the gap of capacitor plates and for a given geometry.

$$\therefore \frac{V}{Q} = \text{ constant } = \frac{1}{C} = \frac{10^6}{2}$$

The above relation holds fore for option (1)

Hence (1) is correct option.

Q. 11 Two charged particles, having same kinetic energy, are allowed to pass through a uniform magnetic field perpendicular to the direction of motion. If the ratio of radii of their circular paths is $6:5$ and their respective masses ratio is $9:4$. Then, the ratio of their charges will be:

Option 1:

$8:5$

Option 2:

$5:4$

Option 3:

$5:3$

Option 4:

$8:7$

Correct Answer:

$5:4$

Solution:

$$\frac{r_1}{r_2} = \frac{6}{5} \rightarrow (1)$$

$$\frac{m_1}{m_2} = \frac{9}{4} \rightarrow (2)$$

$$\frac{q_1}{q_2} = ?$$

It is given that,

$$KE_1 = KE_2$$

$$\frac{P_1^2}{2m_1} = \frac{P_2^2}{2m_2}$$

$$\frac{P_1}{P_2} = \frac{3}{2} \rightarrow 3$$

We know that,

$$r = \frac{mv}{qB} = \frac{p}{qB}$$

$$\frac{r_1}{r_2} = \frac{p_1}{p_2} \times \frac{q_2}{q_1}$$

$$\frac{q_1}{q_2} = \frac{p_1}{p_2} \times \frac{r_1}{r_2} = \frac{3}{2} \times \frac{5}{6} = \frac{5}{4}$$

Hence (2) is correct option.

Q. 12 To increase the resonant frequency in series LCR circuit,

Option 1:
source frequency should be increased.

Option 2:
another resistance should be added in series with the first resistance.

Option 3:
another capacitor should be added in series with the first capacitor.

Option 4:
the source frequency should be decreased.

Correct Answer:
another capacitor should be added in series with the first capacitor.

Solution:

Resonant frequency $= f = \frac{1}{2\pi\sqrt{LC}}$

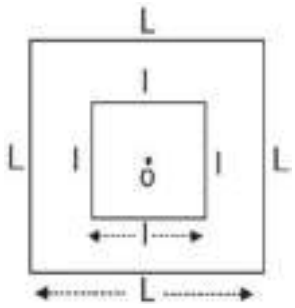
If we add another capaitor in series with the first capaitor, C decreases

$\because \frac{1}{c_{eq}} = \frac{1}{c_1} + \frac{1}{c_2}$

$\therefore$ The resonant frequency in series LCR circuit increases with the addition of capaitor in Series with the first capaitor

Hence (3) is correct option.

Q. 13 A small square loop of wire of side l is placed inside a large square loop of wire $\mathrm{L}(\mathrm{L} \gg l)$. Both loops are coplanar and their centres coincide at point O as shown in figure. The mutual inductance of the system is :



Option 1:
$\frac{2\sqrt{2}\mu_0 \mathrm{L}^2}{\pi l}$

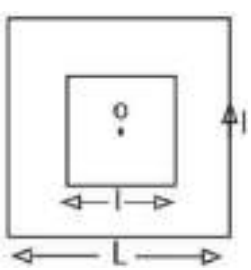
Option 2:
$\frac{\mu_0 l^2}{2\sqrt{2}\pi \mathrm{L}}$

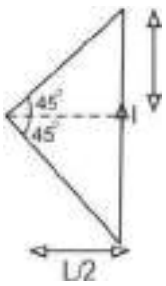
Option 3:
$\frac{2\sqrt{2}\mu_0 l^2}{\pi \mathrm{L}}$

Option 4:
$\frac{\mu_0 \mathrm{L}^2}{2\sqrt{2}\pi l}$

Correct Answer:
$\frac{2\sqrt{2}\mu_0 l^2}{\pi \mathrm{L}}$

Solution:





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If current I is flowing through loop of of length L then the flux associated with the smaller loop is

$\Phi = \mathrm{Mi} = (4\mathrm{B})\mathrm{A}$

$$\mathrm{Mi} = 4\left[\frac{\mu_0 \mathrm{i}}{4\pi(\mathrm{L}/2)}\left(\sin 45^\circ + \sin 45^\circ\right)\right] \mathrm{l}^2$$

$$\mathrm{M} = \frac{\mu_0 \mathrm{l}^2}{\pi \mathrm{L}}(2\sqrt{2})$$

Hence (3) is correct option.

Q. 14 The rms value of conduction current in a parallel plate capacitor is $6.9\mu\mathrm{A}$. The capacity of this capacitor, if it is connected to $230\ \mathrm{V}$ ac supply with an angular frequency of $600\ \mathrm{rad/s}$, will be:

Option 1:
$5\mathrm{pF}$

Option 2:
$50\mathrm{pF}$

Option 3:
$100\mathrm{pF}$

Option 4:
$200\mathrm{pF}$

Correct Answer:
$50\mathrm{pF}$

Solution:

$(\mathrm{i_c})_{\mathrm{rms}} = 6.9\mu\mathrm{A}$

$\mathrm{V_{rms}} = 230\ \mathrm{V}$

$$\omega = 600\left(\frac{\mathrm{rad}}{\mathrm{s}}\right)$$

$$\mathrm{V_{rms}} = \mathrm{I_{rms}}\left(\mathrm{X_c}\right)$$

$$230 = 6.9 \times 10^{-6}\left(\frac{1}{600 \times \mathrm{c}}\right)$$

$$C = \frac{6.9 \times 10^{-6}}{2.3 \times 6 \times 10^{4}}$$

$$= 0.5 \times 10^{-10}$$

$$C = 50 \times 10^{-12}\ F$$

$$C = 50pF$$

Hence (2) is correct option.

Q. 15 Which of the following statement is correct?

Option 1:
In primary rainbow, observer sees red colour on the top and violet on the bottom

Option 2:
In primary rainbow, observer sees violet colour on the top and red on the bottom

Option 3:
In primary rainbow, light wave suffers total internal reflection twice before coming out of water drops.

Option 4:
Primary rainbow is less bright than secondary rainbow.

Correct Answer:
In primary rainbow, observer sees red colour on the top and violet on the bottom

Solution:
→ In primany raintow, observer sees red colour on the top and violet on the bottom
→ In secondary raintow, observer sees violet colour on the top and red on the bottom
→ The light of the secondary bow is one-tenth the intensity of that of the primary bow.
Hence (1) is correct option.

Q. 16 Time taken by light to travel in two different materials A and B of refractive indices μ_{A} and μ_B of same thickness is T_1 and T_2 respectively. If $\mathrm{t}_2 - \mathrm{t}_1 = 5 \times 10^{-10}\mathrm{s}$ and the ratio of μ_{A} to μ_{B} is $1 : 2$. Then, the thickness of material, in meter is : (Given v_{A} and v_{B} are velocities of light in A and B materials respectively.)

Option 1:
$5 \times 10^{-10} v_{\mathrm{A}} \mathrm{m}$

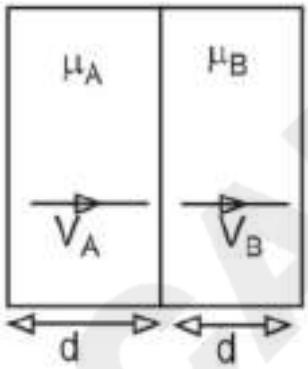
Option 2:
$5 \times 10^{-10}\ \mathrm{m}$

Option 3:
$1.5 \times 10^{-10}\ \mathrm{m}$

Option 4:
$5 \times 10^{-10} v_{\mathrm{B}} \mathrm{m}$

Correct Answer:
$5 \times 10^{-10} v_{\mathrm{A}} \mathrm{m}$

Solution:
Let the thickness of material be d



From Snell's law,

$$\frac{\mu_{\mathrm{A}}}{\mu_{\mathrm{B}}} = \frac{\mathrm{V_B}}{\mathrm{V_A}} = \frac{1}{2}$$

$$\frac{\mathrm{V_B}}{\mathrm{V_A}} = \frac{\mathrm{d}/\mathrm{t}_2}{\mathrm{d}/\mathrm{t}_1} = \frac{1}{2}$$

$$\frac{\mathrm{t}_1}{\mathrm{t}_2} = \frac{1}{2}$$

$$\frac{t_2}{t_1} = \frac{2}{1}$$

$$\frac{t_2 - t_1}{t_1} = 1$$

$$t_1 = 5 \times 10^{-10}\ s$$

$$V_A = \frac{d}{t_1}$$

$$d = V_A t_1 = (5 \times 10^{-10}) V_A$$

Hence (1) is correct option

Q. 17 A metal exposed to light of wavelength $800\ nm$ and and emits photoelectrons with a certain kinetic energy. The maximum kinetic energy of photo-electron doubles when light of wavelength $500\ nm$ is used. The workfunction of the metal is : (Take $hc = 1230eV - nm$).

Option 1:
$1.537eV$

Option 2:
$2.46eV$

Option 3:
$0.615eV$

Option 4:
$1.23eV$

Correct Answer:
$0.615eV$

Solution:
$\lambda = 800\ nm$

$$E\,(eV) = \frac{hc}{\lambda}$$

$$\lambda = \frac{1230}{E} nm$$

$$E_1 = \frac{1230}{\lambda_1} = \frac{1230}{800}eV \rightarrow (1)$$

$$E_2 = \frac{1230}{\lambda_2} = \frac{1230}{500}eV \rightarrow (2)$$

From Einstein's photoelectric equation,

$$E = \phi_0 + (kE_{max})$$

Let,

$$(KE_{max})_1 = K_0$$

$$E_1 = \phi_0 + k_0 \rightarrow \ (3)$$

$$E_2 = \phi_0 + 2k_0 \rightarrow \ (4) \ (\text{Given})$$

$$2 \times (3) - (4)$$

$$2E_1 - E_2 = \phi_0$$

$$\frac{1230}{400} - \frac{1230}{500} = \phi_0$$

$$\phi_0 = \frac{1230}{2000} = \frac{123}{200}$$

Work Function $= \phi_0 = 0.615eV$

The work function of thermal is 0.615 eV

Hence (3) is correct option

Q. 18 The momentum of an electron revolving in n^{th} orbit is given by : (Symbols have their usual meanings)

Option 1:
$\frac{nh}{2\pi r}$

Option 2:
$\frac{nh}{2r}$

Option 3:
$\frac{nh}{2\pi}$

Option 4:
$\frac{2\pi r}{nh}$

Correct Answer:
$\frac{nh}{2\pi r}$

Solution:
The angular momentum of an electron in n^{th} orbit is.

$$L_n = mv_n r_n = \frac{nh}{2\pi}$$

$$P_n = mv_n = \frac{nh}{2\pi r_n}$$

$$P_n = \frac{nh}{2\pi r}$$

Hence (1) is correct option

Q. 19 The magnetic moment of an electron (e) revolving in an orbit around nucleus with an orbital angular momentum is given by :

Option 1:

$\tilde{\mu}_{\mathrm{L}=\frac{e\vec{\mathrm{L}}}{2\ \mathrm{m}}}$

Option 2:

$$\tilde{\mu}_{\mathrm{L}}=-\frac{e\ \overrightarrow{\mathrm{L}}}{2\ \mathrm{m}}$$

Option 3:

$$\overrightarrow{\mu_l}=-\frac{\mathrm{e}\overrightarrow{\mathrm{L}}}{\mathrm{m}}$$

Option 4:

$$\tilde{\mu}_{\mathrm{l}}=\frac{2\mathrm{e}\overrightarrow{\mathrm{L}}}{\mathrm{m}}$$

Correct Answer:

$$\tilde{\mu}_{\mathrm{L}}=-\frac{e\ \overrightarrow{\mathrm{L}}}{2\ \mathrm{m}}$$

Solution:

We know that,

$$\overline{\mu}_{\mathrm{L}}=(\text{Gyromagnetic ratio})\overline{\mathrm{L}}$$

$$\overline{\mu}_{\mathrm{L}}=\left(\frac{\mathrm{q}}{2\mathrm{m}}\right)\overline{\mathrm{L}}$$

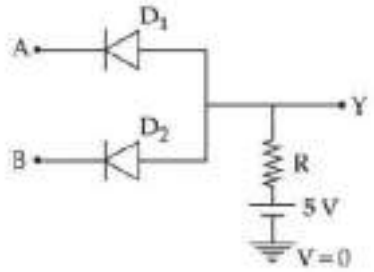
For electron,

$$\mathrm{q}=-\mathrm{e}$$

$$\therefore \overline{\mu}_{\mathrm{L}}=\left(\frac{-\mathrm{e}}{2\mathrm{m}}\right)\overline{\mathrm{L}}$$

Hence (2) is correct option.

Q. 20 In the circuit, the logical value of $A = 1$ or $B = 1$ when potential at A or B is $5\ V$ and the logical value of $A = 0$ or $B = 0$ when potential at A or B is $0\ V$.



The truth table of the given circuit will be :

Option 1:

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

Option 2:

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	1

Option 3:

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	0

Option 4:

A	B	Y
0	0	1
1	0	1
0	1	1
1	1	0

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Correct Answer:

A	B	Y
0	0	0
1	0	0
0	1	0
1	1	1

Solution:
The given circuit is AND gate realisation.

$$Y = A.B$$

$\therefore$ The truth table of given Circuit will be as given in option (1)
Hence (1) is correct option.

Q. 21 A car is moving with speed of $150\ \mathrm{km/h}$ and after applying the break it will move $27\ \mathrm{m}$ before it stops. If the same car is moving with a speed of one third the reported speed then it will stop after travelling_____ m distance.

Correct Answer:
3

Solution:
We Know that,

Stopping distance $= s = \dfrac{u^2}{2a}$

For case 1: $S_1 = \dfrac{u_1^2}{2a}$

For case 2: $S_2 = \dfrac{u_2^2}{2a}$

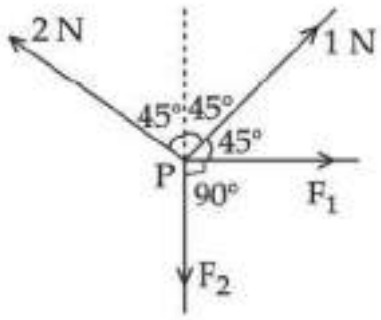
$$\frac{S_1}{S_2} = \frac{u_1^2}{u_2^2}$$

$$\frac{27}{S_2} = \frac{9}{1}$$

$$S_2 = 3\ \mathrm{m}$$

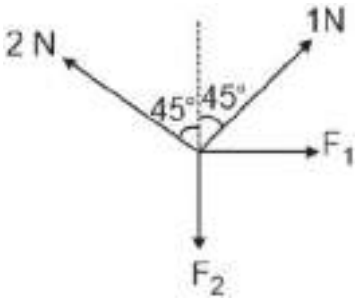
In the later case, the car will stop after travelling $3\ \mathrm{m}$ distance

Q. 22 Four forces are acting at a point P in equilibrium as shown in figure. The ratio of force $\mathrm{F_1}$ to $\mathrm{F_2}$ is $1 : \mathrm{x}$ where $\mathrm{x} =$_____.



Correct Answer:
3

Solution:



For equilibrium,

$$\Sigma \mathrm{F_x} = 0 \ \& \ \Sigma \mathrm{F_y} = 0$$

$$\Sigma \mathrm{F_x} = \mathrm{F_1} + 1\cos 45^\circ - 2\cos 45^\circ$$

$$\mathrm{F_1} = 1\cos 45^\circ = \frac{1}{\sqrt{2}} \rightarrow 1$$

$$\sum \mathrm{F_y} = 1\cos 45^\circ + 2\cos 45^\circ - \mathrm{F_2}$$

$$0 = \frac{3}{\sqrt{2}} - \mathrm{F_2}$$

$$\mathrm{F_2} = \frac{3}{\sqrt{2}} \rightarrow (2)$$

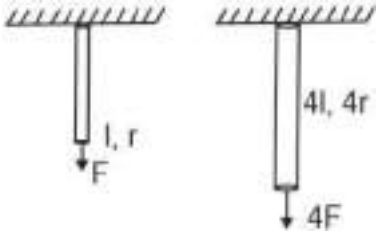
$$\frac{\mathrm{F_1}}{\mathrm{F_2}} = \frac{1}{3} = \frac{1}{\mathrm{x}}$$

$\therefore x = 3$

Q. 23 A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force f, its length increases by $5\ cm$. Another wire of the same material of length $4\ L$ and radius $4r$ is pulled by a force $4\ F$ under same conditions. The increase in length of this wire is _____cm.

Correct Answer:
5

Solution:



For the same material young's modules (Y) will be same

$$\because Y = \frac{F}{A} \times \frac{l}{\Delta l}$$

$$\therefore \frac{F_1}{\pi r^2} \times \frac{l}{\Delta l_1} = \frac{F_2}{\pi (4r)^2} \times \frac{4l}{\Delta l_2}$$

$$\frac{\Delta l_1}{\Delta l_2} = \frac{F_1}{F_2} \times \frac{16r^2}{l\,(r^2)} \times \frac{l}{4l}$$

$$\frac{\Delta l_1}{\Delta l_2} = 1$$

$$\Delta l_2 = \Delta l_1 = 5\ cm$$

The increase in the length of wire of length $4L$ is $5\ cm$

Q. 24 A unit scale is to be prepared whose length does not change with temperature and remains $20\ cm$, using a bimetallic strip made of brass and iron each of different length. The length of both components would change in such a way that difference between their lengths remains constant. If length of brass is $40\ cm$ and length of iron will be _____cm.
$(\alpha_{iron} = 1.2 \times 10^{-5}\ K^{-1}$ and $\alpha_{brass} = 1.8 \times 10^{-5}\ K^{-1})$.

Correct Answer:
60

Solution:
Let,

$$l = l_{Brass} - l_{Iron}$$

$$\Delta l = \Delta l_{Brass} - \Delta l_{Iron}$$

$$0 = l_B \alpha_B \Delta T - l_I \alpha_I \Delta T$$

$$l_B \dot{\alpha}_B. = l_I \alpha_I$$

$$40 \times 1.8 = l_I \times 1.2$$

$$l_I = 60 \text{ cm}$$

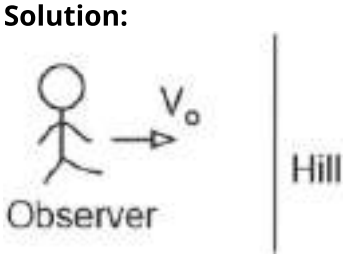
The length of iron will be 60 cm

\end{aligned}

Q. 25 An observer is riding on a bicycle and moving towards a hill at 18kmh^{-1}. He hears a sound from a source at some distance behind him directly as well as after its reflection from the hill. If the original frequency of the sound as emitted by source is 640 Hz and velocity of the sound in air is 320 m/s, the beat frequency between the two sounds heard by observer will be _____Hz.

Correct Answer:
10

Solution:



$$V_0 = 18\frac{km}{hr} = 5\frac{m}{s}$$

$$f = 640\ Hz$$

$$V_{sound} = V = 320\ m/s$$

$$f_{app} = f\left(\frac{v + V_0}{V}\right) = 650\ Hz$$

$$f_{beats} = |f_{app} - f|$$

$$= 10\ Hz$$

The beat frequency between the two Sounds heard by observer will be $10\ Hz$

Q. 26 The volume charge density of a sphere of radius $6\ m$ is $2\ \mu\ C\ cm^{-3}$. The number of lines of force per unit surface area coming out from the surface of the sphere is _____ $\times 10^{10} NC^{-1}$.
[Given : Permittivity of vacuum $\epsilon_0 = 8.85 \times 10^{-12} C^2\ N^{-1} - m^{-2}$]

Correct Answer:
45

Solution:

$$S = \frac{Q}{V} = 2\frac{\mu c}{(cm)^3} = 2\frac{c}{m^3}$$

The number of lines of force per unit surface area coming out from the surface of the sphere is

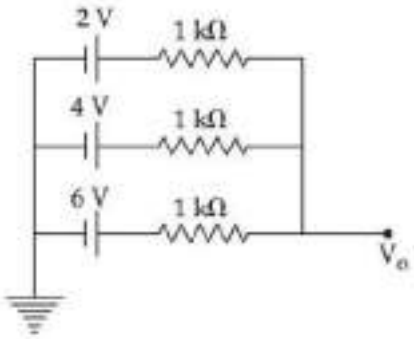
$$E = \frac{SR}{3\varepsilon_0}$$

$$= \frac{2 \times 6}{3 \times 8.85 \times 10^{-12}}$$

$$= \frac{4}{8.85} \times 10^{12}$$

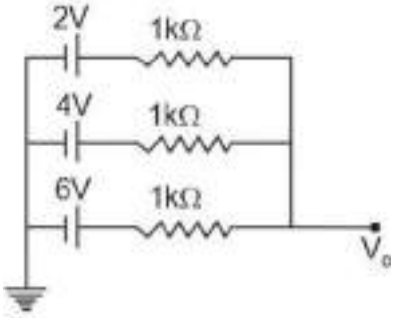
$$E = 45 \times 10^{10}\ N/C$$

Q. 27 In the given figure, the value of V_0 will be_____ V.



Correct Answer:
4

Solution:

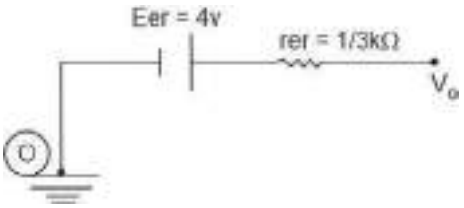


$$\varepsilon_{eq} = \frac{\frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} + \frac{\varepsilon_3}{r_3}}{\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}}$$

$$= \frac{\frac{2}{1} + \frac{4}{1} + \frac{6}{1}}{\frac{3}{1}} = 4$$

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{3}{1}$$

$$r_{eq} = \frac{1}{3}\ k\Omega$$



$$V_0 - 0 = \varepsilon_{eq}$$

$V_0 - 0 = 4V$

Q. 28 Eight copper wire of length L and diameter d are joined in parallel to form a single composite conductor of resistance R. If a single copper wire of length $2l$ have the same resistance (R) then its diameter will be_____d.

Correct Answer:
2

Solution:
Case I: Eight copper wire in parallel,

$$\frac{1}{R} = \frac{1}{R_{eq}} = \frac{8}{R_0} = \frac{8\left(\pi d^2/4\right)}{sl}$$

$$R = \frac{sl}{2\pi d^2} \rightarrow (1)$$

Case II.

$$R = \frac{\rho(2l)}{\pi\left(d'\right)^2} = \frac{sl}{2\pi d^2}$$

$$d' = 2d$$

$\therefore$ The diameter will be $2d$

Q. 29 The energy band gap of semiconducting material to produce violet (wavelength $= 4000\text{Å}$) LED is _____eV. (Round off to the nearest integer).

Correct Answer:
3

Solution:

Energy of band gap $= E(ev) = \frac{12400\text{Å}}{\lambda(\text{Å})}$

$= 3.1e\ V$

Energy of band gap $\simeq 3eV$

Q. 30 The required height of a TV tower which can cover the population of 6.03 lakh is H. If the average population density is 100 per square km and the radius of earth is $6400\ \mathrm{km}$, then the value of H will be_____ m.

Correct Answer:
150

Solution:

$$\frac{\text{Population}}{(\text{Area of Range})} = \text{population density}$$

$$\frac{6.03 \times 10^5}{\pi(2\mathrm{R\ h})} = 100 \quad (\because \text{Range} = \sqrt{2\mathrm{Rh}})$$

$$\mathrm{h} = 0.15\ \mathrm{km} = 150\ \mathrm{m}$$

Chemistry

Q. 1 $\mathrm{SO_2Cl_2}$ on reaction with excess of water results into acidic mixture

$$\mathrm{SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl}$$

16 moles of NaOH is required for the complete neutralisation of the resultant acidic mixture. The number of moles of $\mathrm{SO_2Cl_2}$ used is :

Option 1:
16

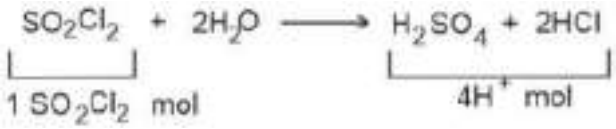
Option 2:
8

Option 3:
4

Option 4:
2

Correct Answer:
4

Solution:
Reaction given,



$\Rightarrow$ 4 mole of H^{+} are produced from 1 mol of $\mathrm{SO_2Cl_2}$

$\Rightarrow$ We know For 1 mole of H^{+} required 1 mol of NaOH for complete neutralisation.

$\Rightarrow$ For 16 mol $\mathrm{H}^{+} = 16$ mole of NaOH

Now,

16 mole of H^{+} are produced from $\left(\frac{16}{4} \times 1\right)$ mole $\mathrm{SO_2Cl_2}$

So, 4 mole of $\mathrm{SO_2Cl_2}$ is used.

Hence, the correct answer is **option (3).**

Q. 2 Which of the following sets of quantum numbers is not allowed?

Option 1:
$\mathrm{n} = 3, 1 = 2,\ \mathrm{m_l} = 0,\ \mathrm{s} = +\frac{1}{2}$

Option 2:
$\mathrm{n} = 3, 1 = 2,\ \mathrm{m_l} = -2,\ \mathrm{s} = +\frac{1}{2}$

Option 3:
$\mathrm{n} = 3, 1 = 3,\ \mathrm{m_l} = -3,\ \mathrm{s} = -\frac{1}{2}$

Option 4:
$\mathrm{n} = 3, 1 = 0,\ \mathrm{m_l} = 0,\ \mathrm{s} = -\frac{1}{2}$

Correct Answer:

$$n = 3, 1 = 3,\ m_l = -3,\ s = -\frac{1}{2}$$

Solution:

We know, Azimuthal quantum number $(\ell) = 0, 1, 2, ...$

So, $n = 3, \ell = 0, 1, 2$

Hence, option (3) has $\ell = 3$, for $n = 3$ which is not allowed.

Hence, the correct answer is **Option (3).**

Q. 3 The depression in freezing point observed for a formic acid solution of concentration $0.5\ mL\ L^{-1}$ is $0.0405°C$. Density of formic acid is $1.05\ g\ mL^{-1}$. The Van't Hoff factor of the formic acid solution is nearly :
(Given for water $k_f = 1.86\ k\ kg\ mol^{-1}$)

Option 1:

0.8

Option 2:

1.1

Option 3:

1.9

Option 4:

2.4

Correct Answer:

1.9

Solution:

We know, $\Delta T = ik_f m.....(1)$

given, $k_f = 1.86\ k\ Kg\ mol^{-1}\ (solvent = water)$

$$molality(m) = \frac{\text{mole of solute}}{\text{Solvent(kg)}}$$

Now, concentration of formic acid $[HCOOH] = 0.5\ mL\ L^{-1}$ means $0.5\ mL$ of $[HCOOH]$ is present in 1 litre H_2O

So, Mass $HCOOH = 0.5\ ml \times Density = 0.5 \times 1.05\ g = 0.525\ g$

So, 0.525 g of $HCOOH$ is present in 1 litre of water

then $\text{Molality} = \dfrac{\text{Mass of HCOOH}}{\text{Molar mass of HCOOH} \times \text{Solvent(kg)}}$

$m = \dfrac{0.525}{4.6 \times 1} \qquad [\because 1L, H_2O = 1kg \text{ of } H_2O]$

Put all value in (1)

$\Delta T_f = ik_f m = i \times 1.86 \times \dfrac{0.525}{46 \times 1}$

$0.0405 = i \times 1.86 \times \dfrac{0.525}{46}$

$i = 1.9(\text{Van's Hoff factor})$

Hence, the correct answer is **Option (3).**

Q. 4 20 mL of 0.1 M NH_4OH is mixed with 40 mL of 0.05 M HCl. The pH of the mixture is nearest to :

$(\text{Given} : K_b\,(NH_4OH) = 1\times10^{-5}, \log 2 = 0.30, \log 3 = 0.48, \log 5 = 0.69, \log 7 = 0.84, \log 11 = 1.04)$

Option 1:
3.2

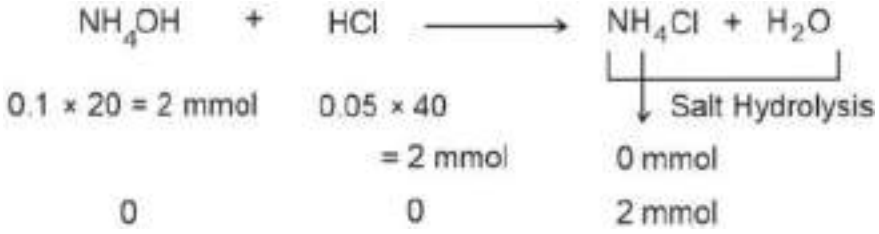
Option 2:
4.2

Option 3:
5.2

Option 4:
6.2

Correct Answer:
5.2

Solution:



NH_4OH is weak base and HCl is strong acid.

Salt NH_4OH is forming with H_2O.

We know,

$$pH = 7 - \frac{P_{K_b}}{2} - \frac{1}{2}\log C$$

$$pH = 7 - \frac{1}{2} \times 5 \quad - \frac{1}{2}\log [NH_4Cl]$$

Now,

$$[NH_4Cl] = \frac{\text{mole of } NH_4Cl}{\text{volume}} = \frac{2 \text{ mmol}}{(40+20)\text{ml}} = \frac{1}{20} \text{ mol/L}$$

$$pH = 7 - \frac{1}{2} \times 5 - \frac{1}{2}\log\left(\frac{1}{30}\right)$$

$$pH = 7 - \frac{5}{2} + \frac{1}{2}[\log 5 + \log 3 + \log 2]$$

$$pH = 7 - \frac{5}{2} + \frac{1}{2}[0.65 + 0.48 + 0.30]$$

$$pH = 7 - 2.5 + 0.735$$

$$pH = 5.235$$

Hence, the correct answer is **Option (3).**

Q. 5 Match **List- I** with **List -II**

List - I	List-II
(A) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	(I) Cu
(B) $CO(g) + 3H_2(g) \rightarrow CH_4(g) + H_2O(g)$	(II) $Cu/ZnO - Cr_2O_3$
(C) $CO(g) + H_2(g) \rightarrow HCHO(g)$	(III) $Fe_xO_y + K_2O + Al_2O_3$
(D) $CO(g) + 2H_2(g) \rightarrow CH_3OH$	(IV) Ni

Choose the correct answer from the options given below:

Option 1:
(A) - (II), (B) - (IV), (C) - (I), (D) - (III)

Option 2:
(A) - (II), (B) - (I), (C) - (IV), (D) - (III)

Option 3:
(A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Option 4:
(A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Correct Answer:
(A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Solution:

$$(A)\ N_2(g) + 3H_2(g) \rightarrow 2NH_3(g) \Rightarrow (iii) FexO_y + K_2O + Al_2O_3$$

$$(B)\ CO(g) + 3H_2(g) \rightarrow CH_4(g) + 2H_2O \Rightarrow (iv)\ Ni$$

$$(C)\ CO(g) + H_2(g) \rightarrow HCHO(g) \Rightarrow (i)\ Cu$$

$$(D)\ CO(g) + 2H_2(g) \rightarrow CH_3OH \Rightarrow (ii)\ Cu/ZnO - Cr_2O_3$$

So, (A) - (iii) , (B) - (iv), (C) - (i), (D) - (ii)

Hence, the correct answer is **Option (3)**

Q. 6 The IUPAC nomenclature of an element with electronic configuration $[\mathrm{Rn}]5\mathrm{f}^{14}6\ \mathrm{d}^{1}7\ \mathrm{s}^{2}$ is :

Option 1:
Unnilbium

Option 2:
Unnilunium

Option 3:
Unnilquadium

Option 4:
Unniltrium

Correct Answer:
Unniltrium

Solution:

$[\mathrm{Rn}]\ 5\mathrm{f}^{14}6\mathrm{d}^{1}7\mathrm{s}^{2} : 86 + 14 + 1 + 2 = 103\ \mathrm{Lr}$

$\text{Atomi number} = 103$

$103 = \text{Unniltrium}$

Hence, the correct answer is **option (4).**

Q. 7 The compound(s) that is(are) removed as slag during the extraction of copper is :

(A) CaO
(B) FeO
(C) $\mathrm{Al_2O_3}$
(D) ZnO
(E) NiO

Choose the correct answer from the options given below :

Option 1:
(C), (D) only

Option 2:
(A),(B),(E) only

Option 3:
(A), (B) only

Option 4:
(B) Only

Correct Answer:
(A),(B),(E) only

Solution:

$$FeO + SiO_2 \rightarrow \underset{(Slag)}{FeSiO_3}$$

FeO is removed as slag during the extraction of copper.

Hence, the correct answer is **option (2).**

Q. 8 The reaction of H_2O_2 with potassium permanganate in acidic medium leads to the formation of mainly :

Option 1:
Mn^{2+}

Option 2:
Mn^{4+}

Option 3:
Mn^{3+}

Option 4:
Mn^{6+}

Correct Answer:

Mn^{2+}

Solution:

The Reaction will be -

$$2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O_2 + SO_2$$

in acidic medium

$$Mn(+7) \rightarrow Mn(+2)$$

Hence, the correct answer is **option (1).**

Q. 9 Choose the correct order of density of the alkali metals :

Option 1:

$Li < K < Na < Rb < Cs$

Option 2:

$Li < Na < K < Rb < Cs$

Option 3:

$Cs < Rb < K < Na < Li$

Option 4:

$Li < Na < K < Cs < Rb$

Correct Answer:

$Li < K < Na < Rb < Cs$

Solution:

The correct order of density is :

$$\begin{array}{rc} & Li < k < Na < Rb < Cs \\ \text{density} \Rightarrow & 0.53 \;\; 0.86 \;\; 0.97 \;\; 1.53 \;\; 1.90 \end{array}$$

Hence, the correct answer is **option (1).**

Q. 10 The geometry around boron in the product 'B' formed from the following reaction is

$$BF_3 + NaH \xrightarrow{450\ K} A + NaF$$
$$A + NMe_3 \rightarrow B$$

Option 1:
trigonal planar

Option 2:
tetrahedral

Option 3:
pyramidal

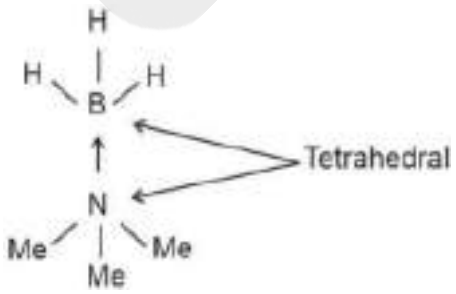
Option 4:
square planar

Correct Answer:
tetrahedral

Solution:
The Reaction will be

$$2BF_3 + 6NaH \longrightarrow B_2H_6 + 6NaF$$
$$(A)$$

$$B_2H_6 + 2NMe_3 \rightarrow 2BH_3 \cdot NMe_3$$



The geometry around Boron. In B is tetrahedral.

Hence, the correct answer is **option (2).**

Q. 11 The interhalogen compound formed from the reaction of bromine with excess of fluorine is a :

Option 1:
hypohalite

Option 2:
halate

Option 3:
perhalate

Option 4:
halite

Correct Answer:
halate

Solution:
We know,

$$\text{hypohalite} - \text{xx}'$$

$$\text{halite} - \text{xx}_3^1$$

$$\text{halate} - \text{xx}_5^1$$

$$\text{Perhalate} - \text{xx}_7^1$$

Reaction,

$$\underset{\text{(excess)}}{\text{Br}_2} + 5\ \text{F}_2 \rightarrow \underset{\substack{[\text{xx's}] \\ \text{halate}}}{2\text{BrF}_5}$$

Hence, the correct answer is **Option (2).**

Q. 12 The photochemical smog does not generally contain :

Option 1:
NO

Option 2:
NO_2

Option 3:
SO_2

Option 4:
$HCHO$

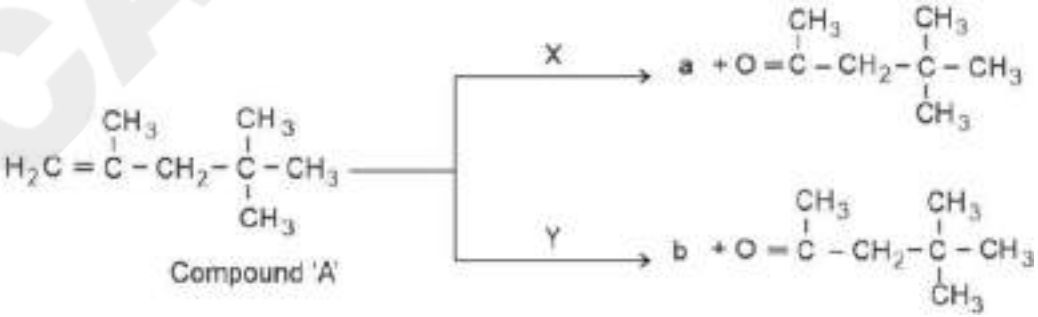
Correct Answer:
SO_2

Solution:
The common components of photochemical smog are
$\text{Ozone}\,(O_3)\,,\text{nitricoxlde}(NO),\text{acrolein},HCHO\text{ and Peroxyacetyl nitrate}(PAN).$
During the formation of smog NO_2 is also present.

But SO_2 is not component of photochemical smog.

Hence, the correct answer is **option (3).**

Q. 13 A compound 'A' on reaction with 'X' and 'Y' produces the same major product but different by product ' a ' and ' b '. Oxidation of ' a ' gives a substance produced by ants.



'X' and 'Y' respectively are

Option 1:
$KMnO_4/H^+$and dil. $KMnO_4, 273\ K$

Option 2:
$KMnO_4$ (dilute), 273 K and $KMnO_4/H^+$

Option 3:
$KMnO_4/H^+$and $O_3, H_2O/Zn$

Option 4:
$O_3, H_2O/Zn$ and $KMnO_4/H^+$

Correct Answer:
$O_3, H_2O/Zn$ and $KMnO_4/H^+$

Solution:
Ants produces formic acid $(HCOOH)$.

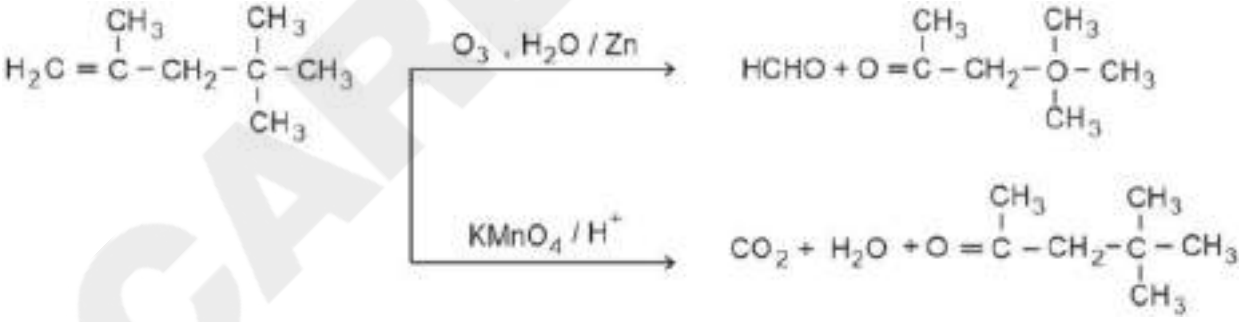
$HCOOH$ is formed from the oxidation of 'a' .

So, a should be $HCHO$

(1) $KMnO_4/H^+$ will give CO_2 $[(CH_2 = C + R)] \rightarrow CO_2 + O{=}C\ R$

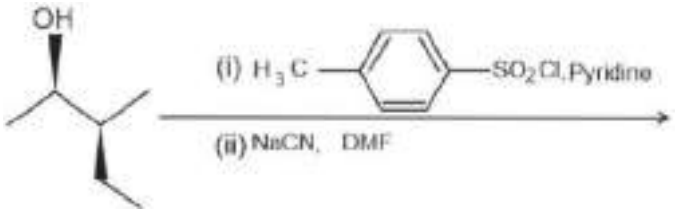
(2) $dil.KMnO_4, 273K$ will give alcohol.

(3) $O_3, H_2O/Zn$ will give $HCHO$ and other compound.

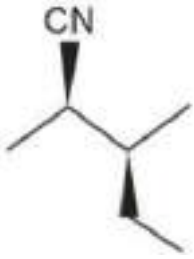


Hence, the correct answer is **option (4)**

Q. 14 Most stable product of the following reaction is :

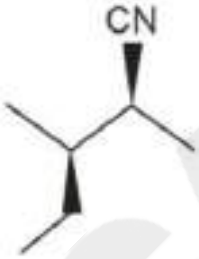


Option 1:

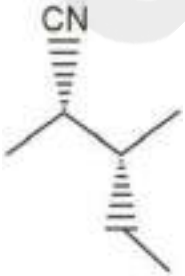


Option 2:

Option 3:



Option 4:



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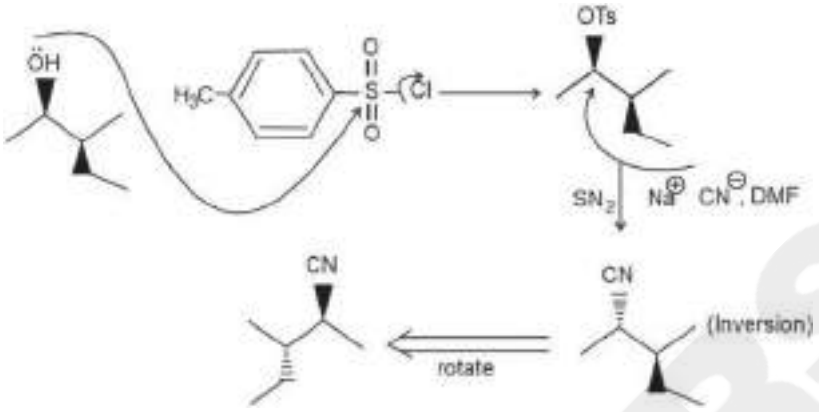
Correct Answer:

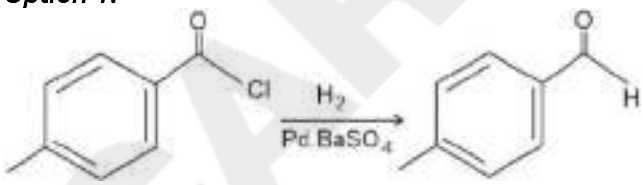
Solution:
The reaction will be -



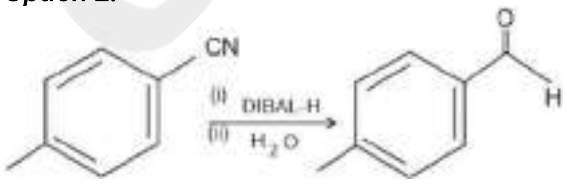
Hence, the correct answer is **option (2).**

Q. 15 Which one of the following reactions does not represent correct combination of substrate and product under the given conditions?

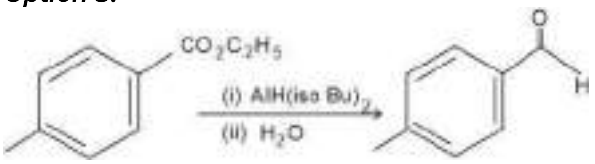
Option 1:



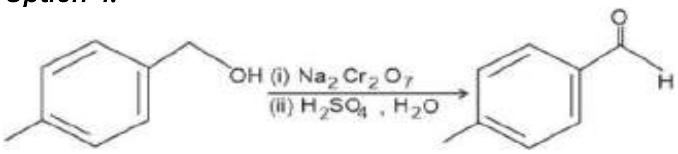
Option 2:



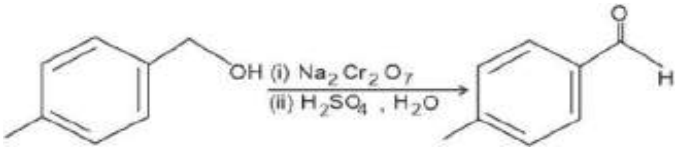
Option 3:



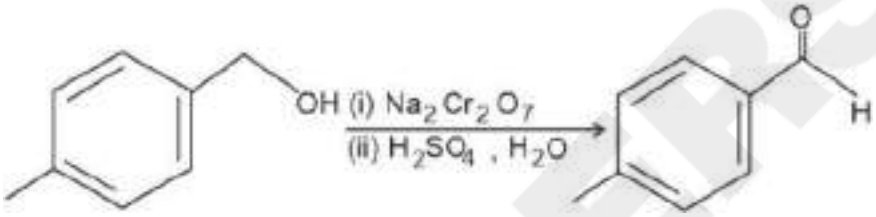
Option 4:



Correct Answer:



Solution:
Option (4) is method of preparation of carboxylic acid.

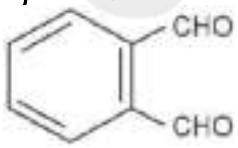


Option (1), (2) and (3) reactions are correct but (4) is incorrect.

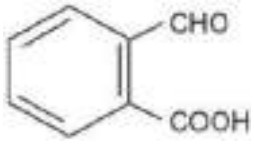
So, the correct answer is **Option (4).**

Q. 16 An organic compound ' A ' on reaction with NH_3 followed by heating gives compound B. Which on further strong heating gives compound C $(C_8H_5NO_2)$. Compound C on sequential reaction with ethanolic KOH, alkyl chloride and hydrolysis with alkali gives a primary amine. The compound A is :

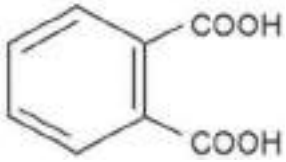
Option 1:



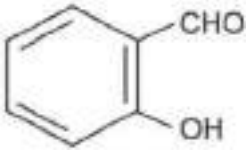
Option 2:



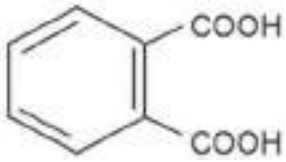
Option 3:



Option 4:

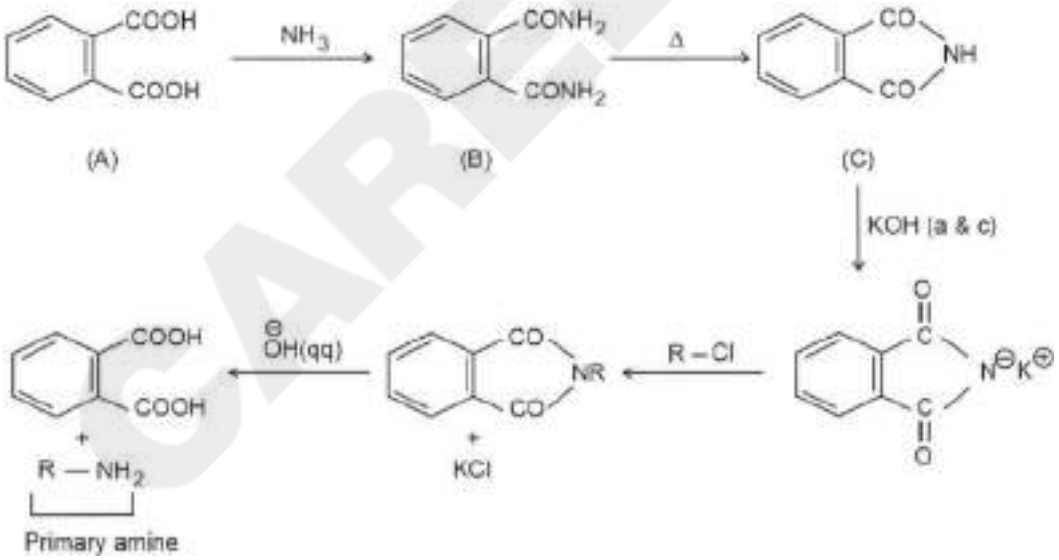


Correct Answer:



Solution:

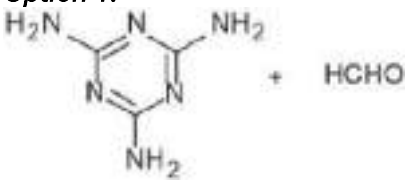
The reaction will be (Gabriel phthalimide synthesis)



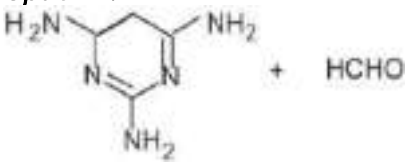
So, the correct answer is **Option (3).**

Q. 17 Melamine polymer is formed by the condensation of

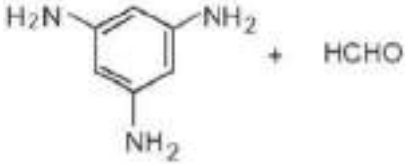
Option 1:



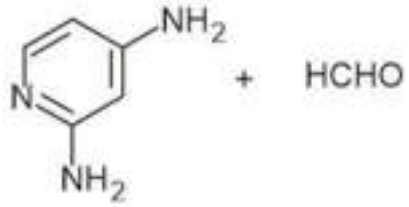
Option 2:



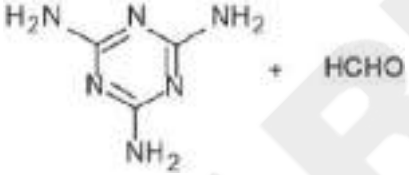
Option 3:



Option 4:

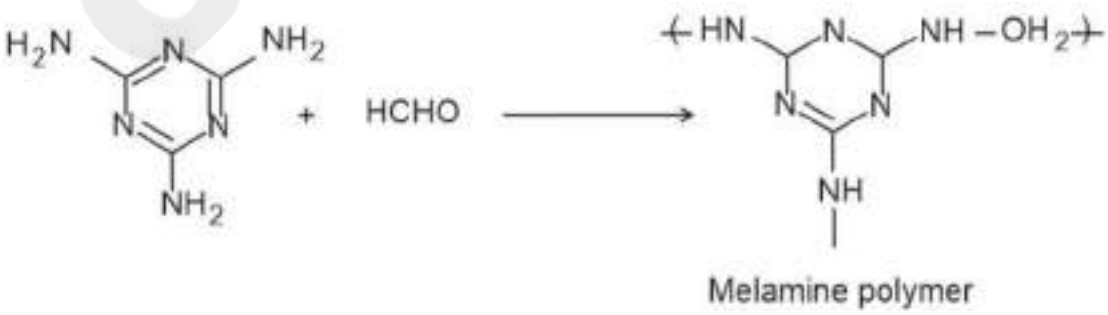


Correct Answer:



Solution:

Melamine - formaldehyde polymer is by the condensation polymerisation of melamine and formaldehyde.



Hence, the correct answer is **Option (1).**

Q. 18 During the denaturation of proteins, which of these structures will remain intact?

Option 1:
Primary

Option 2:
Secondary

Option 3:
Tertiary

Option 4:
Quaternary

Correct Answer:
Primary

Solution:
During the denaturation of proteins, 2° and 3° structures are destroyed but 1° structure remain intact.

Hence, the correct answer is **option (1).**

Q. 19 Drugs used to bind to receptors, inhibiting its natural function and blocking a message are called :

Option 1:
Agonists

Option 2:
Antagonists

Option 3:
Allosterists

Option 4:
Anti histaminists

Correct Answer:
Antagonists

Solution:
Drugs that bind to the receptor site and inhibit its natural function are called antagonists. These are useful when blocking of message is required.

Hence, the correct answer is **option (2).**

Q. 20 Given below are two statements:

Statement I : On heating with $KHSO_4$, glycerol is dehydrated and acrolein is formed.

Statement II : Acrolein has fruity odour and can be used to test glycerol's presence.

Choose the **correct** option.

Option 1:
Both **Statement I** and **Statement II** are correct.

Option 2:
Both **Statement I** and **Statement II** are incorrect.

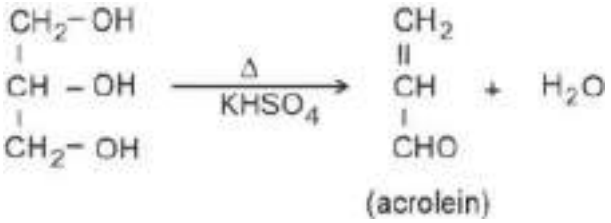
Option 3:
Statement I is correct but **Statement II** is incorrect.

Option 4:
Statement I is incorrect but **Statement II** is correct.

Correct Answer:
Statement I is correct but **Statement II** is incorrect.

Solution:

On heating with $KHSO_4$, glycerol is dehydrated and acrolein is formed which has a pungent odour.



(acrolein)

This is the test for glycerol present either free or combined as an ester.

Acrolein has pugent odour not fruity. So, statement I is correct but statement II Incorrect.

Hence, the correct answer is **Option (3).**

Q. 21 Among the following species

$N_2,\ N_2^+, N_2^-, N_2^{2-}, O_2, O_2^+, O_2^-, O_2^{2-}$

the number of species showing diamagnesim is _____.

Correct Answer:
2

Solution:
According to MOT , if all electrons are paired , then that species will show diamagnesim property.

$M_2 : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2P_x^2 = \pi 2P_y^2, \sigma 2P_z^2$

$N_2^+ : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2P_2^2 = \pi 2P_4^2, \sigma 2P_z^1$

$N_2^- : \sigma 2s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_2^2, \pi^* 2$

$N_2^{2-} = \sigma 2s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_z^2,$

$\pi^* 2P_x^1 = \pi^* 2P_y^1$

$O_2 : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2P_z^2; \pi 2P_x^2 = \pi 2P_y^2, \pi^* 2P_x^1 = \pi^* 2P_y^1$

$O_2^- : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2P_z^2, \pi 2P_x^2 = \pi 2P_y^2, \pi^* 2P_x^2 = \pi^* 2P_y^2,$

$O_2^+ : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma^2 P_2^2, \pi 2P_x^2 = \pi 2P_y^2, \pi^* 2P_x^2 = \pi^* 2P_y^0$

$O_2^{2-} : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, 6^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2P_y^2, \pi^* 2P_x^2 = \pi^* 2P_y^2$

N_2 and O_2^{2-} all electrons are paired. Both of them will show diamagnetism.

Q. 22 The enthalpy of combustion of propane, graphite and dihydrogen at 298 K are $-2220.0\ \text{kJ mol}^{-1}, -393.5\ \text{kJ mol}^{-1}$ and $-285.8\ \text{kJ mol}^{-1}$ respectively. The magnitude of enthalpy of formation of propane (C_3H_8) is _____kJ mol^{-1}. (Nearest integer)

Correct Answer:
103.7

Solution:
Given the combustion enthalpy -

$$(1)\ C_3H_8(g) + 5\ O_2 \longrightarrow 3CO_2 + 4H_2O, \Delta H_1 = -2220.0\ \text{kJ/ mol}$$

$$(2)\ C_{(s)} + O_2 \longrightarrow CO_2 \quad \Delta_c H_2 = -393.5\ \text{kJ/mol}$$

$$(3)\ H_2(\ g) + \frac{1}{2}O_2 \rightarrow H_2O \quad \Delta_c H_3 = -285.8\ \text{kJ/mol}$$

After doing $3 \times (2) + 4 \times (3) - - - - - (1)$

$$(4)\ 3C_{(s)} + 4H_{2(g)} \rightarrow C_3H_{8(g)} \qquad \Delta H = 103.7\ \text{kJ/mol}$$

(4) is the enthalpy of formation propane (C_3H_6)

Answer = 103.7

Q. 23 The pressure of a moist gas at $27^\circ C$ is $4\ \text{atm}$. The volume of the container is doubled at the same temperature. The new pressure of the moist gas is_____$\times 10^{-1}\ \text{atm}$. (Nearest integer)
(Given : The vapour pressure of water at $27^\circ C$ is $0.4\ \text{atm}$.)

Correct Answer:
22

Solution:
We know,

$$P_{\text{moist}} = P_{\text{gas}} + P_{\text{water}}$$

Given, vapour pressure of water = 0.4 atm.

If volume of the container is doubled at same temperature.

Vapour pressure of water will be same but pressure of gas will change.

$$PV = nRT,$$

$$PV = C \qquad (\because T = C, n = c, R = C)$$

$$P_1V_1 = P_2V_2$$

$$P_2 = \frac{V_1}{V_2}P_1 = \frac{V_1}{2V_1}P_1 = \frac{P_1}{2}$$

$$P_2 = \frac{3.6}{2} = 1.8 \text{ atm } (\because P_{gas} = 4 - 0 - 4 = 3.6)$$

New pressure of the moist gas will be

(New) $P_{moist} = P_{gas} + P_{water} = 1.8 + 0.4$

$$P_2(\text{moist}) = 2.2 \text{ atm}$$

$$P_2(\text{moist}) = 22 \times 10^{-1} \text{ atm}$$

Answer = 22

Q. 24 The cell potential for $Zn\left|Zn^{2+}(aq)\right|\left|Sn^{x+}\right|Sn$ is 0.801 V at 298 K. The reaction quotient for the above reaction is 10^{-2}. The number of electrons involved in the given electrochemical cell reaction is_____.

$$\left(\text{Given: } E^{O}_{Zn^{2+}|Zn} = -0.763 \text{ V}, E^{o}_{Sn^{x+}|Sn} = +0.008 \text{ V and } \frac{2.303RT}{F} = 0.06 \text{ V}\right)$$

Correct Answer:
4

Solution:
We know,

$$E_{eu} = E^{o} - \frac{2.303\,RT}{nF}\log Q.....(1)$$

and $E^{o} = E^{o}_{R} - E^{o}_{L} = 0.008 - (0.763)$

$$E^{o} = 0.771 \text{ V}$$

Given,

$$E_{cell} = 0.801 \text{ V}$$

So, from .....(1)

$$0.801 = 0.771 - \frac{2.303\ RT}{nxF}\log(10^{-2})$$

$$0.801 = 0.771 - \frac{0.06}{n}(-2)$$

$$n = \frac{0.06 \times 2}{0.03} = 4$$

Answer is (4)

Q. 25 The half life for the decomposition of gaseous compound A is 240 s when the gaseous pressure was 500 Torr initially. When the pressure was 250 Torr, the half life was found to be 4.0 min. The order of the reaction is_____ . (Nearest integer)

Correct Answer:
1

Solution:
Given, $t_{1/2} = 240\ sec = 4\ min$

$P_{initial} = 500\ torr$ and $P_2 = 250\ torr$

t_{v_2} at $P_2 = 4\ min$

Now, $t_{1/2}$ at $P_i = t_{1/2}$ at P_2

So, $t_{1/2}$ is not dependent on pressure.

Hence, it will be first order reaction because

Answer = 1

Q. 26 Consider the following metal complexes :

$[Co\,(NH_3)_6]^{3+}$
$[CoCl\,(NH_3)_5]^{2+}$
$[Co(CN)_6]^{3-}$
$[Co\,(NH_3)_5\,(H_2O)]^{3+}$

The spin-only magnetic moment value of the complex that absorbes light with shortest wavelength is_____ B.M. (Nearest integer)

Correct Answer:
0

Solution:
The shortest wavelength mean largest Δ_o (splitting) value.

them $CN^{\ominus}$ is SFL which has maximum CFSE.

So, $[Co(CN)_6]^{3-}$ will be d^2sp^3 and $3d^6 4s^o$

Due to SFL all electron will be paired.

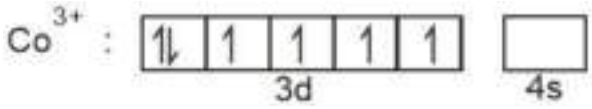
So, $\mu = 0$ due to $n = 0$

Answer = 0

Q. 27 Among Co^{3+}, Ti^{2+}, V^{2+} and Cr^{2+} ions, one if used as a reagent cannot liberate H_2 from dilute mineral acid solution, its spin-only magnetic moment in gaseous state is_____ B.M. (Nearest integer)

Correct Answer:
5

Solution:
Among given ions, CO^{3+} cannot liberate H_2



$n = 4(\text{unpaired } e^{\ominus})$

So, spin only magnetic moment $= \sqrt{n(n+2)}$ BM

$$\mu = \sqrt{4 \times 6} = 4.92 \text{ BM}$$

Answer = 5

Q. 28 While estimating the nitrogen present in an organic compound by Kjeldahl's method, the ammonia evolved from 0.25 g of the compound neutralized 2.5 mL of $2\ M\ H_2SO_4$. The percentage of nitrogen present in organic compound is_____.

Correct Answer:
56

Solution:
NH_3 is neutralized by 2.5 ml of 2M H_2SO_4

mole of $NH_3 \times$ n-factor of $NH_2 = 2.5 \times 2 \times 2$ meq

$$n_{NH_3} \times 1 = 10 \text{ meq}$$

$$n_{NH_3} = 10\text{meq} = 10 \times 10^{-3} \text{ mole}$$

$$\text{mass of nitrogen } = 14 \times 10 \times 10^{-3} = 0.14 \text{ g}$$

$[\text{ mass } = \text{mole} \times \text{molar mass}]$

$$\% \text{ N} = \frac{\text{mass of nitrogen in compound}}{\text{mass of compound}} \times 100 = \frac{0.14}{0.25} \times 100 = 56\%$$

Answer = 56

Q. 29 The number of sp^3 hybridised carbons in an acyclic neutral compound with molecular formula $C_4H_5\ N$ is _____.

"Only $-C \equiv N$ contain compound"

Correct Answer:
1

Solution:

Degree of unsaturation $= \frac{2c + 2 + N - H - X}{2}$

$$DOU = \frac{2 \times 4 + 2 + 1 - 5 - 0}{2} = 3$$

Possible structures -

1) $H_3C - CH = CH - C \equiv N$ (SP^3)

2) $CH_2 = CH - CH_2 - C \equiv N$ (SP^3)

3) $CH_2 = CH - CH = C = NH$

4) $CH_2 = C = CH - CH = NH$

5) $CH_3 - CH = C = C = NH$ (SP^3)

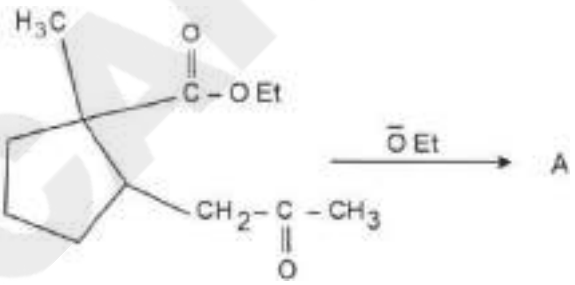
6) $CH_3 - C(=CH_2) - C \equiv N$ (SP^3)

So, in 1,2,5,6 all have 1 Sp^3 carbon and (3), (4) have zero.

Possible Sp^2 carbon is 0 or 1 but which has $-C \equiv N$, has at least 1 Sp^3 carbon.

Answer = 1

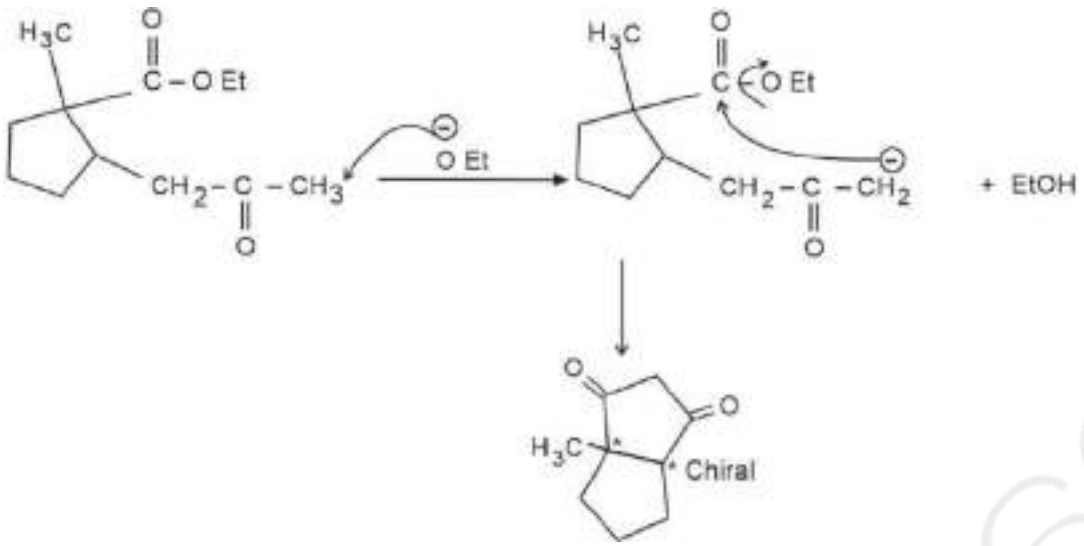
Q. 30 In the given reaction



The number of chiral carbon/s in product A is _____.

Correct Answer:
2

Solution:



Answer = 1

Maths

Q. 1 The total number of functions, $f : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4, 5, 6\}$ such that $f(1) + f(2) = f(3)$, is equal to :

Option 1:
60

Option 2:
90

Option 3:
108

Option 4:
126

Correct Answer:
90

Solution:
$A = \{1, 2, 3, 4\}$

$B = \{1, 2, 3, 4, 5, 6\}$

Here $f(3)$Can be $2, 3, 4, 5, 6$

$f(3) = 2, (f(1), f(2)) \rightarrow (1, 1) \rightarrow (1, 2), (2, 1)$

$f(3) = 3, (f(1), f(2)) \rightarrow$ Cases

$\rightarrow 2 \times 6 = 12$ Cases

$f(3) = 4, (f(1), f(2)) \rightarrow (1,3), (3,1), (2,2)$

$\rightarrow 3 \times 6 = 18$ cases

$f(3) = 5, (f(1), f(2)) \rightarrow (1,4), (4,1), (2,3), (3,2)$

$\rightarrow 4 \times 6 = 24$ cases

$f(3) = 6, (f(1), f(2)) \rightarrow (1,5), (5,1), (2,4), (4,2), (3,3)$

$\rightarrow 5 \times 5 = 30$ cases

Total number of cases $= 6 + 12 + 18 + 24 + 30 = 90$

Hence correct option is 2

Q. 2 If $\alpha, \beta, \gamma, \delta$ are the roots of the equation $x^4 + x^3 + x^2 + x + 1 = 0$, then $\alpha^{2021} + \beta^{2021} + \gamma^{2021} + \delta^{2021}$ is equal to :

Option 1:
-4

Option 2:
-1

Option 3:
1

Option 4:
4

Correct Answer:
-1

Solution:

Let $x^4 + x^3 + x^2 + x - 1 = 0$

$\frac{x^5 - 1}{x - 1} = 0 \quad (x \neq 1)$

$x^5 - 1 = 0$

$$x = (1)^{1/5}$$

$$x = (\cos 2r\pi + i \sin 2r\pi)^{1/5}$$

$$x = \cos \frac{2r\pi}{5} + i \sin \frac{2r\pi}{5}$$

Putting $r = 0, 1, 2, 3, 4$ the five roots are given by

$$\cos 0 + i \sin 0 = 1, \quad \alpha = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$\beta = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}, \gamma = \cos \frac{6\pi}{5} + i \sin \frac{6\pi}{5}$$

$$\delta = \cos \frac{\theta\pi}{5} + i \sin \frac{8\pi}{5}$$

$$\text{then } \alpha^{2021} + \beta^{2021} + y^{2021} + \delta^{2021} = \alpha + \beta + \gamma + \delta$$

$$= -1$$

using De-morves's theroem

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

hence correct option is 2

Q. 3 For $n \in \mathbf{N}$, let $S_n = \left\{z \in C : |z - 3 + 2i| = \frac{n}{4}\right\}$ and $T_n = \left\{z \in C : |z - 2 + 3i| = \frac{1}{n}\right\}$. Then the number of elements in the set $\{n \in \mathbf{N} : S_n \cap T_n = \phi\}$ is :

Option 1:
0

Option 2:
2

Option 3:
3

Option 4:

4

Correct Answer:

2

Solution:

$s_n : |z-(3-2i)| = \frac{n}{4}$ is a circle centre $c_1(3,-2)$ and radius $\left(\frac{n}{4}\right)$.

$T_n : |z-(2-3i)| = \frac{1}{n}$ is a circle centre $c_2(2,-3)$ and radius $\left(\frac{1}{n}\right)$

Here $S_n \cap T_n = \phi$

Both circles do not intersect each other

$$\text{case-1} \ : C_1C_2 > \frac{n}{4}+\frac{1}{n}$$

$$\sqrt{2} > \frac{n}{4}+\frac{1}{n}$$

$$\text{then } n = 1,2,3,4$$

$$\text{Case}-2 : C_1C_2 < \left|\frac{n}{4}-\frac{1}{n}\right|$$

$$\Rightarrow \sqrt{2} < \left|\frac{n^2-4}{4n}\right|.$$

$\Rightarrow n$ has infinite solutions for $n \in N$

Hence correct option is 2

Q. 4 The number of $\theta \in (0, 4\pi)$ for which the system of linear equations

$$3(\sin 3\theta)x - y + z = 2$$

$$3(\cos 2\theta)x + 4y + 3z = 3$$

$$6x + 7y + 7z = 9$$

has no solution, is :

Option 1:
6

Option 2:
7

Option 3:
8

Option 4:
9

Correct Answer:
7

Solution:
The system of equation has no solution

$$D = \begin{vmatrix} 3\sin 3\theta & -1 & 1 \\ 3\cos 2\theta & 4 & 3 \\ 6 & 7 & 7 \end{vmatrix} = 0$$

$$21\sin 3\theta + 42\cos 2\theta - 42 = 0$$

$$\sin 3\theta + 2\cos 2\theta - 2 = 0$$

Number of solution is 7 in $(0, 4\pi)$

Hence correct option is 2

Q. 5 If $\lim\limits_{n\to\infty}\left(\sqrt{n^2-n-1}+n\alpha+\beta\right)=0$ then $8(\alpha+\beta)$ is equal to :

Option 1:
4

Option 2:
-8

Option 3:
-4

Option 4:
8

Correct Answer:

-4

Solution:

$$\lim_{n\to\infty} n\left(1-\frac{n+1}{n^2}\right)^{1/2}+\alpha n+\beta=0$$

$$\lim_{n\to\infty} n\left\{1-\frac{1}{2}\left(\frac{n+1}{n^2}\right)+\frac{\left(\frac{1}{2}\right)\left(\frac{-1}{2}\right)}{2!}\left(\frac{n+1}{n^2}\right)^2+\cdots\right\}+\alpha n+\beta=0$$

$$\lim_{n\to\infty} n-\frac{1}{2}+\frac{1}{n}+\cdots+n\alpha+\beta=0$$

$$\alpha=-1,\quad \beta=\frac{1}{2}$$

$$\beta(\alpha+\beta)=-4$$

Hence correct option is 3

Q. 6 If the absolute maximum value of the function $f(x)=\left(x^2-2x+7\right)e^{\left(4x^3-12x^2-180x+31\right)}$ in the interval $[-3,0]$ is, $f(\alpha)$ then :

Option 1:

$\alpha=0$

Option 2:

$\alpha=-3$

Option 3:

$\alpha\in(-1,0)$

Option 4:

$\alpha\in(-3,-1]$

Correct Answer:

$\alpha=-3$

Solution:

$$f'(x)=e^{\left(4x^3-12x^2-180x+31\right)}\cdot\left(12\left(x^2-2x+7\right)(x+3)(x-5)+2(x-1)\right)$$

for $x\in[-3,0]$

$\Rightarrow \quad f'(x) < 0$

$f(x)$ is decreasing function on $[-3, 0]$

The absolute maximum value of the function $f(x)$ is at $x = -3$

$\Rightarrow \alpha = -3$

Hence correct option is 2

Q. 7 The curve $y(x) = ax^3 + bx^2 + cx + 5$ touches the x-axis at the point $P(-2, 0)$ and cuts the y-axis at the point Q where y' is equal to 3. Then the local maximum value of $y(x)$ is :

Option 1:
$\frac{27}{4}$

Option 2:
$\frac{29}{4}$

Option 3:
$\frac{37}{4}$

Option 4:
$\frac{9}{2}$

Correct Answer:
$\frac{27}{4}$

Solution:
$y(x) = ax^3 + bx^2 + cx + 5$ is passing through $(-2, 0)$

then $8a - 4b + 2c = 5$ ...(1)

$y'(x) = 3ax^2 + 2bx + c$ touches x-axis at $(-3, 0)$

$12a - 4b + c = 0$ ...(2)

again, for $x = 0, y'(x) = 3$

$c = 3$ ...(3)

Solving eq.(1), (2)&(3) $a = \frac{-1}{2}, b = -\frac{3}{4}$

$y'(x) = -\frac{3}{2}x^2 - \frac{3}{2}x + 3$

$y(x)$has local maxima at $x = 1$

$y(1) = \frac{27}{4}$

Hence correct option is 1

Q. 8 The area of the region by $A = \{(x, y) : x^2 \leq y \leq \min\{x + 2, 4 - 3x\}\}$ is :

Option 1:
$\frac{31}{8}$

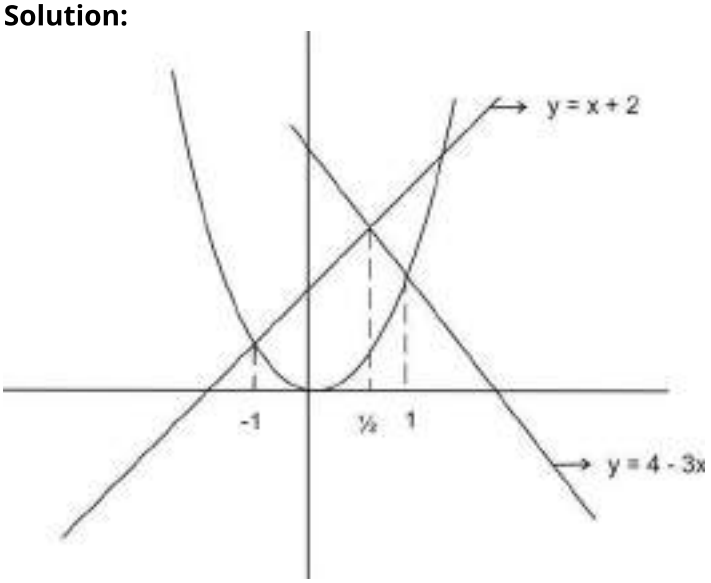
Option 2:
$\frac{17}{6}$

Option 3:
$\frac{19}{6}$

Option 4:
$\frac{27}{8}$

Correct Answer:
$\frac{17}{6}$

Solution:



$$A=\int_{-1}^{\frac{1}{2}}\left(x+2-x^{2}\right)dx+\int_{\frac{1}{2}}^{1}\left(4-3x-x^{2}\right)dx=\frac{17}{6}$$

Hence correct option is 2

Q. 9 For any real number x, let $[x]$ denote the largest integer less than equal to x. Let f be a real valued function defined on the interval $[-10, 10]$ by

$$f(x)=\begin{cases} x-[x], & \text{if } [x] \text{ is odd} \\ 1+[x]-x, & \text{if } [x] \text{ is even} \end{cases}.$$

Then the value of $\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos \pi x dx$ is :

Option 1:
4

Option 2:
2

Option 3:
1

Option 4:
0

Correct Answer:
4

Solution:

$f(x)$ is periodic function where period is 2

$$\frac{\pi^2}{10}\int_{-10}^{10} f(x)\cos \pi x dx = \frac{\pi^2}{10}\times 10\int_{0}^{2} f(x)\cos \pi x dx$$

$$= \pi^2\left(\int_0^1 (1-x)\cos \pi x dx + \int_1^2 (x-1)\cos \pi x dx\right)$$

using by parts

$$= \pi^2 \times \frac{4}{\pi^2} = 4$$

Hence correct option is 1

Q. 10 The slope of the tangent to a curve $C : y = y(x)$ at ay point (x, y) on it is $\frac{2e^{2x} - 6e^{-x} + 9}{2 + 9e^{-2x}}$. If C passes through the points $\left(0, \frac{1}{2} + \frac{\pi}{2\sqrt{2}}\right)$ and $\left(\alpha, \frac{1}{2}e^{2\alpha}\right)$, then e^{α} is equal to :

Option 1:

$\frac{3+\sqrt{2}}{3-\sqrt{2}}$

Option 2:

$\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

Option 3:

$\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$

Option 4:

$\frac{\sqrt{2}+1}{\sqrt{2}-1}$

Correct Answer:

$\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

Solution:

$$\frac{dy}{dx}=\frac{2e^{2x}-6x^{-x}+9}{2+9e^{-2x}}$$

$$\frac{dy}{dx}=e^{2x}-\frac{6e^{x}}{2e^{2x}+9}$$

$$\int dy=\int\cdot\left(e^{2x}-\frac{6e^{x}}{2e^{2x}+9}\right)dx$$

$$y=\frac{e^{2x}}{2}-\tan^{-1}\left(\frac{\sqrt{2}e^{x}}{3}\right)+c$$

If c passes through the point $\left(0,\frac{1}{2}+\frac{\pi}{2\sqrt{2}}\right)$

$$c=\frac{-\pi}{4}-\tan^{-1}\frac{\sqrt{2}}{3}$$

Again c passes through the point $\left(\alpha,\frac{1}{2}e^{2\alpha}\right)$

then $e^{\alpha}=\frac{3}{\sqrt{2}}\left(\frac{3+\sqrt{2}}{3-\sqrt{2}}\right)$

Hence correct option is 2

Q. 11 Let $\beta=\lim_{x\to0}\frac{\alpha x-(e^{2x}-1)}{\alpha x(e^{3x}-1)}$ for some $\alpha\in\mathbb{R}$. Then the value of $\alpha+\beta$ is:

Option 1:

$\frac{14}{5}$

Option 2:

$\frac{3}{2}$

Option 3:

$\frac{5}{2}$

Option 4:

$\frac{7}{2}$

Correct Answer:

$\frac{5}{2}$

Solution:

$$\lim_{x\to 0}\frac{\alpha x-(e^{3x}-1)}{\alpha x\,(e^{3x}-1)}$$

$$=\lim_{x\to 0}\frac{\alpha x-\left(1+3x+\frac{9x^2}{2}+\cdots-1\right)}{\alpha x\quad\left(\frac{e^{3x}-1}{3x}\cdot 3x\right.}$$

$$=\lim_{x\to 0}\frac{(\alpha-3)x-\frac{9x^2}{2}+\cdots}{3\alpha x^2}$$

For this lmit to exist $\alpha-3=0$

$$\Rightarrow \alpha=3 \text{ and } -\frac{9}{2\cdot(3\alpha)}=\beta$$

$$\Rightarrow \beta=\frac{-9}{6\cdot 3}=-\frac{1}{2}.$$

$$\therefore \alpha+\beta=3-\frac{1}{2}=\frac{5}{2}$$

Hence correct option is 3

Q. 12 A line, with the slope greater tha one, passes through the point $A(4,3)$ and inersects the line $x-y-2=0$ at the point B. If the length of the line segment AB is $\frac{\sqrt{29}}{3}$, then B also lies on the line :

Option 1:

$2x+y=9$

Option 2:

$3x-2y=7$

Option 3:

$x+2y=6$

Option 4:
$2x - 3y = 3$

Correct Answer:
$x + 2y = 6$

Solution:
Let $B(x_1, x_1 - 2)$

$$\sqrt{(x_1 - 4)^2 + (x_1 - 2 - 3)^2} = \frac{\sqrt{29}}{3}$$

Squaring on both sides

$$18x_1^2 - 162x_1 + 340 = 0$$

$$x_1 = \frac{51}{9} \quad \text{or} \quad x_1 = \frac{10}{3}$$

$$y_1 = \frac{33}{9} \quad \text{or} \quad y_1 = \frac{4}{3}$$

option 3 will satisfy $\left(\frac{10}{3}, \frac{4}{3}\right)$

Hence correct option is 3

Q. 13 Let the locus of the cetre $(\alpha, \beta), \beta > 0$, of the circle which touches the circle $x^2 + (y - 1)^2 = 1$ externally and also touches the x-axis be L. Then the area bounded by L and the line $y = 4$ is :

Option 1:
$\frac{32\sqrt{2}}{3}$

Option 2:
$\frac{40\sqrt{2}}{3}$

Option 3:
$\frac{64}{3}$

Option 4:

$\frac{32}{3}$

Correct Answer:

$\frac{64}{3}$

Solution:

$(\alpha-0)^2+(\beta-1)^2=(\beta+1)^2$

$\alpha^2=4\beta$

$x^2=4y$

$A=2\int_0^4\left(4-\frac{x^2}{4}\right)dx$

$=2\left[4x-\frac{x^3}{12}\right]_0^4$

$=\frac{64}{3}$

Hence correct option is 3

Q. 14 Let P be the plane containing the traight line $\frac{x-3}{9}=\frac{y+4}{-1}=\frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2}=\frac{y}{3}=\frac{z}{5}$ and $\frac{x}{3}=\frac{y}{7}=\frac{z}{8}$. If d is thed distance of P from the point $(2,-5,11)$, then d^2 is equal to :

Option 1:

$\frac{147}{2}$

Option 2:

96

Option 3:

$\frac{32}{3}$

Option 4:

54

Correct Answer:

$\frac{32}{3}$

Solution:

$a(x-3)+b(y+4)+c(z-7)=0$

$p: 9a-b-5c=0$

$-11a-b+5c=0$

After solving DR's $\alpha(1,-1,2)$

Equation of plane

$x-y+2z=21$

$d=\frac{8}{\sqrt{6}}$

$d^2=\frac{32}{3}$

Hence correct option is 3

Q. 15 Let ABC be triangle such that $\overrightarrow{BC}=\vec{a}, \overrightarrow{CA}=\vec{b}, \overrightarrow{AB}=\vec{c}, |\vec{a}|=6\sqrt{2}, |\vec{b}|=2\sqrt{3}$ and $\vec{b}\cdot\vec{c}=12$.

Consider the statements :

$(S1): |(\vec{a}\times\vec{b})+(\vec{c}\times\vec{b})|-|\vec{c}|=6(2\sqrt{2}-1)$

$(S2): \angle ACB=\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Then

Option 1:

both (S1) and (S2) are true

Option 2:

only (S1) is true

Option 3:

only (S2) is true

Option 4:
both (S1) and (S2) are false

Correct Answer:
only (S2) is true

Solution:

$\vec{a}+\vec{b}+\vec{c}=0$

$\vec{b}+\vec{c}=-\vec{a}$

$|\vec{b}|^2+|\vec{c}|^2+2\vec{b}\cdot\vec{c}=|\vec{a}|^2$

$|\vec{c}|^2=36\Rightarrow|\vec{c}|=6$

S1: $|\vec{a}\times\vec{b}+\vec{c}\times\vec{b}|-|\vec{c}|$

$|\ (\vec{a}+\vec{c}|\times\vec{b}|-|\vec{c}|$

$|-\vec{b}\times\vec{b}|-|\vec{c}|$

$0-6=-6$

$S2:\vec{a}+\vec{b}+\vec{c}=0$

$\vec{b}+\vec{c}=-\vec{a}$

$|\vec{a}|^2+|\vec{b}|^2-2|\vec{a}||\vec{b}|\cos(\angle ACB)=|\vec{c}|^2$

$\cos\ |\ \angle ACB)=\sqrt{\frac{2}{3}}$

hence correct option is 3

Q. 16 If the sum and the product of mean and variance of a binomial distribution are 24 and 128 respectively, the the probability of one or two successes is :

Option 1:
$\frac{33}{2^{32}}$

Option 2:
$\frac{33}{2^{29}}$

Option 3:

$\frac{33}{2^{28}}$

Option 4:

$\frac{33}{2^{27}}$

Correct Answer:

$\frac{33}{2^{28}}$

Solution:

$np + npq = 24$ .....(1)

$np \cdot npq = 128$ .....(2)

Solving (1) and (2)

we get $p = \frac{1}{2}, q = \frac{1}{2}, n = 32$

Now,

$P(x = 1) + P(x = 2)$

$= {}^{32}c_1 pq^{31} + {}^{32}c_2 p^2 q^{30}$

$= \frac{33}{2^{28}}$

Hence correct option is 3

Q. 17 If the numbers appeared on the two throwsof a fair six faced die are α and β, then the probability that $x^2 + \alpha x + \beta > 0$, for all $x \in \mathbf{R}$ is :

Option 1:

$\frac{17}{36}$

Option 2:

$\frac{4}{9}$

Option 3:

$\frac{1}{2}$

Option 4:

$\frac{19}{36}$

Correct Answer:

$\frac{17}{36}$

Solution:

$x^2 + \alpha x + \beta > 0, \forall x \in R$

$D = \alpha^2 - 4\beta < 0$

$\alpha^2 < 4\beta$

Total Cases $= 6 \times 6 = 36$

fav. cases $= \beta = 1, \alpha = 1$

$\beta = 2, \alpha = 1, 2$

$\beta = 3, \alpha = 1, 2, 3$

$\beta = 4, \alpha = 1, 2, 3$

$\beta = 5, \alpha = 1, 2, 3, 4$

$\beta = 6, \alpha = 1, 2, 3, 4$

Total Favourite cases $= 17$

$P(x) = \left(\frac{17}{36}\right)$

Hence correct option is 1

Q. 18 The number of solutions of $|\cos x| = \sin x$, such that $-4\pi \leq x \leq 4\pi$ is :

Option 1:

4

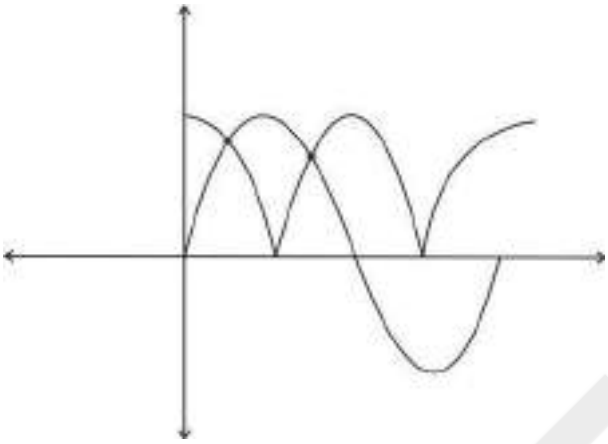
Option 2:
6

Option 3:
8

Option 4:
12

Correct Answer:
8

Solution:



2 solutions in $(0, 2\pi)$

So 8 Solutions in $[-4\pi, 4\pi]$

Hence correct option is 3

Q. 19 A tower PQ stands on a horizontal ground with base Q on the ground. The point Rdivides the tower in two parts such that $\mathrm{QR} = 15\mathrm{m}$. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is :

Option 1:
$5(2\sqrt{3}+3)\mathrm{m}$

Option 2:
$5(\sqrt{3}+3)\mathrm{m}$

Option 3:
$10(\sqrt{3}+1)\mathrm{m}$

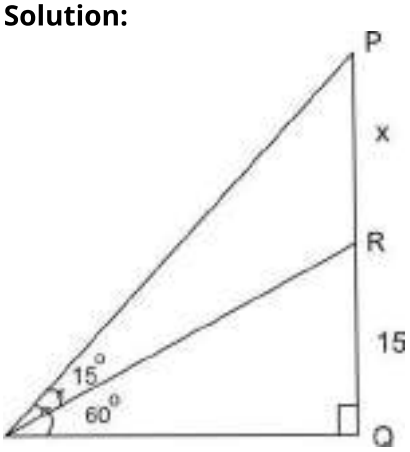
Option 4:

$10(2\sqrt{3}+1)\text{m}$

Correct Answer:

$5(2\sqrt{3}+3)\text{m}$

Solution:



$\frac{15}{\text{AQ}} = \tan 60^\circ$ .....(1)

$\frac{15+\text{x}}{\text{AQ}} = \tan 75^\circ$ .....(2)

$\frac{(1)}{(2)} \Rightarrow \text{x}^1 = 10\sqrt{3}$

$\text{so, PQ} = 5(2\sqrt{3}+3)\text{m}$

Hence correct option is 1

Q. 20 Which of the following statements is a tautology ?

Option 1:

$((\sim \text{p}) \vee \text{q}) \Rightarrow \text{p}$

Option 2:

$\text{p} \Rightarrow ((\sim \text{p}) \vee \text{q})$

Option 3:

$((\sim \text{p}) \vee \text{q}) \Rightarrow \text{q}$

Option 4:

$\text{q} \Rightarrow ((\sim \text{p}) \vee \text{q})$

Correct Answer:

$q \Rightarrow ((\sim p) \vee q)$

Solution:

p	q	$\sim p$	$\sim q$	$\sim p \vee q$
T	T	F	F	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

Options

1	2	3	4
T	T	T	T
T	F	T	T
F	T	T	T
F	T	F	T

Hence correct option is 4

Q. 21

Let $A = \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$ and $B = A - I$. If $\omega = \frac{\sqrt{3}i - 1}{2}$, then the numberof elements in the set $\{n \in \{1, 2, \ldots, 100\} : A^n + (\omega B)^n = A + B\}$ is equal to _____

Correct Answer:

17

Solution:

$$A = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \Rightarrow A^2 = A \Rightarrow A^n = A$$

$$\forall n \in \{1, 2, \ldots, 100\}$$

$$\text{Now, } B = A - I = \begin{bmatrix} 1 & -1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{bmatrix}$$

$$B^2 = -B$$

$$\Rightarrow B^3 = -B^2 = B$$

$$\Rightarrow B^5 = B$$

$\Rightarrow B^{99} = B$

Also, $w^{3k} = 1$

so, n = common of $\{1, 3, 5, \ldots .99\}$ and $\{3, 6, 9, \ldots 99\} = 17$

Hence correct answer is 17

Q. 22 The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in English dictionary . Then the serial number of the word 'MANKIND' is _____.

Correct Answer:
1492

Solution:

M	A	N	K	I	N	D

$$\left(\frac{4 \times 6!}{2!}\right) + (5! \times 0) + \left(\frac{4! \times 3}{2!}\right) + (3! \times 2) + (2! \times 1) + (1! \times 1) + (0! \times 0) + 1 = 1492$$

Hence answer is 1492

Q. 23 If the maximum value of the term independent of t in the expansion of $\left(t^2 x^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}$, $x \geqslant 0$ is K, then $8K$ is equal to _____.

Correct Answer:
6006

Solution:

$$\left(t^2 x^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t}\right)^{15}$$

$$T_{r+1} = {}^{15}C_r \left(t^2 x^{\frac{1}{5}}\right)^{15-r} \cdot \frac{(1-x)^{\frac{r}{10}}}{t^r}$$

for independent of t,

$30 - 2r - r = 0$

$\Rightarrow r = 10$

So maximum value of ${}^{15}c_{10}x(1-x)$ will be at $x = \frac{1}{2}$

i.e. 6006

Hence answer is 6006

Q. 24 Let A, B be two non-zero real numbers. If p and r are the roots of the equation $x^2 - 8ax + 2a = 0$ and q and s are the roots of the equation $x^2 + 12\,bx + 6\,b = 0$, such that $\frac{1}{p}, \frac{1}{q}, \frac{1}{r}, \frac{1}{s}$ are in A.P, then $a^{-1} - b^{-1}$ is equal to _____.

Correct Answer:
38

Solution:

$x^2 - 8ax + 2a = 0$ $\qquad$ $x^2 + 12bx + 6b = 0$

$p + r = 8a$ $\qquad$ $q + s = +2b$

$pr = 2a$ $\qquad$ $qs = 6b$

$\frac{1}{p} + \frac{1}{r} = 4$ $\qquad$ $\frac{1}{q} + \frac{1}{s} = -2$

$\frac{2}{q} = 4$ $\qquad$ $\frac{2}{r} = -2$

$q = \frac{1}{2}$ $\qquad$ $r = -1$

$p = \frac{1}{5}$ $\qquad$ $s = \frac{-1}{4}$

Now, $\frac{1}{a} - \frac{1}{b} = \frac{2}{pr} - \frac{6}{qs} = 38$

Hence answer is 38

Q. 25 Let $a_1 = b_1 = 1, a_n = a_{n-1} + 2$ and $b_n = a_n + b_{n-1}$ for every natural number $n \geq 2$. Then $\sum_{n=1}^{15} a_n \cdot b_n$ is equal to _____.

Correct Answer:
27560

Solution:

$\Rightarrow \quad a_1 = b_1 = 1$

$a_2 = a_1 + 2 = 3$

$a_3 = a_2 + 2 = 5$

$a_4 = a_2 + 2 = 7$

$\Rightarrow a_n = 2n - 1$

$b_2 = a_1 + b_1 = 4$

$b_3 = a_3 + b_2 = 9$

$b_4 = a_4 + b_3 = 16$

$b_n = n^2$

$$\sum_{n=1}^{15}(2n-1)n^2 = \sum_{n=1}^{15}\left(2n^3 - n^2\right)$$

$$= 2\frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{6}$$

Put $n = 15$

$$= \frac{2 \times 225 \times 16 \times 16}{4} - \frac{15 \times 16 \times 31}{6}$$

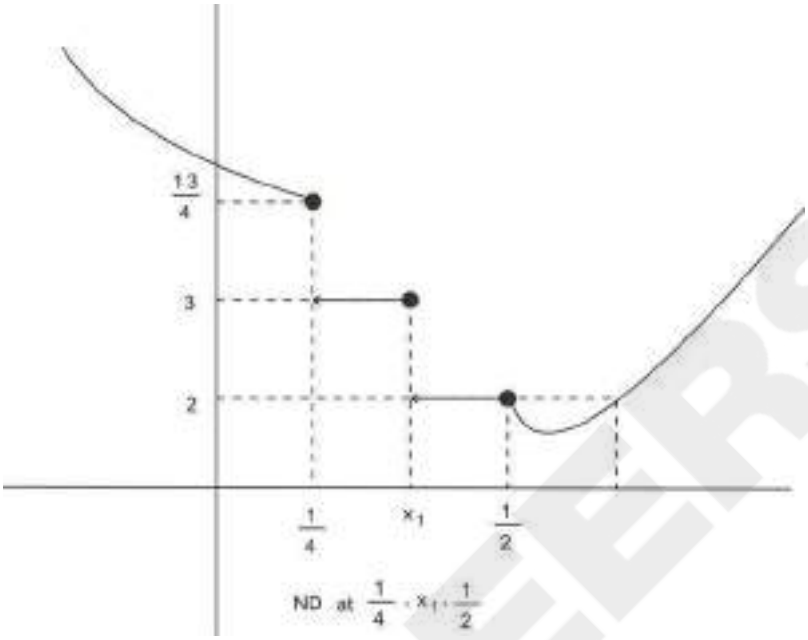
$= 27560$

Hence answer is 27560

Q. 26 Let $f(x)=\begin{cases}\left[4x^2-8x+5\right], & \text{if } 8x^2-6x+1 \geqslant 0\\ \left[4x^2-8x+5\right], & \text{if } 8x^2-6x+1<0\end{cases}$, where $[\alpha]$ denotes the greatest integer less than or equal to α. Then the numbers of points in $\mathbf{R}$ where f is not differentialble is _____.

Correct Answer:
3

Solution:



Hence answer is 3

Q. 27 If $\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}}{n^{k+1}}[(nk+1)+(nk+2)+\ldots+(nk+n)]$
$=33\cdot\lim\limits_{n\to\infty}\frac{1}{n^{k+1}}\cdot\left[1^k+2^k+3^k+\ldots+n^k\right]$,

then the integral value of K is equal to _____.

Correct Answer:
5

Solution:
$\lim\limits_{n\to\infty}\frac{(n+1)^{k-1}}{nk+1}[nk\cdot n+1+2+\cdots+n]$

$$= \lim_{n\to\infty} \frac{(n+1)^{k-1}}{n^{k+1}} \cdot \left[n^2 k + \frac{(n(n+1)}{2}\right]$$

$$\lim_{n\to\infty} \frac{(n+1)^{k-1} \cdot n^2 \left(k + \frac{\left(1+\frac{1}{n}\right)}{2}\right)}{nk+1}$$

$$\lim_{n\to\infty} \left(1+\frac{1}{n}\right)\left(k + \frac{\left(1+\frac{1}{n}\right)}{2}\right)$$

$$\Rightarrow \left(k+\frac{1}{2}\right)$$

RHS

$$\Rightarrow \lim_{n\to\infty} \frac{1}{nk+1}\left(1^k + 2^k + \cdots + h^k\right) = \frac{1}{k+1}$$

LHS=RHS

$$\Rightarrow k + \frac{1}{2} = 33 \cdot \frac{1}{k+1}$$

$$\Rightarrow (2k+1)(k+1) = 66$$

$$\Rightarrow (k-5)(2k+13) = 0$$

$$\Rightarrow k = 5 \text{ or } \frac{13}{2}$$

Hence answer is 5

Q. 28 Let the equation of two diameters of a circle $x^2 + y^2 - 2x + 2fy + 1 = 0$ be $2\,p\,x - y = 1$ and $2x + py = 4p$. Then the slope $m \in (0, \infty)$ of the tangent to the hyperbola $3x^2 - y^2 = 3$ passing through the centre of the circle is equal to _____.

Correct Answer:
2

Solution:
$2p + f - 1 = 0$ .....(1)

$2 - pf - 4p = 0$ .....(2)

$2 = p(f + 4)$

$p = \frac{2}{(f+4)}$

$2P = (1 - f)$

$\frac{4}{f+4} = 1 - f$

$f^2 + 3f = 0$

$f = 0 \text{or} -3 f = 0 or -3$

$\text{Hyperbola } 3x^2 - y^2 = 3, \quad x^2 - \frac{y^2}{3} = 1$

$y = mx \pm \sqrt{m^2 - 3}$

$\text{if passes}(1, 0)$

$0 = m \pm \sqrt{m^2 - 3}$

$m \text{ tends} \infty$

$\text{it passes}(1, 3)$

$3 = m \pm \sqrt{m^2 - 3}$

$(3 - m)^2 = m^2 - 3$

$m = 2$

hence answer is 2

Q. 29 The sum of diameters of the circles that touch (i) the parabola $75x^2 = 64(5y - 3)$ at the point $\left(\frac{8}{5}, \frac{6}{5}\right)$ and (ii) the y-axis , is equal to _____.

Correct Answer:
10

Solution:

$$\Rightarrow \quad x^2 = \frac{64 \times 5}{75}\left(4 - \frac{3}{5}\right)$$

eqn of tangent at $\left(\frac{8}{5}, \frac{6}{5}\right)$

$$x \cdot x_1 = \frac{64 \times 5}{75}\left(\frac{y + y_1}{2} - \frac{3}{5}\right)$$

$$x \times \frac{8}{5} = \frac{64}{15}\left(\frac{4 + \frac{6}{5}}{2} - \frac{3}{5}\right)$$

$$3x - 4y = 0$$

it touches y axis so $f^2 = c$

$$x^2 + y^2 + x\left(3\lambda - \frac{16}{5}\right) + y\left(-4\lambda - \frac{12}{5}\right) + 4 = 0$$

$$\frac{\left(4\lambda + \frac{12}{5}\right)^2}{4} = 4$$

$$\lambda = \frac{2}{5} \text{ or } \lambda = \frac{-8}{5}$$

$$\lambda = \frac{2}{5}, r = 1$$

$$\lambda = \frac{-8}{5}, r = 4$$

$$d_1 + d_2 = 10$$

hence answer is 10

Q. 30 The line of shortest distance between the lines $\frac{x-2}{0} = \frac{y-1}{1} = \frac{z}{1}$ and $\frac{x-3}{2} = \frac{y-5}{2} = \frac{z-1}{1}$ makes an angle of $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ with the plane $P : ax - y - z = 0, (a > 0)$. If the image of the point $(1, 1, -5)$ in the plane P is (α, β, γ), then $\alpha + \beta - \gamma$ is equal to _____.

Correct Answer:

-

Solution:

DR's of line of shortest distance

$$\begin{vmatrix} \hat{i}\cdot\hat{j} & \hat{k} & \\ 0 & 1 & 1 \\ 2 & 2 & 1 \end{vmatrix} = -\hat{\mathrm{i}} + 2\hat{\mathrm{J}} - 2\mathrm{k}^{\mathrm{r}}$$

angle between line and plane is $\cos^{-1}\sqrt{\frac{2}{27}} = \alpha$

$$\cos\alpha = \sqrt{\frac{2}{27}}, \quad \sin\alpha = \frac{5}{3\sqrt{3}}$$

DR's normal to plane $(1, -1, -1)$

$$\sin\alpha = \left|\frac{-a-2+2}{\sqrt{4+4+1}\sqrt{a^2+1+1}}\right| = \frac{5}{3\sqrt{3}}$$

$$\sqrt{3}|\mathrm{a}| = 5\sqrt{\mathrm{a}^2+2}$$

$$3\mathrm{a}^2 = 25\mathrm{a}^2 + 50$$

No value of a

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25 July - Evening Shift

Physics

Q. 1 In AM modulation, a signal is modulated on a carrier wave such that maximum and minimum amplitudes are found to be $6\ \mathrm{V}$ and $2\ \mathrm{V}$ respectively. The modulation index is :

Option 1:
100%

Option 2:
80%

Option 3:
60%

Option 4:
50%

Correct Answer:
50%

Solution:

$$\mathrm{V_{max}} = \mathrm{V_c} + \mathrm{V_m} = 6\ \mathrm{V}$$

$$\mathrm{V_{min}} = \mathrm{V_C} - \mathrm{V_m} = 2\ \mathrm{V}$$

$$\mathrm{V_c} = 4\ \mathrm{V}, \mathrm{V_m} = 2\ \mathrm{V}$$

$$\text{Modulation index } = \mu = \frac{\mathrm{V_m}}{\mathrm{V_c}}$$

$$\mu = \frac{1}{2}$$

or

$$\mu = 50\%$$

Hence (4) is correct option.

Q. 2 The electric current in a circular coil of 2 turns produces a magnetic induction B_1 at its centre. The coil is unwound and is rewound into a circular coil of 5 turns and the same current produces a magnetic induction B_2 at its centre. The ratio of $\frac{B_2}{B_1}$ is

Option 1:
$\frac{5}{2}$

Option 2:
$\frac{25}{4}$

Option 3:
$\frac{5}{4}$

Option 4:
$\frac{25}{2}$

Correct Answer:
$\frac{25}{4}$

Solution:
$l = n_1 (2\pi r_1) = n_2 (2\pi r_2)$

$=\text{Constant}$

$$\frac{n_1}{n_2} = \frac{r_2}{r_1} = \frac{2}{5}$$

Magnetic field due to circular coil is

$$B = \frac{\mu_0 n_i}{2r}$$

$$\frac{B_1}{B_2} = \frac{n_1}{n_2} \times \frac{r_2}{r_1} = \frac{4}{25}$$

$$\therefore \frac{B_2}{B_1} = \frac{25}{4}$$

Hence (2) is correct option.

Q. 3 A drop of liquid of density ρ is floating half immersed in a liquid of density σ and surface tension $7.5 \times 10^{-4} \mathrm{Nem}^{-1}$. The radius of drop in cm will be : $\left(\mathrm{g} = 10\ \mathrm{ms}^{-2}\right)$

Option 1:
$\frac{15}{\sqrt{(2\rho - \sigma)}}$

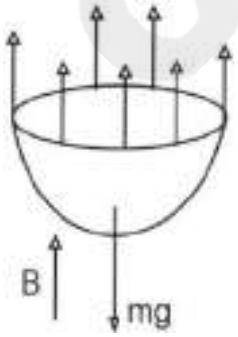
Option 2:
$\frac{15}{\sqrt{(\rho - \sigma)}}$

Option 3:
$\frac{3}{2\sqrt{(\rho - \sigma)}}$

Option 4:
$\frac{3}{20\sqrt{(2\rho - \sigma)}}$

Correct Answer:
$\frac{3}{2\sqrt{(\rho - \sigma)}}$

Solution:



$$\mathrm{B} + \mathrm{T}(2\pi\mathrm{R}) = \mathrm{mg}$$

$$\sigma \times \frac{2\pi R^3}{3} g + (2\pi R) = \varrho \times \frac{2\pi R^3}{3} g$$

$$\frac{\sigma R^2 g}{3} + T = \frac{\varrho R^2 g}{3}$$

$$7.5 \times 10^{-2} = \frac{R^2 g}{3} (\varrho - \sigma)$$

$$R = \frac{3}{2\sqrt{(\rho - \sigma)}}$$

Hence (3) is correct option

Q. 4 Two billiard balls of mass $0.05\ \text{kg}$ each moving in opposite directions with $10\ \text{ms}^{-1}$ collide and rebound with the same speed. If the time duration of contact is $t = 0.005\ \text{s}$, then what is the force exerted on the ball due to each other?

Option 1:
100 N

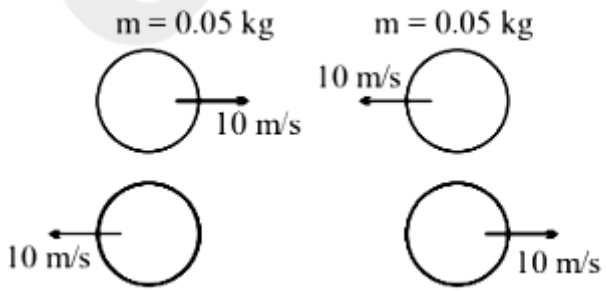
Option 2:
200 N

Option 3:
300 N

Option 4:
400 N

Correct Answer:
200 N

Solution:



$$\text{Force exerted} = \frac{\Delta P}{\Delta t} = \frac{(mv) - (-mv)}{\Delta t}$$

$$= \frac{2mv}{\Delta t}$$

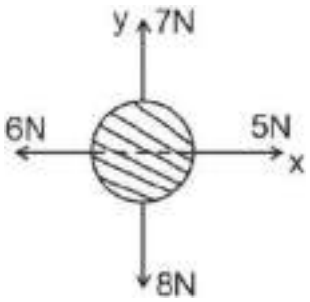
$$F = \frac{2mv}{\Delta t}$$

$$F = \frac{2\,(0.05)\,10}{0.005} = 200\ N$$

The force exerted on the ball due to each other is $200\ N$

Hence (2) is correct option.

Q. 5 For a free body diagram shown in the figure, the four forces are applied in the $'x'$ and $'y'$directions. What additional force must be applied and at what angle with positive $x-$ axis so that the net acceleration of body is zero ?



Option 1:
$\sqrt{2}N, 45^0$

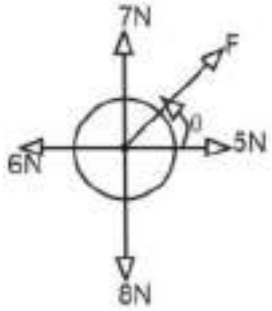
Option 2:
$\sqrt{2}N, 135^\circ$

Option 3:
$\frac{2}{\sqrt{3}}N, 30^\circ$

Option 4:
$2N, 45^\circ$

Correct Answer:
$\sqrt{2}N, 45^0$

Solution:



For net acceleration of body to be zero,

$$\sum \bar{F} = 0$$

$$\text{i.e., } \sum F_X = 0 \text{ and } \sum F_Y = 0$$

$$\sum F_x = (+5) + (-6) + F\cos\theta = 0$$

$$F\cos\theta = 1 \rightarrow (1)$$

$$\sum F_y = +7 + (-8) + F\sin\theta = 0$$

$$F\sin\theta = 1 \rightarrow (2)$$

$$(1) \div (2)$$

$$\tan\theta = 1$$

$$\theta = 45^\circ$$

$$F = \frac{1}{\cos\theta} = \sqrt{2}$$

Hence (1) is correct option

Q. 6 Capacitance of an isolated conducting sphere of radius R_1 becomes n times when it is enclosed by a concentric conducting sphere of radius R_2 connected to earth. The ratio of their radii $\left(\frac{R_2}{R_1}\right)$ is :

Option 1:

$$\frac{n}{n-1}$$

Option 2:

$\frac{2n}{2n+1}$

Option 3:

$\frac{n+1}{n}$

Option 4:

$\frac{2n+1}{n}$

Correct Answer:

$\frac{n}{n-1}$

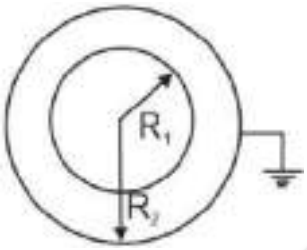
Solution:

Case I

Capacitance of an isolated conducting sphere of radius R_1

$$C_1 = 4\pi\varepsilon_0 R_1 \rightarrow (1)$$

Case II



$$C_2 = \frac{4\pi\varepsilon_0}{(\frac{1}{R_1} - \frac{1}{R_2})} \rightarrow (2)$$

It is given that ,

$$C_2 = nC_1$$

$$\frac{4\pi\varepsilon_0}{\frac{1}{R_1} - \frac{1}{R_2}} = n4\pi\varepsilon_0 R_1$$

$$\frac{R_1 R_2}{R_2 - R_1} = nR_1$$

$$R_2 = nR_2 - nR_1$$

$$nR_1 = (n-1)R_2$$

$$\frac{R_2}{R_1} = \frac{n}{n-1}$$

Hence (1) is correct option

Q. 7 The ratio of wavelengths of proton and deuteron accelerated by potential V_p and V_d is $1 : \sqrt{2}$. Then, the ratio of V_p to V_d will be :

Option 1:
$1 : 1$

Option 2:
$\sqrt{2} : 1$

Option 3:
$2 : 1$

Option 4:
$4 : 1$

Correct Answer:
$4 : 1$

Solution:

$$\lambda_p = \frac{h}{p} = \frac{h}{m_d v_p} = \frac{h}{\sqrt{2 m_p q_p V_p}}$$

$$\lambda_d = \frac{h}{p} = \frac{h}{m_d v_d} = \frac{h_1}{\sqrt{2 m_d q_d V_d}}$$

$$\frac{\lambda_p}{\lambda_d} = \frac{1}{\sqrt{2}} = \frac{\sqrt{m_d q_d V_d}}{\sqrt{m_p q_p V_p}}$$

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{1} \times \frac{\sqrt{V_d}}{\sqrt{v_p}}$$

$$\frac{V_p}{V_d} = \frac{4}{1}$$

Hence (4) is correct option

Q. 8 For an object placed at a distance $2.4\ \text{m}$ from a lens, a sharp focused image is observed on a screen placed at a distance $12\ \text{cm}$ form the lens. A glass plate of refractive index 1.5 and thickness $1\ \text{cm}$ is introduced between the lens and screen such that the glass plate plane faces parallel to the screen. By what distance should the object be shifted so that a sharp focused image is observed again on the screen?

Option 1:
$0.8\ \text{m}$

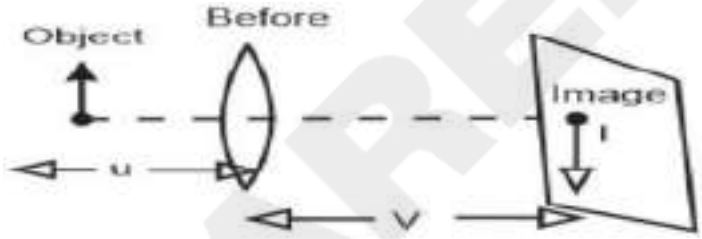
Option 2:
$3.2\ \text{m}$

Option 3:
$1.2\ \text{m}$

Option 4:
$5.6\ \text{m}$

Correct Answer:
$3.2\ \text{m}$

Solution:



$\text{u} = -2.4\ \text{m} = -240\ \text{cm}$

$\text{v} = +12\ \text{cm}$

$$\frac{1}{\text{v}} - \frac{1}{\text{u}} = \frac{1}{\text{f}}$$

$$\frac{1}{+12} - \frac{1}{(-240)} = \frac{1}{\text{f}}$$

$$-\frac{240}{21} = \frac{80}{7}\ \text{cm}$$

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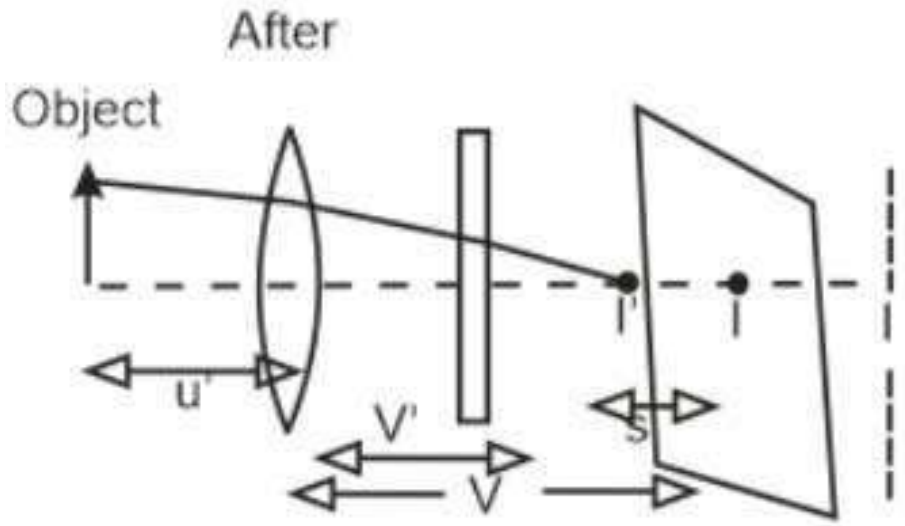


Normal shift due to glass plate

$$s = t\left(1 - \frac{1}{\mu}\right)$$

$$= 1\ \text{cm}\left(1 - \frac{1}{3/2}\right)$$

$$s = \frac{1}{3}\ \text{cm}$$

$$v' = v - s = 12 - \frac{1}{3} = \frac{35}{3}\ \text{cm}$$

$$\frac{1}{v'} - \frac{1}{u'} = \frac{1}{f}$$

$$\frac{1}{+\left(\frac{35}{3}\right)} - \frac{1}{u'} = \frac{1}{\frac{80}{7}}$$

$$\frac{1}{u'} = \frac{3}{35} - \frac{7}{80}$$

$$= \frac{24}{280} - \frac{24.5}{280}$$

$$\frac{1}{u'} = \frac{-1}{560}$$

$$|u'| = 560\,\text{cm} = 5.6\ \text{m}$$

The object is displaced by=5.6 - 2.4

$$= 3.2\,\text{m}$$

Hence (2) is correct option

Q. 9 Light wave traveling in air along x-direction is given by $E_y = 540 \, Sin \, \pi \times 10^4 (x - ct) Vm^{-1}$. Then, the peak value of magnetic field of wave will be (Given $c = 3 \times 10^8 \ ms^{-1}$)

Option 1:
$18 \times 10^{-7} T$

Option 2:
$54 \times 10^{-7} T$

Option 3:
$54 \times 10^{-8} T$

Option 4:
$18 \times 10^{-8} T$

Correct Answer:
$18 \times 10^{-7} T$

Solution:

$$E_y = 540 \sin\left(\pi \times 10^4 (x - ct)\right)$$

$$\frac{E_0}{B_0} = C$$

$$B_0 = \frac{E_0}{C} = \frac{540}{3} \times 10^{-8}$$

$$= 180 \times 10^{-8}$$

$$B_0 = 18 \times 10^{-7} \ T$$

Hence (1) is correct option.

Q. 10 When you walk through a metal detector carrying a metal object in your pocket, it raises an alarm. This phenomenon works on :

Option 1:
Electromagnetic induction

Option 2:
Resonance in ac circuits

Option 3:
Mutual induction in ac circuits

Option 4:
Interference of electromagnetic waves

Correct Answer:
Resonance in ac circuits

Solution:
Metal detector work on the phenomenon of resonance in AC circuits.

Hence (2) is correct option.

Q. 11 An electron with energy $0.1\mathrm{keV}$ moves at right angle to the earth's magnetic field of $1 \times 10^{-4}\mathrm{Wbm}^{-2}$. The frequency of revolution of the electron will be
(Take mass of electron $= 9.0 \times 10^{-31}\ \mathrm{kg}$)

Option 1:
$1.6 \times 10^{5}\ \mathrm{Hz}$

Option 2:
$5.6 \times 10^{5}\ \mathrm{Hz}$

Option 3:
$2.8 \times 10^{6}\ \mathrm{Hz}$

Option 4:
$1.8 \times 10^{6}\ \mathrm{Hz}$

Correct Answer:

2.8×10^6 Hz

Solution:

$\text{KE of etelectron } = 0.1\text{KeV}$

$$\text{B} = 10^{-4}\frac{\text{wb}}{\text{m}^2}$$

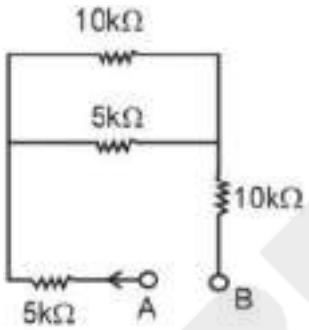
$$\text{f} = \frac{\text{qB}}{2\pi\text{m}} = \frac{1.6 \times 10^{-19} \times 10^{-4}}{6.28 \times 9.1 \times 10^{-31}}$$

$$= \frac{1.6}{6.28 \times 9.1} \times 10^{+8}$$

$$\text{f} = 2.8 \times 10^6 \text{ Hz}$$

Hence (3) is correct option

Q. 12 A current of 15 mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be :



Option 1:

50V

Option 2:

75 V

Option 3:

150 V

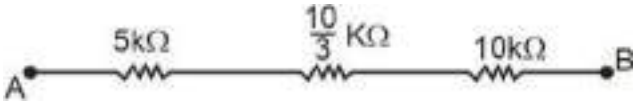
Option 4:

275 V

Correct Answer:

275 V

Solution:



$$R_{AB} = 5 + \frac{10}{3} + 10$$

$$R_{AB} = \frac{55}{3}$$

$$V_{AB} = 15 \times 10^{-3} \times \frac{55}{3}$$

$$V_{AB} = 275 \text{ V}$$

Hence (4) is correct option

Q. 13 The length of a seconds pendulum at a height $h = 2R$ from earth surface will be : (Given $R =$ Radius of earth and acceleration due to gravity at the surface of earth, $g = \pi^2 \text{ ms}^{-2}$)

Option 1:
$\frac{2}{9}\text{m}$

Option 2:
$\frac{4}{9}\text{m}$

Option 3:
$\frac{8}{9}\text{m}$

Option 4:
$\frac{1}{9}\text{m}$

Correct Answer:
$\frac{1}{9}\text{m}$

Solution:
For Seconds pendulam ,

$$T = 2s = 2\pi\sqrt{\frac{l}{9}} \rightarrow (1)$$

$$g_n = \frac{gR^2}{(R+h)^2} = \frac{9}{9}$$

$$T' = 2\pi\sqrt{\frac{l'}{g_h}} = 2s \rightarrow (2)$$

From eq (1) & (2)

$$2\pi\sqrt{\frac{l}{g}} = 2\pi\sqrt{\frac{l'}{g/9}}$$

$$l' = \frac{l}{9}$$

$$(\because l = 1m)$$

$$\therefore l' = \frac{1}{9}m$$

Hence (4) is correct option

Q. 14 Sound travels in a mixture of two moles of helium and n moles of hydrogen. If rms speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound, then the value of n will be :

Option 1:
1

Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
2

Solution:
For non-reacting gaseous mixture

$$\frac{n_{mix}}{v_{mix}-1}=\frac{n_1}{v_1-1}+\frac{n_2}{r_2-1}$$

$$\frac{2+n}{V_{mix}-1}=\frac{2}{\left(\frac{5}{3}-1\right)}+\frac{n}{\left(\frac{7}{5}-1\right)}$$

$$\frac{2+n}{V_{mix}-1}=3+\frac{5}{2}n\rightarrow(1)$$

$$R_{ms}\text{ speed of gas molecules }=\sqrt{\frac{3RT}{M}}$$

$$\text{Speed of sound }=\sqrt{\frac{VRT}{M}}$$

$$\sqrt{\frac{3RT}{M}}=\sqrt{\frac{VRT}{M}}\times\sqrt{2}(\text{ Given })$$

$$2V=3$$

$$V_{mix}=V^2=\frac{3}{2}\rightarrow\ (2)$$

$$\frac{2+n}{\frac{1}{2}}=3+\frac{5n}{2}\text{(from eq 1 \& 2)}$$

$$4+2n=3+\frac{5}{2}n$$

$$1=\frac{n}{2}$$

$$n=2$$

Hence (2) is correct option.

Q. 15 Let η_1 is the efficiency of an engine at $T_1 = 447^\circ C \text{ and } T_2 = 147^\circ C$ while η_2 is the efficiency at $T_1 = 947^\circ C \text{ and } T_2 = 47^\circ C$ The ratio $\frac{\eta_1}{\eta_2}$ will be :

Option 1:
0.41

Option 2:
0.56

Option 3:
0.73

Option 4:
0.70

Correct Answer:
0.56

Solution:

$$\eta_1 = 1 - \frac{T_2}{T_1} = 1 - \frac{420}{720}$$

$$\eta_1 = \frac{300}{720} \rightarrow (1)$$

$$\eta_2 = 1 - \frac{T_2}{T_1} = 1 - \frac{320}{1220}$$

$$\eta_2 = 1 - \frac{900}{1220} \rightarrow (2)$$

$$\frac{\eta_1}{\eta_2} = \frac{300/720}{900/1220} = \frac{1220}{2160}$$

$$= \frac{122}{216} = 0.56$$

Hence (2) is correct option.

Q. 16 An object is taken to a height above the surface of earth at a distance $\frac{5}{4}\mathrm{R}$ from the centre of the earth. Where radius of earth, $\mathrm{R} = 6400\ \mathrm{km}$. The percentage decrease in the weight of the object will be :

Option 1:
36%

Option 2:
50%

Option 3:
64%

Option 4:
25%

Correct Answer:
36%

Solution:

$\mathrm{R} + \mathrm{h} = \frac{5}{4}\mathrm{R}$(from the centre of earth)

$$\mathrm{h} = \frac{\mathrm{R}}{4}$$

$$\frac{\mathrm{g_n}}{\mathrm{g}} = \frac{\mathrm{R}^2}{(\mathrm{R}+\mathrm{h})^2} = \frac{16\mathrm{R}^2}{25\mathrm{R}^2}$$

$$\mathrm{g_n} = \frac{16}{25}\mathrm{g}$$

Weight on the eanth surface $= \mathrm{mg}$

Weight of the object at height $\mathrm{h} = \mathrm{mg_n}$

$$\frac{\mathrm{w}}{\mathrm{w_n}} = \frac{\mathrm{mg}}{\mathrm{mg_n}} = \frac{25}{16}$$

$$\frac{\mathrm{w} - \mathrm{w_n}}{\mathrm{W}} = \frac{9}{25}$$

$$\frac{\Delta W}{W} = 36\%$$

The percentage decrease in the weight of the object will be 36%

Hence (1) is correct option.

Q. 17 A bag of sand of mass $9.8\ \text{kg}$ is suspended by a rope. A bullet of $200\ \text{g}$ travelling with speed $10\ \text{ms}^{-1}$ gets embedded in it, then loss of kinetic energy will be :

Option 1:
$4.9\ \text{J}$

Option 2:
$9.8\ \text{J}$

Option 3:
$14.7\ \text{J}$

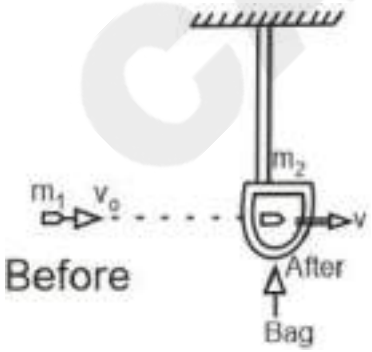
Option 4:
$19.6\ \text{J}$

Correct Answer:
$9.8\ \text{J}$

Solution:

$\text{M}_1 = 9.8\text{kg} \qquad \text{v}_0 = 10\ \text{m/s}$

$\text{M}_2 = 0.2\text{kg}$



By linear momentum conservation ,

$$\text{m}_1\text{v}_0 + \text{m}_2(0) = (\text{m}_1 + \text{m}_2)\,\text{v}$$

$$v = \frac{m_1 v_0}{(m_1 + m_2)} = 0.2 \text{ m/s}$$

The loss of kinetic energy $= KE_i - KE_f$

$$= \left[\frac{1}{2} m_1 v_0^2 - \frac{1}{2}(m_1 + m_2) v^2\right]$$

$$= \left[\frac{1}{2} \times 0.2 \times 100 \frac{-1}{2} \times 10 \times \frac{4}{100}\right]$$

$$= 10 - 0.2$$

$$= 9.8 \text{ J}$$

Hence (2) is correct option

Q. 18 A ball is projected from the ground with a speed 15 ms^{-1} at an angle θ with horizontal so that its range and maximum height are equal, then $\tan\theta$ will be equal to:

Option 1:
$\frac{1}{4}$

Option 2:
$\frac{1}{2}$

Option 3:
2

Option 4:
4

Correct Answer:
4

Solution:
$u = 15 \text{ m/s}$

$R = H$

$$\frac{u^2 \times 2\sin\theta\cos\theta}{g} = \frac{u^2(\sin\theta)^2}{2g}$$

$\tan\theta = 4$

Hence (4) is correct option.

Q. 19 The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are $1\%, 2\%$ and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be :

Option 1:
2

Option 2:
4

Option 3:
6

Option 4:
8

Correct Answer:
8

Solution:

$$\frac{\Delta R}{R} \times 100 = 1\%$$

$$\frac{\Delta I}{I} \times 100 = 2\%$$

$$\frac{\Delta t}{t} \times 100 = 3\%$$

Heat dissipated $= H = I^2Rt$

$$\frac{\Delta H}{H} \times 100 = 2\frac{\Delta I}{H} \times 100 + \frac{\Delta R}{R} \times 100 + \frac{\Delta t}{t} \times 100$$

$$= 2(2\%) + 1\% + 3\%$$

$$\frac{\Delta H}{H} \times 100 = 8\%$$

Hence (4) is correct option.

Q. 20 Hydrogen atom from excited state comes to the ground state by emitting a photon of wavelength λ. The value of principal quantum number 'n' of the excited state will be : (R : Rydberg constant)

Option 1:

$$\sqrt{\frac{\lambda R}{\lambda - 1}}$$

Option 2:

$$\sqrt{\frac{\lambda R}{\lambda R - 1}}$$

Option 3:

$$\sqrt{\frac{\lambda}{\lambda R - 1}}$$

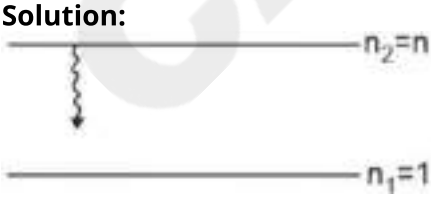
Option 4:

$$\sqrt{\frac{\lambda R^2}{\lambda R - 1}}$$

Correct Answer:

$$\sqrt{\frac{\lambda R}{\lambda R - 1}}$$

Solution:



$$\frac{1}{\lambda} = Rz^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

For hydrogen atom, $z = 1$

$$\frac{1}{\lambda} = R \left[\frac{1}{1} - \frac{1}{n^2}\right]$$

$$\frac{1}{\lambda R} = 1 - \frac{1}{n^2}$$

$$\frac{1}{n^2} = 1 - \frac{1}{\lambda R} = \frac{(\lambda R - 1)}{\lambda R}$$

$$n = \sqrt{\frac{\lambda R}{(\lambda R - 1)}}$$

Hence (2) is correct option

Q. 21 A particle is moving in a straight line such that its velocity is increasing at $5\ ms^{-1}$ per meter. The acceleration of the particle is_____ms^{-2} at a point where its velocity is $20\ ms^{-1}$.

Correct Answer:
5

Solution:

$$a = \frac{\Delta v}{\Delta t} = \frac{5\ m/s}{1\ s}$$

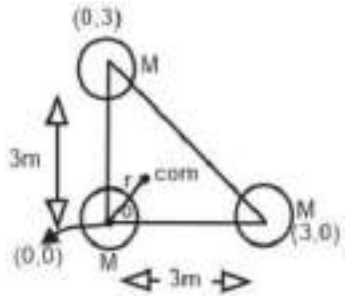
$$a = 5\frac{m}{s^2}$$

The acceleration of the particle is $5\ m/s^2$

Q. 22 Three identical spheres each of mass M are placed at the corners of a right angled triangle with mutually perpendicular sides equal to $3\ m$ each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be $\sqrt{x}\ m$. The value of x is_____.

Correct Answer:
2

Solution:



$$x_{cm} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$= \frac{M(0) + M(3) + M(0)}{3M}$$

$$x_{cm} = 1$$

$$Y_{cm} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3}$$

$$= \frac{M(0) + M(0) + M(3)}{3M}$$

$$Y_{cm} = 1$$

Co-ordination of centre of mass $=(x_{cm}, y_{cm}) = (1, 1)$

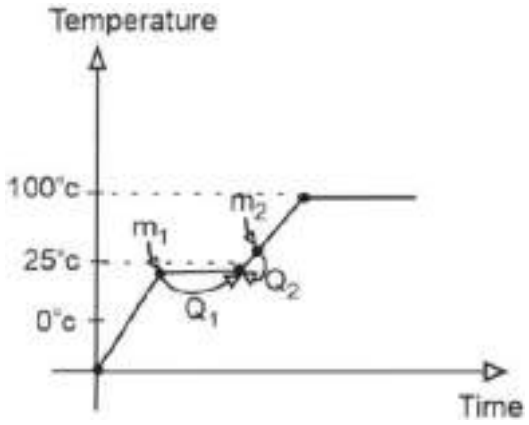
$$r = \sqrt{1^2 + 1^2} = \sqrt{2}$$

The value of $x = 2$

Q. 23 A block of ice of mass $120\ g$ at temperature $0°C$ is put in $300\ g$ of water at $25°C$. The xg of ice melts as the temperature of the water reaches $0°C$. The value of x is_____ [Use specific heat capacity of water $= 4200 Jkg^{-1}\ K^{-1}$, Latent heat of ice $= 3.5 \times 10^5 Jkg^{-1}$]

Correct Answer:
90

Solution:



$M_1 = 120g$

$M_2 = 300g$

By the principle of calorimeter

Heat gain =Heat Lost

$Q_1 = Q_2$

$xLf = m_2 \Delta T_{lost}$

$x \times 3.5 \times 10^5 = 300 \times 4200 \times 25$

$x = 90g$

Q. 24 $\frac{x}{x+4}$ is the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its
(i) third permitted energy level to the second level and
(ii) the highest permitted energy level to the second permitted level.
The value of x will be_____.

Correct Answer:
5

Solution:
Let the energy in case (1) and case (2) be $E_1 \& E_2$ respectively

$$\frac{E_1}{E_2} = \frac{x}{x+4} ----(\text{Given})$$

$$E_1 = -13.6\,eV\left[\frac{1}{2^2} - \frac{1}{3^2}\right] \rightarrow (1)$$

$$E_2 = -13.6ev\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$$

$$\frac{E_1}{E_2} = \frac{\left[\frac{1}{4} - \frac{1}{9}\right]}{\left[\frac{1}{4} - \frac{1}{n^2}\right]} = \frac{x}{x+4}$$

$n \to \infty$ (highest permitted level)

$$\frac{E_1}{E_2} = \frac{\frac{5}{36}}{\frac{1}{4}} = \frac{5}{9} = \frac{x}{x+4}$$

$$x = 5$$

The value of x will be 5

Q. 25 In a potentiometer arrangement, a cell of emf $1.20\ V$ gives a balance point at $36cm$ length of wire. This cell is now replaced by another cell of emf $1.80\ V$. The difference in balancing length of potentiometer wire in above conditions will be_____ $cm.$

Correct Answer:
18

Solution:
For potentiometer,

$$E = \phi l$$

$\phi \to$potential gradient

$l \to$Balancing length

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$\frac{1.2}{1.8} = \frac{36}{l_2}$$

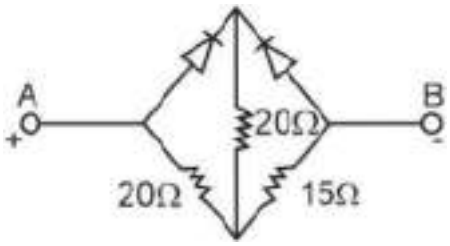
$$l_2 = \frac{36 \times 1.8}{1.2}$$

$l_2 = 54$ cm

The difference in balancing length of potentiometer wire in above conditions will be

$l_2 - l_1 = 18$ cm

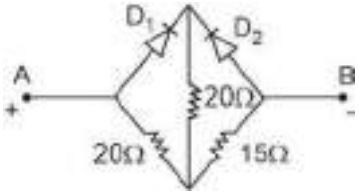
Q. 26 Two ideal diodes are connected in the network as shown is figure. The equivalent resistance between A and B is_____ Ω.



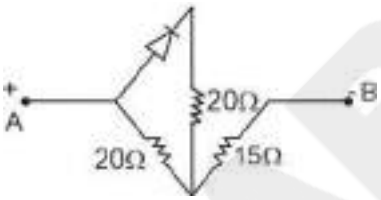
Correct Answer:

25

Solution:



Diode D_1 is forwards biased whereas diode D_2 is reversed biased



$R_{AB} = 10 + 15 = 25\Omega$

The equivalent resistance between A and B is 25Ω

Q. 27 Two waves executing simple harmonic motions travelling in the same direction with same amplitude and frequency are superimposed. The resultant amplitude is equal to the $\sqrt{3}$ times of amplitude of individual motions. The phase difference between the two motions is_____. (degree).

Correct Answer:

60

Solution:
For two waves executing simple harmonic motion travelling in the same direction, the resultant amplitude is

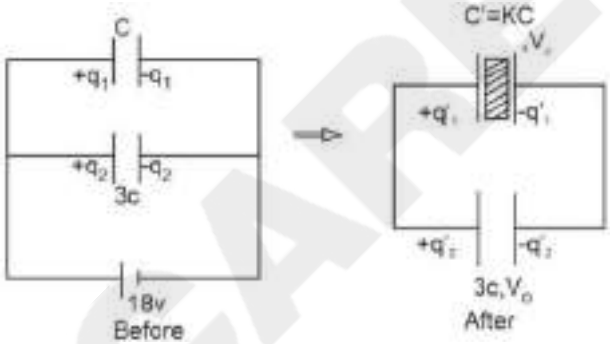
$$R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}$$
$$A_1 = A_2 \quad \text{(Given)}$$
$$R = \sqrt{3}A_1 = \sqrt{3}A_2 \quad \text{(Given)}$$
$$\therefore \sqrt{3}A_1 = \sqrt{A_1^2 + A_1^2 + 2A_1^2\cos\phi}$$
$$\phi = 60^\circ$$

The phase difference betwen the two motions is 60°

Q. 28 Two parallel plate capacitors of capacity C and $3C$ are connected in parallel combination and charged to a potential difference $18\ V$. The battery is then disconnected and the space between the plates of the capacitor of capacity C is completely filled with a material of dielectric constant 9 . The final potential difference across the combination of capacitors will be_____V.

Correct Answer:
6

Solution:



Initially, both the capacitor are changed such that the potential drop across each capacitor is $18\ V$
Affer the insertion of dielectric, the capacitance of first capaitor (c) changes to KC
$V_0 \rightarrow$ common potential at steady state
By charge conservation,

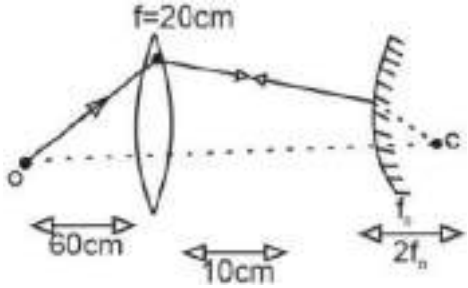
$$q_1 + q_2 = q_1' + q_2'$$
$$C(18) + 3C(18) = kC\,(v_0) + 3C\,(v_0)$$

$18 + 54 = 9v_0 + 3v_0$
$v_0 = 6\ V$
The final potential difference across the combination of capacitors will be $6\ V$

Q. 29 A convex lens of focal length $20\ cm$ is placed in front of a convex mirror with principal axis coinciding each other. The distance between the lens and mirror is $10\ cm$. A point object is placed on principal axis at a distance of $60\ cm$ from the convex lens. The image formed by combination coincides the object itself. The focal length of the convex mirror is _____ cm.

Correct Answer:
10

Solution:



Let the focal length of convex nirror be f_0
If the object and image coincides then the normal incidence must occur on the mirror
For convex lens,

$u = -60\ cm$

$v = 10 + 2f_0$

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{(10 + 2f_0)} - \frac{1}{(-60)} = \frac{1}{+20}$

$\frac{1}{(10 + 2f_0)} = \frac{1}{20} - \frac{1}{60} = \frac{40}{1200}$

$2f_0 = 20\ cm$
$f_0 = 10\ cm$
The focal length of the convex mirror is $10\ cm$.

Q. 30 Magnetic flux (in weber) in a closed circuit of resistance 20Ω varies with time $\mathrm{t}(\ \mathrm{s})$ as $\phi = 8\mathrm{t}^2 - 9\mathrm{t} + 5$.The magnitude of the induced current at $\mathrm{t} = 0.25\ \mathrm{s}$ will be_____ mA.

Correct Answer:
250

Solution:

$\mathrm{R} = 20\Omega$

$\phi = 8\mathrm{t}^2 - 9\mathrm{t} + 5$

$$\mathrm{e} = \left|\frac{\mathrm{d}\phi}{\mathrm{dt}}\right| = |16\mathrm{t} - 9|$$

$$\text{at } \mathrm{t} = 0.25\ \mathrm{s} = \frac{1}{4}\ \mathrm{s}$$

$$\mathrm{e} = |4 - 9| = 5\ \mathrm{V}$$

$$\mathrm{i} = \frac{\mathrm{e}}{\mathrm{k}} = \frac{5}{20} = \frac{1}{4}\ \mathrm{A}$$

$$\mathrm{i} = 0.25\ \mathrm{A} = 250\ \mathrm{mA}$$

The magnitude of the induced current at $\mathrm{t} = 0.25\mathrm{s}$ will be $250\ \mathrm{mA}$.

Chemistry

Q. 1 Match **List I** with **List II :**

List I (molecule)	**List II (hybridization ; shape)**
A. XeO_3	I. $sp^3\ d;\ linear$
B. XeF_2	II. $sp^3;\ pyramidal$
C. $XeOF_4$	III. $sp^3\ d^3;\ distorted\ octahedral$
D. XeF_6	IV. $sp^3\ d^2;\ square\ pyramidal$

Choose the correct answer from the options given below:

Option 1:
A-II, B-I, C-IV, D-III

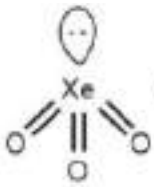
Option 2:
A-II, B-IV, C-III, D-I

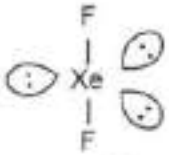
Option 3:
A-IV, B-II, C-III, D-I

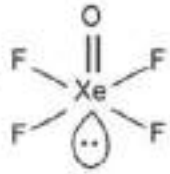
Option 4:
A-IV, B-II, C-I, D-III

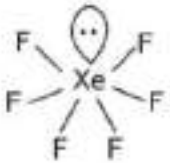
Correct Answer:
A-II, B-I, C-IV, D-III

Solution:

(A) XeO_3  (Pyramidal) SP^3

(B) XeF_2  (Linear) SP^3d

(C) $XeOF_4$  (Square Pyramidal) SP^3d^2

(D) XeF_6  (Distorted octahedral) SP^3d^3

So, A-(II), B-(I),C-(IV),D-(III)

Hence, the correct answer is **option (1).**

Q. 2 Two solutions A and B are prepared by dissolving $1\ g$ of non-volatile solutes x and Y, respectively in $1\ kg$ of water. The ratio of depression in freezing points for A and b is found to be $1:4$.The ratio of molar masses of X and Y is

Option 1:
$1:4$

Option 2:
$1:0.25$

Option 3:
$1:0.20$

Option 4:
$1:5$

Correct Answer:
$1:0.25$

Solution:
We know,

$\Delta T_f \propto m$, if solvent is same

$$\frac{(\Delta T_f)_1}{(\Delta T_{f2})_2} = \frac{m_1}{m_2} = \frac{\omega_{x/M_x}}{\omega_{water}} \times \frac{w_{vater}}{\omega_y/M_y}$$

$$\frac{1}{4} = \frac{\omega_x x}{M_x} \times \frac{M_y}{\omega_y}$$

$$\frac{1}{4} = \frac{M_y}{M_x} \quad (\because \omega_x = 1 \text{ g and } \omega_y = 1 \text{ g})$$

$$\frac{M_x}{M_y} = \frac{4}{1} = \frac{1}{1/4} = \frac{1}{0.25}$$

Hence, the correct answer is **Option (2).**

Q. 3 K_{a_1} , K_{a_2} and K_{a_3} are the respective ionization constants for the following reactions $(a), (b),$ and (c).

(a) $H_2C_2O_4 = H^+ + HC_2O_4^-$

(b) $HC_2O_4^- = H^+ + HC_2O_4^{2-}$

(c) $H_2C_2O_4 = 2H^+ + C_2O_4^{2-}$

The relationship between K_{a_1} , K_{a_2} and K_{a_3} is given as

Option 1:
$K_{a_3} = K_{a_1} + K_{a_2}$

Option 2:
$K_{a_3} = K_{a_1} - K_{a_2}$

Option 3:
$K_{a_3} = K_{a_1}/K_{a_2}$

Option 4:
$K_{a_3} = K_{a_1} \times K_{a_2}$

Correct Answer:
$K_{a_3} = K_{a_1} \times K_{a_2}$

Solution:

$$(a)\ H_2C_2O_4 \rightleftharpoons H^+ + HC_2O_4^- \quad : K_{a_1} = \frac{[H^+]\left[HC_2O_4^-\right]}{[H_2C_2O_4]}$$

$$(b)\ HC_2O_4^- \rightleftharpoons H^+ + C_2O_4^{2-} \quad : Ka_2 = \frac{[H^+]\left[C_2O_4^{2-]}\right]}{\left[HC_2O_4^-\right]}$$

$$(C)\ H_2C_2O_4 \rightleftharpoons 2H^+ + C_2O_4^{2-} : Ka_3 = \frac{[H^+]^2\left[C_2O_4^{2-}\right]}{[H_2C_2O_4]}$$

$$c = a + b,\ so,\ k_{a_3} = k_{a_1} \times ka_2$$

Hence, the correct answer is **Option (4).**

Q. 4 The molar conductivity of a conductivity cell filled with 10 moles of $20\ mL$ NaCl solution is Λ_{m1} and that of 20 moles another identical cell heaving $80\ mL$ NaCl solution is Λ_{m2}, The conductivities exhibited by these two cells are same. The relationship between Λ_{m2} and Λ_{m1} is

Option 1:
$\Lambda_{m2} = 2\Lambda_{m1}$

Option 2:
$\Lambda_{m2} = \Lambda_{m1}/2$

Option 3:
$\Lambda_{m2} = \Lambda_{m1}$

Option 4:
$\Lambda_{m2} = 4\Lambda_{m1}$

Correct Answer:
$\Lambda_{m2} = 2\Lambda_{m1}$

Solution:
We know,

$$\Lambda_m = K \times \frac{1000}{M} \qquad \left\{\begin{matrix} M = \text{molarity} \\ K = \text{conductivity} \end{matrix}\right\}$$

If both cell are identical, K will be same.

So,

$$\Lambda_m \times \frac{1000}{M}$$

Then,

$$\frac{\Lambda_{m_1}}{\Lambda_{m_2}} = \frac{M_2}{M_1} = \frac{20}{80} \times \frac{20}{10}$$

$$\frac{\Lambda_{m_1}}{\Lambda_{m_2}} = \frac{1}{2}$$

$$\Lambda_{m_2} = 2\Lambda_{m_1}$$

Hence, the correct answer is **Option (1).**

Q. 5 For micelle formation, which of the following statements are correct?

A. Micelle formation is an exothermic process.
B. Micelle formation is an endothermic process.
C. The entropy change is positive.
D. The entropy change is negative.

Option 1:
A and D only

Option 2:
A and C only

Option 3:
B and C only

Option 4:
B and D only

Correct Answer:
A and C only

Solution:
Micelle formation is an exothermic process.

The entropy change is positive.

Hence, the correct answer is **Option (2).**

Q. 6 The first ionization enthalpies of $\mathrm{Be, B, N}$ and O follow the order

Option 1:
$\mathrm{O < N < B < Be}$

Option 2:
$\mathrm{Be < B < N < O}$

Option 3:
$\mathrm{B < Be < N < O}$

Option 4:
$\mathrm{B < Be < O < N}$

Correct Answer:
$\mathrm{B < Be < O < N}$

Solution:
The correct order is :-

$$\begin{array}{cccccc} & \mathrm{B} & < & \mathrm{Be} & < & \mathrm{O} & < & \mathrm{N} \\ & 2p^1 & & 2s^2 & & 2p^4 & & 2p^3 \\ \mathrm{IE}_1 & (801) & & (899) & & (1314) & & (1402) \end{array}$$

Hence, the correct answer is **Option (4).**

Q. 7 Given below are two statements.
Statement I: Pig iron is obtained by heating cast iron with scrap iron.
Statement II: Pig iron has a relatively lower carbon content than that of cast iron. In the light of the above statements, choose the *correct answer* from the options given below.

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are **not** correct.

Option 3:
Statement I is correct but Statement II is **not** correct.

Option 4:
Statement I is **not** correct but Statement II is correct.

Correct Answer:
Both Statement I and Statement II are **not** correct.

Solution:
The iron obtained from Blast furnace contains about 4% carbon and many impurities in smaller amount $(\mathrm{e.g.S, P, Si, Mn})$. This is known as pig iron. Cast iron is different from pig iron and coke using hot air blust. It has slightly lower carbon content about 3% and is extremely hard and brittle.

So, Both statement I and II are not correct.

Hence, correct answer is **Option (2).**

Q. 8 High purity $(> 99.95\%)$ dihydrogen is obtained by

Option 1:
reaction of zinc with aqueous alkali.

Option 2:
electrolysis of acidified water using platinum electrodes.

Option 3:
electrolysis of warm aqueous barium hydroxide solution between nickel electrodes.

Option 4:
reaction of zinc with dilute acid.

Correct Answer:
electrolysis of warm aqueous barium hydroxide solution between nickel electrodes.

Solution:
High purity (>99.95 %) dihydrogen is obtained by electrolysing warm aqueous barium hydroxide solution between nickel electrodes.

Hence, the correct answer is **Opton (3).**

Q. 9 The correct order of density is

Option 1:
$\mathrm{Be} > \mathrm{Mg} > \mathrm{Ca} > \mathrm{Sr}$

Option 2:
$\mathrm{Sr} > \mathrm{Ca} > \mathrm{Mg} > \mathrm{Be}$

Option 3:
$\mathrm{Sr} > \mathrm{Be} > \mathrm{Mg} > \mathrm{Ca}$

Option 4:
$\mathrm{Be} > \mathrm{Sr} > \mathrm{Mg} > \mathrm{Ca}$

Correct Answer:
$\mathrm{Sr} > \mathrm{Be} > \mathrm{Mg} > \mathrm{Ca}$

Solution:
The correct order of density is

	Sr	$>$	Br	$>$	Mg	$>$	Ca
$\mathrm{Density/g\ cm^{-3}}$	(2.63)		(1.84)		(1.74)		(1.55)

Hence, the correct answer is **Option (3).**

Q. 10 The total number of acidic oxides from the following list is
$\mathrm{NO}, \mathrm{N_2O}, \mathrm{B_2O_3}, \mathrm{N_2O_5}, \mathrm{CO}, \mathrm{SO_3}, \mathrm{P_4O_{10}}$

Option 1:
3

Option 2:
4

Option 3:
5

Option 4:
6

Correct Answer:
4

Solution:
$CO,\ NO, N_2O$ are neutral oxides.

$B_2O_3,\ \ N_2O_5, SO_3, P_4O_{10}$ are acidic oxides.

Hence, the correct answer is **Option (2).**

Q. 11 The correct order of energy of absorption for the following metal complexes is

$$A : [Ni(en)_3]^{2+}, B : [Ni\ (NH_3)_6]^{2+}, C : [Ni\ (H_2O)_6]^{2+}$$

Option 1:
$C < B < A$

Option 2:
$B < C < A$

Option 3:
$C < A < B$

Option 4:
$A < C < B$

Correct Answer:
$C < B < A$

Solution:
Stronger the ligands the splitting will be large. If splitting is large the energy of absorption will high.

Increasing field strength : $H_2O < NH_3 < en$

So, energy of absorption : $C < B < A$

Hence, the correct answer is **Option (1).**

Q. 12 Match **List I** with **List II.**

List I	List II
A. Sulphate	I. Pesticide
B. Fluoride	II. Bending of bones
C. Nicotine	III. Laxative effect
D. Sodium arsinite	IV. Herbicide

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-IV, B-III, C-II, D-I

Option 3:
A-III, B-II, C-I, D-IV

Option 4:
A-III, B-II, C-IV, D-I

Correct Answer:
A-III, B-II, C-I, D-IV

Solution:
Sulphate! Excessive sulphate (>500 ppm) in drinking water causes laxative effect.

Fluaride: Excess fluoride (>10 ppm) causes harmful effect to bones and teeth.

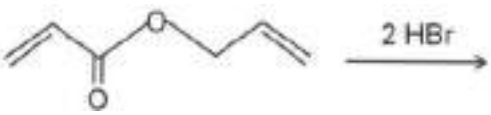
Nicotine: It was used as pest controlling substance for major crops in agaicultural practices. It is a pesticides.

Sodium arsinite: It is a herbicides.

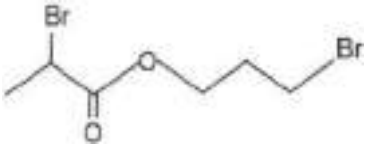
So, A-III, B-II, C-I, D-IV

Hence, the correct answer is **Option (3).**

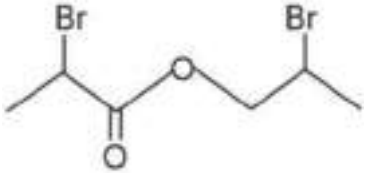
Q. 13 Major product of the following reaction is



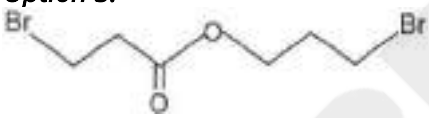
Option 1:



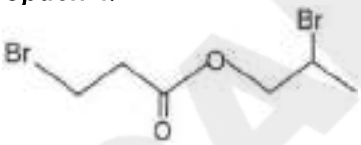
Option 2:



Option 3:

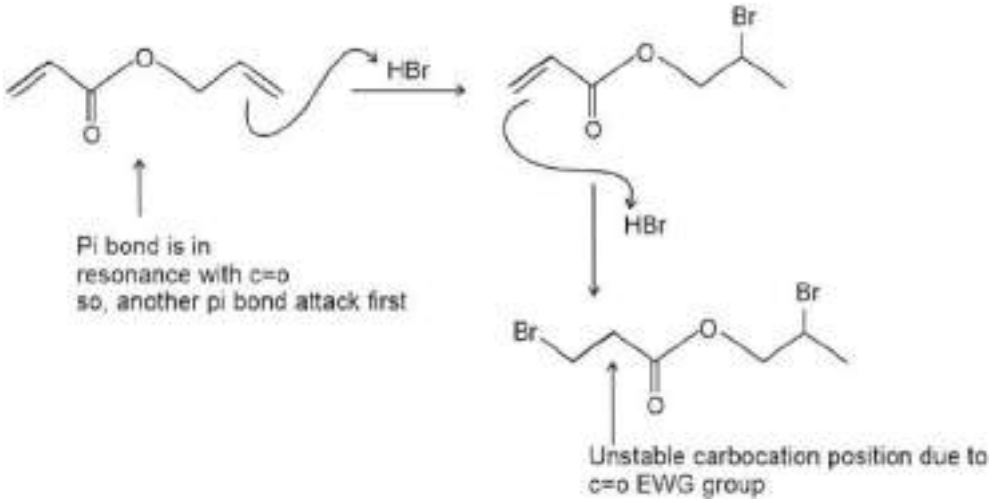


Option 4:



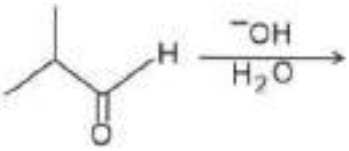
Correct Answer:

Solution:
The reaction will be :-

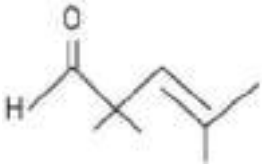


Hence, the correct answer is **Option (4).**

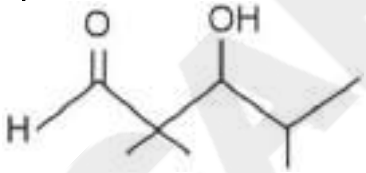
Q. 14 What is the major product of the following reaction?



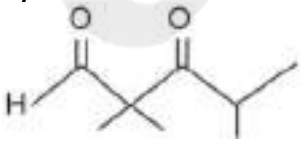
Option 1:



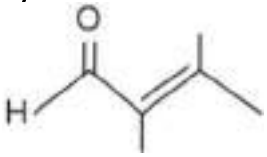
Option 2:



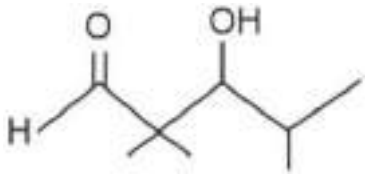
Option 3:



Option 4:

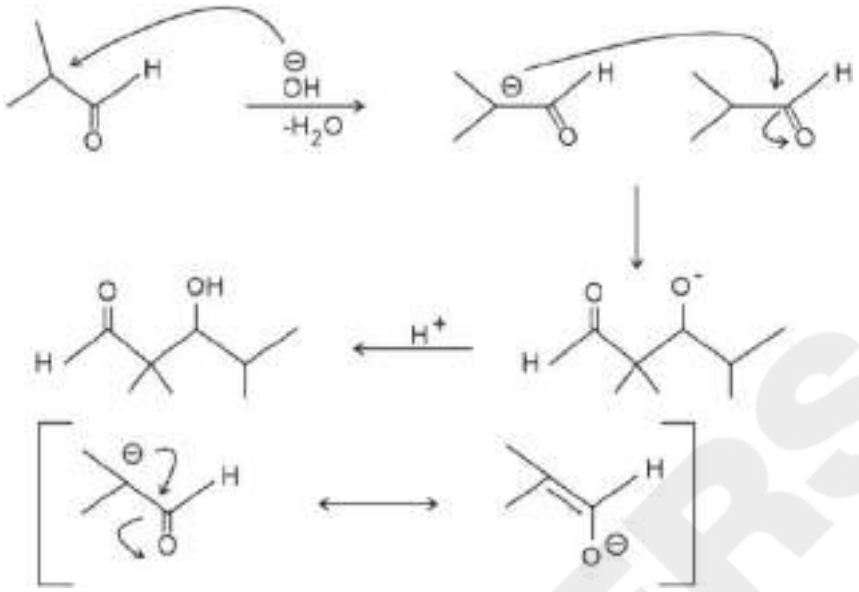


Correct Answer:



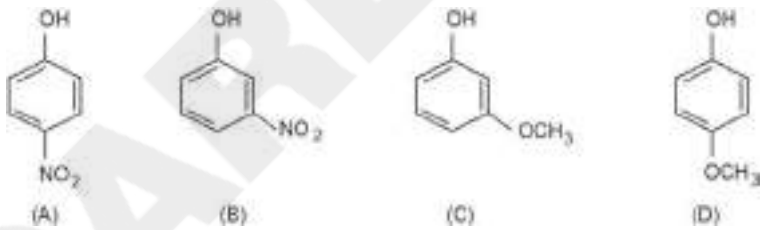
Solution:

The reaction will be (Aldol condensaton) -



Hence, the correct answer is **Option (2).**

Q. 15 Arrange the following in decreasing acidic strength.



Option 1:

$A > B > C > D$

Option 2:

$B > A > C > D$

Option 3:

$D > C > A > B$

Option 4:

$D > C > B > A$

Correct Answer:

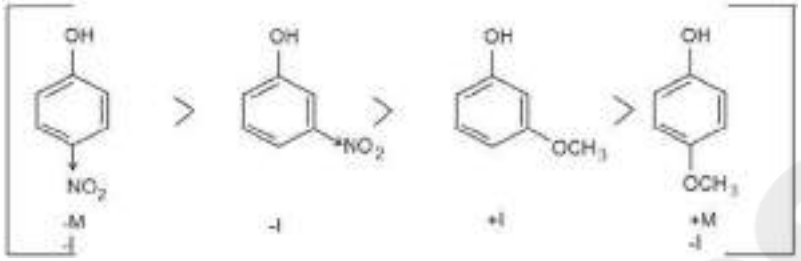
$A > B > C > D$

Solution:

We know, $-NO_2$ is EWG α acidic

$$-OCH_3 \text{ is EDG } \alpha \text{ basic} \propto \frac{1}{\text{acidic}}$$

EGG meta directing and EDG -O, P directing



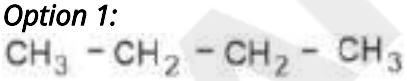
So, $A > B > C > D$

Hence, the correct answer is **Option (1).**

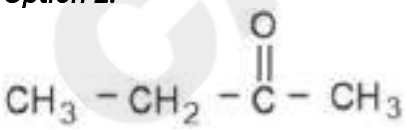
Q. 16 $CH_3 - CH_2 - CN \xrightarrow[\text{Ether}]{CH_3MgBr} A \xrightarrow{H_3O^+} B \xrightarrow[Zn-Hg]{} C$

The correct structure of $\mathbf{C}$ is

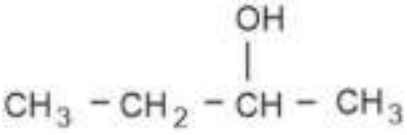
Option 1:



Option 2:



Option 3:



Option 4:

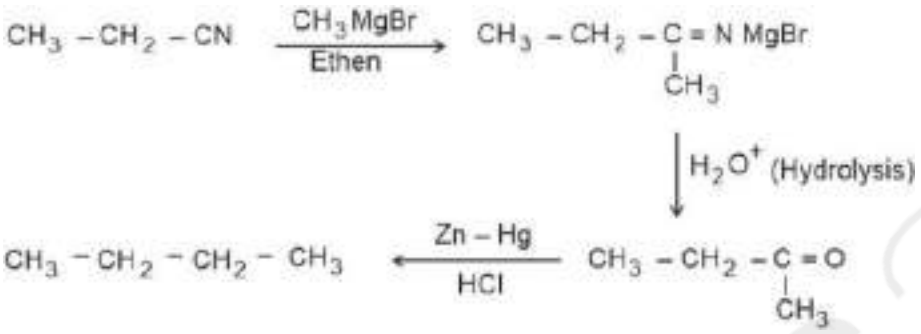
$CH_3 - CH_2 - CH = CH_2$

Correct Answer:

$CH_3 - CH_2 - CH_2 - CH_3$

Solution:

The reaction will be -



Hence, the correct answer is **Option (1).**

Q. 17 Match **List I** with **List II :**

List I **Polymer**	**List II** **used for items**
A. Nylon 6,6	I. Buckets
B. Low density polythene	II. Non-stick utensils
C. High density polythene	III. Bristles of brushes
D. Teflon	IV. Toys

Choose the correct answer from the options given below:

Option 1:
A-III, B-I, C-IV, D-II

Option 2:
A-III, B-IV, C-I, D-II

Option 3:
A-II, B-I, C-IV, D-III

Option 4:
A-II, B-IV, C-I, D-III

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
A. Nylon 6,6 - III. Bristles of brushes
B. Low density polythene - IV. Toys
C. High density polythene - I. Buckets.
D. Teflon - II Non - stick utensils

A - III, B-IV, C-I, D-II

Hence, the correct answer is **Option (2).**

Q. 18 Glycosidic linkage between $C1$ of a-glucose and $C2$ of β-fructose is found in

Option 1:
maltose

Option 2:
sucrose

Option 3:
lactose

Option 4:
amylose

CAREERS360

Correct Answer:
sucrose

Solution:
Glycosidic linkage between $\mathrm{Cl}\ \text{of}\ \alpha-\text{glucose}$ and $\mathrm{C_2}\ \text{of}\ \beta-\text{fructose}$ is found in sucrose.

Hence, the coreect answer is **option (2).**

Q. 19 Some drugs bind to a site other than the active site of an enzyme. This site is known as

Option 1:
non-active site

Option 2:
allosteric site

Option 3:
competitive site

Option 4:
therapeutic site

Correct Answer:
allosteric site

Solution:
Some drugs do need bind to the enzyme's active site. These bind to a different site of enzyme which is called allosteric site.

Hence, the correct answer is **Option(2).**

Q. 20 In base vs. acid titration, at the end point methyl orange is present as

Option 1:
quinonoid form

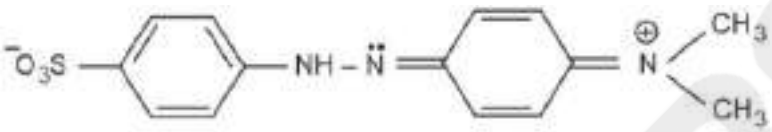
Option 2:
heterocyclic form

Option 3:
phenolic form

Option 4:
benzenoid form

Correct Answer:
quinonoid form

Solution:
In base vs. acld titration, at the end point methyl Orange is present as quinonoid form.



(Quinonoid form of the anion)

Hence, the correct answer is **Option (1).**

Q. 21 $56.0\ \text{L}$ of nitrogen gas is mixed with excess of hydrogen gas and it is found that $20\ \text{L}$ of ammonia gas is produced. The volume of unused nitrogen gas is found to be ___ L.

Correct Answer:
46

Solution:

$$N_2 + H_2 \longrightarrow 2NH_3$$
$$(\text{excess})$$

2 mol of NH_3 formed from 1 mol of N_2

$2 \times 22.4\ \text{L of } NH_3 = 22.4\ \text{L of } N_2 \text{ required}$

$20\ \text{L of } NH_3 = 10\ \text{L of } N_2 \text{ required}$

So, from 56.0 L of NH_3 is used , so 46 L is unused.

Answer = 46

Q. 22 A sealed flask with a capacity of $2\mathrm{dm}^3$ contains $11\ \mathrm{g}$ of propane gas. The flask is so weak that it will burst if the pressure becomes $2\mathrm{MPa}$. The minimum temperature at which the flask will burst is _____ $^\circ\mathrm{C}$. [Nearest integer]

(Given: $\mathrm{R} = 8.3\ \mathrm{J\ K^{-1}\ mol^{-1}}$, Atomic masses of C and H are 12u and 1u, respectively.)
(Assume that propane behaves as an ideal gas.)

Correct Answer:
1655

Solution:
We know, $\mathrm{PY} = \mathrm{nRT}$

$\mathrm{P} = 2\ \mathrm{MPa}, \mathrm{V} = 2\mathrm{dm}^3 = 2 \times 10^{-3}\ \mathrm{m}^3$

$\mathrm{n} = \frac{11}{44} = \frac{1}{4}\ \mathrm{mol\ of\ Propane\ C_3H_8}$

Hence,

$$2 \times 10^6(\ \mathrm{Pa}) \times 2 \times 10^{-3}\ (\ \mathrm{m}^3) = \frac{1}{4}\ \mathrm{mol} \times 8.3\mathrm{JK}^{-1}\ \mathrm{mol}^{-1} \times \mathrm{T}$$

$$4 \times 10^3 = \frac{1}{4} \times 8.3 \times \mathrm{T}$$

$$\mathrm{T} = \frac{16000}{8.3}\mathrm{K} \qquad [\because 1\mathrm{Pam}^3 = 1\mathrm{J}]$$

$$\mathrm{T} = 1927.71\ \mathrm{K}$$

$$\mathrm{T} = 1654.56^\circ\ \mathrm{C}$$

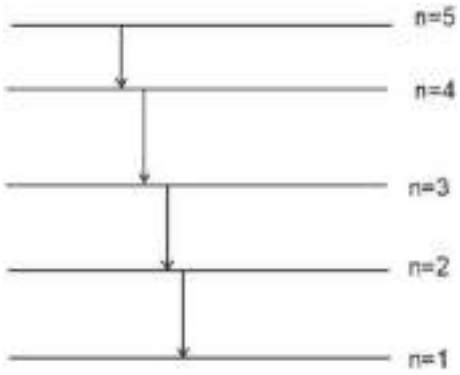
$$\mathrm{T} = 1655^\circ\ \mathrm{C}$$

Answer = 1655

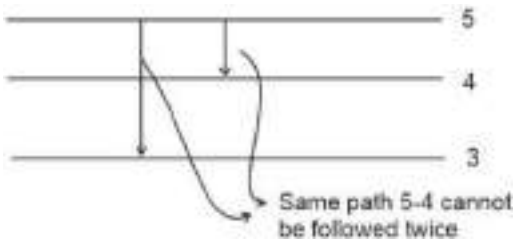
Q. 23 When the excited electron of a H atom from $\mathrm{n} = 5$ drops to the ground state, the maximum number of emission lines observed are _____.

Correct Answer:
4

Solution:
The maximum number of emission lines observed are -



In Hydrogen atom only one electrons is present. For the maximum number of emission lines need to Jump (n-1) orbit step wire because same path cannot be repeated.



Answer = 4

Q. 24 While performing a thermodynamics experiment, a student made the following observations.

$$HCl + NaOH \rightarrow NaCl + H_2O \Delta H = -57.3 \text{ kJ mol}^{-1}$$
$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O \Delta H = -55.3 \text{ kJ mol}^{-1}$$

The enthalpy of ionization of CH_3COOH as calculated by the student is _____kJ mol^{-1}. (nearest integer)

Correct Answer:
2

Solution:
The ionization of CH_3COOH -

$$CH_3COOH \longrightarrow CH_3COO^{\ominus} + H^+ \quad \Delta H^- = ?$$

Given,

$$(1)\ HCl + NaOH \longrightarrow NaCl, \Delta H = -57.3 KJ/mol$$

$(2)CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O, \Delta H = -55.3KJ/mol$

By doing (2) - (1), will become ionisation of CH_3COOH

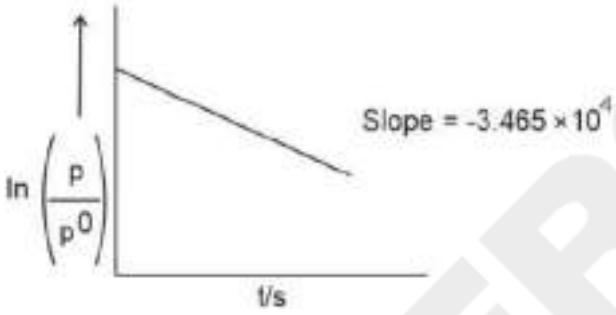
$CH_3COOH + NaCl \rightarrow CH_3COO^{\ominus}Na^{\oplus} + HCl$

So, $\Delta H = -55.3 - (-57.3)$

$\Delta H = 2\ kJ/mol$

Answer = 2

Q. 25 For the decomposition of azomethane, $CH_3\ N_2CH_3(\ g) \rightarrow CH_3CH_3(\ g) + N_2(\ g),$ a first order reaction, the variation in partial pressure with time at $600\ K$ is given as



The half life of the reaction is _____ $\times 10^{-5}$ s. [Nearest integer]

Correct Answer:
2

Solution:
We know,

$$k = \ln\left(\frac{P_0}{P}\right) \times \frac{1}{t}$$

$$\ln\left(\frac{P_0}{P}\right) = kt$$

$$\ln\left(\frac{P}{P_0}\right) = -kt$$

Comparing $y = mx + c$ with above equation

Slope = $-k = -3.465 \times 10^4$

So, $k = 3.465 \times 10^4$

Now, Half life of first order reaction.

$$t_{k2} = \frac{\ln 2}{k} = \frac{0.3010 \times 2.303}{3.465 \times 10^4}$$

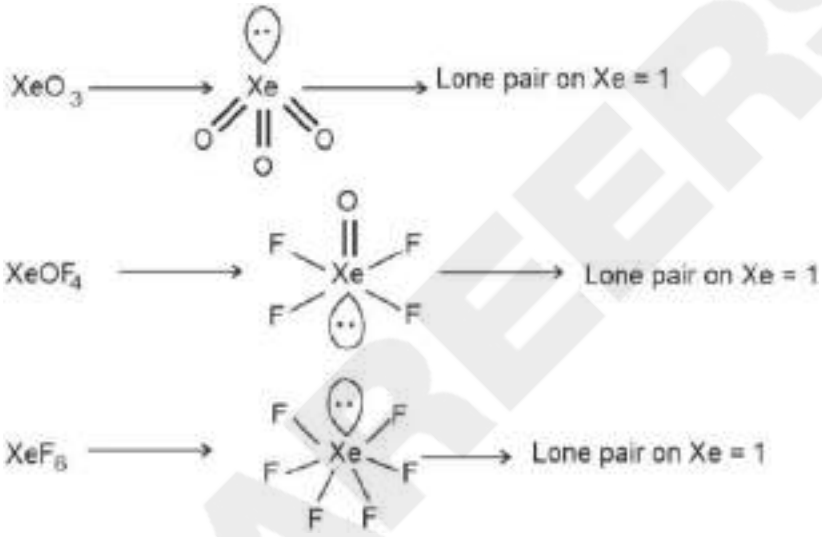
$$tk_2 = 2 \times 10^{-5} \text{sec}$$

Answer = 2

Q. 26 The sum of number of lone pairs of electrons present on the central atoms of $XeO_3, XeOF_4$ and XeF_6, is _____.

Correct Answer:
3

Solution:



Total lone pairs = 1+1+1 = 3

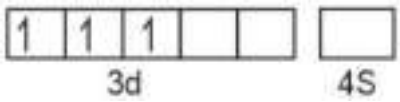
Q. 27 The spin-only magnetic moment value of M^{3+} ion (in gaseous state) from the pairs $Cr^{3+}/Cr^{2+}, Mn^{3+}/Mn^{2+}, Fe^{3+}/Fe^{2+}$ and Co^{3+}/Co^{2+} that has negative standard electrode potential, is ___ B.M. [Nearest integer]

Correct Answer:
4

Solution:
Cr^{3+}/Cr^{2+} has negative standard electrode potential among them.

Spin - only magnetic moment $(\mu) = \sqrt{n(n+2)}$

$Cr^{3+} : 4s^0 3d^3$, only 3 unpaired electron



So, n = 3

$$\text{so, } \mu = \sqrt{3(3+2)} = \sqrt{15}\ \text{BM}$$
$$\mu \approx 4\text{BM}$$

Answer = 4

Q. 28 A sample of 4.5mg of an unknown monohydric alcohol, R-OH was added to methylmagnesium iodide. A gas is evolved and is collected and its volume measured to be $3.1\ \text{mL}$.The molecular weight of the unknown alcohol is ___ g/mol. [Nearest integer]

Correct Answer:
33

Solution:
The reaction will be -

$$R - OH + CH_3MgI \rightarrow ROMgI^+ + CH_4(g)$$
$$3.1\ \text{ml}$$

From above reaction,

1 mole CH_4 is collected from 1 mole of ROH

$22.4\ \text{L}\ CH_4 \longrightarrow 1\ \text{mole of ROH}$

$3.1\ \text{L}\ CH_4 \rightarrow \frac{3.1}{22.4}\ \text{mole of ROH}$

$3.1\ \text{mL of}\ CH_4 \rightarrow 0.138\ \text{mmole of ROH}$

We Know,

$$\text{Mole} = \frac{\text{Mass}}{\text{Molar Mass}}$$

$$\text{Molar Mass} = \frac{4.5\ \text{mg}}{0.138\ \text{mmol}} = \frac{4.5 \times 10^{-3}}{0.138 \times 10^{-3}}\ \text{g/mol}$$

$$\text{Molar Mass} = 32.6\ \text{g/mol}$$

$$\text{Molar Mass} \approx 33\ \text{g/mol}$$

Q. 29 The separation of two coloured substances was done by paper chromatography. The distances travelled by solvent front, substance A and substance B from the base line are $3.25\ \text{cm}, 2.08\ \text{cm}$ and $1.05\ \text{cm}$, respectively. The ratio of $\mathrm{R_f}$ values of A to B is _____.

Correct Answer:
2

Solution:
We know,

$$\mathrm{R_f} = \frac{\text{Distance moved by the substance from base line}}{\text{Distance moved by the solvent from base line}}$$

So,

$$\mathrm{R_{fA}} = \frac{2.08}{3.25} \text{ and } \mathrm{R_{f_B}} = \frac{1.05}{3.25}$$

Then ratio of $\mathrm{R_f}$ values of A to B is

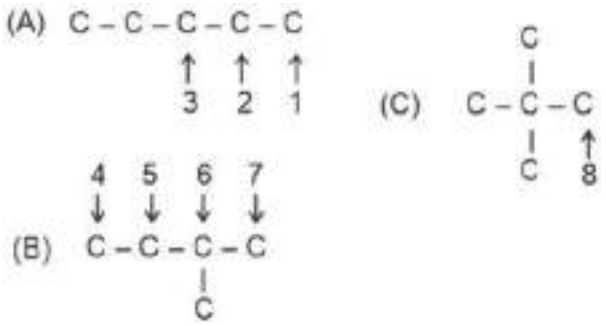
$$\frac{\mathrm{R_{fA}}}{\mathrm{R_{f_B}}} = \frac{2.08/3.25}{1.05/3.25} = 2$$

Answer = 2

Q. 30 The total number of monobromo derivatives formed by the alkanes with molecular formula $\mathrm{C_5H_{12}}$ is (excluding stereo isomers) _____.

Correct Answer:
8

Solution:
The possible position at carbon chain of Bromine



Total at 8 position one Br (monobromo) derivatives are possible.

Maths

Q. 1 For $z \in \mathbb{C}$ if the minimum value of $(|z-3\sqrt{2}|+|z-p\sqrt{2}i|)$ is $5\sqrt{2}$, then a value of P is_____.

Option 1:
3

Option 2:
$\frac{7}{2}$

Option 3:
4

Option 4:
$\frac{9}{2}$

Correct Answer:
4

Solution:
Using triangle law

$$|z-3\sqrt{2}|+|z-p\sqrt{2}i| \geqslant |(z-3\sqrt{2})-(z-p\sqrt{2}i)$$

$$\geqslant |-3\sqrt{2}+p\sqrt{2}i|$$

$$\geqslant \sqrt{18+2p^2}$$

Given that maximum value is $5\sqrt{z}$

$$\therefore \quad \sqrt{18+2p^2}=5\sqrt{2}$$

$$18+2p^2=50$$

$$p=\pm 4$$

Hence correct option is 3

Q. 2 The number of real values of λ, such that the system of linear equations

$$2x-3y+5z=9$$

$$x+3y-z=-18$$

$$3x-y+\left(\lambda^2-|\lambda|\right)z=16$$

has no solutions, is

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
4

Correct Answer:
2

Solution:
For no solution

$$\Delta=0$$

$$\begin{vmatrix} 2 & -3 & 5 \\ 1 & 3 & -1 \\ 3 & -1 & \lambda^2-1\lambda \end{vmatrix}=0$$

$\Rightarrow 2\left(3\lambda^2-3|\lambda|-1\right)+3\left(\lambda^2-|\lambda|+3\right)+5(-1-9)=0$

$\Rightarrow 9\lambda^2-9|\lambda|-43=0$

As $f(0)<0$

$\Rightarrow |\lambda|$ has a positive ad a negative root,which gives z real values of λ.

Hence correct option is 3

Q. 3 The number of bijective functions $f:\{1,3,5,7,\ldots,99\}\rightarrow\{2,4,6,8,\ldots,100\}$,such that $f(3)\geq f(9)\geq f(15)\geq f(21)\geq\ldots\geq f(99)$, is

Option 1:
$^{50}P_{17}$

Option 2:
$^{50}P_{33}$

Option 3:
$33!\times 17!$

Option 4:
$\frac{50!}{2}$

Correct Answer:
$^{50}P_{33}$

Solution:
For bijective function

$f(3)>f(9)>\ldots>f(99)$

First select 17 images for these 17 inputs $^{50}C_{33}$ ways ad divide them in 1 way

Now selelct images for rest of the 33 inputs in 33! ways

$\therefore\ \text{Total}\ =^{50}C_{17}\cdot 33!$

$=\frac{50!}{17!}$

$= 50P_{33}$

Hence correct option is 2

Q. 4 The remainder when $(11)^{1011} + (1011)^{11}$ is divided by 9 is

Option 1:
1

Option 2:
4

Option 3:
6

Option 4:
8

Correct Answer:
8

Solution:
$(11)^{1011}$

$= (9+2)^{1011}$

$= 92 + 2^{1011}$

$= 92 + (8)^{337}$

$= 9\lambda + (9-1)^{337}$

$= 9\lambda + 9k - 1$

$= 9k' + 8$

And $(1011)^{11}$

$= (9k+3)^{11} = 9\lambda + 3^{11}$

$= 9\lambda + 9k$

$= 9\mu$

$\therefore$ Required remainder $= 8$

Hence correct option is 4

Q. 5 The sum $\sum_{n=1}^{21} \frac{3}{(4n-1)(4n+3)}$ is equal to

Option 1:
$\frac{7}{87}$

Option 2:
$\frac{7}{29}$

Option 3:
$\frac{14}{87}$

Option 4:
$\frac{21}{29}$

Correct Answer:
$\frac{7}{29}$

Solution:
Required sum

$$= \frac{3}{3.7} + \frac{3}{7.11} + \cdots + \frac{3}{83.87}$$

$$= \frac{3}{4}\left(\frac{4}{3.7} + \frac{4}{7.11} + \cdots + \frac{4}{83.87}\right)$$

$$= \frac{3}{4}\left(\frac{7-3}{3.7} + \frac{11-7}{7.11} + \cdots + \frac{87-85}{83.87}\right)$$

$$= \frac{3}{4}\left(\frac{1}{3} - \frac{1}{7} + \frac{1}{7} - \frac{1}{11} + \cdots + \frac{1}{83} - \frac{1}{87}\right)$$

$$= \frac{3}{4}\left(\frac{1}{3} - \frac{1}{87}\right)$$

$$= \frac{3}{4}\left(\frac{29-1}{87}\right)$$

$$= \frac{3.28}{4 \cdot 87}$$

$$= \frac{7}{29}$$

Hence correct option is 2

Q. 6 $\lim_{x \to \frac{\pi}{4}} \frac{8\sqrt{2} - (\cos x + \sin x)^7}{\sqrt{2} - \sqrt{2}\sin 2x}$ is equal to

Option 1:
14

Option 2:
7

Option 3:
$14\sqrt{2}$

Option 4:
$7\sqrt{2}$

Correct Answer:
14

Solution:

$$\lim_{x \to \frac{\pi}{4}} \frac{8\sqrt{2} - (\cos x + \sin x)^7}{\sqrt{2}(1 - \sin 2x)}$$

$$\text{Let } x = \frac{\pi}{4} + h$$

$$= \lim_{h \to 0} \frac{8\sqrt{2} - \left(\cos\left(\frac{\pi}{4} + h\right) + \sin\left(\frac{\pi}{4} + h\right)\right)^7}{\sqrt{2}\left(1 - \sin 2\left(\frac{\pi}{4} + h\right)\right)}$$

$$= \lim_{h \to 0} \frac{8\sqrt{2} - (\sqrt{2}\cos h)^7}{\sqrt{2}(1 - \cos 2h)}$$

$$= \lim_{h \to 0} \frac{8\sqrt{2} - 8\sqrt{2}\cos^7 h}{\sqrt{2} \cdot 2\sin^2 h}$$

$$= \lim_{h\to 0} \frac{4\left(1-\cos^7 h\right)}{\sin^2 h}$$

$$= \lim_{h\to 0} \frac{4(-7)\cos^6 h(-\sin h)}{2\sin h \cdot \cos h}$$

$$= 14$$

Hence correct option is 1

Q. 7 $\lim_{n\to\infty} \frac{1}{2^n}\left(\frac{1}{\sqrt{1-\frac{1}{2^n}}}+\frac{1}{\sqrt{1-\frac{2}{2^n}}}+\frac{1}{\sqrt{1-\frac{3}{2^n}}}+\ldots.+\frac{1}{\sqrt{1-\frac{2^n-1}{2^n}}}\right)$ is equal to

Option 1:
$\frac{1}{2}$

Option 2:
1

Option 3:
2

Option 4:
-2

Correct Answer:
2

Solution:
Let $2^n = m$

As $n \to \infty$, so $m \to \infty$

$$\lim_{m\to\infty} \frac{1}{m}\left(\frac{1}{\sqrt{1-\frac{1}{m}}}+\frac{1}{\sqrt{1-\frac{2}{m}}}+\cdots\right)$$

$$= \lim_{m\to\infty} \sum_{r=1}^{m-1} \frac{1}{m\sqrt{1-\frac{r}{m}}}$$

$$= \int_0^1 \frac{1}{\sqrt{1-x}} dx$$

$$= \int_0^1 \frac{1}{\sqrt{x}} dx \quad \text{(King's Rule)}$$

$$= 2\sqrt{x}\Big|_0^1$$

$$= 2$$

Hence correct option is 3

Q. 8 If A and B are two events such that $P(A) = \frac{1}{3}, P(B) = \frac{1}{5}$ and $P(A \cup B) = \frac{1}{2}$, then $P(A \mid B') + P(B \mid A')$ is equal to

Option 1:
$\frac{3}{4}$

Option 2:
$\frac{5}{8}$

Option 3:
$\frac{5}{4}$

Option 4:
$\frac{7}{8}$

Correct Answer:
$\frac{5}{8}$

Solution:

$$P(A) = \frac{1}{3}, P(B) = \frac{1}{5}, P(A \cup B) = \frac{1}{2}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\Rightarrow \frac{1}{2} = \frac{1}{3} + \frac{1}{5} - P(A \cap B)$$

$$\Rightarrow P(A \cap B) = \frac{1}{3} + \frac{1}{5} - \frac{1}{2} = \frac{1}{30}$$

$$\therefore P(A \mid B') + P(B \mid A')$$

$$= \frac{P(A \cap B')}{P(B')} + \frac{P(B \cap A')}{P(A')}$$

$$= \frac{P(A) - P(A \cap B)}{1 - P(B)} + \frac{P(B) - P(A \cap B)}{1 - P(A)}$$

$$= \frac{\frac{1}{3} - \frac{1}{30}}{1 - \frac{1}{5}} + \frac{\frac{1}{5} - \frac{1}{30}}{1 - \frac{1}{3}}$$

$$= \frac{5}{8}$$

Hence correct option is 2

Q. 9 Let $[t]$ denote the greatest integer less than or equal to $[t]$. Then the value of the integral $\int_{-3}^{101} \left([\sin(\pi x)] + e^{[\cos(2\pi x)]}\right) dx$ is equal to

Option 1:
$\frac{52(1-e)}{e}$

Option 2:
$\frac{52}{e}$

Option 3:
$\frac{52(2+e)}{e}$

Option 4:
$\frac{104}{e}$

Correct Answer:
$\frac{52}{e}$

Solution:

$$\text{Period of } [\sin \pi x] = \frac{2\pi}{\pi} = 2$$

$$\text{Period of } e^{[\cos 2\pi x]} = \frac{2\pi}{2\pi} = 1$$

$$\therefore \int_{-3}^{101} \left([\sin \pi x] + e^{[\cos 2\pi x]}\right) dx$$

$$= 52 \left[\int_0^2 \left([\sin \pi x] + e^{[\cos 2\pi x]}\right) dx\right]$$

$$= 52 \left[\left(\int_0^1 0dx + \int_1^2 -1dx\right) + \left[\int_0^{\frac{1}{4}} e^0 dx + \int_{0/4}^4 e^{-1} dx + \int_{3/4}^{5/4} e^0 dx + \int_{5/4}^{7/4} e^{-1} dx + \int_{7/4}^2 e^0 dx\right)\right]$$

$$= \frac{52}{e}$$

Hence correct option is 2

Q. 10 Let the point $P(\alpha, \beta)$ be at a unit distance from each of the two lines $L_1 : 3x - 4y + 12 = 0$, and $L_2 : 8x + 6y + 11 = 0$. If P lies below L_1 and above L_2, then $100(\alpha + \beta)$ is equal to

Option 1:
-14

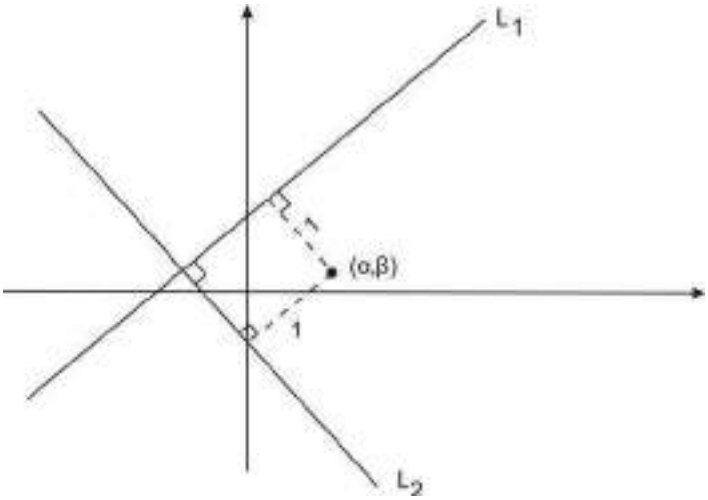
Option 2:
42

Option 3:
-22

Option 4:
14

Correct Answer:
14

Solution:



P and origin lie on same side of L_1

$$\Rightarrow \frac{3\alpha - 4\beta + 12}{12} > 0$$

$$\Rightarrow 3\alpha - 4\beta + 12 > 0 \quad \text{.....(i)}$$

Similarly for L_2

$$\frac{8\alpha + 6\beta + 11}{11} > 0$$

$$\Rightarrow 8\alpha + 6\beta + 11 > 0 \quad \text{.....(ii)}$$

Also distance from L_1 and $\mathrm{L}_2 = 1$

$$\Rightarrow \frac{|3\alpha - 4\beta + 12|}{5} = 1$$

$$\Rightarrow 3\alpha - 4\beta + 12 = 5 \quad \text{(using (ii))}$$

And $\dfrac{|8\alpha + 6\beta + 11|}{10} = 1$

$$8\alpha + 6\beta + 11 = 10$$

Solving these 2 equations

$$\alpha = \frac{-23}{25}, \quad \beta = \frac{53}{50}$$

$$\therefore 100\,(\alpha_1\beta) = 14$$

Hence correct option is 4

Q. 11 Let a smooth curve $y = f(x)$ be such that the slope of the tangent at any point (x, y) on it is directly proportional to $\left(\frac{-y}{x}\right)$. If the curve passes through the points $(1, 2)$ and $(8, 1)$, then $\left|y\left(\frac{1}{8}\right)\right|$ is equal to

Option 1:
$2\,log_e\,2$

Option 2:
4

Option 3:
1

Option 4:
$4\,log_e\,2$

Correct Answer:
4

Solution:

$$\frac{dy}{dx} = k\left(-\frac{y}{x}\right)$$

$$\Rightarrow \frac{dy}{y} = -k\frac{dx}{x}$$

$$\Rightarrow \ln y = -k\ln x + \ln C$$

$$\Rightarrow y = cx^{-k}$$

$$(1, 2) \Rightarrow 2 = c$$

$$(8, 1) \Rightarrow 1 = 2 \cdot (8)^{-k}$$

$$\Rightarrow 8^k = 2$$

$$\Rightarrow k = \frac{1}{3}$$

$\therefore y = 2x^{-\frac{1}{3}}$

$\therefore |y\left(\frac{1}{8}\right)) = 2 \cdot \left(\frac{1}{8}\right)^{-\frac{1}{3}} = 2 \cdot 2 = 4$

Hence correct option is 2

Q. 12 If the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the line $\frac{x}{7} + \frac{y}{2\sqrt{6}} = 1$ on the x-axis and the line $\frac{x}{7} - \frac{y}{2\sqrt{6}} = 1$ on the y-axis, then the eccentricity of the ellipse is

Option 1:
$\frac{5}{7}$

Option 2:
$\frac{2\sqrt{6}}{7}$

Option 3:
$\frac{3}{7}$

Option 4:
$\frac{2\sqrt{5}}{7}$

Correct Answer:
$\frac{5}{7}$

Solution:
Line 1 and ellipse cut x-axis at $x = 7$ and line 2 and ellipse cut y-axis at $y = 2\sqrt{6}$

$\therefore \quad a = 7, \quad b = 2\sqrt{6}$

$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{24}{49} = \frac{25}{49}$

$\Rightarrow e = \frac{5}{7}$

Hence correct option is 1

Q. 13 The tangents at the points $A(1, 3)$ and $B(1, -1)$ on the parabola $y^2 - 2x - 2y = 1$ meet at the point P. Then the area (in unit^2) of the triangle PAB is:

Option 1:
4

Option 2:
6

Option 3:
7

Option 4:
8

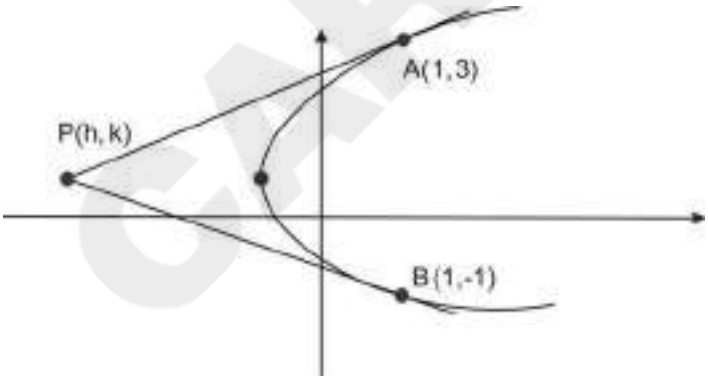
Correct Answer:
8

Solution:

$y^2 - 2x - 2y = 1$

$\Rightarrow y^2 - 2y = 2x + 1$

$\Rightarrow (y - 1)^2 = 2(x + 1)$



Let P be (h, k)

Chord of Contact : $T = 0$

$$ky - 2\left(\frac{x + h}{2}\right) - 2\left(\frac{y + k}{2}\right) = 1$$

$\Rightarrow \quad ky - x - h - y - k = 1$

$\Rightarrow \quad -x + y(k-1) - h - k - 1 = 0$

This should be same as $x = 1$

$\therefore k - 1 = 0 \Rightarrow k = 1$

and $-h - k - 1 = 1$

$\Rightarrow -h - 1 - 1 = 1$

$\Rightarrow h = -3$

$\therefore \quad P(-3, 1)$

$$\text{Area of PAB} = \left| \frac{1}{2} \begin{vmatrix} -3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & -1 & 1 \end{vmatrix} \right|$$

$= 8$

Hence correct option is 4

Q. 14 Let the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{7} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{\alpha} = \frac{1}{25}$ coincide. Then the length of the latus rectum of the hyperbola is:

Option 1:
$\frac{32}{9}$

Option 2:
$\frac{18}{5}$

Option 3:
$\frac{27}{4}$

Option 4:
$\frac{27}{10}$

Correct Answer:

$\frac{27}{10}$

Solution:

For the eclipse are

$= 4\sqrt{1-\frac{7}{16}}$

$= 4 \times \frac{3}{4}$

$= 3$

For hyperbola ,ae should be 3

$\frac{x^2}{\left(\frac{144}{25}\right)} - \frac{y^2}{\left(\frac{\alpha}{25}\right)} = 1$

$\Rightarrow \quad \frac{12}{5} \cdot \sqrt{1+\frac{\alpha}{144}} = 3$

$\Rightarrow \quad \sqrt{\frac{144+\alpha}{144}} = \frac{15}{12}$

$\Rightarrow \quad 144 + \alpha = 225$

$\Rightarrow \quad \alpha = 81$

Lotus rectum $= \frac{2b^2}{a}$

$= \frac{2 \cdot 81}{25 \cdot \frac{12}{5}} = \frac{27}{10}$

Hence correct option is 4

Q. 15 A plane E is perpendicular to the two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$, and passes through the point $P(1, -1, 1)$. If the distance of the plane E from the point $Q(a, a, 2)$ is $3\sqrt{2}$, then $(PQ)^2$ is equal to

Option 1:

9

Option 2:
12

Option 3:
21

Option 4:
33

Correct Answer:
21

Solution:
Normal $\tilde{n}$ to plane is $\tilde{n}_1 \times \vec{n_2}$

$$= \begin{vmatrix} i & j & k \\ 2 & -2 & 1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= (-4+1)\mathrm{i} - (4-1)\mathrm{j} + (-2+2)\mathrm{k}$$

$$= -3\mathrm{i} - 3\mathrm{j}$$

Equation of required plane through $\mathrm{P}(1,-1,1)$ is

$$-3(\mathrm{x}-1) - 3(\mathrm{y}+1) + 0(\mathrm{z}-1) = 0$$

$$\Rightarrow \quad \mathrm{x} - 1 + \mathrm{y} + 1 = 0$$

$$\Rightarrow \mathrm{x} + \mathrm{y} = 0$$

Distance from $\mathrm{Q}(\mathrm{a},\mathrm{a},2)$

$$\frac{|\mathrm{a}+\mathrm{a}|}{\sqrt{2}} = 3\sqrt{2}$$

$$\Rightarrow (\mathrm{a}) = 3$$

$$\Rightarrow \mathrm{a} = \pm 3$$

$$\therefore \quad \mathrm{Q} \text{ is } (3,3,2) \text{ or } (-3,-3,2)$$

$$\therefore \mathrm{PQ}^2 = (3-1)^2 + (3+1)^2 + (2-1)^2$$

$$= 4 + 16 + 1$$

$$= 21$$

Hence correct option is 3

Q. 16 The shortest distance between the lines $\frac{x+7}{-6}=\frac{y-6}{7}=z$ and $\frac{7-x}{2}=y-2=z-6$ is

Option 1:
$2\sqrt{29}$

Option 2:
1

Option 3:
$\sqrt{\frac{37}{29}}$

Option 4:
$\frac{\sqrt{29}}{2}$

Correct Answer:
$2\sqrt{29}$

Solution:

$L_1: \frac{x+7}{-6}=\frac{v-6}{7}=\frac{z-0}{1}$

$\vec{a_1}=-7i+6j, \quad \vec{b_1}=-6i+7j+k$

$L_2: \frac{7-x}{2}=y-2=z-6$

$\Rightarrow \frac{x-7}{-2}=\frac{y-2}{1}=\frac{z-6}{1}$

$\vec{a}_2=7i+2j+6k, \quad \vec{b}_2=-2i+j+k$

$$\text{Shortest distance} = \left|\frac{(\vec{a_2}-\vec{a_1})\cdot\left(\vec{b}_1\times\vec{b}_2\right)}{\left|\vec{b}_1\times\vec{b}_2\right|}\right|$$

Now $\vec{b}_1\times\vec{b}_2=3i+2j+4k$

$$\therefore \quad \mathrm{S \cdot D \cdot} = \left| \frac{(-14\mathrm{i} + 4\mathrm{j} - 6\mathrm{k}) \cdot (3\mathrm{i} + 2\mathrm{j} + 4\mathrm{k})}{\sqrt{9+4+16}} \right|$$

$$= 2\sqrt{29}$$

Hence correct option is 1

Q. 17 Let $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and let $\vec{b}$ be a vector such that $\vec{a} \times \vec{b} = 2\hat{i} - \hat{k}$ and $\vec{a} \cdot \vec{b} = 3$. Then the projection of $\vec{b}$ on the vector $\vec{a} - \vec{b}$ is :

Option 1:

$\frac{2}{\sqrt{21}}$

Option 2:

$2\sqrt{\frac{3}{7}}$

Option 3:

$\frac{2}{3}\sqrt{\frac{7}{3}}$

Option 4:

$\frac{2}{3}$

Correct Answer:

$\frac{2}{\sqrt{21}}$

Solution:

$|\vec{a} \times \vec{b}| = \sqrt{4+1} = \sqrt{5}$

Also $\quad |\vec{a}|^2 |\vec{b}|^2 = |\vec{a} \cdot \vec{b}|^2 + |\vec{a} \times \vec{b}|^2$

$\Rightarrow 6|\vec{b}|^2 = 9 + 5$

$\Rightarrow |\vec{b}|^2 = \frac{7}{3}$

Required projection $= \frac{\vec{b} \cdot (\vec{a} - \vec{b})}{|\vec{a} - \vec{b}|}$ .....(i)

$$\text{Now} |\vec{a}-\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a}\cdot\vec{b} = 6 + \frac{7}{3} - 2\cdot 3$$

$$|\vec{a}-\vec{b}| = \sqrt{\frac{7}{3}}$$

Using (i), projection $= \dfrac{\vec{a}\cdot\vec{b} - |\vec{b}|^2}{\sqrt{\frac{7}{3}}}$

$$= \frac{3-\frac{7}{3}}{\sqrt{\frac{7}{3}}} = \frac{2}{\sqrt{21}}$$

Hence correct option is 1

Q. 18 If the mean deviation about median for the numbers $3, 5, 7, 2k, 12, 16, 21, 24$, arranged in the ascending order, is 6 then the median is

Option 1:
11.5

Option 2:
10.5

Option 3:
12

Option 4:
11

Correct Answer:
11

Solution:
There are 8 (eve) number of terms

$$\therefore \text{ Median } = \frac{2k+12}{2} = k+6$$

Mean deviation about median

$$6 = \frac{|k+3| + |k+1| + |k-1| + |k-6| + |6-k| + |10-k| + |15-k| + |18-k!}{8}$$

Now acseding order, so

$7 < 2k < 12 \Rightarrow 3.5 < k < 6$

$\Rightarrow \quad 48 = 58 - 2k$

$\Rightarrow k = 5$

$\text{Median } = k + 6 = 11$

Hence correct option is 4

Q. 19 $2\sin\left(\frac{\pi}{22}\right)\sin\left(\frac{3\pi}{22}\right)\sin\left(\frac{5\pi}{22}\right)\sin\left(\frac{7\pi}{22}\right)\sin\left(\frac{9\pi}{22}\right)$ is equal to :

Option 1:
$\frac{3}{16}$

Option 2:
$\frac{1}{16}$

Option 3:
$\frac{1}{32}$

Option 4:
$\frac{9}{32}$

Correct Answer:
$\frac{1}{16}$

Solution:
Given expression can be converted to cosine terms

$= 2\cos\left(\frac{10\pi}{22}\right)\cos\left(\frac{8\pi}{22}\right)\cos\left(\frac{6^2\pi}{22}\right)\cos\left(\frac{4\pi}{22}\right)\cdot\cos\left(\frac{2\pi}{22}\right)$

$= 2\cos\left(\frac{\pi}{11}\right)\cdot\cos\left(\frac{2\pi}{11}\right)\cdot\cos\left(\frac{3\pi}{11}\right)\cdot\cos\left(\frac{4\pi}{11}\right)\cdot\cos\left(\frac{5\pi}{11}\right)$

$$=2\cos\left(\frac{\pi}{11}\right)\cdot\cos\left(\frac{2\pi}{11}\right)\cdot\cos\left(\frac{4\pi}{11}\right)\cdot\left[-\cos\left(\pi-\frac{3\pi}{11}\right)\right]\cdot\left[-\cos\left(\pi+\frac{5\pi}{11}\right)\right]$$

$$=2\cos\left(\frac{\pi}{11}\right)\cdot\cos\left(\frac{2\pi}{11}\right)\cdot\cos\left(\frac{4\pi}{11}\right).\cos\left(\frac{8\pi}{11}\right)\cdot\cos\left(\frac{16\pi}{11}\right)$$

$$=\frac{2\cdot\sin\left(\frac{32\pi}{11}\right)}{2^5\cdot\sin\left(\frac{\pi}{11}\right)}$$

$$=\frac{1}{16}$$

hence correct option is 2

Q. 20 Consider the following statements :
P : Ramu is intelligent.
Q : Ramu is rich.
R :Ramu is not honest.
The negation of the statement "Ramu is intelligent and honest if and only if Ramu is not rich" can be expressed as :

Option 1:
$((P\wedge(\sim R))\wedge Q)\wedge((\sim Q)\wedge((\sim P)\vee R))$

Option 2:
$((P\wedge R)\wedge Q)\vee((\sim Q)\wedge((\sim P)\vee(\sim R)))$

Option 3:
$((P\wedge R)\wedge Q)\wedge((\sim Q)\wedge((\sim P)\vee(\sim R)))$

Option 4:
$((\mathrm{P}\wedge(\sim\mathrm{R}))\wedge\mathrm{Q})\vee((\sim\mathrm{Q})\wedge((\sim\mathrm{P})\vee\mathrm{R}))$

Correct Answer:
$((\mathrm{P}\wedge(\sim\mathrm{R}))\wedge\mathrm{Q})\vee((\sim\mathrm{Q})\wedge((\sim\mathrm{P})\vee\mathrm{R}))$

Solution:
Given statement is $(\mathrm{p}\wedge\sim\mathrm{r})\longleftrightarrow\sim\mathrm{q}$

$\equiv((\mathrm{p}\wedge\sim\mathrm{r})\rightarrow\sim\mathrm{q})\wedge(\sim\mathrm{q}\rightarrow(\mathrm{p}\wedge\sim\mathrm{r}))$

$\equiv(\sim(\mathrm{p}\wedge\sim\mathrm{r})\vee\sim\mathrm{q})\wedge(\mathrm{q}\vee(\mathrm{p}\wedge\sim\mathrm{r}))$

It's negation is

$$\sim(\sim(p\wedge\sim r)\vee\wedge q)\vee\sim(q\vee(p\wedge\sim r))$$

$$\equiv((p\wedge\sim r)\wedge q)\vee(\sim q\wedge\sim(p\wedge\sim r))$$

Hence correct option is 4

Q. 21 Let $A=\{1,2,3,4,5,6,7\}$. Define $B=\{T\subseteq A:either 1\notin T or 2\in T\}$ and $C=\{T\subseteq A:$ the sum of all the elements of t is a prime number $\}$. Then the number of elements in the set $B\cup C$ is_____

Correct Answer:
107

Solution:
Let us first find $(B\cup C)'=B'\cap C'$

$\Rightarrow$ Susets having 1 and not having 2 and whose sum of elements is not prime.

$\Rightarrow$ Subsets of $\{3,4,5,6,7\}$ such that 1^+ sum of elements is it a prime number

Such subsets with 0 number of elements = 1

Such subsets with 1 number of elements $=3(\{3\},\{5\},\{7\})$

Such elements with 2 number of elements = 7

With 3 elements = 6

with 4 elements = 3

with 5 elements = 1

$$\therefore \text{Total in } (B\cup C)'=1+3+7+6+3+1=21$$

$$\text{Required } =2^7-21=107$$

Hence answer is 107

Q. 22 Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0) = p, p \neq 0, \text{and } f(1) = \frac{1}{3}$. If the equations $f(x) = 0$ and $fofofof(x) = 0$ have a common real root, then $f(-3)$ is equal to_____

Correct Answer: 25

Solution:

Let polynomial be $x^2 + ax + p$

$$f(1) = \frac{1}{3} \Rightarrow 1 + a + p = \frac{1}{3} \Rightarrow a = -p - \frac{2}{3}$$

$$\therefore f(x) = x^2 - \left(p + \frac{2}{3}\right)x + p$$

Let α be the common root

$$\therefore \alpha^2 - \left(p + \frac{2}{3}\right)\alpha + p = 0 \qquad \text{.....(i)}$$

and

$$f(f(f(f(\alpha)))) = 0$$

$$\Rightarrow f(f(f(0))) = 0$$

$$\Rightarrow f(f(p)) = 0$$

$$\therefore f(p) \text{ is a root of } f(x) = 0$$

$$\text{If } f(p) = \alpha \Rightarrow p^2 - \left(p + \frac{2}{3}\right)p + p = \alpha$$

$$\Rightarrow \frac{p}{3} = \alpha$$

Using (i)

$$\frac{p^2}{9} - \left(p + \frac{2}{3}\right)\frac{p}{3} + p = 0$$

$$\Rightarrow p = \frac{7}{2} \quad (\text{as } p \neq 0)$$

$$\therefore f(x) = x^2 - \frac{25}{6}x + \frac{7}{2}$$

$\therefore f(-3) = 25$

Hence correct answer is 25

Q. 23 Let $A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}, a, b \in \mathbb{R}$.

If for some $n \in N, A^n = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 96 \\ 0 & 0 & 1 \end{bmatrix}$ then $n + a + b$ is equal to_____.

Correct Answer:
24

Solution:

$$A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix}$$

$= I + B$ (let)

where $B = \begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix}$

Now $B^2 = \begin{bmatrix} 0 & 0 & ab \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

and $B^3 = 0$

So $A^n = (I + B)^n$

$= I^n + {}^nC_1 \cdot B + {}^nC_2 \cdot B^2 + 0 + 0 + \cdots$

$$= \mathrm{I} + \mathrm{n}\begin{bmatrix} 0 & a & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{bmatrix} + \frac{\mathrm{n}(\mathrm{n}-1)}{2}$$

$$= \begin{bmatrix} 0 & 0 & ab \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & na & na + \frac{n(n-1)}{2}ab \\ 0 & 1 & nb \\ 0 & 0 & 1 \end{bmatrix}$$

Comparing with give matrix

$$\mathrm{na} = 48, \mathrm{nb} = 96, \frac{\mathrm{n}(\mathrm{n}-1)}{2}\mathrm{ab} = 2112$$

$$\Rightarrow \mathrm{a} = 4, \mathrm{b} = 8, \mathrm{n} = 12$$

$$\therefore \mathrm{n} + \mathrm{a} + \mathrm{b} = 24$$

Hence answer is 24

Q. 24 The sum of the maximum and minimum values of the function $\mathrm{f}(\mathrm{x}) = |5\mathrm{x} - 7| + \left[\mathrm{x}^2 + 2\mathrm{x}\right]$ in the interval $\left[\frac{5}{4}, 2\right]$, where $[\mathrm{t}]$ is the greatest integer $\leq \mathrm{t}$, is_____

Correct Answer: 15

Solution:

$$\mathrm{f}(\mathrm{x}) = |5\mathrm{x} - 7| + \left[\mathrm{x}^2 + 2\mathrm{x}\right]$$

$$= |5\mathrm{x} - 7| + \left[\mathrm{x}^2 + 2\mathrm{x} + 1 - 1\right]$$

$$= |5\mathrm{x} - 7| + \left[(\mathrm{x}+1)^2\right] - 1$$

$$\text{Critical points are } \mathrm{x} = \frac{5}{4}, \frac{7}{5}, \sqrt{6} - 1,$$

$$\sqrt{7} - 1, \sqrt{8} - 1, 2$$

$$\mathrm{f}\left(\frac{5}{4}\right) = \frac{19}{4}, \quad \mathrm{f}\left(\frac{7}{5}\right) = 4,$$

Also for $x > \frac{7}{5}, f(x)$ is increasing

$\therefore f(2) = 3 + 8 = 11$

$\therefore$ Minimum $= 4$, Maximum $= 11$

sum $= 15$

Hence answer is 15

Q. 25 Let $y = y(x)$ be the solution of the differential equation
$\frac{dy}{dx} = \frac{4y^3 + 2yx^2}{3xy^2 + x^3}, y(1) = 1.$
If for some $n \in N, y(2) \in [n-1, n)$, then n is equal to_____.

Correct Answer: 3

Solution:

$$\frac{dy}{dx} = \frac{4\left(\frac{y}{x}\right)^3 + 2\left(\frac{y}{x}\right)}{3\left(\frac{y}{x}\right)^2 + 1}$$

Let $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$\therefore v + x\frac{dv}{dx} = \frac{4v^3 + 2v}{3v^2 + 1}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{4v^3 + 2v}{3x^2 + 1} - v$$

$$\Rightarrow x\frac{dv}{dx} = \frac{v^3 + v}{3v^2 + 1}$$

$$\Rightarrow \frac{(3v^2 + 1)\,dv}{v^3 + v} = \frac{dx}{x}$$

Integrating

$$\Rightarrow \ln|v^3 + v| = \ln x + \ln k$$

$$\Rightarrow \left(\frac{y}{x}\right)^3 + \left(\frac{y}{x}\right) = kx$$

Uising $y(1) = 1$

$\Rightarrow k = 2$

Put $x = 2$

$\frac{y^3}{8} + \frac{y}{2} = 2 \times 2$

$\Rightarrow y^3 + 4y = 32$

$\Rightarrow y^3 + 4y - 32 = 0$

$\Rightarrow 2 \leq y < 3 \quad (\text{as } f(2) < 0, f(3) > 0)$

$\therefore n = 3$

Hence answer is 3

Q. 26 Let f be a twice differentiable function on R. If $f'(0) = 4$ and $f(x) + \int_0^x (x - t)f'(t)dt = \left(e^{2x} + e^{-2x}\right)\cos 2x + \frac{2}{a}x$, then $(2a + 1)^5 a^2$ is equal to_____

Correct Answer: 8

Solution:

$f(x) + x\int_0^x f'(t)dt - \int_0^x tf'(t)dt$

$= \left(e^{2x} + e^{-2x}\right)\cos 2x + \frac{2}{a}x$

Differentiating the equation w.r.t.x

$f'(x) + \int_0^x f'(t)dt + x \cdot f'(x) - x \cdot f'(x)$

$= 2\left(e^{2x} - e^{-2x}\right)\cos 2x - 2\left(e^{2x} + e^{-2x}\right)\sin 2x + \frac{2}{a}$

$\Rightarrow f'(x) + f(x) - f(0) = 2\left(e^{2x} - e^{-2x}\right)\cos 2x - 2\left(e^{2x} + e^{-2x}\right)\sin 2x + \frac{2}{a}$

Put $x = 0$

$f'(0) = \frac{2}{a}$

$$\Rightarrow 4 = \frac{2}{a} \Rightarrow a = \frac{1}{2}$$

$$\therefore (2a+1)^5 \cdot a^2$$

$$= 2^5 \cdot \left(\frac{1}{2}\right)^2$$

$$= 8$$

Hence answer is 8

Q. 27 Let $a_n = \int_{-1}^{n} \left(1 + \frac{x}{2} + \frac{x^2}{3} + \ldots\ldots + \frac{x^{n-1}}{n}\right) dx$ for every $n \in N$. Then the sum of all at the elements of the set $\{n \in N : a_n \in (2, 30)\}$ is_____

Correct Answer:
5

Solution:

$$a_n = x + \frac{x^2}{4} + \frac{x^3}{9} + \cdots + \frac{x^n}{n^2}\Big|_{-1}^{n}$$

$$= (n+1) + \frac{n^2-1}{4} + \frac{n^3+1}{9} + \cdots\cdots + \frac{n^n - (-1)^n}{n^2}$$

$$\therefore \quad a_1 = 2$$

$$a_2 = 2 + 1 + \frac{4-1}{4} = \frac{15}{4}$$

$$a_3 = 4 + \frac{8}{4} + \frac{28}{9} = \frac{100}{9}$$

$$a_4 = 5 + \frac{15}{4} + \frac{65}{9} + \cdots > 30$$

$$\therefore \text{ Sum } = 2 + 3 = 5$$

Hence answer is 5

Q. 28 If the circles $x^2 + y^2 + 6x + 8y + 16 = 0$ and $x^2 + y^2 + 2(3 - \sqrt{3})x + 2(4 - \sqrt{6})y = k + 6\sqrt{3} + 8\sqrt{6}, k > 0$, touch internally at the point $P(\alpha, \beta)$, then $(\alpha + \sqrt{3})^2 + (\beta + \sqrt{6})^2$ is equal to_____

Correct Answer: 25

Solution:

C_1 : Centre $C_1(-3, -4)$, Radius $r_1 = \sqrt{9 + 16 - 16} = 3$

C_2 : Centre$C_2(\sqrt{3} - 3, \sqrt{6} - 4)$and

radies $r_2 = \sqrt{(\sqrt{3} - 3)^2 + (\sqrt{6} - 4)^2 + k + 6\sqrt{3} + 8\sqrt{6}}$

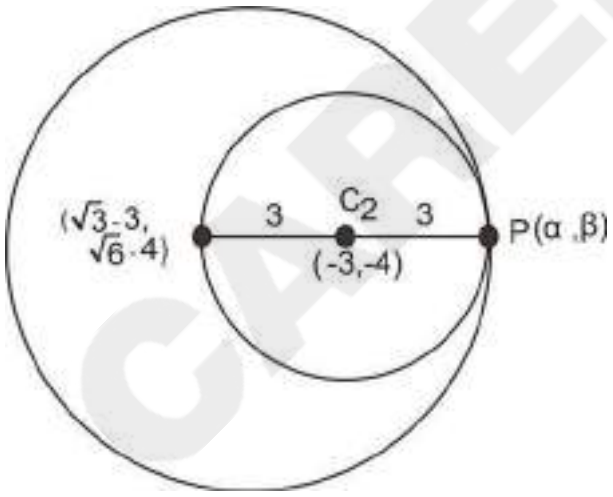
$= \sqrt{k + 34}$

Touch internally

$C_1C_2 = |r_1 - r_2|$

$\Rightarrow \sqrt{3 + 6} = |\sqrt{k + 34} - 3|$

$\Rightarrow k = 2$



Clearly C_2 is mid − point of P&C_1

$\frac{\alpha + \sqrt{3} - 3}{2} = -3$ and $\frac{\beta + \sqrt{6} - 4}{2} = -4$

$\Rightarrow \alpha = -3 - \sqrt{3}, \quad \beta = -4 - \sqrt{6}$

$\therefore$ Required expression

$(\alpha+\sqrt{3})^2+(\beta+\sqrt{6})^2=9+16=25$

Hence answer is 25

Q. 29 Let the area enclosed by the x-axis, and the tangent and normal drawn to the curve $4x^3-3xy^2+6x^2-5xy-8y^2+9x+14=0$ at the point $(-2,3)$ be A. Then $8A$ is equal to_____.

Correct Answer: 170

Solution:
Differentiating the given equation

$12x^2-6xyy'-3y^2+12x-5xy'-5y-16yy'+9=0$

$\text{Put } x=-2, y=3$

$\Rightarrow y'=-\frac{9}{2}$

$\Rightarrow m_N=\frac{2}{9}$

Equation of tangent

$y-3=-\frac{9}{2}(x+2)$

$\text{Put } y=0 \Rightarrow x=-\frac{4}{3}$

Equation of normal

$y-3=\frac{2}{9}(x+2)$

$\text{Put } y=0 \Rightarrow x=-\frac{31}{2}$

$\therefore \text{ Area } A=\left|\frac{1}{2}\times\left(-\frac{31}{2}+\frac{4}{3}\right)\times 3\right|$

$=\frac{85}{4}$

$\Rightarrow 8A=170$

Hence answer is 170

Q. 30 Let $x = \sin\left(2\tan^{-1}\alpha\right)$ and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$. If $S = \left\{\alpha \in \mathbb{R} : y^2 = 1 - x\right\}$, then $\sum_{\alpha\in S} 16\alpha^3$ is equal to_____

Correct Answer:
130

Solution:
$L = \sin\left(2\tan^{-1}\alpha\right)$

Let $\tan^{-1}\alpha = \theta \Rightarrow \tan\theta = \alpha.$

$\therefore \quad x = \sin(2\theta)$

$= \frac{2\tan\theta}{1+\tan^2\theta}$

$= \frac{2\alpha}{1+\alpha^2}$ .....(i)

And $y = \sin\left(\frac{1}{2}\tan^{-1}\left(\frac{4}{3}\right)\right)$

Let $\tan^{-1}\left(\frac{4}{3}\right) = \beta$

$\tan\beta = \frac{4}{3}$

$\Rightarrow \cos\beta = \frac{3}{5}$

$\Rightarrow 1 - 2\sin^2\left(\frac{\beta}{2}\right) = \frac{3}{5}$

$\Rightarrow 2\sin^2\left(\frac{\beta}{2}\right) = \frac{2}{5}$

$\Rightarrow \quad \sin^2\left(\frac{\beta}{2}\right) = \frac{1}{5}$

$$\Rightarrow \quad y^2 = \frac{1}{5}$$

$$\therefore \quad y^2 = 1 - x$$

$$\frac{1}{5} = 1 - \frac{2\alpha}{1+\alpha^2}$$

$$\Rightarrow \frac{2\alpha}{1+\alpha^2} = \frac{4}{5}$$

$$\Rightarrow 5\alpha = 2 + 2\alpha^2$$

$$\Rightarrow \quad \alpha = 2, \frac{1}{2}$$

$$16\alpha^3 = 16 \cdot 8 + 16 \cdot \frac{1}{8} = 130$$

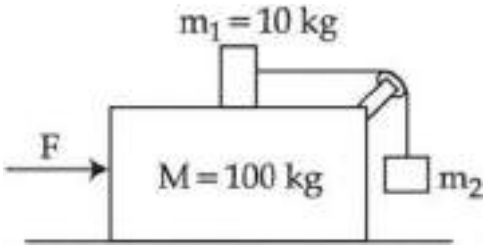
Hence answer is 130

CAREERS360

26 July - Morning Shift

Physics

Q. 1 Three masses $M = 100\ kg,\ m_1 = 10\ kg$ and $m_2 = 20\ kg$ are arranged in a system as shown in figure. All the surfaces are frictionless and strings are inextensible and weightless. The pulleys are also weightless and frictionless. A force f is applied on the system so that the mass m_2 moves upward with an acceleration of $2\ ms^{-2}$. The value of f is : $(\text{Take } g = 10\ ms^{-2})$



Option 1:
$3360\ N$

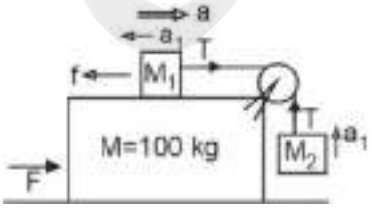
Option 2:
$3380\ N$

Option 3:
$3120\ N$

Option 4:
$3240\ N$

Correct Answer:
$3380\ N$

Solution:



$f \rightarrow$ Pseudo force on m_1 in the frame of block of mass M

$f = m_1 a \rightarrow (1)$

Let the acceleration of system and block of mass m_1 be a and a_1 respectively

$f - T = m_1 a_1 \rightarrow (2)$

$T - m_2 g = m_2 a_2 \rightarrow (3)$

Common acceleration $= a = \dfrac{F}{m_1 + m_2 + m}$

$a = \dfrac{F}{130} \rightarrow (4)$

from eq^n 1,2 & 4

$\dfrac{F}{13} - T = 10 a_1, = 10(2) = 20$

$(a_1 = 2 m/s^2) - - - - (\text{Given})$

$T - 200 = 20(2)$

$T = 240$

$\dfrac{F}{13} - 240 = 20$

$F = 260 \times 13$

$F = 3380 N$

Hence (2) is correct option

Q. 2 A radio can tune to any station in 6 MHz to 10 MHz band. The value of corresponding wavelength bandwidth will be :

Option 1:
4 m

Option 2:
20 m

Option 3:
30 m

Option 4:
50 m

Correct Answer:

$20\ \mathrm{m}$

Solution:

We know that,

$$c = f\lambda$$

$$\lambda_1 = \frac{c}{f_1} = \frac{3 \times 10^8}{6 \times 10^6}$$

$$\lambda_1 = 50\ \mathrm{m}$$

$$\lambda_2 = \frac{c_2}{f_2} = \frac{3 \times 10^8}{10 \times 10^6}$$

$$\lambda_2 = 30\ \mathrm{m}$$

$$\text{Wavelength band} = \Delta\lambda = \lambda_1 - \lambda_2$$

$$= 20\ \mathrm{m}$$

Hence (2) is correct option.

Q. 3 The disintegration rate of a certain radioactive sample at any instant is 4250 disintegrations per minute. 10 minutes later, the rate becomes 2250 disintegrations per minute. The approximate decay constant is : $(\text{Take } \log_{10} 1.88 = 0.274)$

Option 1:

$0.02\ \mathrm{min}^{-1}$

Option 2:

$2.7\ \mathrm{min}^{-1}$

Option 3:

$0.063\ \mathrm{min}^{-1}$

Option 4:

$6.3\ \mathrm{min}^{-1}$

Correct Answer:

$0.02\ \mathrm{min}^{-1}$

Solution:

$A_0 = 4250$

$A_t = 2250$ disintegration per min

$A_t = A_0 2^{-t/T_{v2}}$

$2250 = 42502^{-10/T_{1/2}}$

$\frac{9}{17} = 2^{-10/T_{1/2}}$

$-\log_{10}\left(\frac{9}{17}\right) = \frac{10}{T_{1/2}}\log_{10} 2$

$\because \frac{\ln(2)}{T_{1/2}} = \lambda$

$(\log_e 10)\frac{\log_{10}(1.88)}{10} = \lambda$

$\lambda = 0.0274 \times 2.303$

$= 0.063 \text{ min}^{-1}$

Hence (3) is correct option.

Q. 4 A parallel beam of light of wavelength 900 nm and intensity 100 Wm^{-2} is incident on a surface perpendicular to the beam. The number of photons crossing 1 cm^2 area perpendicular to the beam in one second is :

Option 1:

3×10^{16}

Option 2:

4.5×10^{16}

Option 3:

4.5×10^{17}

Option 4:

4.5×10^{20}

Correct Answer:

3×10^{16}

Solution:

$\lambda = 900 \text{ nm}$

$\mathrm{I} = 100 \text{ W/m}^2$

$$\mathrm{I} = \frac{\text{Energy}}{(\text{Area})(\text{time})} = \frac{\mathrm{n(nf)}}{(\text{ Area})(\text{time})}$$

$$100\frac{\mathrm{w}}{\mathrm{m}^2} = \left(\frac{\mathrm{n}}{\mathrm{t}}\right) \times \left(\frac{6.63 \times 10^{-34} \times 3 \times 10^8}{900 \times 10^{-9} \times 10^{-4}}\right)$$

$\frac{\mathrm{n}}{\mathrm{t}} = \text{no.of photons per unit time crossing area } 1\text{cm}^2$

$$\frac{\mathrm{n}}{\mathrm{t}} = \frac{9 \times 10^{-9}}{19.89 \times 10^{-26}}$$

$$= \frac{9}{19.89} \times 10^{17} \quad = \frac{9}{19.89} \times 10^{17}$$

$$\frac{\mathrm{n}}{\mathrm{t}} = 4.5 \times 10^{16}$$

Hence (2) is correct option

Q. 5 In Young's double slit experiment, the fringe width is $12\ \text{mm}$. If the entire arrangement is placed in water of refractive index $\frac{4}{3}$, then the fringe width becomes (in mm) :

Option 1:
16

Option 2:
9

Option 3:
48

Option 4:
12

Correct Answer:
9

Solution:

$$\beta = \frac{\lambda D}{d} = 12 \times 10^{-3}\ \text{m}$$

Fringe width in water $= \beta' = \frac{\beta}{\mu}$

$$\beta' = \frac{12 \times 10^{-3}}{4/3} = 9 \times 10^{-3}\ \text{m}$$

$$= 9\ \text{mm}$$

Hence (2) option is correct

Q. 6 The magnetic field of a plane electromagnetic wave is given by :

$$\vec{B} = 2 \times 10^{-8} \sin\left(0.5 \times 10^{3} x + 1.5 \times 10^{11} t\right) \hat{j}\ \text{T}$$

The amplitude of the electric field would be :

Option 1:
$6\ \text{Vm}^{-1}$ along x-axis

Option 2:
$3\ \mathrm{Vm}^{-1}$ along z-axis

Option 3:
$6\ \mathrm{Vm}^{-1}$ along z-axis

Option 4:
$2 \times 10^{-8}\ \mathrm{Vm}^{-1}$ along z-axis

Correct Answer:
$6\ \mathrm{Vm}^{-1}$ along z-axis

Solution:
$\bar{\mathrm{B}} = \left(2 \times 10^{-8}\right) \sin\left(0.5 \times 10^{3}\mathrm{x} + 1.5 \times 10^{11}\mathrm{t}\right) \hat{j}$

We know that,

$$\frac{\mathrm{E}_0}{\mathrm{B}_0} = \mathrm{C}$$

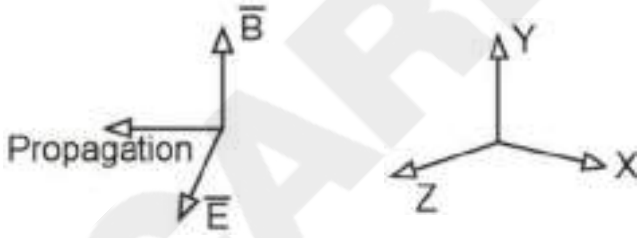
Eo & B_0 are the amplitude of electric field and magntic field

$$\mathrm{E}_0 = \mathrm{C} \times \mathrm{B}_0 = 3 \times 10^{8} \times 2 \times 10^{-8}$$

$$\mathrm{E} = 6\ \mathrm{V/m}$$

From the eq^{n} of $\bar{B}$, it can be said that the propaget of wave is taking place along negative x-axis

Also, $\bar{\mathrm{E}} \times \bar{\mathrm{B}}$ points along the direction of propagation



$\bar{\mathrm{E}}$ points along +z -axis

Hence (3) is correct option.

Q. 7 In a series LR circuit $\mathrm{X_L} = \mathrm{R}$ and power factor of the circuit is P_1. When capacitor with capacitance C such that $\mathrm{X_L} = \mathrm{X_C}$ is put in series, the power factor becomes P_2. The ratio $\frac{\mathrm{P}_1}{\mathrm{P}_2}$ is :

Option 1:

$\frac{1}{2}$

Option 2:

$\frac{1}{\sqrt{2}}$

Option 3:

$\frac{\sqrt{3}}{\sqrt{2}}$

Option 4:

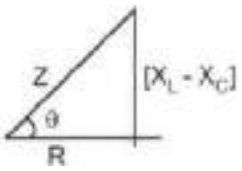
$2:1$

Correct Answer:

$\frac{1}{2}$

Solution:

$$\text{Power factor} = \cos\phi = \frac{\mathrm{R}}{\mathrm{Z}}$$



$$\cos\phi = \frac{\mathrm{R}}{\sqrt{\mathrm{R}^2 + |\mathrm{X_L} - \mathrm{x_C}|^2}}$$

$$\mathrm{P_1} = \cos\phi_1 = \frac{\mathrm{R}}{\sqrt{\mathrm{R}^2 + (\mathrm{X_L})^2}}$$

$$\mathrm{P_1} = \frac{\mathrm{R}}{\sqrt{\mathrm{R}^2 + \mathrm{R}^2}} = \frac{1}{\sqrt{2}}$$

$$\mathrm{P_2} = \cos\phi_2 = \frac{\mathrm{R}}{\sqrt{\mathrm{R}^2 + (\mathrm{x_L} - \mathrm{x_C})^2}}$$

$$\mathrm{P_2} = \frac{\mathrm{R}}{\mathrm{R}} = 1$$

$$\mathrm{P_1} = \frac{1}{\sqrt{2}}$$

Hence (2) is correct option.

Q. 8 A charge particle is moving in a uniform magnetic field $(2\hat{i}+3\hat{j})\ \mathrm{T}$. If it has an acceleration of $(\alpha\hat{i}-4\hat{j})\ ^{\mathrm{m}}/\mathrm{s}^2$, then the value of α will be :

Option 1:
3

Option 2:
6

Option 3:
12

Option 4:
2

Correct Answer:
3

Solution:

$\bar{B}=2\hat{i}+3\hat{j}$

$\bar{a}=\alpha\hat{i}-4\hat{j}$

In magnetic field, the force vector and magnetic field vector are perpendicular to each other i.e. $\bar{F}\perp\bar{B}$

$\therefore\quad \bar{a}\perp\bar{B}$

$\bar{a}\rightarrow$ acceleration vector

$\bar{a}\cdot\bar{B}=a\,B\cos 90^\circ=0$

$2\alpha-12=0$

$\alpha=6$

Hence (2) is correct option.

Q. 9 B_X and B_Y are the magnetic fields at the centre of two coils X and y respectively each carrying equal current. If coil X has 200 turns and $20\ cm$ radius and coil y has 400 turns and $20\ cm$ radius, the ratio of B_X and B_Y is :

Option 1:
$1:1$

Option 2:
$1:2$

Option 3:
$2:1$

Option 4:
$4:1$

Correct Answer:
$1:2$

Solution:

$$B_x = \frac{\mu_0 n_x i}{2r_x}$$

$$B_y = \frac{\mu_0 n_y i}{2r_y}$$

$$\frac{B_x}{B_y} = \frac{n_x}{r_x} \times \frac{r_y}{n_y}$$

$$= \frac{200}{20} \times \frac{20}{400}$$

$$\frac{B_x}{B_y} = \frac{1}{2}$$

Hence (2) is correct option.

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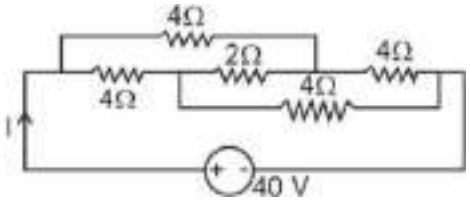
Q. 10 The current I in the given circuit will be :



Option 1:
$10\ \mathrm{A}$

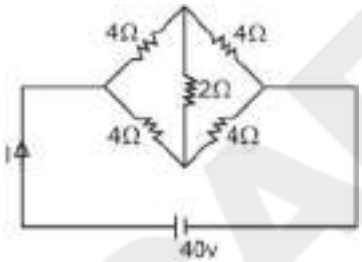
Option 2:
$20\ \mathrm{A}$

Option 3:
$4\ \mathrm{A}$

Option 4:
$40\ \mathrm{A}$

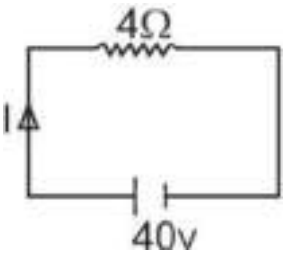
Correct Answer:
$10\ \mathrm{A}$

Solution:
Redrawing the circuit



There will be no current through 2Ω as the network is balanced

$\mathrm{R_{eq}} = 4\Omega$



$$I = \frac{40}{4} = 10A$$

Hence (1) is correct option.

Q. 11 The total charge on the system of capacitors $C_1 = 1\ \mu F, C_2 = 2\ \mu F, C_3 = 4\ \mu F$ and $C_4 = 3\ \mu F$ connected in parallel is :

(Assume a battery of $20\ V$ is connected to the combination)

Option 1:
$200\ \mu C$

Option 2:
$200\ C$

Option 3:
$10\ \mu C$

Option 4:
$10\ C$

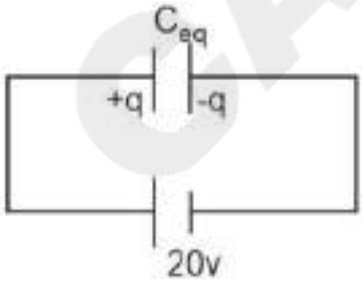
Correct Answer:
$200\ \mu C$

Solution:

$$C_{eq} = C_1 + C_2 + C_3 + C_4$$

$$= 1 + 2 + 4 + 3$$

$$C_{eq} = 10\mu F$$



$$q = (C_{eq})\,V$$

$$= (10mF) \times 20\ V$$

$$q = 200\mu C$$

Hence (1) is correct option.

Q. 12 When a particle executes Simple Hormonic Motion, the nature of graph of velocity as a function of displacement will be :

Option 1:
Circular

Option 2:
Elliptical

Option 3:
Sinusoidal

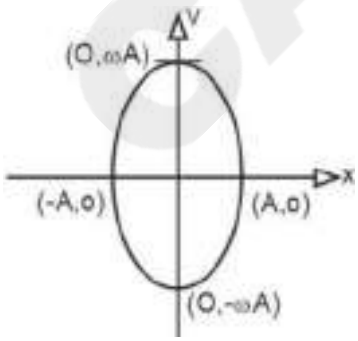
Option 4:
Straight line

Correct Answer:
Elliptical

Solution:
We know that,

$$V^2 = \omega^2 \left(A^2 - x^2\right)$$

$$\frac{V^2}{\omega^2 A^2} + \frac{x^2}{A^2} = 1$$



The nature of graph of velocity as a function of displacement will be elliptical
Hence (2) is correct option

Q. 13 $7\ \text{mol}$ of a certain monoatomic ideal gas undergoes a temperature increase of 40K at constant pressure. The increase in the internal energy of the gas in this process is :

$(\text{Given R} = 8.3\ \text{JK}^{-1}\ \text{mol}^{-1}\ )$

Option 1:
$5810\ \text{J}$

Option 2:
$3486\ \text{J}$

Option 3:
$11620\ \text{J}$

Option 4:
$6972\ \text{J}$

Correct Answer:
$3486\ \text{J}$

Solution:

$\text{n} = 7 \quad \text{V} = 5/3$

$\text{p} = \text{ constant}$

$\Delta\text{T} = 40\ \text{K}$

$$\Delta\text{U} = \frac{\text{nR}\Delta\text{T}}{\text{V} - 1}$$

$$= \frac{7(8.3) \times 40}{2/3}$$

$$= 420 \times 8.3$$

$$\Delta\text{U} = 3486\ \text{J}$$

Hence (2) is correct option.

Q. 14 A monoatomic gas at pressure P and volume V is suddenly compressed to one eighth of its original volume. The final pressure at constant entropy will be :

Option 1:
P

Option 2:
$8\mathrm{P}$

Option 3:
$32\mathrm{P}$

Option 4:
$64\mathrm{P}$

Correct Answer:
$32\mathrm{P}$

Solution:
For monoatomic gas,

$$\mathrm{V} = \frac{5}{3}$$

Since the gas is suddenly compressed it is an adiabatic process

$$\mathrm{V_f} = \frac{\mathrm{Vi}}{8}$$

$$\mathrm{P_i} = \mathrm{P\,(Given)}$$

For adiabatic process

$$\mathrm{P_i v_i^v} = \mathrm{P_f V_f^v}$$

$$\mathrm{P_f} = \mathrm{P}\left(\frac{\mathrm{V_i}}{\mathrm{V_f}}\right)^{\mathrm{v}}$$

$$= \mathrm{P}(8)^{5/3}$$

$$\mathrm{P_f} = 32\mathrm{P}$$

Hence (3) is correct option.

Q. 15 A water drop of radius $1\ \text{cm}$ is broken into 729 equal droplets. If surface tension of water is $75\ \text{dyne/cm}$, then the gain in surface energy upto first decimal place will be :

$(\text{Given } \pi = 3.14\ )$

Option 1:
$8.5 \times 10^{-4}\ \text{J}$

Option 2:
$8.2 \times 10^{-4}\ \text{J}$

Option 3:
$7.5 \times 10^{-4}\ \text{J}$

Option 4:
$5.3 \times 10^{-4}\ \text{J}$

Correct Answer:
$7.5 \times 10^{-4}\ \text{J}$

Solution:

$$\text{T} = 75\frac{\text{dyne}}{\text{cm}}$$

$$= 75 \times \frac{10^{-5}\ \text{N}}{10^{-2}\ \text{m}}$$

$$\text{T} = 75 \times 10^{-3}\ \text{N/m}$$

$$1\ \text{Drop} \rightarrow 729\ \text{Droplets}$$

$$\frac{4}{3}\pi\text{R}^3 = 729 \times \left(\frac{4\pi\text{r}^3}{3}\right)$$

$$\text{R} = 9\text{r} = 1\ \text{cm} = 10^{-2}\ \text{m}$$

$$r = \frac{1}{9}\ cm = \frac{10^{-2}}{9}$$

The gain in surface energy $= S_f - S_i$

$$= T\Delta A$$

$$\Delta S = T(\Delta A)$$

$$= 75 \times 10^{-3}\left(729 \times 4\pi r^2 - 4\pi R^2\right)$$

$$= 75 \times 10^{-3}(36\pi - 4\pi) \times 10^{-4}$$

$$= 75 \times 32\pi \times 10^{-7}$$

$$\Delta S = 7.5 \times 10^{-4}\ J$$

Hence (3) is correct option

Q. 16 The percentage decrease in the weight of a rocket, when taken to a height of $32\ km$ above the surface of earth will, be :

$(\text{Radius of earth } = 6400\ km)$

Option 1:
1%

Option 2:
3%

Option 3:
4%

Option 4:
0.5%

Correct Answer:
1%

Solution:
At,

$$h = 32\ km$$

$$g_n = g(1 - \frac{2h}{R})$$

$$= g(1 - \frac{64}{6400})$$

$$= g(1 - \frac{64}{6400})$$

$$g_n = \frac{99}{100}g$$

Weight at height $h = mg_n$

$$= \frac{99}{100}mg$$

Weight on earth surface $=mg$

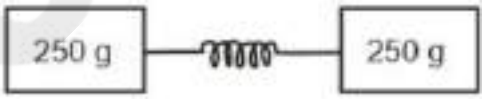
Percentage decrease in the weight

$$= \left(\frac{mg - mg_n}{mg}\right) \times 100$$

$$= 1\%$$

Hence (1) is correct option

Q. 17 As per the given figure, two blocks each of mass $250\ g$ are connected to a spring of spring constant $2\ Nm^{-1}$. If both are given velocity V in opposite directions, then maximum elongation of the spring is:



Option 1:

$$\frac{v}{2\sqrt{2}}$$

Option 2:

$\frac{v}{2}$

Option 3:

$\frac{v}{4}$

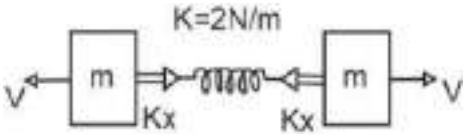
Option 4:

$\frac{v}{\sqrt{2}}$

Correct Answer:

$\frac{v}{2}$

Solution:



$m = 250g = 0.25\ kg$

As the spring extends, the retarding force acts on the block

At maximum elongation, the speed & each block is zero By energy conservation,

$$TE_{pritial} = TE_{final}$$

$$2\left(\frac{1}{2}mv^2\right) = \frac{1}{2}kx_m^2$$

$$0.25 \times v^2 = \frac{1}{2} \times 2x_m^2$$

$$x_m = \frac{v}{2}$$

Hence (2) is correct option.

Q. 18 A monkey of mass $50\ kg$ climbs on a rope which can withstand the tension (T) of $350\ N$. If monkey initially climbs down with an acceleration of $4\ m/s^2$ and then climbs up with an acceleration of $5\ m/s^2$. Choose the correct option $(g = 10\ m/s^2)$.

Option 1:

$T = 700\ N$ while climbing upward

Option 2:

$T = 350$ N while going downward

Option 3:

Rope will break while climbing upward

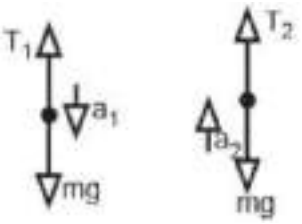
Option 4:

Rope will break while going downward

Correct Answer:

Rope will break while climbing upward

Solution:



$mg - T_1 = ma_1 \rightarrow (1)$

$T_2 - mg = ma_2 \rightarrow (2)$

$a_1 = 4 m/s^2 (given)$

$500 - T_1 = 50(4)$

$T_1 = 300N$

In upwand motion, the tension will more than the downward motion

So, the maximum value of T_2 possible is 350 N

but,

$a_2 = 5$ m/s^2

$T_2 - 500 = 50 \times 5$

$T_2 = 750$ N

$\therefore$ The rope will break while climbing upward

Hence (3) is correct option

Q. 19 Two projectiles thrown at $30°$and $45°$ with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is :

Option 1:
$1 : \sqrt{2}$

Option 2:
$2 : 1$

Option 3:
$\sqrt{2} : 1$

Option 4:
$1 : 2$

Correct Answer:
$\sqrt{2} : 1$

Solution:
$\theta_1 = 30°\text{and } \theta_2 = 45°$
It is given that the time of ascent is same for both of them

$$\mathrm{T_1 = T_2}$$

$$\frac{\mathrm{u_1 \sin\theta_1}}{\mathrm{g}} = \frac{\mathrm{u_2 \sin\theta_2}}{\mathrm{g}}$$

$$\frac{\mathrm{u_1}}{\mathrm{u_2}} = \frac{\sin\theta_2}{\sin\theta_1} = \frac{2}{\sqrt{2}} = \sqrt{2}$$

Hence (3) is correct option

Q. 20 A screw gauge of pitch $0.5\ \mathrm{mm}$ is used to measure the diameter of uniform wire of length $6.8\ \mathrm{cm}$, the main scale reading is $1.5\ \mathrm{mm}$ and circular scale reading is 7. The calculated curved surface area of wire to appropriate significant figures is :
[Screw gauge has 50 divisions on its circular scale]

Option 1:

$6.8\ \text{cm}^2$

Option 2:

$3.4\ \text{cm}^2$

Option 3:

$3.9\ \text{cm}^2$

Option 4:

$2.4\ \text{cm}^2$

Correct Answer:

$3.4\ \text{cm}^2$

Solution:

$\text{pitch} = 0.5\ \text{mm}$

$\text{l} = 6.8\ \text{cm}$

$\text{MSR} = 1.5\ \text{mm}$

$\text{CSD} = 7\ \text{mm}$

$$\text{Reading (Diameter)} = \text{MSR} + \text{CSD} \times \text{LC}$$

$$= 1.5\ \text{mm} + 7 \times \left(\frac{0.5\ \text{mm}}{50}\right)$$

$$\because \text{Least count} = \frac{\text{pitch}}{\text{No of circular scale division}}$$

$$= 1.5\ \text{mm} + 0.07\ \text{mm}$$

$$\text{Diameter(D)} = 1.57\ \text{mm} = 1.57 \times 10^{-1}\text{cm}$$

$$\text{Curved surface area} = 2\pi\text{Rl}$$

$$= \pi\text{Dl}$$

$$= 3.14 \times 1.57 \times 6.8 \times 10^{-1}$$

$$= 3.4\ \text{cm}^2$$

Hence (2) is correct option

Q. 21 If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y = 5x(1-x)$.The y component vector of the initial velocity is _____ $\hat{j}$.

$(\text{ Take } g = 10 \text{ m/s}^2 \ )$

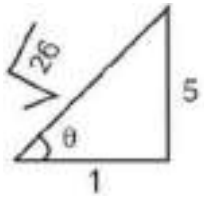
Correct Answer:
5

Solution:
$\bar{u}_x = \hat{i}$
Equation of trajectory : $y = 5x(1-x)$

Companing with;

$$y = x\tan\left(1 - \frac{x}{R}\right)$$

$R = 1$



$$\tan\theta = 5$$

$$R = \frac{2u_x u_y}{g}$$

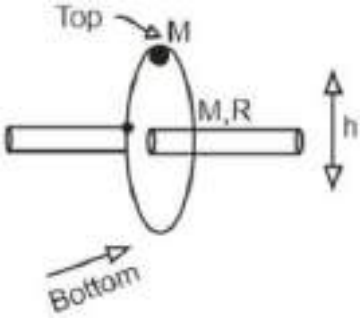
$$1 = \frac{2(1) \times u_y}{10}; u_y = 5 \text{ m/s}$$

The $y$component vector of the initial velocity is $5\hat{j}$

Q. 22 A disc of mass 1 kg and radius r is free to rotate about a horizontal axis passing through its centre and perpendicular to the plane of disc. A body of same mass as that of disc is fixed at the highest point of the disc. Now the system is released, when the body comes to the lowest position, its angular speed will be $4\sqrt{\frac{x}{3R}}$ rad s^{-1} where $x =$ _____. $(g = 10 \text{ ms}^{-2})$

Correct Answer:
5

Solution:



As the system is released and comes to the lowest position

$$\mathrm{TE_{initial}} = \mathrm{TE_{final}}$$

$$\mathrm{Mgh} + 0 = \frac{1}{2}\mathrm{I}\omega^2$$

$$\mathrm{Mg(2R)} = \frac{1}{2}\left(\frac{\mathrm{MR}^2}{2} + \mathrm{MR}^2\right)\omega^2$$

$$8\mathrm{MgR} = 3\mathrm{MR}^2\omega^2$$

$$\sqrt{\frac{8\mathrm{g}}{3\mathrm{R}}} = \omega$$

$$16\mathrm{x} = 8\mathrm{g} = 80$$

$$\mathrm{x} = 5$$

The value of x is 5

Q. 23 In an experiment to determine the Young's modulus of wire of a length exactly $1\ \mathrm{m}$, the extension in the length of the wire is measured as $0.4\ \mathrm{mm}$ with an uncertainty of $\pm 0.02\ \mathrm{mm}$ when a load of $1\ \mathrm{kg}$ is applied. The diameter of the wire is measured as $0.4\ \mathrm{mm}$ with an uncertainty of $\pm 0.01\ \mathrm{mm}$. The error in the measurement of Young's modulus $(\Delta \mathrm{Y})$ is found to be $\mathrm{x} \times 10^{10}\mathrm{Nm}^{-2}$. The value of x is _____.

$(\ \text{Take g} = 10\ \mathrm{ms}^{-2})$

Correct Answer:
2

Solution:

$L = 1\ m$

$\text{Extension} = l = 0.4mm$

$= 4 \times 10^{-4} m$

$\Delta L = 0.02\ mm$

$= 2 \times 10^{-5} m$

$m = 1\ kg$

$d = 0.4\ mm = 4 \times 10^{-4}\ m$

$\Delta d = 1 \times 10^{-5}\ m$

$$Y = \frac{mg}{A} \times \frac{L}{l}$$

$$\frac{\Delta Y}{Y} = \frac{\Delta m}{m} + \frac{\Delta L}{L} + \frac{2\Delta d}{d} + \frac{\Delta l}{l}$$

$$= 0 + 0 + 2\left(\frac{0.01}{0.4}\right) + \left(\frac{0.02}{0.4}\right)$$

$$= \frac{1}{20} + \frac{1}{20} = \frac{2}{20}$$

$$\Delta Y = \frac{Y}{10} = \frac{mg \times L}{(Al)10}$$

$$\Delta Y = \frac{1 \times 10 \times 1}{3.14 \times \frac{16 \times 10^{-8}}{4} \times 0.4 \times 10^{-3} \times 10}$$

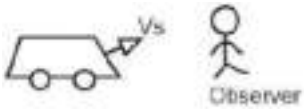
$$\Delta Y = 2 \times 10^{10}$$

The Value of x is 2

Q. 24 When a car is approaching the observer, the frequency of horn is $100\ \text{Hz}$. After passing the observer, it is $50\ \text{Hz}$. If the observer moves with the car, the frequency will be $\frac{x}{3}\ \text{Hz}$ where $x =$ _____.

Correct Answer:
200

Solution:



Let the frequency of horn be fo

Case I

$$f_1 = f_0\left(\frac{v}{v - v_s}\right)$$

$$100 = f_0\left(\frac{v}{v - v_s}\right) \rightarrow (1)$$

Case II

$$f_2 = 50 = f_0\left(\frac{v}{v + v_s}\right) \rightarrow (2)$$

Case III

When the observer moves with car,he will hear the original frequency f

$$f_0 = \frac{x}{3}$$

From eq (1) & (2)

$$\frac{100}{50} = \frac{V + V_s}{V - V_s}$$

$$2(v - v_s) = v + v_s$$

$$V = 3V_s$$

$$100 = f_0 \left(\frac{3V_s}{3V_s - V_s} \right)$$

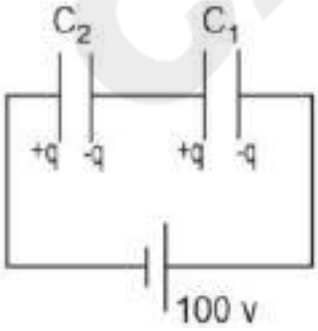
$$f_0 = \frac{200}{3} \text{ Hz}$$

$$\therefore x = 200$$

Q. 25 A composite parallel plate capacitor is made up of two different dielectric materials with different thickness $(t_1 \text{ and } t_2)$ as shown in figure. The two different dielectric materials are separated by a conducting foil f. The voltage of the conducting foil is _____ V.

Correct Answer:
60

Solution:



$$C_2 = \frac{A\varepsilon_0(4)}{1}$$

$$C_1 = \frac{A\varepsilon_0(3)}{(0.5)}$$

$$\frac{C_1}{C_2} = \frac{6}{4} = \frac{3}{2}$$

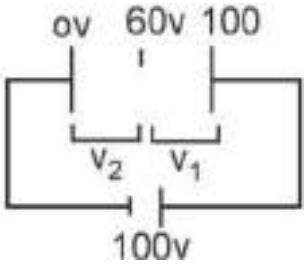
$$\frac{V_2}{V_1} = \frac{q/c_2}{q/c_1} = \frac{3}{2}$$

Let,

$$V_2 = 3K \quad V_1 = 2K$$

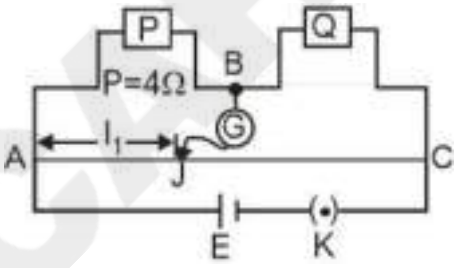
$$V_1 + V_2 = 5K = 100v$$

$$k = 20\ v$$



The voltage of conducting foil is $60\ v$

Q. 26 Resistances are connected in a meter bridge circuit as shown in the figure. The balancing length l_1 is $40\ cm$. Now an unknown resistance x is connected in series with P and new balancing length is found to be $80\ cm$ measured from the same end. Then the value of x will be _____ Ω.



Correct Answer:
20

Solution:
Initial

$$\frac{P}{Q} = \frac{l_1}{100 - l_1 = \frac{4}{6}} \rightarrow (1)$$

Final

$$\frac{\mathrm{P+x}}{\mathrm{Q}} = \frac{\mathrm{l_2}}{100 - \mathrm{l_2}} = \frac{4}{1} \rightarrow (2)$$

$$\frac{\mathrm{P}}{\mathrm{p+x}} = \frac{4/6}{4/1} = \frac{1}{6}$$

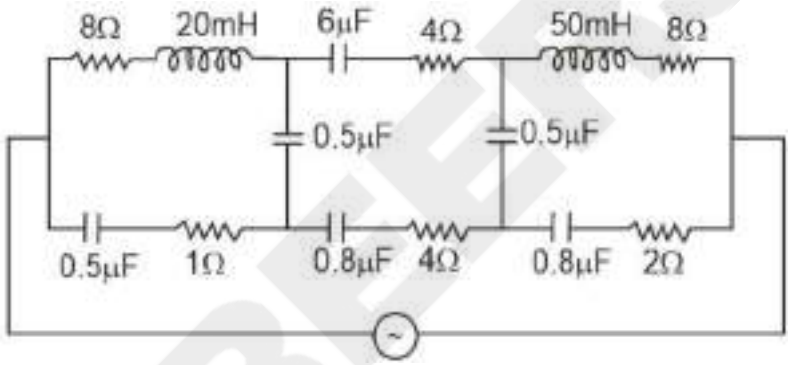
$$6\mathrm{P} = \mathrm{P} + \mathrm{x}$$

$$5\mathrm{P} = \mathrm{x}$$

$$\mathrm{x} = 5\mathrm{p} = 20\Omega$$

The value of x will be 20Ω

Q. 27 The effective current I in the given circuit at very high frequencies will be _____ A.

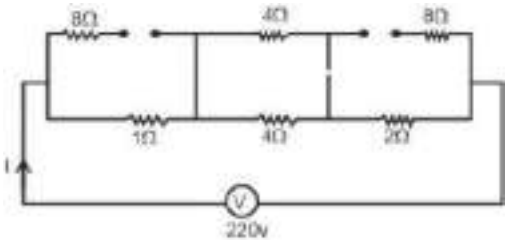


Correct Answer:
44

Solution:
For very high frequency,
$\mathrm{X_L} = 2\pi\ \mathrm{F\ L} \rightarrow \infty$
Inductor will provide infinite opposition. So, we can treat conductor as open circuit

$$\mathrm{X_C} = \frac{1}{2\pi \mathrm{fc}} \rightarrow \mathrm{o}$$

Capacitor will provide no oppositior so, we can treat capacitor as short circuit



$$I_{effertive} = I_{rms} = \frac{V_{rms}}{Z}$$

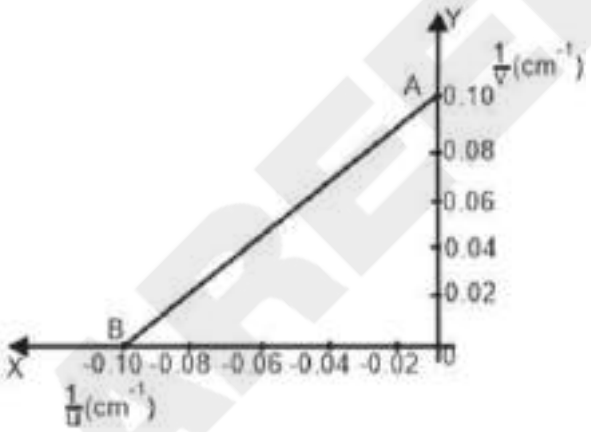
$$\text{here } z = R = 5\Omega$$

$$\therefore I_{effective} = \frac{220}{5} = 44\ A$$

The effective current I in the given circuit at very high frequencies will be $44\ A$

Q. 28 The graph between $\frac{1}{u}$ and $\frac{1}{v}$ for a thin convex lens in order to determine its focal length is plotted as shown in the figure. The refractive index of lens is 1.5 and its both the surfaces have same radius of curvature R. The value of R will be _____ cm.

$(\text{where } u = \text{ object distance}, v = \text{ image distance})$



Correct Answer:
10

Solution:
By lens maker formula,

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$R = R_1 = R_2, \mu = 1.5$$

$$\therefore \frac{1}{f} = (\mu - 1) \times \frac{2}{R_1} = \frac{1}{R}$$

Or

$$\frac{1}{f} = (\mu - 1) \times \frac{2}{R_2} = \frac{1}{R}$$

From thr graph,

For, $\frac{1}{\mu} = -0.10$

$$\frac{1}{v} = -0$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} = \frac{1}{R}$$

$$0 - (0.1) = \frac{1}{R}$$

$$0 \cdot 1 = \frac{1}{R}$$

$$\Rightarrow R = 10\ cm$$

The value of R will be $10\ cm$

Q. 29 In the hydrogen spectrum, λ be the wavelength of first transition line of Lyman series. The wavelength difference will be "$a\lambda$" between the wavelength of 3^{rd} transition line of Paschen series and that of 2^{nd} transition line of Balmer series where $a =$ _____.

Correct Answer:
5

Solution:
First transition line of lyman series

$$\frac{1}{\lambda} = R\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$$

$$\lambda = \frac{4}{3R} \rightarrow (1)$$

Third transition line of paschen

$$\frac{1}{\lambda_1} = \mathrm{R}\left(\frac{1}{3^2} - \frac{1}{6^2}\right)$$

$$\frac{1}{\lambda_1} = \frac{3\mathrm{R}}{36}$$

$$\lambda_1 = \frac{12}{\mathrm{R}} \rightarrow (2)$$

2nd transition line of balmer series,

$$\frac{1}{\lambda_2} = \mathrm{R}\left(\frac{1}{2^2} - \frac{1}{4^2}\right)$$

$$\lambda_2 = \frac{16}{3\mathrm{R}} \rightarrow (3)$$

$$\lambda_1 - \lambda_2 = \frac{12}{\mathrm{R}} - \frac{16}{3\mathrm{R}}$$

$$\mathrm{a}\lambda = \frac{20}{3\mathrm{R}} \text{ (Given)}$$

from eq (1)

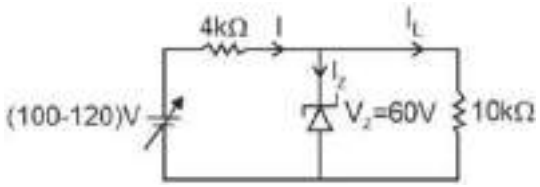
$$\mathrm{a}\left(\frac{4}{3\mathrm{R}}\right) = \frac{20}{3\mathrm{R}}$$

$$\mathrm{a} = 5$$

The value of $\mathrm{a} = 5$

CAREERS360

Q. 30 In the circuit shown below, maximum zener diode current will be _____ mA.



Correct Answer:
9

Solution:
For maximum zener diode current, the variable voltage must be max. i.e., $120\ \mathrm{V}$

$\therefore \mathrm{I}(4000) = 60\ \mathrm{V}$

$\mathrm{I} = 15 \times 10^{-3}\ \mathrm{A} = 15\ \mathrm{mA}$

$\mathrm{I_L} = 6\ \mathrm{mA}$

$\mathrm{I} = \mathrm{I_2} + \mathrm{I_L} \Rightarrow \mathrm{I_2} = 9\ \mathrm{mA}$

Chemistry

Q. 1 Match **List - I** with **List - II.**

List - I **(Compound)**	**List- II** **(Shape)**
(A) BrF_5	(I) bent
(B) $[CrF_6]^{3-}$	(II) square pyramidal
(C) O_3	(III) trigonal bipyramidal
(D) PCl_5	(IV) octahedral

Choose the **correct** answer from the options given below:

Option 1:
$(A)-(I),(B)-(II),(C)-(III),(D)-(IV)$

Option 2:
$(A)-(IV),(B)-(III),(C)-(II),(D)-(I)$

Option 3:
$(A)-(II),(B)-(IV),(C)-(I),(D)-(III)$

Option 4:
$(A)-(III),(B)-(IV),(C)-(II),(D)-(I)$

Correct Answer:
$(A)-(II),(B)-(IV),(C)-(I),(D)-(III)$

Solution:

$$(A)\ \text{Steric Number} = \frac{\text{Valency} + \text{No.of Surrounding atoms} - \text{Charge}}{2}$$

$$1)\ BrF_5\ \ S.N = \frac{7+5}{2} = 6\ \ [5\ \text{bond Pair} + 1\ \text{lone pair}]$$

Shape - Square Pyramidal

$2)[CrF_6]^{3-}\ S\cdot N=\frac{3+6+3}{2}=6\quad[6\text{ bond pair }]$

Shape - Octahedral

$3)O_3\quad S\cdot N=\frac{6+0}{2}=3\quad[2\text{ bond Pair}+1\text{ lone pair}]$

Shape - bent

$4)PCl_5\quad S.N=\frac{5+5}{2}=5\quad[\ 5\text{ bond pair }]$

Shape - Trigonal bipyromidal

$A-II, B-IV, C-I, D-III$

Hence, the correct answer is **Option (3).**

Q. 2 Match **List - I** with **List - II.**

List - I **(Processes/Reactions)**	**List - II** **(Catalyst)**
(A) $2SO_2(g)+O_2(g)\rightarrow 2SO_3(g)$	(I) $Fe(s)$
(B) $4NH_3(g)+5O_2(g)\rightarrow 4NO(g)+6H_2O(g)$	(II) $Pt(s)-Rh(s)$
(C) $N_2(g)+3H_2(g)\rightarrow 2NH_3(g)$	(III) V_2O_5
(D) Vegetable oil $(l)+H_2\rightarrow$ Vegetable ghee(s)	(IV) $Ni(s)$

Choose the **correct** answer from the options given below :

Option 1:
$(A)-(III),(B)-(I),(C)-(II),(D)-(IV)$

Option 2:
$(A)-(III),(B)-(II),(C)-(I),(D)-(IV)$

Option 3:
$(A)-(IV),(B)-(III),(C)-(I),(D)-(II)$

Option 4:
$(A)-(IV),(B)-(II),(C)-(III),(D)-(I)$

Correct Answer:
$(A)-(III),(B)-(II),(C)-(I),(D)-(IV)$

Solution:

List - I **(Processes/Reactions)**	**Process**	**List - II** **(Catalyst)**
(A) $2SO_2(g)+O_2(g)\rightarrow 2SO_3(g)$	Contoct's Process	V_2O_5
(B) $4NH_3(g)+5O_2(g)\rightarrow 4NO(g)+6H_2O(g)$	Ostwald's Process	$Pt(s)-Rh(s)$
(C) $N_2(g)+3H_2(g)\rightarrow 2NH_3(g)$	Hober's Process	$Fe(s)$
(D) $\text{Vegetable oil (l)}+H_2\rightarrow \text{Vegetable ghee(s)}$	Hydrogenation	$Ni(s)$

Hence, the correct answer is **Option (2).**

Q. 3 Given two statements below :
Statement I : In Cl_2 molecule the covalent radius is double of the atomic radius of chlorine.
Statement II : Radius of anionic species is always greater than their parent atomic radius.
Choose the **most appropriate** answer from options given below :

Option 1:
Both **Statement I** and **Statement II** are correct.

Option 2:
Both **Statement I** and **Statement II** are incorrect.

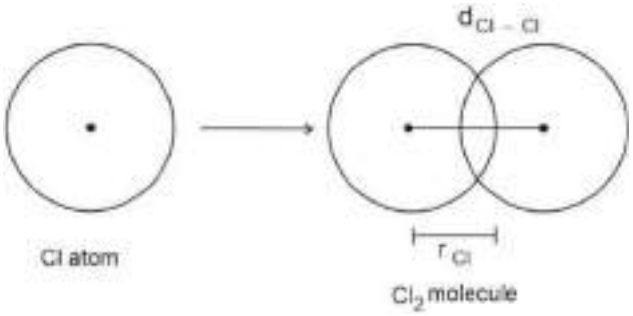
Option 3:
Statement I is correct but **Statement II** is incorrect.

Option 4:
Statement I is incorrect but **Statement II** is correct.

Correct Answer:
Statement I is incorrect but **Statement II** is correct.

Solution:
Size of anion is always greater than Size of parent atom as Zeff decreases.

Covalent Radii is little less than the atomic Radii of Cl because while formation of covalent bond atoms overlap hence interatomic distances decreases in a covalent bond.



Hence, the correct answer is **Option(4).**

Q. 4 Refining using liquation method is the most suitable for metals with :

Option 1:
Low melting point

Option 2:
High boiling point

Option 3:
High electrical conductivity

Option 4:
Less tendency to be soluble in melts than impurities

Correct Answer:
Low melting point

Solution:

Liquation process is used for metals having lesser melting point than impurities.
The metals melts and flows down to collector While impurities are left behind behind and later discarded.

This process is applied only when impurities are higher in concentration.

Hence, the correct answer is **Option (1).**

Q. 5 Which of the following can be used to prevent the decomposition of H_2O_2 ?

Option 1:
Urea

Option 2:
Formaldehyde

Option 3:
Formic acid

Option 4:
Ethanol

Correct Answer:
Urea

Solution:
Compound like urea, Acetanilide act as negative catalyst for decomposition of H_2O_2.

Hence, the correct answer is **Option (1).**

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Q. 6 Reaction of $BeCl_2$ with $LiAlH_4$ gives:

(A) $AlCl_3$

(B) BeH_2

(C) LiH

(D) $LiCl$

(E) $BeAlH_4$

Choose the **correct** answer from options given below :

Option 1:
(A), (D) and (E)

Option 2:
(A), (B) and (D)

Option 3:
(D) and (E)

Option 4:
(B), (C) and (D)

Correct Answer:
(A), (B) and (D)

Solution:
Reaction of $BeCl_2$ with $LiAlH_4$ is a method for formation of BeH_2 giving $LiCl$ and $AlCl_3$ as biproducts.

$$2BeCl_2 + LlAlH_4 \longrightarrow 2BeH_2 + LiCl + AlCl_3$$

Hence, the correct answer is **Option (2).**

Q. 7 Borazine, also known as inorganic benzene, can be prepared by the reaction of 3-equivalents of "X" with 6-equivalents of ""Y". "X" and"Y", respectively are :

Option 1:
$B(OH)_3$ and NH_3

Option 2:
B_2H_6 and NH_3

Option 3:
B_2H_6 and HN_3

Option 4:
NH_3 and B_2O_3

Correct Answer:
B_2H_6 and NH_3

Solution:
Borazine can be prepared by -

$$3B_2H_6 + 6NH_3 \xrightarrow{200^\circ C} 2B_3N_3H_6(g) + 12H_2(g)$$

Hence, X is B_2H_6 and Y is NH_3

Hence, the correct answer is **Option (2).**

Q. 8 Which of the given reactions is not an example of disproportionation reaction ?

Option 1:
$2H_2O_2 \rightarrow 2H_2O + O_2$

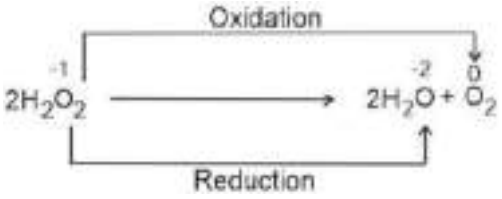
Option 2:
$2NO_2 + H_2O \rightarrow HNO_3 + HNO_2$

Option 3:
$MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$

Option 4:
$3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

Correct Answer:
$MnO_4^- + 4H^+ + 3e^- \rightarrow MnO_2 + 2H_2O$

Solution:
Dispropotion Reaction - It's ortype of redox reaction involving simultaneous oxidation and reduction of the atom of same element into two different oxidation states.



$$2\overset{+4}{N}O_2 + H_2O \longrightarrow H\overset{+5}{N}O_3 + \overset{+3}{H}NO_2$$

$$\overset{+7}{Mn}O_4^- + 4H^+ + 3e^- \longrightarrow \overset{+4}{Mn}O_2 + 2H_2O$$

$$3\overset{+6}{Mn}O_4^{2-} + 4H^+ \longrightarrow 2\overset{+7}{Mn}O_4^- + \overset{+4}{Mn}O_2 + 2H_2O$$

Hence, the correct answer is **Option (3).**

Q. 9 The dark purple colour of $KMnO_4$ disappears in the titration with oxalic acid in acidic medium. The overall change in the oxidation number of manganese in the reaction is :

Option 1:
5

Option 2:
1

Option 3:
7

Option 4:
2

Correct Answer:
5

Solution:
In Acidic medium permagnate ion (MnO_4^-) act as oxidizing agent reducing itself to Mn^{2+} ion.

Balanced reaction is

$$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mm^{2+} + 4H_2O$$

$$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \longrightarrow 2Mm^{2+} + 10CO_2 + 8H_2O$$

Hence, the correct answer is **Option (1).**

Q. 10 $\dot{C} + CH_4 \rightarrow A + B$

A and B in the above atmospheric reaction step are :

Option 1:
C_2H_6 and Cl_2

Option 2:
$\dot{C}HCl_2$ and H_2

Option 3:
$\dot{C}H_3$ and HCl

Option 4:
C_2H_6 and HCl

Correct Answer:
$\dot{C}H_3$ and HCl

Solution:
Chlorination of Alkanes follow free radical mechanism.

1) Initiation Step

$$Cl_2 \xrightarrow{hv} 2\,\dot{C}l$$

2) Propogation Step

$$CH_4 + \dot{C}l \longrightarrow \dot{C}H_3 + HCl$$

$$\dot{C}H_3 + Cl_2 \longrightarrow CH_3Cl + \dot{C}l$$

3) Chain Termination Step

$$\dot{C}H_3 + \dot{C}l \longrightarrow CH_3Cl$$

Hence, the correct answer is **Option (3).**

Q. 11 Which technique among the following, is most appropriate in separation of a mixture of $100\mathrm{mg}$ of P-nitrophenol and picric acid ?

Option 1:
Steam distillation

Option 2:
2-5 ft long column of silica gel

Option 3:
Sublimation

Option 4:
Preparative TLC (Thin Layer Chromatography)

Correct Answer:
Preparative TLC (Thin Layer Chromatography)

Solution:
The method that can be used for separation for 100 mg mixture of picric acid and p-nitrophenal is TLC (Thin layer Chromatography) using a appropriate salvent. Since the retention factor is different for both hence can to reparated.

Hence, **Option (4)** is correct.

Q. 12 The difference in the reaction of phenol with bromine in chloroform and bromine in water medium is due to:

Option 1:
Hyperconjugation in substrate

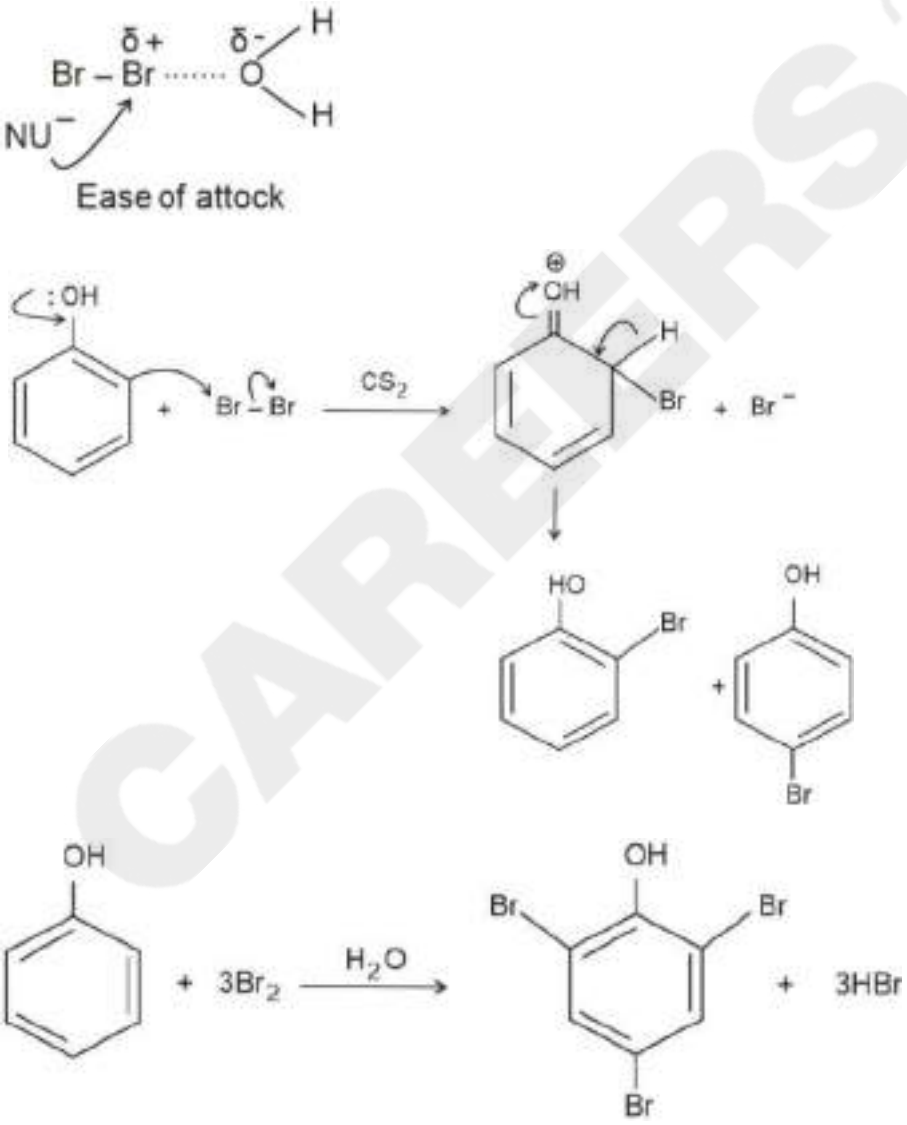
Option 2:
Polarity of solvent

Option 3:
Free radical formation

Option 4:
Electromeric effect the substrate

Correct Answer:
Polarity of solvent

Solution:
On bromination of phenol gives tri substiuted product while in non-polar solvent soch as CS_2 or $CHCl_3$ or mono substituted product. This is for each case of formation of bromium ion in polar solvents than non-polar solvents.



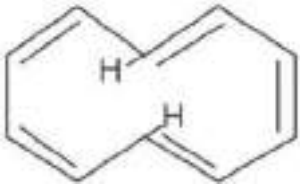
Hence, **Option (2)** is correct.

Q. 13 Which of the following compounds is **not** aromatic?

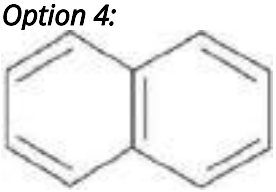
Option 1:

Option 2:

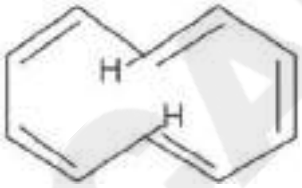
Option 3:



Option 4:



Correct Answer:



Solution:

For a specific to be Aromatic in nature, it must be

1) Planar

2) Cyclic

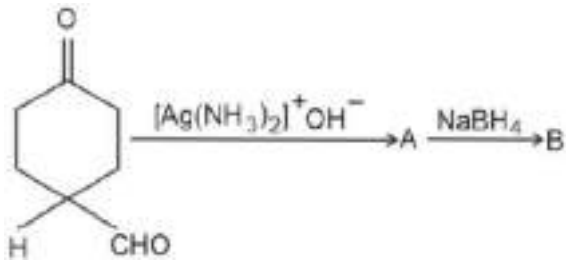
3) Completely Conjugated

4) Follow huckel $(4\mathrm{m}+2)\pi\mathrm{e}^{-}$ role.

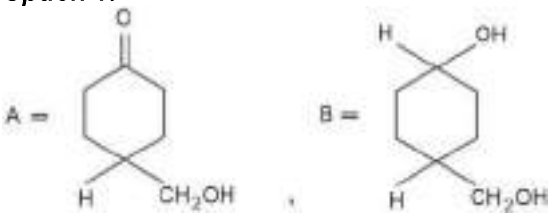
Compound (C) is not planar due to $H - H$ interactions while cis - [10]-Annulene is planar and tran s - [10] Annulane is non planar.

Hence, **option (3)** is correct.

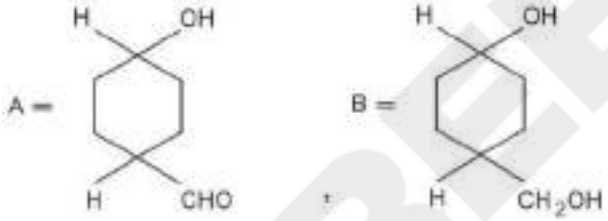
Q. 14 The products formed in the following reaction, **A** and **B** are



Option 1:



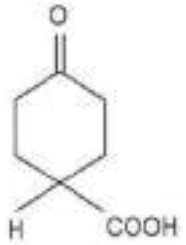
Option 2:



Option 3:

Option 4:

Correct Answer:

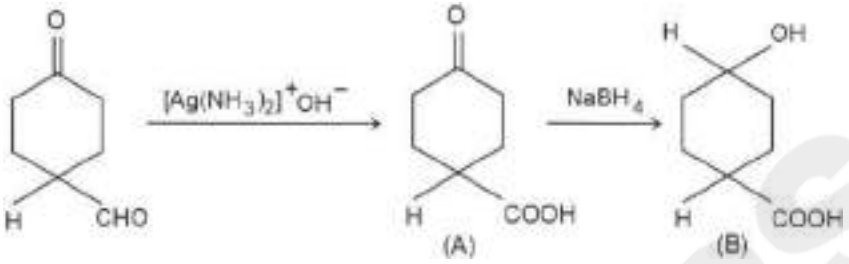
A = 

B =

Solution:

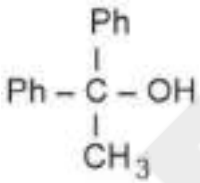
$[Ag\ (NH_3)_2]^+\ OH^-$ is Tollen's reagent and Tollen's test is given by aldehydes.

$NaBH_4$ is a reducing agent which can reduce aldehydes, ketones, acid anhydrides and acid chlorides to alcohol.

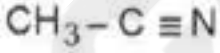


Hence, **Option (3)** is correct.

Q. 15 Which reactant will give the following alcohol on reaction with one mole of phenyl magnesium bromide $(PhMgBr)$ followed by acidic hydrolysis?

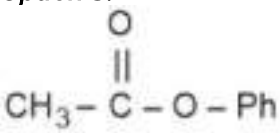


Option 1:

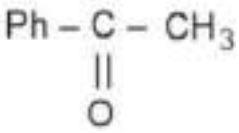


Option 2:

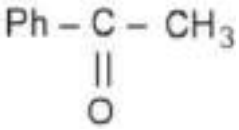
Option 3:



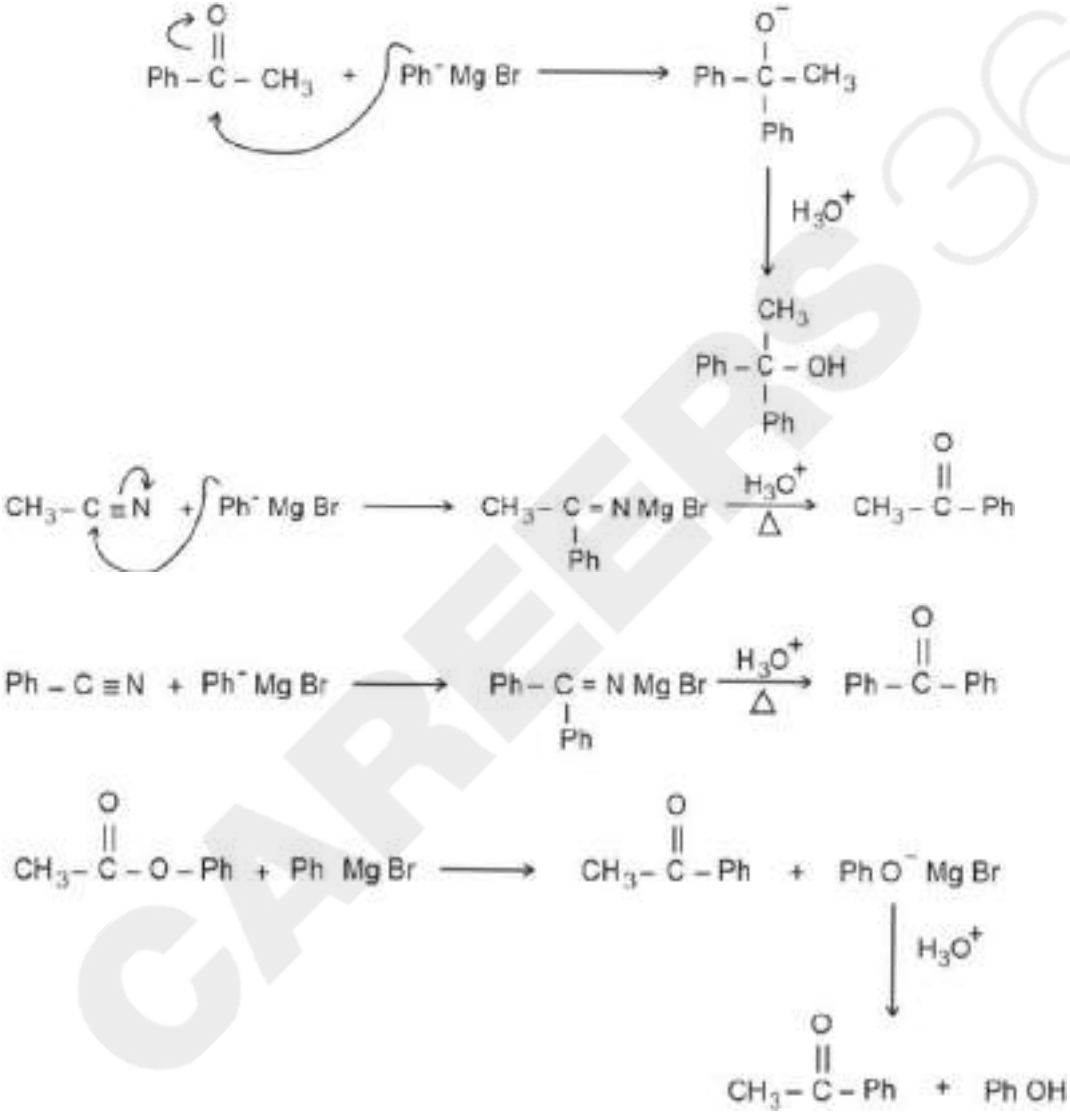
Option 4:



Correct Answer:



Solution:



Hence, **Option (4)** is correct.

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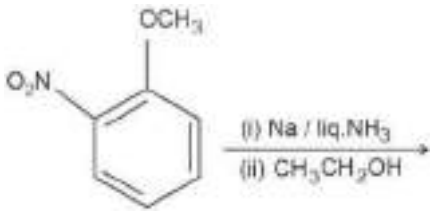
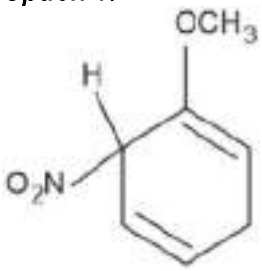
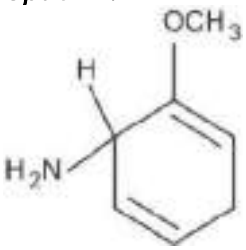
Q. 16 The major product of the following reaction is



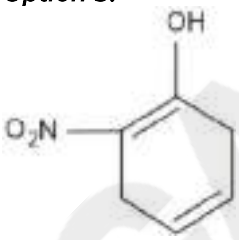
Option 1:



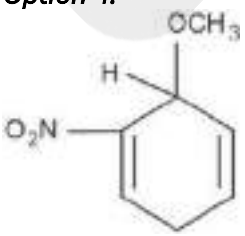
Option 2:



Option 3:



Option 4:



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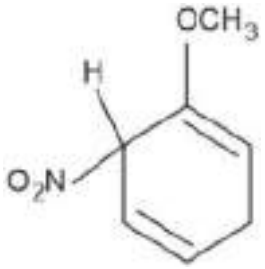
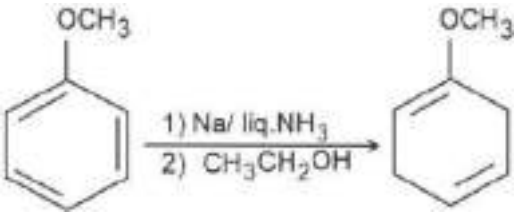
Correct Answer:

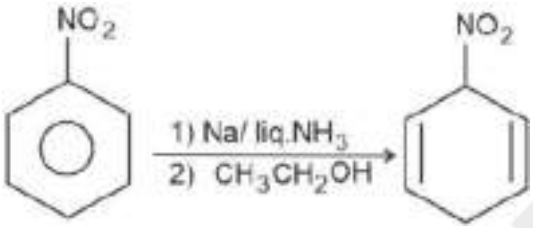


Solution:

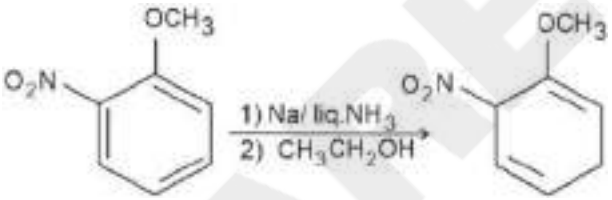
On birch reduction of Amisale (or EDG)



On birch reduction of Nitro benzene

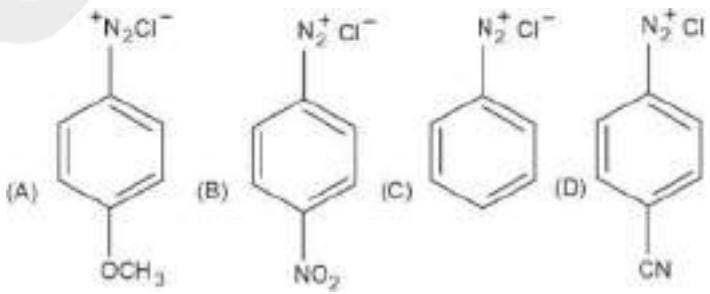


So,



Hence, **Option (1)** is correct.

Q. 17 The correct stability order of the following diazonium salt is



Option 1:
(A) > (B) > (C) > (D)

Option 2:
(A) > (C) > (D) > (B)

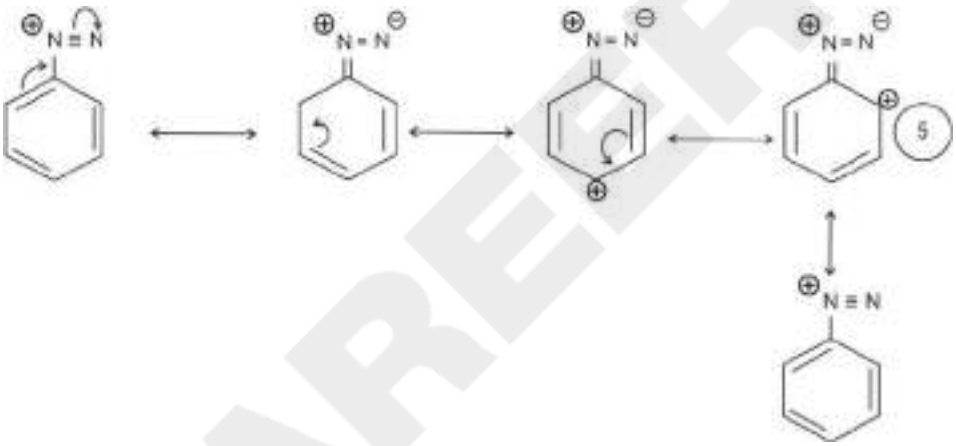
Option 3:
(C) > (A) > (D) > (B)

Option 4:
(C) > (D) > (B) > (A)

Correct Answer:
(A) > (C) > (D) > (B)

Solution:
Stability of dizonium salt α E D G

$$\alpha \frac{1}{\mathrm{E}\ \omega\ \mathrm{G}}$$



Hence, EDG will stabilize dizonium salt while EWG will destabilize.

Order of stability A > C > D > B

Hence, **option (2)** is correct.

Q. 18 Stearic acid and polyethylene glycol react to form which one of the following soap/s detergents?

Option 1:
Cationic detergent

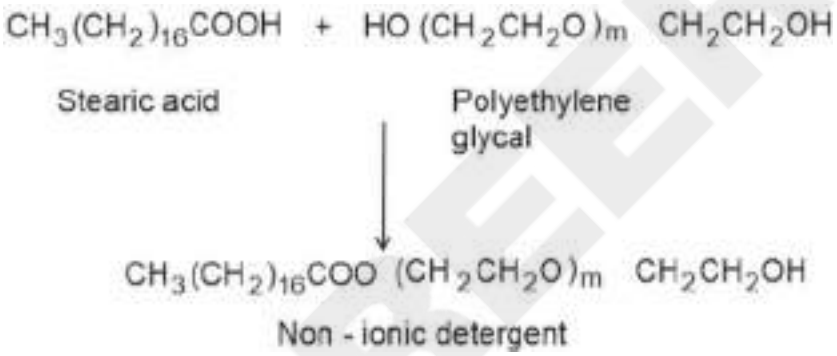
Option 2:
Soap

Option 3:
Anionic detergent

Option 4:
Non-ionic detergent

Correct Answer:
Non-ionic detergent

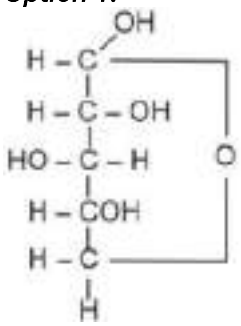
Solution:
On Reaction of Stearic acid with ethylene glycal produces at non-ionic detergants.



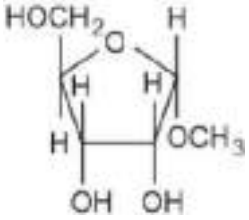
Hence, **Option (4)** is correct.

Q. 19 Which one of the following is a reducing sugar?

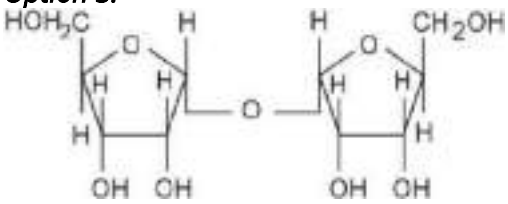
Option 1:



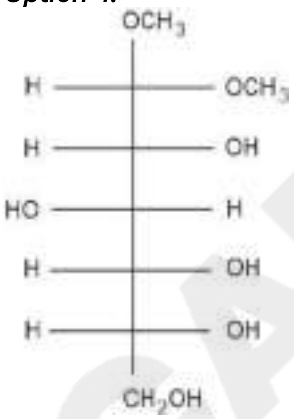
Option 2:



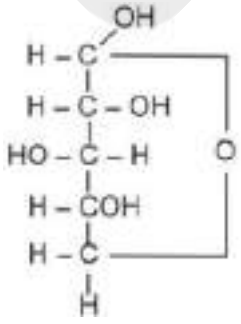
Option 3:



Option 4:



Correct Answer:



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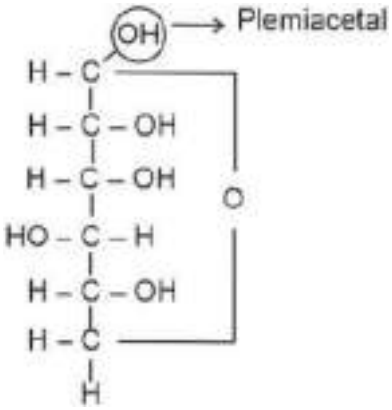
Solution:
Reducing sugars are those which give positive Tollen's test or Benedict's test.
Reducing sugars have a aldehyde group or ketone Or hemiacetol group.



Hence, **Option (1)** is correct.

Q. 20 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A): Experimental reaction of CH_3Cl with aniline and anhydrous $AlCl_3$ does **not** give O and p-methylaniline.

Reason (R): The $-NH_2$ group of aniline becomes deactivating because of salt formation with anhydrous $AlCl_3$ and hence yields m-methyl aniline as the product.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

Option 1:
Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A).**

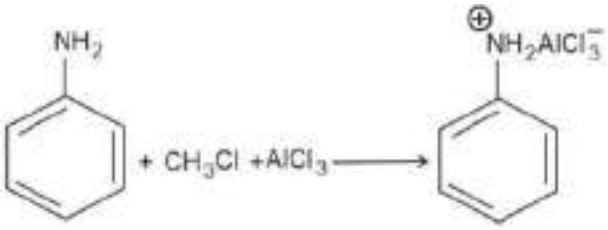
Option 2:
Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A).**

Option 3:
(A) is true, but **(R)** is false.

Option 4:
(A) is false, but **(R)** is true.

Correct Answer:
(A) is true, but **(R)** is false.

Solution:
on Reaction of Amleine with CH_3Cl in presence of $AlCl_3$, insted of friedel Craft reaction it forms Salt and does not forms $-O$ or $-P$ or $-m$ product.



Hence Assertion is correct but not reason.

Hence, **Option (3)** is correct.

Q. 21 Chlorophyll extracted from the crushed green leaves was dissolved in water to make $2\ L$ solution of Mg of concentration $48\ ppm$. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of X is _____. (Nearest Integer)
$(\text{Given : Atomic mass of Mg is } 24\ g\ mol^{-1}; N_A = 6.02 \times 10^{23}\ mol^{-1})$

Correct Answer:
24

Solution:
Concentration of mg is 48 ppm that means

48 mg of mg in 1000 ml or 1 L of H_2O solution hence 2 L of solution will contain 96 mg of mg

$$\text{No. of moles of mg} = \frac{\text{weight of mg}}{\text{Molar mass}}$$

$$= \frac{96 \times 10^{-3}}{24\ g/mol} = 4 \times 10^3 \text{mole}$$

$$\text{No.of atoms} = 4 \times 10^{-3} \times 6 \times 10^{23} = 24 \times 10^{20}$$

Value of x = 24

Q. 22 A mixture of hydrogen and oxygen contains 40% hydrogen by mass when the pressure is 2.2 bar. The partial pressure of hydrogen is _____ bar. (Nearest Integer)

Correct Answer:
2

Solution:
Let the mass of mixture be 100 g

Mass of H_2 gas = 40 g

Mass of O_2 gas = 60 g

$$NH_2 = \frac{40\text{ g}}{2\text{ g/mol}} = 20\text{ mole}$$

$$NO_2 = \frac{60\text{ g}}{32\text{ g/mol}} = 1.875\text{ mole}$$

$$X_{H_2} = \frac{20}{20 + 1.875} = 0.91$$

$$P_{H_2} = X_{H_2} \qquad P_{\text{Total}} = 0.91 \times 2.2\text{ bar}$$

$$\cong 26\text{ ar}$$

Answer = 2

Q. 23 The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is _____ . (Nearest Integer)

$$(\text{Mass of electron is } 9.1 \times 10^{-31}\text{ kg and mass of neutron is } 1.6 \times 10^{-27}\text{ kg })$$

Correct Answer:
1758

Solution:
According to De-broglie hypothesis

$$\lambda = \frac{h}{mv}$$

If $\lambda n = \lambda e^{-}$

$$\frac{h}{m_n V_n} = \frac{h}{m_e V_e}$$

$$V_e = \frac{m_n V_n}{me}$$

$$= \frac{1.6 \times 10^{-27} kg}{9.11 \times 10^{-31} kg}$$

$$V_e = 1758\ V_n$$

Answer is 1758

Q. 24 $2.4\ g$ coal is burnt in a bomb calorimeter in excess of oxygen at $298\ K$ and $1\ atm$ pressure. The temperature of the calorimeter rises from $298\ K$ to $300\ K$. The enthalpy change during the combustion of coal is $-x\ kJ\ mol^{-1}$. The value of x is_____. (Nearest Integer)

(Given: Heat capacity of bomb calorimeter 20.0 kJ K^{-1}. Assume coal to be pure carbon)

Correct Answer:
200

Solution:
On burning 2.4 g of coal temperature inc. from 298 to 300 K.

$$q = C\Delta T$$

$$= 20\ KJK^{-1} \times 2 = 40KJ$$

2.4 g of coal evolves 40 KJ of heat

1 mole (12 g will evolve) $= \frac{40KJ \times 12g/mole}{2.4g}$

$$= 200\ KJ/mole$$

Answer = 200

Q. 25 When $800\ \text{mL}$ of $0.5\ \text{M}$ nitric acid is heated in a beaker, its volume is reduced to half and $11.5\ \text{g}$ of nitric acid is evaporated. The molarity of the remaining nitric acid solution is $\text{x} \times 10^{-2}\text{M}$. (Nearest Integer)

$(\text{Molar mass of nitric acid is } 63\ \text{g mol}^{-1}\ )$

Correct Answer:
54

Solution:
Initially Solution is 0.5 m 800 ml HNO_3

$$\text{M HNO}_3 = 0.5 \times 0.8\text{L} = 0.4 \text{ mole}$$

$$\text{W HNO}_3 = 0.4 \times 63\ \text{g/mole} = 25.2\ \text{g}$$

On heating

Valume reduced to 400 ml and 11.5 g of HNO_3 is evaporated.

Wt.of $\text{HNO}_3 = 25.2\ \text{g} - 11.5\ \text{g} = 13.7\ \text{g}$

$$\text{n HNO}_3 = \frac{13.7}{63\ \text{g/mole}} = 0.21 \text{ mole}$$

$$\text{M HNO}_3 = \frac{0.217}{0.4\ \text{L}} = 0.543\ \text{M}$$

$$= 54.3 \times 10^{-2}\ \text{m}$$

Answer = 54

Q. 26 At $298\ \text{K}$, the equilibrium constant is 2×10^{15} for the reactio:

$$\text{Cu(s)} + 2\text{Ag}^{+}\text{(aq)} \rightleftharpoons \text{Cu}^{2+}\text{(aq)} + 2\text{Ag(s)}$$

The equilibrium constant for the reaction

$\frac{1}{2}\text{Cu}^{2+}\text{(aq)} + \text{Ag(s)} \rightleftharpoons \frac{1}{2}\text{Cu(s)} + \text{Ag}^{+}\text{(aq)}$ is $\text{x} \times 10^{-8}$. The value of x is _____.(Nearest Integer)

Correct Answer:
2

Solution:
For the reaction

$$Co + 2Ag^{+} \longrightarrow CO^{2+} + 2Ag \quad K_{eq} = 2 \times 10^{15}$$

So for,

$$\frac{1}{2}Cu^{2+} + Ag \rightarrow \frac{1}{2}Cu + Ag^{+} \quad K_{eq} = \frac{1}{(K_{eq})^{\frac{1}{2}}}$$

$$K_{eq}^{1} = \sqrt{5 \times 10^{-16}}$$

$$= 2.24 \times 10^{-8}$$

$$x = 2$$

Q. 27 The amount of charge in F (Faraday) required to obtain one mole of iron from Fe_3O_4 is _____.
(Nearest Integer)

Correct Answer:
3

Solution:
Fe_3O_4 exists as $FeO.Fe_2O_3$

Average oxidation state is 2.67 or $\frac{8}{3}$ hence for 1 mole 2.67 F of charge is required closests integer will be 3

Answer is 3

Q. 28 For a reaction $A \rightarrow 2\ B + C$ the half lives are $100\ s$ and $50\ s$ when the concentration of reactant A is 0.5 and $1.0\ mol\ L-1$ respectively. The order of the reaction is _____. (Nearest Integer)

Correct Answer:
2

Solution:
The half life depends on concentration raise to power $(1-\mathrm{n})$ where n is order $\mathrm{t}_{1/2}\ \alpha, \mathrm{a}^{(1-\mathrm{m})}$

$$\frac{(t_{112})_1}{(t_{1/2})_2} = \left(\frac{a_1}{a_2}\right)^{1-n}$$

$$\frac{100}{50} = \left(\frac{0.5}{1}\right)^{1-n}$$

$2^1 = 2^{n-1}$ $\quad$ $n-1=1$ $\quad$ $n=2$

Order of reaction is 2

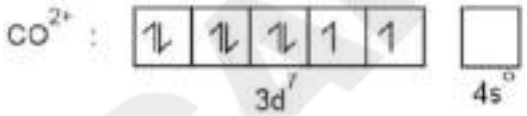
Answer is 2

Q. 29 The difference between spin only magnetic moment values of $[\mathrm{Co}\ (\mathrm{H_2O})_6]\ \mathrm{Cl_2}$ and $[\mathrm{Cr}\ (\mathrm{H_2O})_6]\ \mathrm{Cl_3}$ is _____.

Correct Answer:
0

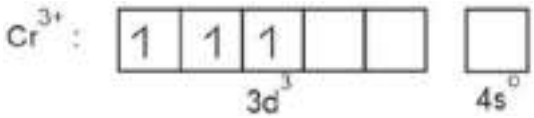
Solution:
In,

$[\mathrm{CO(H_2O)_6}]\ \mathrm{Cl_2}, \mathrm{O.S\ of\ CO\ is} + 2$



No. of unpaired $\mathrm{e^-} = 3$

In $[\mathrm{Cr(H_2O)_6}]\ \mathrm{Cl_3}, \mathrm{O.S\ of\ Cr\ is}\ +3$



No. of unpaired $\mathrm{e^-} = 3$

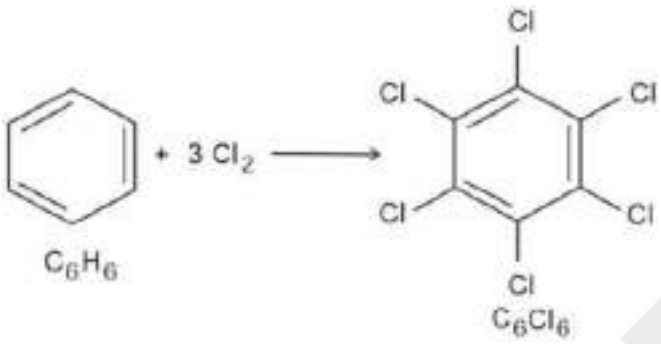
Since both $\mathrm{CO^{2+}\ and\ Cr^{3+}}$ has equal no. of unpaired $\mathrm{e^-}$, difference in spin only magnetic moment is 0.

Answer is 0.

Q. 30 In the presence of sunlight, benzene reacts with Cl_2 to give product, X. The number of hydrogens in x is _____.

Correct Answer:
6

Solution:
The reaction of benzene with Cl_2 in presence of sunlight is



Answer is 6

Maths

Q. 1 Let $f : \mathbf{R} \to \mathbf{R}$ be a continuous function such that $f(3x) - f(x) = x$. If $f(8) = 7$, then $f(14)$ is equal to:

Option 1:
4

Option 2:
10

Option 3:
11

Option 4:
16

Correct Answer:
10

Solution:

$$f(x) - f\left(\frac{x}{3}\right) = \frac{2x}{3}$$

$$f\left(\frac{x}{3}\right) - f\left(\frac{x}{3}\right) = \frac{x}{3^2} \quad \text{.....on adding}$$

$$f(x) = \lim_{n\to\infty} f'\left(\frac{x}{3^n}\right) = x\left(\frac{1}{3} + \frac{1}{3^2} + \cdots, \infty\right)$$

$$f(x) - f(0) = \frac{x}{2}$$

$$f(8) = 7, f(0) = 3$$

$$f(x) = \frac{x}{2} + 3$$

$$f(14) = 10$$

Hence correct option is 2

Q. 2 Let O be the origin and A be the point $z_1 = 1 + 2i$. If B is the point z_2, $\mathrm{Re}(z_2) < 0$, such that OAB is a right angled isosceles triangle with OB as hypotenuse, then which of the following is NOT true?

Option 1:

$$\arg z_2 = \pi - \tan^{-1} 3$$

Option 2:

$$\arg(z_1 - 2z_2) = -\tan^{-1}\frac{4}{3}$$

Option 3:

$$|z_2| = \sqrt{10}$$

Option 4:

$$|2z_1 - z_2| = 5$$

Correct Answer:

$$|2z_1 - z_2| = 5$$

Solution:

$$AB = AO \quad Z^{-i\pi/2} = -2 + i$$

$\text{so OB} = (-2+\text{i}) + (1+2\text{i})$

$Z_2 = -1 - 3\text{i}$

$\therefore |2Z_1 - Z_2| = \sqrt{10}$

Hence correct option is 4

Q. 3 If the system of linear equations.

$$8x + y + 4z = -2$$

$$x + y + z = 0$$

$$\lambda x - 3y = \mu$$

has infinitely many solutions, then the distance of the point $\left(\lambda, \mu, -\frac{1}{2}\right)$ from the plane $8x + y + 4z + 2 = 0$ is :

Option 1:
$3\sqrt{5}$

Option 2:
4

Option 3:
$\frac{26}{9}$

Option 4:
$\frac{10}{3}$

Correct Answer:
$\frac{10}{3}$

Solution:

$$D = \begin{vmatrix} 8 & 1 & 4 \\ 1 & 1 & 1 \\ \lambda & -3 & 0 \end{vmatrix} = 0 \Rightarrow \lambda = 4$$

$\text{Also } D_1 = D_2 = D_3 = 0$

$\text{So } \mu = -2$

$\text{point } \left(4, -2, -\frac{1}{2}\right)$

$\text{Distance from plane} = \left(\frac{10}{3}\right)$

Hence correct option is 4

Q. 4 Let A be a 2×2 matrix with $\det(A) = -1$ and $\det((A + I)(\text{Adj}(A) + I)) = 4$. Then the sum of the diagonal elements of A can be :

Option 1:
-1

Option 2:
2

Option 3:
1

Option 4:
$-\sqrt{2}$

Correct Answer:
2

Solution:

$\text{Let } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ ad } - \text{bc} = -1$

$|A + I||\text{adj}A + I| = 4$

$\Rightarrow \text{ad} - \text{bc} + a + d + 1 = 2 \quad \text{or } -2$

$a + d = 2 \quad \text{or } -2$

Hence correct option is 2

Q. 5 The odd natural number a, such that the area of the region bounded by $y = 1, y = 3, x = 0, x = y^a$ is $\frac{364}{3}$, is equal to:

Option 1:
3

Option 2:
5

Option 3:
7

Option 4:
9

Correct Answer:
5

Solution:

$$A = \int_1^3 y^a \cdot dy = \frac{y^{a+1}}{a+1}\Big|_1^3 = \frac{364}{3}$$

$a = 5$

Hence correct option is 2

Q. 6 Consider two G.Ps. $2, 2^2, 2^3, \ldots$ and $4, 4^2, 4^3, \ldots$ of 60 and n terms respectively. If the geometric mean of all the $60 + n$ terms is $(2)^{\frac{225}{8}}$, then $\sum_{k=1}^{n} k(n-k)$ is equal to :

Option 1:
560

Option 2:
1540

Option 3:
1330

Option 4:

2600

Correct Answer:

1330

Solution:

$\left(\left(2^1 \cdot 2^2 \cdot \ldots \cdot 2^{60}\right)\left(4^1 \cdot 4^2 \cdots 4^n\right)\right)^{\frac{1}{60+n}} = 2^{\frac{225}{8}}$

$\left(2^{30\times 61} \cdot 4^{\frac{n(n+1)}{2}}\right)^{\frac{1}{60+n}} = 2^{\frac{225}{8}}$

$2^{1830+n^2+n} = 2^{\frac{(225)(60+n)}{8}}$

$= 8n^2 - 217n + 1140 = 0$

$n = 20, \frac{57}{8}$

$\sum_{k=1}^{n} nk - k^2 = \frac{n^2(n+1)}{2} - \frac{n(n+1)(2n+1)}{6}$

$= 1330$

Hence correct option is 3

Q. 7

If the function $f(x) = \begin{cases} \frac{\log_e(1-x+x^2)+\log_e(1+x+x^2)}{\sec x - \cos x} & , \quad x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) - \{0\} \\ k & , \quad x = 0 \end{cases}$

is continuous at $x = 0$, then K is equal to:

Option 1:

1

Option 2:

-1

Option 3:

E

Option 4:

0

Correct Answer:

1

Solution:

$$\lim_{x\to 0}\frac{(\ln(1+x^2+x^4))\cos x}{1-\cos^2 x}$$

$$\lim_{x\to 0}\frac{\left(\frac{\ln\left(1+x^2+x^4\right)}{x^2+x^4}\right)x^2(1+x^2)\cos x}{\left(\frac{\sin^2 -x}{x^2}\right)x^2}=1$$

$$\therefore k=1$$

Hence correct option is 1

Q. 8 If $f(x)=\begin{cases}x+a, & x\le 0\\ |x-4|, & x>0\end{cases}$ and $g(x)=\begin{cases}x+1 & ,x<0\\ (x-4)^2+b & , \quad x\geqslant 0\end{cases}$ are continuous on R, then $(gof)(2)+(fog)(-2)$ is equal to:

Option 1:
-10

Option 2:
10

Option 3:
8

Option 4:
-8

Correct Answer:
-8

Solution:

$$f(x)=\begin{cases}x+a; x\le 0\\ |x-4|; x>0\end{cases}$$

$$g(x)=\begin{cases}x+1 & x<0\\ (x-4)^2+b & x\ge 0\end{cases}$$

For continuity $a=4$ and $b=-15$

$$g(f(2))+f(g(-2))$$

$$=g(2)+f(-1)=-8$$

Hence correct option is 4

Q. 9 Let $f(x)=\begin{cases} x^3-x^2+10x-7, & x\le 1 \\ -2x+\log_2\left(b^2-4\right), & x>1. \end{cases}$

Then the set of all values of b, for which $f(x)$ has maximum value at $x=1$, is :

Option 1:
$(-6,-2)$

Option 2:
$(2,6)$

Option 3:
$[-6,-2)\cup(2,6]$

Option 4:
$[-\sqrt{6},-2)\cup(2,\sqrt{6}]$

Correct Answer:
$[-6,-2)\cup(2,6]$

Solution:
$f(1)=3$

$\text{for } x<1, f'(x)=3x^2-2x+10>0$

$\Rightarrow f(x) \text{ is increasing}$

$\text{For } x>1, f'(x)<0$

$\Rightarrow$ Function is decreasing

$\lim\limits_{x\to 1^+} f(x)=-2+\log_2\left(b^2-4\right)$

$\text{For maximum value at } x=1$

$3\ge -2+\log_2\left(b^2-4\right)$

$32\ge b^2-4>0$

$b\in[-6,-2)\cup(2,6]$

Hence correct option is 3

Q. 10 If $a=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{2n}{n^2+k^2}$ and $f(x)=\sqrt{\frac{1-\cos x}{1+\cos x}}, x\in(0,1)$, then :

Option 1:
$2\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$

Option 2:
$f\left(\frac{a}{2}\right)f'\left(\frac{a}{2}\right)=\sqrt{2}$

Option 3:
$\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$

Option 4:
$f\left(\frac{a}{2}\right)=\sqrt{2}f'\left(\frac{a}{2}\right)$

Correct Answer:
$\sqrt{2}f\left(\frac{a}{2}\right)=f'\left(\frac{a}{2}\right)$

Solution:

$$a=\frac{1}{n}\sum_{k=1}^{n}\frac{2}{1+\left(\frac{k}{n}\right)^2}\doteq\int_0^1\frac{2}{1+x^2}dx=\frac{\pi}{2}$$

$$f(x)=\tan\left(\frac{x}{2}\right);\quad x\in(0,1)$$

$$f\left(\frac{\pi}{4}\right)=\sqrt{2}-1$$

$$f'\left(\frac{\pi}{4}\right)=\frac{1}{2}\sec^2\left(\frac{\pi}{8}\right)=\frac{\sqrt{2}}{\sqrt{2}+1}$$

$$f'\left(\frac{\pi}{4}\right)=\sqrt{2}f\left(\frac{\pi}{4}\right).$$

Hence correct option is 3

Q. 11 If $\frac{dy}{dx}+2y\tan x=\sin x, 0<x<\frac{\pi}{2}$ and $y\left(\frac{\pi}{3}\right)=0$, then the maximum value of $y(x)$ is :

Option 1:
$\frac{1}{8}$

Option 2:
$\frac{3}{4}$

Option 3:
$\frac{1}{4}$

Option 4:
$\frac{3}{8}$

Correct Answer:
$\frac{1}{8}$

Solution:

$\frac{dy}{dx}+2y\tan x=\sin x$

$I\cdot F=e^{\int 2\tan x dx}=e^{\ln(\sec x)^2}=\sec^2 x$

$y\left(\sec^2 x\right)=\int \sin x\cdot\sec^2 x dx+c$

$y\cdot\sec^2 x=\sec x+c$

$\text{Put } x=\frac{\pi}{3}, y=0$

$y=\cos x-2\cos^2 x$

$=\frac{1}{8}-2\left(\cos x-\frac{1}{4}\right)^2$

$\therefore y_{max}=\frac{1}{8}$

Hence correct option is 1

Q. 12 A point P moves so that the sum of squares of its distances from the points $(1, 2)$ and $(-2, 1)$ is 14. Let $f(x, y) = 0$ be the locus of P, which intersects the x-axis at the points A, B and the y-axis at the points C, D. Then the area of the quadrilateral $ACBD$ is equal to:

Option 1:
$\frac{9}{2}$

Option 2:
$\frac{3\sqrt{17}}{2}$

Option 3:
$\frac{3\sqrt{17}}{4}$

Option 4:
9

Correct Answer:
$\frac{3\sqrt{17}}{2}$

Solution:

$\Rightarrow (x-1)^2 + (y-2)^2 + (x+2)^2 + (y-1)^2 = 14$

$\Rightarrow x^2 + y^2 + x - 3y - 2 = 0$

$\text{Put } x = 0$

$\Rightarrow y^2 - 3y - 2 = 0$

$\Rightarrow y = \frac{3 \pm \sqrt{17}}{2}$

$\Rightarrow x^2 + x - 2 = 0 \Rightarrow (x+2)(x-1) = 0$

$\therefore A(-2, 0), B(1, 0), C\left(0, \frac{3+\sqrt{17}}{2}\right), D\left(0, \frac{3-\sqrt{17}}{2}\right)$

$$\text{Area } = \frac{1}{2} \times 3 \times \sqrt{17} = \frac{3\sqrt{17}}{2}$$

Hence correct option is 2

Q. 13 Let the tangent drawn to the parabola $y^2 = 24x$ at the point (α, β) is perpendicular to the line $2x + 2y = 5$. Then the normal to the hyperbola $\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ at the point $(\alpha + 4, \beta + 4)$ does NOT pass through the point :

Option 1:
$(25, 10)$

Option 2:
$(20, 12)$

Option 3:
$(30, 8)$

Option 4:
$(15, 13)$

Correct Answer:
$(15, 13)$

Solution:
Tangent at (α, p^2) has slope 1

$\beta^2 = 24\alpha$

equation of tangent $4\beta = 12(x - \alpha), \frac{12}{\beta} = 1$

$\Rightarrow \alpha = 6, \beta = 12$

$\therefore (\alpha + 4, \beta + 4) = (10, 16)$

Normal of $(10, 16)$ to $\frac{x^2}{36} - \frac{y^2}{144} = 1$ is

$2x + 5y = 100$

Hence correct option is 4

Q. 14 The length of the perpendicular from the point $(1,-2,5)$ on the line passing through $(1,2,4)$ and parallel to the line $x+y-z=0=x-2y+3z-5$ is:

Option 1:
$\sqrt{\frac{21}{2}}$

Option 2:
$\sqrt{\frac{9}{2}}$

Option 3:
$\sqrt{\frac{73}{2}}$

Option 4:
1

Correct Answer:
$\sqrt{\frac{21}{2}}$

Solution:

d.r's of the line $= \begin{vmatrix} \hat{i} & \hat{J} & \hat{k} \\ 1 & 1 & -1 \\ 1 & -2 & 3 \end{vmatrix} = \hat{i}-4\hat{J}-3\hat{k}$

Equation of line is $\vec{r}=\hat{i}+2\hat{j}+4\hat{k}+\lambda(\hat{i}-4\hat{J}-3\hat{k})$

Let $A(1,2,4)$ and P be $(1+\lambda, 2-4\lambda, 4-3\lambda)$

$\therefore \overrightarrow{PA}\cdot(\hat{i}-4\hat{j}-3\hat{k})=0$

$\lambda=\frac{1}{2}.P\left(\frac{1}{2},2,-\frac{5}{2}\right)$

$|AP|=\sqrt{\frac{21}{2}}$

Hence correct option is 1

Q. 15 Let $\vec{a} = \alpha\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}, \alpha > 0$. If the projection of $\vec{a} \times \vec{b}$ on the vector $-\hat{i} + 2\hat{j} - 2\hat{k}$ is 30, then α is equal to :

Option 1:
$\frac{15}{2}$

Option 2:
8

Option 3:
$\frac{13}{2}$

Option 4:
7

Correct Answer:
7

Solution:
$\Rightarrow \vec{a} \times \vec{b} = (1-\alpha)\hat{i} + (\alpha^2 - 2)\hat{j} + (\alpha - 2)\hat{k}$

Projection of $\vec{a} \times \vec{b}$ on $-\hat{i} + 2\hat{j} - 2\hat{k}$

$= \frac{(\vec{a} \times \vec{b}) \cdot (-\hat{i} + 2\hat{j} - 2\hat{k})}{3} = 30$

$\Rightarrow 2\alpha^2 - \alpha - 91 = 0$

$\Rightarrow \alpha = 7, -\frac{13}{2}$

Hence correct option is 4

Q. 16 The mean and variance of a binomial distribution are α and $\frac{\alpha}{3}$ respectively. If $P(X = 1) = \frac{4}{243}$, then $P(X = 4 \text{ or } 5)$ is equal to:

Option 1:

$\frac{5}{9}$

Option 2:

$\frac{64}{81}$

Option 3:

$\frac{16}{27}$

Option 4:

$\frac{145}{243}$

Correct Answer:

$\frac{16}{27}$

Solution:

$\Rightarrow np = \alpha$ .....(1)

$npq = \frac{\alpha}{3}$ .....(2)

from (1) & (2)

$q = \frac{1}{3} \quad \& \ p = \frac{2}{3}$

$^{n}C_1 q^{n-1} \cdot p^1 = \frac{4}{243}$

$\frac{n}{3^n} = \frac{2}{243}$

$n = 6$

$P(4 \text{ or } 5) = {}^6C_4 \left(\frac{2}{3}\right)^4 \left(\frac{1}{3}\right)^2 + {}^6C_5 \left(\frac{2}{3}\right)^5 \left(\frac{1}{3}\right)^0$

$= \frac{16}{27}$

Hence correct option is 3

Q. 17 Let E_1, E_2, E_3 be three mutually exclusive events such that $P(E_1) = \frac{2+3p}{6}, P(E_2) = \frac{2-p}{8}$ and $P(E_3) = \frac{1-p}{2}$. If the maximum and minimum values of P are p_1 and p_2, then $(p_1 + p_2)$ is equal to:

Option 1:
$\frac{2}{3}$

Option 2:
$\frac{5}{3}$

Option 3:
$\frac{5}{4}$

Option 4:
1

Correct Answer:
$\frac{5}{3}$

Solution:

$\Rightarrow 0 \leq P(E_i) \leq 1 \text{ for } i = 1, 2, 3$

$\Rightarrow -\frac{2}{3} \leq P \leq 1$

$E_1 \& E_2 \& E_3 \text{ are mutually exclusive}$

$P(E_1) + P(\epsilon_2) + P(\epsilon_3) \leq 1$

$\Rightarrow \frac{2}{3} \leq P \leq 1$

$P_1 = 1, P_2 = \frac{2}{3} \quad P_1 + P_2 = \frac{5}{3}$

Hence correct option is 2

Q. 18 Let $S=\left\{\theta \in[0,2 \pi]: 8^{2 \sin ^{2} \theta}+8^{2 \cos ^{2} \theta}=16\right\}$. Then $n(S)+\sum_{\theta \in S}\left(\sec \left(\frac{\pi}{4}+2 \theta\right) \operatorname{cosec}\left(\frac{\pi}{4}+2 \theta\right)\right)$ is equal to:

Option 1:
0

Option 2:
-2

Option 3:
-4

Option 4:
12

Correct Answer:
-4

Solution:
$8^{2 \sin ^{2} \theta}+8^{2-2 \sin ^{2} \theta}=16$

$y+\frac{64}{y}=16$

$\Rightarrow y=8$

$\Rightarrow \sin ^{2} \theta=\frac{1}{2}$

$n(s)+\sum_{\theta \in S} \frac{1}{\cos \left(\frac{\pi}{4}+2 \theta\right) \sin \left(\frac{\pi}{4}+2 \theta\right)}$

$4+(-2) \times 4=-4$

Hence correct option is 3

Q. 19 $\tan \left(2 \tan ^{-1} \frac{1}{5}+\sec ^{-1} \frac{\sqrt{5}}{2}+2 \tan ^{-1} \frac{1}{8}\right)$ is equal to :

Option 1:
1

Option 2:
2

Option 3:
$\frac{1}{4}$

Option 4:
$\frac{5}{4}$

Correct Answer:
2

Solution:

$$\tan\left(2\left(\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{8}\right)+\tan^{-1}\left(\frac{1}{2}\right)\right)$$

$$=\tan\left[2\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{2}\right)\right]$$

$$=\tan\left[\tan^{-1}\frac{3}{4}+\tan^{-1}\frac{1}{2}\right]$$

$$=\tan\left(\tan^{-1}2\right)=2$$

Hence correct option is 2

Q. 20 The statement $(\sim(p\Leftrightarrow\sim q))\wedge q$ is :

Option 1:
a tautology

Option 2:
a contradiction

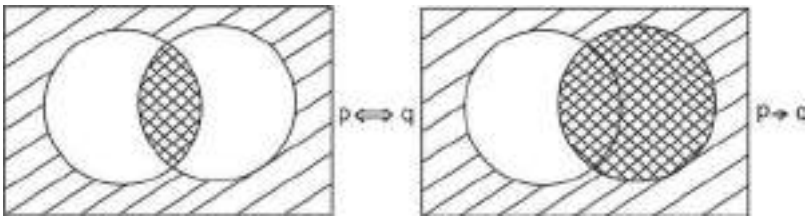
Option 3:
equivalent to $(p\Rightarrow q)\wedge q$

Option 4:
equivalent to $(p \Leftrightarrow q) \wedge q$

Correct Answer:
equivalent to $(p \Leftrightarrow q) \wedge q$

Solution:
$(\sim (\mathrm{p} \Leftrightarrow \sim \mathrm{q})) \wedge \mathrm{q} \equiv (\mathrm{p} \Leftrightarrow \mathrm{q}) \wedge \mathrm{q}$

$(\mathrm{p} \Leftrightarrow \mathrm{q}) \wedge \mathrm{q} \equiv \mathrm{p} \wedge \mathrm{q}$



Hence correct option is 4

Q. 21 If for some $\mathrm{p}, \mathrm{q}, \mathrm{r} \in \mathbf{R}$, not all have same sign, one of the roots of the equation $\left(\mathrm{p}^2+\mathrm{q}^2\right) \mathrm{x}^2-2\mathrm{q}(\mathrm{p}+\mathrm{r})\mathrm{x}+\mathrm{q}^2+\mathrm{r}^2=0$ is also a root of the equation $\mathrm{x}^2+2\mathrm{x}-8=0$, then $\frac{\mathrm{q}^2+\mathrm{r}^2}{\mathrm{p}^2}$ is equal to_____.

Correct Answer:
272

Solution:
$(\mathrm{px}-\mathrm{q})^2+(\mathrm{qx}-\mathrm{r})^2=0$

$\Rightarrow \mathrm{x}=\frac{\mathrm{q}}{\mathrm{p}}=\frac{\mathrm{r}}{\mathrm{q}}=-4$

$\Rightarrow \frac{\mathrm{q}^2+\mathrm{r}^2}{\mathrm{p}^2}=272$

Hence answer is 272

Q. 22 The number of 5 -digit natural numbers, such that the product of their digits is 36 , is _____.

Correct Answer:
180

Solution:
$$3 \times \frac{5!}{2!2!} + \frac{5!}{3! \times 2!} + \frac{5!}{2!} + \frac{5!}{3!} = 180$$

Hence answer is 180

Q. 23 The series of positive multiples of 3 is divided into sets $: \{3\}, \{6, 9, 12\}, \{15, 18, 21, 24, 27\}, \ldots$ Then the sum of the elements in the 11^{th} set is equal to _____.

Correct Answer:
6993

Solution:
$$S_{11} = 3[101 + 102 + \cdots + 121]$$

$$= \frac{3}{2}(222) \times 21 = 6993$$

Hence answer is 6993

Q. 24 The number of distinct real roots of the equation

$x^5\left(x^3 - x^2 - x + 1\right) + x\left(3x^3 - 4x^2 - 2x + 4\right) - 1 = 0$ is _____.

Correct Answer:
3

Solution:
$$x^5\left(x^3 - x^2 - x + 1\right) + x\left(3x^3 - 4x^2 - 2x + 4\right) - 1 = 0$$

$$\Rightarrow (x-1)^2(x+1)\left(x^5 + 3x - 1\right) = 0$$

$$\text{let } f(x) = x^5 + 3x - 1$$

$$f'(x) > 0 \forall x \in R$$

Hence 3 real distinct roots

Q. 25 If the coefficients of x and x^2 in the expansion of $(1+x)^p(1-x)^q, p, q \leq 15,$ are -3 and -5 respectively, then the coefficient of x^3 is equal to_____.

Correct Answer:
23

Solution:

Since coefficient of x is -3

$\Rightarrow {}^{p}c_1 - {}^{q}c_1 = -3$

$\Rightarrow p - q = -3$ .....(1)

Comparing coefficients of x^2

$-{}^{p}c_1 {}^{q}c_1 + {}^{p}c_2 + {}^{q}c_2 = -5$

$-pq + \frac{p(p-1)}{2} + \frac{q(q-1)}{2} = -5$ .....(2)

Solving (1) and (2)

$p = 8, q = 11$

coefficient of x^3 is

$-{}^{q}C_3 + {}^{p}C_3 + {}^{p}C_1 {}^{q}c_2 - {}^{p}C_2 {}^{q}C_1$

$= -{}^{11}C_3 + {}^{8}C_3 + {}^{8}C_1 {}^{11}C_2 - {}^{8}C_2 {}^{11}C_1$

$= 23$

Hence answer is 23

Q. 26 If $n(2n+1)\int_0^1 (1-x^n)^{2n}\,dx = 1177\int_0^1 (1-x^n)^{2n+1}\,dx$, then $n \in \mathbf{N}$ is equal to_____.

Correct Answer: 24

Solution:

$$\text{Let } I_1 = \int_0^1 (1-x^n)^{2n}\,dx,\ I_2 = \int_0^1 (1-x^n)^{2n+1}\,dx$$

$$I_2 = \int_0^1 (1-x^n)^{2n+1}\cdot 1\,dx$$

$$= (1-x^n)^{2n+1}\cdot x\Big|_0^1 - \int_0^1 (2n+1)(1-x^n)^{2n}\left(-nx^{n-1}\right)x\,dx$$

$$I_2 = -n(2n+1)\{I_2 - I_1\}$$

$$\left(2n^2+n+1\right)I_2 = n(2n+1)I_1$$

$$\frac{I_1}{I_2} = \frac{2n^2+n+1}{n(2n+1)} = \frac{1177}{n(2n+1)}$$

$$\Rightarrow 2n^2+n-1176 = 0 \Rightarrow n = 24$$

Hence answer is 24

Q. 27 Let a curve $y = y(x)$ pass through the point $(3,3)$ and the area of the region under this curve, above the x-axis and between the abscissae 3 and $x(>3)$ be $\left(\frac{y}{x}\right)^3$. If this curve also passes through the point $(\alpha, 6\sqrt{10})$ in the first quadrant, then α is equal to _____.

Correct Answer: 6

Solution:

$$x^4 = 3yx\times y' - 3y^2$$

$$\Rightarrow 3xy\frac{dy}{dx} = 3y^2 + x^4$$

$$\text{Put } y^2 = t, y\frac{dy}{dx} = \frac{1}{2}\frac{dt}{dx}$$

$$\frac{dt}{dx} - \frac{2}{x}t = \frac{2}{3}x^3$$

$$\therefore \frac{t}{x^2} = \frac{x^2}{3} + c$$

$$\Rightarrow \frac{y^2}{x^2} = \frac{x^2}{3} - 2$$

$\text{Put } (3,3), c = -2$

$$\therefore \frac{y^2}{x^2} = \frac{x^2}{3} - 2$$

$$3y^2 = x^4 - 6x^2$$

$$x^4 - 6x^2 = 1080$$

$$\therefore x = 6$$

Hence answer is 6

Q. 28 The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0, x + py = 15a$ and $x - y = 3$ respectively. If its orthocentre is $(2, a), -\frac{1}{2} < a < 2$, then P is equal to_____.

Correct Answer:
3

Solution:

$\text{Coordinates of } A(1,-2), B\left(\frac{15a}{1-2p}; \frac{-30a}{1-2P}\right) \text{ and orthocentre } H(2,a)$

$\text{Slope of } AH = p$

$a + 2 = P$ .....(1)

$\text{Slope of } BH = -1$

$31a - 2ab = 15a + 4p - 2$ .....(2)

$\text{from}(1)\text{and}(2)$

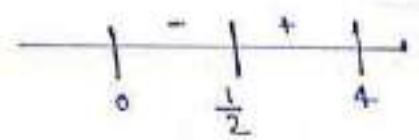
$a = 1 \& P = 3$

Hence answer is 3

Q. 29 Let the function $f(x) = 2x^2 - \log_e x, x > 0$, be decreasing in $(0, a)$ and increasing in $(a, 4)$. A tangent to the parabola $y^2 = 4ax$ at a point P on it passes through the point $(8a, 8a - 1)$ but does not pass through the point $\left(-\frac{1}{a}, 0\right)$. If the equation of the normal at P is $\frac{x}{\alpha} + \frac{y}{\beta} = 1$, then $\alpha + \beta$ is equal to _____.

Correct Answer:
45

Solution:



$f'(x) = 4x - \frac{1}{x}$

$a = \frac{1}{2}$

Let $P(x_1, y_1)$ be any point on $y^2 = 4ax$

$\frac{1}{y_1} = \frac{3 - y_1}{4 - x_1} \Rightarrow y_1^2 - 6y_1 + 8 = 0$

$y_1 = 2, 4$

$\Rightarrow P(8, 4) \text{as } P(2, 2) \text{rejected}$

Equation of normal at P

$y - 4 = -4(x - 8)$

$\frac{x}{9} + \frac{y}{36} = 1$

$\alpha = 9, \beta = 36$

$\alpha + \beta = 45$

Hence answer is 45

Q. 30 Let Q and R be two points on the line $\frac{x+1}{2}=\frac{y+2}{3}=\frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point $\mathrm{P}(4,2,7)$. Then the square of the area of the triangle PQR is _____.

Correct Answer:
153

Solution:
$\mathrm{Let}(2\lambda-1,3\lambda-2,2\lambda+1)\mathrm{be\ any\ point\ on\ the\ line}$

$(2\lambda-5)^2+(3\lambda-4)^2+(2\lambda-6)^2=26$

$\lambda=1,3$

$\mathrm{Q}(1,1,3);\mathrm{R}(5,7,7);\mathrm{P}(4,2,7)$

$\mathrm{Area\ of\ triangle\ PQR}=\frac{1}{2}|\mathrm{P}\vec{Q}\times\mathrm{P}\vec{R}|$

$=\sqrt{153}$

Hence answer is $\sqrt{153}$

CAREERS360

26 July - Evening Shift

Physics

Q. 1 Two projectiles are thrown with same initial velocity making an angle of 45° and 30° with the horizontal respectively. The ratio of their respective ranges will be :

Option 1:
$1 : \sqrt{2}$

Option 2:
$\sqrt{2} : 1$

Option 3:
$2 : \sqrt{3}$

Option 4:
$\sqrt{3} : 2$

Correct Answer:
$2 : \sqrt{3}$

Solution:
$\theta_1 = 45^\circ$

$\theta_2 = 30^\circ$

$$R_1 = \frac{u^2 \sin 2\theta_1}{g}$$

$$R_2 = \frac{u^2 \sin 2\theta_2}{g}$$

$$\frac{R_1}{R_2} = \frac{\sin 2\theta_1}{\sin 2\theta_2} = \frac{\sin 90^\circ}{\sin 60^\circ}$$

$$\frac{R_1}{R_2} = \frac{1}{(\sqrt{3}/2)} = \frac{2}{\sqrt{3}}$$

Hence (3) is correct option.

Q. 2 In a Vernier Calipers, 10 divisions of Vernier scale is equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4^{th} Vernier scale division exactly coincides with the main scale reading. One main scale division is equal to $1\ \text{mm}$.While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6^{th} Vernier scale division exactly coincides with the main scale reading. The diameter of the spherical body will be :

Option 1:
$3.02\ \text{cm}$

Option 2:
$3.06\ \text{cm}$

Option 3:
$3.10\ \text{cm}$

Option 4:
$3.20\ \text{cm}$

Correct Answer:
$3.02\ \text{cm}$

Solution:

$10\text{VSD} = 9\text{MSD}$

$$\text{Least count } (\text{L} \cdot \text{C.}) = 1\ \text{MSD} - 1\ \text{VSD}$$

$$= \frac{1}{10}\text{MSD}$$

$$= 0.1\ \text{mm}$$

$$\text{zero error } = (\text{VSD}) \times \text{LC}$$

$$= (4) \times 0.1$$

$$= 0.4\ \text{mm}$$

$$\text{Reading } = \text{ MSR } + \text{ VSR } + \text{ Zero correction}$$

$$= 30 + 6(0.1) + (-0.4)$$

$$= 30.2 \text{ mm}$$

$$= 3.02 \text{ cm}$$

Hence (1) is correct option.

Q. 3 A ball of mass 0.15 kg hits the wall with its initial speed of 12 ms^{-1} and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is 100 N, calculate the time duration of the contact of ball with the wall.

Option 1:
0.018 s

Option 2:
0.036 s

Option 3:
0.009 s

Option 4:
0.072 s

Correct Answer:
0.036 s

Solution:

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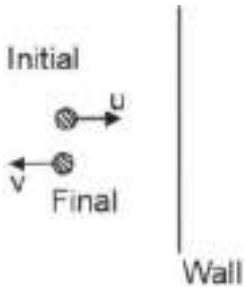


$$\Delta p = p_f - p_i$$

$$= (-mv) - (+mu)$$

$$= (-mv - mu)$$

$$v = u = 12 \text{ m/s} \quad (\text{ Given })$$

$$\Delta p = -0.15(12 + 12)$$

$$\Delta p = -3.6$$

$$F = \frac{\Delta p}{\Delta t}$$

$$100 = \frac{3.6}{\Delta t}$$

$$\Delta t = 0.0369$$

Hence (2) is correct option.

Q. 4 A body of mass 8 kg and another of mass 2 kg are moving with equal kinetic energy. The ratio of their respective momenta will be :

Option 1:
$1 : 1$

Option 2:
$2 : 1$

Option 3:
$1 : 4$

Option 4:
$4 : 1$

Correct Answer:

$2:1$

Solution:

We know that,

$$\mathrm{KE}=\frac{\mathrm{P}^2}{2\mathrm{m}}$$

$$\mathrm{KE}_1=\mathrm{KE}_2$$

$$\frac{\mathrm{P}_1^2}{2\mathrm{m}_1}=\frac{\mathrm{P}_2^2}{2\mathrm{m}_2}$$

$$\frac{\mathrm{P}_1}{\mathrm{P}_2}=\sqrt{\frac{\mathrm{m}_1}{\mathrm{m}_2}}=\sqrt{\frac{8}{2}}=2$$

Hence (2) is correct option

Q. 5 Two uniformly charged spherical conductors A and B of radii $5\ \mathrm{mm}$ and $10\ \mathrm{mm}$ are separated by a distance of $2\ \mathrm{cm}$. If the spheres are connected by a conducting wire, then in equilibrium condition, the ratio of the magnitudes of the electric fields at the surface of the sphere A and B will be :

Option 1:

$1:2$

Option 2:

$2:1$

Option 3:

$1:1$

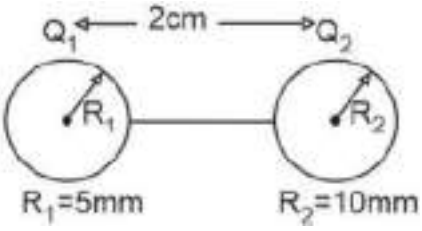
Option 4:

$1:4$

Correct Answer:

$2:1$

Solution:



When the spheres are connated by conducting wire, their potential will be same

$$V_1 = V_2$$

$$\frac{KQ_1}{R_1} = \frac{KQ_2}{R_2}$$

$$\frac{Q_1}{Q_2} = \frac{R_1}{R_2} = \frac{1}{2}$$

Electic field on the surface of the sphere

$$E = \frac{KQ}{R^2}$$

$$\frac{E_1}{E_2} = \frac{Q_1 R_2^2}{Q_2 R_1^2} = \frac{2}{1}$$

Electric field on the surface of the one sphere is independent of other. (as separation is large as compare to size of sphere)

Hence (2) is correct option.

Q. 6 The oscillating magnetic field in a plane electromagnetic wave is given by $B_y = 5 \times 10^{-6} \sin 1000\pi \left(5x - 4 \times 10^8 t\right) T$ The amplitude of electric field will be :

Option 1:

$15 \times 10^2 V_{m^{-1}}$

Option 2:

$5 \times 10^{-6} V m^{-1}$

Option 3:

$16 \times 10^{12} \mathrm{Vm}^{-1}$

Option 4:

$4 \times 10^{2} \mathrm{Vm}^{-1}$

Correct Answer:

$15 \times 10^{2} \mathrm{V}_{\mathrm{m}^{-1}}$

Solution:

$\mathrm{By} = 5 \times 10^{-6} \sin\left(1000\pi\left(5\mathrm{x} - 4 \times 10^{8}\mathrm{t}\right)\right)$

$\mathrm{B}_0 = 5 \times 10^{-6}\mathrm{T}$

We know that,

$\frac{\mathrm{E}_0}{\mathrm{B}_0} = \mathrm{C}$

$\mathrm{E}_0 = 15 \times 10^{2}\frac{\mathrm{V}}{\mathrm{m}}$

Hence (1) is correct option.

Q. 7 Light travels in two media M_1 and M_2 with speeds $1.5 \times 10^{8}\ \mathrm{ms}^{-1}$ and $2.0 \times 10^{8}\ \mathrm{ms}^{-1}$ respectively. The critical angle between them is :

Option 1:

$\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$

Option 2:

$\tan^{-1}\left(\frac{2}{3}\right)$

Option 3:

$\cos^{-1}\left(\frac{3}{4}\right)$

Option 4:

$\sin^{-1}\left(\frac{2}{3}\right)$

Correct Answer:

$\tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$

Solution:

$$\mu_D^R = \frac{1}{\sin i_C} = \frac{\mu_D}{\mu_R} = \frac{V_R}{V_D}$$

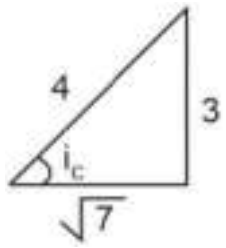
$i_c \rightarrow$ critical angle

$V_R \rightarrow$ velocity in rarer medium

$V_D \rightarrow$ Velocity in denser medium

$$\frac{1}{\sin i_c} = \frac{2 \times 10^8}{1.5 \times 10^8}$$

$$\sin i_c = \frac{3}{4}$$



$$\tan i_c = \frac{3}{\sqrt{7}}$$

$$i_c = \tan^{-1}\left(\frac{3}{\sqrt{7}}\right)$$

Hence (1) is correct option.

Q. 8 A body is projected vertically upwards from the surface of earth with a velocity equal to one third of escape velocity. The maximum height attained by the body will be :

(Take radius of earth $= 6400$ km and $g = 10\ ms^{-2}$)

Option 1:
$800\ \text{km}$

Option 2:
$1600\ \text{km}$

Option 3:
$2133\ \text{km}$

Option 4:
$4800\ \text{km}$

Correct Answer:
$800\ \text{km}$

Solution:

$$u = \frac{V_e}{3}$$

We know that, $V_e = \sqrt{\frac{2GM}{R}}$

By energy conservation, TE at surface = TE at maximum

$$\frac{1}{2}mu^2 + \left[\left(\frac{-GMm}{R}\right)\right] = \frac{1m}{2}(0)^2 + \left(\frac{-GMm}{R+h}\right)$$

$$\frac{8GMm}{9R} = \frac{GMm}{R+h}$$

$$h = \frac{R}{8} = 800\ \text{km}$$

Hence (1) is correct option.

Q. 9 The maximum and minimum voltage of an amplitude modulated signal are $60\ \text{V}$ and $20\ \text{V}$ respectively. The percentage modulation index will be :

Option 1:
0.5%

Option 2:
50%

Option 3:
2%

Option 4:
30%

Correct Answer:
50%

Solution:
$\mathrm{V_{max}} = 60\ \mathrm{V}$

$\mathrm{V_{min}} = 20\ \mathrm{V}$
The percentage modulation index is

$$\mu\% = \frac{\mathrm{V_{max}} - \mathrm{V_{min}}}{\mathrm{V_{max}} + \mathrm{V_{min}}} \times 100$$

$$= \frac{40}{80} \times 100$$

$$= 50\%$$

Hence (2) is correct option.

The correct option is (2)

Q. 10 A nucleus of mass m at rest splits into two parts having masses $\frac{\mathrm{M'}}{3}$ and $\frac{2\mathrm{M'}}{3}\ (\mathrm{M'} < \mathrm{M})$. The ratio of de Broglie wavelength of two parts will be :

Option 1:

$1:2$

Option 2:

$2:1$

Option 3:

$1:1$

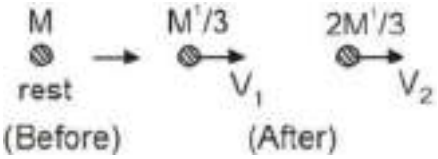
Option 4:

$2:3$

Correct Answer:

$1:2$

Solution:



$$M(o) = \frac{M'}{3} \times V_1 + \frac{2M'}{3} \times V_2$$

$$\frac{V_1}{V_2} = 2$$

De-broglie wavelength $= \frac{h}{mv}$

$$\frac{\lambda_1}{\lambda_2} = \frac{2M'/3 \times V_2}{\frac{M'}{3} \times V_1} = \frac{2}{2}$$

$$= 1$$

Hence (3) is correct option.

Q. 11 An ice cube of dimensions $60\ \text{cm} \times 50\ \text{cm} \times 20\ \text{cm}$ is placed in an insulation box of wall thickness $1\ \text{cm}$. The box keeping the ice cube at 0°C of temperature is brought to a room of temperature 40°C.The rate of melting of ice is approximately:

(Latent heat of fusion of ice is $3.4 \times 10^5\ \text{J kg}^{-1}$ and thermal conducting of insulation wall is $0.05\ \text{Wm}^{-1\circ}\text{C}^{-1}$)

Option 1:
$61 \times 10^{-3}\ \text{kg s}^{-1}$

Option 2:
$61 \times 10^{-5}\ \text{kg s}^{-1}$

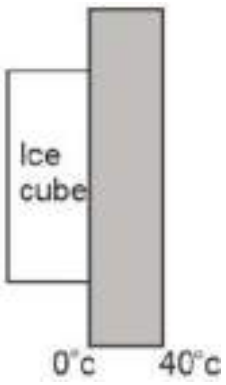
Option 3:
$208\ \text{kg s}^{-1}$

Option 4:
$30 \times 10^{-5}\ \text{kg s}^{-1}$

Correct Answer:
$61 \times 10^{-5}\ \text{kg s}^{-1}$

Solution:
Volume of ice cube $= 6 \times 10^{4}\ (\text{cm})^3$

$$= 6 \times 10^{-2}\text{m}^3$$



$$\text{Rate of fusion} = \frac{\text{dQ}}{\text{dt}} = \left(\frac{\text{dm}}{\text{dt}}\right) \text{L}_\text{f}$$

$$\text{Rate of heat flow} = \frac{\text{KA}(\Delta\text{T})}{\text{t}}$$

$\text{t} \rightarrow$ thickness of wall

At equilibrium,

Rate of fusion = Rate of heat flow

$$\left(\frac{\text{dm}}{\text{dt}}\right) \text{L}_\text{f} = \frac{2\text{KA}_1\Delta\text{T}}{\Delta\text{t}} + \frac{2\text{KA}_2\Delta\text{T}}{\Delta\text{t}} + \frac{2\text{KA}_3\Delta\text{T}}{\Delta\text{t}}$$

$$\left(\frac{dm}{dt}\right) \times 3.4 \times 10^{S} = \left(\frac{2 \times 0.05 \times 40}{10^{-2}}\right)[A_1 + A_2 + A_3]$$

$$\left(\frac{dm}{dt}\right) = \frac{4 \times 10^2}{3.4 \times 10^5} \times [3000 + 1000 + 1200] \times 10^{-4}$$

$$= \frac{4}{3.4} \times 5.2 \times 10^{-4} = 61 \times 10^{-5} \frac{kg}{s}$$

Hence (2) is correct option.

Q. 12 A gas has n degrees of freedom. The ratio of specific heat of gas at constant volume to the specific heat of gas at constant pressure will be :

Option 1:
$\frac{n}{n+2}$

Option 2:
$\frac{n+2}{n}$

Option 3:
$\frac{n}{2n+2}$

Option 4:
$\frac{n}{n-2}$

Correct Answer:
$\frac{n}{n+2}$

Solution:
$\frac{c_v}{c_p} = \frac{1}{v} = \frac{f}{2+f}$
It is given that,
Degree of freedom $= f = n$

$\therefore \frac{c_v}{c_p} = \frac{n}{2+n}$

Hence (1) is correct option.

Q. 13 A transverse wave is represented by $y = 2\sin(\omega t - kx)cm$.The value of wavelength $(in\ cm)$ for which the wave velocity becomes equal to the maximum particle velocity, will be :

Option 1:
4π

Option 2:
2π

Option 3:
π

Option 4:
2

Correct Answer:
4π

Solution:
$y = 2\sin(\omega t - kx)$
Wave velocity = Maximum particle velocity

$$\frac{\omega}{k} = A\omega$$

$$\frac{1}{(2\pi/\lambda)} = A$$

$$\lambda = A(2\pi)$$

$$\lambda = 4\pi$$

Hence (1) is correct option.

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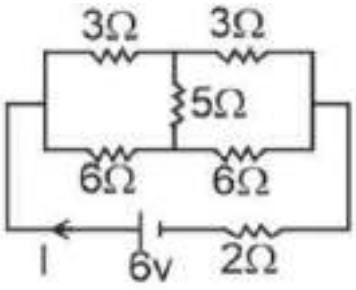
Q. 14 A battery of $6\mathrm{V}$ is connected to the circuit as shown below. The current I drawn from the battery is:



Option 1:
$1\mathrm{A}$

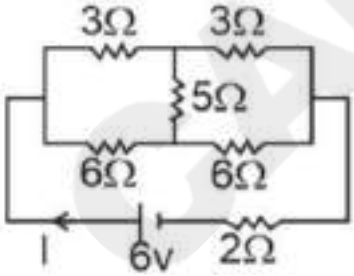
Option 2:
$2\mathrm{A}$

Option 3:
$\frac{6}{11}\mathrm{A}$

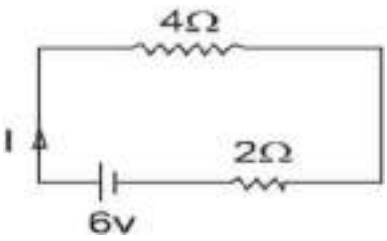
Option 4:
$\frac{4}{3}\mathrm{A}$

Correct Answer:
$1\mathrm{A}$

Solution:



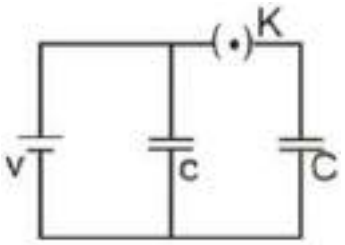
Here,there won't be current through 5Ω resistance as it is balanced wheatstone network



$$I = \frac{6}{6} = 1A$$

Hence (1) is correct option.

Q. 15 A source of potential difference V is connected to the combination of two identical capacitors as shown in the figure. When key $'K'$ is closed, the total energy stored across the combination is E_1. Now key $'K'$ is opened and dielectric of dielectric constant 5 is introduced between the plates of the capacitors. The total energy stored across the combination is now E_2.The ratio E_1/E_2 will be :



Option 1:
$\frac{1}{10}$

Option 2:
$\frac{2}{5}$

Option 3:
$\frac{5}{13}$

Option 4:
$\frac{5}{26}$

Correct Answer:
$\frac{2}{5}$

Solution:
When the key $'K'$ is closed
The potential drop across C is V

$$E_1 = \frac{1}{2}CV^2 + \frac{1}{2}CV^2 = CV^2$$

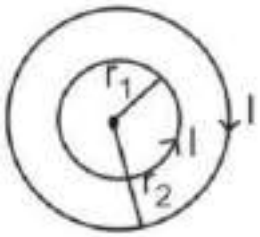
When the key 'K' is open

$$E_2 = \frac{1}{2}C'v^2 = \frac{1}{2}(5c)v^2$$

$$\frac{E_1}{E_2} = \frac{cv^2}{\frac{1}{2}c'v^2} = \frac{2}{5}$$

Hence (2) is correct option.

Q. 16 Two concentric circular loops of radii $r_1 = 30\ cm$ and $r_2 = 50\ cm$ are placed in $X - Y$ plane as shown in the figure. A current $I = 7A$ is flowing through them in the direction as shown in figure. The net magnetic moment of this system of two circular loops is approximately :



Option 1:
$\frac{7}{2}\hat{k}Am^2$

Option 2:
$-\frac{7}{2}\hat{k}Am^2$

Option 3:
$7\hat{k}Am^2$

Option 4:
$-7\hat{k}Am^2$

Correct Answer:
$-\frac{7}{2}\hat{k}Am^2$

Solution:
The net magnetic moment $= \bar{M}_1 + \bar{M}_2$

$$\mathrm{M_{net}} = \mathrm{IA_1}(\hat{\mathrm{k}}) + \mathrm{IA_2}\left(-\hat{\mathrm{K}}\right)$$

$$\vec{\mathrm{M}}_{\mathrm{net}} = \left(7 \times \pi \times 900 \times 10^4 - 7 \times \pi \times 2500 \times 10^{-4}\right)\hat{\mathrm{k}}$$

$$= 7 \times 1600 \times \pi \times 10^{-4}\left(\hat{\mathrm{k}}\right)$$

$$= 35200 \times 10^{-4}\left(\hat{\mathrm{k}}\right)$$

$$\vec{\mathrm{M}}_{\mathrm{net}} \cong -\frac{7}{2}\hat{\mathrm{k}}\mathrm{Am}^2$$

Hence (2) is correct option.

Q. 17 A velocity selector consists of electric field $\vec{E} = \mathrm{E}\,\hat{\mathrm{k}}$ and magnetic field $\vec{B} = \mathrm{B}\,\hat{\mathrm{j}}$ with $\mathrm{B} = 12\ \mathrm{mT}$. The value of E required for an electron of energy $728\ \mathrm{eV}$ moving along the positive $\mathrm{x} - \mathrm{axis}$ to pass undeflected is :

$(\mathrm{Given,\ mass\ of\ electron\ } = 9.1 \times 10^{-31}\ \mathrm{kg}\ )$

Option 1:
$192\ \mathrm{kVm^{-1}}$

Option 2:
$192\ \mathrm{mVm^{-1}}$

Option 3:
$9600\ \mathrm{kVm^{-1}}$

Option 4:
$16\ \mathrm{kVm^{-1}}$

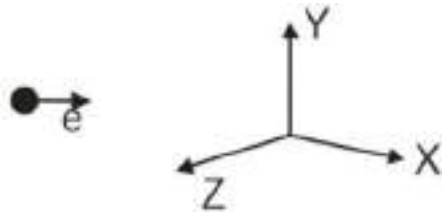
Correct Answer:
$192\ \mathrm{kVm^{-1}}$

Solution:

$$\bar{\mathrm{E}} = \mathrm{E}(\hat{\mathrm{k}})$$

$$\bar{\mathrm{B}} = \mathrm{B}(\hat{\mathrm{j}})$$

$$\mathrm{B} = 12\ \mathrm{mT}$$



$$KE = \frac{1}{2}mv^2 = 728ev \rightarrow (1)$$

For velocity selector,

$$\bar{F}_E = -\bar{F}_m$$

$$q\bar{E} = -q(\bar{V} \times \bar{B})$$

$$E = e(VB)\sin 90^\circ$$

$$(\because q = -e)$$

$$v = \frac{E}{B} \rightarrow (2)$$

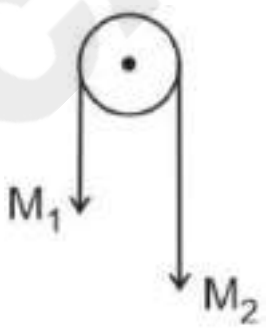
$$E = VB$$

$$E = \left(\frac{728 \times 1.6 \times 10^{-19} \times 2}{9.1 \times 10^{-31}}\right)^{1/2} \times 12 \times 10^{-3}$$

$$E = 192\ kVm^{-1}$$

Hence (1) is correct option

Q. 18 Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1, the acceleration of the system is a_1. When the mass M_2 is thrice that of M_1, the acceleration of the system is a_2.The ratio $\frac{a_1}{a_2}$ will be :



Option 1:
$\frac{1}{3}$

Option 2:
$\frac{2}{3}$

Option 3:
$\frac{3}{2}$

Option 4:
$\frac{1}{2}$

Correct Answer:
$\frac{2}{3}$

Solution:
We know that,

$$a = \frac{(M_2 - M_1)}{(M_1 + M_2)}g$$

For case 1:

$$a_1 = \left(\frac{(2M_2) - (M_1)}{(2M_1) + M_2)}\right)g$$

for case 2:

$$a_2 = \left(\frac{3M_1 - M_1}{3M_1 + M_1}\right)g$$

$$a_2 = \frac{9}{2} \rightarrow (2)$$

From the eq (1) & (2)

$$\frac{a_1}{a_2} = \frac{2}{3}$$

Hence (2) is correct option.

Q. 19 Mass numbers of two nuclei are in the ratio of $4:3$. Their nuclear densities will be in the ratio of

Option 1:
$4:3$

Option 2:
$\left(\frac{3}{4}\right)^{\frac{1}{3}}$

Option 3:
$1:1$

Option 4:
$\left(\frac{4}{3}\right)^{\frac{1}{3}}$

Correct Answer:
$4:3$

Solution:
$\frac{A_1}{A_2}=\frac{4}{3} \quad \cdots \text{(Given)}$
Nuclear density is same for all the nuclei irrespective of their mass number and it is of the order of $10^{17}\ \text{kg/m}^3$
Hence (3) is correct option.

Q. 20 The area of cross section of the rope used to lift a load by a crane is $2.5\times10^{-4}\ \text{m}^2$. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be : $(\ \text{take g} = 10\ \text{ms}^{-2}\ )$

Option 1:
$6.25\times10^{-4}\ \text{m}^2$

Option 2:
$10\times10^{-4}\ \text{m}^2$

Option 3:
$1\times10^{-4}\ \text{m}^2$

Option 4:

$1.67 \times 10^{-4}\ \mathrm{m}^2$

Correct Answer:

$6.25 \times 10^{-4}\ \mathrm{m}^2$

Solution:

For a given material, the breaking stress is same

$$\therefore \frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$\frac{10}{2.5 \times 10^{-4}} = \frac{25}{A_2}$$

$$A_2 = 6.25 \times 10^{-4}\ \mathrm{m}^2$$

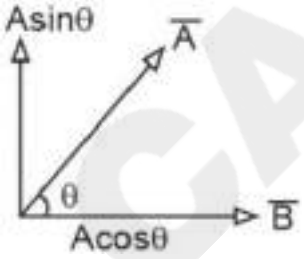
Hence (1) is correct option.

Q. 21 If $\vec{A} = (2\hat{i} + 3\hat{j} - \hat{k})\mathrm{m}$ and $\vec{B} = (\hat{i} + 2\hat{j} + 2\hat{k})\mathrm{m}$. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be _____ m.

Correct Answer:

2

Solution:



The magnitude of component of vector $\bar{A}$ along vector $\bar{B}$ will be

$$A\cos\Theta = \frac{|\bar{A}.\bar{B}|}{|\bar{B}|}$$

$$\bar{A} = 2\hat{i} + 3\hat{j} - \hat{k}$$

$$\bar{B} = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\bar{A} \cdot \bar{B} = 2 + 6 - 2 = 6$$

$$|\bar{B}| = \sqrt{1^2 + 2^2 + 2^2}$$

$$= 3$$

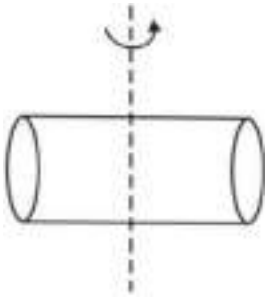
$$A\cos\theta = \frac{\bar{A} \cdot \bar{B}}{|\bar{B}|}$$

$$= \frac{6}{3} = 2$$

Q. 22 The radius of gyration of a cylindrical rod about an axis of rotation perpendicular to its length and passing through the center will be _____ m. Given, the length of the rod is $10\sqrt{3}$ m.

Correct Answer:
5

Solution:



$$I = \frac{Ml^2}{12} = MK^2$$

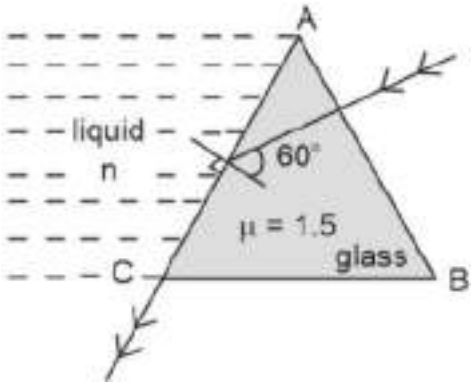
$$K = \frac{\ell}{\sqrt{12}} = \frac{10\sqrt{3}}{2\sqrt{3}}$$

$$K = 5m$$

The radius of gyration is 5m

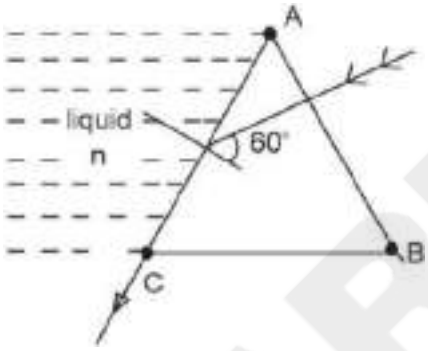
Q. 23 In the given figure, the face AC of the equilateral prism is immersed in a liquid of refractive index 'n'. For incident angle $60°$ at the side AC, the refracteel light beam just grazes along face AC. The refractive index of the liquid $n = \frac{\sqrt{x}}{4}$.

The value of x is _____. $(\text{Given refractive index of glass } = 1.5)$



Correct Answer:
27

Solution:



For refraction on surface AC,

$i = 60°, r = 90°$

By Snell's Law,

$(105)\sin 60° = n \sin 90°$

$n = \frac{3}{2} \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}$

$n = \frac{\sqrt{27}}{4}$

$x = 27$

Q. 24 Two lighter nuclei combine to form a comparatively heavier nucleus by the relation given below : ${}_{1}^{2}\mathrm{X} + {}_{1}^{2}\mathrm{X} = {}_{2}^{4}\mathrm{Y}$

The binding energies per nucleon for ${}_{1}^{2}\mathrm{X}$ and ${}_{2}^{4}\mathrm{Y}$ are $1.1\ \mathrm{MeV}$ and $7.6\ \mathrm{MeV}$ respectively. The energy released in this process is _____ MeV.

Correct Answer:
26

Solution:
${}_{1}^{2}\mathrm{X} + {}_{1}^{2}\mathrm{X} \Rightarrow {}_{2}^{4}\mathrm{Y}$
Disintegration energy is the energy released in the process
Energy released $= \mathrm{TE_i} - \mathrm{TE_f}$

$= (2+2)(-1.1) - 4(-7.6)$

$= 26\ \mathrm{MeV}$

Q. 25 A uniform heavy rod of mass $20\ \mathrm{kg}$, cross sectional area $0.4\ \mathrm{m}^2$ and length $20\ \mathrm{m}$ is hanging from a fixed support. Neglecting the lateral contraction, the elongation in the rod due to its own weight is $\mathrm{x} \times 10^{-9}\ \mathrm{m}$. The value of x is _____.

$(\text{Given, young modulus } \mathrm{Y} = 2 \times 10^{11}\mathrm{Nm}^{-2} \text{ and } \mathrm{g} = 10\ \mathrm{ms}^{-2})$

Correct Answer:
25

Solution:
$\mathrm{M} = 20\ \mathrm{kg}$

$\mathrm{A} = 0.4\ \mathrm{m}^2$

$\ell = 20\ \mathrm{m}$

$\Delta\ell = \mathrm{x} \times 10^{-9}\ \mathrm{m}$

Elongation due to its own weight

$$\Delta\ell = \frac{Mg\ell}{2AY}$$

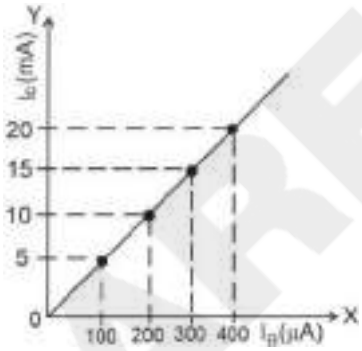
$$= \frac{20 \times 10 \times 20}{2 \times 0.4 \times 2 \times 10^{11}}$$

$$= \frac{4 \times 10^{3}}{16 \times 10^{10}}$$

$$\Delta\ell = 25 \times 10^{-9}\ \text{m}$$

$$\therefore x = 25$$

Q. 26 The typical transfer characteristics of a transistor in CE configuration is shown in figure. A load resistor of $2\ k\Omega$ is connected in the collector branch of the circuit used. The input resistance of the transistor is $0.50\ k\Omega$. The voltage gain of the transistor is _____.



Correct Answer:
200

Solution:
From the graph given,we can say that

$$\text{Slope} = \frac{\Delta I_c}{\Delta I_B} = \frac{20 \times 10^{-3}}{400 \times 10^{-6}}$$

$$= 50$$

$$\text{Voltage gain } = \frac{V_0}{V_i}$$

$$= \frac{\Delta I_c}{\Delta I_B} \times \frac{R_L}{R_{input}}$$

$$\text{voltage gain} = 50 \times \frac{2000}{0.5 \times 1000}$$

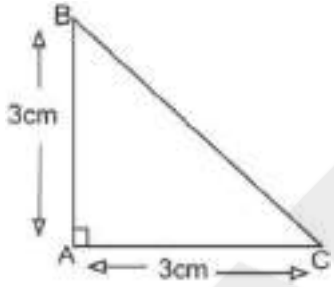
$$= 200$$

The voltage gain of the transistor is 200

Q. 27 Three point charges of magnitude $5\mu C, 0.16\mu C$ and $0.3\mu C$ are located at the vertices A, B, C of a right angled triangle whose sides are $AB = 3 \text{ cm}, BC = 3\sqrt{2} \text{ cm}$ and $CA = 3 \text{ cm}$ and point A is the right angle corner. Charge at point A experiences _____ N of electrostatic force due to the other two charges.

Correct Answer:
17

Solution:



$$q_A = 5 \times 10^{-6} C$$

$$q_B = 0.16 \times 10^{-6} C$$

$$q_C = 0.3 \times 10^{-6} C$$

$$F_C = \frac{k q_A q_C}{(3 \times 10^{-2})^2}$$

$$F_B = \frac{k q_B q_A}{(3 \times 10^{-2})^2}$$

$$F_{net} = \sqrt{F_C^2 + F_B^2}$$

$$= \frac{K}{(3\times10^{-2})^2}\sqrt{(q_A q_c)^2+(q_B q_A)^2}$$

$$= 10^{13}\sqrt{2.25\times10^{-24}+(0.8)^2\times10^{-24}}$$

$$= 10\sqrt{(2.25)+0.64}$$

$$F_{net} = 10\sqrt{2.89}$$

$$F_{net} = 17\ N$$

Change at pt A experiences $17\ N$ electrostatic force due to the other two charges

Q. 28 In a coil of resistance 8Ω, the magnetic flux due to an external magnetic field varies with time as $\phi = \frac{2}{3}\left(9-t^2\right)$. The value of total heat produced in the coil, till the flux becomes zero, will be _____ j.

Correct Answer:
2

Solution:

$$\phi = \frac{2}{3}\left(9-t^2\right) = 0$$

$$t = 3sec$$

$$e = \frac{-d\phi}{dt} = -\frac{2}{3}(0-2t) = \frac{4t}{3}$$

$$\text{Heat produced in 3 sec} = \int \frac{e^2}{r}dt$$

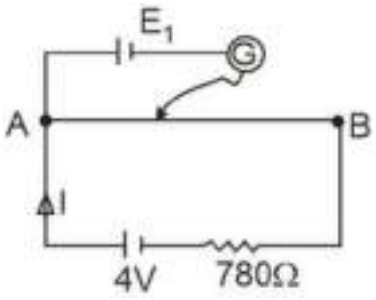
$$= \int_0^3 \frac{16t^2}{9\times8}dt = 2\ J$$

Q. 29 A potentiometer wire of length $300\ cm$ is connected in series with a resistance 780Ω and a standard cell of emf $4\ V$. A constant current flows through potentiometer wire. The length of the null point for cell of emf $20\ mV$ is found to be $60\ cm$. The resistance of the potentiometer wire is _____ Ω.

Correct Answer:
20

Solution:

$\ell_{AB} = 300\ \text{cm}$



$E_1 = \phi \ell_1$

$E_1 = \left(\frac{V_{AB}}{\ell_{AB}}\right) \ell_1$

$E_1 = \left(\frac{IR_{AB}}{\ell_{AB}}\right) \ell_1$

$20 \times 10^{-3} = \left(\frac{4RAB}{R_{AB} + 780}\right) \times \frac{300}{30}$

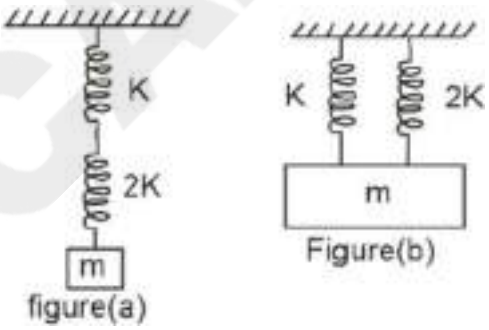
$0.1R_{AB} + 78 = 4R_{AB}$

$78 = 3.9R_{AB}$

$R_{AB} = 20\Omega$

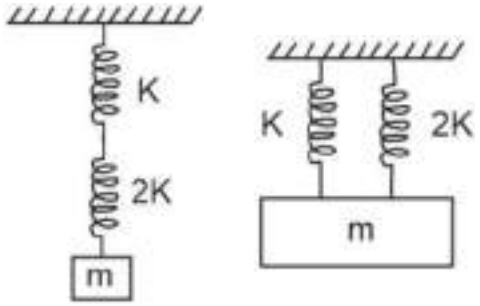
The resistance of the potentiometer wire is 20Ω

Q. 30 As per given figures, two springs of spring constants k and $2k$ are connected to mass m. If the period of oscillation in figure (a) is $3\ s$, then the period of oscillation in figure (b) will be $\sqrt{x}s$. The value of x is _____.



Correct Answer:
2

Solution:



$$K_{eq} = \frac{k(2k)}{k+2k}$$

$$K_{eq} = k_1 + k_2$$

$$= 3K$$

$$K_{eq} = \frac{2k}{3}$$

$$T_1 = 2\pi\sqrt{\frac{m}{(2k/3)}} = 1$$

$$T_2 = 2\pi\sqrt{\frac{m}{3k}}$$

$$\frac{T_1}{T_2} = \frac{3^2}{T_2^2} = \frac{1/(2k/3)}{1/3k}$$

$$\frac{9}{T_2^2} = \frac{3\times 3}{2} = \frac{9}{2}$$

$$T_2 = \sqrt{2s}$$

$$x = 2$$

Chemistry

Q. 1 Hemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 3.3 g of hemoglobin is

$(\text{Given: Atomic mass of Fe is } 56u, N_A = 6.022 \times 10^{23} \text{ mol}^{-1}.)$

Option 1:

1.21×10^5

Option 2:

12.0×10^{16}

Option 3:

1.21×10^{20}

Option 4:

3.4×10^{22}

Correct Answer:

1.21×10^{20}

Solution:

Hemoglobin contains 0.34% of iron by mass,

So, in 100 g Hemoglobin 34 g is iron ,

then in 3.3 g Hemoglobin $\rightarrow 3.3 \times \frac{0.34}{100}$ g of iron

$\Rightarrow$ 3.3 g Hemoglobin $\rightarrow 1.12 \times 10^{-2}$g of iron

Now, $\text{mole of iron} = \frac{1.12 \times 10^{-2}}{56}\text{mole} = 0.01986 \times 10^{-2}\text{mole}$

We know, $1 \text{ mole} = 6.022 \times 10^{23}\text{atoms}$

then,

$$\begin{aligned} 0.019 \times 10^{-2} \text{ mole} &= 0.015 \times 10^{-2} \times 6.022 \times 10^{23} \text{ atoms} \\ &= 1.20 \times 10^{20} \text{ atom} \end{aligned}$$

Hence, **Option (3)** is correct.

Q. 2 Arrange the following in increasing order of their covalent character.

A. CaF_2

B. $CaCl_2$

C. $CaBr_2$

D. CaI_2

Choose the correct answer from the options given below.

Option 1:
$B < A < C < D$

Option 2:
$a < b < c < d$

Option 3:
$A < B < D < C$

Option 4:
$A < C < B < D$

Correct Answer:
$a < b < c < d$

Solution:
According to Fajans rule,

The greater or the larger the size of anion the greater the covalent character of an ionic bond.

Size of anion : $I^{\ominus} > Br^{\ominus} > Cl^{\Theta} > F^{\Theta}$

So, covalent character is increasing order -

So,

$CaF_2 < CaC_2 < CaBr_2 < CaI_2$
$A < B < C < D$

Hence, **Option (2)** is correct.

Q. 3 Class XII students were asked to prepare one litre of buffer solution of $pH\ 8.26$ by their Chemistry teacher. The amount of ammonium chloride to be dissolved by the student in $0.2\ M$ ammonia solution to make one litre of the buffer is

(Given : $pK_b(NH_3) = 4.74$ Molar mass of $NH_3 = 17\ g\ mol^{-1}$ Molar mass of $NH_4Cl = 53.5\ g\ mol^{-1}$)

Option 1:
$53.5\ g$

Option 2:
$72.3\ g$

Option 3:
$107.0\ \text{g}$

Option 4:
$126.0\ \text{g}$

Correct Answer:
$107.0\ \text{g}$

Solution:
Need to prepare one litre of butter solutions of pH 8.26

So, Volume (V) = 11

and pH = 8.26

$$\Rightarrow \text{POH} = 14 - 8.26 = 5.74$$

and we know the formula,

$$\text{POH} = \text{p}^{\text{K}_\text{b}} + \log \frac{[\text{ Salt }]}{[\text{ Base }]}$$

$$\text{POH} = \text{P}^{\text{K}_\text{b}} + \log \frac{[\text{NH}_4^+]}{[\text{NH}_3]}$$

So,

$$5.74 = 4.74 + \log \frac{[\text{NH}_4^+]}{[\text{NH}_3]}$$

$$1 = \log \frac{[\text{NH}_4^+]}{0.2}$$

$$\log\ 10 = \log \frac{[\text{NH}_4^+]}{0.2}$$

$$10 = \frac{[\text{NH}_4^+]}{0.2}$$

$$[\text{NH}_4^+] = 2$$

We know concentration $= \frac{\text{mole}}{\text{v}} = \frac{\omega}{\text{M.V}}$

$$\Rightarrow \quad [NH_4^+] = \frac{\omega}{53.5 \times 1}$$

$$\Rightarrow \quad 2 = \frac{\omega}{53.5}$$

$$\Rightarrow \quad \omega = 107.0 \text{ g}$$

Hence, **Option (3)** is correct.

Q. 4 At $30^\circ C$, the half life for the decomposition of AB_2 is 200 s and is independent of the initial concentration of AB_2. The time required for 80% of the AB_2 to decompose is

$$\text{Given : } \log 2 = 0.30 \log 3 = 0.48$$

Option 1:
200 s

Option 2:
323 s

Option 3:
467 s

Option 4:
532 s

Correct Answer:
467 s

Solution:
The half life is 200 S and it is independent of the initial concentration of AB_2

So, it is first order reaction.

$$tv_2 = \frac{\ln 2}{k} = 200 \text{ sec}$$

$$\Rightarrow \quad k = \frac{\ln 2}{200} \text{ sec}^{-1}$$

Now, $k = \frac{2.303}{t} \log \frac{a}{a-x}$

80% completion , So, $x = 0.89 \Rightarrow a - x = a - 0.89 = 0.29$

Then, $k = \frac{2.303}{t_{80}} \log\left(\frac{9}{0.29}\right)$

$$\frac{\ln 2}{200} = \frac{2.303}{t_{80}} \log\left(\frac{10}{2}\right)$$

$$t_{80} = \frac{\ln 10 - \ln 2}{\ln 2} \times 200$$

$$t_{80} = \frac{\log 10 - \log^2}{\log^2} \times 200$$

$$t_{80} = \left(\frac{1}{\log 2} - 1\right) \times 200 = \left(\frac{1}{0.3} - 1\right) \times 200$$

$$t_{80} = 466.6 \approx 467\text{sec.}$$

Hence, **Option (3)** is correct.

Q. 5 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: Finest gold is red in colour, as the size of the particles increases, it appears purple then blue and finally gold.

Reason R: The colour of the colloidal solution depends on the wavelength of light scattered by the dispersed particles.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both **A** and **R** are true and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are true but **R** is NOT the correct explanation of **A.**

Option 3:
A is true but **R** is false.

Option 4:
A is false but **R** is true.

Correct Answer:
Both **A** and **R** are true and **R** is the correct explanation of **A.**

Solution:
The colour of colloidel solution depends on the wavelength of light scattered by the dispersed particles.

The wavelength of light further depends on the size and nature of the particles. The colour of colloidal solution also changes with the manner in which the observer receives the light. Finest gold is red in colour, as the size of particles increases. It appears purple, than blue and finally golden.

So, Both A and R are true and R is the correct explanation of A.

Hence, **Option (1)** is correct.

Q. 6 The metal that has very low melting point and its periodic position is closer to a metalloid is

Option 1:
Al

Option 2:
Ga

Option 3:
Se

Option 4:
In

Correct Answer:
Ga

Solution:
Melting point of Ga is lowest among given metal

$$\mathrm{Ga} < \mathrm{In} < \mathrm{Tl} < \mathrm{Al}$$

$$\text{Melting Point} \rightarrow 303 \quad 430 \quad 576 \quad 933$$

Hence, **Option (2)** is correct.

Q. 7 The metal that is not extracted from its sulfide ore is

Option 1:
Aluminium

Option 2:
Iron

Option 3:
Lead

Option 4:
Zinc

Correct Answer:
Aluminium

Solution:
For the purpose of extraction, bauxite $\left(AlO_2^{(OH)_{3-22}}\right)$ is chosen for alminium.

So, Al is not extracted from its sulfide ore.

Hence, **Option (1)** is correct.

Q. 8 The products obtained from a reaction of hydrogen peroxide and acidified potassium permanganate are

Option 1:
Mn^{4+}, H_2O only

Option 2:
Mn^{2+}, H_2O only

Option 3:
Mn^{4+}, H_2O, O_2 only

Option 4:
Mn^{2+}, H_2O, O_2 only

Correct Answer:
Mn^{2+}, H_2O, O_2 only

Solution:
The reaction will be

$$2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$$

Products are : Mn^{2+}, H_2O, O_2

Hence, **Option (4)** is correct.

Q. 9 Given below are two statements: one is labelled as **Assertion A** and the other is abelled as **Reason R.**

Assertion A: LiF is sparingly soluble in water.
Reason R : The ionic radius of Li^+ ion is smallest among its group members, hence has least hydration enthalpy.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both **A** and **R** are true and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are true but **R** is NOT the correct explanation of **A.**

Option 3:
A is true but **R** is false.

Option 4:
A is false true but **R** is true.

Correct Answer:
A is true but **R** is false.

Solution:
The Low solubility LiF in water is due to high lattice enthalpy.
So, Lif as sparingly soluble in water.
Li has high lattice enthalpy.
A is true but R is false.

Hence, **Option (3)** is correct.

Q. 10 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: Boric acid is a weak acid.

Reason R: Boric acid is not able to release H^+ ion on its own. It receives OH^+ ion from water and releases H^+ ion.

In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

Option 3:
A is correct but **R** is not correct.

Option 4:
A is not correct but **R** is correct.

Correct Answer:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Solution:
Boric acid is a weak monobasic acid. It is not a protonic acid but acts as a lewis acid by accepting electrons from a hydroxy ion.

$$B(OH)_3 + 240\ H \rightarrow [B(OH)_4)]^- + H_3O^+$$

So, Both A and R are correct and R is the correct explanation of A.

Hence, **Option (1)** is correct.

Q. 11 The metal complex that is diamagnetic is $(\text{Atomic number: Fe, 26; Cu, 29})$

Option 1:
$K_3[Cu(CN)_4]$

Option 2:
$K_2\,[Cu(CN)_4]$

Option 3:
$K_3\,[Fe(CN)_4]$

Option 4:
$K_4\,[FeCl_6]$

Correct Answer:
$K_3\,[Cu(CN)_4]$

Solution:
The metal complex will be diamagnetic if all electrons are paired.

$(1)K_3[(Cu(CN)_4] \rightarrow Cu^{+} \rightarrow 3d'^{o}4s^{o} \rightarrow \text{fully paired} \rightarrow \text{diamagnetic}$

$(2)K_2[(Cu(CN)_4] \rightarrow Cu^{2+} \rightarrow 3d^{s}4s^{o} \rightarrow \text{over unpaired} \rightarrow \text{Paramagnetic}$

$(3)K_3[(Fe(CN)_4] \rightarrow Fe^{+} \rightarrow 3d^{6}4s^{'} \rightarrow 5 \text{ unpaired electrons}$

$(4)K_4[(Fe(Cl)_6] \rightarrow Fe^{2+} \rightarrow 3d^{6}4s^{o} \rightarrow 4 \text{ unpaired electrons}$

Hence, **Option (1)** is correct.

Q. 12 Match **List I** with **List II.**

List I **Pollutant**	**List II** **Source**
A. Microorganisms	I. Strip mining
B. Plant nutrients	II. Domestic sewage
C. Toxic heavy metals	III. Chemical fertilizer
D. Sediment	IV. Chemical factory

Choose the correct answer from the options given below :

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-II, B-I, C-IV D-III

Option 3:
A-I, B-IV, C-II, D-III

Option 4:
A-I, B-IV, C-III, D-II

Correct Answer:
A-II, B-III, C-IV, D-I

Solution:
Major water pollutants -

(A) Microorganism →	(II)	Domestic Sewage
(B) Plant nutrients →	(III)	Chemical Fertilizer

(C) Toxic heavy metal →	(IV)	Chemical Factory
(D) Sediment →	(I)	Strip mining

$A-II, B-III, C-IV, D-I$

Hence, **Option (1)** is correct.

Q. 13 The correct decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is

Option 1:
$-COOH > -CONH_2 > -COCl > -CHO$

Option 2:
$-SO_3H > -COCl > -CONH_2 > -CN$

Option 3:
$-COOR > -COCl > -NH_2 > C=O$

Option 4:
$-COOH > -COOR > -CONH_2 > -COCl$

Correct Answer:
$-COOH > -CONH_2 > -COCl > -CHO$

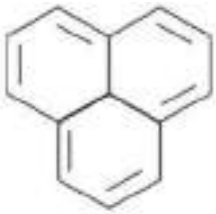
Solution:
The correct decreasing order of priority of functional groups -

$-COOH > -SO_3H > -COOR > -COCl > -CONH_2 > -CN$

$> -CHO > C=O > -OH > -NH_2$

So, **Option (1)** is correct.

Q. 14 Which of the following is not an example of benzenoid compound?

A 

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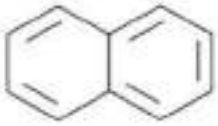
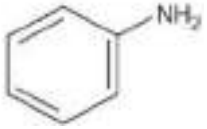
C 

D 

Option 1:
A and C

Option 2:
B and C

Option 3:
Only B

Option 4:
A and B

Correct Answer:
A and B

Solution:
Aromatic compounds containg benzene ring are known as benzolds and those not containing a benzene ring are known as non- benzenoids.

[Note: Put all option iamges in question with A,B,C,D names]

New Options - 1) A and C

2) B and C

3) Only B

4) A and B

Hence, **Option (4)** is correct.

Q. 15 Hydrolysis of which compound will give carbolic acid?

Option 1:
Cumene

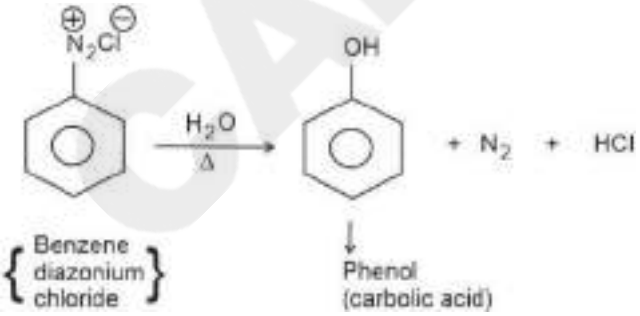
Option 2:
Benzenediazonium chloride

Option 3:
Benzal chloride

Option 4:
Ethylene glycol ketal

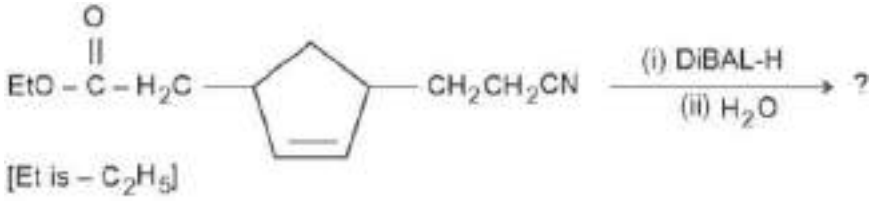
Correct Answer:
Benzenediazonium chloride

Solution:



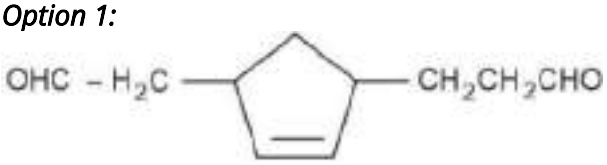
Hence, **Option (2)** is correct.

Q. 16



Consider the above reaction and predict the major product.

Option 1:



Option 2:

Option 3:

Option 4:

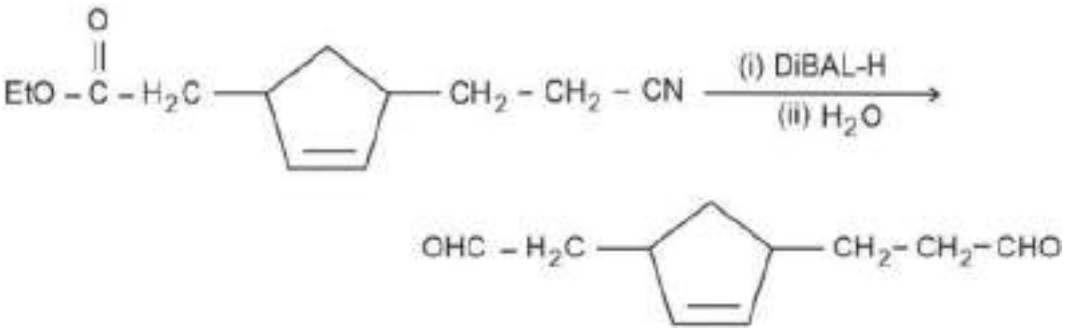
Correct Answer:

Solution:

We know,

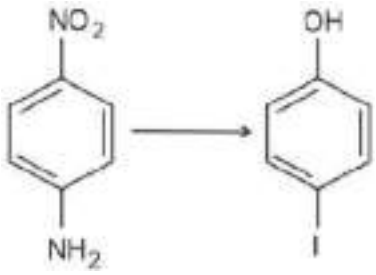
$\mathrm{DiBal-H}$ and $\mathrm{H_2O}$ convert $-\mathrm{COOR}$ to $-\mathrm{CHO}$ and $-\mathrm{CN}$ to $-\mathrm{CHO}$

So, the Reaction will be



Hence, **Option (1)** is correct.

Q. 17 The correct sequential order of the reagents for the given reaction is



Option 1:
$HNO_2, Fe/H^+, HNO_2, KI, H_2O/H^+$

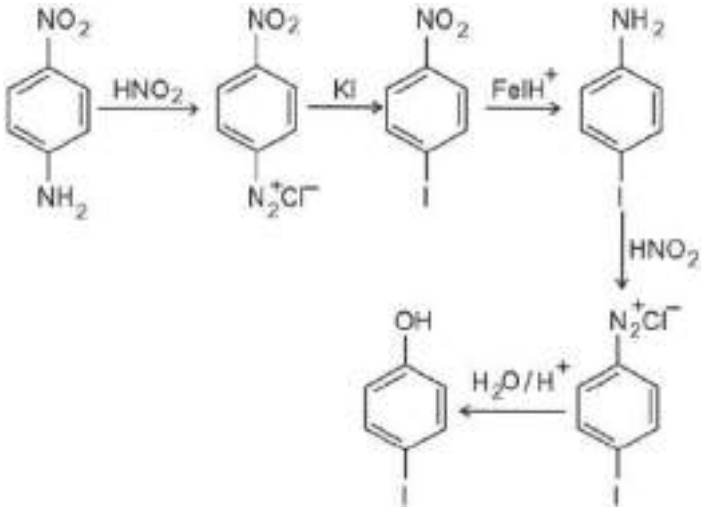
Option 2:
$HNO_2, KI, Fe/H^+, HNO_2, H_2O/ \text{ warm}$

Option 3:
$HNO_2, KI, HNO_2, Fe/H^+, H_2O/H^+$

Option 4:
$HNO_2, Fe/H^+, KI, HNO_2, H_2O/ \text{ warm}$

Correct Answer:
$HNO_2, KI, Fe/H^+, HNO_2, H_2O/ \text{ warm}$

Solution:
The reaction will be -



Hence, **Option (2)** is correct.

Q. 18 Vulcanization of rubber is carried out by heating a mixture of

Option 1:
isoprene and styrene

Option 2:
neoprene and sulphur

Option 3:
isoprene and sulphur

Option 4:
neoprene and styrene

Correct Answer:
isoprene and sulphur

Solution:
Valcunization of rubber is carried out by heating a mixture of isoprene and sulpher.

Hence, **Option (3)** is correct.

Q. 19 Animal starch is the other name of

Option 1:
amylose.

Option 2:
maltose.

Option 3:
glycogen.

Option 4:
amylopectin.

Correct Answer:
glycogen.

Solution:
The carbohydrates are stored in animal body as glycogen. It is also known as animal starch.

Hence, **Option (3)** is correct.

Q. 20 Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A: Phenolphthalein is a pH dependent indicator, remains colourless in acidic solution and gives pink colour in basic medium.

Reason R: Phenolphthalein is a weak acid. It doesn't dissociate in basic medium. In the light of the above statements, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both **A** and **R** are true and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are true but **R** is NOT the correct explanation of **A.**

Option 3:
A is true but **R** is false.

Option 4:
A is false but **R** is true.

Correct Answer:
A is true but **R** is false.

Solution:
Phenolphthalein is a pH dependent indicator, remains colourless in acidic solution and gives pink colour in basic medium.

[Colourless in acid $\rightarrow$ Pink in alkali]

Phenolphthalin is a weak acid, therefore it does not dissociate in the acidic medium and remain in the unionised form. Which is colourless.

So, A is true but R is false.

Hence, **Option (3)** is correct.

Q. 21 A $10\ \text{g}$ mixture of hydrogen and helium is contained in a vessel of capacity $0.0125\ \text{m}^3$ at 6 bar and 27°C. The mass of helium in the mixture is _____ g. (nearest integer)

$\text{Given: R} = 8.3\ \text{J K}^{-1}\ \text{mol}^{-1}$

(Atomic masses of H and He are 1u and 4u, respectively)

Correct Answer:
8

Solution:
There are two gases, so

$$\text{PV} = \text{n}_{\text{mix}}\text{RT}$$

and $\text{n}_{\text{mlx}} = \text{n}_{\text{H}_2} + \text{n}_{\text{He}}$

$$\text{n}_{\text{He}} = \text{x},\ \text{n}_{\text{H}_2} = \text{n}_{\text{mix}} - \text{x}$$

given,

$$\text{P} = 6\ \text{bar} = 6 \times 10^5 \text{Pascal}$$
$$\text{V} = 0.0125\ \text{m}^3$$
$$\text{R} = 8.3\ \text{J K}^{-1}\ \text{mol}^{-1} = 8.3\ \text{m}^3 - \text{Pa mol}^{-1}\text{k}^{-1}$$
$$\text{T} = 27^\circ\text{C} = 300\ \text{K}$$

Now,

$$6 \times 10^5 \times 0.0125 = n_{mix} \times 8.3 \times 300$$

$$n_{mix} = \frac{6 \times 10^5 \times 0.0125}{8.3 \times 300} \approx 3$$

So,

$$n_{He} = x$$
$$n_{H_2} = 3 - x$$
$$w_{He} = x \times 4$$
$$w_{H_2} = 2(3 - x)$$

Now,

$$w_{He} + w_{H_2} = 10$$
$$4x + 2(3 - x) = 10$$
$$x = 2$$

So, $4x = w_{He} = 4 \times 2 = 8$ g

Answer = 8

Q. 22 Consider an imaginary ion $^{48}_{22}X^{3-}$. The nucleus contains ' a' % more neutrons than the number of electrons in the ion. The value of ' a ' is _____. [nearest integer]

Correct Answer:
4

Solution:
$^{48}_{22}X^{3-}$, Mass number = 48 (Prontons+neutrons)

Atomic number = 22 (Protons)

Total number of electron = 22+3

= 25

Now, neutrons = 48-22=26

So, 1 neutron is more in number than electrons.

So,

$$a\% = \frac{1}{25} \times 100 = 4$$

Answer = 4

Q. 23 For the reaction

$$H_2\ F_2(\ g) \rightarrow H_2(\ g) + F_2(\ g)$$
$$\Delta U = -59.6\ \text{kJ mol}^{-1} \text{ at } 27^\circ\text{C}.$$

The enthalpy change for the above reaction is $(-)$ _____ kJmol^{-1} [nearest integer]

Given: $R = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}$.

Correct Answer:
57

Solution:
The Reaction

$$H_2\ F_2(g) \longrightarrow H_2(\ g) + F_2(g) \qquad \Delta H = ?$$

So, $\Delta ng = 2 - 1 = 1$

$$\Delta V = -59.6\ \text{KJ mol}^{-1}$$
$$T = 27^\circ\text{C} = 300\ \text{K}$$
$$R = 8.314 = \text{J/mol} - \text{k}$$

We know,

$$\Delta H = \Delta U + \Delta ngRT$$

$$\Delta H = -50.6 \times 10^3 + 1 \times 8.314 \times 300$$
$$\Delta H = -57.105 \times 10^3\ \text{J/mol}$$
$$\Delta H = -57\ \text{kJ/mol}$$

Answer = 57

Q. 24 The elevation in boiling point for 1 molal solution of non-volatile solute A is $3\ K$. The depression in freezing point for 2 molal solution of A in the same solvent is $6\ K$. The ratio of K_b and K_f i.e., K_b/K_f is $1:\ X$. The value of X is [nearest integer]

Correct Answer:
1

Solution:
We know,

$$\Delta T_b = K_b m \text{ and } \Delta T_f = K_f m$$

Now,

$$\Delta T_b = 3 = K_b m = K_b \times 1$$

$$K_b = 3$$

And

$$\Delta T_f = 6 = K_f \times 2$$

$$K_f = 3$$

So, the ratio of K_b and K_f

$$\frac{k_b}{k_f} = \frac{3}{3} \Rightarrow \frac{1}{x} = \frac{1}{1}$$

So, $x = 1$

Answer = 1

Q. 25 $20\ mL$ of $0.02M$ hypo solution is used for the titration of $10\ mL$ of copper sulphate solution, in the presence of excess of KI using starch as an indicator. The molarity of Cu^{2+} is found to be _____ $\times 10^{-2} M$. [nearest integer]

$$\text{Given : } 2Cu^{2+} + 4I^- \rightarrow Cu_2I_2 + I_2$$
$$I_2 + 2\ S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$$

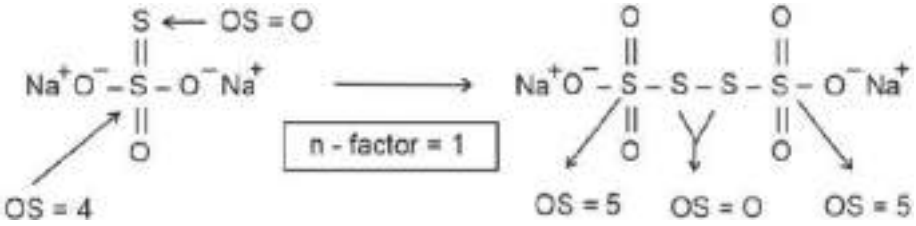
Correct Answer:
4

Solution:
Hypo solution means

$$Na_2\ S_2O_3 + I_2 + 2\ S_2O_3^{2-} \rightarrow 2\ S^- + S_4O_6^{2-}$$

So, meq of $Na_2S_2O_3$ = meq of I_2

$0.02 \times 20 \times 1 =$ meq of $I_2 = 0.4m$ mol



Now, $I_2 \rightarrow 2I^-$

meq of I_2 = mmole $\times$ n $-$ factor

$0.4 =$ mmole $\times 2$

So, mmol of $I_2 = 0.2$

mmol of Cu^{2+} = mol of $I_2 \times 2$

mmol of $Cu^{2+} = 0.4$

$$\text{Molarity of } Cu^{2+} = \frac{\text{mole}}{\text{Volume}} = \frac{0.4 \times 10^{-3}}{10 \times 10^{-3}} = 0.04$$

$M = 4 \times 10^{-2}$

Answer = 4

Q. 26 The number of non-ionisable protons present in the product B obtained from the following reactions is _____.

$C_2H_5OH + PCl_3 \rightarrow C_2H_5Cl + A$

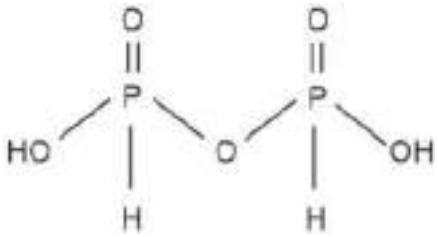
$A + PCl_3 \rightarrow B$

Correct Answer:
2

Solution:
Reaction -

$3C_2H_5OH + PCl_3 \longrightarrow 3C_2H_5Cl + H_3PO_3$

$H_3PO_3 + PCl_3 \longrightarrow H_4P_2O_5$

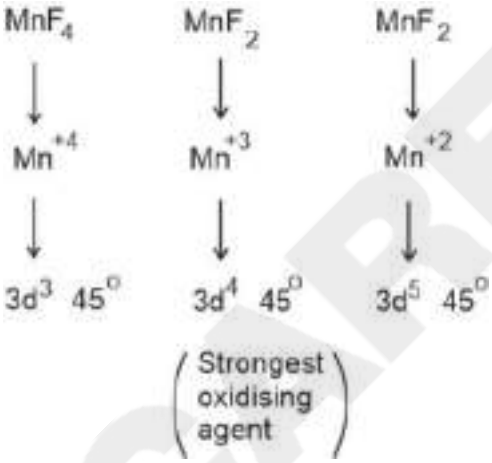


So, number of non-ionisable protons (H) present in $H_4P_2O_5$ is 2. (two P-H bond).

Q. 27 The spin-only magnetic moment value of the compound with strongest oxidizing ability among MnF_4, MnF_3 and MnF_2 is _____. B.M. [nearest integer]

Correct Answer:
5

Solution:
An examination of the $E^{\ominus}$ values for the Redox couple M^{3+}/M^{2+} show that Mn^{3+} ion is strongest oxidising agents in aqueous solution.



$$\mu = \sqrt{n(n+2)} = \sqrt{4(4+2)} = 4.89 \text{ BM.}$$

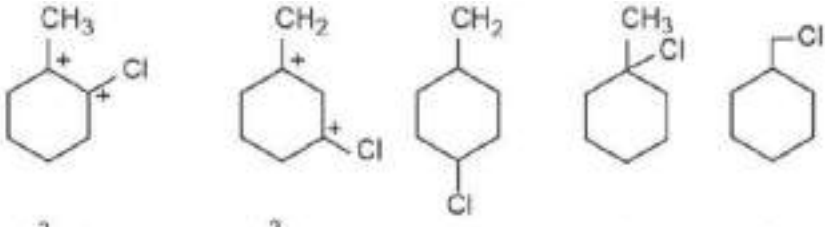
Spin only magnetic moment $\mu \approx 5$ B.M.

Answer = 5

Q. 28 Total number of isomers (including stereoisomers) obtained on monochlorination of methylcyclohexane is _____.

Correct Answer:
12

Solution:
The possible structure monochlorination of methylcyclohexane -



Isomass $2^2=4$ $2^2=4$ 2 (Cis trans) 1 1

Total number of isomers is 12

Answer is 12

Q. 29 A $100\ \text{mL}$ solution of CH_3CH_2MgBr on treatment with methanol produces $2.24\ \text{mL}$ of a gas at STP. The weight of gas produced is _____ mg. [nearest integer]

Correct Answer:
3

Solution:
The reaction will be -

$$CH_3 - CH_2MgBr + CH_3OH \longrightarrow CH_3 - CH_3 + Mg(OH)Br$$

Methanol

Molar mass of $CH_3CH_3 = 12 \times 2 + 1 \times 6 = 30\ \text{g/mol}$

Now, $2.24\text{ml} = 2.24 \times 10^{-3}\ \text{L} = 22.4 \times 10^{-4}\ \text{L}$

We know, $1\ \text{mol} = 22.4\ \text{L}$

So, $22.4 \times 10^{-4}\text{L} = 1 \times 10^{-4}\ \text{mol}$

Then, $\text{mole} = \dfrac{\text{mass}}{\text{Molar mass}}$

$\Rightarrow \text{Mass} = \text{Mole} \times \text{Molar mass}$

$\Rightarrow \text{Mass} = 1 \times 10^{-4} \times 30 \text{ g}$

$\Rightarrow \text{Mass} = 3 \times 10^{-3} \text{ g}$

$\Rightarrow \text{Mass} = 3 \text{ mg}$

Answer = 3

Q. 30 How many of the following drugs is/are example(s) of broad spectrum antibiotics ?
Ofloxacin, Penicillin G, Terpineol, Salvarsan.

Correct Answer:
1

Solution:
Only of loxacin are the broad spectrum antibiotics.

Answer = 1

Maths

Q. 1 The minimum value of the sum of the squares of the roots of

$x^2 + (3 - a)x + 1 = 2a$ is:

Option 1:
4

Option 2:
5

Option 3:
6

Option 4:
8

Correct Answer:
6

Solution:
$x^2 + (3 - a)x + 1 = 2a$

$x^2 + (3 - a)x + (1 - 2a) = 0$

$\alpha + \beta = a - 3$

$\alpha\beta = 1 - 2a.$

$\therefore \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (a - 3)^2 - 2(1 - 2a)$

$= a^2 - 6a + 9 - 2 + 4a$

$= a^2 - 2a + 7$

$= (a - 1)^2 + 6$

$\text{Minimum value} = 6$

Hence correct option is 3

Q. 2 If $z = x + iy$ satisfies $|z| - 2 = 0$ and $|z - i| - |z + 5i| = 0$, then

Option 1:
$x + 2y - 4 = 0$

Option 2:
$x^2 + y - 4 = 0$

Option 3:
$x + 2y + 4 = 0$

Option 4:
$x^2 - y + 3 = 0$

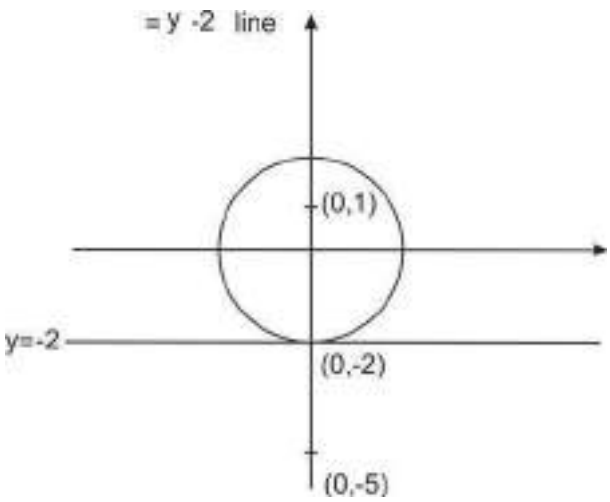
Correct Answer:
$x + 2y + 4 = 0$

Solution:
$|z| = 2$ and $|z - i| = |z + 5i|$

First is circle with cetre at origi ad radius 2 uits

Second is perpendicular isector of lie segmet joining $(0, 1)$ and $(0, -5)$

$\Rightarrow y = -2$ lie



only point $(0, -2)$ lies on both

It satisfies option 3

Hence correct option is 3

Q. 3 Let $\mathrm{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\mathrm{B} = \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$, then the value of $\mathrm{A'BA}$ is:

Option 1:
1224

Option 2:
1042

Option 3:
540

Option 4:
539

Correct Answer:
539

Solution:

$$\mathrm{A'B} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 9^2 & -10^2 & 11^2 \\ 12^2 & 13^2 & -14^2 \\ -15^2 & 16^2 & 17^2 \end{bmatrix}$$

$$= \begin{bmatrix} 9^2 + 12^2 - 15^2 & -10^2 + 13^2 + 16^2 & 11^2 - 14^2 + 17^2 \end{bmatrix}$$

$$(\text{A'B})\ A = \left[9^2 + 12^2 - 15^2 - 10^2 + 13^2 + 16^2 11^2 - 14^2 + 17^2\right] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \left[9^2 + 12^2 - 15^2 - 10^2 + 13^2 + 16^2 + 11^2 - 14^2 + 17^2\right]$$

$$= \left[\left(9^2 - 10^2\right) + \left(11^2 + 12^2\right) + \left(13^2 - 14^2\right) + \left(-15^2 + 16^2\right) + 17^2\right]$$

$$= [-19 + 265 - 27 + 31 + 289]$$

$$= [539]$$

Hence correct option is 4

Q. 4 $\sum_{\substack{i,j=0 \\ i \neq j}}^{n} {}^{n}C_i\, {}^{n}C_j$ is equal to

Option 1:
$2^{2n} - {}^{2n}C_n$

Option 2:
$2^{2n-1} - {}^{2n-1}C_{n-1}$

Option 3:
$2^{2n} - \frac{1}{2}{}^{2n}C_n$

Option 4:
$2^{n-1} + {}^{2n-1}C_n$

Correct Answer:
$2^{2n} - {}^{2n}C_n$

Solution:

$$\sum_{\substack{i,j=0 \\ i \neq j}}^{n} {}^{n}C_i \cdot {}^{n}C_j$$

$$= \sum_{i,j=0}^{n} {}^{n}C_i {}^{n}C_j - \sum_{i=j}^{n} {}^{n}C_i \cdot {}^{n}C_j$$

$$= 2^n \cdot 2^n - {}^{2n}C_n$$

$$= 2^{2n} - {}^{2n}C_n$$

Hence correct option is 1

Q. 5 Let P and Q be any points on the curves $(x-1)^2+(y+1)^2=1$ and $y=x^2$, respectively. The distance between P and Q is minimum for some value of the abscissa of P in the interval

Option 1:
$\left(0, \frac{1}{4}\right)$

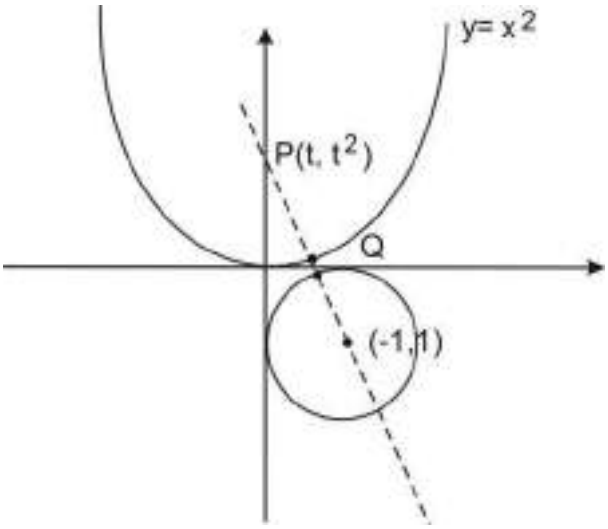
Option 2:
$\left(\frac{1}{2}, \frac{3}{4}\right)$

Option 3:
$\left(\frac{1}{4}, \frac{1}{2}\right)$

Option 4:
$\left(\frac{3}{4}, 1\right)$

Correct Answer:
$\left(\frac{1}{4}, \frac{1}{2}\right)$

Solution:



For maximum distance, normal at $P(t, t^2)$ should also be normal at $Q \Rightarrow$ normal at P should pass through $(1, -1)$

For equation of normal at P

$y = x^2$

$\frac{dy}{dx} = 2x$

$m_N = -\frac{1}{2x} = -\frac{1}{2t}$

$\text{Equation} : y - t^2 = -\frac{1}{2t}(x - t)$

$\Rightarrow 2yt - 2t^3 = -x + t$

$\Rightarrow x + 2ty - t - 2t^3 = 0$

$\text{It passes through} (1; -1)$

$1 - 2t - t - 2t^3 = 0$

$\Rightarrow 2t^3 + 3t - 1 = 0$

$f\left(\frac{1}{4}\right) = \frac{2}{64} + \frac{3}{4} - 1 < 0$

$f\left(\frac{1}{2}\right) = \frac{2}{8} + \frac{3}{2} - 1 > 0$

$\therefore \text{ Root of } 2t^3 + 3t - 1 \text{ lies in interval } \left(\frac{1}{4}, \frac{1}{2}\right)$

Hence correct option is 3

Q. 6 If the maximum value of a, for which the function $f_a(x) = \tan^{-1} 2x - 3ax + 7$ is non-decreasing in $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$, is $\bar{a}$, then $f_{\bar{a}}\left(\frac{\pi}{8}\right)$ is equal to

Option 1:
$8 - \frac{9\pi}{4(9+\pi^2)}$

Option 2:
$8 - \frac{4\pi}{9(4+\pi^2)}$

Option 3:
$8\left(\frac{1+\pi^2}{9+\pi^2}\right)$

Option 4:
$8 - \frac{\pi}{4}$

Correct Answer:
$8 - \frac{9\pi}{4(9+\pi^2)}$

Q. 7 Let $\beta = \lim_{x\to 0} \frac{\alpha x - (e^{2x} - 1)}{\alpha x (e^{3x} - 1)}$ for some $\alpha \in \mathbb{R}$. Then the value of $\alpha + \beta$ is:

Option 1:
$\frac{14}{5}$

Option 2:
$\frac{3}{2}$

Option 3:

$\frac{5}{2}$

Option 4:

$\frac{7}{2}$

Correct Answer:

$\frac{5}{2}$

Solution:

$$\lim_{x\to 0}\frac{\alpha x-(e^{3x}-1)}{\alpha x\,(e^{3x}-1)}$$

$$=\lim_{x\to 0}\frac{\alpha x-\left(1+3x+\frac{9x^2}{2}+\cdots-1\right)}{\alpha x\quad\left(\frac{e^{3x}-1}{3x}\cdot 3x\right.}$$

$$=\lim_{x\to 0}\frac{(\alpha-3)x-\frac{9x^2}{2}+\cdots}{3\alpha x^2}$$

For this lmit to exist $\alpha-3=0$

$$\Rightarrow \alpha=3 \text{ and } -\frac{9}{2\cdot(3\alpha)}=\beta$$

$$\Rightarrow \beta=\frac{-9}{6\cdot 3}=-\frac{1}{2}.$$

$$\therefore \alpha+\beta=3-\frac{1}{2}=\frac{5}{2}$$

Hence correct option is 3

Q. 8 The value of $\log_e 2\frac{d}{dx}(\log_{\cos x}\operatorname{cosec} x)$ at $x=\frac{\pi}{4}$ is

Option 1:

$-2\sqrt{2}$

Option 2:

$2\sqrt{2}$

Option 3:

-4

Option 4:

4

Correct Answer:

4

Solution:

$$\log 2 \cdot \frac{\mathrm{d}}{\mathrm{dx}}\left(\frac{\log(\operatorname{cosec} \mathrm{x})}{\log(\cos \mathrm{x})}\right)$$

$$= -\log 2 \cdot \frac{\mathrm{d}}{\mathrm{dx}}\left(\frac{\log(\sin \mathrm{x})}{\log(\cos \mathrm{x})}\right)$$

$$= -\log 2 \cdot \frac{\log(\cos \mathrm{x}) \cdot \frac{\cos \mathrm{x}}{\sin \mathrm{x}} - \log(\sin \mathrm{x}) \frac{(-\sin \mathrm{x})}{\cos \mathrm{x}}}{(\log \cos \mathrm{x})^2}$$

$$\text{At } \mathrm{x} = \frac{\pi}{4}$$

$$= -\log 2 \cdot \frac{\log\left(\frac{1}{\sqrt{2}}\right) + \log\left(\frac{1}{\sqrt{2}}\right)}{\left(\log\left(\frac{1}{\sqrt{2}}\right)\right)^2}$$

$$= -\log 2 \cdot \frac{\left(2\log\left(\frac{1}{\sqrt{2}}\right)\right)}{\left(\log\left(\frac{1}{\sqrt{2}}\right)\right)^2}$$

$$= \frac{-\log 2 \cdot 2}{\log\left(\frac{1}{\sqrt{2}}\right)}$$

$$= \frac{-2\log 2}{-\log(\sqrt{2})}$$

$$= \frac{2\log 2}{\frac{1}{2}\log 2}$$

$$= 4$$

Hence correct option is 4

Q. 9 $\int_0^{20\pi} (|\sin x| + |\cos x|)^2 dx$ is equal to

Option 1:
$10(\pi + 4)$

Option 2:
$10(\pi + 2)$

Option 3:
$20(\pi - 2)$

Option 4:
$20(\pi + 2)$

Correct Answer:
$20(\pi + 2)$

Solution:
$|\sin x| + |\cos x|$ is periodic with period $\frac{\pi}{2}$

$\therefore$ Required expression

$$= 40 \int_0^{\pi/2} (|\sin x| + |\cos x|)^2 dx$$

$$= 40 \int_0^{\pi/2} (\sin x + \cos x)^2 dx$$

$$= 40 \int_0^{\pi/2} (1 + \sin 2x) dx$$

$$= 40 \left[x - \frac{\cos 2x}{2} \Big|_0^{\pi/2} \right]$$

$$= 40 \left[\frac{\pi}{2} - \frac{1}{2}(-1 - 1) \right]$$

$$= 40 \left[\frac{\pi}{2} + 1 \right]$$

$$= 20(\pi + 2)$$

Hence correct option is 4

Q. 10 Let the solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2-1} = \frac{x^4+2x}{\sqrt{1-x^2}}, x \in (-1,1)$ pass through the origin. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is equal to

Option 1:
$\frac{\pi}{3} - \frac{1}{4}$

Option 2:
$\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

Option 3:
$\frac{\pi}{6} - \frac{\sqrt{3}}{4}$

Option 4:
$\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Correct Answer:
$\frac{\pi}{3} - \frac{\sqrt{3}}{4}$

Solution:
Linear differential equation

To get I.F

$$\int \frac{x}{x^2-1}dx$$

$$= \frac{1}{2}\int \frac{2x}{x^2-1}dx$$

$$= \frac{1}{2}\log|x^2-1|$$

$$= \log\sqrt{1-x^2} \quad (\text{as } x \in (-1,1))$$

$$\text{I.F.} = \sqrt{1-x^2}$$

$\therefore$ Solution is

$$y \cdot \sqrt{1-x^2} = \int (x^4 + 2x)\,dx$$

$$y\sqrt{1-x^2} = \frac{x^5}{5} + x^2 + c$$

$\therefore$ It passes through origin

$$0 = 0 + 0 + c \Rightarrow c = 0$$

$$\therefore \quad y = \left(\frac{x^5}{5} + x^2\right) \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\text{Now } \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$$

$$= \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{x^5}{5\sqrt{1-x^2}} + \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}}dx$$

$$= 0 + 2\int_0^{\frac{\sqrt{3}}{2}} \frac{x^2}{\sqrt{1-x^2}}dx$$

(Using odd and even function properties)

$$\text{Put } x = \sin\theta \Rightarrow dx = \cos\theta d\theta$$

$$= 2\int_0^{\pi/3} \frac{\sin^2\theta \cdot \cos\theta d\theta}{\cos\theta}$$

$$= \int_0^{\pi/3} 2\sin^2\theta d\theta$$

$$= \int_0^{\pi/3} (1 - \cos 2\theta)d\theta$$

$$= \theta - \frac{\sin 2\theta}{2}\Big|_0^{\pi/3}$$

$$= \frac{\pi}{3} - \left(\frac{\sqrt{3}}{4}\right)$$

Hence correct option is 2

Q. 11 The acute angle between the pair of tangents drawn to the ellipse $2x^2 + 3y^2 = 5$ from the point $(1, 3)$ is

Option 1:
$\tan^{-1}\left(\frac{16}{7\sqrt{5}}\right)$

Option 2:
$\tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$

Option 3:
$\tan^{-1}\left(\frac{32}{7\sqrt{5}}\right)$

Option 4:
$\tan^{-1}\left(\frac{3+8\sqrt{5}}{35}\right)$

Correct Answer:
$\tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$

Solution:

$2x^2 + 3y^2 = 5$

$\Rightarrow \frac{x^2}{\left(\frac{5}{2}\right)} + \frac{y^2}{\left(\frac{5}{3}\right)} = 1$

$\Rightarrow a^2 = \frac{5}{2}, \quad b^2 = \frac{5}{3}$

Let equation of tangent be

$y = mx \pm \sqrt{a^2m^2 + b^2}$

$y = mx \pm \sqrt{\frac{5}{2}m^2 + \frac{5}{3}}$

It passes through $(1,3)$

$$\therefore 3 = m \pm \sqrt{\frac{5}{2}m^2 + \frac{5}{3}}$$

$$\Rightarrow (3-m)^2 = \frac{5}{2}m^2 + \frac{5}{3}$$

$$\Rightarrow 9m^2 + 36m - 44 = 0$$

$$\therefore m_1 + m_2 = -4, \quad m_1 m_2 = -\frac{44}{9}$$

$$\text{Now } (m_1 - m_2)^2 = (m_1 + m_2)^2 - 4m_1 m_2$$

$$= 16 + 4\left(\frac{44}{9}\right)$$

$$= \frac{320}{9}$$

$$\therefore |m_1 - m_2| = \frac{8\sqrt{5}}{3}$$

$$\text{Angle between tagents} = \tan^{-}\left(\frac{m_1 - m_2}{1 + m_1 m_2}\right)$$

$$= \tan^{-1}\left(\frac{\frac{8\sqrt{5}}{3}}{1 - \frac{44}{9}}\right)$$

$$= \tan^{-1}\left(\frac{24}{7\sqrt{5}}\right)$$

Hence correct option is 2

Q. 12 The equation of a common tangent to the parabolas $y = x^2$ and $y = -(x-2)^2$ is

Option 1:
$y = 4(x-2)$

Option 2:
$y = 4(x-1)$

Option 3:
$y = 4(x+1)$

Option 4:

$y = 4(x + 2)$

Correct Answer:

$y = 4(x - 1)$

Solution:

$y = x^2 \Rightarrow a = \frac{1}{4}$

Let tangent to it be

And let tangent to $y = -(x - 2)^2$

$y = m(x - 2) + \frac{1}{4}m^2$

$\Rightarrow y = mx + 2m + \frac{1}{4}m^2$

For commo tangent

$-\frac{1}{4}m^2 = -2m + \frac{1}{4}m^2$

$\Rightarrow m = 0, 4$

For $m = 4$ equation of common tangent is $y = 4x - 4$

$\Rightarrow y = 4(x - 1)$

Hence correct option is 2

Q. 13 Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is _____.

Option 1:

12

Option 2:
13

Option 3:
14

Option 4:
16

Correct Answer:
12

Solution:
Equation of circle with PQ as diameter is

$$x^2 - 4x - 6 + y^2 + 2y - 7 = 0$$

$$\Rightarrow x^2 + y^2 - 4x + 2y - 13 = 0$$

$$\therefore a = -2, b = 1, c = -13$$

$$\therefore a + b - c = 12$$

Hence correct option is 1

Q. 14 If the line $x - 1 = 0$ is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point

Option 1:
$(-2\sqrt{5}, 6)$

Option 2:
$(-\sqrt{5}, 3)$

Option 3:
$(\sqrt{5}, -2)$

Option 4:
$(2\sqrt{5}, 3\sqrt{6})$

Correct Answer:
$(\sqrt{5}, -2)$

Solution:

$$kx^2 - y^2 = 6$$

$$\Rightarrow \frac{x^2}{\left(\frac{6}{k}\right)} - \frac{y^2}{6} = 1$$

$$a^2 = \frac{6}{k}, \quad b^2 = 6$$

$$e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{6k}{6} = 1 + k$$

$$\text{Directrix} \Rightarrow x = \frac{a}{e}$$

$$\Rightarrow x = \frac{\sqrt{6}}{\sqrt{k}\sqrt{1+k}}$$

Given that this equals 1

$$\frac{\sqrt{6}}{\sqrt{k(1+k)}} = 1$$

$$\Rightarrow k^2 + k - 6 = 0$$

$$\Rightarrow k = 2$$

$$2x^2 - y^2 = 6$$

$$(\sqrt{5}, -2) \quad \text{satisfies it}$$

Hence correct option is 3

Q. 15 A vector $\vec{A}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}, \hat{i} + \hat{j}$ and the plane determined by the vectors $\hat{i} - \hat{j}, \hat{i} + \hat{k}$. The obtuse angle between $\vec{A}$ and the vector $\vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}$ is

Option 1:

$\frac{3\pi}{4}$

Option 2:

$\frac{2\pi}{3}$

Option 3:

$\frac{4\pi}{5}$

Option 4:

$\frac{5\pi}{6}$

Correct Answer:

$\frac{3\pi}{4}$

Solution:

$\vec{n_1} = \mathrm{i} \times (\mathrm{i} + \mathrm{j}) = \mathrm{i} \times \mathrm{i} + \mathrm{i} \times \mathrm{j} = 0 + \mathrm{k} = \mathrm{k}$

$\vec{n_2} = (\mathrm{i} - \mathrm{j}) \times (\mathrm{i} + \mathrm{k})$

$= \mathrm{i} \times \mathrm{i} + \mathrm{i} \times \mathrm{k} - \mathrm{j} \times \mathrm{i} - \mathrm{j} \times \mathrm{k}$

$= -\mathrm{j} + \mathrm{k} - \mathrm{i}$

$= -\mathrm{i} - \mathrm{j} + \mathrm{k}$

$\therefore \vec{a}$ is parallel to $\vec{n}_1 \times \vec{n}_2$

$$= \begin{vmatrix} i & j & k \\ 0 & 0 & 1 \\ -1 & -1 & 1 \end{vmatrix}$$

$= \mathrm{i} - \mathrm{j}$

Angle between $\vec{a}$ and $\vec{b}$

$$\cos\theta = \frac{\tilde{\mathrm{a}} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$$

$$= \frac{(\mathrm{i} - \mathrm{j}) \cdot (\mathrm{i} - 2\mathrm{j} + 2\mathrm{k})}{\sqrt{2} \cdot 3}$$

$$= \frac{1}{\sqrt{2}}$$

$\therefore$ obtuse angle $= \frac{3\pi}{4}$

Hence correct option is 1

Q. 16 If $0 < x < \frac{1}{\sqrt{2}}$ and $\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$, then a value of $\sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right)$ is

Option 1:
$4\sqrt{(1-x^2)}\,(1-2x^2)$

Option 2:
$4x\sqrt{(1-x^2)}\,(1-2x^2)$

Option 3:
$2x\sqrt{(1-x^2)}\,(1-4x^2)$

Option 4:
$4\sqrt{(1-x^2)}\,(1-4x^2)$

Correct Answer:
$4x\sqrt{(1-x^2)}\,(1-2x^2)$

Solution:

$$\frac{\sin^{-1} x}{\alpha} = \frac{\cos^{-1} x}{\beta}$$

$$\Rightarrow \frac{\sin^{-1} x}{\alpha} = \frac{\frac{\pi}{2} - \sin^{-1} x}{\beta}$$

$$\Rightarrow \sin^{-1} x\left(\frac{1}{\alpha} + \frac{1}{\beta}\right) = \frac{\pi}{2\beta}$$

$$\Rightarrow \sin^{-1} x\left(\frac{\alpha+\beta}{\alpha\beta}\right) = \frac{\pi}{2\beta}$$

$$\Rightarrow \sin^{+} x = \frac{\pi\alpha}{2(\alpha+\beta)}$$

$$\Rightarrow \frac{2\pi\alpha}{\alpha+\beta} = 4\sin^{-1} x$$

$$\therefore \sin\left(\frac{2\pi\alpha}{\alpha+\beta}\right) = \sin\left(4\sin^{-1} x\right)$$

$$\text{Let } \sin^{-1} x = \theta \Rightarrow \sin\theta = x$$

$$\therefore \sin\left(4\sin^{-1} x\right)$$

$= \sin(4\theta)$

$= 2\sin 2\theta \cos 2\theta$

$= 4\sin\theta\cos\theta \cdot \left(1 - 2\sin^2\theta\right)$

$= 4 \cdot x \cdot \sqrt{1 - x^2}\left(1 - 2x^2\right)$

Hence correct option is 2

Q. 17 Negation of the Boolean expression $p \Leftrightarrow (q \Rightarrow p)$ is

Option 1:
$(\sim p) \wedge q$

Option 2:
$p \wedge (\sim q)$

Option 3:
$(\sim p) \vee (\sim q)$

Option 4:
$(\sim p) \wedge (\sim q)$

Correct Answer:
$(\sim p) \wedge (\sim q)$

Solution:
$\sim [p \Leftrightarrow (q \Rightarrow p)]$

$= p \Leftrightarrow \sim (q \Rightarrow p)$

$= p \Leftrightarrow (q \wedge \sim p)$

$= (p \rightarrow (q \wedge \sim p)) \wedge ((q \wedge \sim p) \rightarrow p)$

$= (\sim p \vee (q \wedge \sim p)) \wedge (\sim (q \wedge \sim p) \vee p)$

$= (\sim p) \wedge ((\sim q \vee p) \vee p)$

$= \sim p \wedge (\sim q \vee p)$

$= (\sim p \wedge \sim q) \vee (\sim p \wedge p)$

$= (\sim p \wedge \sim q) \vee t$

$= \sim p \wedge \sim q$

Hence correct option is 4

Q. 18 Let X be a binomially distributed random variable with mean 4 and variance $\frac{4}{3}$. Then, $54\ P(X \leq 2)$ is equal to

Option 1:
$\frac{73}{27}$

Option 2:
$\frac{146}{27}$

Option 3:
$\frac{146}{81}$

Option 4:
$\frac{126}{81}$

Correct Answer:
$\frac{146}{27}$

Solution:

$np = 4$

$npq = \frac{4}{3}$

$\Rightarrow q = \frac{1}{3} \Rightarrow 1 - p = \frac{1}{3} \Rightarrow p = \frac{2}{3}$

$\Rightarrow n = 6$

$\therefore\ 54P(x \leq 2)$

$= 54[P(0) + P(1) + P(2)]$

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$$= 54\left[{}^{6}C_0 \cdot \left(\frac{1}{3}\right)^6 + {}^{6}C_1\left(\frac{2}{3}\right)^5\left(\frac{1}{3}\right) + {}^{6}C_2\left(\frac{2}{3}\right)^4\left(\frac{1}{3}\right)^2\right]$$

$$= 54\left[\frac{1}{3^6}(1 + 6 \cdot 32 + 15 \cdot 16)\right]$$

$$= \frac{146}{27}$$

Hence correct option is 2

Q. 19

The integral $\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x - \sin x)}{\left(1+\frac{2}{\sqrt{3}}\sin 2x\right)}dx$ is equal to

Option 1:

$$\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}\right| + C$$

Option 2:

$$\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{3}\right)}\right| + C$$

Option 3:

$$\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}\right| + C$$

Option 4:

$$\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}-\frac{\pi}{6}\right)}\right| + C$$

Correct Answer:

$$\frac{1}{2}\log_e\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}\right| + C$$

Solution:

$$\int \frac{\left(1-\frac{1}{\sqrt{3}}\right)(\cos x - \sin x)}{\left(1+\frac{2}{\sqrt{3}}\sin 2x\right)}dx$$

$$= \int \frac{(\sqrt{3}-1)\cdot\sqrt{2}\sin\left(\frac{\pi}{4}-x\right)}{2\left(\frac{\sqrt{3}}{2}+\sin 2x\right)}dx$$

$$= \int \frac{(\sqrt{3}-1)\sqrt{2}\sin\left(\frac{\pi}{4}-x\right)dx}{2\left(\sin\left(\frac{\pi}{3}\right)+\sin 2x\right)}$$

$$= \int \frac{(\sqrt{3}-1)\sin\left(\frac{\pi}{4}-x\right)dx}{2\sqrt{2}\left[\sin\left(x+\frac{\pi}{6}\right)\cos\left(x-\frac{\pi}{6}\right)\right]}$$

$$= \frac{1}{2}\int \frac{2\sin\left(\frac{\pi}{12}\right)\sin\left(\frac{\pi}{4}-x\right)dx}{\sin\left(x+\frac{\pi}{6}\right)\cos\left(x-\frac{\pi}{6}\right)}$$

$$= \frac{1}{2}\int \frac{\cos\left(\frac{\pi}{6}-x\right)-\cos\left(\frac{\pi}{3}-x\right)}{\sin\left(x+\frac{\pi}{6}\right)\cos\left(x-\frac{\pi}{6}\right)}dx$$

$$= \frac{1}{2}\left[\int \operatorname{cosec}\left(\frac{\pi}{6}+x\right)dx - \int \sec\left(\frac{\pi}{6}-x\right)dx\right]$$

$$= \frac{1}{2}\left[\ln\left|\tan\left(\frac{\pi}{12}+\frac{x}{2}\right)\right| - \ln\left|\tan\left(\frac{\pi}{6}+\frac{x}{2}\right)\right|\right]$$

$$= \frac{1}{2}\ln\left|\frac{\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)}{\tan\left(\frac{x}{2}+\frac{\pi}{6}\right)}\right| + c$$

Hence correct option is 1

Q. 20 The area bounded by the curves $y = |x^2 - 1|$ and $y = 1$ is

Option 1:
$\frac{2}{3}(\sqrt{2}+1)$

Option 2:
$\frac{4}{3}(\sqrt{2}-1)$

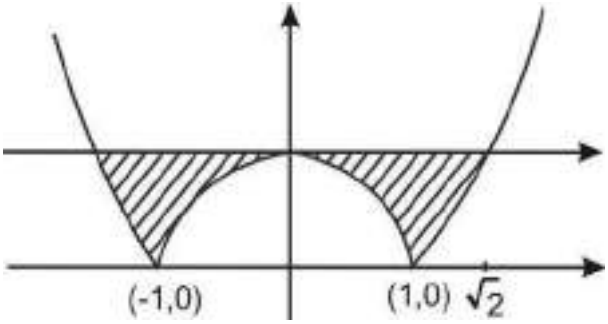
Option 3:
$2(\sqrt{2}-1)$

Option 4:
$\frac{8}{3}(\sqrt{2}-1)$

Correct Answer:

$\frac{8}{3}(\sqrt{2}-1)$

Solution:



$$\text{Area } = 2\left[\int_0^1 \left(1-\left(1-x^2\right)\right)dx + \int_1^{\sqrt{2}} \left(1-\left(x^2-1\right)dx\right)\right]$$

$$= 2\left[\int_0^1 x^2 dx + \int_1^{\sqrt{2}} \left(2-x^2\right)dx\right]$$

$$= 2\left[\frac{x^3}{3}\Big|_0^1 + 2x|_1^{\sqrt{2}} - \frac{x^3}{3}\Big|_1^{\sqrt{2}}\right]$$

$$= 2\left[\frac{1}{3} + 2(\sqrt{2}-1) - \frac{1}{3}(2\sqrt{2}-1)\right]$$

$$= \frac{8}{3}(\sqrt{2}-1)$$

Hence correct option is 4

Q. 21 Let $A=\{1,2,3,4,5,6,7\}$ and $B=\{3,6,7,9\}$. Then the number of elements in the set $\{C \subseteq A : C \cap B \neq \phi\}$ is_____.

Correct Answer:

112

Solution:

Total subsets of $A = 2^7 = 128$

Subsets of A such that $C \cap B = \phi$

$=$ Subsets of $\{1,2,4,5\} = 2^4 = 16$

Required Subsets $= 128 - 16 = 112$

Hence answer is 112

Q. 22 The largest value of a, for which the perpendicular distance of the plane containing the lines $\vec{r} = (\hat{i}+\hat{j}) + \lambda(\hat{i}+a\hat{j}-\hat{k})$ and $\vec{r} = (\hat{i}+\hat{j}) + \mu(-\hat{i}+\hat{j}-a\hat{k})$ from the point $(2,1,4)$ is $\sqrt{3}$, is _____.

Correct Answer:
2

Solution:
Normal vector $\vec{n}$ of the plane

$$= \begin{vmatrix} i & j & k \\ 1 & a & -1 \\ -1 & 1 & -a \end{vmatrix}$$

$$= \left(-a^2+1\right)i - (-a-1)j + (1+a)k$$

$$= \left(1-a^2\right)i + (a+1)j + (a+1)k$$

$$= (a+1)[(1-a)i + j + k]$$

$\therefore \vec{n}$ is parallel to $(1-a)i + j + k$

$\therefore$ Equation of the plane is

$$(1-a)(x-1) + (y-1) + (z-0) = 0$$

$$(1-a)x + y + z - 1 - (1-a) = 0$$

$$(1-a)x + y + z + a - 2 = 0$$

Distance from $(2,1,4)$

$$\frac{|2(|-a) + 1 + 4 + a - 2|}{\sqrt{(1-a)^2+1+1}} = \sqrt{3}$$

$$\Rightarrow \frac{|-a+5|}{\sqrt{1+a^2-2a+2}} = \sqrt{3}$$

$$\Rightarrow |5-a| = \sqrt{3}\sqrt{a^2-2a+3}$$

$$\Rightarrow a^2 + 25 - 10a = 3\left(a^2-2a+3\right)$$

$\Rightarrow a^2 - 10a + 25 = 3a^2 - 6a + 9$

$\Rightarrow 2a^2 + 4a - 16 = 0$

$\Rightarrow a^2 + 2a - b = 0$

$\Rightarrow a = 2, -4$

$\therefore \text{Maximum } a = 2$

Hence answer is 2

Q. 23 Numbers are to be formed between 1000 and 3000, which are divisible by 4, using the digits $1, 2, 3, 4, 5$ and 6 without repetition of digits. Then the total number of such numbers is_____.

Correct Answer:
30

Solution:
Case I : 1.....

Last two digits can be $24, 32, 36, 52, 56, 64$

Total such numbers $= 6 \times 3 = 18$

Case II : 2.....

Last two digits can be $16, 36, 56, 64$

Total such numbers $= 4 \times 3 = 12$

$\therefore$ Total $= 18 + 12 = 30$

Hence answer is 30

Q. 24 If $\sum_{k=1}^{10} \frac{k}{k^4 + k^2 + 1} = \frac{m}{n}$, where m and n are co-prime, then $m + n$ is equal to _____.

Correct Answer:
166

Solution:

$$\frac{k}{k^4+k^2+1}=\frac{k}{k^4+2k^2+1-k^2}$$

$$=\frac{k}{\left(k^2+1\right)^2-k^2}$$

$$=\frac{k}{\left(k^2-k+1\right)\left(k^2+k+1\right)}$$

$$=\frac{1}{2}\left[\frac{2k}{\left(k^2-k+1\right)\left(k^2+k+1\right)}\right]$$

$$=\frac{1}{2}\left[\frac{\left(k^2+k+1\right)-\left(k^2-k+1\right)}{\left(k^2-k+1\right)\left(k^2+k+1\right)}\right]$$

$$=\frac{1}{2}\left[\frac{1}{k^2-k+1}-\frac{1}{k^2+k+1}\right]$$

$\therefore$ Required sum

$$=\frac{1}{2}\left[\frac{1}{1}-\frac{1}{3}+\frac{1}{3}-\frac{1}{7}+\cdots \quad \cdots+\frac{1}{91}-\frac{1}{111}\right]$$

$$=\frac{1}{2}\left[1-\frac{1}{111}\right]$$

$$=\frac{1}{2}\cdot\frac{110}{111}$$

$$=\frac{55}{111}$$

$$\Rightarrow m=55, n=111$$

$$\Rightarrow m+n=166$$

Hence answer is 166

Q. 25 If the sum of solutions of the system of equations $2\sin^2\theta-\cos 2\theta=0$ and $2\cos^2\theta+3\sin\theta=0$ in the interval $[0,2\pi]$ is $k\pi$, then K is equal to _____.

Correct Answer: 3

Solution:

$2\sin^2\theta-\cos 2\theta=0$

$\Rightarrow 2\sin^2\theta - (1 - 2\sin^2\theta) = 0$

$\Rightarrow 4\sin^2\theta = 1$

$\Rightarrow \sin^2\theta = \frac{1}{4} \Rightarrow \sin\theta = \frac{1}{2}, -\frac{1}{2}$

And $2\cos^2\theta + 3\sin\theta = 0$

$\Rightarrow 2 - 2\sin^2\theta + 3\sin\theta = 0$

$\Rightarrow 2\sin^2\theta - 3\sin\theta - 2 = 0$

$\Rightarrow 2\sin^2\theta - 4\sin\theta + \sin\theta - 2 = 0$

$\Rightarrow 2\sin\theta(\sin\theta - 2) + 1(\sin\theta - 2) = 0$

$\Rightarrow \sin\theta = \frac{-1}{2}, \sin\theta = 2$

Common Solution have

$\sin\theta = -\frac{1}{2}$

In $[0, 2\pi]$, solutions are $\mathrm{x} = \frac{7\pi}{6}, \frac{11\pi}{6}$

$\text{Sum } = \frac{18\pi}{6} = 3\pi \Rightarrow \mathrm{k} = 3$

Hence answer is 3

Q. 26 The mean and standard deviation of 40 observations are 30 and 5 respectively. It was noticed that two of these observations 12 and 10 were wrongly recorded. If σ is the standard deviation of the data after omitting the two wrong observations from the data, then $38\sigma^2$ is equal to_____.

Correct Answer:
238

Solution:
$\text{Mean } = 30$

$\Rightarrow \frac{\sum \mathrm{x}}{40} = 30 \Rightarrow \sum \mathrm{x} = 1200$

$$\text{Variance } = 25$$

$$\Rightarrow \frac{\sum x^2}{40} - 30^2 = 25$$

$$\Rightarrow \sum x^2 = 37000$$

$$\text{Correct } \sum x = 1200 - 12 - 10 = 1178$$

$$\text{Correct } \sum x^2 = 37000 - 144 - 100 = 36756$$

$$\therefore 38\sigma^2 = 38\left[\frac{36756}{38} - \left(\frac{1178}{38}\right)^2\right]$$

$$= 238$$

Hence answer is 238

Q. 27 The plane passing through the line $L : lx - y + 3(1-l)z = 1, x + 2y - z = 2$ and perpendicular to the plane $3x + 2y + z = 6$ is $3x - 8y + 7z = 4$. If θ is the acute angle between the line L and the y-axis, then $415\cos^2\theta$ is equal to _____.

Correct Answer:
125

Solution:
Let plane be

$$lx - y + 3(1-l)z - 1 + \lambda(x + 2y - z - 2) = 0$$

$$\Rightarrow (l+h)x + y(-1+2\lambda) + z[3 - 3l - 2] - 1 - 2\lambda = 0$$

It is perpendicular to $3x + 2y + z = 6$

$$\therefore 3(l+\lambda) + 2(-1+2h) + 3 - 3l - h = 0$$

$$\Rightarrow 3l + 3h - 2 + 4h + 3 - 3l - h = 0$$

$$\Rightarrow 6\lambda + 1 = 0$$

$$\Rightarrow \lambda = -\frac{1}{6}$$

$\therefore$ Plane is

$$\left(l-\frac{1}{6}\right)x+y\left(-1-\frac{1}{3}\right)+z\left(3-3l+\frac{1}{6}\right)-1+\frac{1}{3}=0$$

$$\Rightarrow \frac{(6l-1)x}{6}-\frac{4}{3}y+z\left(\frac{19-18l}{6}\right)-\frac{2}{3}=0$$

$$\Rightarrow (6l-1)x-8y+z(19-18l)-4=0$$

$$\text{Comparing with } 3x-8y+7z=4$$

$$6l-1=3\Rightarrow l=\frac{2}{3}$$

For direction ratios of line L

$$=\begin{vmatrix} i & j & k \\ l & -1 & 3(1-l) \\ 1 & 2 & -1 \end{vmatrix}$$

$$=(6l-5)i+(3-2l)j+(2l+1)k$$

$$\text{Put } l=\frac{2}{3}$$

$$=-i+\frac{5}{3}j+\frac{7}{3}k$$

Angle with y -axis

$$\cos\theta=\frac{\frac{5}{3}}{\sqrt{1}\sqrt{1+\frac{25}{9}+\frac{49}{9}}}=\frac{5}{\sqrt{83}}$$

$$\therefore 415\cos^2\theta=415\cdot\frac{25}{83}=125$$

Hence answer is 125

Q. 28 Let $x=\sin\left(2\tan^{-1}\alpha\right)$ and $y=\sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$. If $S=\left\{\alpha\in\mathbb{R}:y^2=1-x\right\}$, then $\sum_{\alpha\in S}16\alpha^3$ is equal to_____

Correct Answer:
130

Solution:

$$L = \sin\left(2\tan^{-1}\alpha\right)$$

$$\text{Let } \tan^{-1}\alpha = \theta \Rightarrow \tan\theta = \alpha.$$

$$\therefore \quad x = \sin(2\theta)$$

$$= \frac{2\tan\theta}{1+\tan^2\theta}$$

$$= \frac{2\alpha}{1+\alpha^2} \qquad \text{.....(i)}$$

$$\text{And } y = \sin\left(\frac{1}{2}\tan^{-1}\left(\frac{4}{3}\right)\right)$$

$$\text{Let } \tan^{-1}\left(\frac{4}{3}\right) = \beta$$

$$\tan\beta = \frac{4}{3}$$

$$\Rightarrow \cos\beta = \frac{3}{5}$$

$$\Rightarrow 1 - 2\sin^2\left(\frac{\beta}{2}\right) = \frac{3}{5}$$

$$\Rightarrow 2\sin^2\left(\frac{\beta}{2}\right) = \frac{2}{5}$$

$$\Rightarrow \quad \sin^2\left(\frac{\beta}{2}\right) = \frac{1}{5}$$

$$\Rightarrow \quad y^2 = \frac{1}{5}$$

$$\therefore \quad y^2 = 1 - x$$

$$\frac{1}{5} = 1 - \frac{2\alpha}{1+\alpha^2}$$

$$\Rightarrow \frac{2\alpha}{1+\alpha^2} = \frac{4}{5}$$

$$\Rightarrow 5\alpha = 2 + 2\alpha^2$$

$$\Rightarrow \quad \alpha = 2, \frac{1}{2}$$

$$16\alpha^3 = 16 \cdot 8 + 16 \cdot \frac{1}{8} = 130$$

Hence answer is 130

Q. 29 Different $\mathrm{A.P.'s}$ are constructed with the first term 100, the last term 199, and integral common differences. The sum of the common differences of all such $\mathrm{A.P.'s}$ having at least 3 terms and at most 33 terms is_____.

Correct Answer:
53

Solution:

$199 = 100 + (\mathrm{n} - 1)\mathrm{d}$

$\Rightarrow (\mathrm{n} - 1)\mathrm{d} = 99$

$\Rightarrow \mathrm{d} = \dfrac{99}{(\mathrm{n} - 1)}$

$\text{Now} \quad 3 \leq \mathrm{n} \leq 33$

$\text{For d to be integer}$

$\mathrm{n} - 1 = 3, 9, 11$

$\mathrm{n} = 4, 10, 12$

$\therefore \mathrm{d} = 33, 11, 9$

$\text{Sum} \ = 33 + 11 + 9 = 53$

Hence answer is 53

Q. 30 The number of matrices $\mathrm{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $\mathrm{a, b, c, d} \in \{-1, 0, 1, 2, 3, \ldots\ldots, 10\}$, such that $\mathrm{A} = \mathrm{A}^{-1}$, is_____.

Correct Answer:
50

Solution:

$$\mathrm{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\mathrm{A} = \mathrm{A}^{-1}$$

$$\Rightarrow \mathrm{AA} = \mathrm{A}^{-1}\mathrm{A}$$

$$\Rightarrow \mathrm{A}^2 = \mathrm{I}$$

$$\Rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore \mathrm{a}^2 + \mathrm{bc} = 1, \mathrm{d}^2 + \mathrm{bc} = 1, \mathrm{b}(\mathrm{a}+\mathrm{d}) = 0, \mathrm{c}(\mathrm{a}+\mathrm{d}) = 0$$

From first two equations,

$$\mathrm{a}^2 = \mathrm{d}^2 \Rightarrow \mathrm{a} = \mathrm{d}, \mathrm{a} = -\mathrm{d}$$

Case I : $\mathrm{a} = -\mathrm{d}$

(a, d) can be $(0,0), (1,-1), (-1,1)$. For these b, c can have $2, 23$ and 23 values $= 48$ values

Case II : $\mathrm{a} = \mathrm{d}$

(a, d) can be $(1,1)$or $(-1,-1)$. For this (b, c) can be $(0,0)$ only

$$\therefore \text{ Total } = 48 + 2 = 50$$

Hence answer is 50

27 July - Morning Shift

Physics

Q. 1 A torque meter is calibrated to reference standards of mass, length and time each with 5% accuracy. After calibration, the measured torque with this torque meter will have net accuracy of:

Option 1:
15%

Option 2:
25%

Option 3:
75%

Option 4:
5%

Correct Answer:
25%

Solution:

$$[\tau] = [\mathrm{M'L^2T^{-2}}]$$

$$\frac{\Delta\tau}{\tau} = \frac{\Delta \mathrm{M}}{\mathrm{M}} + \frac{2\Delta \mathrm{L}}{\mathrm{L}} + 2\frac{\Delta \mathrm{T}}{\mathrm{T}}$$

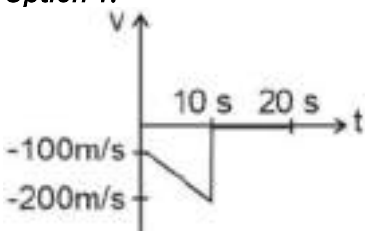
$$\frac{\Delta\tau}{\mathrm{t}} \times 100 = \left(\frac{\Delta \mathrm{M}}{\mathrm{M}} + 2\frac{\Delta \mathrm{L}}{\mathrm{L}} + 2\frac{\Delta \mathrm{T}}{\mathrm{T}}\right) \times 100$$

$$= 25\%$$

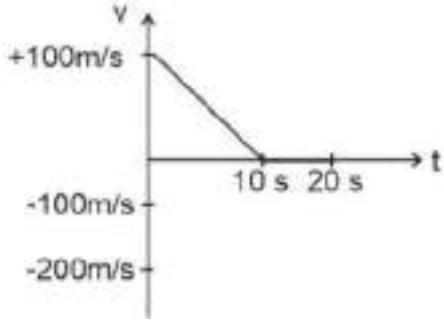
Hence 2 is correct option.

Q. 2 A bullet is shot vertically downwards with an initial velocity of $100\ \mathrm{m/s}$ from a certain height. Within $10\ \mathrm{s}$, the bullet reaches the ground and instantaneously comes to rest due to the perfectly inelastic collision. The velocity-time curve for total time $\mathrm{t} = 20\mathrm{s}$ will be: $\left(\ \mathrm{Take\ g} = 10\ \mathrm{m/s^2}\right)$.

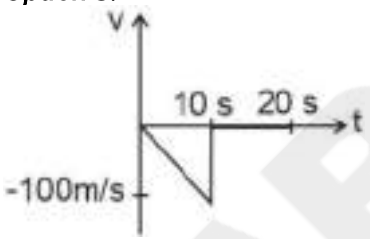
Option 1:



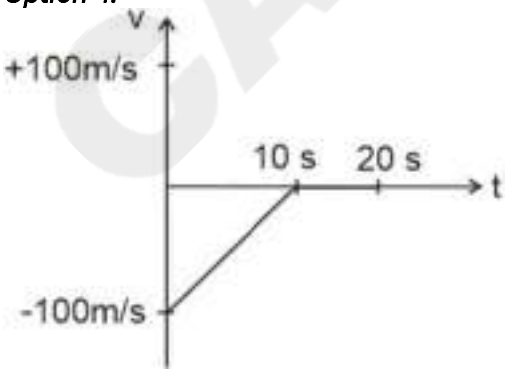
Option 2:



Option 3:

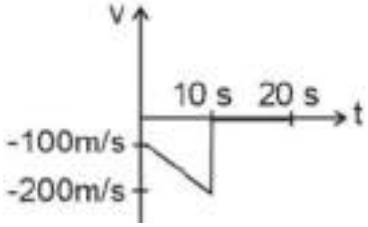


Option 4:



CAREERS360

Correct Answer:



Solution:

$\mathrm{u} \rightarrow$ Initial velocity of bullet

$\mathrm{v} \rightarrow$ Velocity of bullet when it reaches the ground

$$\mathrm{V} = \mathrm{u} + \mathrm{at}$$

$$\mathrm{V} = (-100) + (-10) \times 10$$

$$\mathrm{V} = -200\ \mathrm{m/s}$$

Ball comes to rest at $\mathrm{t} = 10\ \mathrm{s}$ immediately affer the collision.

The first option is the correct depiction of the situation.

Hence 1 is correct option

Q. 3 Sand is being dropped from a stationary dropper at a rate of $0.5\ \mathrm{kg}^{-1}$ on a conveyor belt moving with a velocity of $5\ \mathrm{ms}^{-1}$.The power needed to keep the belt moving with the same velocity will be :

Option 1:

$1.25\ \mathrm{W}$

Option 2:

$2.5\ \mathrm{W}$

Option 3:

$6.25\ \mathrm{W}$

Option 4:

$12.5\ \mathrm{W}$

Correct Answer:

$12.5\ \mathrm{W}$

Solution:

CAREERS 360

The power needed to keep the belt moving with the same velocity will be

$$P = \left(\frac{dm}{dt}\right) v^2$$

$$= (0.5) \times 25$$

$$P = 12.5\ W$$

Hence 4 is correct option.

Q. 4 A bag is gently dropped on a conveyor belt moving at a speed of $2\ m/s$. The coefficient of friction between the conveyor belt and bag is 0.4. Initially, the bag slips on the belt before it stops due to friction. The distance travelled by the bag on the belt during slipping motion, is : $[\text{Take } g = 10\ m/s^{-2}]$

Option 1:
$2\ m$

Option 2:
$0.5\ m$

Option 3:
$3.2\ m$

Option 4:
$0.8\ ms$

Correct Answer:
$2\ m$

Solution:

$$f = \mu mg$$

$$a = \frac{f}{m} = \mu g$$

Bag corne to rest when the speed of the bag is same as the conveyor belt ie.,$4\ m/s$

$$v^2 = u^2 + 2\ as$$

$(4)^2 = 0 + 2(0.4 \times 10)s$

$S = 2\ m$

The distance travelled by the bag on the belt during slipping motion is $2\ m$

Hence 1 is correct option.

Q. 5 Two cylindrical vessels of equal cross-sectional area $16\ cm^2$ contain water upto heights $100\ cm$ and $150\ cm$ respectively. The vessels are interconnected so that the water levels in them become equal. The work done by the force of gravity during the process, is $[Take, density\ of\ water = 10^3\ kg/m^3\ and\ g = 10\ ms^{-2}]$:

Option 1:
$0.25\ J$

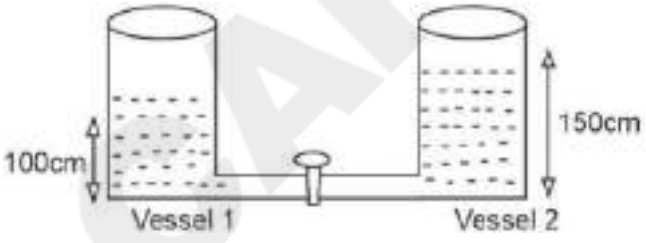
Option 2:
$1\ J$

Option 3:
$8\ J$

Option 4:
$12\ J$

Correct Answer:
$1\ J$

Solution:



The final water level is $125\ cm$ Rise in the water Level of vessel 1 is $25\ cm$ whereas there is corresponding fall in the water level of vessel 2 by $25\ cm$.

Work done by gravity is

$W = -\Delta U$

$= U_i - U_f$

$$U_i = \left[\frac{\rho A h_1^2 g}{2} + \frac{\rho A_2^2 g}{2}\right]$$

$$U_f = 2\left[\frac{\rho A h^2 g}{2}\right]$$

$$W = U_i - U_f$$

$$= \frac{\rho A g}{2}\left[\left(h_1^2 + h_2^2\right) - 2h^2\right]$$

$$= \frac{10^3 \times 16 \times 10^{-4} \times 10}{2}[1 + 2.25 - 2 \times 15625]$$

$$W = 8[3.25 - 3.125]$$

$$W = 1\ J$$

Hence 2 is correct option.

Q. 6 Two satellites A and B having masses in the ratio $4:3$, are revolving in circular orbits of radii $3r$ and $4r$ respectively around the earth. The ratio of total mechanical energy of A to B is:

Option 1:
$9:16$

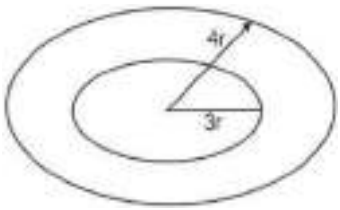
Option 2:
$16:9$

Option 3:
$1:1$

Option 4:
$4:3$

Correct Answer:
$16:9$

Solution:



Total mechanical energy is $\mathrm{TE}=-\frac{\mathrm{PE}}{2}$

for satellite 1 ,

$$\mathrm{TE}_1=\frac{-\mathrm{GMm}_1}{2(3\mathrm{r})}$$

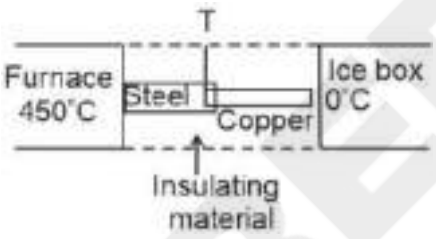
For satellite 2,

$$\mathrm{TE}_2=\frac{-\mathrm{GMm}_2}{2(4\mathrm{r})}$$

$$\frac{\mathrm{TE}_1}{\mathrm{TE}_2}=\frac{8\mathrm{m}_1}{6\mathrm{m}_2}=\frac{4}{3}\times\frac{4}{3}=\frac{16}{9}$$

Hence 2 is correct option.

Q. 7 If K_1 and K_2 are the thermal conductivities, L_1 and L_2 are the lengths and A_1 and A_2 are the cross sectional areas of steel and copper rods respectively such that $\frac{\mathrm{K}_2}{\mathrm{K}_1}=9,\frac{\mathrm{A}_1}{\mathrm{A}_2}=2,\frac{\mathrm{L}_1}{\mathrm{L}_2}=2$. Then, for the arrangement as shown in the figure, the value of temperature T of the steel - copper junction in the steady state will be :



Option 1:
$18^{\circ}\mathrm{C}$

Option 2:
$14^{\circ}\mathrm{C}$

Option 3:
$45^{\circ}\mathrm{C}$

Option 4:
$150^{\circ}\mathrm{C}$

Correct Answer:
$45^{\circ}\mathrm{C}$

Solution:
Let the heat current flowing through steel and Copper

$$H_1 = H_2$$

$$\frac{K_1 A_1 (450 - T)}{\ell_1} = \frac{K_2 A_2 (T - 0)}{\ell_2}$$

$$2(450 - T) = 9 \times 2(T)$$

$$450 - T = 9T$$

$$10T = 450$$

$$T = 45^\circ$$

Hence 3 is correct option.

Q. 8 Read the following statements :
A. When small temperature difference between a liquid and its surrounding is doubled, the rate of loss of heat of the liquid becomes twice.

B. Two bodies P and Q having equal surface areas are maintained at temperature $10^\circ C$ and $20^\circ C$. The thermal radiation emitted in a given time by P and Q are in the ratio $1 : 1.15$.

C. A Carnot Engine working between $100\ K$ and $400\ K$ has an efficiency of 75%.
D. When small temperature difference between a liquid and its surrounding is quadrupled, the rate of loss of heat of the liquid becomes twice.
Choose the correct answer from the options given below :

Option 1:
A, B, C only

Option 2:
A, B only

Option 3:
A. C only

Option 4:
B, C, D only

Correct Answer:
A, B, C only

Solution:

$$\text{Rate of loss of heat}(H) = \frac{KA\Delta T}{\ell}$$

(i) If temperature difference is doubled, the rate of loss of heat of the liquid becomes twice

(ii)

$$\Delta T_1 = T_P - T_0$$

$$\Delta T_2 = T_Q = T_0$$

$T_0 \rightarrow$Temperature of surrounding

$$\frac{H_1}{H_2} = \frac{1}{1.15} = \frac{(10 - T_0)}{(20 - T_0)}$$

$$20 - T_0 = 11.5 - 1.15T_0$$

$$0.15T_0 = -8.5$$

$$T_0 = -\frac{8.5}{0.15} = \frac{-1.7}{0.03}$$

$$T_0 = -\frac{170^\circ}{3}C$$

(iii)

$$\eta = 1 - \frac{T_C}{T_H}$$

$$= 1 - \frac{100}{400}$$

$$\eta = \frac{3}{4}$$

$$\eta\% = 75\%$$

(iv)

$$\Delta T_f = 4\Delta T_i$$

$$H_f = 4H_i$$

If the temperature difference is quadrupled, the rate of loss of heat of the liquid becomes four times.The fourth statement is not correct.

The A, B and C are correct statements.

Hence 1 is correct option.

Q. 9 Same gas is filled in two vessels of the same volume at the same temperature. If the ratio of the number of molecules is $1:4$, then
A. The r.m.s. velocity of gas molecules in two vessels will be the same.
B. The ratio of pressure in these vessels will be $1:4$.
C. The ratio of pressure will be $1:1$.
D The r.m.s. velocity of gas molecules in two vessels will be in the ratio of $1:4$.
Choose the correct answer from the options given below :

Option 1:
A and C only

Option 2:
B and D only

Option 3:
A and B only

Option 4:
C and D only

Correct Answer:
A and B only

Solution:

$$V_{rms}=\sqrt{\frac{3RT}{M}}$$

$$P=\frac{NK_BT}{V}$$

$\Rightarrow$ RMS velocity of gas in independent of the number of molecules

$$\Rightarrow \frac{p_1}{p_2}=\frac{N_1}{N_2}=\frac{1}{4}$$

The ratio of pressure will be $1/4$

$\rightarrow$ Statements A&B are correct

Hence 3 is correct option

Q. 10 Two identical positive charges Q each are fixed at a distance of $'2a'$ apart from each other. Another point charge q_0 with mass $'m'$ is placed at midpoint between two fixed charges. For a small displacement along the line joining the fixed charges, the charge q_0 executes SHM. The time period of oscillation of charge q_0 will be :

Option 1:

$$\sqrt{\frac{4\pi^3\varepsilon_0 ma^3}{q_0 Q}}$$

Option 2:

$$\sqrt{\frac{q_0 Q}{4\pi^3\varepsilon_0 ma^3}}$$

Option 3:

$$\sqrt{\frac{2\pi^2\varepsilon_0 ma^3}{q_0 Q}}$$

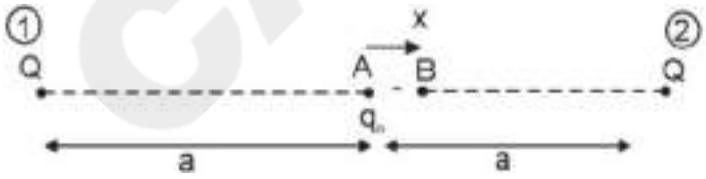
Option 4:

$$\sqrt{\frac{8\pi^3\varepsilon_0 ma^3}{q_0 Q}}$$

Correct Answer:

$$\sqrt{\frac{4\pi^3\varepsilon_0 ma^3}{q_0 Q}}$$

Solution:



If change q_o is displaced by x from point A to B

$$F_2 = \frac{KQq_o}{(a-x)^2}$$

$$F_1 = \frac{KQq_o}{(a+x)^2}$$

$$F_{net} = F_2 - F_1$$

$$= KQq_0\left[\frac{1}{(a-x)^2} - \frac{1}{(a+x)^2}\right]$$

$$F_{net} = kQq_0\left[\frac{4ax}{(a^2-x^2)^2}\right]$$

$$a_{net} = \frac{4KQq_0ax}{m\,(a^2-x^2)^2}$$

$$x << a$$

$$\therefore a_{net} = -\left(\frac{4KQq_0a}{ma^4}\right)x$$

$$\omega^2 = \frac{4KQq_0}{ma^3}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{ma^3}{4kQq_0}}$$

$$T = 2\pi\sqrt{\frac{ma^3 \times 4\pi\varepsilon_0}{4Qq_0}}$$

$$T = \sqrt{\frac{4\pi^3ma^3\varepsilon_0}{Qq_0}}$$

Hence 1 is correct option

Q. 11 Two sources of equal emfs are connected in series. This combination is connected to an external resistance R. The internal resistances of the two sources are r_1 and $r_2\,(r_1 > r_2)$. If the potential difference across the source of internal resistance r_1 is zero, then the value of R will be :

Option 1:

$r_1 - r_2$

Option 2:

$\frac{r_1 r_2}{r_1 + r_2}$

Option 3:

$\frac{r_1 + r_2}{2}$

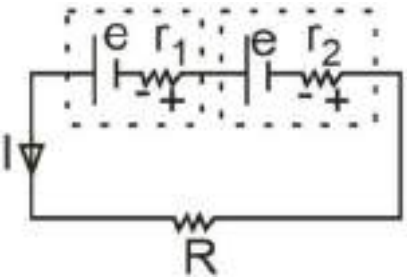
Option 4:

$r_2 - r_1$

Correct Answer:

$r_1 - r_2$

Solution:



Let the source of emf be e

$$I = \frac{2e}{(R + r_1 + r_2)}$$

$$V = e - Ir_1 = 0 \text{ (Given)}$$

$$O = e - \frac{2er_1}{(R + r_1 + r_2)}$$

$$1 = \frac{2r_1}{(R + r_1 + r_2)}$$

$$(R + r_1 + r_2) = 2r_1$$

$$R + r_2 = r_1$$

$$R = r_1 - r_2$$

Hence 1 is correct option.

Q. 12 Two bar magnets oscillate in a horizontal plane in earth's magnetic field with time periods of $3\ s$ and $4\ s$ respectively. If their moments of inertia are in the ratio of $3 : 2$, then the ratio of their magnetic moments will be:

Option 1:

$2 : 1$

Option 2:

$8 : 3$

Option 3:

$1 : 3$

Option 4:

$27 : 16$

Correct Answer:

$8 : 3$

Solution:

$$T = 2\pi\sqrt{\frac{I}{MB}}$$

$$\frac{T_1}{T_2} = \frac{3}{4} \rightarrow (1) \text{ (Given)}$$

$$\frac{I_1}{I_2} = \frac{3}{2} \rightarrow (2) \text{ (Given)}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{I_1}{I_2} \times \frac{M_2}{M_1}}$$

$$\frac{3}{4} = \sqrt{\frac{3}{2} \times \frac{M_2}{M_1}}$$

$$\frac{M_2}{M_1} = \left(\frac{3}{4}\right)^2 \times \frac{2}{3}$$

$$\frac{M_2}{M_1} = \frac{3}{8}$$

$$\therefore \frac{M_1}{M_2} = \frac{8}{3}$$

Hence 2 is correct option.

Q. 13 A magnet hung at 45° with magnetic meridian makes an angle of 60° with the horizontal. The actual value of the angle of dip is -

Option 1:

$\tan^{-1}\left(\sqrt{\frac{3}{2}}\right)$

Option 2:

$\tan^{-1}(\sqrt{6})$

Option 3:

$\tan^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Option 4:

$\tan^{-1}\left(\sqrt{\frac{1}{2}}\right)$

Correct Answer:

$\tan^{-1}\left(\sqrt{\frac{3}{2}}\right)$

Solution:

Apparent angle of dip $= 60^\circ = \delta'$

Angle of declination $= \theta = 60^\circ$

Actual angle of dip $= \tan\delta^\circ$

$$= \tan\delta' \times \cos\theta$$

$$= (\sqrt{3}) \times \frac{1}{2}$$

$$\tan\delta = \frac{\sqrt{3}}{2}$$

$$\text{Actual angle} = \delta = \tan^{-1}\left(\sqrt{\frac{3}{2}}\right) \text{ of dip}$$

Hence 1 is correct option.

Q. 14 A direct current of $4\ \mathrm{A}$ and an alternating current of peak value $4\ \mathrm{A}$ flow through resistance of $3\ \Omega$ and $2\ \Omega$ respectively. The ratio of heat produced in the two resistances in same interval of time will be :

Option 1:
$3:2$

Option 2:
$3:1$

Option 3:
$3:4$

Option 4:
$4:3$

Correct Answer:
$3:1$

Solution:
$\mathrm{I_{DC}} = 4\mathrm{A}$

$\mathrm{T_0} = 4\mathrm{A}$

$\mathrm{R_1} = 3\Omega$

$\mathrm{R_2} = 2\Omega$

Let the ratio of heat produced in the two resistance in same interval of time be $\mathrm{H_1 \& H_2}$ respectively.

$$\frac{\mathrm{H_1}}{\mathrm{H_2}} = \frac{\mathrm{I_{DC}^2 R_1\,(\Delta t)}}{\mathrm{I_{rms}^2 R_2 \Delta t}} = \frac{16 \times 3}{\left(\frac{4}{\sqrt{2}}\right)^2 \times 2}$$

$$\frac{\mathrm{H_1}}{\mathrm{H_2}} = 3$$

Hence 2 is correct option.

Q. 15 A beam of light travelling along $\mathrm{X-axis}$ is described by the electric field $\mathrm{E_y} = 900 \sin \omega(\mathrm{t - x/c})$. The ratio of electric force to magnetic force on a charge q moving along $\mathrm{Y-axis}$ with a speed of $3 \times 10^7\ \mathrm{ms^{-1}}$ will be :
$(\mathrm{Given\ speed\ of\ light\ } = 3 \times 10^8\ \mathrm{ms^{-1}}$

Option 1:

$1:1$

Option 2:

$1:10$

Option 3:

$10:1$

Option 4:

$1:2$

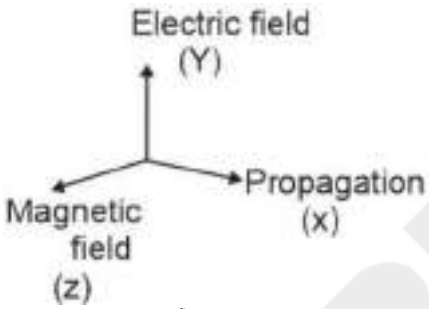
Correct Answer:

$10:1$

Solution:

$E_y = 900\sin(\omega(t - x/c))$

$$\frac{F_E}{F_M} = \frac{qE}{q\ VB\sin\ 90^\circ}$$



$V = 3 \times 10^7\ m/s$

$$\frac{E}{B} = 3 \times 10^8\ m/s$$

$$\frac{F_E}{F_M} = \frac{E}{VB} = \frac{3 \times 10^8}{3 \times 10^7} = \frac{10}{1}$$

Hence 3 is correct option.

Q. 16 A microscope was initially placed in air (refractive index 1). It is then immersed in oil (refractive index 2). For a light whose wavelength in air is λ, calculate the change of microscope's resolving power due to oil and choose the correct option.

Option 1:

Resolving power will be $\frac{1}{4}$ in the oil than it was in the air.

Option 2:

Resolving power will be twice in the oil than it was in the air.

Option 3:

Resolving power will be four times in the oil than it was in the air.

Option 4:

Resolving power will be $\frac{1}{2}$ in the oil than it was in the air.

Correct Answer:

Resolving power will be four times in the oil than it was in the air.

Solution:

$\mu_{\text{air}} = 1.$

$\mu_{\text{oil}} = 2$

$\lambda_{\text{air}} = \lambda$

Resolving power of a microscope is

$$\text{R} \cdot \text{P} \cdot = \frac{2\mu \sin\theta}{\lambda}$$

$$\frac{(\text{R} \cdot \text{P})_{\text{air}}}{(\text{R} \cdot \text{P})_{\text{oil}}} = \frac{\mu_{\text{air}}}{\lambda_{\text{air}}} \times \frac{\lambda_{\text{oil}}}{\mu_{\text{oil}}}$$

$$\frac{(\text{R} \cdot \text{P} \cdot)_{\text{air}}}{(\text{R} \cdot \text{P})_{\text{oil}}} = \left(\frac{\mu_{\text{ais}}}{\mu_{\text{oil}}}\right)^2$$

$$(\text{R} \cdot \text{P})_{\text{oil}} = 4(\text{R} \cdot \text{P})_{\text{air}}$$

Hence 3 is correct option

Q. 17 An electron (mass m) with an initial velocity $\overrightarrow{v} = v_0\hat{i}\,(v_0 > 0)$ is moving in an electric field $\overrightarrow{E} = -E_0\hat{i}\,(E_0 > 0)$ where E_0 is constant. If at $t = 0$ de Broglie wavelength is $\lambda_0 = \frac{h}{mv_0}$, then its de Broglie wavelength after time t is given by

Option 1:
λ_0

Option 2:
$\lambda_0\left(1 + \frac{eE_0t}{mv_0}\right)$

Option 3:
$\lambda_0 t$

Option 4:
$\frac{\lambda_0}{\left(1 + \frac{eE_0t}{mv_0}\right)}$

Correct Answer:
$\frac{\lambda_0}{\left(1 + \frac{eE_0t}{mv_0}\right)}$

Solution:
$\bar{v} = \bar{u} + \bar{a}t$

$\bar{v} = v_0\hat{i} + \frac{(-e)(-E_0)\hat{i}}{m}t$

$\bar{v} = \left(v_0 + \frac{eE_0t}{m}\right)\hat{i} \rightarrow (1)$

$\lambda_0 = \frac{h}{mv_0}$ (Given)

At time t,

$\lambda = \frac{h}{mv}$

$= \frac{h}{m\left(v_0 + \frac{eE_0t}{m}\right)}$

$$\lambda = \frac{\lambda_0 v_0}{\left(v_0 + \frac{eE_0 t}{m}\right)}$$

$$\lambda = \frac{\lambda_0}{\left(1 + \frac{eE_0 t}{mv_0}\right)}$$

Hence 4 is correct option.

Q. 18 What is the half-life period of a radioactive material if its activity drops to $1/16^{\text{th}}$ of its initial value in 30 years?

Option 1:
9.5 years

Option 2:
8.5 years

Option 3:
7.5 years

Option 4:
10.5 years

Correct Answer:
7.5 years

Solution:
Activity of the material at time t is,

$$A_t = A_0 2^{-t/T_{1/2}}$$

$$\frac{A_0}{16} = A_0 2^{-30/T_{1/2}}$$

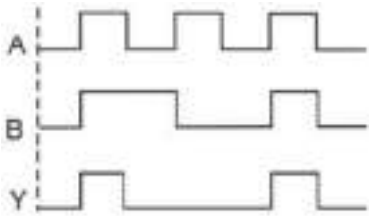
$$\frac{1}{2^4} = \frac{1}{2^{30}/T_{1/2}}$$

$T_{1/2} = 7.5$ years

The half life period of a radioactive material is 7.5 year

Hence 3 is correct option.

Q. 19 A logic gate circuit has two inputs A and B and output Y. The voltage waveforms of A, B and Y are shown below.



The logic gate circuit is:

Option 1:
AND gate

Option 2:
OR gate

Option 3:
NOR gate

Option 4:
NAND gate

Correct Answer:
AND gate

Solution:
Writing the voltage waveforms in tabular form

A	B	Y
0	0	0
1	1	1
0	1	0
1	0	0
0	0	0
1	1	1

AND gate is the output of the above values .

Hence 1 is correct option

Q. 20 At a particular station, the TV transmission tower has a height of $100\ \mathrm{m}$. To triple its coverage range, height of the tower should be increased to

Option 1:
$200\ \mathrm{m}$

Option 2:
$300\ \mathrm{m}$

Option 3:
$600\ \mathrm{m}$

Option 4:
$900\ \mathrm{m}$

Correct Answer:
$900\ \mathrm{m}$

Solution:

$\mathrm{h_1 = 100m}$

$\mathrm{Range} = \sqrt{2\mathrm{hR}}$

$\mathrm{R_1} = \sqrt{2\mathrm{h_1R}}$

$\mathrm{R_2 = 3R_1\ (Given)}$

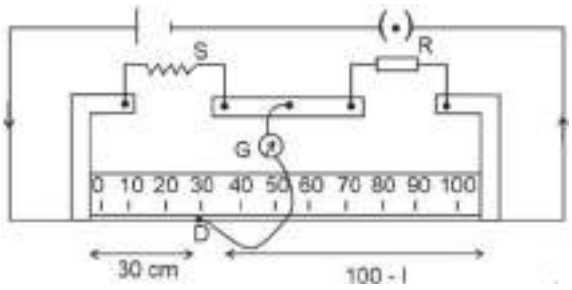
$\sqrt{2\mathrm{h_2R}} = 3\sqrt{2\mathrm{h_1R}}$

$\mathrm{h_2 = 9h_1}$

$\Rightarrow \mathrm{h_2 = 900m}$

Hence 4 is correct option

Q. 21 In a meter bridge experiment, for measuring unknown resistance $'S'$, the null point is obtained at a distance $30\ \text{cm}$ from the left side as shown at point D. If R is $5.6\ \text{k}\Omega$, then the value of unknown resistance $''S'$ will be _____ Ω.



Correct Answer:
2400

Solution:

$$\frac{\text{S}}{\text{R}} = \frac{30}{70}$$

$$\frac{\text{S}}{5.6} = \frac{3}{7}$$

$$\text{S} = \frac{3}{7} \times 5.6$$

$$= 3 \times 0.8$$

$$\text{S} = 2.4\text{k}\Omega$$

$$\therefore \text{S} = 2400\Omega$$

The value of unknown resistance 's' will be 2400Ω

Q. 22 The one division of main scale of Vernier callipers reads $1\ \text{mm}$ and 10 divisions of Vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other, the zero of the Vernier lies to the right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the Vernier scale lies in between $4.1\ \text{cm}$ and $4.2\ \text{cm}$ and 6^{th} Vernier division coincides with a main scale division. The diameter of the bob will be _____ $\times 10^{-2}\ \text{cm}$.

Correct Answer:
412

Solution:

$1\text{MSD} = 1\ \text{mm}$

$10\text{VSD} = 9\text{MSD}$

$1\text{VSD} = \left(\frac{9}{10}\right)\text{MSD}$

$\text{L} \cdot \text{C} = 1\text{MSD} - 1\text{VSD} = \frac{1}{10}\text{MSD}$

$\text{Zero error} = (\text{MSD})\ (\text{LC})$

$= (4)(0.1)\text{mm}$

$= 0.4\ \text{mm}$

$= 0.04\ \text{cm}$

$\text{Zero correction} = -(0.04)\text{cm}$

$\text{Reading} = \text{MSR} + \text{VSR} + \text{Zero correction}$

$= 4.1\ \text{cm} + (6 \times 0.1\ \text{mm}) + (-0.04)\ \text{cm}$

$\text{Reading} = 4.12\ \text{cm}$

$= 412 \times 10^{-2}\ \text{cm}$

\end{aligned}

Q. 23 Two beams of light having intensities I and 4 I interfere to produce a fringe pattern on a screen. The phase difference between the two beams are $\pi/2$ and $\pi/3$ at points A and B respectively. The difference between the resultant intensities at the two points is xI. The value of x will be _____.

Correct Answer:
2

Solution:
$I_1 = I$

$I_2 = 4I$

$I_{resultant} = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$I_A = I + 4I + 4I\cos(\pi/2)$

$I_A = 5I$

$I_B = I + 4I + 4I\cos(\pi/3)$

$I_B = 7I$

The difference between two intensities at a point is,

$I_B - I_A = 2I$

$x = 2$

Q. 24 To light, a $50\ W, 100\ V$ lamp is connected, in series with a capacitor of capacitance $\frac{50}{\pi\sqrt{x}}\mu F$, with $200\ V, 50\ Hz AC$ source. The value of x will be _____.

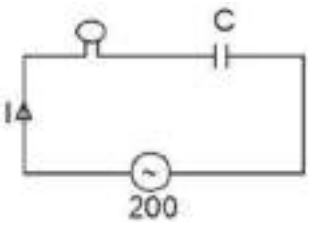
Correct Answer:
3

Solution:
$p = 50w$

$c = \frac{50}{\pi\sqrt{x}}$

$V_{rms} = 200$

$f = 50Hz$



$$\text{current through lamp} = I$$

$$= \frac{P}{V}$$

$$= \frac{1}{2}A$$

$$I = \frac{V_{rms}}{Z}$$

$$\frac{1}{2} = \frac{200}{\sqrt{X_c^2 + R^2}}$$

$$\sqrt{X_c^2 + R^2} = 400$$

$$X_c^2 + \left[\frac{(100)^2}{50}\right]^2 = 160000$$

$$X_c^2 + 40000 = 160000$$

$$X_c^2 = \frac{1}{\left(\frac{2\pi \times 50 \times 50 \times 10^{-6}}{\pi\sqrt{x}}\right)^2} = 120000$$

$$\frac{x}{25 \times 10^6 \times 10^{-12}} = 120000$$

$$x = 25 \times 12 \times 10^{-2}$$

$$x = 3$$

Q. 25 A $1\ m$ long copper wire carries a current of $1\ A$. If the cross section of the wire is $2.0\ mm^2$ and the resistivity of copper is $1.7 \times 10^{-8}\Omega m$, the force experienced by moving electron in the wire is _____$\times 10^{-23}\ N$.
$(\text{charge on electorn} = 1.6 \times 10^{-19} C)$

Correct Answer:
136

Solution:
$\ell = 1\ m$

$I = 1\ A$

$A = 2 \times 10^{-6}\ m^2$

$\rho = 1.7 \times 10^{-8} \Omega m$

The force experienced by the moving electron in the wire is

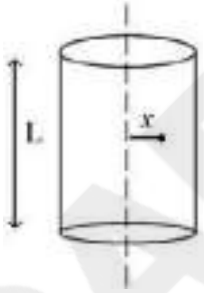
$$F = qE = \frac{qJ}{\sigma} = J\rho q$$

$$F = \left(\frac{1}{2 \times 10^{-6}}\right) \times \left(1.7 \times 10^{-8}\right) \times \left(1.6 \times 10^{-19}\right)$$

$$F = \frac{2.72}{2} \times 10^{-21}$$

$$F = 136 \times 10^{-23}\ N$$

Q. 26 A long cylindrical volume contains a uniformly distributed charge of density $\rho\ Cm^{-3}$. The electric field inside the cylindrical volume at a distance $x = \frac{2\varepsilon_0}{\rho} m$ from its axis is ——Vm^{-1}.



Correct Answer:

1

Solution:

$q_{\text{enc}} = \varrho Al$

$q_{\text{enc}} = \varrho \times (\pi x^2)\, l$

$\oint \bar{E} \cdot d\bar{A} = \frac{q_{\text{enc}}}{\varepsilon_0}$

$E \times 2\pi x l = \frac{5 \times \pi x^2 l}{\varepsilon_0}$

$\mathrm{E} = \frac{\rho \mathrm{x}}{2\varepsilon_0}$

at $\mathrm{x} = \frac{2\varepsilon_0}{\varrho}$

$\mathrm{E} = 1\mathrm{v/m}$

Q. 27 A mass 0.9 kg, attached to a horizontal spring, executes SHM with an amplitude A_1. When this mass passes through its mean position, then a smaller mass of 124 g is placed over it and both masses move together with amplitude A_2. If the ratio $\frac{\mathrm{A}_1}{\mathrm{A}_2}$ is $\frac{\alpha}{\alpha - 1}$, then the value of α will be _____.

Correct Answer:
16

Solution:
By energy conservation,

$\mathrm{TE}_{\text{initial}} = \mathrm{IE}_{\text{final}}$

$\frac{1}{2}\mathrm{m}_1\omega^2\mathrm{A}_1^2 = \frac{1}{2}\mathrm{m}_2\omega^2\mathrm{A}_2^2$

$\mathrm{m}_1 = 0.9\ \mathrm{kg}$

$\mathrm{m}_2 = 0.9\ \mathrm{kg} + 0.124\ \mathrm{kg}$

$\mathrm{m}_2 = 1.024\ \mathrm{kg}$

$$\frac{A_1}{A_2} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{1024}{900}}$$

$$\frac{A_1}{A_2} = \frac{2^5}{30} = \frac{32}{30} = \frac{\alpha}{\alpha - 1}$$

$$\frac{16}{15} = \frac{\alpha}{\alpha - 1}$$

$$\alpha = 16$$

Q. 28 A square aluminum $(\text{shear modulus is } 25 \times 10^9 \text{Nm}^{-2})$ slab of side $60\ \text{cm}$ and thickness $15\ \text{cm}$ is subjected to a shearing force (on its narrow face) of $18.0 \times 10^4 \text{N}$. The lower edge is riveted to the floor. The displacement of the upper edge is _____ μm.

Correct Answer:
48

Solution:

$$\text{A} = 15 \times 16 = 900 \times 10^{-4}\text{m}^2$$

$$\text{F} = 18 \times 10^4$$

$$\eta = \frac{\text{F/A}}{(\text{x}/\ell)}$$

$$25 \times 10^9 = \frac{18 \times 10^4}{900 \times 10^{-4}} \times \frac{\text{x}}{60 \times 10^{-2}}$$

$$\text{x} = \frac{60 \times 10^{-2} \times 10^8}{50 \times 25 \times 10^9}$$

$$= 4.8 \times 10^{-3} \times 10^{-2}$$

$$\text{x} = 48\mu\text{m}$$

Q. 29 A pulley of radius $1.5\ \mathrm{m}$ is rotated about its axis by a force $\mathrm{F} = \left(12\mathrm{t} - 3\mathrm{t}^2\right)\mathrm{N}$ applied tangentially (while t is measured in seconds). If moment of inertia of the pulley about its axis of rotation is $4.5\ \mathrm{kg\ m^2}$, the number of rotations made by the pulley before its direction of motion is reversed, will be $\frac{\mathrm{K}}{\pi}$. The value of K is _____.

Correct Answer:
18

Solution:

$$\mathrm{F} = 12\mathrm{t} - 3\mathrm{t}^2$$

$$\tau = \mathrm{FR} = 18\mathrm{t} - 4.5\mathrm{t}^2$$

$$\alpha = \frac{\tau}{\mathrm{I}} = 4\mathrm{t} - \mathrm{t}^2$$

The time when the direction of motion is reversed, $\omega = 0$

$$\frac{\mathrm{d}\omega}{\mathrm{dt}} = \alpha$$

$$\mathrm{d}\omega = \alpha \mathrm{dt}$$

$$\omega = \left[\frac{4\mathrm{t}^2}{2}\frac{-\mathrm{t}^3}{3}\right]_0^{\mathrm{t}}$$

$$\omega = 2\mathrm{t}^2\ \frac{-\mathrm{t}^3}{3}$$

$$\text{when } \omega = 0, 0 = 2\mathrm{t}^2\left(1 - \frac{\mathrm{t}}{6}\right)$$

$$\mathrm{t} = 0 \text{ or } \mathrm{t} = 6$$

$$\frac{\mathrm{d}\theta}{\mathrm{dt}} = 2\mathrm{t}^2 - \frac{\mathrm{t}^3}{3}$$

$$\mathrm{d}\theta = 2\mathrm{t}^2\mathrm{dt} - \frac{\mathrm{t}^3}{3}\mathrm{dt}$$

$$\theta = \frac{2\mathrm{t}^3}{3} - \frac{\mathrm{t}^4}{12}$$

$$= \frac{2}{3} \times 216 - \frac{36 \times 36}{12}$$

$= 144 - 108$

$\theta = 36$

$\text{No.of rotation} = \frac{\Theta}{2\pi} = \frac{18}{\pi}$

$\mathrm{K} = 18$

Q. 30 A ball of mass m is thrown vertically upward. Another ball of mass $2\ \mathrm{m}$ is thrown at an angle θ with the vertical. Both the balls stay in air for the same period of time. The ratio of the heights attained by the two balls respectively is $\frac{1}{\mathrm{x}}$. The value of x is _____.

Correct Answer:
1

Solution:
$\rightarrow$Tlime of flight depends on vertical speed.

$\mathrm{T} = \frac{2\mathrm{u_y}}{\mathrm{g}} = \frac{2\mathrm{u}\sin\theta}{\mathrm{g}}$

$\rightarrow$As the time of flight is same for both of them, the maximum vertical hight will be same.

$\Rightarrow \because \frac{1}{\mathrm{x}} = 1$

$\mathrm{x} = 1$

Chemistry

Q. 1 $250\ \mathrm{g}$ solution of D-glucose in water contains 10.8% of carbon by weight. The molality of the solution is nearest to
(Given: Atomic Weights are, $\mathrm{H},\ 1\mathrm{u}; \mathrm{C},\ 12\mathrm{u}; \mathrm{O},\ 16\mathrm{u}$)

Option 1:
1.03

Option 2:
2.06

Option 3:
3.09

Option 4:
5.40

Correct Answer:
2.06

Solution:
We need to think of the reverse solution

$$\text{molality} \rightarrow \text{mole of solute} \rightarrow \text{mass of } C_6H_{12}O_6$$
$$\rightarrow \text{weight of solvent} \rightarrow [250g - \text{mass of } C_6H_{12}O_6]$$

$$\text{mass of } C_6H_{12}O_6 \rightarrow \text{moles of } C_6 \text{ present}$$
$$\downarrow$$
$$\text{mass of carbon in solution.}$$

Now, 10.8 % of carbon by weight present in a 250 g solution

$$\text{mass of carbon} = \frac{250}{100} \times 10.8 = 27\ \text{g}$$

$$\text{Hence, moles of 6C } (C_6) \text{ present} = \frac{27}{12 \times 6} = 0.375 \text{ moles}$$

$$\text{mole of } C_6H_{12}O_6 = 0.375 \text{ moles} - (1)$$
$$\text{mass of } C_6H_{12}O_6 = 0.375 \times 180 = 67.5\ \text{g}$$
$$\text{So, mass of solvent} = 250\ \text{g} - 67.5\ \text{g} = 182.5\text{g}$$

$$\text{mass of solvent} = 182.5 \times 10^{-3}\ \text{kg}$$

$$\text{Now, molality} = \frac{\text{moles of } C_6H_{12}O_6}{\text{mass of solvent}} = \frac{0.375}{182.5 \times 10^{-3}}$$

$$\text{molality} = 2.06$$

Option (2) is correct.

Q. 2 Given below are two statements.
Statement I: $O_2, Cu^{2+}, \text{and } Fe^{3+}$ are weakly attracted by magnetic field and are magnetized in the same direction as magnetic field.

Statement II : $NaCl \text{ and } H_2O$ are weakly magnetized in opposite direction to magnetic field.
In the light of the above statements, choose the most appropriate answer from the options given below.

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:
Both Statement I and Statement II are correct.

Solution:
$O_2, Cu^{2+} \text{ and } Fe^{3+}$ are paramagnetic in nature. they are attracted by magnetic field.

$NaCl \text{ and } H_2O$ are weakly repelled by a magnetic field. They are weakly magnetised in a magnetic field in opposite direction.

So, Statement I is correct and Statement II is correct.

Option (1) is correct.

Q. 3 Given below are two statements. One is labelled as **Assertion A** and the other is labelled as **Reason R.**
Assertion A: Energy of $2\ s$ orbital of hydrogen atom is greater than that of $2\ s$ orbital of lithium.
Reason R: Energies of the orbitals in the same subshell decrease with increase in the atomic number.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:
Energy of $2\mathrm{s}$ orbital of Hydrogen atom is greater than that of $2\mathrm{s}$ orbital of Lithium.

(R) Energies of the orbitals in the same subshell decrease with increase in atomic number.

Both A and R are correct and R is the correct explanation of A.

Option (1) is correct.

Q. 4 Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Activated charcoal adsorbs $\mathrm{SO_2}$ more efficiently than $\mathrm{CH_4}$.
Reason R: Gases with lower critical temperatures are readily adsorbed by activated charcoal.
In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both $\mathrm{A\ and\ R}$ are correct and R is the correct explanation of A.

Option 2:
Both $\mathrm{A\ and\ R}$ are correct but R is NOT the correct explanation of A.

Option 3:
A is correct but R is not correct.

Option 4:
A is not correct but R is correct.

Correct Answer:
A is correct but R is not correct.

Solution:
Easily liquefiable gases with higher critical temperature are readily adsorbed as van der Waal's forces are stronger near the critical temperatures. Thus activated charcol adsorb more sulphur dioxide (critical temperatures 630 k) than methane (critical temperature 190 k).

A is correct but R is not correct.

Option (3) is correct.

Q. 5 Boiling point of a 2% aqueous solution of a non-volatile solute A is equal to the boiling point of 8% aqueous solution of a non-volatile solute B. The relation between molecular weights of $\mathrm{A\ and\ B}$ is

Option 1:

$M_A = 4M_B$

Option 2:

$M_B = 4M_A$

Option 3:

$M_A = 8M_B$

Option 4:

$M_B = 8M_A$

Correct Answer:

$M_B = 4M_A$

Solution:

We know that

$\Delta T_b = K_b m, \text{m is molality.}$

ΔT_b will be same for A and B due to solvent is same (water $\rightarrow$ aqeous) and boilling point of aqueous solution of both A and B are equal.

K_b will be due to solvent is same for A and B.

Hence, $(\Delta T_b)_A = (\Delta T_b)_B$

$(K_b m)_A = (K_b m)_B$

$m_A = m_B$

$$\frac{\frac{w_A}{M_A}}{W_{solvent}} = \frac{\frac{w_B}{M_B}}{W_{solvent}}$$

Now , 2% aqueous solution of a non-volatile solute A means in 100 ml or 100 g water 2 g of A is present and 8% aqueous solution means in 100 ml or 100 g of water 8 g of B is present.

So,

$$\frac{2}{\frac{M_A}{100g}} = \frac{8}{\frac{M_B}{100g}}$$

$\Rightarrow M_B = 4\,M_A$

So, option (2) is correct.

Q. 6 The incorrect statement is

Option 1:
The first ionization enthalpy of K is less than that of $\mathrm{Na\ and\ Li.}$

Option 2:
Xe does not have the lowest first ionization enthalpy in its group.

Option 3:
The first ionization enthalpy of element with atomic number 37 is lower than that of the element with atomic number 38 .

Option 4:
The first ionization enthalpy of Ga is higher than that of the d-block element with atomic number 30 .

Correct Answer:
The first ionization enthalpy of Ga is higher than that of the d-block element with atomic number 30 .

Solution:
(1) is correct statement.

(2) is correct statement.

(3) Rb (atomic number 37) $\mathrm{IE_1}$ is lower than Sr (atomic number 38) $\mathrm{IE_1}$. this also correct.

(4) $\mathrm{IE_1}$ of $\mathrm{Ga}\ \left(4s^2 4p^1\right)$ is higher than $\mathrm{IE_1}$ of Zn atomic number $30\ \left(3d^{10}\ 4s^2\right)$, this statement is incorrect.

Due to Zn has fully paired orbitals and Ga has not.

Option (4) is correct.

Q. 7 Which of the following methods are not used to refine any metal?

A. Liquation

B. Calcination

C. Electrolysis

D. Leaching

E. Distillation

Choose the correct answer from the options given below:

Option 1:
B and D only

Option 2:
A, B, D and E only

Option 3:
B, D and E only

Option 4:
A,C,and E only

Correct Answer:
B and D only

Solution:
Calcination and Leaching methods are not used to refine any metal.

Refining methods are distillation , electrolysis , liquation,zone refining , vapour phase refining and chromotographic

Option (1) is correct.

Q. 8 Given below are two statements.
Statement I: Hydrogen peroxide can act as an oxidizing agent in both acidic and basic conditions.

Statement II: Density of hydrogen peroxide at $298\ \mathrm{K}$ is lower than that of $\mathrm{D_2O}$.
In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is true but Statement II is false.

Solution:
Statement I is correct.
Statement II is incorrect.
Density of $\mathrm{H_2O}$ is 1.00 and Density of $\mathrm{D_2O}$ is 1.10 at 298 k.

Option (3) is correct.

Q. 9 Given below are two statements.
Statement I: The chlorides of $\mathrm{Be\ and\ Al}$ have Cl-bridged structure. Both are soluble in organic solvents and act as Lewis bases.
Statement II: Hydroxides of $\mathrm{Be\ and\ Al}$ dissolve in excess alkali to give beryllate and aluminate ions.

In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is false but Statement II is true.

Solution:
The chlorides of Be and Al have Cl—bridged structure. They act as Lewis acid not Lewis base.

So, Statement I is false.
Statement II is true.
Option (4) is correct.

Q. 10 Which oxoacid of phosphorous has the highest number of oxygen atoms present in its chemical formula?

Option 1:
Pyrophosphorus acid

Option 2:
Hypophosphoric acid

Option 3:
Phosphoric acid

Option 4:
Pyrophosphoric acid

Correct Answer:
Pyrophosphoric acid

Solution:

$(1)\,\text{Pyrophosphorus acid} \rightarrow H_4P_2O_5 \rightarrow 5\,\text{oxygen atoms}$

$(2)\,\text{Hypophosphoric acid} \rightarrow H_4P_2O_6 \rightarrow 6\,\text{oxygen atoms}$

$(3)\text{Phosphoric acid} \rightarrow H_3PO_4 \rightarrow 4\,\text{oxygen atoms}$

$(4)\,\text{Pyrophosphoric acid} \rightarrow H_4P_2O_7 \rightarrow 7\,\text{oxygen atoms}$

So, pyrophosphoric acid was the highest number of oxygen atoms present.

Option (4) is correct.

Q. 11 Given below are two statements.
Statement I: Iron (III) catalyst, acidified $K_2Cr_2O_7$ and neutral $KMnO_4$ have the ability to oxidise I^- to I_2 independently.
Statement II: Manganate ion is paramagnetic in nature and involves $p\pi - p\pi$ bonding.
In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both Statement I and Statement II are true.

Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Both Statement I and Statement II are false.

Solution:
Iron (III) catalyst cannot oxidise $\mathrm{I^{-}\,to\,I_2}$ independently.
Iron (III) catalyses the the reaction between iodide and persulphate ions.

$$2\mathrm{I^-} + \mathrm{S_2O_8^{2-}} \rightarrow \mathrm{I_2} + 2\mathrm{SO_4^{2-}}$$

Statement (1) is false
Manganate ion is paramagnetic in nature and involves $\mathrm{p\pi - d\pi}$ bonding not $\mathrm{p\pi - p\pi}$ bonding.
Statement II is false

Option (2) is correct.

Q. 12 The total number of $\mathrm{Mn = O}$ bonds in $\mathrm{Mn_2O_7}$ is_____.

Option 1:
4

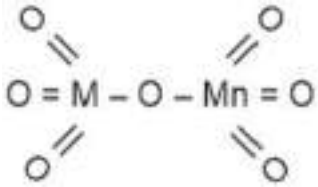
Option 2:
5

Option 3:
6

Option 4:
3

Correct Answer:
6

Solution:



6 Mn = O bonds

1 Mn – O – Mn bond

Option (3) is correct.

Q. 13 Match List I with List II.

List I Pollutant	**List II Disease/sickness**
A.Sulphate(> 500ppm)	I.Methemoglobinemia
B.Nitrate(> 50ppm)	II.Brown mottling of teeth
C.Lead(> 50ppb)	III.Laxative effect
D.Fluoride(> 2ppm)	IV.Kidney damage

Choose the correct answer from the options given below:

Option 1:
A-IV, B-I, C-II, D-III

Option 2:
A-III, B-I, C-IV, D-II

Option 3:
A-II, B-IV, C-I, D-III

Option 4:
A-II, B-IV, C-III, D-I

Correct Answer:
A-III, B-I, C-IV, D-II

Solution:

$$(\mathrm{A})\ \mathrm{Sulphate}\,(>500\,\mathrm{ppm}) \rightarrow \mathrm{III.Laxactive\ effect.}$$

$$(\mathrm{B})\ \mathrm{Nitrate}\,(>50\,\mathrm{ppm}) \rightarrow \mathrm{I.Methemoglobinemia}$$

$$(\mathrm{C})\ \mathrm{Lead}\,(>50\,\mathrm{ppb}) \rightarrow \mathrm{IV.Kidney\ damage}$$

$$(\mathrm{D})\ \mathrm{Fluoride}\,(>2\,\mathrm{ppm}) \rightarrow \mathrm{II.Brown\ mottling\ of\ teeth}$$

$$\mathrm{So,(A)-III,B-I,C-IV,D-II}$$

Option (2) is correct.

Q. 14 Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and, the other is labelled as Reason $\mathbf{R}$.

Assertion A: [6] Annulene, [8] Annulene and cis-[10] Annulene, are respectively aromatic, not-aromatic and aromatic.

[6] Annulene [8] Annulene Cis – [10] Annulene

Reason R: Planarity is one of the requirements of aromatic systems.
In the light of the above statements, choose the **most appropriate** answer from the options given below.

Option 1:
Both $\mathbf{A}$ and $\mathbf{R}$ are correct and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$.

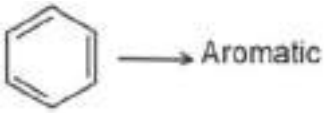
Option 2:
Both $\mathbf{A}$ and $\mathbf{R}$ are correct but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$.

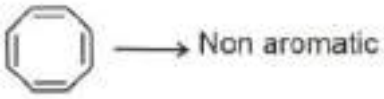
Option 3:
A is correct but R is not correct.

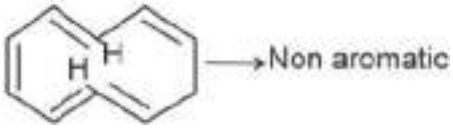
Option 4:
A is not correct but R is correct.

Correct Answer:
A is not correct but R is correct.

Solution:



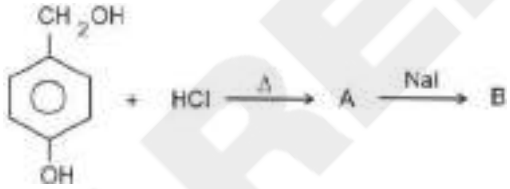




Statement I is incorrect.
Statement II is correct.

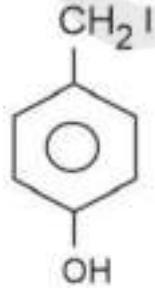
Option (4) is correct.

Q. 15

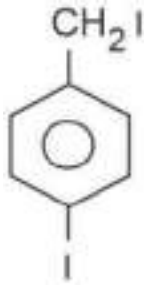


In the above reaction product B is

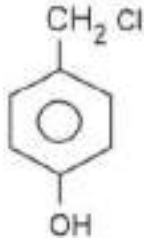
Option 1:



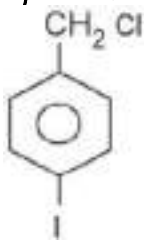
Option 2:



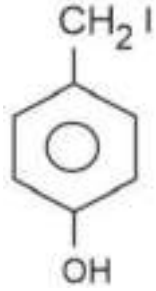
Option 3:



Option 4:

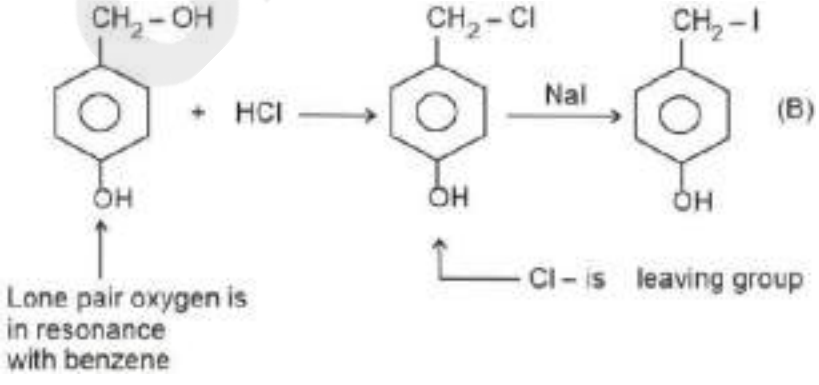


Correct Answer:



Solution:

The reaction will be -



Option (1) is correct.

Q. 16 Match List I with List II.

List I **Polymers**	**List II** **Commercial names**
A. Phenol-formaldehyde resin	I. Glyptal
B. Copolymer of 1,3-butadiene and styrene	II. Novolac
C. Polyester of glycol and phthalic acid	III. Buna-S
D. Polyester of glycol and terephthalic acid	IV. Dacron

Choose the correct answer from the options given below:

Option 1:
A-II, B-III, C-IV, D-I

Option 2:
A-II, B-III, C-I, D-IV

Option 3:
A-II, B-I, C-III, D-IV

Option 4:
A-III, B-II, C-IV, D-I

Correct Answer:
A-II, B-III, C-I, D-IV

Solution:
A. Phenol-formaldehyde resin $\longrightarrow$ II.Norolac

B. Copolymer of 1,3-butadiene and styrene $\longrightarrow$ III.Buna-S

C. Polyester of glycol and phthalic acid $\longrightarrow$ I. Glyptal

D. Polyester of glycol and terephthalic acid $\longrightarrow$ IV. Dacron

So, A-II, B-III, C-I, D-IV

Option (2) is correct.

Q. 17 A sugar $'X'$ dehydrates very slowly under acidic condition to give furfural which on further reaction with resorcinol gives the coloured product after sometime.Sugar $'X'$ is

Option 1:
Aldopentose

Option 2:
Aldotetrose

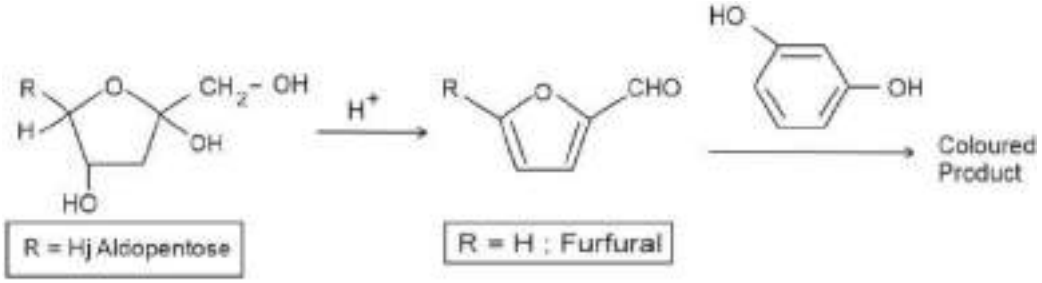
Option 3:
Oxalic acid

Option 4:
Ketotetrose

Correct Answer:
Aldopentose

Solution:

Aldoses dehydrate very slowly. under acidic condition to give furfural. which reacts with resorcinol(1,3- dihydroxy benzene) to given a coloured product.



Option (1) is correct.

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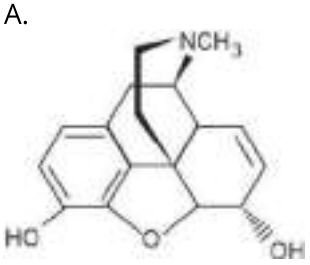
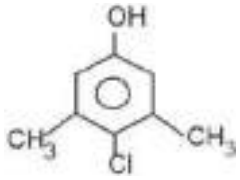
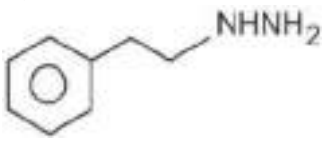
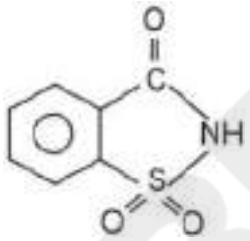
Q. 18 Match **List I** with **List II**

List I	**List II**
A. 	I. Anti-depressant
B. 	II. 550 times sweeter than cane sugar.
C. 	III. Narcotic analgesic
D. 	IV. Antiseptic

Choose the correct answer from the options given below :

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
A-III, B-I, C-II, D-IV

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-III, B-I, C-IV, D-II

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
A. Morphine $\longrightarrow$ III. Narcotic analgesic

B. Chloroxylenol $\longrightarrow$ IV. Antiseptic

C. Phenelzine(Nardi) $\longrightarrow$ I.Anti-depressant

D.Saccharin $\longrightarrow$ II.550 times sweeter than cane sugar.

So, A-III, B-IV, C-I, D-II

Option (3) is correct.

Q. 19 In Carius method of estimation of halogen, $0.45\ \text{g}$ of an organic compound gave $0.36\ \text{g}$ of AgBr. Find out the percentage of bromine in the compound.
(Molar masses : $\text{AgBr} = 188\ \text{g mol}^{-1}; \text{Br} = 80\ \text{g mol}^{-1}$)

Option 1:
34.04%

Option 2:
40.04%

Option 3:
36.03%

Option 4:
38.04%

Correct Answer:
34.04%

Solution:

We know,

$$\text{Percentage of halogen} = \frac{\text{atomic mass of X} \times m_1 \times 100}{\text{molecular mass of AgX} \times m}$$

$$\% \text{ of Br} = \frac{\text{atom k mass of Br} \times m_1 \times 100}{\text{molecular mass of AgBr} \times m}$$

$m_1 = \text{AgBr formed} = 0.36 \text{ g}$
$m = \text{organic compound taken} = 0.45 \text{ g}$

$$\text{So, \% of Br} = \frac{80 \times 0.36 \times 100}{188 \times 0.45}$$
$$\% \text{ of Br} = 34.04\%$$

Option (1) is correct.

Q. 20 Match List I with List II

List I	List II
A. Benzenesulphonyl chloride	I. Test for primary amines
B. Hoffmann bromamide reaction	II. Anti Saytzeff
C. Carbylamine reaction	III. Hinsberg reagent
D. Hoffmann orientation	IV. Known reaction of Isocyanates

Choose the correct answer from the options given below:

Option 1:
A-IV, B-III, C-II, D-I

Option 2:
B-IV, B-II, C-I, D-III

Option 3:
A-III, B-IV, C-I, D-II

Option 4:
A-IV, B-III, C-I, D-II

Correct Answer:
A-III, B-IV, C-I, D-II

Solution:
A. Benzenesulphonyl chloride $\longrightarrow$ III. Hinberg reagent.

B. Hoffmann bromamide reaction $\longrightarrow$ IV. Known reaction of Isocynates

C. Carbylamine reaction $\longrightarrow$ I. Teast for primary amines.

D. Hoffmann orientation $\longrightarrow$ II.Anti saytzeff.

Option (3) is correct.

Q. 21 $20\ \text{mL}$ of $0.02\text{M}\ \text{K}_2\text{Cr}_2\text{O}_7$ solution is used for the titration of $10\ \text{mL}$ of Fe^{2+} solution in the acidic medium.
The molarity of Fe^{2+} solution is _____$\times 10^{-2}\text{M}$. (Nearest Integer)

Correct Answer:
24

Solution:
First balance the reaction with Half Reaction method.

$$6\text{F}_e^{2+} + \text{Cr}_2^{+6}\text{O}_7^{2-} + 14\text{H}^+ \longrightarrow 6\text{F}_e^{3+} + 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$$

$$\text{milliequivalent of } \text{Fe}^{2+} = \text{milliequivalent of } \text{Cr}_2\text{O}_7^{2-}$$

$$\text{M} \times \text{V} \times \text{n} - \text{factor} = \text{M} \times \text{V} \times \text{n} - \text{factor}$$
$$\text{M} \times 10 \times (1 \times (3-2)) = 0.02 \times 20 \times [2\,(6-3)]$$
$$\text{M} \times 10 \times 1 = 0.4 \times 6$$
$$\text{M} = 0.24 = 24 \times 10^{-2}$$

Ans = 24

Q. 22 $2NO + 2H_2 \rightarrow N_2 + 2H_2O$

The above reaction has been studied at $800^{\circ}C$. The related data are given in the table below

Reaction serial number	Initial preesure of H_2/kPa	Initial Pressure of NO/kPa	Initial rate $\left(\frac{-dp}{dt}\right)/(kPa/s)$
1	65.6	40.0	0.135
2	65.6	20.1	0.033
3	38.6	65.6	0.214
4	19.2	65.6	0.106

The order of the reaction with respect to NO is_____.

Correct Answer:

2

Solution:

From reaction number (1) and (2) pressure of this H_2 constant.

$40.0 \longrightarrow 20.1, 2 \text{ times of } 20.1 \text{ is } 4.0$

$0.135 \longrightarrow 0.037, 4 \text{ times of } 0.033 \text{ is } 0.135$

Means when doubling the concentration of NO the rate is getting change 4 times,

This indicates the rate depends upon the square of the concentration of NO.

So, The order of reaction with respect to NO is 2.

Ans = 2.

Q. 23 Amongst the following, the number of oxide(s) which are paramagnetic in nature is

$Na_2O, KO_2, NO_2, N_2O, ClO_2, NO, SO_2, Cl_2O$

Correct Answer:
4

Solution:
Unpaired electron causes the paramagnetic nature
So, KO_2, NO_2, ClO_2, NO are paramagnetic in nature due presense of unpaired electron.

Q. 24 The molar heat capacity for an ideal gas at constant pressure is $20.785\ \text{J K}^{-1}\ \text{mol}^{-1}$. The change in internal energy is $5000\ \text{J}$ upon heating it from $300\ \text{K to } 500\ \text{K}$. The number of moles of the gas at constant volume is ___.
[Nearest integer] (Given: $R = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}$)

Correct Answer:
2

Solution:
We Know,
$$\Delta H = Cp\Delta T \text{ and } \Delta H = \Delta V - R\Delta T$$

$$\text{So, } Cp\Delta T = \Delta V - R\Delta T$$
$$\text{for n mole, } nCp\Delta T = \Delta V - nR\Delta T$$

$$\Delta V \text{ for gas} = 5000\,\text{J},$$
$$\Delta T = 50.0\,\text{K} - 300\,\text{K} = 200\,\text{K}$$
$$Cp = 20.785\,\text{J K}^{-1}\,\text{mol}^{-1}$$
$$R = 8.314\,\text{J K}^{-1}\,\text{mol}^{-1}$$

Now,
$$n \times 200 \times 20.785 = 5000 - n \times 8.314 \times 200$$
$$n = \frac{5000}{200 \times 12.471}$$
$$n = 2\,\text{mol}$$

Ans =2

Q. 25 According to MO theory, number of species/ions from the following having identical bond order is_____.
$CN^-, NO^+, O_2, O_2^+, O_2^{2+}$

Correct Answer:
3

Solution:
We know ,

$$\text{Bond order} = \frac{\text{Nb} - \text{Na}}{2}$$

$$\text{CN}^- \Rightarrow \text{BO} = \frac{6-0}{2} = 3$$

$$\text{NO}^+ \Rightarrow \text{BO} = \frac{6-0}{2} = 3$$

$$\text{O}_2 \Rightarrow \text{BO} = \frac{6-2}{2} = 2$$

$$\text{O}_2^+ \Rightarrow \text{BO} = \frac{6-1}{2} = 2.5$$

$$\text{O}_2^{2+} \Rightarrow \text{BO} = \frac{6-0}{2} = 3$$

$\text{So, CN}^-, \text{NO}^+$ and O_2^{2+} have same bond order.

Ans = 3

Q. 26 At $310\ \text{K}$, the solubility of CaF_2 in water is $2.34 \times 10^{-3}\ \text{g}/100\ \text{mL}$. The solubility product of CaF_2 is _____$\times 10^{-8}(\ \text{mol/L})^3$. (Give molar mass : $\text{CaF}_2 = 78\ \text{g mol}^{-1}$)

Correct Answer:
0

Solution:
The solubility of CaF_2 in water $= 2.34 \times 10^{-3}\ \text{g}/100\ \text{ml}$

$$= \frac{2.34 \times 10^{-3}}{78}\text{mol}/100\text{ml}$$
$$= 3 \times 10^{-5}\text{mol}/100\text{ml}$$
$$= \frac{3 \times 10^{-5}}{0.1}\text{mol/L}$$
$$= 3 \times 10^{-4}\text{mol/L}$$

$$\underset{\text{(s)}}{\text{Now, CuF}_2} \longrightarrow \underset{\text{s}}{\text{Cu}^{2+}} + \underset{\text{2s}}{2\text{F}^-}$$

$$\begin{aligned}\text{solubility product} &= s \times (2s)^2 = 3s^3 \\ &= 3 \times \left(3 \times 10^{-4}\right)^3 \\ &= 108 \times 10^{-12} \\ &= 0.0108 \times 10^{-8}\,(\text{mol/L})^3\end{aligned}$$

Ans = 0.

Q. 27 The conductivity of a solution of complex with formula $CoCl_3\,(NH_3)_4$ corresponds to $1:1$ electrolyte, then the primary valency of central metal ion is_____

Correct Answer:
1

Solution:
The solution conductivity corresponds to 1 : is electrolyte of complex $CoCl_3\,(NH_3)_4$.

Then the formula of complex will be $[CoCl_2\,(NH_3)_4]^+\,Cl^{\Theta}$

The primary valences are normally ionisable and are satisfied by negative ions.

So, primary valencey will be 1 because there is only one positive charge present.

Ans = 1.

Q. 28 In the titration of $KMnO_4$ and oxalic acid in acidic medium, the change in oxidation number of carbon at the end point is_____

Correct Answer:
1

Solution:
Oxalate ion or oxalic acid is oxidised at $333\,k$.
$5C_2O_4^{2-} + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$

Change in oxidation number of Carbon: $[+3 \rightarrow +4]$
So, change is $= 4 - 3 = 1$

Q. 29 Optical activity of an enantiomeric mixture is $+12.6^{\circ}$ and the specific rotation of $^{(+)}$ isomer is $+30^{\circ}$. The optical purity is _____%

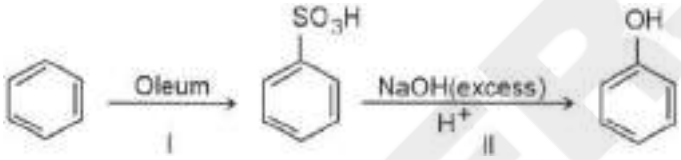
Correct Answer:
42

Solution:

$$\text{The optical purity } = (\%) = \frac{\text{rotation of mixture}}{\text{rotation of a isomer}} \times 100$$
$$= \frac{12.6}{30} \times 100$$
$$= 42\%$$

Ans = 42.

Q. 30 In the following reaction



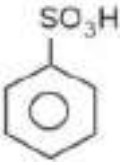
The % yield for reaction I is 60% and that of reaction II is 50%. The overall yield of the complete reaction is _____%. [nearest integer]

Correct Answer:
30

Solution:

Let initial moles are n. of

Reaction I : 60 % of yield

So, $\text{n} \rightarrow 0.6\,\text{n}$ mole of  formed

Reaction II : 50 % yield

$\text{So, } 0.6\,\text{n mole} \rightarrow \frac{0.6}{2} = 0.3\text{n mole of.}$

The overall yield will be $= \frac{0.3\text{n}}{\text{n}} \times 100$

$= 30\%$

Ans = 30.

Maths

Q. 1 Let $\mathrm{R_1}$ and $\mathrm{R_2}$ be two relations defined on $\mathbb{R}$ by $\mathrm{a\ R_1 b} \Leftrightarrow \mathrm{ab} \geq 0$ and $\mathrm{a\ R_2 b} \Leftrightarrow \mathrm{a} \geq \mathrm{b}$ Then,

Option 1:
$\mathrm{R_1}$ is an equivalence relation but not $\mathrm{R_2}$

Option 2:
$\mathrm{R_2}$ is an equivalence relation but not $\mathrm{R_1}$

Option 3:
both $\mathrm{R_1}$ and $\mathrm{R_2}$ are equivalence relations

Option 4:
neither $\mathrm{R_1}$ nor $\mathrm{R_2}$ is an equivalence relation

Correct Answer:
neither $\mathrm{R_1}$ nor $\mathrm{R_2}$ is an equivalence relation

Solution:
$\mathrm{R_1} = \{\mathrm{xy} \geq 0, \mathrm{x}, \mathrm{y} \in \mathrm{R}\}$

$\text{for reflexive } \mathrm{x} \times \mathrm{x} \geq 0 \text{ which is true for Symmetric}$

$\text{if } \mathrm{xy} \geq 0 \Leftrightarrow \mathrm{yx} \geq 0$

$\text{if } \mathrm{x} = 2, \mathrm{y} = 0 \text{ and } \mathrm{z} = -2$

$\text{Then} \mathrm{x} \cdot \mathrm{y} \geq 0 \& \ \mathrm{y.z} \geq 0 \text{ but } \mathrm{x} \cdot \mathrm{z} \geq 0 \text{ is not true}$

$\Rightarrow \text{not transitive relation}$

$\Rightarrow \mathrm{R_1} \text{ is not equivalence}$

R_2 if $a \geq b$ it does not implies $b \geq a$

$\Rightarrow R_2$ is not equivalence relation

Hence correct option is 4

Q. 2 Let $f, g : \mathbb{N} - \{1\} \to \mathbb{N}$ be functions defined by $f(a) = \alpha,$ where α is the maximum of the powers of those primes p such that p^{α} divides $a,$ and $g(a) = a + 1,$ for all $a \in \mathbb{N} - \{1\}$.Then, the function $f + g$ is

Option 1:
one-one but not onto

Option 2:
onto but not one-one

Option 3:
both one-one and onto

Option 4:
neither one-one nor onto

Correct Answer:
neither one-one nor onto

Solution:

$f' : N - \{1\} \to N \quad f(a) = \alpha$

where α is max of powers of prime p such that p^{α}dividesa.Also $g(a) = a + 1$

$\therefore f(2) = 1 \quad g(2) = 3$

$f(3) = 1 \quad g(3) = 4$

$f(4) = 2 \quad g(4) = 5$

$f(5) = 1 \quad g(5) = 6$

$\Rightarrow f(2) + g(2) = 4$

$(f(3) + g(3)) = 5$

$f(4) + g(4) = 7$

$f(5) + g(5) = 7$

$\because$ many one$f(x)+g(x)$does not contain 1

$\Rightarrow$ into function

Hence correct option is 4

Q. 3 Let the minimum value v_0 of $v=|z|^2+|z-3|^2+|z-6i|^2, z\in\mathbb{C}$ is attained at $z=z_0$. Then $\left|2z_0^2-\bar{z}_0^3+3\right|^2+v_0^2$ is equal to

Option 1:
1000

Option 2:
1024

Option 3:
1105

Option 4:
1196

Correct Answer:
1000

Solution:

$$z_0=\left(\frac{0+3+0}{3},\frac{0+6+0}{3}\right)=(1,2)$$

$$v_0=|1+2i|^2+\left(1+2i-3\right|^2+|1+2i-6i|^2=30$$

$$\text{Then } \left|2z_0^2-\bar{z}_0^3+3\right|^2+v_0^2$$

$$=\left|2(1+2i)^2-(1-2i)^3+3\right|^2+900$$

$$=|2(1-4+4i)-(1-4-4i)(1-2i)+3|^2+900$$

$$=|8+6i|^2+900=100+900=1000$$

Hence correct option is 1.

Q. 4 Let $A=\begin{pmatrix}1 & 2\\ -2 & -5\end{pmatrix}$. Let $\alpha,\beta\in\mathbb{R}$ be such that $\alpha A^2+\beta A=2I$. Then $\alpha+\beta$ is equal to

Option 1:
-10

Option 2:
-6

Option 3:
6

Option 4:
10

Correct Answer:
10

Solution:
Characteristic equation of matric A

$$|A-\lambda I|=0$$

$$\begin{vmatrix}1-\lambda & 2\\ -2 & -5-\lambda\end{vmatrix}=0$$

$$\Rightarrow \lambda^2+4\lambda=1$$

$$\Rightarrow A^2+4-A=I$$

$$\Rightarrow 2A^2+8A=2I \quad \text{.....(1)}$$

$$\text{given that } \alpha A^2+\beta A=2I \quad \text{.....(2)}$$

$$\text{comparing equetion } (1)\&\ (2) \text{ we get}$$

$$\alpha=2,\beta=8$$

$$\therefore \alpha+\beta=10$$

Hence correct option is 4

Q. 5 The remainder when $(2021)^{2022} + (2022)^{2021}$ is divided by 7 is

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
6

Correct Answer:
0

Solution:

$(2021)^{2022} + (2022)^{2021}$

$(2023-2)^{2022} + (2023-1)^{2021}$

$= 7n_1 + 2^{202} + 7n_2 - 1$

$= 7(n_1 + n_2) + 8^{674} - 1$

$= 7(n_1 + n_2) + (7-1)^{674} - 1$

$= 7(n_1 + n_2) + 7n_3 + 1 - 1$

$= 7(n_1 + n_2 + n_3)$

$\therefore$ Given number is divisible by 7 hence reamainder is zero.

Q. 6 Suppose $a_1, a_2, \ldots, a_n, \ldots$ be an arithmetic progression of natural numbers. If the ratio of the sum of first five terms to the sum of first nine terms of the progression is $5:17$ and $110 < a_{15} < 120$, then the sum of the first ten terms of the progression is equal to

Option 1:
290

Option 2:
380

Option 3:
460

Option 4:
510

Correct Answer:
380

Solution:

$$\frac{S_5}{S_g} = \frac{5}{17} \Rightarrow \frac{\frac{5}{2}(2a+4d)}{\frac{9}{2}(2a+8d)} = \frac{5}{17}$$

$$\Rightarrow d = 4a$$

$$a_{15} = a + 14d = 57a$$

$$\text{Now } 110 < a_{15} < 120$$

$$\Rightarrow 110 < 57a < 120$$

$$\Rightarrow a = 2 \therefore d = 8$$

$$S_{10} = \frac{10}{2}(2 \times 2 + 9 \times 8) = 380$$

Hence correct option is 2

Q. 7 Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined as $f(x) = a\sin\left(\frac{\pi[x]}{2}\right) + [2 - x], a \in \mathbb{R}$, where $[t]$ is the greatest integer less than or equal to t. If $\lim_{x \to -1} f(x)$ exists, then the value of $\int_0^4 f(x)dx$ is equal to

Option 1:
-1

Option 2:
-2

Option 3:

1

Option 4:

2

Correct Answer:

-2

Solution:

$$\lim_{x\to -1^+} a\ \sin\left(\pi\frac{[x]}{2}\right)+[2-x]=-a+2$$

$$\lim_{x\to -1^-} a\ \sin\left(\pi\frac{[x]}{2}\right)+[2-x]=0+3=3$$

$$\lim_{x\to -1} f(x) \text{exist when } a=-1$$

Now,

$$\int_0^4 f(x)dx=\int_0^1 f(x)dx+\int_1^2 f(x)dx+\int_2^3 f(x)dx+\int_3^4 f(x)dx$$

$$=\int_0^1 (0+i)dx+\int_1^2 (-1+0)dx+\int_1^3 (0-1)dx+\int_3^4 (1-2)dx$$

$$=1-1-1-1=-2$$

Hence correct option is 2

Q. 8 Let $I=\int_{\pi/4}^{\pi/3}\left(\frac{8\sin x-\sin 2x}{x}\right)dx$. Then

Option 1:

$\frac{\pi}{2}<I<\frac{3\pi}{4}$

Option 2:

$\frac{\pi}{5}<I<\frac{5\pi}{12}$

Option 3:

$$\frac{5\pi}{12} < I < \frac{\sqrt{2}}{3}\pi$$

Option 4:

$$\frac{3\pi}{4} < I < \pi$$

Correct Answer:

$$\frac{5\pi}{12} < I < \frac{\sqrt{2}}{3}\pi$$

Solution:

$f(x) = 8\sin x - \sin 2x$

$f'(x) = 8\cos x - 2\cos 2x$

$f''(x) = -8\sin x + 4\sin 2x$

$= -8\sin x(1 - \cos x)$

$\therefore f''(x) < 0 \quad x \in \left(\frac{\pi}{4}, \frac{\pi}{3}\right)$

$\therefore f'(x)\text{is} \downarrow \text{function.}$

$f'\left(\frac{\pi}{3}\right) < f'(x) < f'\left(\frac{\pi}{4}\right)$

$5 < f'(x) < \frac{8}{\sqrt{2}}$

$5 < f'(x) < 4\sqrt{2}$

$5x < f(x) < 4\sqrt{2}x$

$5 < \frac{f(x)}{x} < 4\sqrt{2}$

$\int_{\pi/4}^{\pi/3} 5dx < \int_{\pi/4}^{\pi/3} \frac{f(x)}{x}dx < \int_{\pi/4}^{\frac{\pi}{3}} 4\sqrt{2}dx$

$\int_{\pi/4}^{\pi/3} 5dx < \int_{\frac{\pi}{4}}^{\pi/3} \left(\frac{8\sin x - \sin 2x}{x}\right)dx < \int_{\pi/4}^{\pi/3} \left(\frac{4}{\sqrt{2}}\right)dx$

$\frac{5\pi}{12} < I < \frac{\sqrt{2}\pi}{3}$

Hence correct option is 3

Q. 9 The area of the smaller region enclosed by the curves $y^2 = 8x + 4$ and $x^2 + y^2 + 4\sqrt{3}x - 4 = 0$ is equal to

Option 1:
$\frac{1}{3}(2 - 12\sqrt{3} + 8\pi)$

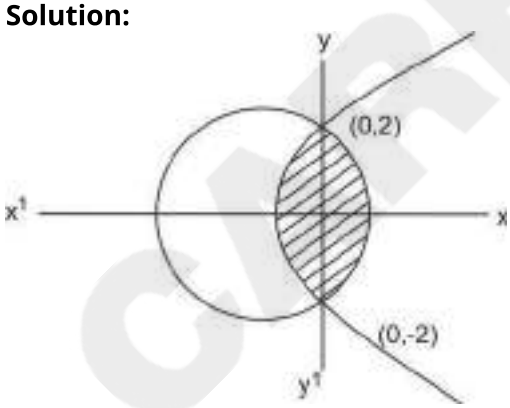
Option 2:
$\frac{1}{3}(2 - 12\sqrt{3} + 6\pi)$

Option 3:
$\frac{1}{3}(4 - 12\sqrt{3} + 8\pi)$

Option 4:
$\frac{1}{3}(4 - 12\sqrt{3} + 6\pi)$

Correct Answer:
$\frac{1}{3}(4 - 12\sqrt{3} + 8\pi)$

Solution:



$x^2 + y^2 + 4\sqrt{3}x - 4 = 0$

$y^2 = 8x + 4$

Point of intersections are $(0, 2)$ & $(0, -2)$both are symmetric about x-axis

$$\text{Area } = 2\int_0^2 \left(\sqrt{16 - y^2} - 2\sqrt{3}\right) - \left(\frac{y^2 - 4}{8}\right) dy$$

$$=\frac{1}{3}[8\pi+4-12\sqrt{3}]$$

Hence correct option is 3

Q. 10 Let $y=y_1(x)$ and $y=y_2(x)$ be two distinct solutions of the differential equation $\frac{dy}{dx}=x+y$, with $y_1(0)=0$ and $y_2(0)=1$ respectively. Then, the number of points of intersection of $y=y_1(x)$ and $y=y_2(x)$ is

Option 1:
0

Option 2:
1

Option 3:
2

Option 4:
3

Correct Answer:
0

Solution:

$$\frac{dy}{dx}=x+y \Rightarrow \frac{dy}{dx}-y=x$$

$$\text{If}=e^{-x}$$

$$\therefore \text{solution is } y\cdot e^{-x}=\int x\cdot e^{-x}dx$$

$$\Rightarrow ye^{-x}=-xe^{-x}-e^{-x}+c$$

$$\Rightarrow y=-x-1+ce^{x}$$

$$y_1(0)=0 \Rightarrow c=1$$

$$\therefore y_1=-x-1+e^{x} \quad \text{.....(1)}$$

$$y_2(0)=1 \Rightarrow c=2$$

$$\therefore y_2=-x-1+2e^{x}\ldots\ (2)$$

Now $y_2 - y_1 = e^x > 0 \quad \therefore y_2 \neq y_1$

$\therefore$ Number of point of intersections of $y_1 \& \ y_2$ is zero.

Hence correct option is 1

Q. 11 Let $P(a, b)$ be a point on the parabola $y^2 = 8x$ such that the tangent at P passes through the centre of the circle $x^2 + y^2 - 10x - 14y + 65 = 0$. Let a be the product of all possible values of a and B be the product of all possible values of b. Then the value of $A + B$ is equal to

Option 1:
0

Option 2:
25

Option 3:
40

Option 4:
65

Correct Answer:
65

Solution:
$P(a, b)$ is point on $y^2 = 8x$, such that tangent at P pass through centre of

$x^2 + y^2 - 10x - 14y + 65 = 0$ i.e $(5, 7)$

Tangent at $P\left(at^2, 2at\right)$ is $ty = x + at^2$

$A = 2 \& \text{ it pass through } (5, 7)$

$7t = 5 + 2t^2$

$\Rightarrow t = 1, t = \frac{5}{2}$

$\therefore P\left(at^2, 2at\right) \Rightarrow (2, 4) \text{ when } t = 1 \ \& \left(\frac{25}{2}, 10\right) \text{ when } t = \frac{5}{2}$

$\therefore A = 2 \times \frac{25}{2} = 25$

$\therefore A+B=65$

Hence correct option is 4

Q. 12 Let $\vec{a}=\alpha\hat{i}+\hat{j}+\beta\hat{k}$ and $\vec{b}=3\hat{i}-5\hat{j}+4\hat{k}$ be two vectors, such that $\vec{a}\times\vec{b}=-\hat{i}+9\hat{i}+12\hat{k}$. Then the projection of $\vec{b}-2\vec{a}$ on $\vec{b}+\vec{a}$ is equal to

Option 1:
2

Option 2:
$\frac{39}{5}$

Option 3:
9

Option 4:
$\frac{46}{5}$

Correct Answer:
$\frac{46}{5}$

Solution:

Let $\vec{a}=\alpha\hat{i}+\hat{j}+\beta k,\ \vec{b}=3\hat{i}-5f+4\hat{k}$

$\vec{a}\times\vec{b}=-\hat{i}+9\hat{j}+12\hat{k}$

$$\Rightarrow\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ \alpha&1&\beta\\ 3&-5&4\end{vmatrix}$$

$\Rightarrow(4+5\beta)\hat{i}+(3\beta-4\alpha)\hat{j}+(-5\alpha-3)\hat{k}$

$=-\hat{i}+9\hat{j}+12\hat{k}$

$\therefore 4+5\beta=+,3\beta-4\alpha=9,-5\alpha-3=12$

$\therefore\beta=-1,\alpha=-3.$

$\therefore\vec{\alpha}=-3\hat{i}+\hat{j}-\hat{k},\ \vec{b}=3\hat{i}-5\hat{j}+4\hat{k}$

$\therefore \vec{a}+\vec{b}=-4\hat{j}+3\hat{k}$

$|\vec{a}|^2=11, |\vec{b}|^2=50$

$\vec{a}\cdot\vec{b}=-9+(-5)-4=-18$

$\therefore$ Projectile of$(\vec{b}-2\vec{a})$on$\vec{a}+\vec{b}$f

$$\frac{(\vec{b}-2\vec{a})\cdot(\vec{a}+\vec{b})}{|\vec{a}+\vec{b}|}=\frac{|\vec{b}|^2-2|\vec{a}|^2-|\vec{a}\cdot\vec{b}\rangle}{|\vec{a}+\vec{b}|}$$

$$=\frac{50-22-(-18)}{-5}=\frac{46}{5}$$

Hence correct option is 4

Q. 13 Let $\vec{a}=2\hat{i}-\hat{j}+5\hat{k}$ and $\vec{b}=\alpha\hat{i}+\beta\hat{j}+2\hat{k}$ If $((\vec{a}\times\vec{b})\times\hat{i})\cdot\hat{k}=\frac{23}{2}$ then $|\vec{b}\times 2j|$ is equal to

Option 1:
4

Option 2:
5

Option 3:
$\sqrt{21}$

Option 4:
$\sqrt{17}$

Correct Answer:
5

Solution:
$\vec{a}=2\hat{i}-\hat{j}+5\hat{k}, \vec{b}=2\hat{i}+\beta\hat{j}+2k$

$((\vec{a}\times\vec{b})\times\hat{i})\cdot\hat{k}=\frac{23}{2}$, then $|\vec{b}\times 2\hat{j}|$ is

$$((\vec{a}\cdot\hat{\mathrm{i}})\vec{b}-(\vec{b}\cdot\hat{\mathrm{i}})\vec{a})-\hat{\mathrm{k}}=\frac{23}{2}$$

$$(\vec{a}\cdot\hat{\mathrm{i}})(\vec{b}\cdot\hat{\mathrm{i}})-(\vec{b}\cdot\hat{\mathrm{i}})(\vec{a}\cdot\hat{\mathrm{k}})=\frac{23}{2}$$

$$2\times 2-2\times 5=\frac{23}{2}\Rightarrow 5\alpha=4\frac{-23}{2}$$

$$\alpha=-\frac{3}{2}$$

$$\vec{b}\times 2\hat{j}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\ \alpha&\beta&2\\ 0&2&0\end{vmatrix}=-4\hat{i}+2\alpha\hat{k}$$

$$\therefore|\tilde{\mathrm{b}}\times 2\hat{\mathrm{j}}|=\sqrt{16+4\alpha^2}=\sqrt{16+4\times\frac{9}{4}}$$

$$=5.$$

Hence correct option is 2

Q. 14 Let S be the sample space of all five digit numbers. It P is the probability that a randomly selected number from S, is a multiple of 7 but not divisible by 5 , then $9\mathrm{p}$ is equal to

Option 1:
1.0146

Option 2:
1.2085

Option 3:
1.0285

Option 4:
1.1521

Correct Answer:
1.0285

Solution:
$\mathrm{n(s)}=$ all 5 digit nos $=9\times 10^4$

A: no is multiple of 7 but not divisible by 5

Smallest 5 digit divisible by 7 is 1003

Largest 5 digit divisible by 7 is 99995

$\therefore 99995 = 10003 + (n-1)7 \Rightarrow n = 12857$

Numbers divisible by 35

$99995 = 10010 + (\mathrm{P}-1)35 \Rightarrow \mathrm{P} = 2572$

$\therefore$ Numbers divisible by 7 but not by 35 are

$12857 - 2572 = 10285$

$\therefore \quad \mathrm{P} = \dfrac{10285}{90000} \qquad \because 9\mathrm{P} = 1.0285$

Hence correct option is 3

Q. 15 Let a vertical tower AB of height $2\mathrm{h}$ stands on a horizontal ground. Let from a point P on the ground a man can see upto height H of the tower with an angle of elevation 2α. When from P, he moves a distance d in the direction of $\overrightarrow{AP}$, he can see the top B of the tower with an angle of elevation α. If $\mathrm{d} = \sqrt{7}\mathrm{h},$ then $\tan\alpha$ is equal to

Option 1:
$\sqrt{5} - 2$

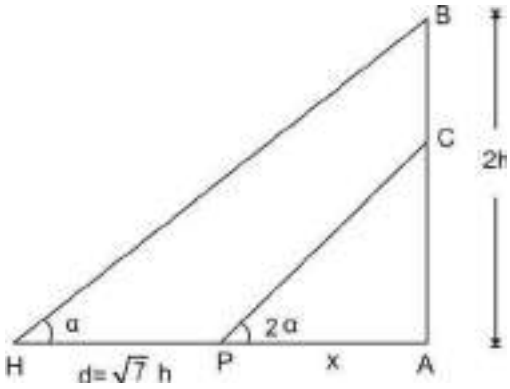
Option 2:
$\sqrt{3} - 1$

Option 3:
$\sqrt{7} - 2$

Option 4:
$\sqrt{7} - \sqrt{3}$

Correct Answer:
$\sqrt{7} - 2$

Solution:



$$\tan 2\alpha = \frac{h}{x}$$

$$\text{and } \tan\alpha = \frac{2h}{x+\sqrt{7}h}$$

$$\tan\alpha = \frac{2h}{h\cot 2\alpha + \sqrt{7}h}$$

$$\tan\alpha = \frac{2}{\frac{(1-\tan^2\alpha)}{2\tan\alpha} + \sqrt{7}}$$

$$\text{Put } \tan\alpha = t \text{ \& simplify}$$

$$\Rightarrow \tan\alpha = \sqrt{7} - 2$$

Hence correct option is 3

Q. 16 $(p \wedge r) \Leftrightarrow (p \wedge (\sim q))$ is equivalent to $(\sim p)$ when r is

Option 1:
p

Option 2:
$\sim p$

Option 3:
q

Option 4:
$\sim q$

Correct Answer:
$\sim p$

Solution:

$\text{Given}(p \cap \gamma) \Leftrightarrow (p \wedge (\sim q)) \equiv (\sim p)$

$\text{Taking } r = q$

p	q	$\sim p$	$\sim q$	$p \wedge q$	$p \wedge \sim q$	$(p \wedge r) \Leftrightarrow (p \wedge (\sim q))$
T	T	F	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	F	F	T
F	F	T	T	F	F	T

So, clear

$(p \wedge \gamma) \Leftrightarrow (p \wedge (\sim q)) \equiv (\sim p)$

Hence correct option is 2

Q. 17 If the plane P passes through the intersection of two mutually perpendicular planes $2x + ky - 5z = 1$ and $3kx - ky + z = 5, k < 3$ and intercepts a unit length on positive X-axis, then the intercept made by the plane P on the y-axis is

Option 1:

$\frac{1}{11}$

Option 2:

$\frac{5}{11}$

Option 3:

6

Option 4:

7

Correct Answer:

7

Solution:

Two given plane mutually perpendicular

$2(3k) + k(-k) + (-5)1 = 0$

$k = 1, 5$

$\text{but } k < 3 \quad \text{so } k = 1$

Plane passing through these phase is

$$2x+4=5z-1+\lambda(3x-y+z-5)=0$$

$$\frac{x}{\frac{5\lambda+1}{2+3\lambda}}+\frac{y}{\frac{5\lambda+1}{1-\lambda}}+\frac{z}{\frac{5\lambda+1}{\lambda-5}}=1$$

$$\text{given } \frac{5\lambda+1}{2+3\lambda}=1\Rightarrow\lambda=\frac{1}{2}$$

$$\text{So intercept on } y-\text{axis}=\frac{5\lambda+1}{1-\lambda}=7$$

Hence correct option is 4

Q. 18 Let $A(1,1), B(-4,3), C(-2,-5)$ be vertices of a triangle ABC, P be a point on side BC, and Δ_1 and Δ_2 be the areas of triangles APB and ABC, respectively. If $\Delta_1 : \Delta_2 = 4:7$, then the area enclosed by the lines AP, AC and the $x-$axis is

Option 1:
$\frac{1}{4}$

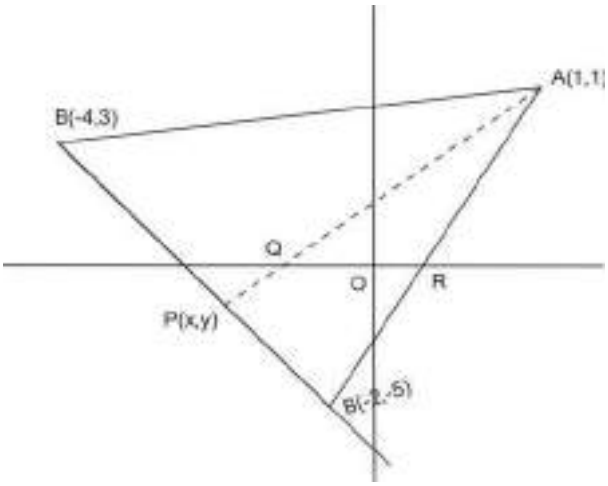
Option 2:
$\frac{3}{4}$

Option 3:
$\frac{1}{2}$

Option 4:
1

Correct Answer:
$\frac{1}{2}$

Solution:



given

$$\Delta_1 = \frac{1}{2}\begin{vmatrix} x & y & 1 \\ 1 & 1 & 1 \\ -4 & 3 & 1 \end{vmatrix}$$

$$\Delta_2 = \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ -4 & 3 & 1 \\ -2 & -5 & 1 \end{vmatrix}$$

$\text{Given } \frac{\Delta_1}{\Delta_2} = \frac{4}{7} \Rightarrow \frac{-2x-5y+7}{36} = \frac{4}{7}$

$\Rightarrow 14x + 35y = -95 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$

$\text{equation of BC is } 4x + y = -13 \ldots\ldots\ldots (2)$

Solve eq (1) & (2)

$\text{Point } P\left(\frac{-20}{7}, \frac{-11}{7}\right)$

$\text{Here Point } Q\left(\frac{-1}{2}, 0\right) \text{ \& } R\left(\frac{1}{2}, 0\right) \text{So Area of } \Delta AQR = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$

Q. 19 If the circle $x^2 + y^2 - 2gx + 6y - 19c = 0, g, c \in \mathbb{R}$ passes through the point $(6, 1)$ and its centre lies on the line $x - 2cy = 8$, then the length of intercept made by the circle on $x - \text{axis}$ is

Option 1:

$\sqrt{11}$

Option 2:

4

Option 3:

3

Option 4:

$2\sqrt{23}$

Correct Answer:

$2\sqrt{23}$

Solution:

$\text{Given circle } x^2+y^2-29x+6y-19c=0$

$\text{passes through}(6,1)$

$12g+19c=43\cdots(1)$

Centre (g , -3) lies on given line

$\text{So, } g+6c=8\cdots(2)$

Solve equation (1) & (2)

$c=1 \ \& \ y=2$

$\text{equation of circle } x^2+y^2-4x+6y-19=0$

Length of intercept on x-axis

$=2\sqrt{g^2-c}=2\sqrt{23}$

Q. 20 Let a function $f:\mathbb{R}\to\mathbb{R}$ be defined as :

$$f(x)=\begin{cases}\int_0^x(5-|t-3|)dt, & x>4\\ x^2+bx & ,x\le 4\end{cases}$$

where $b\in\mathbb{R}$. If f is continuous at $x=4$, then which of the following statements is **NOT** true?

Option 1:

f is not differentiable at $x = 4$

Option 2:

$f'(3) + f'(5) = \frac{35}{4}$

Option 3:

f is increasing in $\left(-\infty, \frac{1}{8}\right) \cup (8, \infty)$

Option 4:

f has a local minima at $x = \frac{1}{8}$

Correct Answer:

f is increasing in $\left(-\infty, \frac{1}{8}\right) \cup (8, \infty)$

Solution:

Given $f(x) \begin{cases} \int_0^x (5 - |t-3|)dt, & x > 4 \\ x^2 + bx, & x \leq 4 \end{cases}$

$f(x)$ is continuous at $x = 4$

So $\lim\limits_{x \to 4^-} f(x) = \lim\limits_{x \to 4^+} f(x) = f(4)$

So $16 + 4b = \int_0^3 (2 - t)dt + \int_3^4 (8 - t)dt$

$\Rightarrow 16 + 4b = 15$

So $b = -\frac{1}{4}$ at $x = 4$

$LHD = 2x + b = \frac{31}{4}$

$RHD = 5 - |x - 3| = 4$

$LHD \neq RHD$

option (A) is true and $f'(3) + f'(5) = \frac{23}{4} + 3 = \frac{35}{4}$

option (B) is true

$\therefore f(x) = x^2 - \frac{x}{4}$ at $x \leq 4$

$f'(x) = 2x - \frac{1}{4}$

This function is not increasing.

In the interval in $x \in \left(-\infty, \frac{1}{8}\right)$

Option (c) is not true \\

This function f(x) is also yocal minina at $x = \frac{1}{8}$

Q. 21 For $k \in \mathbb{R}$, let the solutions of the equation $\cos\left(\sin^{-1}\left(x\cot\left(\tan^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)\right)\right) = k, 0 < |x| < \frac{1}{\sqrt{2}}$ be α and β, where the inverse trigonometric functions take only principal values. If the solutions of the equation $x^2 - bx - 5 = 0$ are $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ and $\frac{\alpha}{\beta}$, then $\frac{b}{k^2}$ is equal to_____.

Correct Answer:

12

Solution:

$\cos\left(\sin^{-1}x\right) = \cos\left(\cos^{-1}\sqrt{1-x^2}\right) = \sqrt{1-x^2}$

$\cos\left(\tan^{-1}\sqrt{1-x^2}\right) = \cot^{-x^{-1}}|\left(\sqrt{\frac{1}{\sqrt{1-x^2}}}\right) = \frac{1}{\sqrt{1-x^2}}$

$\Rightarrow \cos\left(\sin^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)\right) = \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}}$

$\Rightarrow \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}} = k$

$\Rightarrow 1 - 2x^2 = k^2\left(1-x^2\right)$

$\Rightarrow \left(k^2-2\right)x^2 = k^2 - 1$

$x^2 = \frac{k^2-1}{k^2-2}$

$$\alpha = \sqrt{\frac{k^2-1}{k^2-2}} \Rightarrow \alpha^2 = \frac{k^2-1}{k^2-2}$$

$$\beta = \sqrt{\frac{k^2-1}{k^2-2}} \Rightarrow \beta^2 = \frac{k^2-1}{k^2-2}$$

$$\frac{1}{\alpha^2} + \frac{1}{\beta^2} = 2\left(\frac{k^2-2}{k^2-1}\right) \ \& \ \frac{\alpha}{\beta} = -1$$

$$\text{Sum of roots } = \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{\alpha}{\beta} = b$$

$$\Rightarrow \frac{2(k^2-2)}{k^2+} - 1 = b \ldots (1)$$

$$\text{Poduct of roots} = \left(\frac{1}{\alpha^2} + 1 - \frac{1}{\beta^2}\right)\frac{\alpha}{\beta} = -5$$

$$\Rightarrow \frac{2(k^2-2)}{k^2-1}(-1) = -5$$

$$\Rightarrow 2k^2 - 4 = 5k^2 - 5$$

$$\Rightarrow 3k^2 = 1 \Rightarrow k^2 = \frac{1}{3} \cdots, \text{ put in}(1)$$

$$\Rightarrow b = \frac{2(k^2-2)}{k^2-1} - 1 = 5 - 1 = 4$$

$$\frac{b}{k^2} = \frac{4}{\frac{1}{3}} = 12$$

Q. 22 The mean and variance of 10 observations were calculated as 15 and 15 respectively by a student who took by mistake 25 instead of 15 for one observation. Then, the correct standard deviation is_____.

Correct Answer:
2

Solution:

$$n = 10, \bar{x} = \frac{\sum x_i}{10} = 15$$

$$\sigma^2 = \frac{\sum x_1^2}{10} - (\bar{x})^2 = 15$$

$$\Rightarrow \sum_{i=1}^{10} x_i = 150$$

$$\Rightarrow \sum_{i=1}^{9} x_i + 25 = 150$$

$$\Rightarrow \sum_{i=1}^{9} x_i = 125$$

$$\Rightarrow \sum_{j=1}^{9} x_i + 15 = 140$$

$$\text{Aatual mean } = \frac{140}{10} = 14 = \bar{x}_{new}$$

$$\sum_{i=1}^{9} \frac{x_i^2 + 25^2 - 15^2}{10} = 15$$

$$\Rightarrow \sum_{i=1}^{9} x_i^2 + 625 = 2400$$

$$\sum_{i=1}^{9} x_i^2 = 1775$$

$$\sum_{i=1}^{9} x_i^2 + 15^2 = 2000 = \left(\sum x_i^2\right)_{actual}$$

$$\sigma^2_{actual} = \frac{\left(\sum x_i^2\right)_{actual} - (\bar{x}_{new})^2}{10}$$

$$= \frac{2000}{10} - 14^2$$

$$= 200 - 196 = 4$$

$$(S, D)_{actual} = \sigma = 2$$

Q. 23 Let the line $\frac{x-3}{7} = \frac{y-2}{-1} = \frac{z-3}{-4}$ intersect the plane containing the lines $\frac{x-4}{1} = \frac{y+1}{-2} = \frac{z}{1}$ and $4ax - y + 5z - 7a = 0 = 2x - 5y - z - 3, a \in \mathbb{R}$ at the point $P(\alpha, \beta, \gamma)$. Then the value of $\alpha + \beta + \gamma$ equals_____.

Correct Answer:
12

Solution:
Equation of plane
$4ax - y + 5z - 7a + \lambda(2x - 5y - z - 3) = 0 \text{this satisfy } (4, -1, 0)$
$16a + 1 - 7a + \lambda(8 + 5 - 3) = 0$
$9a + 1 + 10d = 0 \cdots (1)$
Normal vector of the plane A is $(4a + 2\lambda, -1 - 5\lambda, 5 - \lambda)$
vector along the line which contained the plane A is $\hat{i} - 2j + k$
$\therefore 4a + 2\lambda + 2 + 10\lambda + 5 - \lambda = 0$
$11\lambda + 4a + 7 = 0 \cdots (a)$
Solve (1) and (2) to get $a = 1, \lambda = -1$
Now equation of plane

$x + 2y + 3z - 2 = 0$

Let the point in the line $\frac{x-3}{7} = \frac{y-2}{-1} = \frac{z-3}{-4} = t$
is $(7t + 3, -t + 2, -4t + 3)$ satisfy the equation of plane A
$7t + 3 - 2t + 4 + 9 - 12t - 2 = 0$
$t = 2$
So $\alpha + \beta + \gamma = 2t + 8 = 12$

Q. 24 An ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the vertices of the hyperbola $H : \frac{x^2}{49} - \frac{y^2}{64} = -1$. Let the major and minor axes of the ellipse E coincide with the transverse and conjugate axes of the hyperbola H, respectively. Let the product of the eccentricities of E and H be $\frac{1}{2}$. If L is the length of the latus rectum of the ellipse E, then the value of $113l$ is equal to_____.

Correct Answer: 1552

Solution:
$\text{Hyp} : \frac{y^2}{64} - \frac{x^2}{49} = 1$

An ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

passes through the vertices of the Hyperbola

$$H : \frac{x^2}{49} - \frac{y^2}{64} = -1$$

so $b^2 = 64$

$$e_H = \sqrt{1 + \frac{a^2}{b^2}} = \sqrt{1 + \frac{49}{64}}$$

elipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$e_E = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{1 - \frac{a^2}{64}}$$

$$b = 8, \sqrt{\frac{1 - a^2}{64}} \times \frac{\sqrt{113}}{8} = \frac{1}{2} \Rightarrow \sqrt{64 - a^2} \times \sqrt{113} = 32$$

$$(64 - a^2) = \frac{32^2}{113}$$

$$\Rightarrow a^2 = 64 - \frac{32^2}{113}$$

$$l = \frac{2a^2}{b} = \frac{2}{8}\left(64 - \frac{32^2}{113}\right) = \frac{1552}{113}$$

$$113\, l = 1552$$

Q. 25 Let $y = y(x)$ be the solution curve of the differential equation $\sin\left(2x^2\right) \log_e\left(\tan x^2\right) dy + \left(4xy - 4\sqrt{2}x \sin\left(x^2 - \frac{\pi}{4}\right)\right) dx = 0, 0 < x < \sqrt{\frac{\pi}{2}}$, which passes through the point $\left(\sqrt{\frac{\pi}{6}}, 1\right)$. Then $\left|y\left(\sqrt{\frac{\pi}{3}}\right)\right|$ is equal to _____.

Correct Answer:
1

Solution:

$$\sin\left(2x^2\right)\ln\left(\tan x^2\right)dy+\left(4xy-4\sqrt{2}x\sin\left(x^2-\frac{\pi}{4}\right)\right)dx=0$$

$$\ln\left(\tan x^2\right)dy+\frac{4xy\,dx}{\sin\left(2x^2\right)}-\frac{4\sqrt{2}x\sin\left(x^2-\frac{\pi}{4}\right)}{\sin\left(2x^2\right)}dx=0$$

$$d\left(y\cdot\ln\left(\tan x^2\right)\right)-4\sqrt{2}x\frac{\left(\sin x^2-\cos x^2\right)}{\sqrt{2}\times 2\sin x^2\cos x^2}dx=0$$

$$d\left(y\ln\left(\tan x^2\right)\right)-\frac{4x\left(\sin x^2-\cos x^2\right)}{\left(\sin x^2+\cos x^2\right)^2-1}dx=0$$

$$\Rightarrow\int d\left(y\ln\left(\tan x^2\right)\right)+2\int\frac{dt}{t^2-1}=\int 0$$

$$\Rightarrow 4\ln\left(\tan x^2\right)+2\cdot\frac{1}{2}\ln\left|\frac{t+1}{t+1}\right)=c$$

$$y\ln\left(\tan x^2\right)+\ln\left(\frac{\sin x^2+\cos x^2-1}{\sin x^2+\cos x^2+1}\right)=c$$

$$\text{Put } y=1 \text{ and } x=\sqrt{\frac{\pi}{6}}$$

$$1\ln\left(\frac{1}{\sqrt{3}}\right)+\ln\frac{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}-1\right)}{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}+1\right)}=c$$

$$x=\sqrt{\frac{\pi}{3}}\Rightarrow y(\ln\sqrt{3})+\ln\frac{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}-1\right)}{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}+1\right)}$$

$$=\ln\left(\frac{1}{\sqrt{3}}\right)+\ln\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$$

$$y(\ln\sqrt{3})=\ln\left(\frac{1}{\sqrt{3}}\right)$$

$$\Rightarrow y=-1\Rightarrow|4|-1\quad\text{Ans }|y|=1$$

Q. 26 Let m and n be the number of points on the curve $y^5-9xy+2x=0$, where the tangents to the curve are parallel to x-axis and y-axis, respectively. Then the value of $m+n$ equals_____.

Correct Answer: 2

Solution:

$$y^5 - 9xy + 2x = 0$$

$$5y^4\frac{dy}{dx} - 9x\frac{dy}{dx} - 9y + 2 = 0$$

$$\frac{dy}{dx}\left(5y^4 - 9x\right) = 9y - 2$$

$$\frac{dy}{dx} = \frac{9y - 2}{5y^4 - 9x} = 0 \text{ (for horizontal tangent)}$$

$$y = \frac{2}{9} \Rightarrow \text{which does rot satisfy the original equation} \Rightarrow m = 0$$

$$\text{Now } 5y^4 - 9x = 0 \text{(for vertical tangent)}$$

$$5y^4(9y - 2) - 9y^5 = 0$$

$$y^4[45y - 10 - 9y] = 0$$

$$y = 0 \text{ or } 36y = 10$$

$$y = \frac{5}{18}$$

$$y = 0 \Rightarrow x = 0 \& y = \frac{5}{18} \Rightarrow x(0,0)\left(x, \frac{5}{18}\right)$$

$$N = 2$$

$$M + N = 0 + 2 = 2$$

Q. 27 Let $f(x) = 2x^2 - x - 1$ and $S = \{n \in \mathbb{Z} : |f(n)| \leq 800\}$. Then, the value of $\sum_{n \in S} f(n)$ is equal to _____.

Correct Answer:
10620

Solution:

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$$f(x) = 2x^2 - x - 1$$
$$|f(x)| \leq 800$$
$$2n^2 - n - 801 \leq 0$$
$$n^2 - \frac{n}{2} - \frac{201}{2} \leq 0$$
$$\left(h - \frac{1}{4}\right)^2 - \frac{801}{2} - \frac{1}{16} \leq 0$$
$$\left(n - \frac{1}{4}\right)^2 - \frac{6409}{16} \leq 0$$
$$\left(n - \frac{1}{4} - \frac{\sqrt{6409}}{4}\right)\left(n - \frac{1}{4} + \frac{\sqrt{6409}}{16}\right) \leq 0$$
$$\frac{1 - \sqrt{6409}}{4} \leq n \leq \frac{1 + \sqrt{6409}}{4}$$
$$n = \{-19, -18, -17, \ldots 0, \ldots, 2, \ldots 20\}$$
$$\sum_{n \in S} f(x) = \sum \left(2x^2 - x - 1\right)$$
$$= 2\left[19^2 + 18^2 + \cdots + 1^2 + 1^2 + 2^2 + \cdots + 20^2\right]$$
$$= 4\left[1^2 + 2^2 + \cdots\cdots + 19^2\right] + 2\left[20^2\right] - 20 - 40$$
$$= \frac{4 \times 19 \times 20 \times (2 \times 19 + 1)}{6} + 2 \times 400 - 60$$
$$= \frac{4 \times 19 \times 20 \times 39}{6} + 800 - 60 - 9880 + 800 - 60$$
$$= 10620$$

Q. 28 Let S be the set containing all 3×3 matrices with entries from $\{-1, 0, 1\}$. The total number of matrices $A \in S$ such that the sum of all the diagonal elements of $A^T A$ is 6 is _____.

Correct Answer:
5376

Solution:
$$\mathrm{Tr}_r\left(AA^T\right) = 6$$

$$AA^{\top} = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

Now given $a^2 + d^2 + y^2 + b^2 + e^2 + h^2 + c^2 + f^2 + i^2 = 6$

$= {}^{9}c_3 \times 2^6$

$= 5376$

Q. 29 If the length of the latus rectum of the ellipse $x^2 + 4y^2 + 2x + 8y - \lambda = 0$ is 4, and L is the length of its major axis, then $\lambda + l$ is equal to _____.

Correct Answer: 75

Solution:

$\lambda + l = 75$

$x^2 + 4y^2 + 2x + 8y - \lambda = 0$

$$\frac{(x+1)^2}{\lambda+5} + \frac{(y+1)^2}{\frac{\lambda+5}{4}} = 1$$

$$\because \frac{2b^2}{a} = 4$$

$$\frac{2(\lambda+5)}{4} = 4(\sqrt{\lambda+5})$$

$\Rightarrow \lambda = 59$

$\lambda \neq -5$

$l = 2a = 2\sqrt{\lambda+5} = 2\sqrt{65} = 16$

$\Rightarrow \lambda + l = 59 + 16 = 75$

Q. 30 Let $S = \{z \in \mathbb{C} : z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} (\mathrm{Re}(z) + \mathrm{Im}(z))$ is equal to _____.

Correct Answer:

-

Solution:

$$s = \{z \in c; z^2 + \bar{z} = 0\}$$

$$\det z = x + iy$$

$$z^2 = x^2 - y^2 + 2ixy$$

$$\bar{z} = x - iy$$

$$z^2 + \tilde{z} = x^2 - y^2 + x + i(2xy - y) = 0$$

$$\Rightarrow x^2 + x - y^2 = 0 \ \& \ 2xy - y = 0$$

$$y = 0 \text{ or } x = \frac{1}{2}$$

$$\text{if } y = 0, x = 0, -1$$

$$\text{if } x = \frac{1}{2}, y = \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}$$

$$\sum_{z \epsilon s} (\text{Re}(z) + \text{In}(z)) = \left(0 - 1 + \frac{1}{2} + \frac{1}{2}\right) + 0 + 0 + \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}\Bigg)$$

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27 July - Evening Shift

Physics

Q. 1 An expression of energy density is given by $u = \frac{\alpha}{\beta}\sin\left(\frac{\alpha x}{kt}\right)$, where α, β are constants, x is displacement, k is Boltzmann constant and t is the temperature. The dimensions of β will be:

Option 1:
$[ML^2\ T^{-2}\theta^{-1}]$

Option 2:
$[M^0\ L^2\ T^{-2}]$

Option 3:
$[M^0\ L^0\ T^0]$

Option 4:
$[M^0\ L^2\ T^0]$

Correct Answer:
$[M^0\ L^2\ T^0]$

Solution:

$$u = \frac{\alpha}{\beta}\sin\left(\frac{\alpha x}{kt}\right)$$

$$[x] = [L']$$

$$[kt] = [\text{ Energy }] = [M^1L^2T^{-2}]$$

$$\left[\frac{\alpha x}{kt}\right] = [M^\circ L^\circ T^0]$$

$$\therefore [\alpha] = [M^1L^1T^{-2}] \rightarrow (1)$$

$$[u] = \left[\frac{\alpha}{\beta}\right] = [M^1L^{-1}T^{-2}]$$

$$[\beta] = [L^2] \rightarrow (2)$$

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Hence 4 is correct option.

Q. 2 A body of mass $10\ \mathrm{kg}$ is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point $(20, 10)$. If T is the time of flight, then its momentum vector, at time $\mathrm{t} = \frac{\mathrm{T}}{\sqrt{2}}$, is_____.

[Take $\mathrm{g} = 10\ \mathrm{m/s^2}$]

Option 1:
$100\hat{\mathrm{i}} + (100\sqrt{2} - 200)\hat{\mathrm{j}}$

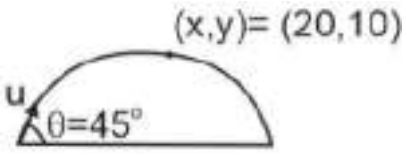
Option 2:
$100\sqrt{2}\hat{\mathrm{i}} + (100 - 200\sqrt{2})\hat{\mathrm{j}}$

Option 3:
$100\hat{\mathrm{i}} + (100 - 200\sqrt{2})\hat{\mathrm{j}}$

Option 4:
$100\sqrt{2}\hat{\mathrm{i}} + (100\sqrt{2} - 200)\hat{\mathrm{j}}$

Correct Answer:
$100\hat{\mathrm{i}} + (100\sqrt{2} - 200)\hat{\mathrm{j}}$

Solution:



$\mathrm{m} = 10\mathrm{kg}$

We know that, the equation of trajectory is

$$y = x\tan\theta\left(1 - \frac{x}{R}\right)$$

$$10 = 20\left(\tan 45^\circ\right)\left(1 - \frac{20}{R}\right)$$

$$\frac{1}{2} = 1 - \frac{20}{R}$$

$$\frac{1}{2} = \frac{20}{R}$$

$$R = 40 \rightarrow (1)$$

Also we know that,

$$R = \frac{u^2 \sin 2\theta}{g}$$

$$40 = \frac{u^2 \sin 90^\circ}{10}$$

$$u^2 = 400$$

$$u = 20 \text{ m/s} \rightarrow (2)$$

$$\text{At } t = \frac{T}{\sqrt{2}}$$

$$\bar{V} = \bar{v}_x + \bar{V}_y$$

$$= u\cos\theta(\hat{i}) + (u\sin\theta - gt)\hat{j}$$

$$= 10\sqrt{2}\hat{i} + \left(10\sqrt{2} - 10 \times \frac{T}{\sqrt{2}}\right)\hat{j}$$

$$\text{But } T = \frac{2u\sin\theta}{g} = \frac{2 \times 20 \times}{10}\frac{1}{\sqrt{2}}$$

$$T = 2\sqrt{2}$$

$$\bar{v} = 10\sqrt{2}\hat{i} + (10\sqrt{2} - 20)\hat{j}$$

$\therefore$ Momentum vector $= \bar{P} = m\bar{v}$

$$\bar{P} = 100\sqrt{2}\hat{i} + \left(100\sqrt{2} - 200\right)\hat{j}$$

Hence 1 is correct option.

Q. 3 A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ. The magnitude of the contact force will be:

Option 1:
Mg

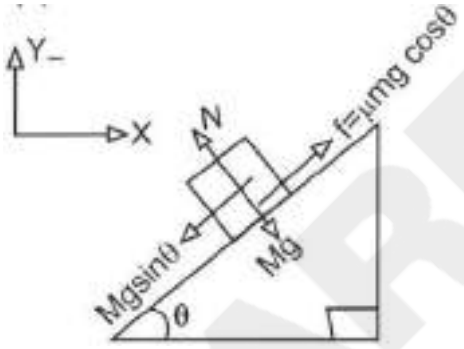
Option 2:
$\mathrm{Mg}\cos\theta$

Option 3:
$\sqrt{\mathrm{Mg}\sin\theta + \mathrm{Mg}\cos\theta}$

Option 4:
$\mathrm{Mg}\sin\theta\sqrt{1+\mu}$

Correct Answer:
Mg

Solution:



For block moving at constant velocity

$$\sum \mathrm{F_x} = 0$$

$$-\mathrm{mg}\sin\Theta + \mu\mathrm{Mg}\cos\Theta = 0$$

$$\tan\Theta = \mu$$

$$\mathrm{f} = \mathrm{Mg}\sin\Theta = \mu\mathrm{Mg}\cos\Theta \rightarrow (1)$$

$$\text{Contact force} = \sqrt{\mathrm{f}^2 + \mathrm{N}^2}$$

$$\sqrt{(\mathrm{Mg}\sin\Theta)^2 + (\mathrm{Mg}\cos\Theta)^2}$$

$$\text{Contact force} = \text{Mg}$$

Hence 1 is correct option.

Q. 4 A block $'\text{A}'$ takes $2\ \text{s}$ to slide down a frictionless incline of 30° and length $'\text{l}'$, kept inside a lift going up with uniform velocity $'\text{v}'$. If the incline is changed to 45°, the time taken by the block, to slide down the incline, will be approximately :

Option 1:
$2.66\ \text{s}$

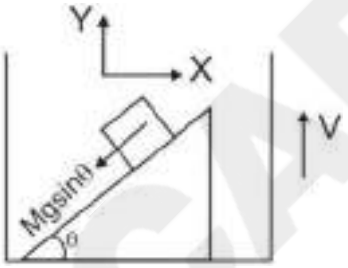
Option 2:
$0.83\ \text{s}$

Option 3:
$1.68\ \text{s}$

Option 4:
$0.70\ \text{s}$

Correct Answer:
$1.68\ \text{s}$

Solution:
For a lift moving with constant velocity there is no pseudo force acting on the block placed inside it due to lifts motion.



from the equation of motion ,

$$\text{S} = \text{ut} + \frac{1}{2}\text{at}^2$$

$$-\ell = 0 + \frac{1}{2}\left(-\text{g}\sin\Theta\right)\text{t}^2$$

$$\text{t} = \sqrt{\frac{2\ell}{\text{g}\sin\Theta}}$$

The above equation gives the time taken by block to slide down a frictionless incline of angle Θ with the horizontal and length ℓ

Case (1) for $\Theta = 30^\circ, \mathrm{t} = 2\mathrm{s}$

$$\mathrm{t}_1 = 2 = \sqrt{\frac{2\ell}{\mathrm{g}\sin 30^\circ}}$$

Case (2)

$$\mathrm{t}_2 = \sqrt{\frac{2\ell}{\mathrm{g}\sin 45^\circ}}$$

$$\frac{\mathrm{t}_1}{\mathrm{t}_2} = \sqrt{\frac{\sin 45^\circ}{\sin 30^\circ}} = \sqrt{\frac{1/\sqrt{2}}{1/2}}$$

$$\frac{2}{\mathrm{t}_2} = 2^{1/4}$$

$$\mathrm{t}_2 = 2^{3/4}\mathrm{s}$$

$$\therefore \mathrm{t}_2 = 1.68\,\mathrm{s}$$

Hence 3 is correct option

Q. 5 The velocity of the bullet becomes one third after it penetrates $4\ \mathrm{cm}$ in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling at $(4+x)\mathrm{cm}$ inside the block. The value of x is :

Option 1:
2.0

Option 2:
1.0

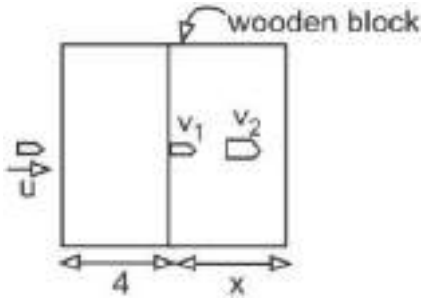
Option 3:
0.5

Option 4:
1.5

Correct Answer:

0.5

Solution:



$$V_1 = \frac{u}{3}\,(\text{Given})$$

$$V_2 = 0$$

* For constant retardation by the equations of motion

$$V_1^2 = u^2 + 2as$$

$$\frac{u^2}{9} - u^2 = 2a\left(4 \times 10^{-2}\right)$$

$$-\frac{4u^2}{9 \times 4 \times 10^{-2}} = a$$

$$a = \frac{-100u^2}{9} \rightarrow (1)$$

$$v_2^2 = u^2 + 2as$$

$$0 = u^2 + 2\left(-\frac{100u^2}{9} \times (4+x) \times 10^{-2}\right)$$

$$u^2 = \frac{2u^2(4+x)}{9}$$

$$x = 0.5\ \text{cm}$$

Hence 3 is correct option

Q. 6 A body of mass m is projected with velocity λv_e in vertically upward direction from the surface of the earth into space. It is given that v_e is escape velocity and $\lambda < 1$. If air resistance is considered to be negligible, then the maximum height from the centre of earth, to which the body can go, will be :
(R : radius of earth)

Option 1:
$\frac{\mathrm{R}}{1+\lambda^2}$

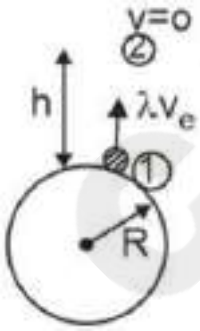
Option 2:
$\frac{\mathrm{R}}{1-\lambda^2}$

Option 3:
$\frac{\mathrm{R}}{1-\lambda}$

Option 4:
$\frac{\lambda^2\mathrm{R}}{1-\lambda^2}$

Correct Answer:
$\frac{\mathrm{R}}{1-\lambda^2}$

Solution:



Applying energy consevation between pt.(1) and (2)

$$\mathrm{TE}_1 = \mathrm{TE}_2$$

$$\frac{-\mathrm{GMm}}{\mathrm{R}} + \frac{1}{2}\mathrm{m}\left(\lambda \mathrm{V}_e\right)^2 = \frac{-\mathrm{GMm}}{(\mathrm{R}+\mathrm{h})} + 0$$

$$\because V_e = \sqrt{\frac{2GM}{R}}$$

$$\frac{-GMm}{R} + \lambda^2 \frac{GMm}{R} = \frac{-GMm}{R+h}$$

The height from the centre of earth is $(R+h)$

$$\frac{1}{R+h} = \frac{1}{R} - \frac{\lambda^2}{R}$$

$$\frac{1}{R+h} = \frac{1-\lambda^2}{R}$$

$$\Rightarrow \frac{1}{R+h} = \frac{R}{1-\lambda^2}$$

Hence 2 is correct option.

Q. 7 A steel wire of length $3.2\ \text{m}\ (Y_s = 2.0 \times 10^{11} \text{Nm}^{-2})$and a copper wire of length $4.4\ \text{m}\ (Y_c = 1.1 \times 10^{11} \text{Nm}^{-2})$, both of radius $1.4\ \text{mm}$ are connected end to end. When stretched by a load, the net elongation is found to be $1.4\ \text{mm}$. The load applied, in Newton, will be: $(\ \text{Given}\ \pi = \frac{22}{7})$

Option 1:
360

Option 2:
180

Option 3:
1080

Option 4:
154

Correct Answer:
154

Solution:
$\ell_s = 3.2\ \text{m}$

$Y_s = 2 \times 10^{11}\ \text{N/m}^2$

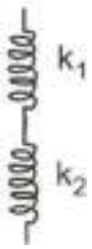
$\ell_c = 4.4$ m

$Y_c = 1.1 \times 10^{11}$ N/m^2

$r = 1.4$ mm $= 1.4 \times 10^{-3}$ m

$\Delta\ell = 1.4$ mm $= 1.4 \times 10^{-3}$ m

$$F = \left(\frac{YA}{\ell}\right)\Delta\ell$$



$$K = \left(\frac{YA}{\ell}\right)\Delta\ell$$

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2}$$

$$\frac{1}{K_{eq}} = \frac{\ell_s}{Y_s A_s} + \frac{\ell_c}{Y_c A_C}$$

$$\frac{1}{K_{eq}} = \frac{3.2}{A \times 2 \times 10^{11}} + \frac{4.4}{A \times 1.1 \times 10^{11}}$$

taking $A_s = A_c = A$

$$\frac{1}{K_{eq}} = \frac{1.6 \times 10^{-11}}{A} + \frac{4 \times 10^{-11}}{A}$$

$$K_{eq} = \frac{A}{5.6 \times 10^{-11}}$$

$$K_{eq} = \frac{A \times 10^{11}}{5.6}$$

$$F = K_{eq}(\Delta\ell)$$

$$= \frac{A \times 10^{11}}{5.6} \times 1.4 \times 10^{-3}$$

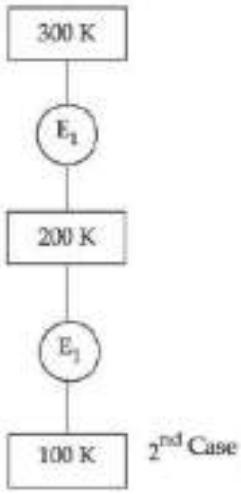
$$= \frac{3.14 \times (1.4 \times 10^{-3})^3 \times 10^{11}}{5.6}$$

$$F = \frac{22 \times (1.4)^2 \times 10^2}{7 \times 4}$$

$$F = 154 \text{ N}$$

Hence 4 is correct option

Q. 8 In 1^{st} case, Carnot engine operates between temperatures $300 \text{ K and } 100 \text{ K}$. In 2^{nd} case, as shown in the figure, a combination of two engines is used. The efficiency of this combination (in 2^{nd} case) will be :



Option 1:
Same as the 1^{st} case

Option 2:
always greater then the 1^{st} case.

Option 3:
always less than the 1^{st} case.

Option 4:
may increase or decrease with respact to the 1^{st} case.

Correct Answer:
always less than the 1^{st} case.

Solution:
Case 1

$$\eta_1 = 1 - \frac{T_1}{T_2} = 1 - \frac{100}{300}$$

$$\eta_1 = \frac{2}{3} \rightarrow (1)$$

Case 2

The efficiency of the combination

$$\eta_2 = \eta' . \eta''$$

$$= \left(1 - \frac{200}{300}\right)\left(1 - \frac{100}{200}\right)$$

$$\eta_2 = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$$

The efficiency in the 2nd case is less than the 1st case

Hence 3 is correct option

Q. 9 Which statements are correct about degrees of freedom?
(A) A molecule with n degrees of freedom has n^2 different ways of storing energy.
(B) Each degree of freedom is associated with $\frac{1}{2}\mathrm{RT}$ average energy per mole.
(C) A monatomic gas molecule has 1 rotational degree of freedom where as diatomic molecule has 2 rotational degrees of freedom.
(D) $\mathrm{CH_4}$ has a total of 6 degrees of freedom.
Choose the correct answer from the options given below :

Option 1:
(B) and (C) only

Option 2:
(B) and (D) only

Option 3:
(A) and (B) only

Option 4:
(C) and (D) only

Correct Answer:
(B) and (D) only

Solution:
$\rightarrow$ A molecule with n degrees of freedom has n different ways of storing energy.
$\rightarrow$ Energy corresponding to each degree of freedom is $\frac{1}{2}\mathrm{RT}$ per mole
$\rightarrow$ Monoatomic gas has just three translational degrees and no rotational degree.
$\rightarrow$ Diatomic gas has thee translational and two rotational degree if freedom
$\rightarrow$ $\mathrm{CH_4}$ has three translational and three rotational degree of freedom

Hence 2 is correct option.

Q. 10 A charge of $4\mu\mathrm{C}$ is to be divided into two. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be :

Option 1:
$1\mu\mathrm{C}$ and $3\mu\mathrm{C}$

Option 2:
$2\mu\mathrm{C}$ and $2\mu\mathrm{C}$

Option 3:
0 and $4\mu\mathrm{C}$

Option 4:
$1.5\mu\mathrm{C}$ and $2.5\mu\mathrm{C}$

Correct Answer:
$2\mu\mathrm{C}$ and $2\mu\mathrm{C}$

Solution:

$$\mathrm{F}=\frac{\mathrm{KQ}\left(4-\mathrm{Q}\right)}{\mathrm{r}^2}$$

For maximum force,

$$\frac{\mathrm{dF}}{\mathrm{dr}}=0$$

$$Q = \frac{4}{2} = 2\mu c$$

The magnitude of the divided charges so that the force between them is maximum,will be $2\mu c \text{ and } 2\mu c$

Hence 2 is correct option.

Q. 11 (A) The drift velocity of electrons decreases with the increase in the temperature of conductor.
(B) The drift velocity is inversely proportional to the area of cross-section of given conductor.
(C) The drift velocity does not depend on the applied potential difference to the conductor.
(D) The drift velocity of electron is inversely proportional to the length of the conductor.
(E) The drift velocity increases with the increase in the temperature of conductor.
Choose the correct answer from the options given below :

Option 1:
(A) and (B) only

Option 2:
(A) and (D) only

Option 3:
(B) and (E) only

Option 4:
(B) and (C) only

Correct Answer:
(A) and (D) only

Solution:

$$\text{Drift velocity} = V_d = \frac{eE\tau}{m}$$

$$\text{Also, } E = \frac{\Delta v}{\ell}$$

With increase in temperature, relaxation time decreases. So the drift velocity also decreases $\rightarrow$ A & D statements are correct

Hence 2 is correct option.

Q. 12 A compass needle of oscillation magnetometer oscillates 20 times per minute at a place P of dip 30°. The number of oscillations per minute become 10 at another place Q of 60° dip. The ratio of the total magnetic field at the two places $(B_Q : B_P)$ is :

Option 1:
$\sqrt{3} : 4$

Option 2:
$4 : \sqrt{3}$

Option 3:
$\sqrt{3} : 2$

Option 4:
$2 : \sqrt{3}$

Correct Answer:
$\sqrt{3} : 4$

Solution:

$\delta_1 = 30^\circ \quad (\text{At pt.P})$

$\delta_2 = 60^\circ \quad (\text{At pt.Q})$

$$T = 2\pi\sqrt{\frac{I}{MB}}$$

$$f = \frac{1}{2\pi}\sqrt{\frac{MB}{I}}$$

$$\frac{f_1}{f_2} = \frac{20}{10} = \sqrt{\frac{(BH)_1}{(BH)_2}} = \sqrt{\frac{B_1 \cos 30^\circ}{B_2 \cos 60^\circ}}$$

$$\frac{2}{1}=\sqrt{\frac{B_1}{B_2}}\times\frac{\sqrt{3}}{2}\times\frac{2}{1}$$

$$\frac{B_P}{B_Q}=\frac{B_1}{B_2}=\frac{4}{\sqrt{3}}$$

Hence 1 is correct option.

Q. 13 A cyclotron is used to accelerate protons. If the operating magnetic field is $1.0\ \mathrm{T}$ and the radius of the cyclotron 'dees' is $60\ \mathrm{cm}$, the kinetic energy of the accelerated protons in MeV will be :
[use $m_p = 1.6\times10^{-27}\ \mathrm{kg}, e = 1.6\times10^{-19}\mathrm{C}$]

Option 1:
12

Option 2:
18

Option 3:
16

Option 4:
32

Correct Answer:
18

Solution:

$B = 1T$

$r = 60cm$

$$r=\frac{mv}{qB}$$

$$P = mv = qBr$$

Kinetic energy of the accelerated protons is

$$KE=\frac{p^2}{2m}=\frac{q^2B^2r^2}{2m}$$

$$=\frac{(1.6\times10^{-19})^2\times(1)^2\times(0.36)}{2\times1.6\times10^{-27}}$$

$$= \frac{1.6 \times 0.36 \times 10^{-38}}{2 \times 10^{-27}}$$

$$\mathrm{KE} = \frac{1.6 \times 0.18 \times 10^{-11}}{1.6 \times 10^{-13}} \mathrm{MeV}$$

$$\mathrm{KE} = 18 \mathrm{MeV}$$

Hence 2 is correct option

Q. 14 A series LCR circuit has $\mathrm{L} = 0.01\mathrm{H}, \mathrm{R} = 10\Omega$ and $\mathrm{C} = 1\mu\mathrm{F}$ and it is connected to ac voltage of amplitude $(\mathrm{V_m})\ 50\ \mathrm{V}$. At frequency 60 % lower than resonant frequency, the amplitude of current will be approximately :

Option 1:
$466\ \mathrm{mA}$

Option 2:
$312\ \mathrm{mA}$

Option 3:
$238\ \mathrm{mA}$

Option 4:
$196\ \mathrm{mA}$

Correct Answer:
$466\ \mathrm{mA}$

Solution:

$$\mathrm{L} = 0.01\mathrm{H}$$

$$\mathrm{R} = 10\Omega$$

$$\mathrm{C} = 10^{-6}\mathrm{F}$$

$$\mathrm{V_m} = 50\mathrm{v}$$

$$\mathrm{f} = \left(\frac{60}{100}\right) \times \mathrm{f_{resonant}}$$

$$= \frac{3}{5} \times \frac{1}{2\pi\sqrt{\mathrm{LC}}}$$

$$f = \frac{3 \times 10^4}{10\pi} \text{ or } \omega = 2\pi f$$

$$\omega = \frac{3}{5} \times 10^4$$

$$X_L = \omega L = \frac{300}{5} = 60$$

$$x_c = \frac{1}{\omega c} = \frac{}{\frac{3}{5} \times 10^4} \times 10^{-6}$$

$$x_c = \frac{500}{3}$$

$$z = \sqrt{R^2 + (x_L - x_C)^2}$$

$$= \sqrt{10^2 + \left(\frac{320}{3}\right)^2}$$

$$= 10\sqrt{1 + \left(\frac{32}{3}\right)^2}$$

$$z = \frac{10}{3} \times \sqrt{1033}$$

$$I_m = \frac{V_m}{z} = \frac{50}{\frac{10}{3} \times \sqrt{1033}}$$

$$= 466 \text{ mA}$$

Hence 1 is correct option

Q. 15 Identify the correct statements from the following descriptions of various properties of electromagnetic waves.
(A) In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.
(B) The energy in electromagnetic wave is divided equally between electric and magnetic fields.
(C) Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.
(D) The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.
(E) The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.
Choose the most appropriate answer from the options given below :

Option 1:
(D) only

Option 2:
(B) and (D) only

Option 3:
(B),(C) and (E) only

Option 4:
(A), (B) and (E) only

Correct Answer:
(B) and (D) only

Solution:
$\rightarrow$ Electric and magnetic field are always perpendicular to each offer
$\rightarrow$ Energy in electromagnetic wave is divided equally between electric and magnetic fields.

$$\text{Energy density} = \frac{1}{2}\Sigma_0 E_0^2 = \frac{1}{2}\frac{B_0^2}{\mu_0}$$

$\rightarrow \frac{E_0}{B_0} = C$

The ratio of amplitude of Electric field and magnetic field is equal to speed of light.
$\rightarrow$ The direction of propagation of wave is mutually perpendicular electric field and magnetic field

Hence 2 is correct option

Q. 16 Two coherent sources of light interfere. The intensity ratio of two sources is 1: 4. For this interference pattern if the value of $\frac{I_{max}+I_{min}}{I_{max}-I_{min}}$ is equal to $\frac{2\alpha+1}{\beta+3}$, then $\frac{\alpha}{\beta}$ will be :

Option 1:
1.5

Option 2:
2

Option 3:
0.5

Option 4:
1

Correct Answer:
2

Solution:

$\frac{I_1}{I_2} = \frac{1}{4}$

$\frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2} = \frac{(3)^2}{(1)^2}$

$\frac{I_{max}}{I_{min}} = \frac{9}{1}$

$\frac{I_{max}+I_{min}}{I_{max}-I_{min}} = \frac{10}{8} = \frac{5}{4} = \frac{2\alpha+1}{\beta+3}$

$2\alpha + 1 = 5$

$\alpha = 2$

$\beta = 1$

$\therefore \frac{\alpha}{\beta} = 2$

Hence 2 is correct option

Q. 17 With reference to the observations in the photo-electric effect, identify the correct statements from below:

(A) The square of the maximum velocity of photoelectrons varies linearly with the frequency of incident light.

(B) The value of saturation current increases on moving the source of light away from the metal surface.
(C) The maximum kinetic energy of photoelectrons decreases on decreasing the power of LED (Light emitting diode) source of light.
(D) The immediate emission of photo-electrons out of metal surface can not be explained by particle nature of light/electromagnetic waves.
(E) Existence of threshold wavelength can not be explained by wave nature of light/electromagnetic waves.

Choose the correct answer from the options given below.

Option 1:
(A) and (B) only

Option 2:
(A) and (E) only

Option 3:
(C) and (E) only

Option 4:
(D) and (E) only

Correct Answer:
(A) and (E) only

Solution:

$\because \frac{1}{2}mv_m^2 = hv - \phi$

$\Rightarrow v_m^2$ varies linearly with frequency.

And, threshold wavelength can be explained by the particle nature of light.

Q. 18 The activity of a radioactive material is 6.4×10^{-4} curie. Its half life is 5 days. The activity will become 5×10^{-6} curie after :

Option 1:
7 days

Option 2:
15 days

Option 3:
25 days

Option 4:
35 days

Correct Answer:
35 days

Solution:
The activity at time t

$$A_t = A_0\, 2^{-t/T_{1/2}}$$

OR

$$A_t = A_0\, \epsilon^{-\lambda t}$$

$$A_t = 5 \times 10^{-6} = 6.4 \times 10^{-4} 2^{-t/5}$$

$$\frac{5}{6.4} \times 10^{-2} = 2^{-t/5}$$

$$\frac{1}{128} = \frac{1}{2^{t/5}}$$

$$t/5 = 7$$

$$t = 35 \text{ days}$$

The activity will become 5×10^{-6} curie after 35 days

Hence 4 is correct option

Q. 19 For a constant collector-emitter voltage of $8\ \text{V}$, the collector current of a transistor reached to the value of $6\ \text{mA}$ from $4\ \text{mA}$, whereas base current changed from $20\mu\text{A}\ to\ 25\mu\text{A}$ value. If transistor is in active state, small signal current gain (current amplification factor) will be :

Option 1:
240

Option 2:
400

Option 3:
0.0025

Option 4:
200

Correct Answer:
400

Solution:
$V_{cc} = 8\ \text{V}$

$\Delta i_c = 6\ \text{mA} - 4\ \text{mA} = 2\ \text{mA}$

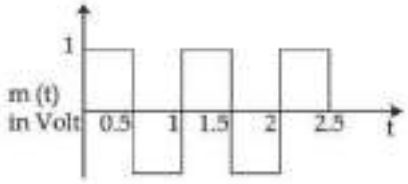
$\Delta i_b = 25\mu\text{A} - 20\mu\text{A} - 5\mu\text{A}$

$$\text{current gain } = \beta_{ac} = \frac{\Delta ic}{\Delta i_b}$$

$$= \frac{2 \times 10^{-3}}{5 \times 10^{-6}}$$
$$= 400$$

Hence 2 is correct option.

Q. 20 A square wave of the modulating signal is shown in the figure. The carrier wave is given by $C(t) = 5\sin(8\pi t)\text{Volt}$. The modulation index is :



Option 1:
0.2

Option 2:
0.1

Option 3:
0.3

Option 4:
0.4

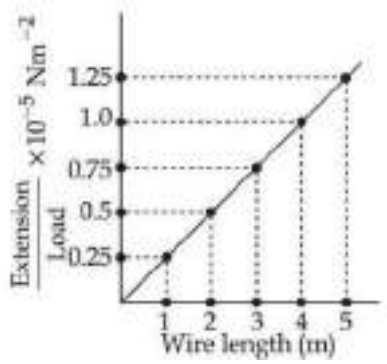
Correct Answer:
0.2

Solution:

$$\text{Modulation index} = \frac{A_m}{A_c}$$
$$= \frac{1}{5}$$
$$= 0.2$$

Hence 1 is correct option

Q. 21 In an experiment to determine the Young's modulus, steel wires of five different lengths $(1, 2, 3, 4, \text{ and } 5 \text{ m})$ but of same cross section (2 mm^2) were taken and curves between extenstion and load were obtained. The slope (extension/load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11} \text{Nm}^{-2}$, then the value of x is_____.



Correct Answer:
2

Solution:

$$A = 2(\text{ mm})^2 = 2 \times 10^{-6} \text{ m}^2$$

$$\text{slope} = \frac{\left(\frac{\text{Extension}}{\text{load}}\right) \times 10^{-5} \text{Nm}^{-2}}{\text{length}} = \frac{\Delta \ell}{F \times \ell}$$

$$\text{But } Y = \frac{F}{A} \times \frac{\ell}{\Delta \ell}$$

$$\frac{1}{AY} = \frac{\Delta \ell}{F \ell}$$

$$\text{Slope } = \frac{1}{AY} = 0.25 \times 10^{-5}$$

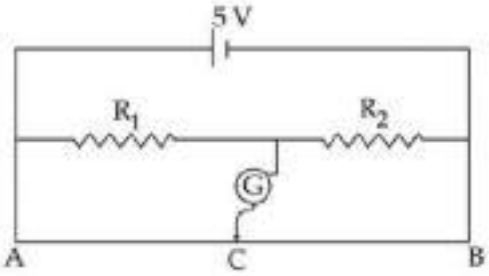
$$Y = \frac{1}{2 \times 0.25 \times 10^{-11}}$$

$$Y = 2 \times 10^{11} \text{ N/m}^2$$

$$x = 2$$

The value of x is 2.

Q. 22 In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is $40\ \mathrm{cm}$. The balancing length, if the radius of the wire AB is doubled, will be _____cm.



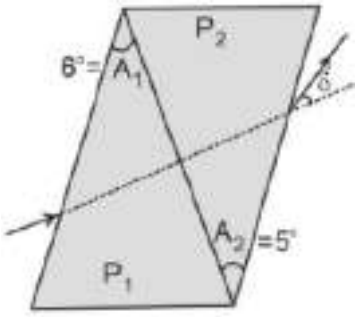
Correct Answer:
40

Solution:
For balanced wheatstone network

$$\frac{\mathrm{R}_1}{\mathrm{R}_2}=\frac{\mathrm{R}_{\mathrm{AC}}}{\mathrm{R}_{\mathrm{CB}}}=\frac{\rho\frac{\ell_{\mathrm{AC}}}{\mathrm{A}}}{\frac{\rho\ell_{\mathrm{CB}}}{\mathrm{A}}}=\frac{\ell_{\mathrm{AC}}}{\ell_{\mathrm{CB}}}$$

The balancing length, if the radius of the wire AB is doubled has no effect
$\therefore$ The balancing length is $40\ \mathrm{cm}$

Q. 23 A thin prism of angle 6° and refractive index for yellow light (n_Y) 1.5 is combined with another prism of angle 5° and $n_Y = 1.55$. The combination produces no dispersion. The net average deviation (δ) produced by the combination is $\left(\frac{1}{x}\right)^\circ$. The value of x is_____.



Correct Answer:
4

Solution:
Let the deviation through prism P_1 and prism P_2 be δ_1 & δ_2 respectively

$$\delta_1 = A_1(\mu_1 - 1)$$

$$= 6^\circ(1.5 - 1)$$

$$\delta_1 = 3^\circ$$

$$\delta_2 = A_2(\mu_2 - 1)$$

$$= 5^\circ(1.55 - 1)$$

$$\delta_2 = 5^\circ(0.55)$$

δ_1 and δ_2 are in opposite direction

$$\delta_1 + \delta_2 = 3^\circ + (-5^\circ(0.55))$$

$$\delta_1 + \delta_2 = (0.25)^\circ$$

$$\text{Net average deviation} = \delta_1 + \delta_2$$

$$= -(0.25)^\circ$$

$$= \frac{1}{4}$$

$\therefore x = 4$

Q. 24 A conducting circular loop is placed in $X - Y$ plane in presence of magnetic field $\vec{B} = \left(3t^3\hat{j} + 3t^2\hat{k}\right)$ in SI unit. If the radius of the loop is $1\ m$, the induced emf in the loop, at time, $t = 2\ s$ is $n\pi V$. The value of n is_____.

Correct Answer:
6

Solution:
$\bar{B} = 3t^3(\hat{j}) + 3t^2(\hat{k})$

$r = 1\ m$

$e = \left|\frac{d}{dt}(\bar{B}\bar{A})\right|$

$\bar{A} = \pi(1)^2\hat{k}$

$\Phi = \bar{B} \cdot \bar{A} = 3\pi t^2$

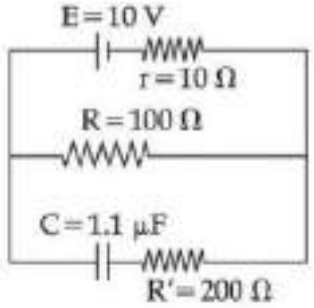
$e = \left|\frac{d(\phi)}{dt}\right| = 3\pi(2t)$

at $t = 2\ s$,

$e = 6\pi V$

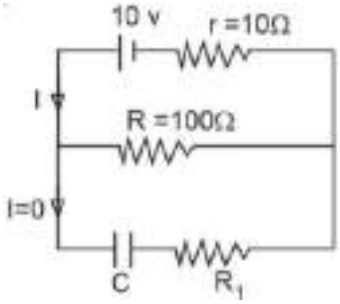
The value of n is 6

Q. 25 As show in the figure, in steady state, the charge stored in the capacitor is_____ $\times 10^{-6}\mathrm{C}$.



Correct Answer:
10

Solution:
At steady state,there is no current through capacitor



Potential drop across capacitor is

$$\mathrm{V} = \mathrm{IR} = \frac{1}{11} \times 100 = \frac{100}{11}\mathrm{V}$$

The change stored in the capacitor is

$$\mathrm{Q} = \mathrm{CV} = 1.1 \times 10^{-6} \times \frac{100}{11}$$

$$\mathrm{Q} = 10\mu c$$

Q. 26 A parallel plate capacitor with width $4\ \mathrm{cm}$, length $8\ \mathrm{cm}$ and separation between the plates of $4\ \mathrm{mm}$ is connected to a battery of $20\ \mathrm{V}$. A dielectric slab of dielectric constant 5 having length $1\ \mathrm{cm}$, width $4\ \mathrm{cm}$ and thickness $4\ \mathrm{mm}$ is inserted between the plates of parallel plate capacitor. The electrostatic energy of this system will be_____ $\epsilon_0\mathrm{J}$. (Where ϵ_0 is the permittivity of free space)

Correct Answer:
240

Solution:

$$\text{Area} = \text{length} \times \text{width} = 32 \times 10^{-4}\text{m}^2$$

$$C_{eq} = C_1 + C_2$$

$$C_{eq} = \frac{7A\varepsilon_0}{8d} + \frac{A\varepsilon_0 k}{8\,(d)}$$

$$= \frac{A\varepsilon_0}{d}\left(\frac{7}{8} + \frac{5}{8}\right)$$

$$C_{eq} = \frac{3A\varepsilon_0}{2d}$$

Eletrostatic energy of the system is

$$U = \frac{1}{2}C_{eq}V^2$$

$$\frac{3}{4} \times \frac{32 \times 10^{-4}\varepsilon_0}{4 \times 10^{-3}} \times 400$$

$$U = 240\varepsilon_0 J$$

The electrostatic energy of this system will be $240\varepsilon_0 J$

Q. 27 A wire of length $30\ \text{cm}$, stretched between rigid supports, has it's n^{th} and $(n+1)^{th}$harmonics at $400\ \text{Hz}$ and $450\ \text{Hz}$, respectively. If tension in the string is $2700\ \text{N}$, it's linear mass density is_____ kg/m

Correct Answer:
3

Solution:

$\ell = 30\ \text{cm} = 0.3\ \text{m}$

$T = 2700\ \text{N}$

$$f_n = \frac{nv}{2\ell} = \frac{n}{2\ell} \times \sqrt{\frac{T}{\mu}}$$

$$f_{n+1} = \frac{(n+1)}{2\ell}\sqrt{\frac{I}{\mu}}$$

$$f_n = \frac{400}{450} = \frac{n}{(n+1)}$$

$$n = 8$$

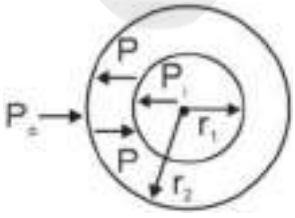
$$400 = \frac{8}{2 \times 0.3} \times 30 \times \frac{\sqrt{3}}{\sqrt{\mu}}$$

$$\mu = 3 \text{ kg/m}$$

Q. 28 A spherical soap bubble of radius 3 cm is formed inside another spherical soap bubble of radius 6 cm. If the internal pressure of the smaller bubble of radius 3 cm in the above system is equal to the internal pressure of the another single soap bubble of radius r cm. The value of r is_____.

Correct Answer:
2 cm

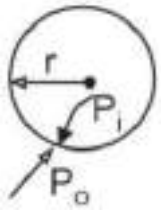
Solution:



$$P_i - P = \frac{4T}{r_1}$$

$$P - P_o = \frac{4T}{r_2}$$

$$P_i - P_o = \frac{4T}{r_1} + \frac{4T}{r_2} \rightarrow (1)$$



$$P_i - P_o = \frac{4T}{r} \rightarrow (2)$$

From eq. (1) and (2)

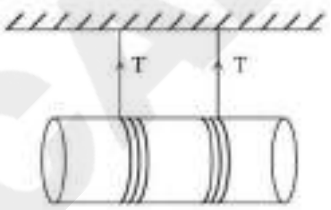
$$\frac{4t}{r} = \frac{4T}{r_1} + \frac{4T}{r_2} \text{(Given)}$$

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2}$$

$$= \frac{1}{3} + \frac{1}{6} = \frac{9}{18}$$

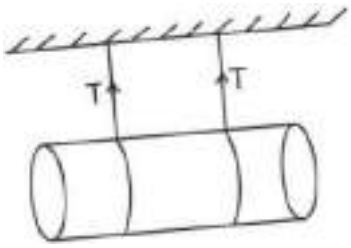
$$r = 2\text{cm}$$

Q. 29 A solid cylinder length is suspended symmetrically through two massless strings, as shown in the figure. The distance from the initial rest position, the cylinder should by unbinding the strings to achieve a speed of $4\ \text{ms}^{-1}$, is _____cm .(take $g = 10\ \text{ms}^{-2}$)



Correct Answer:
120

Solution:



$$2TR = I\alpha = \frac{MR^2}{2}\alpha$$

$$T = \frac{MR\alpha}{4} \rightarrow (1)$$

$$Mg - 2T = Ma_{cm} \rightarrow \ (2)$$

$$a_{cm} = R\alpha \ (\text{for pure rolling motion})$$

$$Mg = \frac{3Ma_{cm}}{2}$$

$$a_{cm} = \frac{29}{3}$$

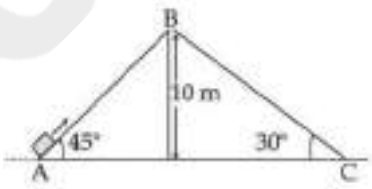
$$v^2 = u^2 + 2as$$

$$16^0 = 0 + 2 \times \frac{29}{3}s$$

$$s = 1.2m$$

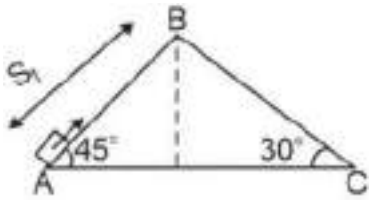
$$s = 120cm$$

Q. 30 Two inclined planes are placed as shown in figure. A block is projected from the Point A of inclined plane AB along its surface with a velocity just sufficient to carry it to the top Point B at a height $10\ m$. After reaching the Point B the block slides down on inclined plane BC. Time it takes to reach to the point C from point A is $t(\sqrt{2}+1)s$. The value of t is _____ (use $g = 10\ m/s^2$)



Correct Answer:
2s

Solution:



For journey from A to B,

$$V_B^2 = V_A^2 + 2aS_1$$

$$0 = V_A^2 + 2(-g\sin 45^\circ) \times \frac{10}{\sin 45^\circ}$$

$$V_A = 10\sqrt{2}\ \text{m/s}$$

$$V_B = V_A + at$$

$$0 = 10\sqrt{2} + (-g\sin 45^\circ)\, t_1$$

$$t_1 = 2\text{s}$$

For journey from B to C

$$S = ut_2 + \frac{1}{2}at_2^2$$

$$\frac{10}{\sin 30^\circ} = \frac{1}{2} \times g\sin 30^\circ t_2^2$$

$$t_2^2 = 8$$

$$t_2 = 2\sqrt{2}\text{s}$$

$$\text{Total time} = t_1 + t_2$$

$$= 2 + 2\sqrt{2}$$

$$t = 2\text{s}$$

CAREERS360

Chemistry

Q. 1 The correct decreasing order of energy for the orbitals having, following set of quantum numbers :

(A) $n = 3, l = 0, \ m = 0$

(B) $n = 4, l = 0, \ m = 0$

(C) $n = 3, l = 1, \ m = 0$

(D) $n = 3, l = 2, \ m = 1$

is :

Option 1:

$(D) > (B) > (C) > (A)$

Option 2:

$(B) > (D) > (C) > (A)$

Option 3:

$(C) > (B) > (D) > (A)$

Option 4:

$(B) > (C) > (D) > (A)$

Correct Answer:

$(D) > (B) > (C) > (A)$

Solution:

$(A)\ n = 3, l = 0, m = 0 \longrightarrow 3s$ orbital and $n + l = 3 + 0 = 3$

$(B)\ n = 4, l = 0, m = 0 \longrightarrow 4s$ orbital and $n + l = 4 + 0 = 4$

$(C)\ n = 3, l = 1, m = 0 \rightarrow 3P_x$ orbital and $n + l = 3 + 1 = 4$

$(D)\ n = 3, l = 2, m = 1 \longrightarrow 3d$ orbital and $n + l = 3 + 2 = 5$

(1) lower the $(n + l)$ value, lower the energy.

(2) lower the value of n will have the lover energy if $(n + l)$ value is same for two orbitals.

$(D)\ n + l = 5 > (B) n + l = 4, n = 4 > (C) n + l = 4, n = 3 > (A) n + l = 3.$

So, (D) > (B) > (C) > (A)
Option (1) is correct.

Q. 2 Match List - I with List - II.

List I	List II
(A) $\psi_{MO} = \psi_A - \psi_B$	(I) Dipole moment
(B) $\mu = Q \times r$	(II) Bonding molecular orbital
(C) $\frac{N_b - N_a}{2}$	(III) Anti-bonding molecular orbital
(D) $\psi_{MO} = \psi_A + \psi_B$	(IV) Bond order

Choose the correct answer from the options given below :

Option 1:
(A) - (II), (B) - (I), (C) - (IV), (D) - (III)

Option 2:
(A) - (III), (B) - (IV) ,(C) - (I) ,(D) - (II)

Option 3:
(A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Option 4:
(A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Correct Answer:
(A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Solution:
(A) $\Psi_{M0} = \Psi_A - \Psi_B \rightarrow$ (III) Anti-Bonding molecular orbital
(B) $\mu = Q \times r \longrightarrow$ (I) Dipole moment
(C) $\frac{N_b - Na}{2} \longrightarrow$ (IV) Bond order

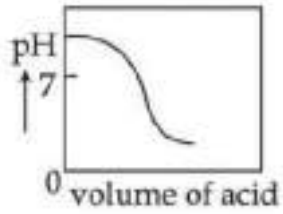
(D) $\Psi_{M_0} = \Psi_A + \Psi_B \rightarrow$ (II) Bonding molecular orbital

So, (A) – (III), (B) – (I), (C) – (IV), (D) – (II)

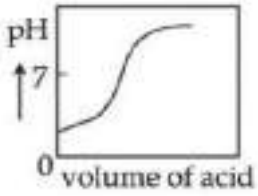
Option (3) is correct.

Q. 3 The plot of pH-metric titration of weak base NH_4OH vs strong acid HCl looks like :

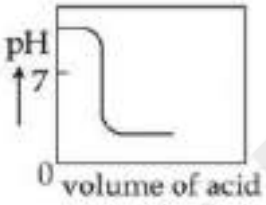
Option 1:



Option 2:

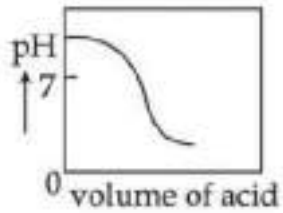


Option 3:



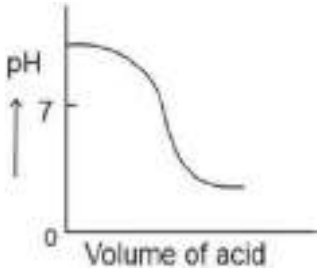
Option 4:

Correct Answer:



Solution:
For weak base NH_4OH vs strong acid HCl, the curve of pH- changes will be-

Due to weak base the pH will change slowly.



Option (1) is correct.

Q. 4 Given below are two statements :
Statement I : For KI, molar conductivity increases steeply with dilution
Statement II : For carbonic acid, molar conductivity increases slowly with dilution In the light of the above statements, choose the correct answer from the options given below:

Option 1:
Both **Statement I** and **Statement II** are true

Option 2:
Both **Statement I** and **Statement II** are false

Option 3:
Statement I is true but **Statement II** is false

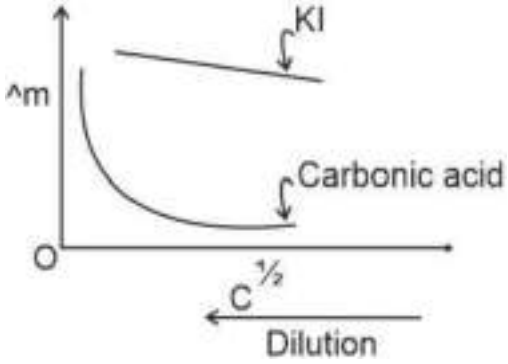
Option 4:
Statement I is false but **Statement II** is true

Correct Answer:
Both **Statement I** and **Statement II** are false

Solution:

For strong electrolydes, Λ increases slowly with dilution and can be represended by the equation $\Lambda_m = \Lambda_m^0 - AC^{v_2}$

KI is sxrong electrolytes. So, molar conductivity increases slowly with dilution.
For weak electrolytes corbonic acid, molar conductiyity increases sleeply with dilution.



So Both statement I and II are false.

So, option (2) is correct.

Q. 5 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A): Dissolved substances can be removed from a colloidal solution by diffusion through a parchment paper.

Reason (R): Particles in a true solution cannot pass through parchment paper but the colloidal particles can pass through the parchment paper.

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

Option 2:
Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

Option 3:
(A) is correct but **(R)** is not correct

Option 4:
(A) is not correct but **(R)** is correct

Correct Answer:
(A) is correct but **(R)** is not correct

Solution:
Dissolved can be aemoved from a colloidal solution by diffusion through a parchment paper.
So, (A) is correct.

Particles in a true solution can pass thorough parchment paper but the colloidal partictes cannot pass through the parchment paper.

So,$(\mathbb{R})$ is not correct.

Option (3) is correct.

Q. 6 Outermost electronic configurations of four elements A, B, C, D are given below :
(A) $3\ s^2$
(B) $3\ s^2 3p^1$
(C) $3s^2 3p^3$
(D) $3\ s^2 3p^4$
The correct order of first ionization enthalpy for them is :

Option 1:
$(A) < (B) < (C) < (D)$

Option 2:
$(\mathrm{B}) < (\mathrm{A}) < (\mathrm{D}) < (\mathrm{C})$

Option 3:
$(B) < (\mathrm{D}) < (\mathrm{A}) < (\mathrm{C})$

Option 4:
$(B) < (A) < (C) < (D)$

Correct Answer:
$(\mathrm{B}) < (\mathrm{A}) < (\mathrm{D}) < (\mathrm{C})$

Solution:
(A) $3s^2 \rightarrow Mg$
(B) $3s^2 3p^1 \rightarrow Al$
(C) $3s^2 3P^3 \rightarrow P(\text{Half filled})(\text{more stable than S})$
(D) $3s^2 3p^4 \rightarrow S(\text{Partially filled})$

The penetration of a $3s$ -electron to the nucleus is more than that of a $3p$-electron. of Al.

Order of First ionization enthalpy will be -
$(B) < (A) < (D) < (C)$

Option(2) is correct.

Q. 7 An element A of group 1 shows similarity to an element B belonging to group 2 . If A has maximum hydration enthalpy in group 1 then b is :

Option 1:
Mg

Option 2:
Be

Option 3:
Ca

Option 4:
Sr

Correct Answer:
Mg

Solution:
In group 1, Li has maximum hydration enthalpy.

Li shows diagonal relation with Mg.

Option (1) is correct.

Q. 8 Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : Boron is unable to form BF_6^{3-}.
Reason (R) : Size of B is very small.
In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both (A) and (R) are true and (R) is the correct explanation of (A)

Option 2:
Both (A) and (R) are true but (R) is not the correct explanation of (A)

Option 3:
(A) is true but (R) is false

Option 4:
(A) is false but (R) is true

Correct Answer:
Both (A) and (R) are true but (R) is not the correct explanation of (A)

Solution:
(A) is true

(B) is true but (R) is not correct explanation of (A).

Boron is unable to form BF_6^{3-} due to absence of d-orbital or unable to expand octet in Boron.

Option (2) is correct.

Q. 9 In neutral or alkaline solution, MnO_4^- oxidises thiosulphate to :

Option 1:
$S_2O_7^{2-}$

Option 2:
$S_2O_8^{2-}$

Option 3:
SO_3^{2-}

Option 4:
SO_4^{2-}

Correct Answer:
SO_4^{2-}

Solution:
In neutral or faintly alkaline solutions,

MnO_4^- oxidises thiosulphate to sulphate.

$$8MnO_4^- + 3\ \underset{\text{thiosulphate}}{S_2O_3^{2-}} + H_2O \longrightarrow 8\,MnO_2 + 6\,\underset{\text{sulphate}}{SO_4^{2-}} + 2\,OH^{\Theta}$$

Option (4) is correct.

Q. 10 Low oxidation state of metals in their complexes are common when ligands :

Option 1:
have good π-accepting character

Option 2:
have good σ-donor character

Option 3:
are having good π-donating ability

Option 4:
are having poor σ-donating ability

Correct Answer:
have good π-accepting character

Solution:
Low oxidation state of metals in their complexes are common when ligands have good π—accepting character.

The Metal-ligand π—bond is formed by the. donation of pair of electrons from a filled d-orbital of a metal into the vacant antibonding π orbtial of ligand.

Option (1) is correct.

Q. 11 Given below are two statements :
Statement I : The non bio-degradable fly ash and slag from steel industry can be used by cement industry.
Statement II : The fuel obtained from plastic waste is lead free.
In the light of the above statements, choose the most appropriate answer from the options given below :

Option 1:
Both Statement I and Statement II are correct

Option 2:
Both Statement I and Statement II are incorrect

Option 3:
Statement I is correct but Statement II is incorrect

Option 4:
Statement I is incorrect but Statement II is correct

Correct Answer:
Both Statement I and Statement II are correct

Solution:

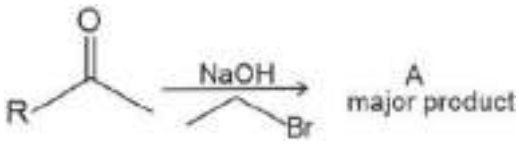
The non-bio-degrable fly ash and slag from steel industry can be used by cement industry.

Fuel obtained from plastic waste has high octane. reting and it cantains no lead and is known as "green fuel".

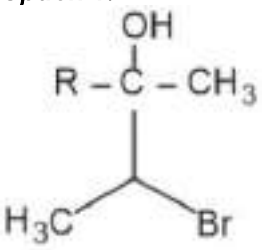
So, Both Statement I and II are correct.

Option (1) is correct.

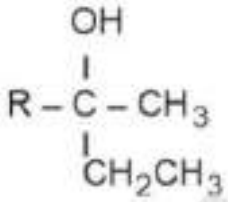
Q. 12 The structure of A in the given reaction is :



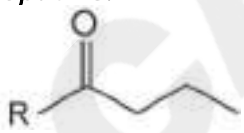
Option 1:



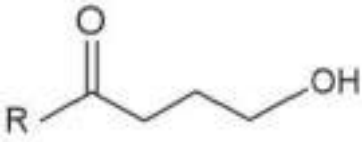
Option 2:



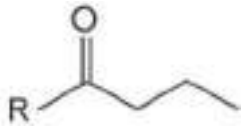
Option 3:



Option 4:

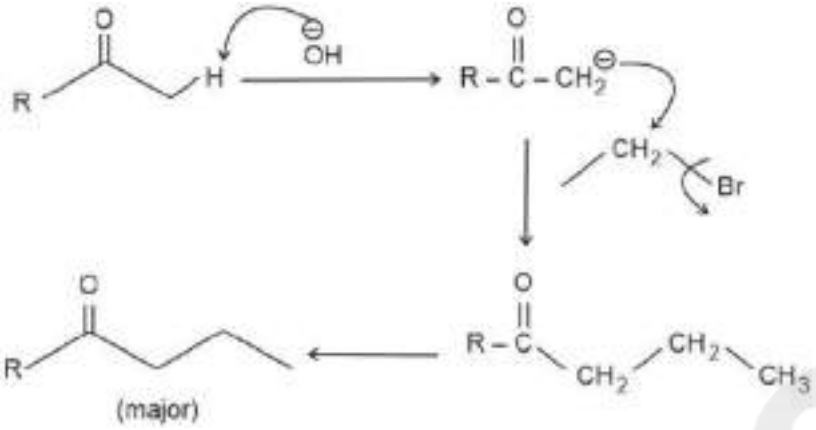


Correct Answer:



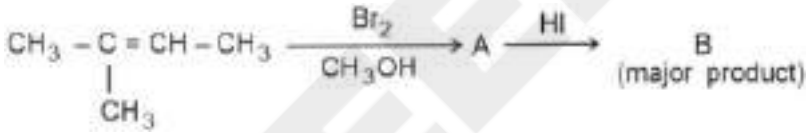
Solution:

The reaction will be -

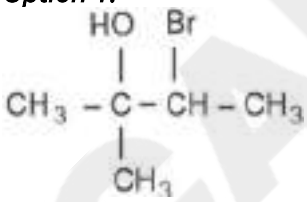


Option (3) is correct.

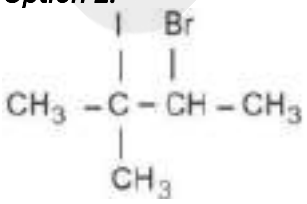
Q. 13 Major product 'B' of the following reaction sequence is :



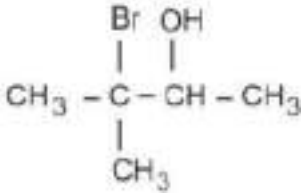
Option 1:



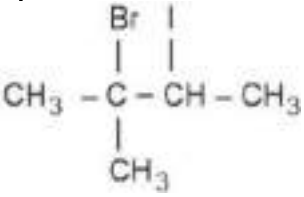
Option 2:



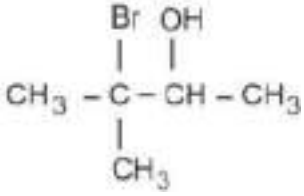
Option 3:



Option 4:

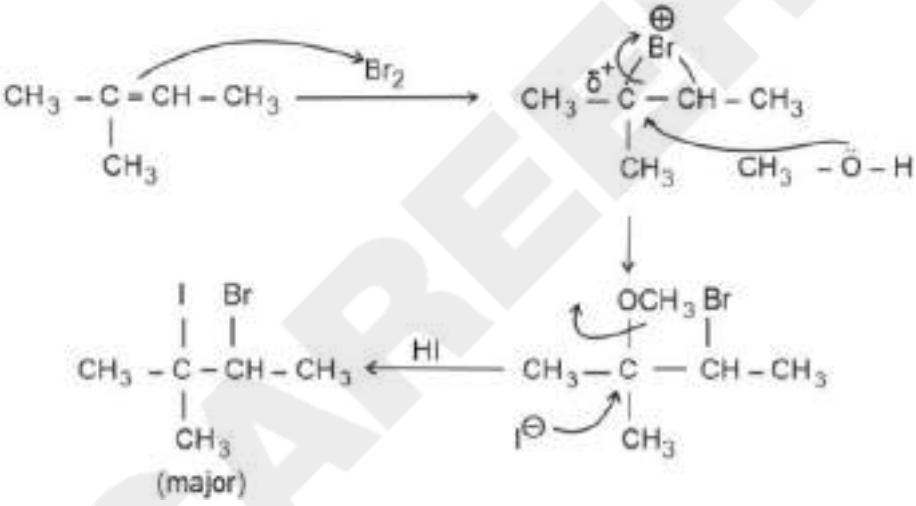


Correct Answer:



Solution:

The reaction will be -



Option (3) is correct.

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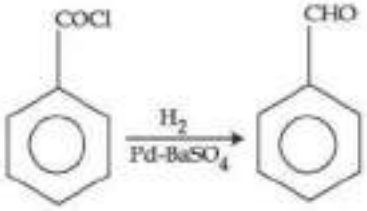
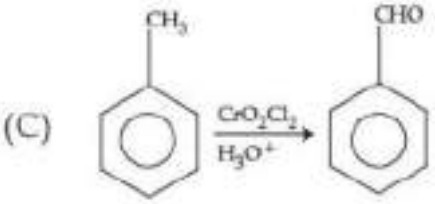
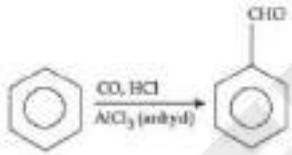
Q. 14 Match List -I with List -II.

List-I	List-II
(A) 	(I) Gatterman Koch reaction
(B) $CH_3 - CN \xrightarrow[H_3O^+]{Sncl_2/HCl} CH_3 - CHO$	(II) Etard reaction
(C) 	(III) Stephen reaction
(D) 	(IV) Rosenmund reaction

Choose the correct answer from the options given below :

Option 1:
$(A) - (IV), (B) - (III), (C) - (II), (D) - (I)$

Option 2:
$(A) - (I), (B) - (II), (C) - (III), (D) - (IV)$

Option 3:
$(A) - (II), (B) - (III), (C) - (IV), (D) - (I)$

Option 4:
$(A) - (III), (B) - (II), (C) - (I), (D) - (IV)$

Correct Answer:
$(A) - (IV), (B) - (III), (C) - (II), (D) - (I)$

Solution:

(A) $\rightarrow$ (IV) Rosenmund reaction

(B) $\rightarrow$ (III) Stephon reaction

(C) $\rightarrow$ (II) Etard reaction

(D) $\rightarrow$ (I) Gatterman Koch reaction

So, option (1) is correct.

Q. 15 Match **List - I** with **List - II.**

List-I (Polymer)	**List-II (Monomer)**
(A) Neoprene	(I) Acrylonitrile
(B) Teflon	(II) Chloroprene
(C) Acrilan	(III) Tetrafluoroethene
(D) Natural rubber	(IV) Isoprene

Choose the correct answer from the options given below :

Option 1:
(A) – (II), (B) – (III), (C) – (I), (D) – (IV)

Option 2:
(A) – (II), (B) – (I), (C) – (III), (D) – (IV)

Option 3:
(A) – (II), (B) – (I), (C) – (IV), (D) – (III)

Option 4:
$(A) - (I), (B) - (II), (C) - (III), (D) - (IV)$

Correct Answer:
$(A) - (II), (B) - (III), (C) - (I), (D) - (IV)$

Solution:
(A) Neoprene $\longrightarrow$ (II) Chloroprene

(B) Teflon $\longrightarrow$ (III) Tetrafluoroethene

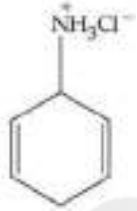
(C) Acrilan $\longrightarrow$ (I) Acrylonitrile

(D) Natural rubber $\longrightarrow$ (IV) Isoprene.

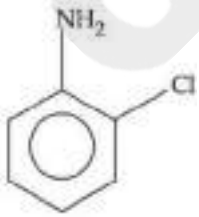
So, option (1) is correct.

Q. 16 An organic compound $'A'$ contains nitrogen and chlorine. It dissolves readily in water to give a solution that turns litmus red. Titration of compound $'A'$ with standard base indicates that the molecular weight of $'A'$ is 131 ± 2. When a sample of $'A'$ is treated with aq.$NaOH$, a liquid separates which contains n but not Cl]. Treatment of the obtained liquid with nitrous acid followed by phenol gives orange precipitate. The compound $'A'$ is:

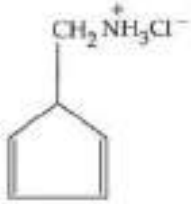
Option 1:



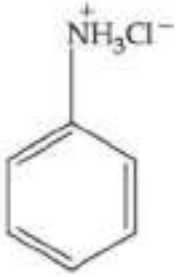
Option 2:



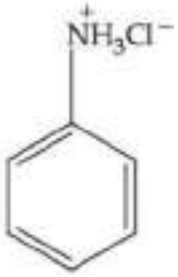
Option 3:



Option 4:

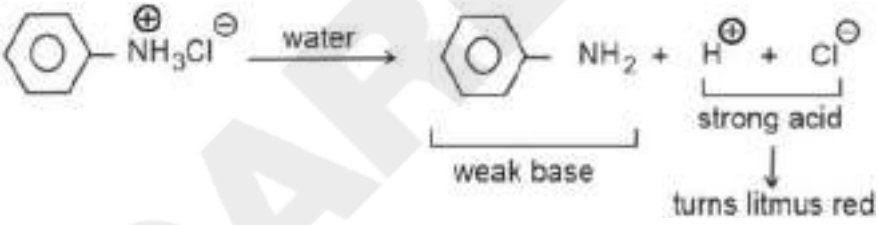


Correct Answer:



Solution:

We know acid turns litmus red.



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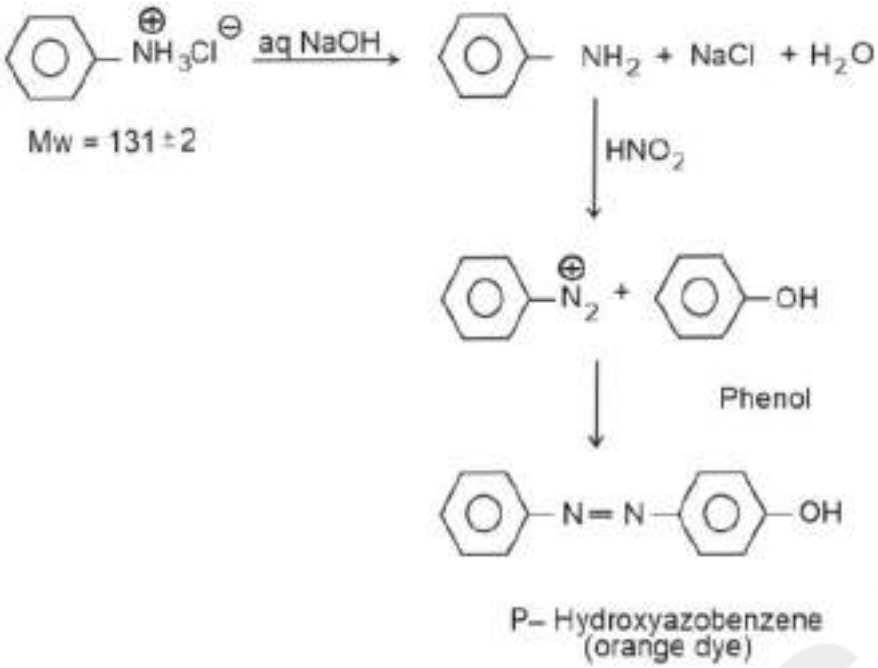


Option (4) is correct.

Q. 17 Match **List - I** with **List - II.**

List-I	**List-II**
$(A)\text{Glucose} + HI$	(I) Gluconic acid
$(B)\text{Glucose} + Br_2 \text{ water}$	(II) Glucose pentacetate
(C) Glucose + acetic anhydride	(III) Saccharic acid
$(D)\text{Glucose} + HNO_3$	(IV) Hexane

Choose the correct answer from the options given below :

Option 1:

(A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Option 2:
(A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Option 3:
(A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Option 4:
(A)- (I), (B) - (III), (C) - (IV), (D) - (II)

Correct Answer:
(A) - (IV), (B) - (I), (C) - (II), (D) - (III)

Solution:

$$\text{(A) Grueose} + \text{HI} \rightarrow \text{(IV) Hexane}$$

$$\text{(B) Glucose} + \text{Br}_2 \text{ water} \rightarrow \text{(I) Gluconic acid}$$

$$\text{(C) Glucose} + \text{acetic anhydride} \rightarrow \text{(II) Glucose pentacetate}$$

$$\text{(D) Gslucose} + \text{HNO}_3 \rightarrow \text{(III) Saccharic acid}$$

$$\text{So, (A)} - \text{(IV), (B)} - \text{(I), (C)} - \text{(II), (D)} - \text{(III)}$$

Option (1) is correct.

Q. 18 Which of the following enhances the lathering property of soap?

Option 1:
Sodium stearate

Option 2:
Sodium carbonate

Option 3:
Sodium rosinate

Option 4:
Trisodium phosphate

Correct Answer:
Sodium rosinate

Solution:
A Gum called called, rosin is added while making them. It forms Sodium rosinate which lathers well.

Option (3) is correct.

Q. 19 **Match List - I with List - II.**

List-I(Mixture)	**List-II (Purification Process)**
(A) Chloroform & Aniline	(I) Steam distillation
(B) Benzoic acid & Napthalene	(II) Sublimation
(C) Water & Aniline	(III) Distillation
(D) Napthalene & Sodium chloride	(IV) Crystallisation

Choose the correct answer from the options given below :

Option 1:
(A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Option 2:
(A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Option 3:
(A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Option 4:
(A) - (III),(B) - (IV),(C) - (I),(D) - (II)

Correct Answer:
(A) - (III),(B) - (IV),(C) - (I),(D) - (II)

Solution:
$(A)\ \text{Chloroform \& Aniline} \longrightarrow (III)\ \text{Distillation}$

$(B)\ \text{Benzoic acid \& Napthalene} \longrightarrow (IV)\ \text{Crystallisation}$

$(C)\ \text{Water\&Aniline} \longrightarrow (I)\ \text{Steam Distillation}$

$(D)\ \text{Napthalene \& Sodium chloride} \longrightarrow (II)\ \text{Sublimation}$

$\text{So}, (A)-(III), (B)-(IV), (C)-(I), (D)-(II)$

Option (4) is correct.

Q. 20 Fe^{3+} cation gives a prussian blue precipitate on addition of potassium ferrocyanide solution due to the formation of :

Option 1:
$[Fe\,(H_2O)_6]_2\,[Fe(CN)_6]$

Option 2:
$Fe_2\,[Fe(CN)_6]_2$

Option 3:
$Fe_3\,[Fe(OH)_2(CN)_4]_2$

Option 4:
$Fe_4\,[Fe(CN)_6]_3$

Correct Answer:
$Fe_4\,[Fe(CN)_6]_3$

Solution:
The reaction will be

$$4Fe^{3+} + 3\,[FeC(N)_6]^{4-} \longrightarrow Fe_4\,[Fe(CN)_6]_3$$

(prussian blue)

Option (4) is correct.

Q. 21 The normality of H_2SO_4 in the solution obtained on mixing $100\ mL$ of $0.1M H_2SO_4$ with $50\ mL$ of $0.1M\ NaOH$ is _____ $\times 10^{-1}\ N$. (Nearest Integer)

Correct Answer:
1

Solution:

$$\text{Normality} = \frac{\text{Gram equivalents of solute (gmeq)}}{\text{Litres of Solution } (V_L)}$$

$$\text{Normality} = \text{Molarity} \times \text{n} - \text{factor}$$

This is acid and base reaction. So need to find lef acid or bare solution.

$$\text{meq of base} = 50 \times 0.1 \times 1 = 5\,\text{meq}$$
$$\text{meq of acid} = 100 \times 0.1 \times 2 = 20\,\text{meq}$$

meq of actd is more than meq of base.

$$H_2SO_4 \text{ solute (acid) Left meq} = 20 - 5 = 15\,\text{meq}$$

$$\text{Normality} = \frac{\text{meq of } H_2SO_4}{\text{total volume}} = \frac{15\,\text{meq}}{(100+50)\text{ml}}$$

$$\text{Normality} = \frac{15 \times 10^{-3}\text{eq}}{150 \times 10^{-3}\ L} = 0.1\ N = 1 \times 10^{-1}\ N$$

Ans = 1.

Q. 22 For a real gas at $25^{\circ}C$ temperature and high pressure (99 bar) the value of compressibility factor is 2, so the value of Vander Waal's constant 'b' should be_____ $\times 10^{-2}\ L\ mol^{-1}$(Nearest integer)
(Given $R = 0.083\ L\ bar\ K^{-1}\ mol^{-1}$)

Correct Answer:
25

Solution:
we know, at high pressure the compressibility factor will be -

$$z = 1 + \frac{Pb}{RT} \Rightarrow 2 = 1 + \frac{99 \times b}{0.083 \times 298}$$
$$2 = 1 + \left(\frac{99}{0.083 \times 298}\right)_b$$
$$b = \frac{0.083 \times 298}{99} = 0.249$$
$$b = 25 \times 10^{-2}\,L\,mol^{-1}$$

Ans = 25.

Q. 23 A gas (Molar mass $= 280\ g\ mol^{-1}$) was burnt in excess O_2 in a constant volume calorimeter and during combustion the temperature of calorimeter increased from $298.0\ K$ to $298.45\ K$. If the heat capacity of calorimeter is $2.5\ kJ\ K^{-1}$ and enthalpy of combustion of gas is $9\ kJ\ mol^{-1}$ then amount of gas burnt is_____ g. (Nearest Integer)

Correct Answer:
35

Solution:
We know,
$$c_m = \frac{C}{n} = \frac{\text{Heat capacity}}{\text{mole}}$$
$$\text{and } \Delta H = C_P \Delta T = \frac{C}{n}\Delta T$$

$$\text{So, } g = \frac{2.5}{n} \times (258.45 - 298)$$
$$n = \frac{2.5}{8} \times 0.45$$
$$\frac{w}{280} = \frac{2.5}{9} \times 0.45 \Rightarrow w = 35\ g$$

Ans = 35.

Q. 24 When a certain amount of solid A is dissolved in $100\ \mathrm{g}$ of water at $25^{\circ}\mathrm{C}$ to make a dilute solution, the vapour pressure of the solution is reduced to one-half of that of pure water. The vapour pressure of pure water is $23.76\mathrm{mm\ Hg}$. The number of moles of solute A added is_____. (Nearest Integer)

Correct Answer:
3

Solution:
We know the formulas

$$\frac{\mathrm{p_1^0 - p_1}}{\mathrm{p_1^0}} = \frac{\mathrm{n_2}}{\mathrm{n_1 + n_2}} \quad \left\{ \begin{array}{l} n_2 = \text{ mole of solute} \\ n_1 = \text{ mole of solvent} \end{array} \right\}$$

due to dilute solution $\mathrm{n_1 \gg n_2}\ \ \mathrm{so, n_1 + n_2 = n_1}$

then $\dfrac{\mathrm{p_1^0 - p_1}}{\mathrm{p_1^0}} = \dfrac{\mathrm{n_2}}{\mathrm{n_1}},$

$\mathrm{given\ p_1^0 = 23.76\ mmHg}$

$\mathrm{and\ p_1 = \dfrac{p_1^0}{2}}$

So,

$$\frac{\mathrm{p_1^0 - \frac{p_1^0}{2}}}{\mathrm{p_1^0}} = \frac{\mathrm{n_2}}{\mathrm{n_1}}$$

$$\frac{1}{2} = \frac{\mathrm{n_2}}{\mathrm{w_0/M_1}}$$

$$\mathrm{n_2} = \frac{\mathrm{w_1}}{\mathrm{M_1}} \times \frac{1}{2} = \frac{100}{18 \times 2}$$

$$\mathrm{n_2} = 2.77 \approx 3$$

Ans = 3

Q. 25 $[\mathrm{A}] \rightarrow [\mathrm{B}]$
Reactant Product
If formation of compound [B] follows the first order of kinetics and after 70 minutes the concentration of $[\mathrm{A}]$ was found to be half of its initial concentration. Then the rate constant of the reaction is $\mathrm{x} \times 10^{-6}\ \mathrm{s}^{-1}$. The value of x is_____. (Nearest Integer)

Correct Answer:
165

Solution:
we know for the first order reaction,

$$\text{Half life } t_{1/2} = \frac{\ln 2}{k}$$
$$70 \text{ min} = \frac{\ln 2}{x \times 10^{-6}} \text{ sec}$$
$$70 \times 60 = \frac{\ln 2}{x \times 10^{-6}}$$
$$x = 165$$

Ans=165.

Q. 26 Among the following ores Bauxite, Siderite, Cuprite, Calamine, Haematite, Kaolinite, Malachite, Magnetite, Sphalerite, Limonite, Cryolite, the number of principal ores if iron is_____.

Correct Answer:
4

Solution:
1) Bauxite – $AlOx(OH)_{3-2x}$
2) Siderite – $FeCO_3$
3) Cuprite – Cu_2O
4) Calamine – $ZnCO_3$
5) Haematite – Fe_2O_3
6) Kaolinite – $[Al_2\ (OH_4Si_2O_5]$
7) Malachite – $[CuCO_3 \cdot Cu(OH)_2]$
8) Magentite – Fe_3O_4
9) Sphalerite – zns(zinc Blende)
10) Limonite – $2Fe_2O_3.3H_2O$
11) Crydite – Na_3AlF_6

Among above total 4 ores has iron as principal ores.

Ans = 4.

Q. 27 The oxidation state of manganese in the product obtained in a reaction of potassium permanganate and hydrogen peroxide in basic medium is_____.

Correct Answer:
4

Solution:
The reaction will be

$$2MnO_4^- + 3H_2O_2 \longrightarrow 2MnO_2 + 3O_2 + 2H_2O + 2OH^-$$

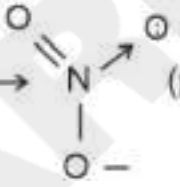
$$\text{in } MnO_2 \rightarrow Mn \text{ is in } +4 \text{ oxidation state.}$$

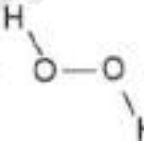
Ans = 4.

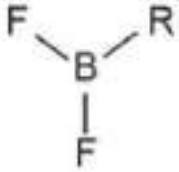
Q. 28 The number of molecule(s) or ion(s) from the following having non-planar structure is_____.
$NO_3^-, H_2O_2, BF_3, PCl_3, XeF_4, SF_4, XeO_3, PH_4^+, SO_3, [Al(OH)_4]^-$

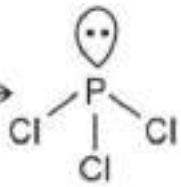
Correct Answer:
6

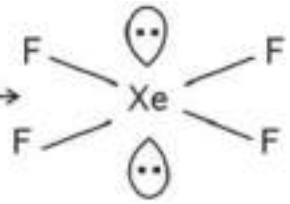
Solution:

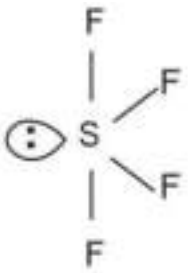
1) NO_3^-  (planar – SP^2)

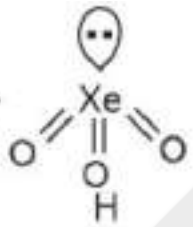
2) H_2O_2  (non-planar - SP^3) (open-book structure)

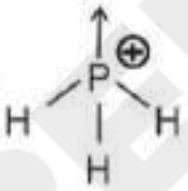
3) BF_3  (SP^2 – planar)

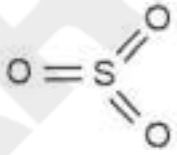
4) PCl_3  (SP^3 – non-planar)

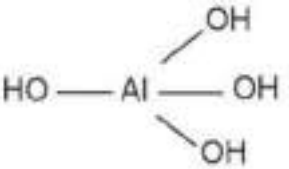
5) XeF_4  (SP^3d^2 – planar)

6) SF_4  (SP^3d – non-planar)

7) XeO_3  (SP^3 – non-planar)

8) PH_4^+  (SP^3 – non-planar)

9) SO_3  (SP^2 – planar)

10) $[Al(OH)_4]^-$  (SP^3 – non-planar)

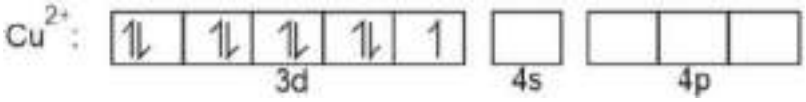
So, Total 6 molecular/ions having non-planar structure.

Ans = 6.

Q. 29 The spin only magnetic moment of the complex present in Fehling's reagent is_____ B.M. (Nearest integer).

Correct Answer:
2

Solution:
The Fehling's reagent is Cu^{2+}.

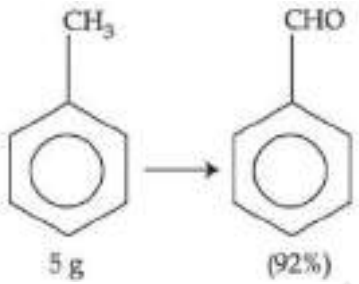


Total 1 electron is unpaired.

$$\text{So, } \mu = \sqrt{n(n+2)}$$
$$\mu = \sqrt{1(1+2)} = \sqrt{3} = 1.73$$
$$\mu = 2$$

Ans = 2.

Q. 30



In the above reaction, 5 g of toluene is converted into benzaldehyde with 92% yield. The amount of benzaldehyde produced is _____$\times 10^{-2}$ g. (Nearest integer)

Correct Answer:
530

Solution:
From the reaction 1 mol of toluene is converted into 1 mol of benzaldehyde but yield is 92% so 0.92 mol of benzaldehyde form.

$$1 \text{ mole Ph} - CH_3 \longrightarrow 0.92 \text{ mole Ph} - CHO$$
$$92 \text{ gram Ph} - CH_3 \longrightarrow 0.92 \text{ mol Ph} - CHO$$
$$5 \text{ gram Ph} - CH_3 \longrightarrow 0.05 \text{ mol Ph} - CHO$$
$$5 \text{ gram Ph} - CH_3 \longrightarrow 5.30 \text{ gram Ph} - CHO$$

$$\text{Amount of benzaldehyde} = 530 \times 10^{-2} \text{ g.}$$
Ans = 530.

Maths

Q. 1 The domain of the function $f(x)=\sin^{-1}\left[2x^2-3\right]+\log_2\left(\log_{\frac{1}{2}}\left(x^2-5x+5\right)\right)$, where $[t]$ is the greatest integer function, is :

Option 1:
$\left(-\sqrt{\frac{5}{2}}, \frac{5-\sqrt{5}}{2}\right)$

Option 2:
$\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$

Option 3:
$\left(1, \frac{5-\sqrt{5}}{2}\right)$

Option 4:
$\left[1, \frac{5+\sqrt{5}}{2}\right)$

Correct Answer:
$\left(1, \frac{5-\sqrt{5}}{2}\right)$

Solution:

CAREERS360

$\sin^{-1}\left[2x^2-3\right]+\log_2\left(\log_{\frac{1}{2}}\left(x^2-5x+5\right)\right)$

For domain of$\sin^{-1}\left[2x^2-3\right]$

$-1\le\left[2x^2-3\right]\le 1$

$-1\le\left[2x^2\right]-3\le 1$

$2\le\left[2x^2\right]\le 4$

$\left[2x^2\right]=2,3,4$

$2\le 2x^2<5$

$1\le x^2<\frac{5}{2}$

$x\in\left(-\sqrt{\frac{5}{2}},-1\right]\cup\left[1,\sqrt{\frac{5}{2}}\right)$

For domain of $\log_2\left(\log_{\frac{1}{2}}\left(x^2-5x+5\right)\right)$

$\log_{\frac{1}{2}}\left(x^2-5x+5\right)>0$

$\Rightarrow x^2-5x+5<1$

$\Rightarrow \quad x^2-5x+4<0$

$\Rightarrow \quad (x-1)(x-4)<0$

$\Rightarrow \quad x\in(1,4)$

Also $x^2-5x+5>0$

$\left(x-\left(\frac{5+\sqrt{5}}{2}\right)\right)\left(x-\left(\frac{5-\sqrt{5}}{2}\right)\right)>0$

$x\in\left(-\infty,\frac{5-\sqrt{5}}{2}\right)\cup\left(\frac{5+\sqrt{5}}{2},\infty\right)$

Common part is $x\in\left(1,\frac{5-\sqrt{5}}{2}\right)$

Q. 2 Let S be the set of all $(\alpha,\beta),\pi<\alpha,\beta<2\pi$, for which the complex number $\frac{1-i\sin\alpha}{1+2i\sin\alpha}$ is purely imaginary and $\frac{1+i\cos\beta}{1-2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta}=\sin 2\alpha+i\cos 2\beta,(\alpha,\beta)\in S$. Then $\sum_{(\alpha,\beta)\in S}\left(iZ_{\alpha\beta}+\frac{1}{i\bar{Z}_{\alpha\beta}}\right)$ is equal to :

Option 1:
3

Option 2:
$3\,i$

Option 3:
1

Option 4:
$2-i$

Correct Answer:
1

Solution:

$\pi<\alpha,\beta<2\pi$

$$\frac{1-\mathrm{i}\sin\alpha}{1+\mathrm{i}(2\sin\alpha)}=\text{ Purely imaginary}$$

$$\Rightarrow\frac{(1-\mathrm{i}\sin\alpha)(1-\mathrm{i}(2\sin\alpha))}{1+4\sin^2\alpha}=\text{ Purely imaginary}$$

$$\Rightarrow\frac{1-2\sin^2\alpha}{1+4\sin^2\alpha}=0$$

$$\Rightarrow\sin^2\alpha=\frac{1}{2}$$

$$\Rightarrow\alpha=\left\{\frac{5\pi}{4},\frac{7\pi}{4}\right\}$$

$$\&\frac{1+\mathrm{i}\cos\beta}{1+\mathrm{i}(-2\cos\beta)}=\text{ Purely real}$$

$$\Rightarrow\frac{(1+\mathrm{i}\cos\beta)(1+2\ \mathrm{i}\cos\ \beta)}{1+4\cos^2\beta}=\text{ Purely real}$$

$$\Rightarrow 3\cos\beta=0$$

$$\Rightarrow\beta=\frac{3\pi}{2}$$

$$\Rightarrow Z_{\alpha\beta}=\sin\frac{5\pi}{2}+\mathrm{i}\cos 3\pi=1-\mathrm{i}$$

$$Z_{\alpha\beta} = \sin\frac{7\pi}{2} + i\cos 3\pi = -1 - i$$

$$or \text{ Required value } = \left[i(1-i) + \frac{1}{i(1+i)}\right] + \left[i(-1-i) + \frac{1}{i(-1+i)}\right]$$

$$= i(-2i) + \frac{1}{i}\frac{i}{(-2)} \Rightarrow 2 - 1 = 1$$

Q. 3 If α, β are the roots of the equation

$$x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right) + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0,$$

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$,, is :

Option 1:

$3x^2 - 20x - 12 = 0$

Option 2:

$3x^2 - 10x - 4 = 0$

Option 3:

$3x^2 - 10x + 2 = 0$

Option 4:

$3x^2 - 20x + 16 = 0$

Correct Answer:

$3x^2 - 20x - 12 = 0$

Q. 4 Let $A = \begin{pmatrix} 4 & -2 \\ \alpha & \beta \end{pmatrix}$.

If $A^2 + \gamma A + 18I = O$, then $\det(A)$ is equal to_____

Option 1:

-18

Option 2:
18

Option 3:
-50

Option 4:
50

Correct Answer:
18

Solution:

Using caylay Hamiltion theorem

$|A-2I|=0$

$\Rightarrow \begin{vmatrix} 4-2 & -2 \\ \alpha & \beta-2 \end{vmatrix}=0$

$\text{Rightarrow}(4-2)(\beta-2)+2\alpha=0$

$\Rightarrow 2^2-\beta\lambda-4\lambda+4\beta+2\alpha=0$

$\therefore \quad A^2-(4+\beta)A+(4\beta+2\alpha)I=0$

$\therefore \quad 4\beta+2\alpha=18$

$|A|=4\beta+2\alpha=18$

$\therefore \quad \text{option (B)}$

Q. 5 If for $p \neq q \neq 0$, the function $f(x)=\dfrac{\sqrt[7]{p(729+x)}-3}{\sqrt[3]{729+qx}-9}$ is continuous at $x=0$, then :

Option 1:
$7pqf(0)-1=0$

Option 2:
$63qf(0)-p^2=0$

Option 3:
$21qf(0)-p^2=0$

Option 4:

$7pqf(0) - 9 = 0$

Correct Answer:

$63qf(0) - p^2 = 0$

Solution:

$$f(0) = \lim_{x\to 0} \frac{\sqrt[7]{p(729+x)} - 3}{\sqrt[3]{729+qx} - 9}$$

For linit to exist, numeratea $\to 0$

$\therefore p = 3$

$$= \lim_{x\to 0} \frac{(2187+3x)^{\frac{1}{7}} - 3}{(729+qx)^{1/3} - 9}$$

$$= \lim_{x\to 0} \frac{3\left[\left(1+\frac{x}{q_{29}}\right)^{\frac{1}{7}} - 1\right]}{9\left[\left(1+\frac{q^x}{729}\right)^{1/3} - 1\right]}$$

$$= \lim_{x\to 0} \frac{1\left(\frac{x}{7\times 729} + \cdots\right)}{3\left(\frac{9x}{3.729} + \cdots\right)}$$

$$= \frac{1\times 3\times 729}{3\times 7\times 729\times 9}$$

$$= \frac{1}{7q}$$

$$\Rightarrow f(0) = \frac{1}{7q}$$

$$\Rightarrow \quad 7q\, f(0) = 1$$

$$\Rightarrow 7p^2q\, f(0) = p^2$$

$$\Rightarrow 63q\, f(0) = p^2$$

$\therefore$ option (B)

Q. 6 Let $f(x) = 2 + |x| - |x-1| + |x+1|, x \in \mathbf{R}$.

Consider

$$(S_1) : f'\left(-\frac{3}{2}\right) + f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) = 2$$

$$(S_2) : \int_{-2}^{2} f(x)dx = 12$$

Then,

Option 1:
both (S_1) and (S_2) are correct

Option 2:
both (S_1) and (S_2) are wrong

Option 3:
only (S_1) is correct

Option 4:
only (S_2) is correct

Correct Answer:
only (S_2) is correct

Solution:

$$f(x) = 2 + |x| - |x-1| + |x+1|$$

For $x < -1$

$$f(x) = 2 - x + x - 1 - x - 1$$

$$= -x$$

$$f(x) = 2 - x + x$$

$$= x + 2$$

For $0 \leqslant x < 1$

$$f(x) = 2 + x + x - 1 + x + 1$$

$$= 3x + 2$$

For $x \geqslant 1$

$$f(x) = 2 + x - x + 1 + x + 1$$

$$= x + 4$$

S_1 :

$$f'\left(-\frac{3}{2}\right) = -1$$

$$f'\left(-\frac{1}{2}\right) = 1$$

$$f'\left(\frac{1}{2}\right) = 3$$

$$f'\left(\frac{3}{2}\right) = 1$$

$$\text{sum} = 4$$

$\therefore S_1$ is incorrect

S_2 :

$$\int_{-2}^{2} f(x)dx$$

$$= \int_{-2}^{-1} xdx + \int_{-1}^{0} (x+2)dx + \int_{0}^{1} (3x+2)dx + \int_{1}^{2} (x+4)dx$$

$$= \frac{x^2}{2}\int_{-2}^{-1} + \frac{x^2}{2} + 2x\int_{-1}^{0} \frac{3x^2}{2} + 2x\int_{0}^{1} + \frac{x^2}{2} + 4x\int_{1}^{2}$$

$$= \frac{3}{2} + \frac{3}{2} + \frac{7}{2} + \frac{11}{2} = 12$$

$\therefore$ option (D)

Q. 7 Let the sum of an infinite G.P., whose first term is a and the common ratio is r, be 5 . Let the sum of its first five terms be $\frac{98}{25}$. Then the sum of the first 21 terms of an AP, whose first term is $10ar$, n^{th} term is a_n and the common difference is $10a^2$, is equal to :

Option 1:
$21a_{11}$

Option 2:
$22a_{11}$

Option 3:
$15a_{16}$

Option 4:
$14a_{16}$

Correct Answer:
$21a_{11}$

Solution:

$$\text{Sum of 21 terms}$$
$$= \frac{21}{2}\left[2(10ar) + 20 \cdot \text{ loar }^2\right]$$
$$= 21r\left(10ar + 1.00a^2\right)$$
$$= 21\left[a_1 + (11-1)\cdot 10a^2\right)$$
$$= 21\left(a_1 + (11-1)d\right)$$
$$= 21 \cdot a_{11}$$
$$\therefore \text{ option (A)}$$

Q. 8 The area of the region enclosed by $y \leq 4x^2, x^2 \leq 9y$ and $y \leq 4$, is equal to :

Option 1:
$\frac{40}{3}$

Option 2:
$\frac{56}{3}$

Option 3:
$\frac{112}{3}$

Option 4:
$\frac{80}{3}$

Correct Answer:
$\frac{80}{3}$

Solution:

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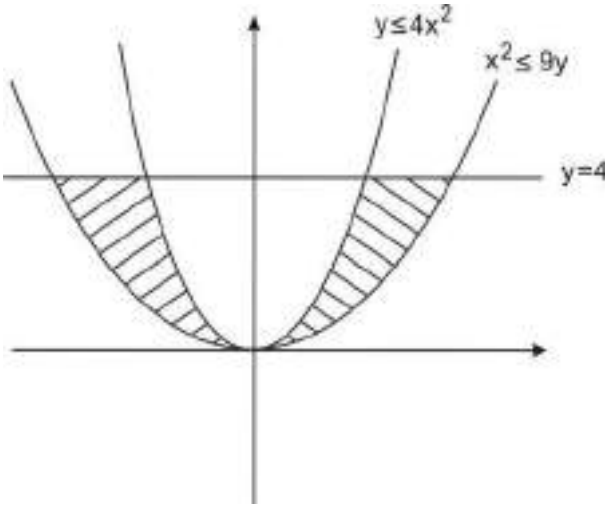


$$\text{Area} = 2\int_0^4 \left(3\sqrt{y} - \frac{\sqrt{y}}{2}\right) dx$$

$$= 2 \times \frac{5}{2} \cdot \left.\frac{y^{\frac{3}{2}}}{\frac{3}{2}}\right|_0^4$$

$$= \frac{10}{3} \cdot (4)^{\frac{3}{2}}$$

$$= \frac{80}{3}$$

$\therefore$ option (D)

Q. 9 $\int_0^2 \left(|2x^2 - 3x| + \left[x - \frac{1}{2}\right]\right) dx$, where $[t]$ is the greatest integer function, is equal to :

Option 1:
$\frac{7}{6}$

Option 2:
$\frac{19}{12}$

Option 3:
$\frac{31}{12}$

Option 4:
$\frac{3}{2}$

Correct Answer:

$\frac{19}{12}$

Solution:

$$I=\int_0^2 \left|2x^2-3x\right| dx+\int_0^2 \left[x-\frac{1}{2}\right] dx$$

$$=\int_0^{3/2}\left(3x-2x^2\right)+\int_{0/2}^2 \left(2x^2-3x\right) dx+\int_0^{\frac{1}{2}}(-1)dx+\int_{\frac{3}{2}}^{\frac{3}{2}} 0dx+\int_1^2 1dx$$

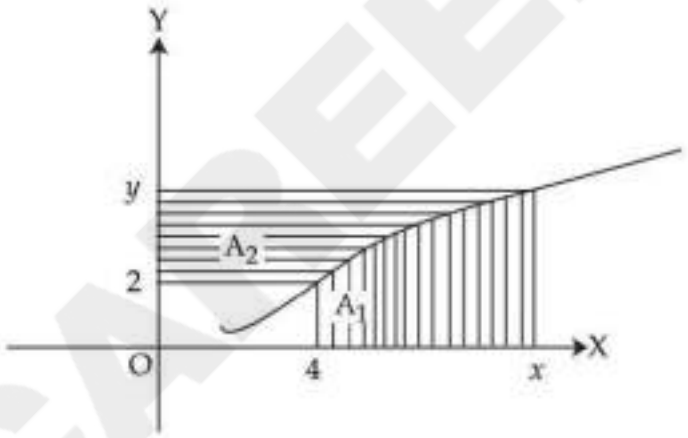
$$=\frac{3x^2}{2}-\frac{2x^3}{3}\Big|_0^{3/2}+\frac{2x^3}{3}-\frac{3x^2}{2}\Big|_{3/2}^2-x|_0^{\frac{1}{2}}+x|_{\frac{3}{2}}^2$$

$$=\left(\frac{27}{8}-\frac{27}{12}\right)+\left(\frac{16}{3}-6-\frac{27}{12}+\frac{27}{8}\right)-\left(\frac{1}{2}\right)+\left(2-\frac{3}{2}\right)$$

$$=\frac{19}{12}$$

$\therefore$ Option (B)

Q. 10 Consider a curve $y=y(x)$ in the first quadrant as shown in the figure. Let the area A_1 is twice the area A_2. Then the normal to the curve perpendicular to the line $2x-12y=15$ does NOT pass through the point.



Option 1:

$(6,21)$

Option 2:

$(8,9)$

Option 3:

$(10,-4)$

Option 4:
$(12, -15)$

Correct Answer:
$(10, -4)$

Solution:
$A_1 + A_2 = xy - 8$

$\Rightarrow A_1 + \frac{A_1}{2} = xy - 8$

$\Rightarrow \quad A_1 = \frac{2}{3}(xy - 8)$

$\Rightarrow \int_4^x f(x)dx = \frac{2}{3}x \cdot f(x) - \frac{16}{3}$

Differentiate wir.t. x

$f(x) = \frac{2}{3}f(x) + \frac{2}{3}xf'(x)$

$\Rightarrow \quad \frac{f(x)}{3} = \frac{2}{3}xf'(x)$

$\Rightarrow \quad \frac{f'(x)}{f(x)} = \frac{1}{2x}$

$\Rightarrow 2\int \frac{f'(x)}{f(x)}dx = \int \frac{dx}{x}$

$\therefore$ option C

Q. 11 The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0, x + py = 39$ and $x - y = 3$ respectively and $P(2, 3)$ is its circumcentre. Then which of the following is NOT true ?

Option 1:
$(AC)^2 = 9p$

Option 2:
$(AC)^2 + p^2 = 136$

Option 3:
$32 < \text{area}\,(\Delta ABC) < 36$

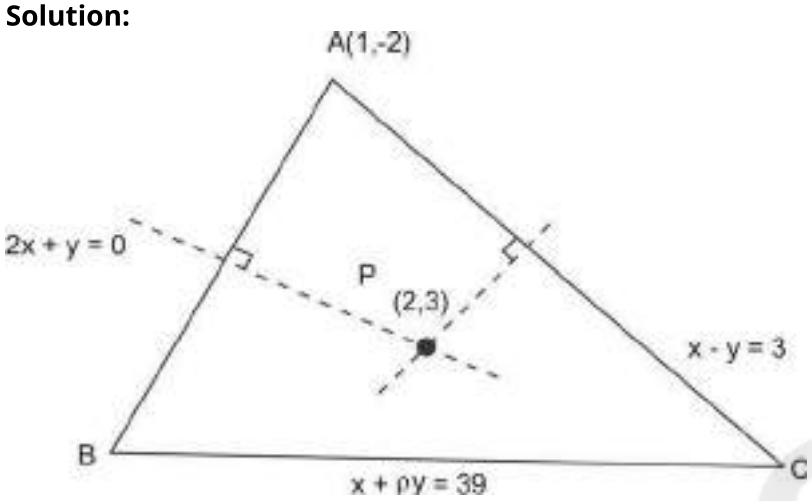
Option 4:

$34 < \text{area}(\triangle ABC) < 38$

Correct Answer:

$34 < \text{area}(\triangle ABC) < 38$

Solution:



perpendicular bisector of AB:

Line perpendicular to AB and passing therough (2,3)

$$y - 3 = \frac{1}{2}(x - 2)$$

$$\Rightarrow 2y - 6 = x - 2$$

$$\Rightarrow x - 2y + 4 = 0$$

B is image of A(1,-2) in $x - 2y + 4y = 0$

$$\frac{x-1}{1} = \frac{y+2}{-2} = \frac{-2(1+4+4)}{1+4}$$

$$\frac{x-1}{1} = \frac{y+2}{-2} = \frac{-18}{5}$$

$$x = \frac{-13}{5}, y = \frac{26}{5}$$

$$\therefore B\left(-\frac{13}{5}, \frac{26}{5}\right)$$

Similary C(7,4)

$$\text{Area of } \Delta ABC = \frac{1}{2}\begin{vmatrix} 1 & -2 & 1 \\ \frac{-13}{5} & \frac{26}{5} & 1 \\ 7 & 4 & 1 \end{vmatrix}$$

$$= 32.4$$

option (D) is wrong.

Q. 12 A circle C_1 passes through the origin O and has diameter 4 on the positive x-axis. The line $y = 2x$ gives a chord OA of circle C_1. Let C_2 be the circle with OA as a diameter. If the tangent to C_2 at the point A meets the x-axis at P and y-axis at Q, then $QA : AP$ is equal to:

Option 1:
$1 : 4$

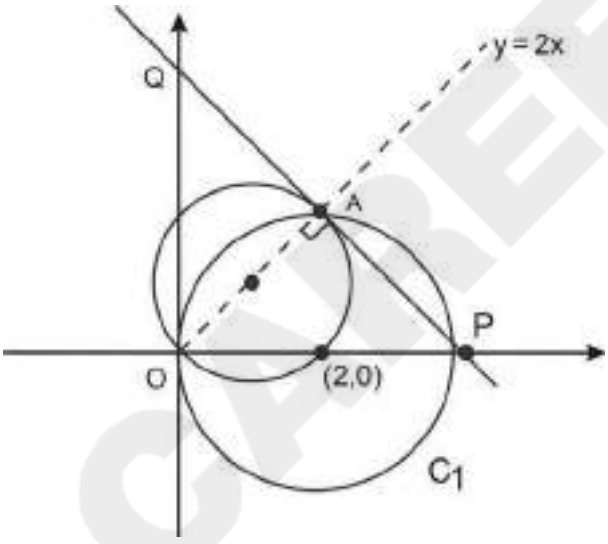
Option 2:
$1 : 5$

Option 3:
$2 : 5$

Option 4:
$1 : 3$

Correct Answer:
$1 : 4$

Solution:



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Equation of C_1 is

$(x-2)^2+y^2=2^2$

$\Rightarrow \quad x^2+y^2-4x=0$

For point A

$x^2+(2x)^2-4x=0$

$5x^2-4x=0$

$x=\frac{4}{5}$

$\therefore A\left(\frac{4}{5},\frac{8}{5}\right)$

$QP \perp OA$

$\therefore Equation of QP$

$\left(y-\frac{8}{5}\right)=-\frac{1}{2}\left(x-\frac{4}{5}\right)$

$\Rightarrow 5y-8=-\frac{1}{2}(5x-4)$

$\Rightarrow 10y-16=-5x+4$

$\Rightarrow 5x+10y-20=0$

$\Rightarrow x+2y-4=0$

$\therefore P(4,0), Q(0,2)$

Let A divide QP in ratio h: 1

$\frac{4\lambda+0}{R+1}=\frac{4}{5}$

$\Rightarrow 5\lambda=\lambda+1$

$\Rightarrow \lambda=\frac{1}{4}$

$\therefore$ Ratio $=1:4$

$\therefore$ option (A)

Q. 13 If the length of the latus rectum of a parabola, whose focus is (a, a) and the tangent at its vertex is $x+y=a$, is 16 , then $|a|$ is equal to :

Option 1:

$2\sqrt{2}$

Option 2:
$2\sqrt{3}$

Option 3:
$4\sqrt{2}$

Option 4:
4

Correct Answer:
$4\sqrt{2}$

Solution:
Distance between focus and tangent at vector

$$= \frac{|a + a - a|}{\sqrt{1+1}}$$

$$= \frac{|a|}{\sqrt{2}}$$

$$\text{Length of L} \cdot \text{R} \cdot = \frac{4|a|}{\sqrt{2}} = 16$$

$$\Rightarrow |a| = 4\sqrt{2}$$

$\therefore$ option (c)

Q. 14 If the length of the perpendicular drawn from the point $P(a, 4, 2),\ a > 0$ on the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1_3}$ is $2\sqrt{6}$ units and $Q(\alpha_1, \alpha_2, \alpha_3)$ is the image of the point P in this line, then $a + \sum_{i=1} \alpha_i$ is equal to:

Option 1:
7

Option 2:
8

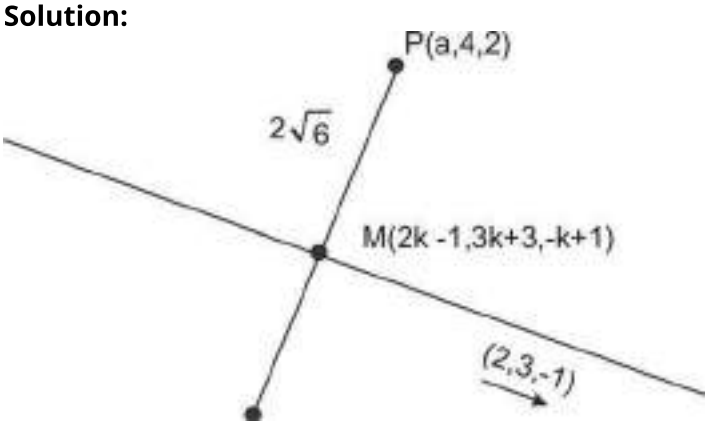
Option 3:
12

Option 4:
14

Correct Answer:

8

Solution:



DR's of P M= $(2k-1-a, 3k-1, -k-1)$

As PM $\perp$ line

$\Rightarrow 2(2k-1-a)+3(3k-1)-1(-k-1)=0$

$\Rightarrow 4k-2-2a+9)+3(3k-1)-1(-k-1)=0$

$\Rightarrow 14k-4-2a=0$

$\Rightarrow 7k-2=a$

Also $PM=2\sqrt{6}$

$\Rightarrow PM^2=24$

$(-5k+1)^2+(3k-1)^2+(-k-1)^2=24$

$\Rightarrow k=1$

$\Rightarrow a=5$

$\therefore P(5,4,2), M(1,6,0)$

Using mid-point formula

$\frac{5+\alpha_1}{2}=1, \frac{4+\alpha_2}{2}=6, \frac{2+\alpha_3}{2}=0$

$\alpha_1=-3, \alpha_2=8, \alpha_3=-2$

$\therefore a+\sum\alpha$

$=5-3+8-2$

$=8$

$\therefore$ option (B)

Q. 15 If the line of intersection of the planes $ax + by = 3$ and $ax + by + cz = 0, a > 0$ makes an angle 30° with the plane $y - z + 2 = 0$, then the direction cosines of the line are :

Option 1:
$\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$

Option 2:
$\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0$

Option 3:
$\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$

Option 4:
$\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

Correct Answer:
$\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$

Solution:

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Direction ratios of the line of intersection

$$= \begin{vmatrix} i & j & k \\ a & b & 0 \\ a & b & c \end{vmatrix}$$

$= bci - acj$

$= c(bi - aj)$

$\therefore$ Direction ratios of b,-a, 0

Angle between line and $y - z + 2 = 0$ is 30°

$$\therefore \sin(30^\circ) = \left|\frac{(bi - aj)\cdot(j - k)}{\sqrt{a^2 + b^2}\sqrt{1+1}}\right|$$

$$\Rightarrow \quad \frac{1}{2} = \left|\frac{-a}{\sqrt{a^2 + b^2}\sqrt{2}}\right|$$

$$\Rightarrow \sqrt{a^2 + b^2} = \sqrt{2}|a|$$

$$\Rightarrow a^2 + b^2 = 2a^2$$

$$\Rightarrow a^2 = b^2$$

option (A) and (B) satisfy these conditions

Q. 16 Let x have a binomial distribution $B(n, p)$ such that the sum and the product of the mean and variance of x are 24 and 128 respectively. If $P(X > n - 3) = \frac{k}{2^n}$, then K is equal to :

Option 1:
528

Option 2:
529

Option 3:
629

Option 4:
630

Correct Answer:
529

Solution:

$$np + npq = 24 \Rightarrow mp(1+q) = 24$$
$$\Rightarrow np(2-p) = 24$$
$$\Rightarrow np = \frac{24}{2-p}$$
$$\text{Also} np \cdot npq = 128$$
$$\Rightarrow n^2p^2q = 128$$
$$\Rightarrow n^2p^2(1-p) = 128$$
$$\Rightarrow \frac{24^2}{(2-p)^2}(1-p) = 128$$
$$\Rightarrow \frac{(1-p)}{(2-p)^2} = \frac{2}{9}$$
$$\Rightarrow 9 - 9p = 2\left(p^2 + 4 - 4p\right)$$
$$\Rightarrow 2p^2 - 8p + 8 + 9p - 9 = 0$$
$$\Rightarrow 2p^2 + p - 1 = 0$$
$$\Rightarrow p = \frac{1}{2} \quad (\text{as } p > 0)$$
$$\Rightarrow n = 32$$
$$\therefore P(X > n-3)$$
$$= P(X > 29)$$
$$= P(30) + P(31) + P(32)$$
$$= {}^{32}C_{30} \cdot \frac{1}{2^{32}} + {}^{32}C_{31} \cdot \frac{1}{2^{32}} + {}^{32}C_{32} \cdot \frac{1}{2^{32}}$$
$$= \frac{529}{2^{32}}$$
$$\therefore \quad k = 529$$
$$\therefore \text{ option (B)}$$

Q. 17 A six faced die is biased such that
$3 \times P(\text{a prime number}) = 6 \times P(\text{a composite number}) = 2 \times P(1)$.
Let x be a random variable that counts the number of times one gets a perfect square on some throws of this die. If the die is thrown twice, then the mean of x is

Option 1:
$\frac{3}{11}$

Option 2:
$\frac{5}{11}$

Option 3:
$\frac{7}{11}$

Option 4:

$\frac{8}{11}$

Correct Answer:

$\frac{8}{11}$

Solution:

$\text{Let } P(1) = k$

$\therefore \quad P(2) = P(3) = P(5) = \frac{2k}{3}$

$\text{and } P(4) = P(6) = \frac{k}{3}$

As Sum=1

$k + (2k) + \frac{2k}{3} = 1$

$\Rightarrow \quad k = \frac{3}{11}$

$\text{Now } n = 2$

$\text{and } P = P(1) + P(4) = k + \frac{k}{3} = \frac{4k}{3}$

$= \frac{4}{11}$

$\therefore \text{ Mean } = np = 2 \times \frac{4}{11} = \frac{8}{11}$

$\therefore \text{ Option(D)}$

Q. 18 The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is $45°$. Let R be a point on AQ and from a point B, vertically above, the angle of elevation of P is $60°$. If $\angle BAQ = 30°, AB = d$ and the area of the trapezium $PQRB$ is α, then the ordered pair (d, α) is :

Option 1:

$(10(\sqrt{3} - 1), 25)$

Option 2:

$\left(10(\sqrt{3}-1), \frac{25}{2}\right)$

Option 3:

$(10(\sqrt{3}+1), 25)$

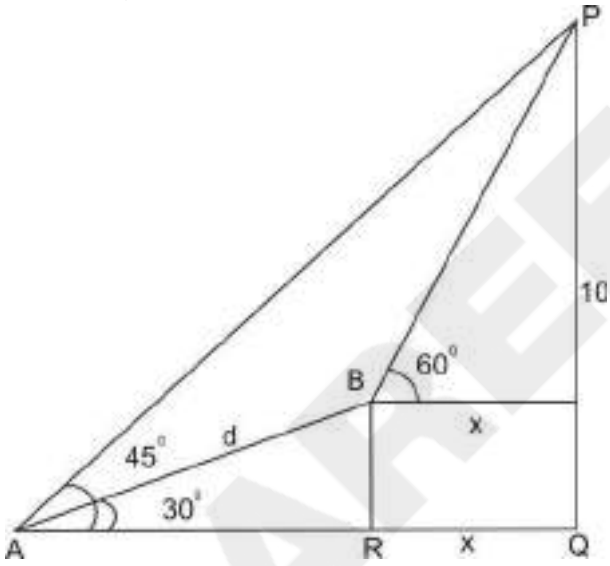
Option 4:

$\left(10(\sqrt{3}+1), \frac{25}{2}\right)$

Correct Answer:

$(10(\sqrt{3}-1), 25)$

Solution:



$\frac{PQ}{AQ} = \tan 45^\circ$

$\Rightarrow AQ = PQ = 10$

Now $AR + RQ = AQ$

$d\cos 30^\circ + x = 10 \ldots$ i

Also, $\frac{10 - d\sin 30^\circ}{x} = \tan 60^\circ$

$x = \frac{10 - \frac{d}{2}}{\sqrt{3}}$

Put the in (i)

$$\frac{\sqrt{3}d}{2}+\frac{10-\frac{d}{2}}{\sqrt{3}}=10$$

$$\Rightarrow\frac{3d+20-d}{2\sqrt{3}}=10$$

$$\Rightarrow 2d+20=20\sqrt{3}$$

$$\Rightarrow 2d=20(\sqrt{3}-1)$$

$$\Rightarrow d=10(\sqrt{3}-1)$$

Now area of PQRB

$$=\frac{1}{2}(d\sin 30+10)\cdot x$$

$$=\frac{1}{2}(5(\sqrt{3}-1)+10)\cdot\frac{1}{\sqrt{3}}(10-5(\sqrt{3-1}))$$

$$=\frac{1}{2}(5\sqrt{3}+5)\cdot\frac{1}{\sqrt{3}}(15-5\sqrt{3})$$

$$=\frac{1}{2\sqrt{3}}\cdot 5(\sqrt{3}+1)\cdot 5\sqrt{3}(\sqrt{3}-1)$$

$$=\frac{5.5}{2}(3-1)$$

$$=25$$

$\therefore$ option (A)

Q. 19

Let $S=\left\{\theta\in\left(0,\frac{\pi}{2}\right):\sum_{m=1}^{9}\sec\left(\theta+(m-1)\frac{\pi}{6}\right)\sec\left(\theta+\frac{m\pi}{6}\right)=-\frac{8}{\sqrt{3}}\right\}$.

Then :

Option 1:

$S=\left\{\frac{\pi}{12}\right\}$

Option 2:

$$S=\left\{\frac{2\pi}{3}\right\}$$

Option 3:

$$\sum_{\theta\in S}\theta=\frac{\pi}{2}$$

Option 4:

$$\sum_{\theta\in S}\theta=\frac{3\pi}{4}$$

Correct Answer:

$$\sum_{\theta\in S}\theta=\frac{\pi}{2}$$

Solution:

$$\sec\left(\theta+\frac{m\pi}{6}-\frac{\pi}{6}\right)\cdot\sec\left(\theta+\frac{m\pi}{6}\right)$$

$$=\frac{1}{\sin\left(\frac{\pi}{6}\right)}\frac{\sin\left(\frac{\pi}{6}\right)}{\left[\cos\left(\theta+\frac{m\pi}{6}-\frac{\pi}{6}\right)\cos\left(\theta+m\frac{\pi}{6}\right)\right]}$$

$$=2\left[\frac{\sin\left[\left(\theta+\frac{m\pi}{6}\right)-\left(\theta+\frac{m\pi}{6}-\frac{\pi}{6}\right)\right]}{\cos\left(\theta+\frac{m\pi}{6}-\frac{\pi}{6}\right)\cos\left(\theta+\frac{m\pi}{6}\right)}\right]$$

$$=2\left[\tan\left(\theta+\frac{m\pi}{6}\right)-\tan\left(\theta+\frac{m\pi}{6}-\frac{\pi}{6}\right)\right]$$

$$\sum_{m=1}^{9} 2\left[\tan\left(\theta+\frac{9\pi}{6}\right)-\tan(\theta)\right]$$

$$\frac{-2}{\sin\theta\cos\theta}=\frac{-4}{\sin 2\theta}$$

$$\frac{-4}{\sin 2\theta}=-\frac{8}{\sqrt{3}}$$

$$\Rightarrow \sin 2\theta=\frac{\sqrt{3}}{2}$$

$$\Rightarrow 2\theta=60^{\circ},120^{\circ}$$

$$\Rightarrow \theta=30^{\circ},60^{\circ}$$

$$S=\left\{\frac{\pi}{6},\frac{\pi}{3}\right\}$$

Q. 20 If the truth value of the statement $(\mathrm{P}\wedge(\sim \mathrm{R}))\rightarrow((\sim \mathrm{R})\wedge \mathrm{Q})$ is F, then the truth value of which of the following is f ?

Option 1:
$\mathrm{P}\vee \mathrm{Q}\rightarrow\sim \mathrm{R}$

Option 2:
$\mathrm{R}\vee \mathrm{Q}\rightarrow\sim \mathrm{P}$

Option 3:
$\sim(\mathrm{P}\vee \mathrm{Q})\rightarrow\sim \mathrm{R}$

Option 4:
$\sim(\mathrm{R}\vee \mathrm{Q})\rightarrow\sim \mathrm{P}$

Correct Answer:
$\sim(\mathrm{R}\vee \mathrm{Q})\rightarrow\sim \mathrm{P}$

Solution:

$$P\wedge \sim R \rightarrow (\sim R \wedge Q) \text{ is false}$$
$\therefore P\wedge \sim R$ is true and $((\sim R) \wedge Q)$ is false
$\Rightarrow$ P is true R is false , Q is false
$(A) : P \vee Q \rightarrow \sim R \Rightarrow T \rightarrow T \Rightarrow T$
$(B) : R \vee Q \rightarrow \sim P \Rightarrow F \rightarrow F \Rightarrow T$
$(C) : \sim (P \vee Q) \rightarrow \sim R \Rightarrow F \rightarrow T \Rightarrow T$
$(D) \sim (R \vee Q) \rightarrow \sim P \Rightarrow T \rightarrow F \Rightarrow F$
$\therefore$ option (D)

Q. 21

Consider a matrix $A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{bmatrix}$, where α, β, γ are three distinct natural numbers.

If $\frac{\det(\text{adj}(\text{adj}(\text{adj}(\text{adj } A))))}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}} = 2^{32} \times 3^{16}$, then the number of such 3 - tuples (α, β, γ) is_____.

Correct Answer:
42

Solution:

$$|A| = \begin{vmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{vmatrix}$$

$R_3 \rightarrow R_1 + R_3$

$$= \begin{vmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \alpha+\beta+\gamma & \alpha+\beta+\gamma & \alpha+\beta+\gamma \end{vmatrix}$$

$$= (\alpha+\beta+\gamma^*) \begin{vmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ 1 & 1 & 1 \end{vmatrix}$$

$= (\alpha+\beta+\gamma)(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$

Now $\dfrac{|\operatorname{adj}(\operatorname{adj}(\operatorname{adj}(\operatorname{adj} A)))|}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}} = 2^{32}\cdot 3^{16}$

$\Rightarrow \dfrac{|A|^{(n-1)^4}}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}} = 2^{3^2\cdot 3^{16}}$

$\Rightarrow \left(\alpha+\beta^{\beta+\gamma}\right)^{16} = 2^{3^2}\cdot 3^{16}$

$\Rightarrow \quad \alpha+\beta+\gamma = 4\cdot 3 = 12$

$\Rightarrow \alpha+\beta+\gamma = 12$

Totel Natural number solutions of this equation ${}^{12-1}C_{3-1} = {}^{11}C_2$

$= 55$

Removing cases where α, β, γ are not distinat
$\alpha = \beta = \gamma = 4$

$\alpha = \beta \Rightarrow 4$ cases $(1,1,10), (2,2,8), (3,3,6), (5,5,2)$

Similarly 4 cases each for $\beta = \gamma$ and $\alpha = \gamma$

$\therefore$ Answer is $55 - 1 - 4 - 4 - 4 = 42$

Ans: (42)

Q. 22 The number of functions f, from the set $A = \{x \in \mathbf{N} : x^2 - 10x + 9 \le 0\}$ to the set $B = \{n^2 : n \in \mathbf{N}\}$ such that $f(x) \le (x-3)^2 + 1$, for every $x \in A$, is_____.

Correct Answer:
1440

Solution:

$x^2 - 10x + 9 \leqslant 0$

$(x-1)(x-9) \leqslant 0$

$x \in [1, 9]$

$A = \{1, 2, 3, \ldots, 9\}$

$f(1) \le (1-3)^2 + 1 \Rightarrow f(1) \leqslant 5 \Rightarrow$ 2 choices in B which are 1,4

similarly f(2), f(3), ..., f(9) have 1,1,1,2,3,4,5,6 options

$\therefore$ Required value $= 2\times 1\times 1\times 1\times 2\times 3\times 4\times 5\times 6$

$= 1440$

Ans: 1440

Q. 23 Let for the 9^{th} term in the binomial expansion of $(3+6x)^n$, in the increasing powers of $6x$, to be the greatest for $x=\frac{3}{2}$, the least value of n is n_0. If K is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k+n_0$ is equal to :

Correct Answer:
24

Solution:

$$\frac{n+1}{1+\left|\frac{a}{b}\right|}=\frac{n+1}{1+\left|\frac{3}{6\times\frac{3}{2}}\right|}=\frac{3(n+1)}{4}$$

As g^{th} term is greatest,

$\therefore 8<\frac{3(n+1)}{4}<9$

$32<3(n+1)<36$

$10.66<n+1<12$

$9.66<n<11$

$\therefore n_0=10$

$k=\frac{{}^{10}C_6\cdot(3)^4(6)^6}{{}^{10}C_3\cdot 3^7(6)^3}=14$

$\therefore \quad k+n_0=24$

Q. 24 $\frac{2^3-1^3}{1\times 7}+\frac{4^3-3^3+2^3-1^3}{2\times 11}+\frac{6^3-5^3+4^3-3^3+2^3-1^3}{3\times 15}+\ldots+\frac{30^3-29^3+28^3-27^3+\ldots+2^3-1^3}{15\times 63}$ is equal to_____.

Correct Answer:
120

Solution:

$2^3 - 1^3 = 7$

$4^3 - 3^3 + 2^3 - 1^3 = 37 + 7 = 44$

$6^3 - 5^3 + 4^3 - 3^3 + 2^3 - 1^3 = 91 + 44 = 135$

Required exprexion

$= 1 + \frac{4}{2} + \frac{9}{3} + \cdots - \cdots$

$= 1 + 2 + 3 + \ldots + 15$

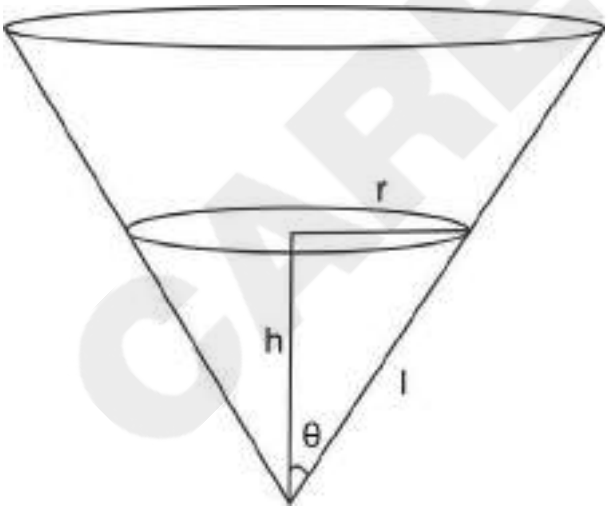
$= 120$

Ans: 120

Q. 25 A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1}\frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is_____

Correct Answer:
5

Solution:
Let height of water be h, radius be r.



$$\tan\theta = \frac{3}{4}$$

$$\Rightarrow \frac{r}{h} = \frac{3}{4} \Rightarrow r = \frac{3h}{4}$$

$$\text{Also } l = \sqrt{r^2 + h^2} = \sqrt{\frac{9}{16}h^2 + h^2} = \frac{5h}{4}$$

$$\text{Wet area} = \pi rl$$

$$= \pi \times \frac{3h}{4} \times \frac{5h}{4}$$

$$A = \frac{15\pi}{16}h^2$$

$$\frac{dA}{dt} = \frac{15\pi}{8}h \cdot \frac{dh}{dt} \cdots \text{(i)}$$

$$\text{Now Volume } V = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi \times \frac{9}{16}h^3$$

$$V = \frac{3\pi}{16}h^3$$

$$\frac{dV}{dt} = \frac{9\pi h^2}{16}\frac{dh}{dt}$$

$$6 = \frac{9\pi}{16} \times 16 \times \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{6}{9\pi} = \frac{2}{3\pi}$$

$$\text{Putting } h = 4 \text{ and } \frac{dh}{dt} = \frac{2}{3\pi} \text{in(i)}$$

$$\frac{dA}{dt} = \frac{15\pi}{8} \times 4 \times \frac{2}{3\pi} = 5$$

Ans: 5

Q. 26 For the curve $C : (x^2 + y^2 - 3) + (x^2 - y^2 - 1)^5 = 0$, the value of $3y' - y^3y''$, at the point $(\alpha, \alpha), \alpha > 0$, on C, is equal to_____.

Correct Answer: 16

Solution:

$$(\alpha^2+\alpha^2-3)+(\alpha^2-\alpha^2-1)^5=1$$

(α,α) lies on given curve, so $\Rightarrow 2\alpha^2=4$

$$\Rightarrow \alpha=\sqrt{2} \quad (\text{as } \alpha>0)$$

Differentiating the curve

$2x+2yy'+5\left(x^2-y^2-1\right)^4(2x-2yy')=0$

Put $x=y=\sqrt{2}$.

$2\sqrt{2}+2\sqrt{2}y'+5\cdot\left(2\sqrt{2}-2\sqrt{2}y'\right)=0$

$\Rightarrow 12\sqrt{2}-8\sqrt{2}y'=0$

$\Rightarrow y'=\frac{3}{2}$

Differentiating again

$2+2\left(yy''+(y')^2\right)+20\left(x^2-y^2-1\right)(2x-2yy')(2x-2yy')+5\left(x^2-y^2-1\right)^4\left(2-2y'^2-2yy'\right)=0$

Put $x=y=\sqrt{2}$ and $y'=\frac{3}{2}$

$y''=-\frac{23}{4\sqrt{2}}$

$\therefore \quad 3y'-y^3y''=3\times\frac{3}{2}-(2\sqrt{2})\times\left(-\frac{23}{4\sqrt{2}}\right)$

$=\frac{9}{2}+\frac{23}{2}=16.$

Ans'.16

Q. 27 Let $f(x)=\min\{[x-1],[x-2],\ldots,[x-10]\}$ where $[t]$ denotes the greatest integer $\leq t$. Then $\int_0^{10} f(x)dx+\int_0^{10}(f(x))^2\,dx+\int_0^{10}|f(x)|dx$ is equal to_____.

Correct Answer: 385

Solution:

clearly $f(x) = [x - 10] = [x] - 10$

For $0 < x < 10 \Rightarrow f(x) < 0$

$\therefore f(x) + |f(x)| = 0$

$$\therefore \int_0^{10} f(x)dx + \int_0^{10} (f(x))^2 dx + \int_0^{10} |f(x)\Big) dx$$

$$= \int_0^{10} (f(x))^2 dx$$

$$= \int_0^1 (-10)^2 dx + \int_1^2 (-9)^2 dx + \ldots + \int_9^{10} (-1)^2 dx$$

$$= 1^2 + 2^2 + \cdots + 10^2$$

$$= \frac{10(11)(21)}{6}$$

$$= 385$$

Ans: 385

Q. 28 Let f be a differentiable function satisfying $f(x) = \frac{2}{\sqrt{3}} \int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right) d\lambda, x > 0$ and $f(1) = \sqrt{3}$. If $y = f(x)$ passes through the point $(\alpha, 6)$, then α is equal to_____.

Correct Answer: 12

Solution:

$$f(x) = \frac{2}{\sqrt{3}} \int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right) d\lambda$$

Differentiating with respect to x

$$f'(x) = \frac{2}{\sqrt{3}} \int_0^{\sqrt{3}} f'\left(\frac{\lambda^2 x}{3}\right) \cdot \frac{\lambda^2}{3} d\lambda$$

$$f'(x) = \frac{2}{\sqrt{3}} \cdot \frac{1}{2} \int_0^{\sqrt{3}} \left[f'\left(\frac{h^2 x}{3}\right) \frac{2\lambda}{3}\right] \lambda d\lambda$$

Using integration by parts

$$f'(x) = \frac{1}{\sqrt{3}}\left[\frac{2}{x}f\left(\frac{x\lambda^2}{3}\right)\Big|_0^{\sqrt{3}} - \int_0^{\sqrt{3}} \frac{-}{x}f\left(\frac{x\lambda^2}{3}\right) d2\right]$$

$$\Rightarrow f'(x) = \frac{1}{\sqrt{3}}\left[\frac{\sqrt{3}}{x}f(x) - \frac{1}{x}\cdot\frac{\sqrt{3}}{2}f(x)\right]$$

$$\Rightarrow \sqrt{3}f'(x) = \frac{\sqrt{3}f(x)}{x} - \frac{\sqrt{3}f(x)}{2x}$$

$$\Rightarrow xf'(x) = f(x) - \frac{f(x)}{2}$$

$$\Rightarrow xf'(x) = \frac{f(x)}{2}$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{1}{2x}$$

$Integrating, \ln f(x) = \frac{1}{2}\ln x + \ln c$

$Using f(1) = \sqrt{3}, \quad c = \sqrt{3}$

$Put y = 6 \Rightarrow \alpha = 12$

$Ans : 12$

Q. 29

A common tangent T to the curves $C_1 : \frac{x^2}{4} + \frac{y^2}{9} = 1$ and $C_2 : \frac{x^2}{42} - \frac{y^2}{143} = 1$ does not pass through the fourth quadrant.

If T touches C_1 at (x_1, y_1) and C_2 at (x_2, y_2), then $|2x_1 + x_2|$ is equal to_____

Correct Answer: 20

Solution:

Let tangent to C_1 be

$y = mx \pm \sqrt{4m^2 + 9}$

and for C_2, tangent be

$y = mx \pm \sqrt{42m^2 - 143}$

For common tangent

$4m^2 + 9 = 42m^2 - 143$

$\Rightarrow 38m^2 = 152$

$\Rightarrow m^2 = 4$

$\Rightarrow m = 2, -2$

For m=2, tangent is $y = 2x + 5$

$x_1 = -\frac{8}{5}, x_2 = \frac{-84}{5}$

$|2x_1 + x_2| = 20$

Ans: 20

Q. 30 Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors such that $\vec{a} \times \vec{b} = 4\vec{c}, \vec{b} \times \vec{c} = 9\vec{a}$ and $\vec{c} \times \vec{a} = \alpha \vec{b}, \alpha > 0$. If $|\vec{a}| + |\vec{b}| + |\vec{c}| = \frac{1}{36}$, then $\sqrt{\alpha}$ is equal to_____.

Correct Answer:

$-\frac{43}{36}$

Solution:

$\vec{a} \times \vec{b} = 4 \cdot \vec{c} \cdots (i)$

$\vec{b} \times \vec{c} = 9 \cdot \vec{a} \cdots (ii)$ *Taking dot products with* $\vec{c}, \vec{a}, \vec{b}$, *we get* $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} \times \vec{a} = \alpha \cdot \vec{b} \cdots (iii)$

$\vec{c} = \vec{c} \cdot \vec{a} = 0 Hence (i) \Rightarrow |\vec{a}| \cdot |\vec{b}| = 4 \cdot |\vec{c}| \cdots (iv)(ii) \Rightarrow |\vec{b}| \cdot |\vec{c}| = 9 \cdot |\vec{a}| \cdots (v)(iii) \Rightarrow |\vec{c}| \cdot |\vec{a}| = \alpha \cdot |\vec{b}| \cdots (vi) Multiplying (iv), (v) and (vi)$

$\Rightarrow |\vec{a}| \cdot |\vec{b}| \cdot |\vec{c}| = 36\alpha \cdots (\text{ vii}$
$Dividing (vii) by (iv)$
$\Rightarrow |\vec{c}^2 = 9\alpha \Rightarrow |\vec{c}| = 3\sqrt{\alpha} \cdots (\text{ viii })$
$Dividing (vii) by (v)$
$\Rightarrow |\vec{a}|^2 = 4\alpha \Rightarrow |\vec{a}| = 2\sqrt{\alpha}$
$Dividing (viii) by (vi)$
$\Rightarrow |\vec{b}|^2 = 36 \Rightarrow |\vec{b}| = 6$
$Now \ , as \ \ given$
$, 3\sqrt{\alpha} + 2\sqrt{\alpha} + 6 = \frac{1}{36} \Rightarrow \sqrt{\alpha} = \frac{-43}{36}$

CAREERS360

28 July - Morning Shift

Physics

Q. 1 The dimensions of $\left(\frac{B^2}{\mu_0}\right)$ will be:
(if μ_0: permeability of free space and B : magnetic field)

Option 1:
$[ML^2\ T^{-2}]$

Option 2:
$[MLT^{-2}]$

Option 3:
$[ML^{-1}\ T^{-2}]$

Option 4:
$[ML^2\ T^{-2}\ A^{-1}]$

Correct Answer:
$[ML^{-1}\ T^{-2}]$

Solution:
$\left[\frac{B^2}{\mu_0}\right] = [\text{ Energy density }]$

$= [M'L^{-1}T^{-2}]$

Hence 3 is correct option.

Q. 2 A NCC parade is going at a uniform speed of $9\ \text{km/h}$ under a mango tree on which a monkey is sitting at a height of $19.6\ \text{m}$. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is : (Given $g = 9.8\ \text{m/s}^2$)

Option 1:
$5\ \text{m}$

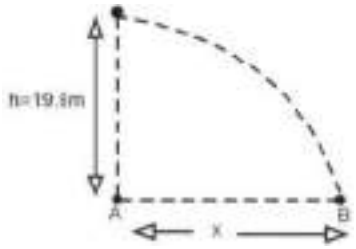
Option 2:
$10\ \text{m}$

Option 3:
$19.8\ \text{m}$

Option 4:
$24.5\ \text{m}$

Correct Answer:
$5\ \text{m}$

Solution:



$$\text{Time to fall} = \sqrt{\frac{2h}{g}} = 2\ \text{s}$$

$$\text{Velocity at parade} = 9\ \frac{\text{km}}{\text{hr}}$$

$$= 9 \times \frac{5}{18}\ \text{m/s}$$

$$= 2.5\ \text{m/s}$$

Cadet will receive mango at a distance x

$$x = v_x t = 2.5 \times 2 = 5\ \text{m}$$

Hence 1 is correct option.

Q. 3 In two different experiments, an object of mass $5\ \text{kg}$ moving with a speed of $25\ \text{ms}^{-1}$ hits two different walls and comes to rest within (i) 3 second, (ii) 5 seconds, respectively. Choose the correct option out of the following :

Option 1:
Impulse and average force acting on the object will be same for both the cases.

Option 2:
Impulse will be same for both the cases but the average force will be different.

Option 3:
Average force will be same for both the cases but the impulse will be different.

Option 4:
Average force and impulse will be different for both the cases.

Correct Answer:
Impulse will be same for both the cases but the average force will be different.

Solution:
$\text{Impulse} = \overline{\text{P}_\text{f}} - \overline{\text{P}_\text{i}} = \Delta\overline{\text{P}}$

$$\text{Avg.Force} = \frac{\Delta\bar{\text{P}}}{\Delta\text{t}} = \frac{\text{Impulse}}{\Delta\text{t}}$$

Impulse is same in both the case as the change in momentum is same in both the case but average force is different as the time is different .

Hence 2 is correct option.

Q. 4 A balloon has mass of $10\ \text{g}$ in air. The air escapes from the balloon at a uniform rate with velocity $4.5\ \text{cm/s}$. If the balloon shrinks in 5 s completely. Then, the average force acting on that balloon will be (in dyne).

Option 1:
3

Option 2:
9

Option 3:
12

Option 4:
18

Correct Answer:
9

Solution:
$\text{m} = \log = 10^{-2}\ \text{kg}$

$$\text{F}_{\text{avg}} = \left(\frac{\Delta\text{m}}{\Delta\text{t}}\right)\text{V}$$

$$= \frac{10^{-2}}{5} \times 4.5 \times 10^{-2}$$

$$= \frac{9}{10} \times 10^{-4}$$

$$= 9 \times 10^{-5}\ \mathrm{N}$$

$$\mathrm{F_{avg}} = 9\ \mathrm{dyne}$$

Hence 2 is correct option.

Q. 5 If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately :

Option 1:
decrease by 2%

Option 2:
decrease by 4%

Option 3:
increase by 2%

Option 4:
increase by 4%

Correct Answer:
increase by 4%

Solution:

$$\mathrm{g} = \frac{\mathrm{GM}}{\mathrm{R}^2}$$

$$\text{as } \mathrm{R} \downarrow, 9 \uparrow$$

$$\frac{\Delta \mathrm{g}}{\mathrm{g}} = -2\frac{\Delta \mathrm{R}}{\mathrm{R}}$$

$\frac{\Delta g}{g} \times 100 = 4\%$

The acceleration due to gravity on the earth's surface will approximately increase by 4%

Hence 4 is correct option.

Q. 6 The force required to stretch a wire of cross-section $1\ \mathrm{cm}^2$ to double its length will be : (Given Yong's modulus of the wire $= 2 \times 10^{11}\ \mathrm{N/m^2}$)

Option 1:
$1 \times 10^7\ \mathrm{N}$

Option 2:
$1.5 \times 10^7\ \mathrm{N}$

Option 3:
$2 \times 10^7\ \mathrm{N}$

Option 4:
$2.5 \times 10^7\ \mathrm{N}$

Correct Answer:
$2 \times 10^7\ \mathrm{N}$

Solution:

$\mathrm{A} = 10^{-4}\ \mathrm{m}^2$

$\Delta\ell = \ell$

$\mathrm{Y} = 2 \times 10^{11}$

$\mathrm{Y} = \frac{\mathrm{F}}{\mathrm{A}} \times \frac{\ell}{\Delta\ell}$

$\mathrm{F} = \mathrm{AY}$

$= 2 \times 10^7\ \mathrm{N}$

Hence 3 is correct option.

Q. 7 A Carnot engine has efficiency of 50%. If the temperature of sink is reduced by $40^\circ\mathrm{C}$, its efficiency increases by 30%. The temperature of the source will be :

Option 1:
$166.7\ \mathrm{K}$

Option 2:
$255.1\ \mathrm{K}$

Option 3:
$266.7\ \mathrm{K}$

Option 4:
$367.7\ \mathrm{K}$

Correct Answer:
$266.7\ \mathrm{K}$

Solution:

$$\eta = 0.5 = 1 - \frac{\mathrm{T_1}}{\mathrm{T_2}} \to 1$$

$$\eta' = 0.5 + \frac{30}{100}\,\eta = 0.65$$

$$\eta' = 0.65 = 1 - \left(\frac{\mathrm{T_1} - 40}{\mathrm{T_2}}\right)$$

$$0.65 = 1 - \frac{\mathrm{T_1}}{\mathrm{T_2}} + \frac{40}{\mathrm{T_2}}$$

$$0.15 = \frac{40}{\mathrm{T_2}}$$

$$\mathrm{T_2} = \frac{4000}{15} = \frac{800}{3}$$

$$\mathrm{T_2} = 266.7\ \mathrm{K}$$

The temperature of the source is $266.7\ \mathrm{K}$

Hence 3 is correct option.

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Q. 8 Given below are two statements :

Statement I : The average momentum of a molecule in a sample of an ideal gas depends on temperature.

Statement II : The rms speed of oxygen molecules in a gas is v. If the temperature is doubled and the oxygen molecules dissociate into oxygen atoms, the rms speed will become $2v$

In the light of the above statements, choose the correct answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are true

Solution:

$\longrightarrow$ The average momentum of a molecule in a sample of an ideal gas is zero

$$\rightarrow V_{rms}=\sqrt{\frac{3RT}{M}}$$

$$\text{if } T_f=\frac{T_i}{2}$$

$$M_f=\frac{M_i}{2} \quad (\text{ after dissociation})$$

$$(V_{rms})_f=2\,(V_{rms})_i$$

$\therefore$ Statement I is false but Statement II is true

Hence 4 is correct option.

Q. 9 In the wave equation

$$y = 0.5 \sin \frac{2\pi}{\lambda}(400t - x)m$$

the velocity of the wave will be:

Option 1:
$200\ m/s$

Option 2:
$200\sqrt{2}\ m/s$

Option 3:
$400\ m/s$

Option 4:
$400\sqrt{2}\ m/s$

Correct Answer:
$400\ m/s$

Solution:

$$y = 0.5 \sin \left[\frac{2\pi}{\lambda}(400t - x)\right]$$

$$\text{velocity of the wave} = \frac{\omega}{k} = v$$

$$= 400\ m/s$$

Hence 3 is correct option.

Q. 10 Two capacitors, each having capacitance $40\mu F$ are connected in series. The space between one of the capacitors is filled with dielectric material of dielectric constant K such that the equivalence capacitance of the system became $24\mu F$. The value of K will be :

Option 1:
1.5

Option 2:
2.5

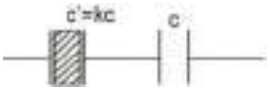
Option 3:
1.2

Option 4:
3

Correct Answer:

1.5

Solution:



$$C_{eq} = \frac{c \times c'}{c + c'} = \frac{kc}{(k+1)}$$

$$24\mu F = \frac{k(40\mu F)}{k+1}$$

$$\frac{3}{5} = \frac{k}{k+1}$$

$$3k + 3 = 5k$$

$$3 = 2k$$

$$k = \frac{3}{2} = 1.5$$

The value of k is 1.5

Hence 1 is correct option.

Q. 11 A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is:

Option 1:

$9 : 1$

Option 2:

$1 : 9$

Option 3:

$4 : 1$

Option 4:

$3 : 1$

Correct Answer:

$9 : 1$

Solution:
For stretaching of wire

$$R \alpha \ell^2$$

$$\frac{R_{New}}{R_{original}} = \frac{\ell^2_{new}}{\ell^2_{original}}$$

$$\ell_{new} = 3\ell_{original} \text{ (Given)}$$

$$\frac{R_{new}}{R_{original}} = \frac{9}{1}$$

Hence 1 is correct option.

Q. 12 The current sensitivity of a galvanometer can be increased by :
(A) decreasing the number of turns
(B) increasing the magnetic field
(C) decreasing the area of the coil
(D) decreasing the torsional constant of the spring
Choose the most appropriate answer from the options given below :

Option 1:
(B) and (C) only

Option 2:
(C) and (D) only

Option 3:
(A) and (C) only

Option 4:
(B) and (D) only

Correct Answer:
(B) and (D) only

Solution:

$$\text{Current sensitivity} = \frac{\phi}{I}$$

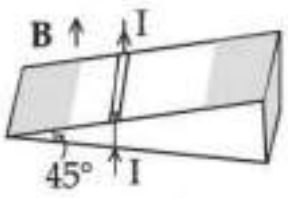
$$= \frac{NAB}{K}$$

From the abare relation, it can be said that the current sensitivity is increased by increasing the magnetic field and

decreasing the torsional constant of the spring
The correct statements are B & D

Hence 4 is correct option.

Q. 13 As shown in the figure, a metallic rod of linear density $0.45\ \text{kg m}^{-1}$ is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The minimum current flowing in the rod required to keep it stationary, when $0.15\ \text{T}$ magnetic field is acting on it in the vertical upward direction, will be :
{ Use $g = 10\ \text{m/s}^2$}



Option 1:
$30\ \text{A}$

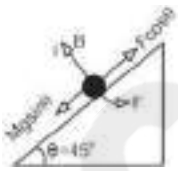
Option 2:
$15\ \text{A}$

Option 3:
$10\ \text{A}$

Option 4:
$3\ \text{A}$

Correct Answer:
$30\ \text{A}$

Solution:



Force on current carrying conductor
is $\text{F} = \text{BI}\ell \sin 90^\circ$

$\text{F} = (0.15) \times \text{i} \times \ell \rightarrow (1)$
For stationary rod,

$\text{mg} \sin\theta = \text{F} \cos\theta = \text{Bi}\ell \cos\theta$

$\frac{\text{mg}}{\ell} = \text{Bi}$

$0.45 \times 10 = 0.15\text{i}$

$i = 30\ A$

Hence 1 is correct option.

Q. 14 The equation of current in a purely inductive circuit is $5\sin(49\pi t - 30^\circ)$. If the inductance is 30mH then the equation for the voltage across the inductor, will be :
$\{\ \text{Let}\ \pi = \frac{22}{7}\}$

Option 1:
$1.47\sin(49\pi t - 30^\circ)$

Option 2:
$1.47\sin(49\pi t + 60^\circ)$

Option 3:
$23.1\sin(49\pi t - 30^\circ)$

Option 4:
$23.1\sin(49\pi t + 60^\circ)$

Correct Answer:
$23.1\sin(49\pi t + 60^\circ)$

Solution:
$I_L = 5\sin(49\pi t - 30^\circ)$

$L = 30\text{mH}$

$X_L = \omega L = 2\pi fL$

$= 49\pi \times 30 \times 10^{-3}$

$= 49 \times \frac{22}{7} \times 30 \times 10^{-3}$

$X_L = 210 \times 22 \times 10^{-3}$

$X_L = 462 \times 10^{-2}$

$V_L = (I_{L_0})\, X_L$

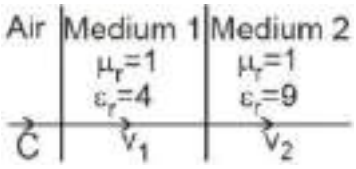
$V_L = (23.1)\sin(49\pi t - 30^\circ + 90^\circ)$

Voltage is ahead of current by 90° in pure inductive ckt

$$V_L = 23.1 \sin(49\pi t + 60^\circ)$$

Hence 4 is correct option.

Q. 15 As shown in the figure, after passing through the medium 1 . The speed of light v_2 in medium 2 will be :



(Given $c = 3 \times 10^8\ ms^{-1}$)

Option 1:
$1.0 \times 10^8\ ms^{-1}$

Option 2:
$0.5 \times 10^8\ ms^{-1}$

Option 3:
$1.5 \times 10^8\ ms^{-1}$

Option 4:
$3.0 \times 10^8\ ms^{-1}$

Correct Answer:
$1.0 \times 10^8\ ms^{-1}$

Solution:

$$V = \frac{c}{\sqrt{\mu_r \varepsilon_r}}$$

$$V_1 = \frac{c}{\sqrt{(u_r)_1 (\varepsilon_r)_1}} = 1.5 \times 10^8\ m/s$$

$$V_2 = \frac{c}{\sqrt{(u_r)_2 (\varepsilon_r)_2}} = \frac{3 \times 10^8}{3}$$

$$= 10^8\ m/s$$

Hence 1 is correct option.

Q. 16 In normal adujstment, for a refracting telescope, the distance between objective and eye piece is $30\ \text{cm}$. The focal length of the objective, when the angular magnification of the telescope is 2 , will be:

Option 1:
$20\ \text{cm}$

Option 2:
$30\ \text{cm}$

Option 3:
$10\ \text{cm}$

Option 4:
$15\ \text{cm}$

Correct Answer:
$20\ \text{cm}$

Solution:
In normal adjustment final image is at infinity

$$\mu_e = f_e$$

Distance between objective and eyepiece$= f_o + f_e$

$$30 = f_o + f_e \rightarrow (1)$$

Angular magnification $= \frac{f_o}{f_e}$

$$2 = \frac{f_o}{f_e}$$

$$f_e = \frac{f_o}{f_e} \rightarrow (2)$$

From eq (1) & (2)

$$30 = \frac{3f_o}{2}$$

$$f_0 = 20\text{cm}$$

Hence 1 is correct option

Q. 17 The equation $\lambda = \frac{1.227}{x}\ \text{nm}$ can be used to find the de-Brogli wavelength of an electron. In this equation x stands for :

Where m= mass of electron

P= momentum of electron

K=Kinetic energy of electron

V=Accelerating potential in volts for electron

Option 1:

$\sqrt{\mathrm{mK}}$

Option 2:

$\sqrt{\mathrm{P}}$

Option 3:

$\sqrt{\mathrm{K}}$

Option 4:

$\sqrt{\mathrm{V}}$

Correct Answer:

$\sqrt{\mathrm{V}}$

Solution:

$\lambda = \frac{1.227}{\mathrm{x}}\ \text{nm}$

For electron, de-broglie wavelength is

$\lambda = \frac{12.27}{\sqrt{\mathrm{V}}}\text{Å}$

$= \frac{1.227}{\sqrt{\mathrm{V}}}\ \text{nm}$

$\mathrm{x} = \sqrt{\mathrm{V}}$

Hence 4 is correct option

Q. 18 The half life period of a radioactive substance is 60 days. The time taken for $\frac{7}{8}$th of its original mass to disintegrate will be :

Option 1:
120 days

Option 2:
130 days

Option 3:
180 days

Option 4:
20 days

Correct Answer:
180 days

Solution:

$$T_{1/2} = 60 \text{days}$$

$$N_t = N_0 - \frac{7}{8}N_0 = \frac{1}{8}N_0 \ \text{(Given)}$$

$$N_t \rightarrow \text{No of nuclei at time t}$$

$$N_0 \rightarrow \text{No of nuclei initially}$$

$$N_t = N_0 2^{-t/T_{1/2}}$$

$$\frac{1}{8} = \frac{1}{2^{t/T_{1/2}}}$$

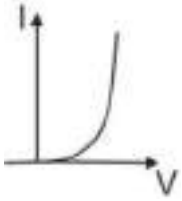
$$t_{T_{1/2}} = 3$$

$$t = 180 \text{ days}$$

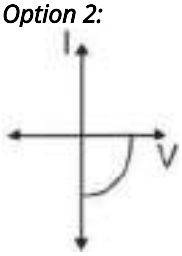
Hence 3 is correct option.

Q. 19 Identify the solar cell characteristics from the following options :

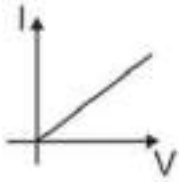
Option 1:



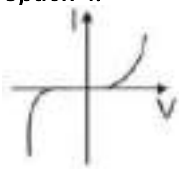
Option 2:



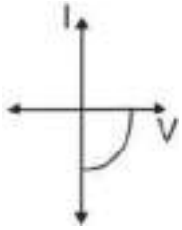
Option 3:



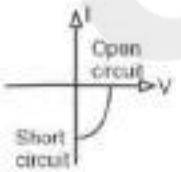
Option 4:



Correct Answer:



Solution:
The solar cell characteristics,



Hence 2 is correct option.

Q. 20 In the case of amplitude modulation to avoid distortion the modulation index (μ) should be :

Option 1:
$\mu \leq 1$

Option 2:
$\mu \geqslant 1$

Option 3:
$\mu = 2$

Option 4:
$\mu = 0$

Correct Answer:
$\mu \leq 1$

Solution:
To avoid distortion, the amplitude modulation should be less than 1 Becouse if $\mu > 1$then there will be distortion of signal due to inferfenence between carrier and messave wave.

Hence 1 is correct option.

Q. 21 If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, the value of α will be _____.

Correct Answer:
5

Solution:
let ,

$\bar{\mathrm{A}} = 2\hat{\mathrm{i}} + 4\hat{\mathrm{j}} - 2\hat{\mathrm{k}}$ and

$\bar{\mathrm{B}} = \hat{\mathrm{i}} + 2\hat{j} + \alpha\hat{\mathrm{k}}$
The projection of $\bar{\mathrm{A}}$ on $\bar{\mathrm{B}}$ is zero

i.e., $\mathrm{A}\cos\theta = 0$

$$\frac{\bar{\mathrm{A}} \cdot \bar{\mathrm{B}}}{|\bar{\mathrm{B}}|} = 0$$

$$2 + 8 - 2\alpha = 0$$

$$\alpha = 5$$

Q. 22 A freshly prepared radioactive source of half life 2 hours 30 minutes emits radiation which is 64 times the permissible safe level. The minimum time, after which it would be possible to work safely with source, will be_____ hours.

Correct Answer:
15

Solution:
Let the permissible level be A

$$\mathrm{A_0} = 64\ \mathrm{A}$$

$$\mathrm{A_t} = \mathrm{A} \quad (\text{ at time t})$$

$$\mathrm{A_t} = \mathrm{A_0} 2^{-\mathrm{t}/\mathrm{T}_{1/2}}$$

$$\frac{1}{64} = \frac{1}{2^6} = \frac{1}{2^{\mathrm{t}/\mathrm{T}_{1/2}}}$$

$$\mathrm{t}/\mathrm{T}_{1/2} = 6$$

$$\mathrm{t} = 6 \times \mathrm{T}_{1/2} = 15\mathrm{hrs}$$

Q. 23 In a Young's double slit experiment, a laser light of $560\ \mathrm{nm}$ produces an interference pattern with consecutive bright fringes' separation of $7.2\ \mathrm{mm}$. Now another light is used to produce an interference pattern with consecutive bright fringes' separation of $8.1\ \mathrm{mm}$. The wavelength of second light is _____nm.

Correct Answer:
630 nm

Solution:

$$\beta_1 = 7.2 \times 10^{-3} = \frac{\lambda_1 \mathrm{D}}{\mathrm{d}} \rightarrow (1)$$

$$\beta_2 = 8.1 \times 10^{-3} = \frac{\lambda_2 \mathrm{D}}{\mathrm{d}} \rightarrow (2)$$

$$\frac{\beta_1}{\beta_2} = \frac{\lambda_1}{\lambda_2} = \frac{7.2}{8.1}$$

$$\frac{560}{\lambda_2} = \frac{8}{9}$$

$$\lambda_2 = 630 \text{ nm}$$

The wavelength of second light is 630 nm

Q. 24 The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are 212 rads^{-1} and 232 rads^{-1}. The value of resistance in the circuit is $R = 5\Omega$. The self inductance in the circuit is _____mH.

Correct Answer:
250

Solution:

$$I_0 = \frac{1}{\sqrt{2}} I_{max}, \text{at}$$

$$\omega = 212 \frac{\text{rad}}{\text{s}} \text{ or } 232 \text{ rad}$$

$$\text{Let } \omega_1 = 212 \frac{\text{rad}}{\text{s}}$$

$$\omega_2 = 232 \frac{\text{rad}}{\text{s}}$$

$$R = 5\Omega$$

We know that,

Resonant frequency $= \omega_0 = \sqrt{\omega_1 \omega_2}$

$$\frac{1}{\sqrt{LC}} = \omega_0 = \sqrt{(212)(232)} \rightarrow (1)$$

$$\text{At resonance, } z = R$$

$$\omega_2 - \omega_1 = \frac{R}{L}$$

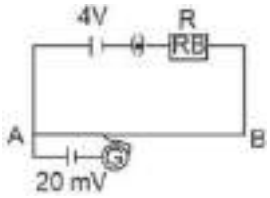
$$20 = \frac{5}{L}$$

$$L = \frac{1}{4} = 0.25H$$

$$L = 250mH$$

The self inductance in the circuit is $250\mathrm{mH}$

Q. 25 As shown in the figure, a potentiometer wire of resistance 20Ω and length $300\ \mathrm{cm}$ is connected with resistance box (R.B.) and a standard cell of emf $4\ \mathrm{V}$. For a resistance $'\mathrm{R}'$ of resistance box introduced into the circuit, the null point for a cell of $20\mathrm{mV}$ is found to be $60\ \mathrm{cm}$. The value of $'\mathrm{R}'$ is_____Ω.



Correct Answer:
780

Solution:

$$\mathrm{I} = \frac{4}{\mathrm{R}+20} \rightarrow (1)$$

$$\frac{\mathrm{V_{AB}}}{\ell_{\mathrm{AB}}} = \phi$$

Let the null point is at P,

$$\mathrm{V_{AP}} = \phi(\ell_{\mathrm{AP}})$$

$$20 \times 10^{-3} = \left(\frac{\mathrm{V_{AB}}}{\ell_{\mathrm{AB}}}\right) \times (60\ \mathrm{cm})$$

$$\mathrm{V_{AB}} = 0.1\mathrm{V} = \mathrm{I}(20)$$

$$0.1 = \left(\frac{4}{\mathrm{R}+20}\right) 20$$

$$\mathrm{R}+20 = 800$$

$$\mathrm{R} = 780\Omega$$

Q. 26 Two electric dipoles of dipole moments $1.2 \times 10^{-30}\text{Cm}$ and $2.4 \times 10^{-30}\text{Cm}$ are placed in two different uniform electric fields of strengths $5 \times 10^{4}\text{NC}^{-1}$ and $15 \times 10^{4}\text{NC}^{-1}$ respectively. The ratio of maximum torque experienced by the electric dipoles will be $\frac{1}{\text{x}}$. The value of x is_____.

Correct Answer:
6

Solution:

$\text{P}_1 = 1.2 \times 10^{-30}\ \text{cm}$

$\text{P}_2 = 2.4 \times 10^{-30}\text{cm}$

$\text{E}_1 = 5 \times 10^{4}\ \text{N/C}$

$\text{E}_2 = 15 \times 10^{4}\ \text{N/C}$

$\tau_{\text{max}} = \text{pE}\sin 90^\circ = \text{pE}$

$\frac{\tau_1}{\tau_2} = \frac{\text{P}_1\text{E}_1}{\text{P}_2\text{E}_2} = \frac{1}{6}$

$\text{x} = 6$

The value of x is 6

Q. 27 The frequency of echo will be_____ Hz if the train blowing a whistle of frequency $320\ \text{Hz}$ is moving with a velocity of $36\ \text{km/h}$ towards a hill from which an echo is heard by the train driver. Velocity of sound in air is $330\ \text{m/s}$.

Correct Answer:
340

Solution:

$\text{f}_o = 320\ \text{Hz}$

Velocity of train $= 36\frac{\text{km}}{\text{hr}}$

$= 36 \times \frac{5}{8}\frac{\text{m}}{\text{s}}$

$$= 10\text{m/s}$$

$$= f_0\left(\frac{V}{V - V_s}\right)$$

$$= 320 \times \left(\frac{330}{330 - 10}\right)$$

$$= 330\text{Hz}$$

Frequency of echo heard by train driver is

$$f_2 = f_1\left(\frac{V + V_0}{V}\right)$$

$$= 330\left(\frac{330 + 10}{330}\right)$$

$$f_2 = 340\text{Hz}$$

Q. 28 The diameter of an air bubble which was initially $2\ \text{mm}$, rises steadily through a solution of density $1750\ \text{kg m}^{-3}$ at the rate of $0.35\ \text{cms}^{-1}$. The coefficient of viscosity of the solution is_____ poise (in nearest integer). (the density of air is negligible).

Correct Answer:
11

Solution:
If it rises steadily that implies the net force on the bubble is zero

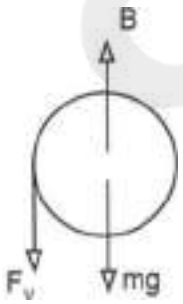
$$\text{mg} = 0(\text{air density is negligible})$$

$$F_v = \text{viscous drag force}$$

$$= 6\pi\eta rv.3\pi\eta dv$$

$$d \rightarrow \text{diameter of bubble}$$

$$B = mg + F_v$$



$$B = mg + F_v$$

$$\frac{4}{3}\pi r^3 g\rho = 0 + 3\pi\eta dv$$

$$r = \frac{d}{2}$$

$$\frac{\pi d^3 g \rho}{6} = 3\pi \eta dv$$

$$\frac{d^2 \rho g}{18} = \eta v$$

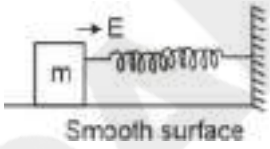
$$\eta = \frac{d^2 \rho 9}{18V}$$

$$= \frac{(2 \times 10^{-3})^2 \times 1750 \times 10}{18 \times 35 \times 10^{-4}}$$

$$= \frac{2000}{18} \times 10^{-1} \frac{NS}{m^2}$$

$$\eta = 11 \frac{Ns}{m^2}$$

Q. 29 A block of mass $'m'$ (as shown in figure) moving with kinetic energy E compresses a spring through a distance $25\ cm$ when, its speed is halved. The value of spring constant of used spring will $nE\ Nm^{-1}$ for $n =$_____.



Correct Answer:
24

Solution:

If the speed is halved, the kinetic energy will be $\frac{1}{4}^{th}$

$$E = PE + KE$$

$$E = PE + \frac{E}{4}$$

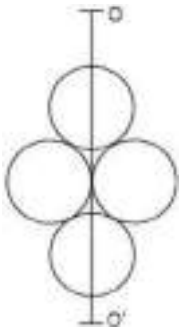
$$\frac{3E}{4} = \frac{1}{2}kx^2$$

$$k = \frac{3E}{2x^2} = \frac{3E}{2 \times \left(\frac{1}{4}\right)^2}$$

$$K = 24E$$

$$\therefore n = 24$$

Q. 30 Four identical discs each of mass $'M'$ and diameter $'a'$ are arranged in a small plane as shown in figure. If the moment of inertia of the system about OO' is $\frac{x}{4} Ma^2$. Then, the value of x will be_____.



Correct Answer:
3

Solution:

$$I = 2\left(\frac{MR^2}{4}\right) + 2\left(\frac{5MR^2}{4}\right)$$

$$= \frac{12MR^2}{4}$$

$$\text{But } R = a/2$$

$$\therefore I = \frac{3Ma^2}{4}$$

The value of x is 3

Chemistry

Q. 1 Identify the incorrect statement from the following.

Option 1:
A circular path around the nucleus in which an electron moves is proposed as Bohr's orbit.

Option 2:
An orbital is the one electron wave function (Ψ) in an atom.

Option 3:
The existence of Bohr's orbits is supported by hydrogen spectrum.

Option 4:
Atomic orbital is characterised by the quantum numbers n and l only.

Correct Answer:
Atomic orbital is characterised by the quantum numbers n and l only.

Solution:
Atomic orbital is characterised by n, l, m not only n and l.

Hence option (D) is correct.

Q. 2 Which of the following relation is not correct?

Option 1:
$\Delta \mathrm{H}=\Delta \mathrm{U}-\mathrm{P} \Delta \mathrm{V}$

Option 2:
$\Delta \mathrm{U}=\mathrm{q}+\mathrm{W}$

Option 3:
$\Delta S_{sys} + \Delta S_{surr} \geqslant 0$

Option 4:
$\Delta G = \Delta H - T\Delta S$

Correct Answer:
$\Delta H = \Delta U - P\Delta V$

Solution:
We know that -

$H = U + PV$
$\Delta H = \Delta V + \Delta (PV)$, at constant pressure -
$\Delta H = \Delta U + P\Delta V$

Hence option (A) is correct.

Q. 3 Match **List - I with List - II.**

List - I	**List - II**
(A) $Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$	(I) Primary battery
(B) $Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$	(II) Discharging of secondary battery
(C) $2PbSO_4(s) + 2H_2O(l) \rightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)$	(III) Fuel cell
(D) $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$	(IV) Charging of secondary battery

Choose the correct answer from the options given below :

Option 1:
$(A) - (I), (B) - (II), (C) - (III), (D) - (IV)$

Option 2:
$(A) - (IV), (B) - (I), (C) - (II), (D) - (III)$

Option 3:
$(A)-(II),(B)-(I),(C)-(IV),(D)-(III)$

Option 4:
$(A)-(II),(B)-(I),(C)-(III),(D)-(IV)$

Correct Answer:
$(A)-(II),(B)-(I),(C)-(IV),(D)-(III)$

Solution:
a) $Cd(s)+2Ni(OH)_3 \rightarrow CdO+2Ni(OH)_2+H_2O$
Discharge of secondary battery

(b) $Zn(Hg)+HgO(s) \rightarrow ZnO(s)+Hg(l)$
This is primary battery

(c) $2PbSO_4(s)+2H_2O(l) \rightarrow Pb(s)+PbO_2(s)+2H_2SO_4(aq)$
This represents charging of secondary battery

(d) $2H_2(g)+O_2(g) \rightarrow 2H_2O(l)$
This is an example of fuel cell.

Hence option (C) is correct.

Q. 4 Match **List - I with List - II.**

List - I **Reaction**	**List - II** **Catalyst**
(A) $4NH_3(g)+5O_2(g) \rightarrow 4NO(g)+6H_2O(g)$	(I) $NO(g)$
(B) $N_2(g)+3H_2(g) \rightarrow 2NH_3(g)$	(II) $H_2SO_4(l)$
(C) $C_{12}H_{22}O_{11}(aq)+H_2O(\ell) \rightarrow C_6H_{12}O_6+C_6H_{12}O_6$ Glucose Fructose	(III) $Pt(s)$
(D) $2SO_2(g)+O_2(g) \rightarrow 2SO_3(g)$	(IV) $Fe(s)$

Choose the correct answer from the options given below :

Option 1:
(A) - (II) - (B) - (III) - (C) - (I) - (D) - (IV)

Option 2:
(A) - (III) - (B) - (II) - (C) - (I) - (D) - (IV)

Option 3:
(A) - (III) - (B) - (IV) - (C) - (II) - (D) - (I)

Option 4:
(A) - (III) - (B) - (II) - (C) - (IV) - (D) - (I)

Correct Answer:
(A) - (III) - (B) - (IV) - (C) - (II) - (D) - (I)

Solution:

$(a)\ 4NH_3(g) + 5O_2(g) \overset{Pt(s)}{\rightarrow} 4NO(g) + 6H_2O(g)$

Catalyst used is Pt(s) which is known as Ostwald process.

$(b)\ N_2(g) + 3H_2(g) \overset{Fe(s)}{\rightarrow} 2NH_3(g)$

In presence of Fe(s) , as catalyst, this process is known as Born Haber process.

$(c)\ C_{12}H_{22}O_{11}(aq) + H_2O(\ell) \overset{H^+}{\rightarrow} C_6H_{12}O_6(\text{Glucose}) + C_6H_{12}O_6(\text{Fructose})$

H_2SO_4 is used in the Inversion of Sugar process.

$(d)\ 2SO_2(g) + O_2(g) \overset{NO(g)}{\rightarrow} 2SO_3(g)$

$NO(g)$ is used as a Catalyst.

Hence option (C) is correct.

Q. 5 In which of the following pairs, electron gain enthalpies of constituent elements are nearly the same or identical ?

(A) Rb and Cs
(B) Na and K
(C) Ar and Kr
(D) I and At

Choose the correct answer from the options given below :

Option 1:
(A) and (B) only

Option 2:
(B) and (C) only

Option 3:
(A) and (C) only

Option 4:
(C) and (D) only

Correct Answer:
(A) and (C) only

Solution:
From the given option -

(Rb and Cs) have nearly same electron gain enthalpy which is nearly $-46\ \mathrm{KJ/mol}$.

Whereas (Ar and Kr) have nearly same electron gain enthalpy of $+96\ \mathrm{KJ/mol}$.

Hence option (C) is correct.

Q. 6 Which of the reaction is suitable for concentrating ore by leaching process?

Option 1:
$2\mathrm{Cu_2\ S} + 3\mathrm{O_2} \rightarrow 2\mathrm{Cu_2O} + 2\mathrm{SO_2}$

Option 2:
$\mathrm{Fe_3O_4} + \mathrm{CO} \rightarrow 3\mathrm{FeO} + \mathrm{CO_2}$

Option 3:
$\mathrm{Al_2O_3} + 2\mathrm{NaOH} + 3\mathrm{H_2O} \rightarrow 2\mathrm{Na}\left[\mathrm{Al(OH)_4}\right]$

Option 4:
$\mathrm{Al_2O_3} + 6\mathrm{Mg} \rightarrow 6\mathrm{MgO} + 4\mathrm{Al}$

Correct Answer:
$\mathrm{Al_2O_3} + 2\mathrm{NaOH} + 3\mathrm{H_2O} \rightarrow 2\mathrm{Na}\left[\mathrm{Al(OH)_4}\right]$

Solution:
The leaching process is -

$$\mathrm{Al_2O_3} + 2\mathrm{NaOH} + 3\mathrm{H_2O} \rightarrow 2\mathrm{Na}\left[\mathrm{Al(OH)_4}\right]$$

Hence option (1) is correct.

Q. 7 The metal salts formed during softening of hardwater using Clark's method are :

Option 1:
$\mathrm{Ca(OH)_2}$ and $\mathrm{Mg(OH)_2}$

Option 2:
$\mathrm{CaCO_3}$ and $\mathrm{Mg(OH)_2}$

Option 3:
$\mathrm{Ca(OH)_2}$ and $\mathrm{MgCO_3}$

Option 4:
$\mathrm{CaCO_3}$ and $\mathrm{MgCO_3}$

Correct Answer:
$\mathrm{CaCO_3}$ and $\mathrm{Mg(OH)_2}$

Solution:
Clark's method for softening of hard water -

$$\mathrm{Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O}$$
$$\mathrm{Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow 2CaCO_3 + Mg(OH)_2 + 2H_2O}$$

Hence metal salts formed are $\mathrm{CaCO_3}$ and $\mathrm{Mg(OH)_2}$.

Option (B) is correct.

Q. 8 Which of the following statement is incorrect?

Option 1:
Low solubility of LiF in water is due to its small hydration enthalpy.

Option 2:
$\mathrm{KO_2}$ is paramagnetic.

Option 3:
Solution of sodium in liquid ammonia is conducting in nature.

Option 4:
Sodium metal has higher density than potassium metal.

Correct Answer:
Low solubility of LiF in water is due to its small hydration enthalpy.

Solution:
The statement I is incorrect as because LiF shows low solubility in water due to high lattice enthalpy not hydration enthalphy.

Hence option (A) is correct .

Q. 9 Match **List - I with List - II**, match the gas evolved during each reaction.

List - I	**List - II**
(A) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$	(I) H_2
(B) $KMnO_4 + HCl \rightarrow$	(II) N_2
(C) $Al + NaOH + H_2O \rightarrow$	(III) O_2
(D) $NaNO_3 \xrightarrow{\Delta}$	(IV) Cl_2

Choose the correct answer from the options given below :

Option 1:
$(A)-(II), (B)-(III), (C)-(I), (D)-(IV)$

Option 2:
$(A)-(III), (B)-(I), (C)-(IV), (D)-(II)$

Option 3:
$(A)-(II), (B)-(IV), (C)-(I), (D)-(III)$

Option 4:
$(A)-(III), (B)-(IV), (C)-(I), (D)-(II)$

Correct Answer:
$(A)-(II), (B)-(IV), (C)-(I), (D)-(III)$

Solution:
a) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$
Hence N_2 gas is evolved.

$KMnO_4 + HCl \rightarrow MnCl_2 + KCl + Cl_2 + H_2O$

$\rightarrow Cl_2$ gas is evolved.

$Al + NaOH + H_2O \rightarrow H_2 + Na\,[Al(OH)_4]$
$\rightarrow H_2$ gas is evolved

$NaNO_3 \rightarrow NaNO_2 + O_2$
$\rightarrow O_2$ gas is evolved.

Hence option (C) is correct.

Q. 10 Which of the following has least tendency to liberate H_2 from mineral acids?

Option 1:
Cu

Option 2:
Mn

Option 3:
Ni

Option 4:
Zn

Correct Answer:
Cu

Solution:
In reactivity series , the elements which are below Hydrogen , have neglisible tendency to liberate H_2 from mineral acids.

As Copper is least electropositive among the given metals, also it lies below Hydrogen in rectivity series,

Copper has least tendency to liberate H_2 from mineral acids.

Q. 11 Given below are two statements :
Statement I : In polluted water values of both dissolved oxygen and BOD are very low.

Statement II : Eutrophication results in decrease in the amount of dissolved oxygen.
In the light of the above statements, choose the **most appropriate** answer from the options given below :

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

Solution:
Eutrophication is a process due to which excessive growth of weed in water bodies takes place going to increase in pollution levels.

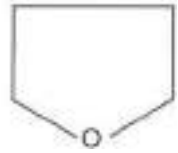
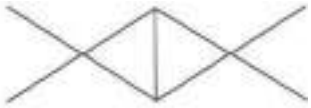
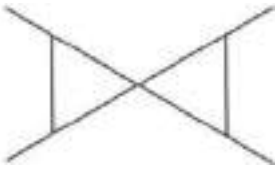
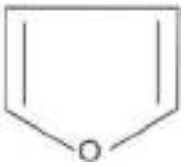
Eutrophication leads to decrease in amount of dissolved oxygen in water bodies.

As the amount of dissolved oxygen decreases , value of BOD [Biological oxygen demand] increase.
So statement I is false and statement II is true.

Hence option (D) is correct.

CAREERS360

Q. 12 Match **List-I with List-II**

List-I	List-II
(A) 	(I) Spiro compound
(B) 	(II) Aromatic compound
(C) 	(III) Non-planar Heterocyclic compound
(D) 	(IV) Bicyclo compound

Choose the correct answer from the options given below :

Option 1:
$(A)-(II),(B)-(I),(C)-(IV),(D)-(III)$

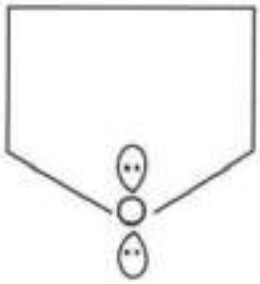
Option 2:
$(A)-(IV),(B)-(III),(C)-(I),(D)-(II)$

Option 3:
$(A)-(III),(B)-(IV),(C)-(I),(D)-(II)$

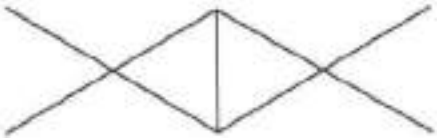
Option 4:
$(A)-(IV),(B)-(III),(C)-(II),(D)-(I)$

Correct Answer:
$(A)-(III),(B)-(IV),(C)-(I),(D)-(II)$

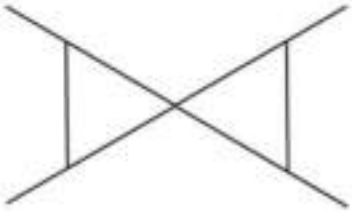
Solution:



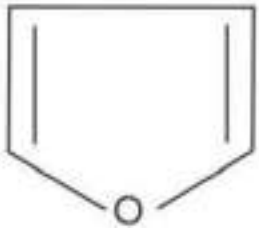
due to sp^3 carbon atoms , this is non planer heterocylic Compound.



Bicyclo Compound



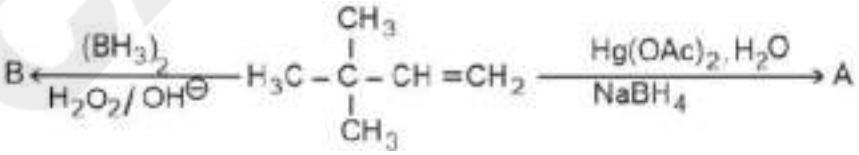
Spiro Compound



Aromatic Compound

Hence option (C) is correct.

Q. 13 Choose the correct option for the following reactions.



Option 1:

'A' and 'B' are both Markovnikov addition products.

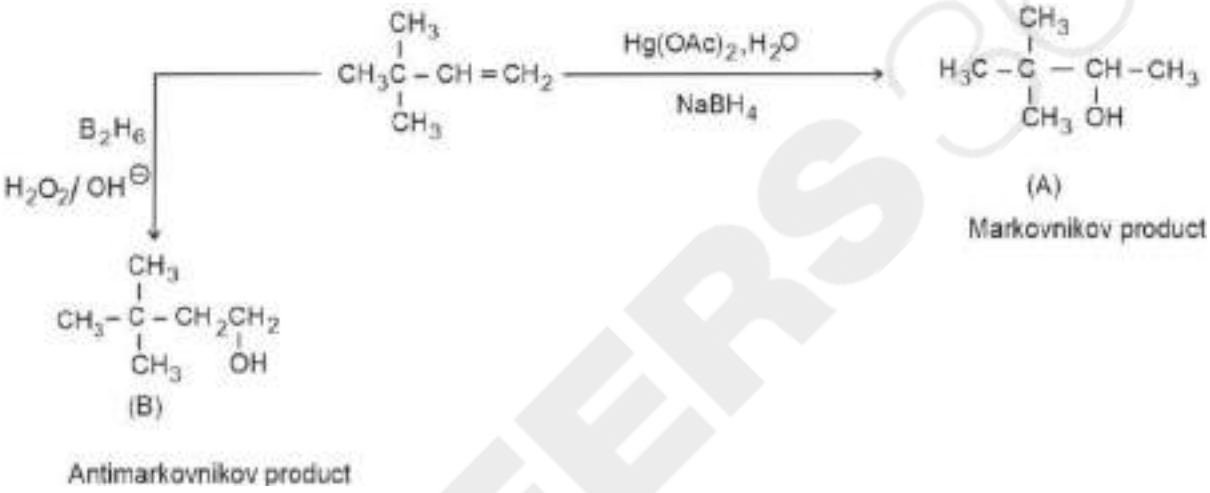
Option 2:
$'A'$ is Markovnikov product and $'B'$ is anti-Markovnikov product.

Option 3:
$'A'\ and\ 'B'$ are both anti-Markovnikov products.

Option 4:
$'B'$ is Markovnikov and $'A'$ is anti-Markovnikov product.

Correct Answer:
$'A'$ is Markovnikov product and $'B'$ is anti-Markovnikov product.

Solution:

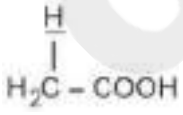


In the reaction, product (A) formed will be Markovnikov product and product (B) will be anti Markovnikov product.

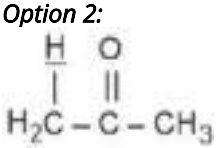
Hence option (B) is correct.

Q. 14 Among the following marked proton of which compound shows lowest pK_a value ?

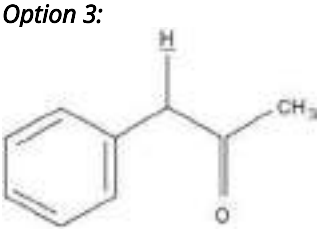
Option 1:



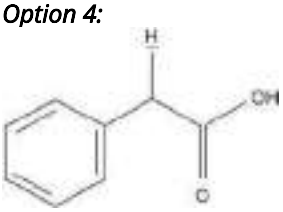
Option 2:



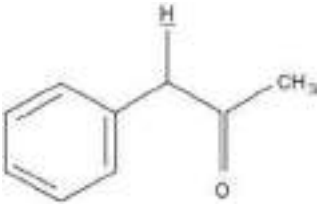
Option 3:



Option 4:



Correct Answer:

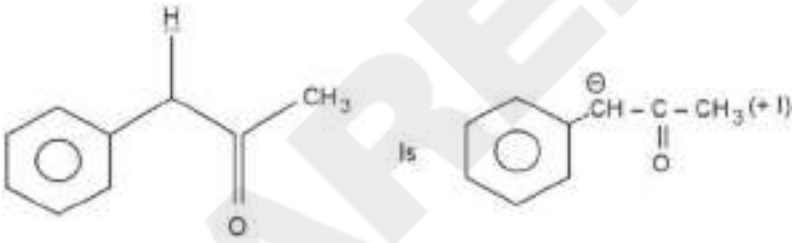


Solution:

pK_a value is determined by the acidic character of a molecule and acidic character is determined by stability of conjugate base.

Lowest pK_a value will be shown by highest acidic compound means most stable conjugate base.

Conjugate base of



Which is most stable because of Resonance through benzene ring.

Hence option (C) is correct.

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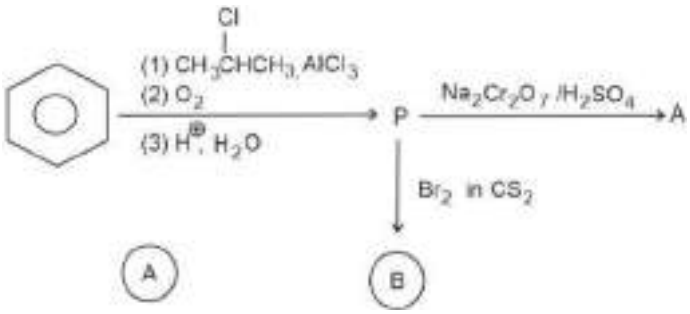
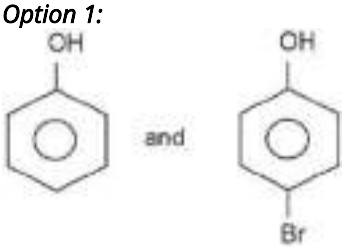
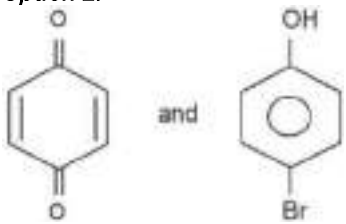
Q. 15 Identify the major products A and B for the below given reaction sequence.



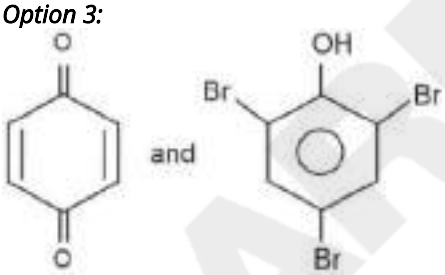
Option 1:



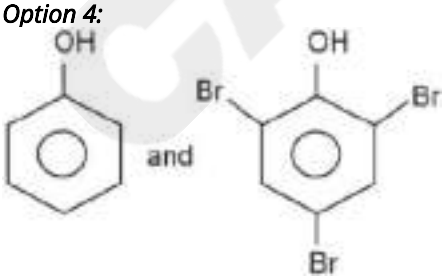
Option 2:



Option 3:



Option 4:



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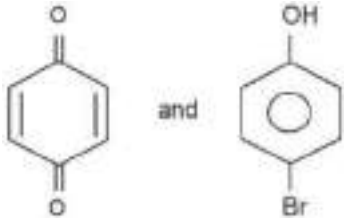
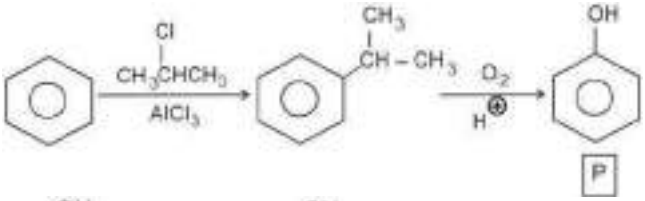
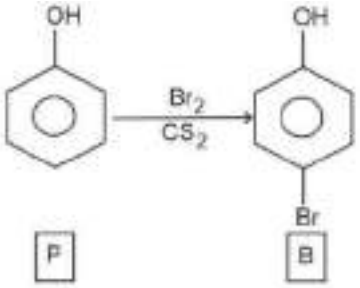
Correct Answer:

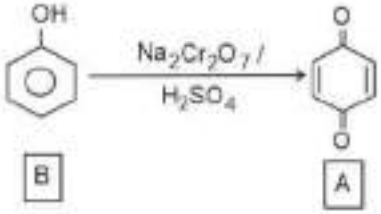


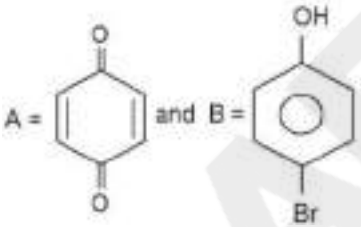
Solution:

From the given question, loc have -



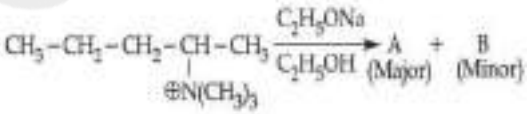






Option (B) is correct.

Q. 16 Identify the correct statement for the below given transformation.



Option 1:

$A-CH_3CH_2CH=CH-CH_3,\ B-CH_3CH_2CH_2CH=CH_2,$ Saytzeff products

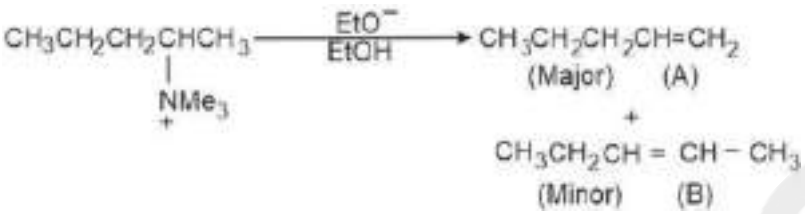
Option 2:
$A-CH_3CH_2CH=CH-CH_3,\ B-CH_3CH_2CH_2CH=CH_2,$ Hofmann products

Option 3:
$A-CH_3CH_2CH_2CH=CH_2,\ B-CH_3CH_2CH=CHCH_3,$ Hofmann products

Option 4:
$A-CH_3CH_2CH_2CH=CH_2,\ B-CH_3CH_2CH=CHCH_3,$ Saytzeff products

Correct Answer:
$A-CH_3CH_2CH_2CH=CH_2,\ B-CH_3CH_2CH=CHCH_3,$ Hofmann products

Solution:
The reaction belongs to elimination reaction -



This is derived by the factor of steric hinderance.

Hence option (C) is correct.

Q. 17 Terylene polymer is obtained by condensation of :

Option 1:
Ethane-1,2-diol and Benzene-1,3 dicarboxylic acid

Option 2:
Propane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

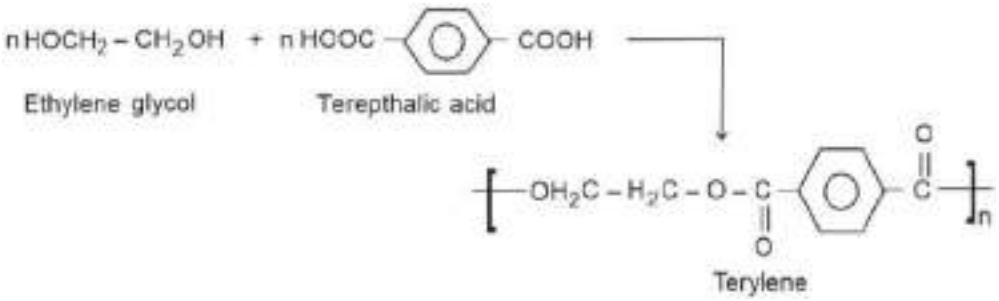
Option 3:
Ethane-1,2-diol and Benzene-1, 4 dicarboxylic acid

Option 4:
Ethane-1,2-diol and Benzene-1, 2 dicarboxylic acid

Correct Answer:
Ethane-1,2-diol and Benzene-1, 4 dicarboxylic acid

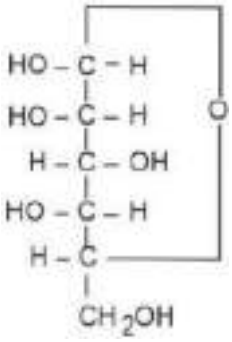
Solution:

Terlylene -



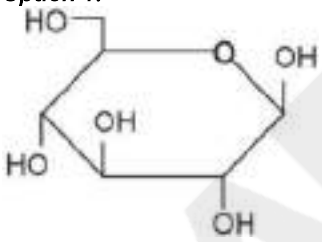
Hence option (C) is correct.

Q. 18 For the below given cyclic hemiacetal (X), the correct pyranose structure is :

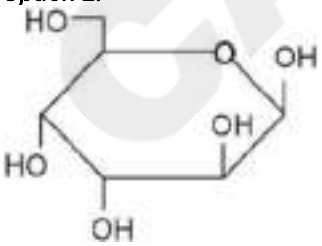


(X)

Option 1:



Option 2:



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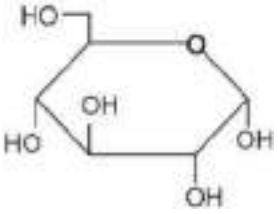
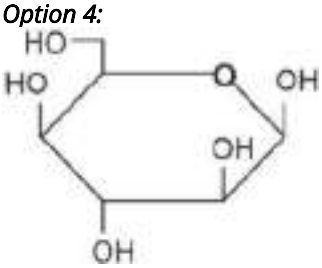
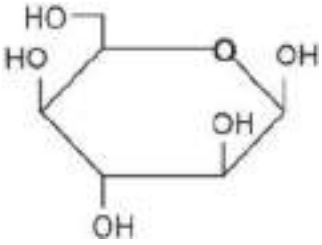
Option 3:



Option 4:

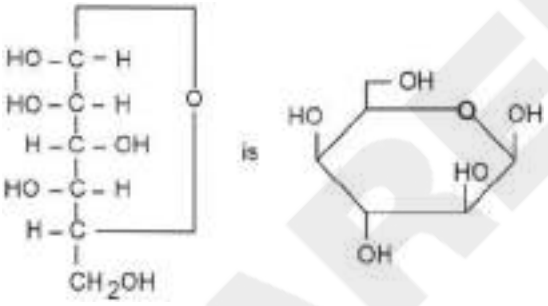


Correct Answer:



Solution:

The pyranose structure for -



Hence option (D) is correct.

Q. 19 Statements about Enzyme Inhibitor Drugs are given below :

(A) There are Competitive and Non-competitive inhibitor drugs.

(B) These can bind at the active sites and allosteric sites.

(C) Competitive Drugs are allosteric site blocking drugs.

(D) Non-competitive Drugs are active site blocking drugs.

Choose the correct answer from the options given below :

Option 1:

(A), (D) only

Option 2:
(A), (C) only

Option 3:
(A),(B) only

Option 4:
(A),(B),(C) only

Correct Answer:
(A),(B) only

Solution:
Enzyme inhibitor drugs are competitive and drugs which can bind at active and allosteric sites.

Hence option (C) is correct.

Q. 20 For kinetic study of the reaction of iodide ion with H_2O_2 at room temperature :
(A) Always use freshly prepared starch solution.
(B) Always keep the concentration of sodium thiosulphate solution less than that KI solution.
(C) Record the time immediately after the appearance of blue colour.
(D) Record the time immediately before the appearance of blue colour.
(E) Always keep the concentration of sodium thiosulphate solution more than that of KI solution.
Choose the correct answer from the options given below :

Option 1:
(A), (B), (C) only

Option 2:
(A),(D),(E) only

Option 3:
(D), (E) only

Option 4:
(A),(B),(E) only

Correct Answer:
(A), (B), (C) only

Solution:

For Kinetic study of reaction of iodide ion with H_2O_2 at room temperature , always freshly prepared strach solution is used.

$\rightarrow$ Concentration of sodium thiosulphate solution is taken less than that of KI solution.
and time is recorded immediately after the appearance of blue colour.

Q. 21 In the given reaction,
$X + Y + 3Z \leftrightarrows X_YZ_3$
if one mole of each of X and Y with 0.05 mol of Z gives compound XYZ_3. (Given : Atomic masses of X, Y and Z are 10,20 and 30 amu, respectively.) The yield of XYZ_3 is _____ g .(Nearest integer)

Correct Answer:
2

Solution:
Given -

$X + Y + 3Z \leftrightarrows X_YZ_3$
1 mole i mol 0.05 mol

From the given Reaction -

3 moles of Z yields 1 mole of XYZ_3.

$$1 \text{ Mole of Z yields} \rightarrow \frac{1}{3} \text{ mole of } XYZ_3$$

$$0.05 \text{ Mole of Z yields} \rightarrow \frac{0.05}{3} \text{ mole of } XYZ_3.$$

$$\therefore \text{mass of } XYZ_3 \text{ formal} = \frac{0.05}{3}[10 + 20 + 30 \times 3]$$
$$= 2g$$

Q. 22 An element M crystallises in a body centred cubic unit cell with a cell edge of $300pm$. The density of the element is 6.0 g cm^{-3}. The number of atoms present in 180 g of the element is_____ $\times 10^{23}$. (Nearest integer)

Correct Answer:
22.22

Solution:
Given in question -

M = nody centered cubic
$\Rightarrow Z = 2.$

a = edge length = 300 pm
denisty = 6 g/cm^3

from the formula of density-

$$d = \frac{Z \times A}{a^3} \ (A = \text{mass of 1 atom of A})$$

$$6\,g/cm^3 = \frac{2 \times A}{(300 \times 10^{-10})^3}$$

$$A = 81 \times 10^{-24}\,g$$

$$\text{atomic mass} = 81 \times 10^{-24} \times 6.022 \times 10^{23}$$

$$\text{no.of moles in } 180g = \frac{180}{48.7\,g/mol} = 3.7\,\text{moles}.$$

$$\therefore \text{atoms of A} = 3.7 \times 6.022 \times 10^{23} = 22.22 \times 10^{23}\ \text{atoms}$$

Q. 23 The number of paramagnetic species among the following is_____.

$B_2, Li_2, C_2, C_2^-, O_2^{2-}, O_2^+$ and He_2^+

Correct Answer:

4

Solution:

Paramagnetic species among following are -

$B_2, C_2^-, O_2^+, He_2^+$

Hence the correct answer is (4).

Q. 24 150 g of acetic acid was contaminated with 10.2 g ascorbic acid $(C_6H_8O_6)$ to lower down its freezing point by $(x \times 10^{-1})^\circ C$. The value of x is_____.(Nearest integer)

[Given $K_f = 3.9\ K\ kg\ mol^{-1}$;

molar mass of ascorbic acid $= 176\ g\ mol^{-1}$]

Correct Answer:

15

Solution:

Given , 150 g of CH_3COOH

10.2 g of ascorbic acod having 0.058 moles.

$K_f = 3.9\,K\,kg\,mol^{-1}$.

M. mass of ascorbic acid = 176 g/mol.

$$\text{no.of moles of ascorbic acid} = \frac{10.2}{176} = 0.058\,\text{moles}.$$

$\Delta T_f = K_f.m \quad m = \text{molality}$

$$= 3.9 \times \frac{0.058}{150} \times 1000$$
$$= 1.5^{\circ}\mathrm{C}$$
$$= \left(15 \times 10^{-1}\right)^{\circ}\mathrm{C}$$

Hence answer is 15

Q. 25 K_a for butyric acid (C_3H_7COOH) is 2×10^{-5}. The pH of $0.2M$ solution of butyric acid is_____$\times 10^{-1}$. (Nearest integer)
[Given $\log 2 = 0.30$]

Correct Answer:
27

Solution:
$\mathrm{Ka}\,(C_3H_7COOH) = 2 \times 10^{-5}$
$\mathrm{pH\ of\ 0.2\ M\ solution\ of\ butyric\ acid} = - \times 10^{-1}.$

Log 2 = 0.3

$$\mathrm{pH} = \frac{1}{2}\,\mathrm{PKa} - \frac{1}{2}\log \mathrm{C} \qquad \mathrm{PKa} = -\log \mathrm{Ka}$$
$$= \frac{1}{2}\,(4.7) - \frac{1}{2}\log\,(0.2) \qquad = -\log\left(2 \times 10^{-5}\right)$$
$$= 2.35 + 0.35 = 2.7 \qquad = 4.7$$

$\mathrm{pH} = 27 \times 10^{-1} \Rightarrow 27$ is answer

Q. 26 For the given first order reaction
$A \rightarrow B$
the half life of the reaction is $0.3010\ \mathrm{min}$. The ratio of the initial concentration of reactant to the concentration of reactant at time $2.0\ \mathrm{min}$ will be equal to _____.(Nearest integer)

Correct Answer:
100

Solution:
Half Life = 0.3010 min

$$A \longrightarrow B$$

From first order Reaction -
$\mathrm{A} = \mathrm{Ao\,e{-}kt} \Rightarrow \ln \mathrm{A} = \ln \mathrm{Ao} - \mathrm{kt}$

$$K = \frac{2.303}{t}\log\frac{[Ao]}{[At]} \qquad t_{\frac{1}{2}} = \frac{k}{0.6}\,\frac{0.693}{k}$$

$$\Rightarrow \frac{0.693}{t_{\frac{1}{2}}} = \frac{2.303}{2}\log\left(\frac{Ao}{At}\right)$$

$$\log\frac{Ao}{At} = 2$$

$$\therefore \frac{Ao}{At} = 10^2 = 100$$

Answer is 100

Q. 27 The number of interhalogens from the following having square pyramidal structure is :
$ClF_3, IF_7, BrF_5, BrF_3, I_2Cl_6, IF_5, ClF, ClF_5$

Correct Answer:
3

Solution:
Compounds BrF_5, IF_5, ClF_5 are square pyramidal structure.

Q. 28 The disproportionation of MnO_4^{2-} in acidic medium resulted in the formation of two manganese compounds A and B. If the oxidation state of Mn in B is smaller than that of A, then the spin-only magnetic moment (μ) value of B in BM is_____ . (Nearest integer)

Correct Answer:
4

Solution:

$$\overset{+7}{\rightarrow MnO_4^{-2}} \xrightarrow{H^+} \overset{+4}{MnO_4^-} + MnO_2$$

- no.of unpaired electrons = 3 [$3d^3$ configuration of Mn^{+4}].

Hence, Magnetic moment $(\mu) = \sqrt{n(n+2)}\,B.M$

$$= \sqrt{3(2+3)} = \sqrt{15}\,B.M$$

$$= 3.877\,B.M$$

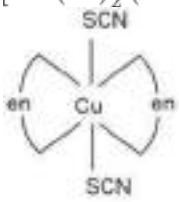
Nearest integer = 4 which is the answer.

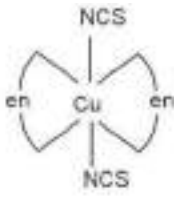
Q. 29 Total number of relatively more stable isomer(s) possible for octahedral complex $[Cu(en)_2(SCN)_2]$ will be_____.

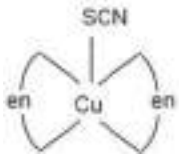
Correct Answer:
3

Solution:
Total no.of relatively more stable isomeric possible-
for octahedral complex -

$[\mathrm{Cu\,(en)_2\,(SCN)_2}] -$







$\rightarrow$ More stable isomeric belongs to the isomers which are trans in nature having low steric hinderance and more stability.

Hence in the given compound we have 3 isomers which are relatively more stable.

Q. 30 On complete combustion of $0.492\ \mathrm{g}$ of an organic compound containing $\mathrm{C, H\ and\ O}$, $0.7938\ \mathrm{g}$ of $\mathrm{CO_2}$ and $0.4428\ \mathrm{g}$ of $\mathrm{H_2O}$ was produced. The % composition of oxygen in the compound is_____.

Correct Answer:
46

Solution:
Given,
Organic compound containing $\mathrm{C, H, O} = 0.492\,\mathrm{g}.$

$\mathrm{CO_2\,produced} = 0.7938\,\mathrm{g}$
$\mathrm{H_2O\,produced} = 0.4428\,\mathrm{g}$

Let the organic compound we have us $\mathrm{C_xH_yO_z}$.

no.of moles of $\mathrm{CO_2\,in}\ 0.7938\,\mathrm{g} = 0.018\ \mathrm{moles}$
no.of moles of $\mathrm{H_2O\,in}\ 0.4428\,\mathrm{g} = 0.0246\ \mathrm{moles}.$

So moles of $\mathrm{C} = 0.018$ which is equal to $0.216\,\mathrm{g}$
moles of $\mathrm{H} = 0.0492$ which is equal to $0.0492\,\mathrm{g}$

weight of oxygen = Total weight - weight of C - weight of H

$$= 0.492 - 0.216 - 0.049$$
$$= 0.227\,\mathrm{g}.$$

$$\% \text{ of oxygen in compound} = \frac{0.227}{0.492} \times 100\% \cong 46\% \text{ approx}$$

Hence answer is 46.

Maths

Q. 1 Let the solution curve of the differential equation $x\ dy = \left(\sqrt{x^2+y^2}+y\right)dx, x>0,$ intersect the line $x=1$ at $y=0$ at $y=0$ and the line $x=2$ at $y=\alpha$. Then the value of α is:

Option 1:
$\frac{1}{2}$

Option 2:
$\frac{3}{2}$

Option 3:
$-\frac{3}{2}$

Option 4:
$\frac{5}{2}$

Correct Answer:
$\frac{3}{2}$

Solution:

CAREERS360

$$x\,dy = \left(\sqrt{x^2+y^2}+y\right)dx$$

$$x\,dy - y\,dx = \sqrt{x^2+y^2}dx$$

$$\frac{x\,dy - y\,dx}{x^2} = \sqrt{1+\frac{y^2}{x^2}}\cdot\frac{dx}{x}$$

$$\frac{d\left(\frac{y}{x}\right)}{\sqrt{1+\left(\frac{y}{x}\right)^2}} = \frac{dx}{x}$$

$$\ln\left(\frac{y}{x}+\sqrt{\left(\frac{y}{x}\right)^2+1}\right) = \ln x + \ln c$$

$$\frac{y+\sqrt{y^2+x^2}}{x} = cx$$

$$y+\sqrt{y^2+x^2} = cx^2$$

$$x=1, y=0 \Rightarrow 0+1=c \Rightarrow c=1$$

$$\text{curve is } y+\sqrt{x^2+y^2} = x^2$$

$$x=2, y=\alpha$$

$$2+\sqrt{4+\alpha^2} = 4$$

$$4+\alpha^2 = 16+\alpha^2 = 8\alpha$$

$$\alpha = \frac{3}{2}$$

Q. 2 Considering only the principal values of the inverse trigonometric functions, the domain of the function $f(x) = \cos^{-1}\left(\frac{x^2-4x+2}{x^2+3}\right)$ is:

Option 1:
$\left(-\infty, \frac{1}{4}\right]$

Option 2:
$\left[-\frac{1}{4}, \infty\right)$

Option 3:
$(-1/3, \infty)$

Option 4:
$\left(-\infty, \frac{1}{3}\right]$

Correct Answer:
$\left[-\frac{1}{4}, \infty\right)$

Solution:

$$\left|\frac{x^2+4x+2}{x^2+3}\right| \leq 1$$
$$\left(x^2-4x+2\right)^2 \leq \left(x^2+3\right)^2$$
$$\left(x^2-4x+2\right)^2 - \left(x^2+3\right)^2 \leq 0$$
$$\left(2x^2-4x+5\right)(-4x-1) \leq 0$$
$$-4x-1 \leq 0 \rightarrow x \geq -\frac{1}{4}$$
$$\left[-\frac{1}{4}, \infty\right)$$

Q. 3 Let the vectors $\vec{a} = (1+t)\hat{i} + (1-t)\hat{j} + \hat{k}$, $\vec{b} = (1-t)\hat{i} + (1+t)\hat{j} + 2\hat{k}$ and $\vec{c} = t\hat{i} - t\hat{j} + \hat{k}, t \in \mathbf{R}$ be such that for $\alpha, \beta, \gamma \in \mathbf{R}, \alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \vec{0} \Rightarrow \alpha = \beta = \gamma = 0$. Then, the set of all value of t is:

Option 1:
a non-empty finite set

Option 2:
equal to $\mathbf{N}$

Option 3:
equal to $\mathbf{R} - \{0\}$

Option 4:
equal to $\mathbf{R}$

Correct Answer:
equal to $\mathbf{R} - \{0\}$

Solution:
By its given condition
$\vec{a}, \vec{b}, \vec{c}$ are lineardy independent vectors
$\Rightarrow \left[\vec{a}\,\vec{b}\,\vec{c}\right] \neq 0 \cdots\cdots (1)$

Now $\left[\vec{a}\,\vec{b}\,\vec{c}\right]$

$$= \begin{vmatrix} 1+t & 1-t & 1 \\ 1-t & 1+t & 2 \\ t & -t & 1 \end{vmatrix}$$

$c_2 \rightarrow c_1 + c_2$

$$\begin{vmatrix} 1+t & 2 & 1 \\ 1-t & 2 & 2 \\ t & 0 & 1 \end{vmatrix} = 2 \begin{vmatrix} 1+t & 1 & 1 \\ 1-t & 1 & 2 \\ t & 0 & 1 \end{vmatrix}$$

$$= 2[(1+t)-(1-t)+t]$$

$$= 2[3t] = 6t$$

$$[\tilde{a}\ \tilde{b}\ \tilde{c}] \neq 0 \Rightarrow t \neq 0$$

Q. 4 Considering the principal values of the inverse trigonometric functions, the sum of all the solutions of the equation $\cos^{-1}(x) - 2\sin^{-1}(x) = \cos^{-1}(2x)$ is equal to :

Option 1:

0

Option 2:

1

Option 3:

$\frac{1}{2}$

Option 4:

$-\frac{1}{2}$

Correct Answer:

0

Solution:

$$\cos^{-1} x = 2\sin^{-1} x = \cos^{-1} 2x$$

$$\cos^{-1} x - 2\left(\frac{\pi}{2} - \cos^{-1} x\right) = \cos^{-1} 2x$$

$$\cos^{-1} x - \pi + 2\cos^{-1} x = \cos^{-1} 2x$$

$$3\cos^2 x = \pi + \cos^{-1} 2x - \cdots$$

$$\cos\left(3\cos^{-1} x\right) = \cos\left(\pi + \cos^{-1} 2x\right)$$

$$4x^3 - 3x = -2x$$

$$4x^3 = x \Rightarrow x = 0 \pm \frac{1}{2}$$

All satisfy the original equation

$$\text{Sum} = -\frac{1}{2} + 0 + \frac{1}{2} = 0$$

Q. 5 Let the operations $*, \odot \in \{\wedge, \vee\}$. If $(p * q) \odot (p \odot \sim q)$ is a tautology, then the ordered pair $(*, \odot)$ is:

Option 1:
$(\vee, \wedge)$

Option 2:
$(\vee, \vee)$

Option 3:
$(\wedge, \wedge)$

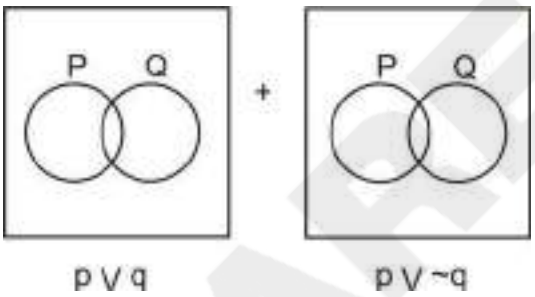
Option 4:
$(\wedge, \vee)$

Correct Answer:
$(\vee, \vee)$

Solution:
well check each option

$$\text{for } A\pi = V \text{ of } 0 = A$$
$$(p \vee q) \wedge (p \vee \sim q)$$
$$\equiv p \vee (q \wedge \sim q)$$
$$\equiv P \vee (c) \equiv P$$
$$\text{for } B : x = V, 0 = V$$
$$(p \vee q) \vee (p \vee \sim q) \equiv t$$

using venn diagrams



Q. 6 Let A vector $\overrightarrow{a}$ has magnitude 9. Let a vector $\overrightarrow{b}$ be such that for every $(x, y) \in \mathbf{R} \times \mathbf{R} - \{(0, 0)\}$, the vector $(x\overrightarrow{a} + y\overrightarrow{b})$ is perpendicular to the vector $(6y\overrightarrow{a} - 18x\overrightarrow{b})$. Then the value of $|\overrightarrow{a} \times \overrightarrow{b}|$ is equal to:

Option 1:
$9\sqrt{3}$

Option 2:
$27\sqrt{3}$

Option 3:
9

Option 4:
81

Correct Answer:
$27\sqrt{3}$

Solution:
$|\tilde{a}| = 9 \quad \&(x\tilde{a} + y\tilde{b}) \cdot (6y\tilde{a} - 18x\tilde{b}) = 0$

$\Rightarrow 6xy|\tilde{a}|^2 - 18x^2(\tilde{a} \cdot \tilde{b}) + 6y^2(\tilde{a} \cdot \tilde{b}) - |8xy|\tilde{b}\Big|^2 = 0$ This should hold $\forall x, y \in RXR$

$\Rightarrow 6xy\left(|\tilde{a}|^2 - 3\left|\tilde{b}\right|^2\right) + (\tilde{a} \cdot \tilde{b})\left(y^2 - 3x^2\right) = 0$

$\therefore |\tilde{a}|^2 = 3|\tilde{b}|^2 \& (\tilde{a} \cdot \tilde{b}) = 0$

Now $|\tilde{a} \times \tilde{b}|^2 = |\tilde{a}|^2|\tilde{b}|^2 - (\tilde{a} \cdot \tilde{b})^2$

$= |\tilde{a}|^2 \cdot \frac{|\tilde{a}|^2}{3}$

$\therefore |\tilde{a} \times \tilde{b}| = \frac{|\tilde{a}|^2}{\sqrt{2}} = \frac{81}{\sqrt{3}} = 27\sqrt{3}$

Q. 7 For $t \in (0, 2\pi)$, if ABC is an equilateral triangle with vertices $A(\sin t, -\cos t), B(\cos t, \sin t)$ and $C(a, b)$ such that its orthocentre lies on a circle with centre $\left(1, \frac{1}{3}\right)$, then $\left(a^2 - b^2\right)$ is equal to:

Option 1:
$\frac{8}{3}$

Option 2:
8

Option 3:
$\frac{77}{9}$

Option 4:
$\frac{80}{9}$

Correct Answer:
8

Solution:
$s \equiv \sin t, c \equiv \cos t$
Let orthocentre be (h, k)
since it if an equilateral triangle hence orthocentre cor coincides with centroid

$\therefore a+s+c=3h\ ,\ b+s-c=3k$

$\therefore (3h-a)^2+(3k-b)^2=(s+c)^2+(s-c)^2=2\left(s^2+c^2\right)=2$

$\therefore \left(k-\frac{a}{3}\right)^2+\left(k-\frac{b}{3}\right)^2=\frac{2}{4}$

$circle\ center\ at\ \left(\frac{a}{3},\frac{b}{3}\right)$

$gives, \frac{a}{3}=1, \frac{b}{3}=\frac{1}{3}\Rightarrow a=3, b=1$

$\therefore a^2+a^2-b^2=8$

Answer - B

Q. 8 For $\alpha \in \mathbf{N}$, consider a relation R on $\mathbf{N}$ given by $\mathrm{R}=\{(\mathrm{x}, \mathrm{y}): 3\mathrm{x}+\alpha\mathrm{y}$ is a multiple of $7\}$. The relation r is an equivalence relation if and only if :

Option 1:
$\alpha=14$

Option 2:
α is a multiple of 4

Option 3:
4 is the remainder when α is divided by 10

Option 4:
4 is the remainder when α is divided by 7

Correct Answer:
4 is the remainder when α is divided by 7

Solution:
for R to be reflexive$\Rightarrow$ xRx

$\Rightarrow 3\mathrm{x}+\alpha\mathrm{x}=7\mathrm{x}\Rightarrow(3+\alpha)\mathrm{x}=7\mathrm{k}$

$\Rightarrow 3+\alpha=7\lambda\Rightarrow\alpha=7\lambda-3=7\mathrm{N}+4,$

$\mathrm{k}, \lambda, \mathrm{N}\in \mathrm{I}$

$\therefore$ whenαdivided by 7 , remainder is 4.

R to be symmetricxRy $\Rightarrow$ yRx

$3\mathrm{x}+\alpha\mathrm{y}=7\mathrm{N_i}, 3\mathrm{y}+\alpha\mathrm{x}=7\mathrm{N_2}$

$\Rightarrow(3+\alpha)(\mathrm{x}+\mathrm{y})=7\left(\mathrm{N_1}+\mathrm{N_2}\right)=7\mathrm{N_3}$

which holds. when$3+\alpha$ is multiole of 7

$\therefore \alpha=7N+4$(as did earlier)

R to be transitive

xRy & yRz $\Rightarrow$ xRz.

$3x + 2y = 7N_1$ &f $3y + 2z = 7N_2$ and
$3x + \alpha z = 7N_3$
$\therefore 3x + 7N_2 - 3y = 7N_3$
$\therefore 7N_1 - \alpha y + 7N_2 - 3y = 7N_3$
$\therefore 7(N_1 + N_2) - (3 + \alpha)y = 7N_3$
$\therefore (3 + \alpha)y = 7N$

which is true again when $3 + \alpha$ divisib'e by. 7;
i.e. when α divided by 7 , remainder is 4.

Answer is D

Q. 9

Out of 60% female and 40% male candidates appearing in an exam, 60% candidates qualify it. The number of females qualifying the exam is twice the number of males qualifying it. A candidate is randomly chosen from the qualified candidates. The probability, that the chosen candidate is a female, is :

Option 1:
$\frac{2}{3}$

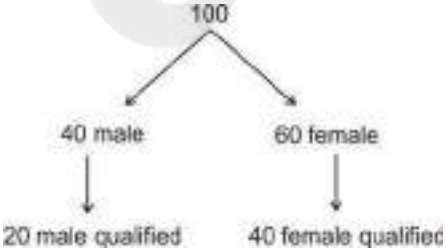
Option 2:
$\frac{11}{16}$

Option 3:
$\frac{23}{32}$

Option 4:
$\frac{13}{16}$

Correct Answer:
$\frac{2}{3}$

Solution:



Rrobability that chosen candidate is Fermale =
$\frac{40}{60} = \frac{2}{3}$

Q. 10 If $y = y(x), x \in (0, \pi/2)$ be the solution curve of the differential equation $(\sin^2 2x)\frac{dy}{dx} + (8\sin^2 2x + 2\sin 4x)y = 2e^{-4x}(2\sin 2x + \cos 2x)$, with $y(\pi/4) = e^{-\pi}$, then $y(\pi/6)$ is equal to:

Option 1:
$\frac{2}{\sqrt{3}}e^{-2\pi/3}$

Option 2:
$\frac{2}{\sqrt{3}}e^{2\pi/3}$

Option 3:
$\frac{1}{\sqrt{3}}e^{-2\pi/3}$

Option 4:
$\frac{1}{\sqrt{3}}e^{2\pi/3}$

Correct Answer:
$\frac{2}{\sqrt{3}}e^{-2\pi/3}$

Solution:
Given differential equation can be re-usritton as

$$\frac{dy}{dx} + (8 + 4\cot 2x)4 = \frac{2e^{-4x}}{\sin^2 2x}(2\sin x + \cos 2x)$$

which is a lénear diff equation

$$\text{I.f} = e^{\int (8+4\cot 2x)dx} = e^{8x} \times \sin^2 2x$$

$\therefore$ Solution is

$$y\left(e^{8x} \cdot \sin^2 2x\right) = \int 2e^{4x}(2\sin 2x + \cos 2x)dx + c$$

$$= e^{4x} \cdot \sin 2x + c$$

$$\text{Given } y\left(\frac{\pi}{4}\right) = e^{-\pi} \Rightarrow c = 0$$

$$\therefore y = \frac{e^{-4x}}{\sin 2x}$$

$$\therefore y\left(\frac{\pi}{6}\right) = \frac{e^{-4\times\frac{\pi}{6}}}{\sin\left(2\cdot\frac{\pi}{6}\right)} = \frac{2}{\sqrt{3}}e^{-\frac{2\pi}{3}}$$

Q. 11 If the tangents drawn at the points P and Q on the parabola $y^2 = 2x - 3$ intersect at the point $R(0, 1)$, then the orthocentre of the triangle PQR is :

Option 1:
$(0, 1)$

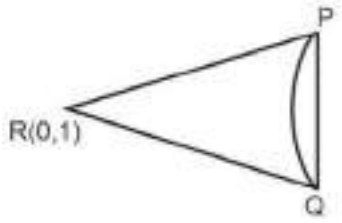
Option 2:
$(2, -1)$

Option 3:
$(6, 3)$

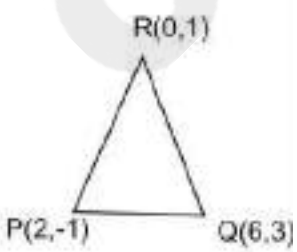
Option 4:
$(2, 1)$

Correct Answer:
$(2, -1)$

Solution:
$y^2 = 2x - 3 \ldots (1)$
equation of chord of Contact
$PQ : \gamma = 0$
$y \times 1 = (x + 0) - 3$
$y = x - 3 \ldots (2)$



$from (1) \& (2)$
$(x \cdot 3)^2 = 2x - 3$
$x^2 - 8x + 12 = 0$
$(x - 2)(x - 6) = 0$
$x = 2 \text{ or } 6$
$y = -1 \text{ or } 3$



$\therefore (3h-a)^2+(3k-b)^2=(s+c)^2+(s-c)^2=2\left(s^2+c^2\right)=2$

$\therefore \left(x-\frac{a}{3}\right)^2+\left(k-\frac{b}{3}\right)^2=\frac{2}{9}$

circle centre at $\left(\frac{a}{3},\frac{b}{3}\right)$

gives, $\frac{a}{3}=1,\frac{b}{3}=\frac{1}{3}\Rightarrow a=3,b=1$

$\therefore a^2-b^2=8$

$\mathrm{MPO}=\frac{1}{4}=1$

$\mathrm{MPP}=\frac{2}{6}=\frac{1}{3}$

$\mathrm{MPP}=\frac{2}{6}=\frac{1}{3}$

$\mathrm{MPR}=\frac{2}{-2}=-1$

$\mathrm{MPQ}\sim\mathrm{MPR}=-1\Rightarrow \mathrm{PQ}\perp\mathrm{PR}$

orthocentre $=\mathrm{P}(2,-1)$

Q. 12 Let C be the centre of the circle $x^2+y^2-x+2y=\frac{11}{4}$ and P be a point on the circle. A line passes through the point C, makes an angle of $\frac{\pi}{4}$ with the line CP and intersects the circle at the points Q and r. Then the area of the triangle PQR (in unit^2) is :

Option 1:
2

Option 2:
$2\sqrt{2}$

Option 3:
$8\sin\left(\frac{\pi}{8}\right)$

Option 4:
$8\cos\left(\frac{\pi}{8}\right)$

Correct Answer:
$2\sqrt{2}$

Solution:

$$x^2 + y^2 - x + 2y = \frac{11}{4}$$

$$\left(x - \frac{1}{2}\right)^2 + (y+1)^2 = 2^2$$

or $\triangle PQR$

$$PR = 2k \times \sin 2 \geq \frac{1}{3}$$

$$= 4.6 \sin \frac{\pi}{8}$$

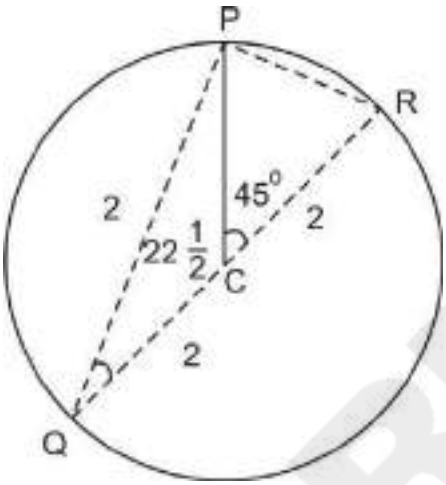
$$PQ = QR \cos 22\frac{1}{2}$$

$$= 4 \cos \frac{\pi}{8}$$

$$\text{As } \triangle P2R = \frac{1}{2} PR \times PQ$$

$$= \frac{1}{2}\left(4^2 \sin \frac{\pi}{6}\right)\left(4 \cos \frac{\pi}{8}\right)$$

$$4 \sin \frac{\pi}{4} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$



Q. 13 The remainder when $7^{2022} + 3^{2022}$ is divided by 5 is:

Option 1:
0

Option 2:
2

Option 3:
3

Option 4:

4

Correct Answer:

3

Solution:

$7^{2022}+3^{2022}$

$=(49)^{1011}+9^{1011}$

$=(50-1)^{1011}+(18-1)^{1011}$

$=51-1+5k-1$

$=5m-2$

$\text{Remainder } =5-2=3$

Q. 14 Let the matrix $A=\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$ and the matrix $B_0=A^{49}+2\,A^{98}$. If $B_n=\text{Adj}\,(B_{n-1})$ for all $n\geq 1$, then $\det(B_4)$ is equal to:

Option 1:

3^{28}

Option 2:

3^{30}

Option 3:

3^{32}

Option 4:

3^{36}

Correct Answer:

3^{32}

Solution:

$A^2=\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$

$=\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$

$a\leftrightarrow R_2$

$$-\begin{bmatrix}1&0&0\\0&0&1\\0&1&0\end{bmatrix}$$

$R_2 \leftrightarrow R_3$

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix} = I$$

$B_0 = A^{49} + 2A^{98}$

$= A + 2I$

$B_n = \text{Ad J}\,(B_{n-1})$

$= |B_0|^{(n-1)^4}$

$= |B_0|'$

$$B_0 = \begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix} + \begin{bmatrix}2&0&0\\0&2&0\\0&0&2\end{bmatrix}$$

$$= \begin{bmatrix}2&1&0\\0&2&1\\1&0&2\end{bmatrix}$$

$= 2(4-0) - 1(0-1) = 9$

$B_4 = (9)^{16} = 3^{3^2}$

Q. 15 Let $S_1 = \left\{z_1 \in C : |z_1 - 3| = \frac{1}{2}\right\}$ and $S_2 = \{z_2 \in \mathbf{C} : |z_2 - |z_2 + 1|| = |z_2 + |z_2 - 1||\}$.
Then, for $z_1 \in S_1$ and $Z_2 \in S_2$, the least value of $|z_2 - z_1|$ is:

Option 1:
0

Option 2:
$\frac{1}{2}$

Option 3:
$\frac{3}{2}$

Option 4:
$\frac{5}{2}$

Correct Answer:
$\frac{3}{2}$

Solution:

$|z_2+|z_2-1||^2 = |z_2-|z_2+1||^2$

$\Rightarrow |z_2+|z_2-1||\,(\bar{z}_2+|z_2-1|) = (z_2-|z_2+1|)\,(\bar{z}_2-(z_2+1)$

$\Rightarrow z_2\,|\bar{z}_2+\bar{z}_2-1|-(\overline{z_2}-|z_2+1|\,|+\bar{z}_2\,(|z_2-1|+|z_2+1|)$

$= |z_2+1|^2 = |z_2-1|^2$

$\Rightarrow (z_2+\overline{z_2})\,((z_2-11)+(z_2+11) = 2\,(z_2+\overline{z_2})$

$\Rightarrow (z_2+\bar{z}_2)\,(|z_2-1|+|z_2+1|-2) = 0$

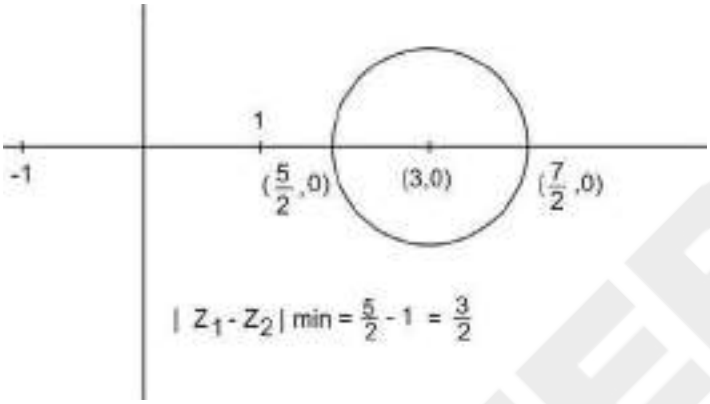
$\therefore z_2+\bar{z}_2 = 0$ or $|z_2-1|+|z_2+1|-2 = 0$

$\therefore z_2$ lie on imagmary axis. or on real axis

with in $[-1,1]$

AISO $|z-3| = \dfrac{1}{2}$ ie on circle having centre

3 and radius $\dfrac{1}{2}$.



$|z_1-z_2|\ \min\ = \dfrac{5}{2}-1 = \dfrac{3}{2}$

Q. 16 The foot of the perpendicular from a point on the circle $x^2+y^2=1, z=0$ to the plane $2x+3y+z=6$ lies on which one of the following curves ?

Option 1:
$(6x+5y-12)^2+4(3x+7y-8)^2=1, z=6-2x-3y$

Option 2:
$(5x+6y-12)^2+4(3x+5y-9)^2=1, z=6-2x-3y$

Option 3:
$(6x+5y-14)^2+9(3x+5y-7)^2=1, \quad z=6-2x-3y$

Option 4:
$(5x+6y-14)^2+9(3x+7y-8)^2=1, z=6-2x-3y$

Correct Answer:

$(5x+6y-12)^2+4(3x+5y-9)^2=1, z=6-2x-3y$

Solution:

$$\frac{h-\cos\theta}{2}=\frac{k-\sin\theta}{3}=\frac{\omega-0}{1}=\frac{-1(2\cos\theta+3\sin\theta-6)}{14}$$

$$\frac{h-\cos\theta}{2}=\frac{-(2\cos\theta+3\sin\theta-6)}{14}$$

$$h=\frac{\cos\theta-2(2\cos\theta+3\sin\theta-6)}{14}$$

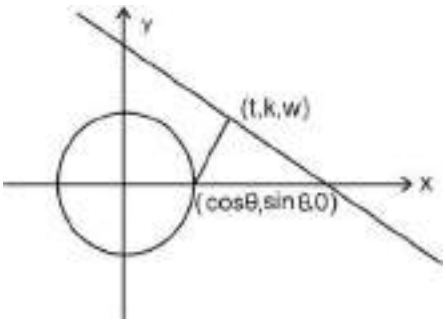
$$n=\frac{10\cos\theta-6\sin\theta+12}{14}$$

$$1<=\sin\theta-\frac{3}{14}(2\cos\theta+3\sin\theta-6)$$

$$k=\frac{5\sin\theta-6\cos\theta+18}{14}$$

Elementary $\text{Sin}\,\theta$ and $\text{Cos}\,\theta$

$$(5h+6k-12)^2+4(3h+5k-9)^2=1$$



Q. 17 If the minimum value of $f(x)=\frac{5x^2}{2}+\frac{\alpha}{x^5}, x>0$, is 14, then the value of α is equal to :

Option 1:

32

Option 2:

64

Option 3:

128

Option 4:

256

Correct Answer:

128

Solution:

$$\frac{x^2}{2}+\frac{x^2}{2}+\frac{x^2}{2}+\frac{x^2}{2}+\frac{x^2}{2}+\frac{\alpha}{2x^5}+\frac{\alpha^1}{2x^5}\geq 7\left(\frac{\alpha^2}{2^7}\right)^{\frac{1}{7}}$$

$$\frac{7\cdot(\alpha)^{2/7}}{2}=14$$

$$\left(\alpha^2\right)^{1/7}=2^2$$

$$\alpha=\left(2^2\right)^{7/2}=2^7$$

$$\alpha=128$$

Q. 18 Let α, β and γ be three positive real numbers. Let $f(x)=\alpha x^5+\beta x^3+\gamma x, x\in \mathbf{R}$ and $g:\mathbf{R}\to\mathbf{R}$ be such thattt $g(f(x))=x$ for all $x\in\mathbf{R}$. If $a_1, a_2, a_3, \ldots, a_n$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{n}\sum_{i=1}^{n} f(a_i)\right)\right)$ is equal to:

Option 1:
0

Option 2:
3

Option 3:
9

Option 4:
27

Correct Answer:
0

Solution:
Consider a case when $\alpha=\beta=0$ then

$$f(x)=4x$$

$$g(x)=\frac{x}{y}$$

$$\frac{1}{n}\sum_{i=1}^{n} f(a_i)\Rightarrow \frac{3}{n}(a_1+a_2+\cdots+a_n)$$

$$\Rightarrow f(g(0))\Rightarrow f(0)$$

$$\Rightarrow 0$$

Q. 19 Consider the sequence $a_1, a_2, a_3, \ldots$ such that $a_1=1, a_2=2$ and $a_{n+2}=\frac{2}{a_{n+1}}+a_n$ for $n=1,2,3,\ldots$ If $\left(\frac{a_1+\frac{1}{a_2}}{a_3}\right)\cdot\left(\frac{a_2+\frac{1}{a_3}}{a_4}\right)\cdot\left(\frac{a_3+\frac{1}{a_4}}{a_5}\right)\cdots\left(\frac{a_{30}+\frac{1}{a_{31}}}{a_{32}}\right)=2^{\alpha}(61C_{31})$, then α is equal to:

Option 1:
-30

Option 2:
-31

Option 3:
-60

Option 4:
-61

Correct Answer:
-60

Solution:
$a_{n+2}a_{n+1} - a_{n+1} \cdot a_n = 2$

series will satisfy

$$\frac{a_n + \frac{1}{a_{n+1}}}{a_{n+2}} = \frac{a_{n+2} - \frac{1}{a_{n+1}}}{a_{n+2}}$$

$$= 1 - \frac{1}{a_{n+1} \cdot a_{n+2}} = 1 - \frac{1}{2(r+1)}$$

$$= \frac{2\gamma + 1}{2(\gamma + 1)}$$

Now proof is given by

$$= \prod_{\gamma=1}^{30} \frac{(2\gamma + 1)}{2(\gamma + 1)}$$

$$= \frac{(1 \cdot 3 \cdot 5 \cdot \cdots \cdot 61)}{2^{30} \cdot (2 \cdot 3 \cdot \cdots \cdot 31)}$$

$$\Rightarrow \frac{(1.3 \cdot 5 \cdot \cdots \cdot 61)}{L31 \cdot 2^{30}} \times \frac{2^{30}}{2^{30}} \times \frac{L^{30}}{130}$$

$$= \frac{L6)}{2^{\circ}\angle 31 \cdot L^{\circ}}$$

$\alpha = -60$

Q. 20 The minimum value of the twice differentiable function $f(x) = \int_0^x e^{x-t} f'(t)dt - \left(x^2 - x + 1\right) e^x, x \in \mathbf{R}$, is:

Option 1:
$-\frac{2}{\sqrt{e}}$

Option 2:
$-2\sqrt{e}$

Option 3:

$-\sqrt{e}$

Option 4:

$\frac{2}{\sqrt{e}}$

Correct Answer:

$-\frac{2}{\sqrt{e}}$

Solution:

$$f(x)=e^x\cdot\int_0^x\frac{f'(t)}{e^t}dt$$

$$f(x)=e^x\cdot\int_0^x\frac{f'(t)}{e^t}dt+e^x\cdot\frac{f'(x)}{e^x}$$

$$=-\left[(2x-1)\cdot e^x+\left(x^2-x+1\right)\cdot e^x\right]$$

$$\frac{f'(x)}{e^t}dt=x^2+x$$

$$e^x=2x+1$$

$$f'(x)=(2x+1)\cdot e^x$$

$$f'(x)=0$$

$$f'(x)=(2x+1)\cdot e^x-2e^x+c \mid f(0)=-1$$

$$-1=1-2+c$$

$$c=0$$

$$f(x)=e^x(2x-1)$$

$$f\left(\frac{-1}{2}\right)=\frac{-2}{\sqrt{e}}$$

Q. 21 Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from $\{A, B, C, D, E\}$ or a number from $\{1, 2, 3, 4, 5\}$ with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from $\{1, 2, 3, 4, 5\}$ is $\alpha \times 5^6$, then α is equal to _____.

Correct Answer:

-

Q. 22 Let $P(-2, -1, 1)$ and $Q\left(\frac{56}{17}, \frac{43}{17}, \frac{111}{17}\right)$ be the vertices of the rhombus $PRQS$. If the direction ratios of the diagonal RS are $\alpha, -1, \beta$, where both α and β are integers of minimum absolute values, then $\alpha^2+\beta^2$ is equal to _____.

Correct Answer:

450

Solution:

$\mathrm{RS} \equiv (\alpha, -1, \beta)$

$\mathrm{DR\ of\ PQ} \equiv \left(\frac{56}{17}+2, \frac{43}{17}+1, \frac{111}{17}-1\right)$

$\equiv \left(\frac{90}{17}, \frac{60}{17}, \frac{24}{17}\right)$

$\frac{90}{17}\alpha + \frac{60}{17}(-1) + \frac{94}{17}\beta = 0$

$90\alpha + 94\beta = 60$

$\beta = \frac{60 - 9^{\circ}\alpha}{94}$

$\beta = \frac{30(2-3\alpha)}{94}$

$\beta = -30\frac{(3\alpha-2)}{94}$

$\beta = \frac{-15}{47}(3\alpha - 2)$

$\Rightarrow \frac{\beta}{75} = \frac{3\alpha - 2}{47}$

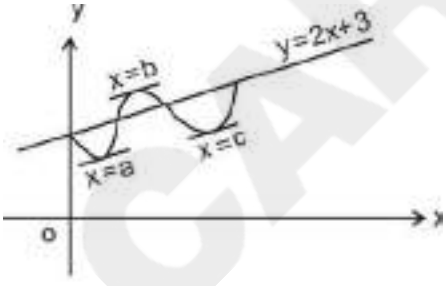
$\Rightarrow \beta = -15, \quad \alpha = -15$

$\alpha^2 + \beta^2 = 225 + 225 = 450$

Q. 23 Let $\mathrm{f} : [0,1] \to \mathbf{R}$ be a twice differentiable function in $(0,1)$ such that $\mathrm{f}(0) = 3$ and $\mathrm{f}(1) = 5$. If the line $\mathrm{y} = 2\mathrm{x} + 3$ intersects the graph of f at only two distinct points in $(0,1)$, then the least number of points $\mathrm{x} \in (0,1)$, at which $\mathrm{f}''(\mathrm{x}) = 0$, is _____.

Correct Answer:
2

Solution:



$\mathrm{f}'(\mathrm{a}) = \mathrm{f}'(\mathrm{b}) = \mathrm{f}'(\mathrm{c}) = 2$

$\Rightarrow \mathrm{f}''(\mathrm{x})$ is zero

for at least $\mathrm{x}_1 \in (\mathrm{a}, \mathrm{b}) \& \mathrm{x}_2 \in (\mathrm{b}, \mathrm{c})$

Hence, two number of points.

Q. 24 If $\int_0^{\sqrt{3}} \frac{15x^3}{\sqrt{1+x^2+\sqrt{(1+x^2)^3}}}\,dx = \alpha\sqrt{2}+\beta\sqrt{3}$, where α, β are integers, then $\alpha+\beta$ is equal to _____.

Correct Answer:
10

Solution:

$\text{Put } 1+x^2=t^2$

$2xdx = 2t2t$

$xdx = tdt$

$\therefore \int_1^2 \frac{15\,(t^2-1)\,tdt}{\sqrt{t^2+t^3}}$

$15\int_1^2 \frac{t\,(t^2-1)}{t\sqrt{1-t}}dt$

$\text{Put } 1+t=u^2$

$dt = 2udu$

$15\int_{\sqrt{2}}^{\sqrt{3}} \frac{(u^2-1)^2-1}{u}\times 2udu$

$30\int_{\sqrt{2}}^{\sqrt{3}} \left(u^4-2u^2\right)du$

$30\left(\frac{u^5}{5}-\frac{2u^3}{3}\right)_{\sqrt{2}}^{\sqrt{3}}$

$30\left[\frac{1}{5}\left((\sqrt{3})^5-(\sqrt{2})^5\right)-\frac{2}{3}\left((\sqrt{3})^3-(\sqrt{2})^3\right)\right]$

$30\left[\frac{1}{5}(9\sqrt{3}-4\sqrt{2})-\frac{2}{3}(3\sqrt{3}-2\sqrt{2})\right]$

$30\left[-\frac{1}{5}\times\sqrt{3}+\frac{8}{15}\sqrt{2}\right]$

$-6\sqrt{3}+16\sqrt{2}=\alpha\sqrt{2}+\beta\sqrt{3}$

$\alpha=16, \beta=-6$

$\therefore \alpha+\beta=10$

Q. 25 Let $A=\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}$ and $B=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}$, $\alpha,\beta\in\mathbf{R}$ Let α_1 be the value of α which satisfies $(A+B)^2=A^2+\begin{bmatrix}2 & 2\\ 2 & 2\end{bmatrix}$ and α_2 be the value of α which satisfies $(A+B)^2=B^2$. Then $|\alpha_1-\alpha_2|$ is equal to _____.

Correct Answer:
2

Solution:

$$A+B=\begin{bmatrix}\beta+1 & 0\\ 3 & \alpha\end{bmatrix}$$

$$(A+\beta)^2=\begin{bmatrix}\beta+1 & 0\\ 3 & \alpha\end{bmatrix}\begin{bmatrix}\beta+1 & 0\\ 3 & \alpha\end{bmatrix}$$

$$=\begin{bmatrix}(\beta+1)^2 & 0\\ 3(\beta+1)+3\alpha & \alpha^2\end{bmatrix}$$

$$A^2=\begin{bmatrix}1 & -1\\ 2 & \alpha\end{bmatrix}\begin{bmatrix}1 & -1\\ 2 & 2\end{bmatrix}$$

$$\therefore\begin{bmatrix}1 & -\alpha+1\\ 2\alpha+4 & \alpha^2\end{bmatrix}=\begin{bmatrix}(\beta+1)^2 & 0\\ 3(\alpha+\beta+1) & \alpha^2\end{bmatrix}$$

$\alpha=1=\alpha_1$

$$\beta^2=\begin{bmatrix}\beta & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}\beta & 1\\ 1 & \theta\end{bmatrix}$$

$$=\begin{bmatrix}\beta^2+1 & \beta\\ \beta & 1\end{bmatrix}=\begin{bmatrix}(\beta+1)^2 & 0\\ 3(\beta+1)+3\alpha & \alpha^2\end{bmatrix}$$

$\therefore \beta=0, \alpha=-1\div\alpha_2$

$|\alpha_1-\alpha_2|=|1-(-1)|=2$

Q. 26 For $p, q \in \mathbf{R}$, consider the real valued function $f(x)=(x-p)^2-q, x \in \mathbf{R}$ and $q>0$. Let a_1, a_2, a_3 and a_4 be in an arithmetic progression with mean P and positive common difference. If $|f(a_i)|=500$ for all $i=1,2,3,4$, then the absolute difference between the roots of $f(x)=0$ is _____.

Correct Answer:
50

Solution:

$f(x)=0 \Rightarrow (x-p)^2-q=0$

Roots are $p+\sqrt{q}, p-\sqrt{q}$ absolute difference between roots $2\sqrt{q}$.

Now, $|f(a_i)|=500$

Let a_1, a_2, a_3, a_4, are $a, a+d, a+2d_1, +3d$

$|f(a_4)|=500$

$\left|(a_1-p)^2-q\right|=500$

$\Rightarrow (a_1-p)^2-q=500$

$\Rightarrow \frac{9}{4}d^2-q=500 \quad -(1)$

and $|f(a_1)|^2=|f(a_2)|^2$

$\left\{(a_1-p)^2-q\right\}^2=\left((a_2-p)^2-q\right)^2$

$\Rightarrow \left((a_1-p)^2-(a_2-p)^2\right)\left((a_1-p)^2-q+(a_2-p)^2-q\right)=0$

$\Rightarrow \frac{9}{4}d^2-q+\frac{d^2}{4}-q=0$

$\Rightarrow d^2=\frac{4q}{5}$

from eq^{n} (1) $\frac{9}{4} \cdot \frac{4 \cdot q}{5} - q = 500$

$$\frac{4q}{5} = 500$$

and $2\sqrt{q} = 2 \times \frac{50}{2} = 50$

Q. 27 For the hyperbola $H : x^2 - y^2 = 1$ and the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b > 0,$ let the

(1) eccentricity of E be reciprocal of the eccentricity of H, and

(2) the line $y = \sqrt{\frac{5}{2}}x + K$ be a common tangent of E and H. Then $4\left(a^2 + b^2\right)$ is equal to _____.

Correct Answer: 3

Solution:

$e_E = \sqrt{1 - \frac{b^2}{a^2}}, e_H = \sqrt{2}$

if $e_E = \frac{1}{e_H}$

$\Rightarrow \frac{a^2 - b^2}{a^2} = \frac{1}{2}$

$2a^{2-2b} \cdot 2 = a^2$

$a^2 = 2b^2$

and $y = \sqrt{\frac{5}{2}}x + k$ is tangent to ellipse then

$k^2 = a^2 \times \frac{5}{2} + b^2 = \frac{3}{2}$

$6b^2 = \frac{3}{2} \Rightarrow b^2 = \frac{1}{4}$ and $a^2 = \frac{1}{2}$

$\therefore 4 \cdot \left(a^2 + b^2\right) = 3$

Q. 28 Let $x_1, x_2, x_3, \ldots, x_{20}$ be in geometric progression with $x_1 = 3$ and the common ratio $\frac{1}{2}$. A new data is constructed replacing each x_i by $(x_i - i)^2$. If $\bar{x}$ is the mean of new data, then the greatest integer less than or equal to $\bar{x}$ is _____.

Correct Answer:
142

Solution:

$$\Sigma x_0' = \frac{3\left(1-\left(\frac{1}{2}\right)^{20}\right)}{1-\frac{1}{2}} = 6\left(1-\frac{1}{2^2}\right)$$

$$= \sum_{i=1}^{20} (x_i + i)^2$$

$$= \sum_{i=1}^{20} (x_i)^2 + (i)^2 - 2x_i i$$

$$\text{Now } \sum_{i=1}^{20} (x_i)^2 = \frac{9\left(1-\left(\frac{1}{4}\right)^{20}\right)}{1-\frac{1}{4}}$$

$$= 12\left(1-\frac{1}{2^{40}}\right)$$

$$\sum_{i=1}^{20} i^2 = \frac{1}{6} \times 20 \times 21 \times 41 = 287^\circ$$

$$\sum_{i=1}^{20} x_i i = s = 3 + 2\cdot 3\cdot\frac{1}{2} + 3\cdot 3\cdot\frac{1}{2^2} + 4\cdot 3\cdot\frac{1}{2^3} + \cdots$$

$$= 6\left(2-\frac{22}{2^{20}}\right)$$

$$\bar{x} = \frac{12 - \frac{12}{2^{40}} + 2870 - 12\left(2-\frac{22}{2^{20}}\right)}{20}$$

$$\bar{x} = \frac{2858}{20} + \left(\frac{-12}{240} + \frac{22}{22^0}\right) \times \frac{1}{20}$$

$$[\bar{x}] = 142$$

Q. 29 $\lim\limits_{x\to 0}\left(\frac{(x+2\cos x)^3 + 2(x+2\cos x)^2 + 3\sin(x+2\cos x)}{(x+2)^3 + 2(x+2)^2 + 3\sin(x+2)}\right)^{\frac{100}{x}}$ is equal to _____.

Correct Answer: 1

Solution:

$$\lim_{x\to 10}\left(\frac{(x+2\cos x)^3 + 2(x+2\cos x)^2 + 3\sin(x+2\cos x)}{(x+2)^3 + 2(x+2)^2 + 3\sin(x+2)}\right)^2 \text{ form } 1^\infty$$

$$= \lim_{x\to 0}\left[\left(\frac{(x+2\cos x)^3+2(x+2\cos x)^2+3\sin(x+2\cos x)}{(x+2)^3+2(x+2)^2+3\sin(x+2)}\right) - 1\right] \times \frac{100}{x}$$

$$e^{\frac{100}{e^{16+3\sin 2}}} \lim_{x\to 0} \frac{3(x+2\cos x)^2 \times (1+2\sin x) - 3(x+2)^-4(x+2\cos x)}{x(1-2\sin x) - 4(x+2) + 3\cos(x+2\cos x) \times (1-2\sin x) - 3\cos(x+2)}$$

$$e^{\frac{100}{e^{16+3\sin 2}}} x\to 0 \frac{12 - 3(4) + 8 \times 1 - 8 + 3\cos 2 - 3\cos 2}{1} \text{using L'H rule}$$

$$= e^0 = 1$$

Q. 30 The sum of all real values of x for which $\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2+5x+12}$ is equal to _____.

Option 1:
No real values possible

Option 2:
0

Option 3:
2

Option 4:
4

Correct Answer:
No real values possible

Solution:

$$\frac{3x^2-9x+17}{x^2+3x+10}=\frac{5x^2-7x+19}{3x^2-5x+12}$$

$$\frac{x^2+3x+10+2x^2-12x+7}{x^2+3x+10}=\frac{3x^2+5x+12+2x^2-12x+7}{3x^2+5x+12}$$

$$1+\frac{2x^2-12x+7}{x^2+3x+10}=1+\frac{2x^2-12x+7}{3x^2+5x+12}$$

$$(2x^2-12x+7)\left(\frac{1}{x^2+3x+10}-\frac{1}{3x^2+5x+12}\right)=0$$

$2x^2-12x+7=0$ or $3x^2+5x+12=x^2+3x+10$

$x=\frac{12\pm\sqrt{2}}{4}$ $\qquad 2x^2+2x+2=0$

$x^2+x+1=0$

Sum of Roots=6 $\qquad$ No solution

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28 July - Evening Shift

Physics

Q. 1 Consider the efficiency of carnot's engine is given by $\eta = \frac{\alpha\beta}{\sin\theta}\log e\frac{\beta x}{kT}$, where α and β are constants. If T is temperature, k is Boltzmann constant, θ is angular displacement and x has the dimensions of length. Then, choose the incorrect option

Option 1:
Dimensions of β is same as that of force.

Option 2:
Dimensions of $\alpha^{-1}\mathrm{x}$ is same as that of energy.

Option 3:
Dimensions of $\eta^{-1}\sin\theta$ is same as that of $\alpha\beta$.

Option 4:
Dimensions of α is same as that of β.

Correct Answer:
Dimensions of β is same as that of force.

Solution:

$$\eta = \frac{\alpha\beta}{\sin\theta}\log_e\left(\frac{\beta\mathrm{x}}{\mathrm{kT}}\right)$$

$$[\mathrm{KT}] = [\text{ Energy }] = \left[\mathrm{M}^1\mathrm{L}^2\mathrm{T}^{-2}\right]$$

$$\left[\frac{\beta\mathrm{x}}{\mathrm{KT}}\right] = [\mathrm{M}^\circ\mathrm{L}^\circ\mathrm{T}^\circ]$$

$$\therefore [\beta\mathrm{x}] = \left[\mathrm{M}^1\mathrm{L}^2\mathrm{T}^{-2}\right]$$

$$\rightarrow [\beta] = \left[\mathrm{M}^1\mathrm{L}^1\mathrm{T}^{-2}\right]$$

$$\because [\mathrm{x}] = \left[\mathrm{L}^1\right]$$

$[\theta] = [\mathrm{M^\circ L^\circ T^\circ}]$

$[\eta] = [\alpha\beta] = [\mathrm{M^\circ L^\circ T^\circ}]$

$\therefore [\alpha] = [\beta^{-1}]$

$\Rightarrow [\alpha] = [\mathrm{M^{-1}L^{-1}T^{+2}}]$

Dimension of α is not same as that of β

$\therefore$ Hence 4 is incorrect option.

Q. 2 At time $\mathrm{t} = 0$ a particle starts travelling from a height $7\hat{z}\ \mathrm{cm}$ in a plane keeping z coordinate constant. At any instant of time it's position along the $\hat{x}\ \text{and}\ \hat{y}$ directions are defined as $3\mathrm{t}\ \text{and}\ 5\mathrm{t}^3$ respectively. At $\mathrm{t} = 1\ \mathrm{s}$ acceleration of the particle will be

Option 1:
$-30\ \hat{\mathrm{y}}$

Option 2:
$30\hat{\mathrm{y}}$

Option 3:
$3\hat{\mathrm{x}} + 15\hat{\mathrm{y}}$

Option 4:
$3\hat{\mathrm{x}} + 15\hat{\mathrm{y}} + 7\hat{\mathrm{z}}$

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Correct Answer:
$30\hat{y}$

Solution:
$\bar{r} = 3t\hat{i} + 5t^3\hat{j} + 7\hat{k}$

$\text{Initial position is } (0, 0, 7)\text{cm}$

$\bar{r} \rightarrow \text{position vector}$

$$\bar{v} = \frac{d\bar{r}}{dt} = 3\hat{i} + 15t^2\hat{j} + 0$$

$$\bar{a} = \frac{d\bar{v}}{dt} = 0 + 30t\hat{j} + 0$$

$\text{at } t = 1, \bar{a} = 30\hat{j} \text{ or}$

$$\bar{a} = 30\hat{y}$$

Hence 2 is correct option

The comect option is (2)

Q. 3 A pressure-pump has a horizontal tube of cross sectional area $10\ \text{cm}^2$ for the outflow of water at a speed of $20\ \text{m/s}$. The force exerted on the vertical wall just in front of the tube which stops water horizontally flowing out of the tube, is :
[given: density of water $= 1000\ \text{kg/m}^3$]

Option 1:
$300\ \text{N}$

Option 2:
$500\ \text{N}$

Option 3:
$250\ \text{N}$

Option 4:

$400\ \mathrm{N}$

Correct Answer:

$400\ \mathrm{N}$

Solution:

$\mathrm{A} = 10\ \mathrm{cm}^2 = 10 \times 10^{-4\ \mathrm{m}^2}$

$\mathrm{V} = 20\ \mathrm{m/s}$

$\Delta \mathrm{p}_{\text{wall}} - \Delta \mathrm{P}_{\text{wall}} = \mathrm{P}_\mathrm{i} - \mathrm{P}_\mathrm{f}$

$\Delta \mathrm{p} = (\Delta \mathrm{m})\mathrm{v} - 0$

$\Delta \mathrm{p} = (\Delta \mathrm{m})\mathrm{v}$

$\Delta \mathrm{p}_{\text{wall}} = \rho(\mathrm{A}\Delta \mathrm{x})\mathrm{v}$

Force exerted on wall is

$$\mathrm{F} = \frac{\Delta \mathrm{P\ wall}}{\Delta \mathrm{t}} = \rho \mathrm{A}\frac{(\Delta \mathrm{x})\mathrm{v}}{\Delta \mathrm{t}}$$

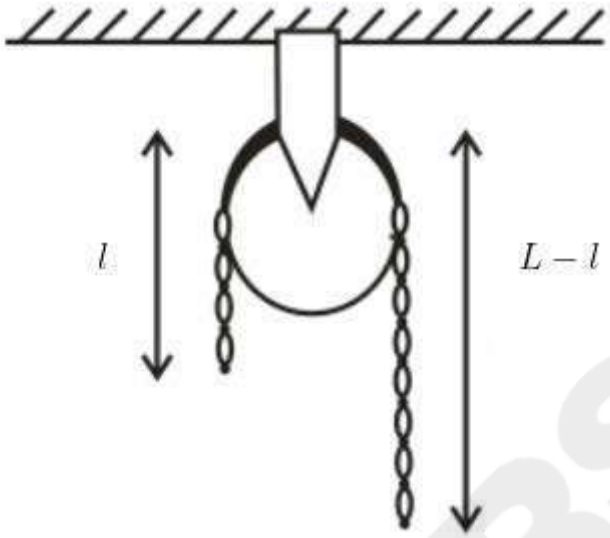
$\mathrm{F} = \rho \mathrm{AV}^2$

$= 10^3 \times 10^{-3} \times (20)^2$

$\mathrm{F} = 400\ \mathrm{N}$

Hence 4 is correct option.

Q. 4 A uniform metal chain of mass m and length $'\mathrm{L}'$ passes over a massless and frictionless pulley. It is released from rest with a part of its length l is hanging on one side and rest of its length $L-l$ is hanging on the other side of the pully. At a certain point of time, when $l=\frac{L}{x}$, the acceleration of the chain is $\frac{\mathrm{g}}{2}$. The value of x is_____.



Option 1:
6

Option 2:
2

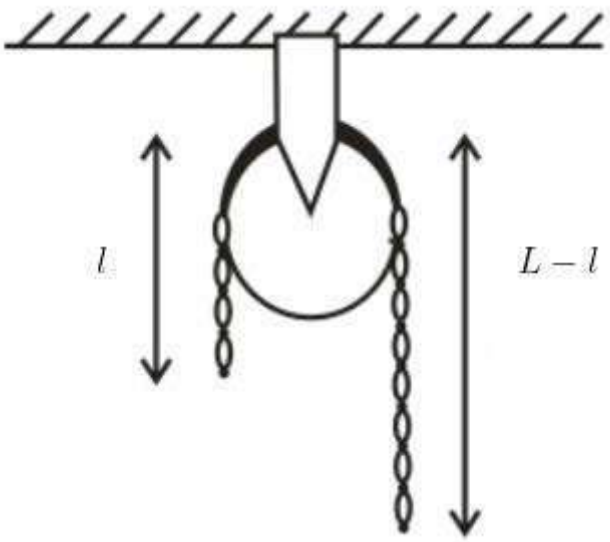
Option 3:
1.5

Option 4:
4

Correct Answer:
4

Solution:

CAREERS 360



Let the mass of complete chain be $'\mathrm{m}'$and mass of chain of length $'\ell'$ and $'\mathrm{L}-\ell'$ be m_1 and m_2 respectively

$$\mathrm{m}_1+\mathrm{m}_2=\mathrm{m}$$

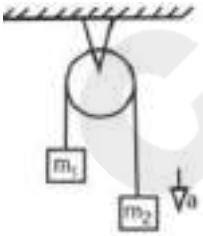
$$\mathrm{m}_1=\left(\frac{\mathrm{m}}{\mathrm{L}}\right)\ell$$

$$\mathrm{m}_2=\left(\frac{\mathrm{m}}{\mathrm{L}}\right)(\mathrm{L}-\ell)$$

$$(\mathrm{m}_2-\mathrm{m}_1)_{\mathrm{g}}=(\mathrm{m}_2+\mathrm{m}_1)\,\mathrm{a}$$

$$\mathrm{a}=\left(\frac{\mathrm{m}_2-\mathrm{m}_1}{\mathrm{m}_2+\mathrm{m}_1}\right)\mathrm{g}$$

$$\frac{\mathrm{g}}{2}=\frac{\left(\frac{\mathrm{m}}{\mathrm{L}}\right)(\mathrm{L}-2\ell)}{\mathrm{m}}\mathrm{g}$$



$$\frac{1}{2}=\frac{\mathrm{L}-2\ell}{\mathrm{L}}$$

$$\mathrm{L}=2\mathrm{L}-4\ell$$

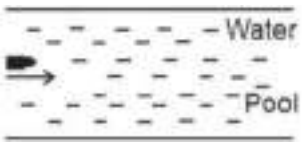
$$4\ell=\mathrm{L}$$

$$\ell = \frac{L}{4}$$

$$\therefore x = 4$$

Hence 4 is the correct option

Q. 5 A bullet of mass $200\ g$ having initial kinetic energy $90\ J$ is shot inside a long swimming pool as shown in the figure. If it's kinetic energy reduces to $40\ J$ within $1\ s$, the minimum length of the pool, the bullet has to travel so that it completely comes to rest is



Option 1:
$45\ m$

Option 2:
$90\ m$

Option 3:
$125\ m$

Option 4:
$25\ m$

Correct Answer:
$45\ m$

Solution:
Assuming that the bullet undergoes constant retardation

$$KE_i = 90\ J = \frac{1}{2}mu^2$$

$$KE_f = 40\ J = \frac{1}{2}mv^2$$

$$\frac{u}{v} = \frac{3}{2} \rightarrow (1)$$

Also, $90 = \frac{1}{2}(0.2)u^2$

$u = 30\ m/s$

$\therefore v = 20\ m/s \quad (\text{from eqn } 1)$

$\text{from } t = 0 \text{ to } t = 1\ s,$

$v = u + at$

$20 = 30 + a(1)$

$a = -10\ m/s^2$

The time at which bullet stops,

$\therefore \quad v = 0 = 30 + (-10)t$

$t = 3s$

$S = ut + \frac{1}{2}at^2$

$= 30(3) + \frac{1}{2}(-10)(3)^2$

$= 90 - 45 = 45m$

Hence 1 is correct option

Q. 6 Assume there are two identical simple pendulum clocks. Clock -1 is placed on the earth and Clock - 2 is placed on a space station located at a height h above the earth surface. Clock -1 and Clock - 2 operate at time periods $4\ s \text{ and } 6\ s$ respectively. Then the value of h is -
(consider radius of earth $R_E = 6400\ km \text{ and g on earth } 10\ m/s^2$)

Option 1:
$1200\ km$

Option 2:
$1600\ km$

Option 3:

$3200\ \text{km}$

Option 4:

$4800\ \text{km}$

Correct Answer:

$3200\ \text{km}$

Solution:

$$T_1 = 4S = 2\pi\sqrt{\frac{l}{g}} \rightarrow (1)$$

$$T_2 = 6S = 2\pi\sqrt{\frac{l}{g_h}} \rightarrow (2)$$

$$g_h = \frac{gR^2}{(R+h)^2} \rightarrow (3)$$

$$\frac{T_1}{T_2} = \frac{4}{6} = \sqrt{\frac{g_h}{g}}$$

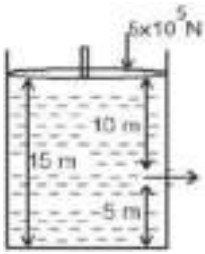
$$\frac{2}{3} = \sqrt{\frac{g_h}{g}}$$

$$\frac{4}{9} = \frac{g_h}{g} = \frac{R^2}{(R+h)^2}$$

$$\therefore h = \frac{R}{2} = 3200\ \text{km}$$

Hence 3 is correct option

Q. 7 Consider a cylindrical tank of radius $1\ \mathrm{m}$ is filled with water. The top surface of water is at $15\ \mathrm{m}$ from the bottom of the cylinder. There is a hole on the wall of cylinder at a height of $5\ \mathrm{m}$ from the bottom. A force of $5 \times 10^5\ \mathrm{N}$ is applied an the top surface of water using a piston. The speed of ifflux from the hole will be : (given atmospheric pressure $\mathrm{P_A} = 1.01 \times 10^5\ \mathrm{Pa}$, density of water $\rho_{\mathrm{W}} = 1000\ \mathrm{kg/m^3}$ and gravitational acceleration $\mathrm{g} = 10\ \mathrm{m/s^2}$)



Option 1:
$11.6\ \mathrm{m/s}$

Option 2:
$10.8\ \mathrm{m/s}$

Option 3:
$17.8\ \mathrm{m/s}$

Option 4:
$14.4\ \mathrm{m/s}$

Correct Answer:
$17.8\ \mathrm{m/s}$

Solution:
$\mathrm{h_1} = 15\ \mathrm{m}$

$\mathrm{h_2} = 5\ \mathrm{m}$

From Eq of Continuity

$$\mathrm{A_1V_1} = \mathrm{A_2V_2}$$

$$\mathrm{V_1} = \left(\frac{\mathrm{A_2}}{\mathrm{A_1}}\right)\mathrm{V_2}$$

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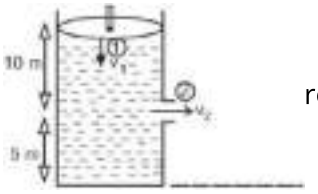


reference level

Applying Bernoulli's equation at pt (1) & (2)

$$\mathrm{P_1} + \frac{1}{2}\rho \mathrm{v_1^2} + \rho \mathrm{g h_1} = \mathrm{P_2} + \frac{1}{2}\rho \mathrm{v_2^2} + \rho \mathrm{g h_2}$$

$$\frac{5 \times 10^5}{\pi (1)^2} + 0 + \rho \mathrm{g h_1} = 1.01 \times 10^5 + \frac{1}{2}\rho \mathrm{v_2^2} + \rho \mathrm{g}\,(\mathrm{h_2})$$

$$\mathrm{v_1} << \mathrm{v_2} \quad (\mathrm{A_1} \gg \mathrm{A_2})$$

$$\left(\frac{5}{\pi} - 1.01\right) \times 10^5 + 10^5 = 500 \mathrm{v_2^2}$$

$$\frac{4.686}{\pi} \times 10^5 = 500\ \mathrm{V_2^2}$$

$$\mathrm{V_2} = 17.8\ \mathrm{m/s}$$

Hence 3 is correct option.

Q. 8 A vessel contains $14\ \mathrm{g}$ of nitrogen gas at a temperature of $27^{\circ}\mathrm{C}$. The amount of heat to be transferred to the gas to double the r.m.s speed of its molecules will be :
Take $\mathrm{R} = 8.32\ \mathrm{J\ mol^{-1} k^{-1}}$.

Option 1:
$2229\ \mathrm{J}$

Option 2:
$5616\ \text{J}$

Option 3:
$9360\ \text{J}$

Option 4:
$13,104\ \text{J}$

Correct Answer:
$9360\ \text{J}$

Solution:

$\text{m} = 149 \qquad \text{T} = 300\,\text{k}$

$$\text{V}_{\text{rms}} = \sqrt{\frac{3\text{RT}}{\text{M}}}$$

To double the rms speed,

$$\text{T}_{\text{f}} = 4\text{T}_{\text{i}} = 1200\ \text{K}$$

From first law of thermodynamice,

$$\text{Q} = \Delta\text{U} + \text{W}$$

$$\text{Q} = \Delta\text{U} + 0 \quad (\text{V} = \text{constant of vessd})$$

$$\text{Q} = \Delta\text{U} = \frac{\text{nR}\Delta\text{T}}{\text{V} - 1}$$

$$\text{Q} = \frac{(1/2)(8.32)}{\left(\frac{7}{5} - 1\right)} \times (900)$$

$$= \frac{5(8.32) \times 900}{4}$$

$$\text{Q} = 9360\ \text{J}$$

Hence 3 is correct option.

Q. 9 A slab of dielectric constant K has the same cross-sectional area as the plates of a parallel plate capacitor and thickness $\frac{3}{4}\mathrm{d}$, where d is the separation of the plates. The capacitance of the capacitor when the slab is inserted between the plates will be : (Given $\mathrm{C_0} =$ capacitance of capacitor with air as medium between plates.)

Option 1:
$\frac{4\mathrm{KC_0}}{3+\mathrm{K}}$

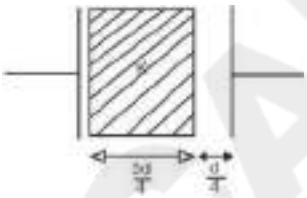
Option 2:
$\frac{3\mathrm{KC_0}}{3+\mathrm{K}}$

Option 3:
$\frac{3+\mathrm{K}}{4\mathrm{KC_0}}$

Option 4:
$\frac{\mathrm{K}}{4+\mathrm{K}}$

Correct Answer:
$\frac{4\mathrm{KC_0}}{3+\mathrm{K}}$

Solution:



$$\mathrm{C} = \frac{\mathrm{A}\Sigma_0}{\frac{\mathrm{t_1}}{\mathrm{k_1}} + \frac{\mathrm{t_2}}{\mathrm{k_2}}} = \frac{\mathrm{A}\Sigma_0}{\frac{3\mathrm{d}/4}{\mathrm{k}} + \frac{\mathrm{d}/4}{1}}$$

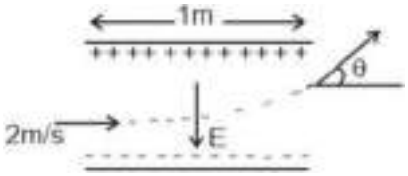
$$= \frac{4\mathrm{A}\Sigma_0\mathrm{K}}{\mathrm{d}(3+\mathrm{k})}$$

$$\mathrm{C_0} = \frac{\mathrm{As_0}}{\mathrm{d}} \ (\text{Given})$$

$$\mathrm{C} = \frac{4\mathrm{C_0k}}{3+\mathrm{k}}$$

Hence 1 is correct option.

Q. 10 A uniform electric field $\mathrm{E} = (8\ \mathrm{m/e})\mathrm{V/m}$ is created between two parallel plates of length $1\ \mathrm{m}$ as shown in figure, (where $\mathrm{m} =$ mass of electron and $\mathrm{e} =$ charge of electron). An electron enters the field symmetrically between the plates with a speed of $2\ \mathrm{m/s}$. The angle of the deviation (θ) of the path of the electron as it comes out of the field will be_____.



Option 1:
$\tan^{-1}(4)$

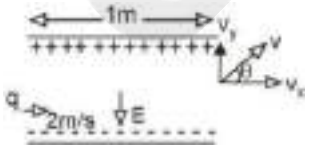
Option 2:
$\tan^{-1}(2)$

Option 3:
$\tan^{-1}\left(\frac{1}{3}\right)$

Option 4:
$\tan^{-1}(3)$

Correct Answer:
$\tan^{-1}(2)$

Solution:



$$\mathrm{V_y} = \mathrm{u_y} + \mathrm{a_y t}$$

$$\mathrm{V_y} = 0 + \frac{(\mathrm{e})(-\mathrm{E})\mathrm{t}}{\mathrm{m}}$$

$$\mathrm{V_y} = 8\mathrm{t} \rightarrow (1)$$

$V_x = 2\ m/s \rightarrow (2)$

Time to cross the plates,

$$t = \frac{x}{V_x} = \frac{1}{2}s$$

$$\therefore V_y = 4\ m/s$$

$$\tan\theta = \frac{V_y}{V_x} = \frac{4}{2} = 2$$

$$\theta = \tan^{-1}(2)$$

Hence 2 is correct option

Q. 11 Given below are two statements:

Statement I : A uniform wire of resistance 80Ω is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be 5Ω.

Statement II: Two resistances $2R \text{ and } 3R$ are connected in parallel in a electric circuit. The value of thermal energy developed in $3R \text{ and } 2R$ will be in the ratio 3: 2.

In the light of the above statements, choose the most appropriate answer from the option given below

Option 1:

Both statement I and statement II are correct

Option 2:

Both statement I and statement II are incorrect

Option 3:

Statement I is correct but statement II is incorrect

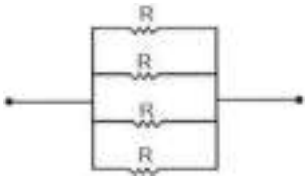
Option 4:
Statement I is incorrect but statement II is correct

Correct Answer:
Both statement I and statement II are correct

Solution:
Statement 1:

Resistance of each part $= \frac{80}{4}$

$= 20\Omega$

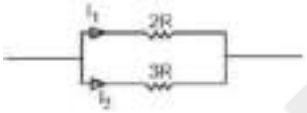


$R_{eq} = \frac{R}{4}$

$= 5\Omega$

The equivalent resistance of the combination will be 5Ω

Statement 2:



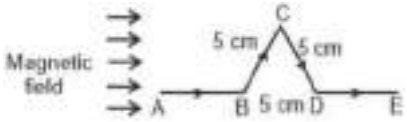
$I_1(2R) = I_2(3R)\,(\text{parallel connection})$

$\frac{I_1}{I_2} = \frac{3}{2} \quad ; \quad \frac{H_1}{H_2} = \frac{I_1^2(2R)t}{I_2^2(3R)t} = \frac{3}{2}$

Both statement I & II are correct

Hence 1 is correct option.

Q. 12 A triangular shaped wire carrying $10\ \mathrm{A}$ current is placed in a uniform magnetic field of $0.5\ \mathrm{T}$, as shown in figure. The magnetic force on segment CD is (Given $\mathrm{BC} = \mathrm{CD} = \mathrm{BD} = 5\ \mathrm{cm}$.)



Option 1:
$0.126\ \mathrm{N}$

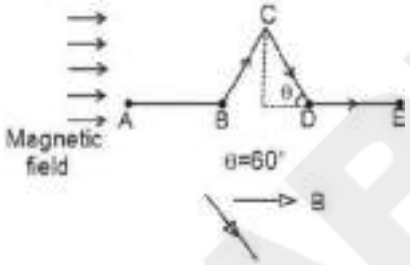
Option 2:
$0.312\ \mathrm{N}$

Option 3:
$0.216\ \mathrm{N}$

Option 4:
$0.245\ \mathrm{N}$

Correct Answer:
$0.216\ \mathrm{N}$

Solution:



$$\bar{\mathrm{F}} = \mathrm{I}(\bar{\ell} \times \bar{\mathrm{B}})$$

$$\mathrm{F} = \mathrm{I}\ell\mathrm{B}\sin\theta$$

$$= 10 \times 5 \times 10^{-2} \times 0.5 \times \frac{\sqrt{3}}{2}$$

$$\mathrm{F} = \frac{25\sqrt{3}}{2} \times 10^{-2}$$

$$\mathrm{F} = 0.216\ \mathrm{N}$$

Hence 3 is correct option.

Q. 13 The magnetic field at the center of current carrying circular loop is B_1. The magnetic field at a distance of $\sqrt{3}$ times radius of the given circular loop from the center on its axis is B_2. The value of B_1/B_2 will be

Option 1:
$9:4$

Option 2:
$12:\sqrt{5}$

Option 3:
$8:1$

Option 4:
$5:\sqrt{3}$

Correct Answer:
$8:1$

Solution:
Magnetic field at the centre of current carrying coil is

$$B_1=\frac{\mu_o i}{2r}\rightarrow(1)$$

Magnetic field at the axial position

$$B_2=\frac{\mu_0 i r^2}{2\left(r^2+x^2\right)^{3/2}}$$

$$x=\sqrt{3}r$$

$$B_2=\frac{\mu_0 i}{16r}$$

$$\frac{B_1}{B_2}=8$$

Hence 3 is correct option.

Q. 14 A transformer operating at primary voltage $8\mathrm{kV}$ and secondary voltage $160\ \mathrm{V}$ serves a load of $80\ \mathrm{kW}$. Assuming the transformer to be ideal with purely resistive load and working on unity power factor, the loads in the primary and secondary circuit would be

Option 1:
800Ω and 1.06Ω

Option 2:
10Ω and 500Ω

Option 3:
800Ω and 0.32Ω

Option 4:
1.06Ω and 500Ω

Correct Answer:
800Ω and 0.32Ω

Solution:

$\mathrm{V_p} = 8000\ \mathrm{V}$

$\mathrm{V_s} = 160\ \mathrm{V}$

$\mathrm{p} = \dfrac{\mathrm{V_s^2}}{\mathrm{R_s}} = 80\ \mathrm{kW}$

$\dfrac{160 \times 160}{\mathrm{R_s}} = 80 \times 10^3$

$\mathrm{R_s} = 320 \times 10^{-3}\Omega$

$= 0.32\Omega$

Hence 3 is correct option.

Q. 15 Sun light falls normally on a surface of area $36\ \mathrm{cm}^2$ and exerts an average force of $7.2 \times 10^{-9}\ \mathrm{N}$ within a time period of 20 minutes. Considering a case of complete absorption, the energy flux of incident light is

Option 1:
$25.92 \times 10^2\ \mathrm{W/cm^2}$

Option 2:
$8.64 \times 10^{-6}\ \mathrm{W/cm^2}$

Option 3:
$6.0\ \mathrm{W/cm^2}$

Option 4:
$0.06\ \mathrm{W/cm^2}$

Correct Answer:
$0.06\ \mathrm{W/cm^2}$

Solution:

$\mathrm{A} = 36 \times 10^{-4}\ \mathrm{m}^2 = 36\ \mathrm{cm}^2$

$\mathrm{F} = 7.2 \times 10^{-9}\ \mathrm{N}$

$\Delta \mathrm{t} = 20\ \mathrm{min} = 1200\ \mathrm{s}$

$$\text{Energy flux} = \frac{\text{Energy}}{(\text{time})\ (\text{Area})}$$

$= (\ \text{Radiation pressure})$

$$= \frac{\mathrm{FC}}{\mathrm{A}}$$

$$= \frac{7.2 \times 10^{-9} \times 3 \times 10^{8}}{36}$$

$$= 0.6 \times 10^{-1} \frac{\mathrm{w}}{\mathrm{cm}^2}$$

$$\text{Energy flux} = 0.06$$

Hence 4 is correct option.

Q. 16 The power of a lens (biconvex) is $1.25\ \mathrm{m}^{-1}$ in particular medium. Refractive index of the lens is 1.5 and radii of curvature are $20\ \mathrm{cm}$ and $40\ \mathrm{cm}$ respectively. The refractive index of surrounding medium:

Option 1:
1.0

Option 2:
$\frac{9}{7}$

Option 3:
$\frac{3}{2}$

Option 4:
$\frac{4}{3}$

Correct Answer:
$\frac{9}{7}$

Solution:
$\mathrm{P} = 1.25\mathrm{m}^{-1}$

$\mu = 1.5$

$$R_1 = 20\text{cm}, R_2 = 40\text{cm}$$

$$P = \frac{1}{f} = (\mu_g^n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\mu_g^m = \frac{\mu_g}{\mu_m} = \frac{1.5}{\mu_m}$$

$$P = \left(\frac{1.5}{\mu_m} - 1\right)\left(\frac{1}{+20} - \frac{1}{(-40)}\right)$$

$$\frac{5}{4} = \left(\frac{1.5}{\mu_m} - 1\right)\left(\frac{3}{40 \times 10^{-2}}\right)$$

$$\frac{1}{6} = \frac{1.5}{\mu_m} - 1$$

$$\frac{7}{6} = \frac{1.5}{\mu_m}$$

$$\mu_m = \frac{9}{7}$$

Hence 2 is correct option.

Q. 17 Two streams of photons, possessing energies equal to five and ten times the work function of metal are incident on the metal surface successively. The ratio of maximum velocities of the photoelectron emitted, in the two cases respectively, will be

Option 1:
$1 : 2$

Option 2:
$1 : 3$

Option 3:

$2:3$

Option 4:

$3:2$

Correct Answer:

$2:3$

Solution:

$\mathrm{hf_1} = 5\phi_o = \text{Energy of 1st stream of photons}$

$\mathrm{hf_2} = 10\phi_0 = \text{Energy of } 2^{\text{nd}} \text{ stream of photons}$

From the Einstain's photoelectric equation,

$$\mathrm{hf} = \phi_0 + \frac{1}{2}\mathrm{mv}^2$$

For first stream of photons,

$$\mathrm{hf_1} = 5\phi_0 = \phi_0 + \frac{1}{2}\mathrm{mv_1^2}$$

$$4\phi_0 = \frac{1}{2}\mathrm{mv_1^2} \rightarrow \ (1)$$

For second stream of photons,

$$\mathrm{hf_2} = 10\phi_0 = \phi_0 + \frac{1}{2}\mathrm{mv_2^2}$$

$$9\phi_0 = \frac{1}{2}\mathrm{mv_2^2} \rightarrow (2)$$

$$\frac{4}{9} = \frac{\mathrm{V_1^2}}{\mathrm{V_2^2}}$$

$$\frac{\mathrm{V_1}}{\mathrm{V_2}} = \frac{2}{3}$$

Hence 3 is correct option.

Q. 18 A radioactive sample decays $\frac{7}{8}$ times its original quantity in 15 minutes. The half life of the sample is

Option 1:
$5\ \text{min}$

Option 2:
$7.5\ \text{min}$

Option 3:
$15\ \text{min}$

Option 4:
$30\ \text{min}$

Correct Answer:
$5\ \text{min}$

Solution:
Amount of radioactive sample remaining at time $\text{t} = 15\ \text{mins}$

$$\text{N}_\text{t} = \text{N}_\text{o} - \frac{7\text{N}_\text{o}}{8} = \frac{\text{N}_\text{o}}{8}$$

$$\text{N}_\text{o} \rightarrow \text{Original quantity}$$

By the law of radioactive decay

$$\text{N}_\text{t} = \text{N}_0 2^{-\text{t}/\text{T}_{1/2}}$$

$$\frac{\text{N}_0}{8} = \text{N}_0\, 2^{-\text{t}/\text{T}_{1/2}}$$

$$\frac{\text{t}}{\text{T}_{1/2}} = 3$$

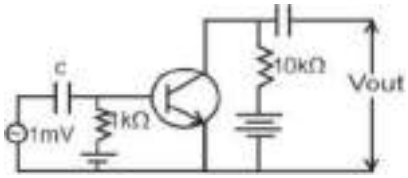
$$\frac{15}{\text{T}_{1/2}} = 3$$

$$\text{T}_{1/2} = 5\ \text{min}$$

The half life of the sample is $5\ \text{min}$

Hence 1 is correct option

Q. 19 An n.p.n transistor with current gain $\beta = 100$ in common emitter configuration is shown in figure. The output voltage of the amplifier will be



Option 1:
$0.1\ \text{V}$

Option 2:
$1.0\ \text{V}$

Option 3:
$10\ \text{V}$

Option 4:
$100\ \text{V}$

Correct Answer:
$1.0\ \text{V}$

Solution:

$$\Delta V_i = \Delta I_B R_B$$

$$I_{mV} = \Delta I_B \times 1000$$

$$\Delta I_B = 10^{-6} A$$

$$B = 100 = \frac{\Delta I_C}{\Delta I_B}; \Delta I_C = 10^{-4}\ A$$

The output voltage of the amplifier is

$$\Delta V_o = \Delta I_c R_c = 10^{-4} \times 10^4$$

$$= 1\text{V}$$

Hence 2 is correct option

Q. 20 A FM Broad cast transmitter, using modulating signal of frequency 20kHz has a deviation ratio of 10 . The Bandwidth required for transmission is:

Option 1:
220kHz

Option 2:
180kHz

Option 3:
360kHz

Option 4:
440kHz

Correct Answer:
440kHz

Solution:

$$\text{Deviation Ratio} = \frac{\text{Maximum frequency deviation}}{\text{Maximum modulating signal}}$$

$$10 = \frac{\Delta f_m}{f_m} = \frac{\Delta f_m}{20}$$

$$\Delta f_m = 200\text{kHz}$$

$$\text{Bandwidth } = 2\left(\Delta f_m + f_m\right)$$

$$= 2(200 + 20)$$

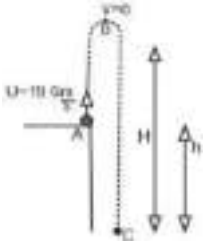
$$= 440\text{kHz}$$

Hence 4 is correct option

Q. 21 A ball is thrown vertically upwards with a velocity of $19.6\ \text{ms}^{-1}$ from the top of a tower. The ball strikes the ground after $6\ \text{s}$. The height from the ground up to which the ball can rise will be $\left(\frac{k}{5}\right)\ \text{m}$. The value of k is_____ (use $\text{g} = 9.8\ \text{m/s}^2$)

Correct Answer:
392

Solution:



$\text{h} \rightarrow$ height of building

$\text{H} \rightarrow$ height from the ground upto which the ball can rise

From A to B

$$\text{v}^2 = \text{u}^2 + 2\text{as}$$

$$0 = (19.6)^2 + 2(-9.8)(\text{H} - \text{h})$$

$$\text{H} - \text{h} = 19.6\ \text{m} \rightarrow\ (1)$$

From A to C

$$\text{s} = \text{ut} + \frac{1}{2}\text{at}^2$$

$$(-\text{h}) = (+19.6) \times 6 + \frac{1}{2}(-9.8)(6)^2$$

$$-\text{h} = -9.8 \times 6$$

$$\text{h} = 58.8\ \text{m} \rightarrow 2$$

$$\text{H} = \text{h} + 19.6$$

$$= 58.8 + 19.6$$

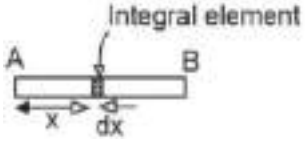
$$\text{H} = 78.4 = \frac{\text{k}}{5}$$

$K = 392$

Q. 22 The distance of centre of mass from end A of a one dimensional rod (AB) having mass density $\varrho = \varrho_0 \left(1 - \frac{x^2}{L^2}\right) kg/m$ and length L (in meter) is $\frac{3L}{\alpha}m$. The value of α is_____ . (where x is the distance from end A)

Correct Answer:
8

Solution:



$$\rho = \rho_0 \left(1 - \frac{x^2}{L^2}\right)$$

$$dm = \rho\, dx$$

$$dm = \rho_0 \left(1 - \frac{x^2}{L^2}\right) dx$$

$$x_{cm} = \frac{3L}{\alpha} = \frac{\int_0^L x dm}{\int_0^L dm}$$

$$= \frac{\rho_o \left(\left[\frac{x^2}{2}\right]_0^L - \left[\frac{x^4}{4L^2}\right]_0^L\right)}{\rho_o \left([x]_0^L - (x^3/3L^2)_0^L\right]}$$

$$\frac{3L}{\alpha} = \frac{\rho_o \left[\left(\frac{L_2}{2}\right) - \left(\frac{L^2}{4}\right)\right]}{\rho_o \left(L - \frac{L}{3}\right)}$$

$$= \frac{\frac{L^2}{4}}{\frac{2L}{3}}$$

$$\frac{3L}{\alpha} = \frac{3L}{8}$$

$$\therefore \alpha = 8$$

The value of α is 8

Q. 23 A string of area of cross-section $4\ \text{mm}^2$ and length $0.5\ \text{m}$ is connected with a rigid body of mass $2\ \text{kg}$. The body is rotated in a vertical circular path of radius $0.5\ \text{m}$. The body acquires a speed of $5\ \text{m/s}$ at the bottom of the circular path. Strain produced in the string when the body is at the bottom of the circle is_____ $\times 10^{-5}$
(use young's modulus $10^{11}\ \text{N/m}^2$ and $\text{g} = 10\ \text{m/s}^2$)

Correct Answer:
30

Solution:

$$\text{A} = 4\ \text{mm}^2 = 4 \times 10^{-6}\ \text{m}^2$$

$$\ell = 0.5\ \text{m}$$

$$\text{m} = 2\ \text{kg}$$

$$\text{T} = \frac{\text{mv}^2}{\text{R}} + \text{mg} = 120\ \text{N}$$

$$\text{Strain} = \frac{\Delta\ell}{\ell} = \frac{\text{F}}{\text{AY}}$$

$$= \frac{120}{4 \times 10^{-6} \times 10^{11}}$$

$$\text{Strain } = 30 \times 10^{-5}$$

Q. 24 At a certain temperature, the degrees of freedom per molecule for gas is 8 . The gas performs $150\ \text{J}$ of work when it expands under constant pressure. The amount of heat absorbed by the gas will be _____J.

Correct Answer:
750

Solution:

$$\text{f} = 8, \text{v} = \frac{2+\text{f}}{\text{f}} = \frac{5}{4}$$

$$\text{W} = \text{p}\Delta\text{v} = 150\ \text{J}$$

From first law of thermodynamic

$$\text{Q} = \Delta\text{U} + \text{W}$$

$$= \frac{\text{p}\Delta\text{V}}{\text{V}-1} + \text{p}\Delta\text{V}$$

$$= \frac{150}{\frac{5}{4}-1} + 150$$

$$\text{Q} = 750\ \text{J}$$

The amount of heat absorbed by the gas will be $750\ \text{J}$

Q. 25 The potential energy of a particle of mass $4\ \text{kg}$ in motion along the x-axis is given by $\text{U} = 4(1-\cos 4x)\text{J}$. The time period of the particle for small oscillation $(\sin\theta \simeq \theta)$ is $\left(\frac{\pi}{\text{K}}\right)$ s. The value of K is_____.

Correct Answer:
2

Solution:
$u = 4(1 - \cos 4x)$

$\frac{du}{dx} = -4(-\sin 4x) \times 4$

$= +16 \sin 4x$

$k_{eff} = \frac{d^2u}{dx^2} = 64 \cos 4x$

for small oscillation,

$x \to 0$

$4x \to 0$

$\cos 4x \to 1$

$K_{eff} = 64$

$T = 2\pi\sqrt{\frac{m}{k_{eff}}} = 2\pi \times \sqrt{\frac{4}{64}}$

$T = \frac{\pi}{2}s$

$\therefore K = 2$

The value of K is 2

Q. 26 An electrical bulb rated $220\ V, 100\ W$, is connected in series with another bulb rated $220\ V, 60\ W$. If the voltage across combination is $220\ V$, the power consumed by the $100\ W$ bulb will be about_____ W

Correct Answer:
14

Solution:
Bulb 1

$$(V_R)_1 = 220\ V \qquad R_1 = \frac{(V_R)_1^2}{(P_R)_1} = 484\Omega$$

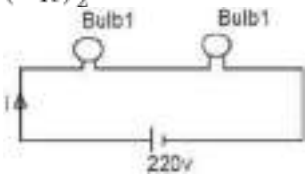
$$(P_R)_1 = 100\ W$$

$(V_{R1})\ \&\ (P_{R2})$ are voltage rating and power rating for the bulb1 respectively

Bulb 2

$$(V_R)_2 = 220\ V \qquad R_2 = \frac{(V_R)_2^2}{(P_R)_2} = \frac{4840}{6}\Omega$$

$$(P_R)_2 = 60\ W$$



$$220 = I\,(R_1 + R_2)$$

$$= I\left(484 + \frac{4840}{6}\right)$$

$$10 = I\left(22 + \frac{220}{6}\right)$$

$$I = \frac{60}{352}\ A$$

Power consumed by bulb 1

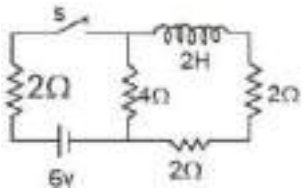
$$P = I^2 R_1$$

$$= \frac{60 \times 60}{352 \times 352} \times (22 \times 22)$$

$$= \frac{60 \times 60}{16 \times 16} = \frac{225}{16}$$

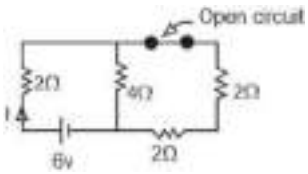
$$P \cong 14\ W$$

Q. 27 For the given circuit the current through battery of $6\ \mathrm{V}$ just after closing the switch 'S' will be _____A.



Correct Answer:
1

Solution:
Just after closing the circuit the inductor behaves as opent circuit (∞ resistance)



$$\mathrm{I} = \frac{6\mathrm{v}}{6\Omega} = 1\mathrm{A}$$

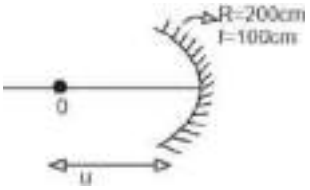
The current through battery of $6\ \mathrm{V}$ is $\mathrm{I} = 1\ \mathrm{A}$

Q. 28 An object 'o' is placed at a distance of $100\ \mathrm{cm}$ in front of a concave mirror of radius of curvature $200\ \mathrm{cm}$ as shown in the figure. The object starts moving towards the mirror at a speed $2\ \mathrm{cm/s}$. The position of the image from the mirror after $10\ \mathrm{s}$ will be at _____cm.

Correct Answer:
400

Solution:
After $10\ \mathrm{s}$,the distance of object from mirror will be

$$u = 100 - 2 \times 10 = 80\text{cm}$$



$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{(-80)} = \frac{1}{-100}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{100} = \frac{20}{8000}$$

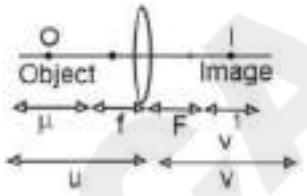
$$v = 400 \text{ cm}$$

The position of the image from mirror after 10 s at 400 cm

Q. 29 In an experiment with a convex lens, The plot of the image distance (v') against the object distance (μ') measured from the focus gives a curve $v'\mu' = 225$. If all the distances are measured in cm. The magnitude of the focal length of the lens is _____cm.

Correct Answer:
15

Solution:



$$u = u' + f \rightarrow \ (1)$$

$$v = v' + f \rightarrow \ (2)$$

By lens equation

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{+(v' + f)} - \frac{1}{-(\mu' + f)} = \frac{1}{f}$$

$$[(\mu' + f) + (v' + f)]\, f = (v' + f)\,(\mu' + f)$$

$$(\mu')\, f + f^2 + v'f + f^2 = v'\mu' + \mu' f + v'f + f^2$$

$$f^2 = \mu' v' = 225 \text{ (Given)}$$

$$f = 15 \text{ cm}$$

Q. 30 In an experiment to find acceleration due to gravity (g) using simple pendulum, time period of 0.5 s is measured from time of 100 oscillation with a watch of 1 s resolution. If measured value of length is 10 cm known to 1 mm accuracy, The accuracy in the determination of g is found to be $x\%$. The value of x is_____.

Correct Answer:
5

Solution:

$T_1 = 0.5$ s for 100 oscillations,

$\Delta T = 1s \quad T = 100 T_1 = 50$ s.

$\Delta \ell = 1\text{mm} = 0.1$ cm

$\ell = 10$ cm

$$\frac{\Delta g}{g} \times 100 = x$$

$$\frac{\Delta g}{g} = \frac{x}{100}$$

$$T = 2\pi\sqrt{\frac{\ell}{9}}$$

$$g \,\alpha\, \frac{\ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta \ell}{\ell} + \frac{2\Delta T}{T}$$

$$\frac{x}{100} = \frac{0.1}{10} + 2 \times \frac{1}{50}$$

$$\frac{x}{100} = \frac{1}{100} + \frac{4}{100}$$

$$x = 5$$

The value of x is 5

Chemistry

Q. 1 Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R
Assertion A : Zero orbital overlap is an out of phase overlap.
Reason R : It results due to different orientation / direction of approach of orbitals.
In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both A and R are true and R is the correct explanation of A

Option 2:
Both A and R are true but R is NOT the correct explanation of A

Option 3:
A is true but R is false

Option 4:
A is false but R is true

Correct Answer:
Both A and R are true and R is the correct explanation of A

Solution:
Zero orbital overlap is an out of phase overlap.
Out of phase overlap results due to different orientation or direction of approach of orbitals.

Hence , Assertion is true and Reason is correct explanation
Option (A) is correct.

Q. 2 The correct decreasing order for metallic character is

Option 1:
$\mathrm{Na} > \mathrm{Mg} > \mathrm{Be} > \mathrm{Si} > \mathrm{P}$

Option 2:
$\mathrm{P} > \mathrm{Si} > \mathrm{Be} > \mathrm{Mg} > \mathrm{Na}$

Option 3:
$\mathrm{Si} > \mathrm{P} > \mathrm{Be} > \mathrm{Na} > \mathrm{Mg}$

Option 4:
$\mathrm{Be} > \mathrm{Na} > \mathrm{Mg} > \mathrm{Si} > \mathrm{P}$

Correct Answer:
$\mathrm{Na} > \mathrm{Mg} > \mathrm{Be} > \mathrm{Si} > \mathrm{P}$

Solution:
Metalic character increases down the Group and decreases along the period as we move from left to right.

Hence correct order of decreasing metalic character is $\mathrm{Na} > \mathrm{Mg} > \mathrm{Be} > \mathrm{Si} > \mathrm{P}$

Oprtion (A) is correct.

Q. 3 Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R

Assertion A: The reduction of a metal oxide is easier if the metal formed is in liquid state than solid state.

Reason R : The value of ΔG^{Θ} becomes more on negative side as entropy is higher in liquid state than solid state.
In the light of the above statements, choose the most appropriate answer from the options given below

Option 1:
Both A and R are correct and R is the correct explanation of A

Option 2:
Both A and R are correct but R is NOT the correct explanation of A

Option 3:
A is correct but R is not correct

Option 4:
A is not correct but R is correct

Correct Answer:
Both A and R are correct and R is the correct explanation of A

Solution:
As we know -
$\Delta \mathrm{G}=\Delta \mathrm{H}-\mathrm{T} \Delta \mathrm{S}$

as entropy of liquid is more than solid so on melting the entropy increases and $\Delta \mathrm{G}$ becomes more negative leading to ease in reducung metal.

Hence it can be said that reduction of a metal oxide is easier if he metal formed is in liquid state than solid state which is explained by $\Delta \mathrm{G}$ factor explained above

Hence option (A) is correct.

Q. 4 The products obtained during treatment of hard water using Clark's method are :

Option 1:
$\mathrm{CaCO_3}$ and $\mathrm{MgCO_3}$

Option 2:
$\mathrm{Ca(OH)_2}$ and $\mathrm{Mg(OH)_2}$

Option 3:
$\mathrm{CaCO_3}$ and $\mathrm{Mg(OH)_2}$

Option 4:
$\mathrm{Ca(OH)_2}$ and $\mathrm{MgCO_3}$

Correct Answer:
$\mathrm{CaCO_3}$ and $\mathrm{Mg(OH)_2}$

Solution:
Clark's method -

$$\mathrm{Ca\,(HCO_3)_2 + 2Ca\,(OH)_2 \rightarrow 2CaCO_3 + 2H_2O}$$
$$\mathrm{Mg\,(HCO_3)_2 + 2Ca\,(OH)_2 \rightarrow 2CaCO_3 + Mg(OH)_2 + 2H_2O}$$

Hence product obtained are $\mathrm{CaCO_3}$ and $\mathrm{Mg(OH)_2}$.

Option (C) is correct.

Q. 5 **Statement I** : An alloy of lithium and magnesium is used to make aircraft plates.
Statement II : The magnesium ions are important for cell-membrane integrity.
In the light the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Both Statement I and Statement II are false

Solution:
Alloy of Lithium and Magnesium is used to make armour plates not aircraft plates.

It's Calcium which plays important roles in neuromuscular function , interneuronal transmission and cell membrane integrity.

Here due to given statements which are false-

Option (B) is correct.

Q. 6 White phosphorus reacts with thionyl chloride to give

Option 1:
PCl_5, SO_2 and S_2Cl_2

Option 2:
PCl_3, SO_2 and S_2Cl_2

Option 3:
PCl_3, SO_2 and Cl_2

Option 4:
PCl_5, SO_2 and Cl_2

Correct Answer:
PCl_3, SO_2 and Cl_2

Solution:
$P_4 + 8\,SOCl_2 \rightarrow 4\,SO_2 + 2\,S_2Cl_2$

White Phosphorous reacts with thinoyl chloride to give PCl_3, SO_2 and S_2Cl_2 as represented by the reaction above.

Option (C) is correct.

Q. 7 Concentrated HNO_3 reacts with Iodine to give

Option 1:
HI, NO_2 and H_2O

Option 2:
HIO_2, N_2O and H_2O

Option 3:
HIO_3, NO_2 and H_2O

Option 4:
HIO_4, N_2O and H_2O

Correct Answer:
HIO_3, NO_2 and H_2O

Solution:
Reaction for concentrated HNO_3 with Iodine -

$I_2 + 10\,HNO_3(\text{conc.}) \longrightarrow 2\,HIO_3 + 10\,NO_2 + 4\,H_2O$

Hence product obtained are HIO_3, NO_2 and H_2O

Option (C) is correct.

Q. 8 Which of the following pair is not isoelectronic species?
(At. no. $\mathrm{Sm}, 62; \mathrm{Er}, 68; \mathrm{Yb}, 70; \mathrm{Lu}, 71; \mathrm{Eu}, 63; \mathrm{Tb}, 65; \mathrm{Tm}, 69$)

Option 1:
Sm^{2+} and Er^{3+}

Option 2:
Yb^{2+} and Lu^{3+}

Option 3:
Eu^{2+} and Tb^{4+}

Option 4:
Tb^{2+} and Tm^{4+}

Correct Answer:
Sm^{2+} and Er^{3+}

Solution:
$\mathrm{Sm}^{+2} = 60$ electrons
$\mathrm{Er}^{+3} = 65$ electrons
$\mathrm{Tb}^{+2} = 63$ electrons
$\mathrm{Tm}^{+4} = 63$ electrons.

Tb^{+2} and Tm^{+4} have different electrons
So not isoelectronic

$\Rightarrow$ Option (D) is correct.
Option (A) is also correct }- Sm^{+2} and Fr^{+3} are not isoelectronic.

Q. 9 Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Permanganate titrations are not performed in presence of hydrochloric acid.

Reason R : Chlorine is formed as a consequence of oxidation of hydrochloric acid.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both A and R are true and R is the correct explanation of A

Option 2:
Both A and R are true but R is NOT the correct explanation of A

Option 3:
A is true but R is false

Option 4:
A is false but R is true

Correct Answer:
Both A and R are true and R is the correct explanation of A

Solution:
Permanganate titrations are not performed in presence of hydrochloric acid.

Reaction

$$2\,\mathrm{KMnO_4} + 16\,\mathrm{HCl} \longrightarrow 2\,\mathrm{MnCl_2} + 2\,\mathrm{KCl} + 8\,\mathrm{H_2O} + \mathrm{Cl_2}$$

HCl gets oxidised by $\mathrm{KMnO_4}$ into $\mathrm{Cl_2}$.

Chloride is formed as a consequence of oxidation of hydrochloride acid.

Hence option (A) is correct.

Q. 10 Match List I with List II

List I (Complex)	List II (Hybridization)
A. $Ni(CO)_4$	I. sp^3
B. $[Ni(CN)_4]^{2-}$	II. sp^3d^2
C. $[Co(CN)_6]^{3-}$	III. d^2sp^3
D. $[CoF_6]^{3-}$	IV. dsp^2

Choose the correct answer from the options given below :

Option 1:
A-IV, B-I, C-III, D-II

Option 2:
A-I, B-IV, C-III, D-II

Option 3:
A-I, B-IV, C-II, D-III

Option 4:
A-IV,B-I,C-II,D-III

Correct Answer:
A-I, B-IV, C-III, D-II

Solution:
From the list of hybridization -

$Ni(CO)_4$ Hybridization is sp^3
$[Ni\ (CN)_4]^{2-}$ Hybridization is dsp^2
$[Co\ (CN)_6]^{3-}$ Hybridization is d^2sp^3

$[Co(F_6)]^{3-}$ Hybridization is sp^3d^2

Hence option (B) is correct.

Q. 11 Dinitrogen and dioxygen, the main constituents of air do not react with each other in atmosphere to form oxides of nitrogen because

Option 1:
N_2 is unreactive in the condition of atmosphere.

Option 2:
Oxides of nitrogen are unstable.

Option 3:
Reaction between them can occur in the presence of a catalyst.

Option 4:
The reaction is endothermic and require very high temperature.

Correct Answer:
The reaction is endothermic and require very high temperature.

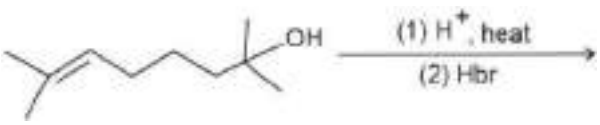
Solution:

$$N_2 + O_2 \overset{\text{High temp.}}{\rightleftharpoons} 2NO$$

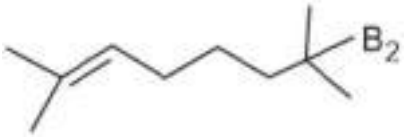
Dinitrogen and dioxygen , the main constituents in air donot react with each other in atomsphere to form oxides of nitrogen because the reaction is endothermic and requires very high temperatue.

Option (D) is correct.

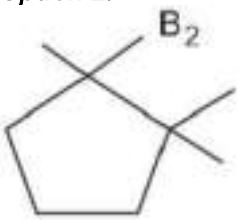
Q. 12 The major products in the given reaction is



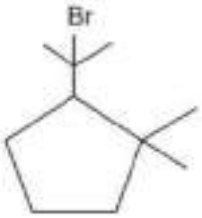
Option 1:



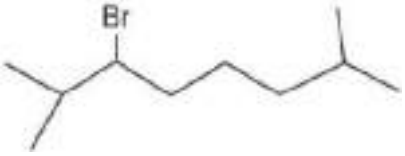
Option 2:



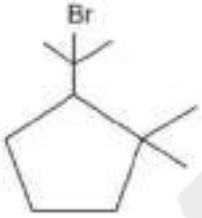
Option 3:



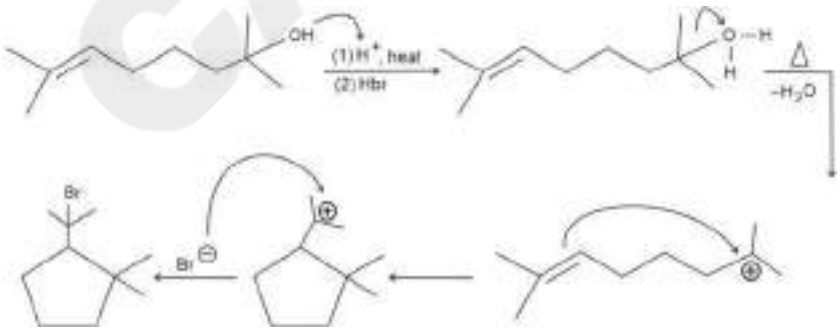
Option 4:



Correct Answer:



Solution:



Hence option (C) is correct.

Q. 13 Arrange the following in increasing order of reactivity towards nitration

A. p-xylene
B. bromobenzene
C. mesitylene
D. nitrobenzene
E. benzene

Choose the correct answer from the options given below

Option 1:

$C < D < E < A < B$

Option 2:

$D < B < E < A < C$

Option 3:

$D < C < E < A < B$

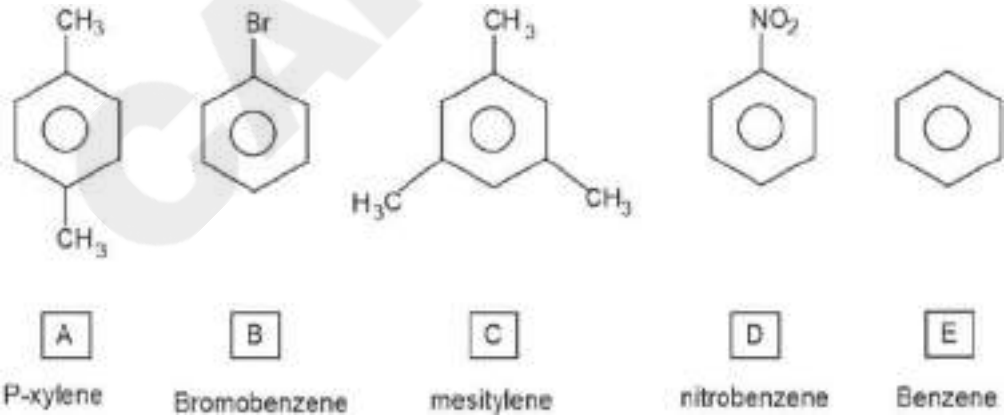
Option 4:

$C < D < E < B < A$

Correct Answer:

$D < B < E < A < C$

Solution:

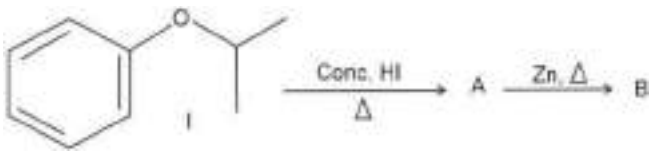


$-NO_2$ is deactivatiing
$-Br$ is deactivating
$-CH_3$ is activating group.

$D < B < E < A < C$ is the increasing order of reactivity towards nitrogen.

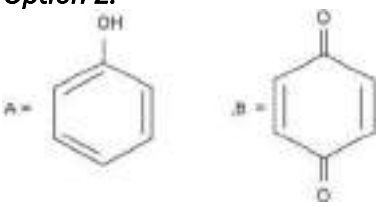
Option (B) is correct.

Q. 14 Compound I is heated with Conc. HI to give a hydroxy compound A which is further heated with Zn dust to give compound B. Identify A and B.

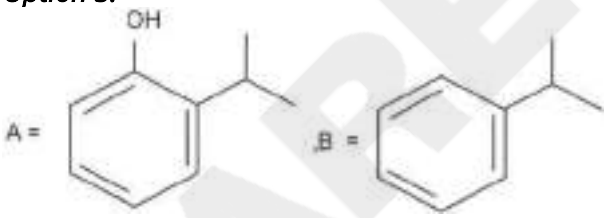


Option 1:

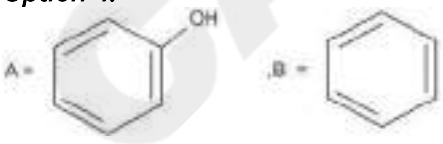
Option 2:



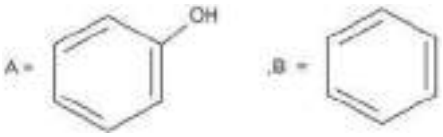
Option 3:



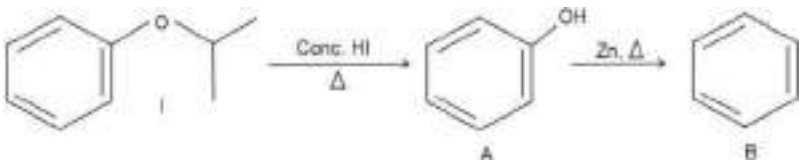
Option 4:



Correct Answer:



Solution:



Hence option (D) is correct.

Q. 15 Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R

Assertion A : Aniline on nitration yields ortho, meta \& para nitro derivatives of aniline.
Reason R : Nitrating mixture is a strong acidic mixture.
In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both A and R are true and R is the correct explanation of A

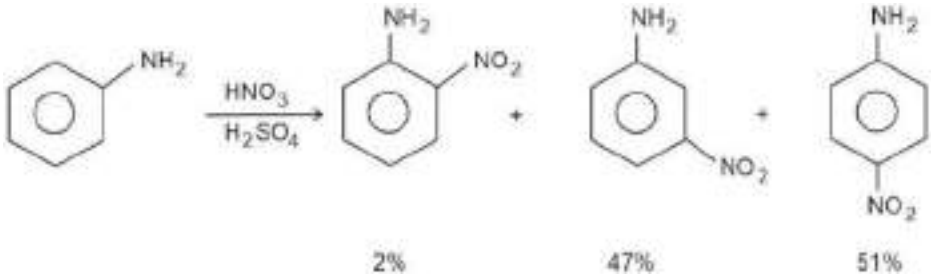
Option 2:
Both A and R are true but R is NOT the correct explanation of A

Option 3:
A is true but R is false

Option 4:
A is false but R is true

Correct Answer:
Both A and R are true and R is the correct explanation of A

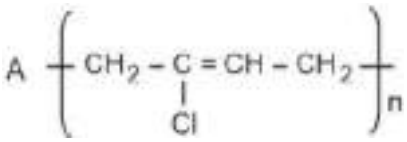
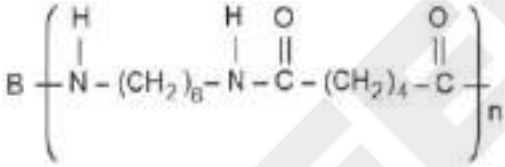
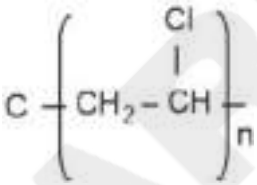
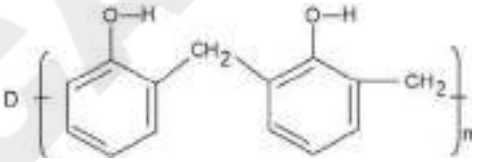
Solution:



Meta product is obtained in significant quantity because anilinium ion is formed in acidic medium which deactivates the ring

Hence option (A) is correct.

Q. 16 Match **List I** with **List II**

List I (polymer)	List-II (Nature)
A 	I.Thermosetting polymer
B 	II.Fibers
C 	III.Elastomer
D 	IV.Thermoplastic polymer

Choose the correct answer from the options given below :

Option 1:
A-II, B-III, C-IV, D-I

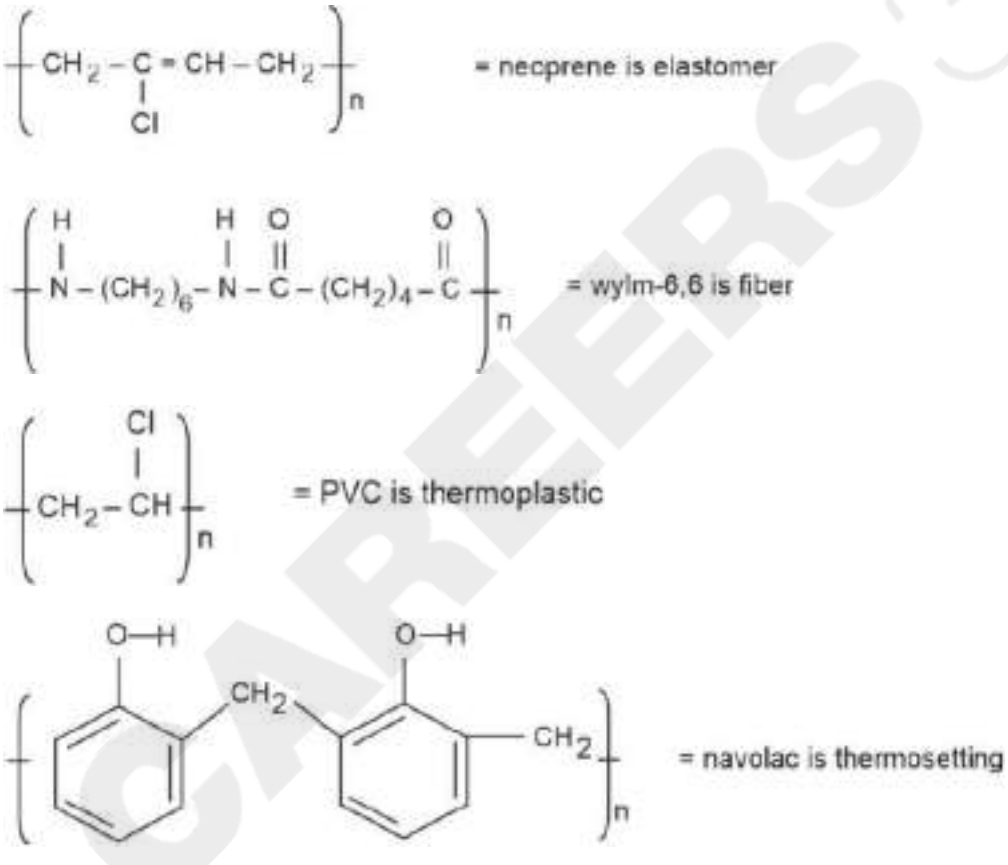
Option 2:
A-III, B-II, C-IV, D-I

Option 3:
A-III, B-I, C-IV, D-II

Option 4:
A-I, B-III, C-IV, D-II

Correct Answer:
A-III, B-II, C-IV, D-I

Solution:



Option (B) is correct.

Q. 17 Two statements in respect of drug-enzyme interaction are given below
Statement I : Action of an enzyme can be blocked only when an inhibitor blocks the active site of the enzyme.
Statement II : An inhibitor can form a strong covalent bond with the enzyme.
In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both Statement I and Statement II are true

Option 2:
Both Statement I and Statement II are false

Option 3:
Statement I is true but Statement II is false

Option 4:
Statement I is false but Statement II is true

Correct Answer:
Statement I is false but Statement II is true

Solution:
The statement "Action of an enzyme can be blocked only when an inhibitor blocks the active site of the enzyme " is wrong whereas reason "An inhibitor can form a strong covalent bond with the enzyme" is correct.

Hence option (D) is correct.

Q. 18 Given below are two statements : One is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion $\mathbf{A}$: Thin layer chromatography is an adsorption chromatography.

Reason $\mathbf{R}$: A thin layer of silica gel is spread over a glass plate of suitable size in thin layer chromatography which acts as an adsorbent.

In the light of the above statements, choose the correct answer from the options given below

Option 1:
Both A and R are true and R is the correct explanation of A

Option 2:
Both A and R are true but R is NOT the correct explanation of A

Option 3:
A is true but R is false

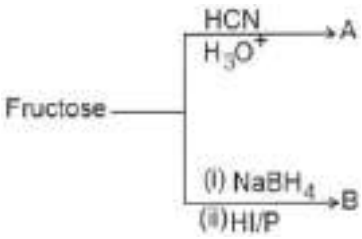
Option 4:
A is false but R is true

Correct Answer:
Both A and R are true and R is the correct explanation of A

Solution:
A Thin layer chromatography is an adsorption chromatography because a thin layer of silica gel is spread over a glass plate of suitable size in thin layer chromatography which acts as an adsorbent.

Hence option (A) is correct.

Q. 19 The formulas of A and B for the following reaction sequence



are

Option 1:

$\mathrm{A} = \mathrm{C_7H_{14}O_8}, \quad \mathrm{B} = \mathrm{C_6H_{14}}$

Option 2:

$\mathrm{A} = \mathrm{C_7H_{13}O_7}, \quad \mathrm{B} = \mathrm{C_7H_{14}O}$

Option 3:

$\mathrm{A} = \mathrm{C_7H_{12}O_8}, \quad \mathrm{B} = \mathrm{C_6H_{14}}$

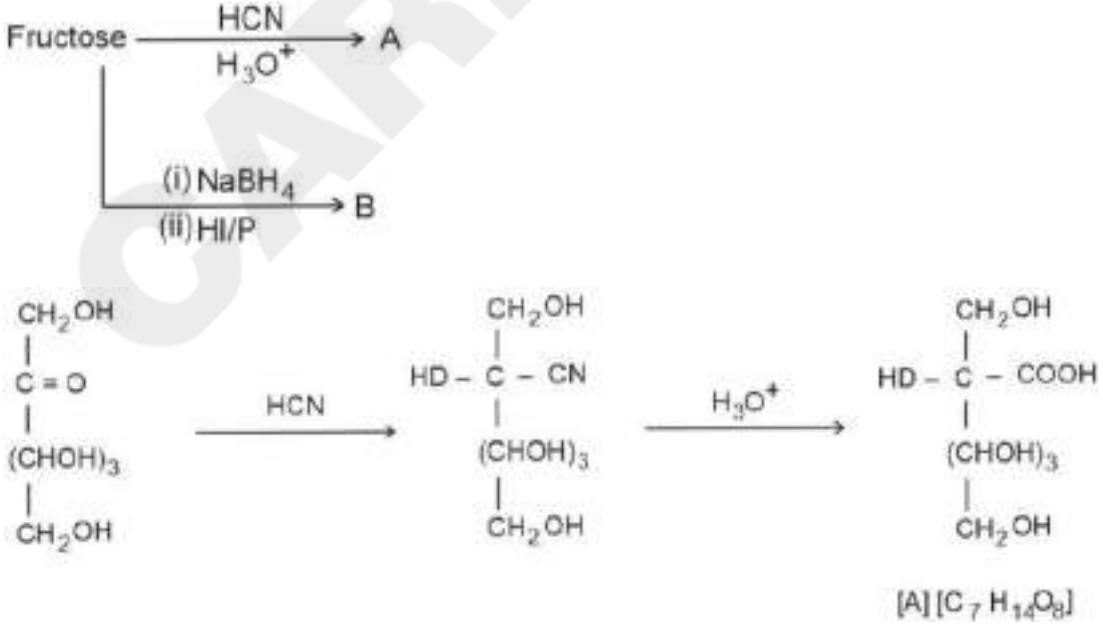
Option 4:

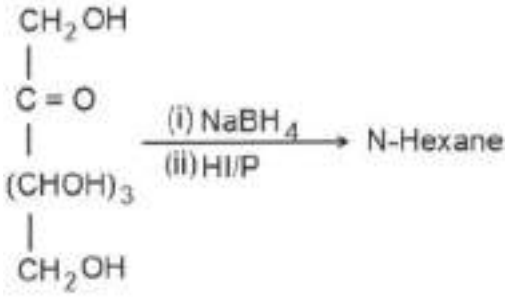
$\mathrm{A} = \mathrm{C_7H_{14}O_8}. \quad \mathrm{B} = \mathrm{C_6H_{14}O_6}$

Correct Answer:

$\mathrm{A} = \mathrm{C_7H_{14}O_8}, \quad \mathrm{B} = \mathrm{C_6H_{14}}$

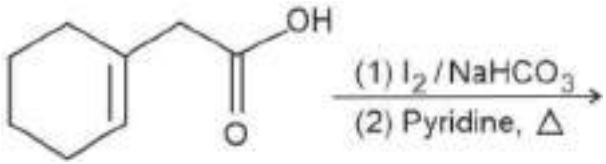
Solution:





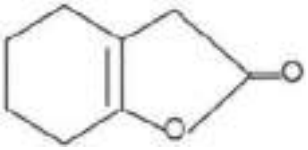
Hence $A = C_7H_{14}O_8$ and $B = C_6H_{14}$
Hence option (A) is correct.

Q. 20

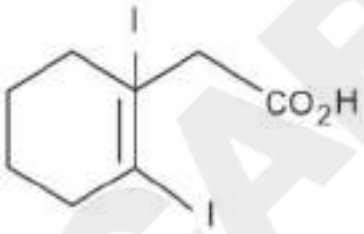


Find out the major product for the above reaction

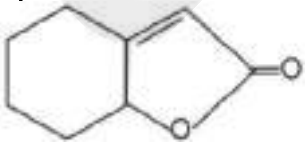
Option 1:



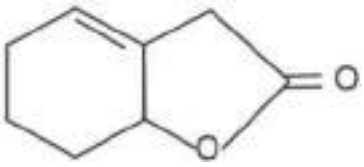
Option 2:



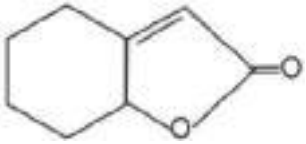
Option 3:



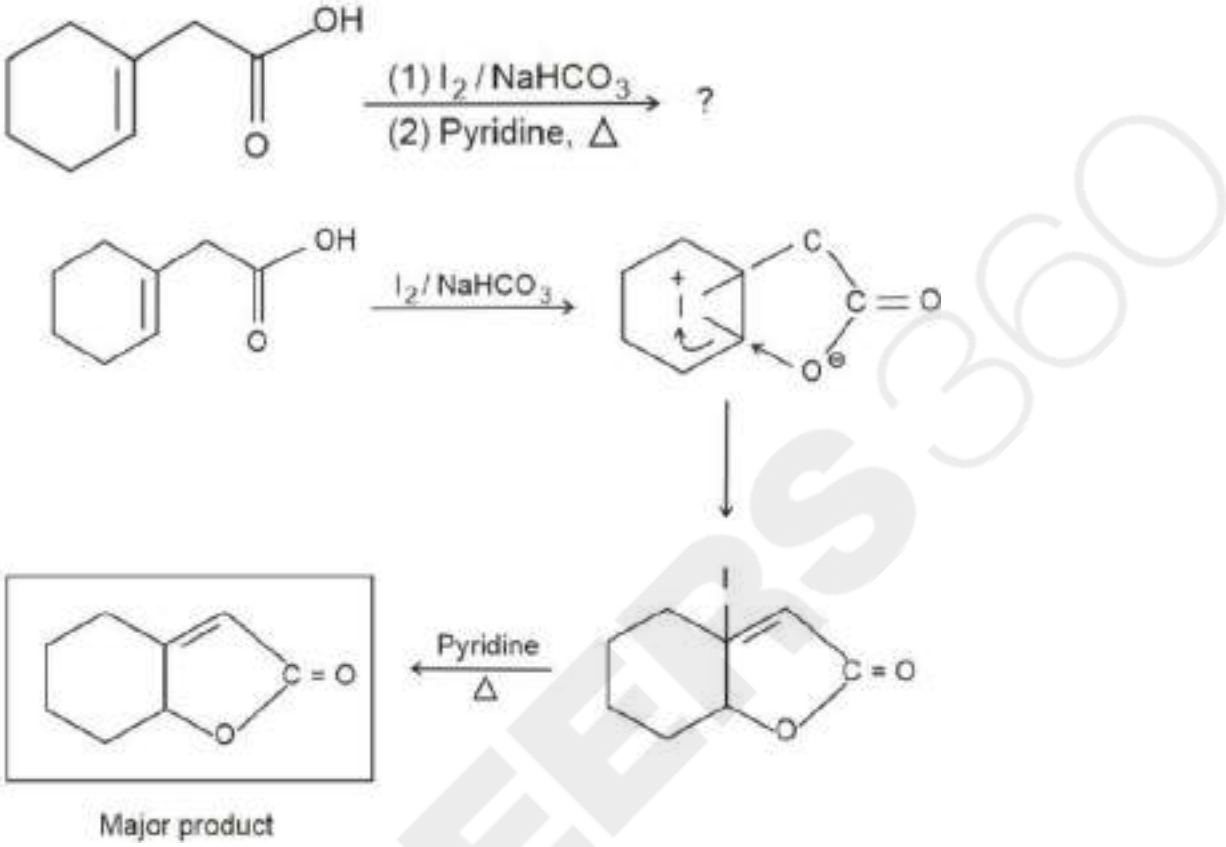
Option 4:



Correct Answer:



Solution:



Hence option (C) is correct.

Q. 21 $2\ \text{L of}\ 0.2\text{M}\ H_2SO_4$ is reacted with $2\ \text{L of}\ 0.1\text{M}\ \text{NaOH}$ solution, the molarity of the resulting product Na_2SO_4 in the solution is_____ millimolar. (Nearest integer)

Correct Answer:

25

Solution:

$$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$$

initially	0.4 mol	0.2 mol	-
	0.3 mol	-	0.1 mol

$$\text{Molarity of } Na_2SO_4 \text{ is } \frac{0.1}{4} = 0.025\,\text{M}$$

Hence molarity of resuting product Na_2SO_4 in the solution is $0.025\ M$
or 25 MM

Q. 22 Metal M crystallizes into a fcc lattice with the edge length of 4.0×10^{-8} cm. The atomic mass of the metal is_____ g/mol. (Nearest integer)

(Use $N_A = 6.02 \times 10^{23}\ mol^{-1}$, density of metal, $M = 9.03\ g\ cm^{-3}$)

Correct Answer:
87

Solution:

$$a = \text{edge length} = 4 \times 10^{-8} cm$$
$$d = 9.03\ g|ml$$
$$d = \frac{Z \times M}{a^3 \times N_A}$$

$$\Rightarrow M = \frac{d \times N_A \times a^3}{Z} = \frac{9.03 \times 6.02 \times 10^{23} \times 64 \times 10^{-24}}{4}$$
$$\simeq 87$$

$$\text{where } Z = 4\ [\text{fcc lattice}]$$
$$a^3 = 64 \times 10^{-24}$$

Hence answer is 87

Q. 23 If the wavelength for an electron emitted from H-atom is 3.3×10^{-10} m, then energy absorbed by the electron in its ground state compared to minimum energy required for its escape from the atom, is_____ times. (Nearest integer)
[Given : $h = 6.626 \times 10^{-34}\ J\ s$]
Mass of electron $= 9.1 \times 10^{-31}\ kg$

Correct Answer:
2

Solution:

$\lambda = 3.3 \times 10^{-10}\mathrm{m}$
$\mathrm{h} = 6.626 \times 10^{-34}\ \mathrm{J\,s}$
$\mathrm{me} = 9.1 \times 10^{-31}$

$$\lambda = \frac{\mathrm{h}}{\sqrt{2\mathrm{mk}}}$$
$$\mathrm{k} = \frac{\mathrm{h}^2}{2\mathrm{m}\lambda^2} = \frac{(6.6 \times 10^{-34})^2}{2 \times 9.1 \times 10^{-31} \times (3.3 \times 10^{-10})^2}$$
$$\mathrm{k} = \frac{43.9 \times 10^{-68}}{2 \times 9.1 \times 10^{-31} \times 10.89 \times 10^{-20}}$$
$$\mathrm{k} = 2.215 \times 10^{-18}$$

Energy absorbed -
$$\mathrm{Eabs} = \mathrm{Ereq} + \mathrm{k} - - - (1)$$

devide eqn (1) by Ereq. we get -
$$\frac{\mathrm{Eabs}}{\mathrm{Ereq}} = \frac{\mathrm{Ereq}}{\mathrm{Ereq}} + \frac{\mathrm{k}}{\mathrm{Ereq}}$$
$$= 1 + \frac{\mathrm{k}}{\mathrm{Ereq}} = 1 + \frac{2.215 \times 10^{-18}}{13.6 \times 1.602 \times 10^{-19}} = 2.0116$$

Hence the answer is 2.

Q. 24 A gaseous mixture of two substances A and B, under a total pressure of $0.8\ \mathrm{atm}$ is in equilibrium with an ideal liquid solution. The mole fraction of substance A is 0.5 in the vapour phase and 0.2 in the liquid phase. The vapour pressure of pure liquid A is_____ atm. (Nearest integer)

Correct Answer:
2

Solution:
Given, $\mathrm{y_A} = 0.5$
Hence $\mathrm{y_B} = 1 - \mathrm{y_A} = 1 - 0.5 = 0.5$

$\mathrm{y_A}, \mathrm{y_B}$ = Vapour pressure in liquid phase for substance A and B respectively

$\mathrm{x_A}, \mathrm{x_B}$ = Vapour pressure in gaseous / vapour phase for element A and B respectively

$\therefore \mathrm{p_A} = \mathrm{p_B} = 0.4\mathrm{atm}.$

We know that -

$x_A = 0.2$

$p_A = p_A^0 x_A$

$\therefore p_A^0 = \frac{p_A}{x_A} = \frac{0.4}{0.2} = 2$

Q. 25 At $600\ K$, $2\ mol$ of NO are mixed with $1\ mol$ of O_2.

$2NO_{(g)} + O_2(g) \rightleftarrows 2NO_2(g)$

The reaction occurring as above comes to equilibrium under a total pressure of 1 atm. Analysis of the system shows that $0.6\ mol$ of oxygen are present at equilibrium. The equilibrium constant for the reaction is_____. (Nearest integer)

Correct Answer:

2

Solution:

Given -

$$2NO + O_2 \longrightarrow 2NO_2$$

$t=t_0$	2	1	-
t=t	2-2x	1-x	2x
eqb^m	1.2	0.6	0.8

$$Kp = \frac{(P_{No2})^2}{(Po_2)\,(P_{NO})^2} = \frac{(0.8/2.6)^2}{\left(\frac{1.2}{2.6}\right)^2\left(\frac{0.6}{2.6}\right)} = 1.925$$

Hence the answer is (2) [nearest integer]

Q. 26 A sample of $0.125\ \text{g}$ of an organic compound when analyzed by Duma's method yields $22.78\ \text{mL}$ of nitrogen gas collected over KOH solution at $280\ \text{K}$ and $759\ \text{mmHg}$. The percentage of nitrogen in the given organic compound is _____
(Nearest integer)
Given :
(a) The vapour pressure of water of $280\ \text{K}$ is $14.2\ \text{mmHg}$.
(b) $\text{R} = 0.082\ \text{L atm K}^{-1}\ \text{mol}^{-1}$

Correct Answer:
22

Solution:
Given -

$$\text{V} = 22.78\,\text{ml}, \text{T} = 280\text{K}$$
$$\text{P}_{\text{Total}} = 759\,\text{mm Hg}$$
$$\text{R} = 0.082\,\text{L atm K}^{-1}\text{mol}^{-1}$$

$$\text{P}_{\text{N2}} = \text{P}_{\text{Total}} - \text{P}_{\text{Water}}$$
$$= 759 - 14.2 = 744.8\,\text{mm Hg}$$

$$\text{PV} = \text{nRT} \Rightarrow \eta_{\text{N2}} = \frac{744.8 \times 22.78}{0.082 \times 1000 \times 760 \times 280} = 0.00097$$

$$\text{W}_{\text{Nitrogen}} = \eta_{\text{N2}} \times \text{M}_{\text{N2}} = \eta_{\text{N2}} \times 14$$
$$= 0.00097 \times 14 = 0.02716$$

$$\%\text{N} = \frac{0.02716}{0.125} \times 100 = 21.728$$

Hence answer is 22

Q. 27 On reaction with stronger oxidizing agent like KIO_4, hydrogen peroxide oxidizes with the evolution of O_2. The oxidation number of I in KIO_4 changes to_____.

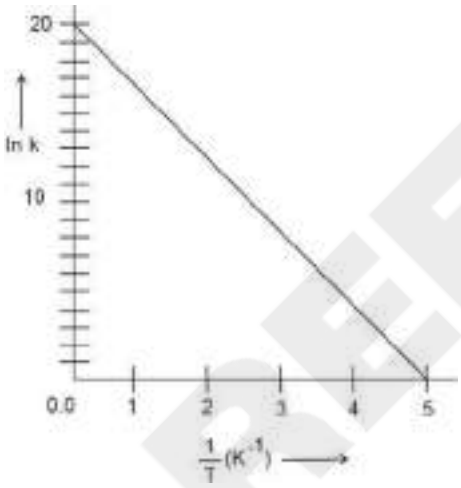
Correct Answer:
5

Solution:
$IO_4^- + H_2O_2 \longrightarrow IO_3^- + O_2$

In $IO_3^- \Rightarrow x - 6 = -1$
$x = 6 - 1 = 5$

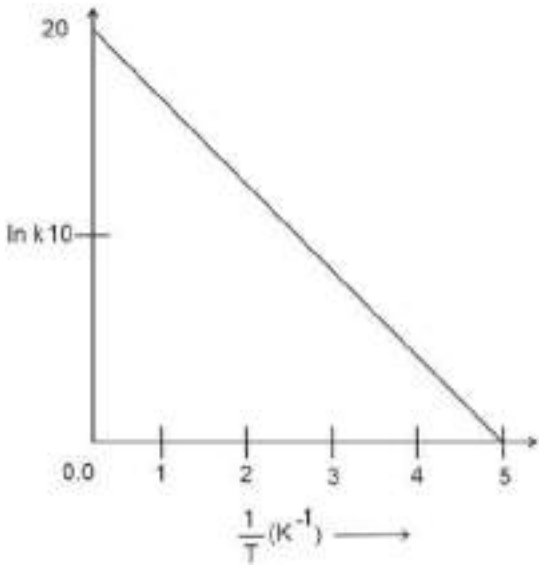
Hence oxidation number of I in $KIO_4 = 5$

Q. 28 For a reaction, given below is the graph of $\ln kvs \frac{1}{T}$. The activation energy for the reaction is equal to_____ $cal\ mol^{-1}$. (nearest integer)
(Given : $R = 2calK^{-1}\ mol^{-1}$)



Correct Answer:
8

Solution:



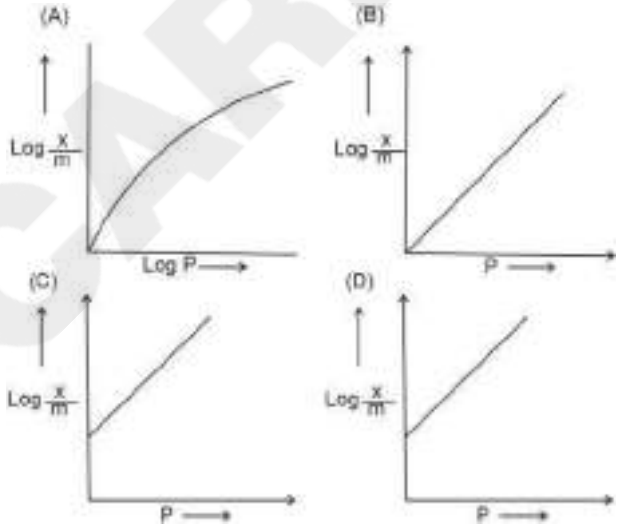
$$K = Ae^{-Ea/RT}$$
$$\ln K = \ln A - \frac{Ea}{RT}$$

$$\text{slope} = \frac{Ea}{R} = \frac{20}{5}$$

$$Ea = 4R = 8\text{Cal/mol.}$$

Hence answer is 8

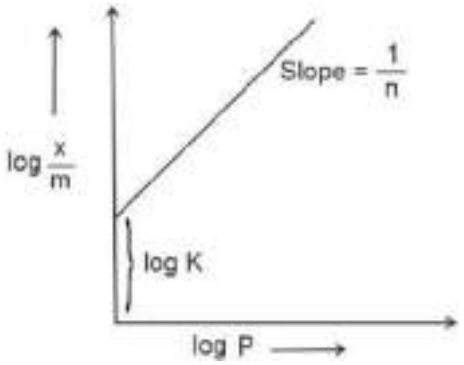
Q. 29 Among the following the number of curves not in accordance with Freundlich adsorption isotherm is_____.



Correct Answer:
3

Solution:
Freundlich adsorption isotherem -

$$\frac{x}{m} = KP^{\frac{1}{n}}$$
$$\log \frac{x}{m} = \frac{1}{n} \log P + \log K$$



Hence only 1 curve is in accordance to Freundlich adsorption isotherem.
Rest 3 are not

$\Rightarrow$ Correct answer is 3.

Q. 30 Among the following the number of state variables is_____
Internal energy (U)
Volume (V)
Heat (q)
Enthalpy (H)

Correct Answer:
3

Solution:
State variable $\rightarrow$ Internal energy
Volume
Enthalpy

Hence correct answer = 3

Maths

Q. 1 $If\ \ e^{f(x)} = \frac{10+x}{10-x}, x \in (-10, 10) and f(x) = \lambda f\left(\frac{200x}{100+x^2}\right), then\ \ \lambda$

Option 1:
$\frac{1}{2}$

Option 2:
$\frac{1}{4}$

Option 3:
4

Option 4:
3

Correct Answer:
3

Solution:
Solution:

$$f(x) = \ln\left\{\frac{10+x}{10-x}\right\}$$

$$\therefore \quad f\left(\frac{200x}{100+x^2}\right) = \ln\left(\frac{10+\frac{200x}{100+x^2}}{10-\frac{200x}{100+x^2}}\right)$$

$$= \ln\left(\frac{1000+200x+10x^2}{1000-200x+10x^2}\right)$$

$$= \ln\left(\frac{10(10+x)^2}{10(10-x)^2}\right)$$

$$= 2\ln\left(\frac{10+x}{10-x}\right) = 2f(x)$$

$$f(x) = \frac{1}{2}f\left(\frac{200x}{100+x^2}\right)$$

$$\lambda = \frac{1}{2}$$

Q. 2 Let α, β be the roots of the equation $x^2 - \sqrt{2}x + \sqrt{6} = 0$ and $\frac{1}{\alpha^2}+1, \frac{1}{\beta^2}+1$ be the roots of the equation $x^2 + ax + b = 0$. Then the roots of the equation $x^2 - (a+b-2)x + (a+b+2) = 0$ are :

Option 1:
non-real complex numbers

Option 2:
real and both negative

Option 3:
real and both positive

Option 4:
real and exactly one of them is positive

Correct Answer:
real and both negative

Solution:

$\alpha+\beta=\sqrt{2}, \quad \alpha\beta=\sqrt{6}$

$\text{Now, a} = -\left(\frac{1}{\alpha^2}+1+\frac{1}{\beta^2}+1\right)$

$= -\left(2+\frac{\alpha^2+\beta^2}{\alpha^2\beta^2}\right)$

$= -\left(2+\frac{(\alpha+\beta\beta^2-2\alpha\beta}{\alpha^2\beta^2}\right)$

$= -\left(2+\frac{2-2\sqrt{6}}{6}\right)$

$= -\left(\frac{14-2\sqrt{6}}{6}\right)$

$\text{b}= \left(\frac{1}{\alpha^2}+1\right)\left(\frac{1}{\beta^2}+1\right)$

$= \frac{1}{\alpha^2\beta^2}+\frac{1}{\alpha^2}+\frac{1}{\beta^2}+1$

$= \frac{1}{6}+\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha^2\beta^2}+1$

CAREERS360

$$= \frac{7}{6} + \frac{2-2\sqrt{6}}{6}$$

$$+ = \frac{3-2\sqrt{6}}{6}$$

$$\therefore a+b = -\frac{14}{6} + \frac{2\sqrt{6}}{6} + \frac{9}{6} - \frac{2\sqrt{6}}{6}$$

$$= -\frac{5}{6}$$

$$\therefore a+b+2 = \frac{7}{6}$$

$$a+b-2 = \frac{-17}{6}$$

Equation is $x^2 + \frac{17}{6}x + \frac{7}{6} = 0$

$\Rightarrow 6x^2 + 17x + 7 = 0$

Both joots δ_t this equation are roal and negative

Q. 3 Let A and B be any two 3×3 symmetric and skew symmetric matrices respectively. Then which of the following is NOT true?

Option 1:

$A^4 - B^4$ is a smmetric matrix

Option 2:

$AB - BA$ is a symmetric matrix

Option 3:

$B^5 - A^5$ is a skew-symmetric matrix

Option 4:

$AB + BA$ is a skew-symmetric matrix

Correct Answer:

$AB - BA$ is a symmetric matrix

Solution:

(A) $(A^4 - B^4)' = (A^4)' - (B^4)' = (A')^4 - (B')^4$
$= A^4 - (-B)^4 = A^4 - B^4$
$\therefore$ It is a symmetric matrix

(B) $(AB - BA)' = (AB)' - (BA)'$
$= B'A' - A'B'$
$= -BA - A(-B)$
$= -BA + AB$
$= AB - BA$
$\therefore$ Symmetrie

(c) $(B^5 - A^5)^1$
$= (B^5)^1 - (A^5)^1$
$= (B^1)^5 - (A^1)^5$
$= (-B)^5 - (A)^5$
$= -B^5 - A^5$
$\therefore$ Neither symmetric nor skew symmetric.
$\therefore$ option (c)

Q. 4 Let $f(x) = ax^2 + bx + c$ be such that $f(1) = 3, f(-2) = \lambda$ and $f(3) = 4$. If $f(0) + f(1) + f(-2) + f(3) = 14$, then λ is equal to:

Option 1:
-4

Option 2:
$\frac{13}{2}$

Option 3:
$\frac{23}{2}$

Option 4:
4

Correct Answer:
4

Solution:

$f(1) = 3 \Rightarrow a + b + c = 3$

$f(3) = 4 \Rightarrow 9a + 3b + c = 4$

$f(0) + f(1) + f(-2) + f(3) = 14$

$\Rightarrow c + 3 + 4a - 2b + c + 4 = 14$

$\Rightarrow 4a - 2b + 2c = 7$

Solving these equations using Cramer's Rule,

$a = \frac{1}{6}; b = -\frac{1}{6}, c = 3$

$\therefore \lambda = f(-2) = 4a - 2b + c$

$= \frac{4}{6} + \frac{2}{6} + 3$

$= \frac{24}{6}$

$= 4$

$\therefore$ option (D)

Q. 5 The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \lim_{n\to\infty} \frac{\cos(2\pi x) - x^{2n}\sin(x-1)}{1 + x^{2n+1} - x^{2n}}$ is continuous for all x in:

Option 1:

$\mathbb{R} - \{-1\}$

Option 2:

$\mathbb{R} - \{-1, 1\}$

Option 3:

$R - \{1\}$

Option 4:

$R - \{0\}$

Correct Answer:

$\mathbb{R} - \{-1\}$

Solution:

For $-1 < x < 1$

$\lim_{n\to\infty} x^{2n} = \lim_{n\to\infty} x^{2n+1} = 0$

$f(x) = \lim_{n\to\infty} \frac{\cos(2\pi x) - x^{2n}\sin(x-1)}{1 + x^{2n+1} - x^{2n}}$

$= \cos(2xx)$

which is a continuous function

For $x > 1$ or $x < -1$

$f(x) = \lim_{n\to\infty} \frac{\frac{1}{x^{2n}}\cos(2\pi x) - \sin(x-1)}{\frac{1}{x^{2n}} + x - 1}$

$= -\frac{\sin(x-1)}{(x-1)}$

RHL at $x = 1$:

$\lim_{h\to 0^+} \frac{-\sin(1+h-1)}{1+h-1} = -1$

LHL of $x = 1$

$\cos(2\pi) = 1$

$\therefore$ Discontinuous at $\neq 1$

RHL at $x = -1$:

$\cos 2\pi = 1$

LHL at $x = -1$

$\lim_{h\to 0^+} \frac{-\sin(1-h-1)}{1-h-1} = -1$

$\therefore$ Discontinues of $x = -1$

$\therefore$ option (B)

Q. 6 The function $f(x) = xe^{x(1-x)}, x \in \mathbb{R},$ is:

Option 1:

increasing in $\left(-\frac{1}{2}, 1\right)$

Option 2:

decreasing in $\left(\frac{1}{2}, 2\right)$

Option 3:

increasing in $\left(-1,-\frac{1}{2}\right)$

Option 4:

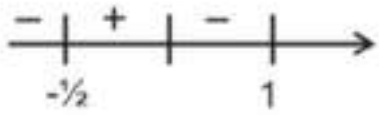
decreasing in $\left(-\frac{1}{2},\frac{1}{2}\right)$

Correct Answer:

increasing in $\left(-\frac{1}{2},1\right)$

Solution:

$$f'(x) = e^{x(1-x)} + xe^{x(1-x)} \cdot (1-2x)$$
$$= e^{x(1-x)}\left[1+x-2x^2\right]$$
$$= -e^{x(1-x)}(2x+1)(x-1)$$



$\therefore$ option (A)

Q. 7 The sum of the absolute maximum and absolute minimum values of the function $f(x) = \tan^{-1}(\sin x - \cos x)$ in the interval $[0,\pi]$ is :

Option 1:

0

Option 2:

$\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) - \frac{\pi}{4}$

Option 3:

$\cos^{-1}\left(\frac{1}{\sqrt{3}}\right) - \frac{\pi}{4}$

Option 4:

$\frac{-\pi}{12}$

Correct Answer:

$\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)-\frac{\pi}{4}$

Solution:

$$f'(x)=\frac{1}{1+(\sin x-\cos x)^2}[\cos x+\sin x]$$

$$=\frac{1}{1+(\sin x-\cos x)^2}\cdot\frac{1}{\sqrt{2}}\sin\left(x+\frac{\pi}{4}\right)$$

$\therefore f'(x)>0$ for $\left(0,\frac{3\pi}{4}\right)$ and

$f'(x)<0$ for $\left(\frac{3\pi}{4},\pi\right)$

$$f(0)=\tan^{-1}(-1)=-\frac{\pi}{4}$$

$$f\left(\frac{3\pi}{4}\right)=\tan^{-1}(\sqrt{2})$$

$$f(\pi)=\tan^{-1}(1)=\frac{\pi}{4}$$

$$\text{Sum}=\tan^{-1}(\sqrt{2})-\frac{\pi}{4}$$

$$=\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)-\frac{\pi}{4}$$

$\therefore Option(c)$

Q. 8 $\lim_{x\to 0}\frac{(1+x)^{1/x}-e+\frac{1}{2}ex}{x^2}$

Option 1:

1

Option 2:

$\frac{e}{2}$

Option 3:

$-\frac{e}{2}$

Option 4:

$\frac{11e}{24}$

Correct Answer:

$\frac{11e}{24}$

Solution:

$$\lim_{x\to 0}\frac{(1+x)^{1/x}-e+\frac{1}{2}ex}{x^2}$$

$$=\lim_{x\to 0}\frac{e\left(1-\frac{x}{2}+\frac{11}{24}x^2+\ldots\right)-e+\frac{1}{2}ex}{x^2}$$

$$=\lim_{x\to 0}\frac{\frac{11}{24}ex^2+\ldots}{x^2}=\frac{11}{24}e$$

Q. 9 Let $I_n(x)=\int_0^x \frac{1}{(t^2+5)^n}dt, n=1,2,3,\ldots$ Then:

Option 1:

$50I_6-9I_5=xI_5'$

Option 2:

$50I_6-11I_5=xI_5'$

Option 3:

$50I_6-9I_5=I_5'$

Option 4:

$50I_6-11I_5=I_5'$

Correct Answer:

$50I_6-9I_5=xI_5'$

Solution:

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$$I_n = \int_0^x \frac{1}{(t^2+5)^n} \cdot 1dt$$

Integrating by parts

$$I_n = \frac{t}{(t^2+5)^n}\Big|_0^x - \int \frac{t \cdot (-n) \cdot 2t}{\left(\frac{1}{t^2}+5\right)^{n+1}}dt$$

$$I_n = \frac{x}{(x^2+5)^n} + 2n\left[\int \frac{t^2+5}{(t^2+5)^{n+1}}dt - 5\int \frac{dt}{(t^2+5)^{n+1}}\right]$$

$$I_n = \frac{x}{(x^2+5)^n} + 2n\left[I_n - 5I_{n+1}\right]$$

Put=5

$$I_5 = \frac{x}{(x^2+5)^5} + 10I_5 - 50I_6$$

$$50I_6 - 9I_5 = \frac{x}{(x^2+5)^5} = xI_5'$$

$\therefore$ Option (A)

Q. 10 The area enclosed by the curves $y = \log_e\left(x+e^2\right), x = \log_e\left(\frac{2}{y}\right)$ and $x = \log_e 2$, above the line $y = 1$ is:

Option 1:
$2 + e - \log_e 2$

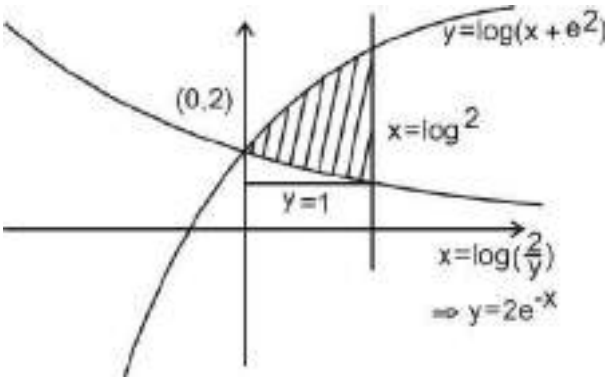
Option 2:
$1 + e - \log_e 2$

Option 3:
$e - \log_e 2$

Option 4:
$1 + \log_e 2$

Correct Answer:
$2 + e - \log_e 2$

Solution:



$$\text{Area} = \int_0^{\log 2} \left(\log\left(x + e^2\right) - 2e^{-x}\right) dx$$
$$= \left(\log 2 + e^2\right) \log\left(\log 2 + e^2\right) - \log 2 - 2e^2 - 1$$

This does not match any option.

Q. 11 Let $y = y(x)$ be the solution curve of the differential equation $\frac{dy}{dx} + \frac{1}{x^2-1}y = \left(\frac{x-1}{x+1}\right)^{1/2}, x > 1$ passing through the point $\left(2, \sqrt{\frac{1}{3}}\right)$. Then $\sqrt{7}y(8)$ is equal to:

Option 1:
$11 + 6\log_e 3$

Option 2:
19

Option 3:
$12 - 2\log_e 3$

Option 4:
$19 - 6\log_e 3$

Correct Answer:
$19 - 6\log_e 3$

Solution:

For $I \cdot F$

$$= \int \frac{1}{(x-1)(x+1)} dx$$

$$= \frac{1}{2}\left[\int \frac{1}{x-1} dx - \int \frac{1}{x+1} dx\right]$$

$$= \frac{1}{2} \ln\left(\frac{x-1}{x+1}\right)$$

$$= \ln \sqrt{\frac{x-1}{x+1}}$$

$$\therefore I \cdot F \cdot = \sqrt{\frac{x-1}{x+1}}$$

$\therefore$ Solution is

$$y \cdot \sqrt{\frac{x-1}{x+1}} = \int \frac{x-1}{x+1} dx$$

$$y\sqrt{\frac{x-1}{x+1}} = x - 2\ln(x+1) + c$$

Using $\left(2, \sqrt{\frac{1}{3}}\right)$

$$\sqrt{\frac{1}{3}} \cdot \sqrt{\frac{1}{3}} = 2 - 2\ln 3 + c$$

$$c = 2\ln 3 - \frac{5}{3}$$

put $x = 8$

$$y\sqrt{\frac{7}{9}} = 8 - 2\ln 9 + 2\ln 3 - \frac{5}{3}$$

$$\sqrt{7}y = 3\left[\frac{19}{3} - 2\ln 3\right]$$

$$= 19 - 6\ln 3$$

$\therefore$ option (D)

Q. 12 The differential equation of the family of circles passing through the points $(0, 2)$ and $(0, -2)$ is:

Option 1:

$2xy\frac{dy}{dx}+\left(x^2-y^2+4\right)=0$

Option 2:

$2xy\frac{dy}{dx}+\left(x^2+y^2-4\right)=0$

Option 3:

$2xy\frac{dy}{dx}+\left(y^2-x^2+4\right)=0$

Option 4:

$2xy\frac{dy}{dx}-\left(x^2-y^2+4\right)=0$

Correct Answer:

$2xy\frac{dy}{dx}+\left(x^2-y^2+4\right)=0$

Solution:

Using family of circles, equation is

$x^2+(y-2)(y+2)+\lambda x=0$

$\Rightarrow\frac{x^2+y^2-4}{x}=-\lambda$

Differentrating

$x\left(2x+2yy'\right)-\left(x^2+y^2-4i\right)=0$

$2xyy'+x^2-y^2+4=0$

$\therefore$ option (A)

Q. 13 Let the tangents at two points A and B on the circle $x^2+y^2-4x+3=0$ meet at origin $O(0,0)$. Then the area of the triangle OAB is:

Option 1:

$\frac{3\sqrt{3}}{2}$

Option 2:

$\frac{3\sqrt{3}}{4}$

Option 3:

$\frac{3}{2\sqrt{3}}$

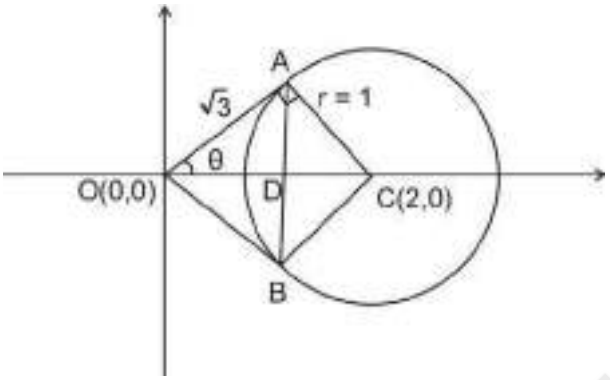
Option 4:

$\frac{3}{4\sqrt{3}}$

Correct Answer:

$\frac{3\sqrt{3}}{4}$

Solution:



$r = \sqrt{4-3} = 1$

$O\hat{A} = \sqrt{s_1} = \sqrt{3}$

$\therefore \tan\theta = \frac{1}{\sqrt{3}}$

$\text{Area of O A B} = 2 \times \left(\frac{1}{2} \times \sqrt{3}\cos\theta \cdot \sqrt{3}\sin\theta\right)$

$= 3\sin\theta \cdot \cos\theta$

$= 3 \times \frac{1}{2} \times \frac{\sqrt{3}}{2}$

$= \frac{3\sqrt{3}}{4}$

$\therefore$ Option (B)

Q. 14 Let the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ pass through the point $(2\sqrt{2}, -2\sqrt{2})$. A parabola is drawn whose focus is same as the focus of H with positive abscissa and the directrix of the parabola passes through the other focus of H. If the length of the latus rectum of the parabola is e times the length of the latus rectum of H, where e is the eccentricity of H, then which of the following points lies on the parabola?

Option 1:
$(2\sqrt{3}, 3\sqrt{2})$

Option 2:
$(3\sqrt{3}, -6\sqrt{2})$

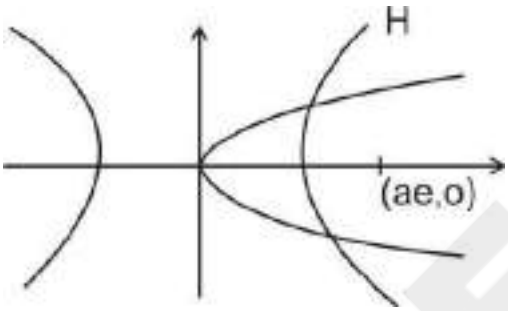
Option 3:
$(\sqrt{3}, -\sqrt{6})$

Option 4:
$(3\sqrt{6}, 6\sqrt{2})$

Correct Answer:
$(3\sqrt{6}, 6\sqrt{2})$

Solution:

$(2\sqrt{2}, -2\sqrt{2}) \Rightarrow \frac{8}{a^2} - \frac{b}{b^2} = 1 \quad \cdots \text{(i)}$



Let Parabola be $y^2 = 4(ae)x$

$\therefore 4ae = e \times \frac{2b^2}{a}$

$2a^2 = b^2$

Put this in (i)

$\frac{8}{a^2} - \frac{b}{2a^2} = 1$

$\Rightarrow \frac{4}{a^2} = 1 \Rightarrow a^2 = 4 \Rightarrow b^2 = 8$

$e^2 = 1 + \frac{b^2}{a^2} = 3 \Rightarrow e = \sqrt{3}$

$\therefore$ Parabola is $y^2 = 8\sqrt{3}x$

option (B) satisfies it

$\therefore$ option (B)

Q. 15 Let the lines $\frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$ and $\frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$ be coplanar and P be the plane containing these two lines. Then which of the following points does **NOT** lie on P ?

Option 1:
$(0,-2,-2)$

Option 2:
$(-5,0,-1)$

Option 3:
$(3,-1,0)$

Option 4:
$(0,4,5)$

Correct Answer:
$(0,4,5)$

Solution:
Using condition for coplanarity

$$\begin{vmatrix} 1-(-26) & 2-(-18) & 3-(-28) \\ 2 & 1 & 2 \\ -2 & 3 & 2 \end{vmatrix}=0$$

$\Rightarrow \quad r_1=3$

Equation of plane

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 3 & 1 & 2 \\ -2 & 3 & 3 \end{vmatrix}=0$$

$\Rightarrow \quad 3x+13y-11z+4=0$

Checking options, option (D) does not satisfy it.

$\therefore$ option (D)

Q. 16 A plane P is parallel to two lines whose direction ratios are $-2, 1, -3$ and $-1, 2, -2$ and it contains the point $(2, 2, -2)$. Let P intersect the co-ordinate axes at the points $\mathrm{A}, \mathrm{B}, \mathrm{C}$ making the intercepts α, β, γ. If V is the volume of the tetrahedron OABC, where O is the origin, and $\mathrm{p} = \alpha + \beta + \gamma$, then the ordered pair (V, p) is :equal to :

Option 1:
$(48, -13)$

Option 2:
$(24, -13)$

Option 3:
$(48, 11)$

Option 4:
$(24, -5)$

Correct Answer:
$(24, -13)$

Solution:

$$\vec{n} = \begin{vmatrix} \mathrm{i} & \mathrm{j} & \mathrm{k} \\ -2 & 1 & -3 \\ -1 & 2 & -2 \end{vmatrix}$$

$$= 4\mathrm{i} - \mathrm{j} - 3\mathrm{k}$$

Equation of plarie is

$$4(\mathrm{x} - 2) - 1(\mathrm{y} - 2) - 3(\mathrm{z} + 2) = 0$$

$$4\mathrm{x} - 8 - \mathrm{y} + 2 - 3\mathrm{z} - 6 = 0$$

$$4\mathrm{x} - \mathrm{y} - 3\mathrm{z} = 12$$

$$\frac{\mathrm{x}}{3} + \frac{\mathrm{y}}{-12} + \frac{\mathrm{z}}{(-4)} = 1$$

$$\alpha = 3, \beta = -12, -4$$

$$\alpha + \beta + \gamma = -13$$

$$\text{Volume} = \frac{1}{6}[\ \alpha \quad \beta\ ]$$

$$= \frac{1}{6} \times 3 \times 12 \times -4$$

$$= 24$$

$\therefore$ option (B)

Q. 17 Let S be the set of all $a \in \mathbb{R}$ for which the angle between the vectors $\vec{u} = a(\log_e b)\hat{i} - 6\hat{j} + 3\hat{k}$ and $\vec{v} = (\log_e b)\hat{i} + 2\hat{j} + 2a(\log_e b)\hat{k}, (b > 1)$ is acute. Then S is equal to:

Option 1:
$\left(-\infty, -\frac{4}{3}\right)$

Option 2:
ϕ

Option 3:
$\left(-\frac{4}{3}, 0\right)$

Option 4:
$\left(\frac{12}{7}, \infty\right)$

Correct Answer:
ϕ

Solution:
$\tilde{u} \cdot \tilde{v} = a(\log b)^2 - 12 + 6a(\log b) > 0$

For $b > 1 \Rightarrow \log b > 0$

Let $\log b = t \Rightarrow t > 0$

$\Rightarrow at^2 + 6at - 12 > 0$ for all $t > 0$ But for $t = 0 \Rightarrow f(t) = -12$

$\therefore at^2 + 6at - 12$ cannot be postive foer all $t > 0$ for any value of a

$\therefore$ option (B)

Q. 18 A horizontal park is in the shape of a triangle OAB with $AB = 16$. A vertical lamp post OP is erected at the point O such that $\angle PAO = \angle PBO = 15^0$ and $\angle PCO = 45°$, where C is the midpoint of AB. Then $(OP)^2$ is equal to

Option 1:
$\frac{32}{\sqrt{3}}(\sqrt{3} - 1)$

Option 2:

$\frac{32}{\sqrt{3}}(2-\sqrt{3})$

Option 3:

$\frac{16}{\sqrt{3}}(\sqrt{3}-1)$

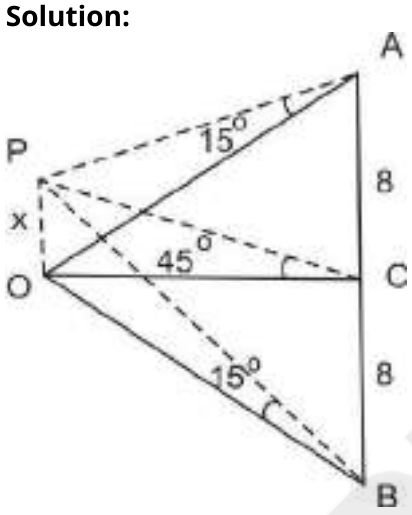
Option 4:

$\frac{16}{\sqrt{3}}(2-\sqrt{3})$

Correct Answer:

$\frac{32}{\sqrt{3}}(2-\sqrt{3})$

Solution:



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$\text{Let } OP = x$

$\therefore OC = x$

$\text{and } \frac{OP}{OA} = \tan 15^\circ$

$OA = \frac{x}{\tan 15^\circ} = \frac{x}{\tan(45^\circ - 30^\circ)}$

$= \frac{x}{\frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}}} = \frac{x(\sqrt{3}+1)}{\sqrt{3}-1}$

$\text{Now} \quad OA^2 = 8^2 + Oc^2$

$\frac{x^2(\sqrt{3}+1)^2}{(\sqrt{3}-1)^2} = 64 + x^2$

$x^2\left[\frac{(\sqrt{3}+1)^2 - (\sqrt{3}-1)^2}{(\sqrt{3}-1)^2}\right] = 64$

$x^2 = \frac{64(3+1-2\sqrt{3})}{4\sqrt{3}}$

$= \frac{32}{\sqrt{3}} \cdot (2-\sqrt{3})$

$\therefore \text{ option (B)}$

Q. 19 Let A and B be two events such that $P(B \mid A) = \frac{2}{5}, P(A \mid B) = \frac{1}{7}$ and $P(A \cap B) = \frac{1}{9}$. Consider $(S1)\ P(A' \cup B) = \frac{5}{6}$,

$(S2) P(A' \cap B') = \frac{1}{18}$

Then

Option 1:
Both (S1) and (S2) are true

Option 2:
Both (S1) and (S2) are false

Option 3:
Only (S1) is true

Option 4:

Only (S2) is true

Correct Answer:

Both (S1) and (S2) are true

Solution:

$$P(B \mid A) = \frac{2}{5}$$

$$\Rightarrow \frac{P(A \cap B)}{P(A)} = \frac{2}{5}$$

$$\Rightarrow P(A) = \frac{5}{2} \times \frac{1}{9} = \frac{5}{18}$$

And $P(A \mid B) = \frac{1}{7}$

$$\Rightarrow \frac{P(A \cap B)}{P(B)} = \frac{1}{7}$$

$$\Rightarrow P(B) = 7 \times \frac{1}{9} = \frac{7}{9}$$

$$S_1 : P(A' \cup B) = P(A') + P(B) - P(A' \cap B)$$

$$= (1 - P(A') + P(B') - (P(B) - P(A \cap B))$$

$$= \frac{13}{18} + \frac{14}{18} - \left(\frac{14}{18} - \frac{2}{18}\right)$$

$$= \frac{15}{18} = \frac{5}{6}$$

$\therefore S_1$ is true

$$S_2 : P(A' \cap B') = P(A \cup B)'$$

$$= 1 - P(A \cup B)$$

$$= 1 - (P(A) + P(B) - P(A \cap B))$$

$$= \frac{1}{18}$$

$\therefore S_2$ is true

$\therefore$ option (A)

Q. 20 Let

p : Ramesh listens to music.

q : Ramesh is out of his village.

r : It is Sunday.

s : It is Saturday.

Then the statement "Ramesh listens to music only if he is in his village and it is Sunday or Saturday" can be expressed as

Option 1:

$((\sim q) \wedge (r \vee s)) \Rightarrow p$

Option 2:

$(q \wedge (r \vee s)) \Rightarrow p$

Option 3:

$p \Rightarrow (q \wedge (r \vee s))$

Option 4:

$p \Rightarrow ((\sim q) \wedge (r \vee s))$

Correct Answer:

$p \Rightarrow ((\sim q) \wedge (r \vee s))$

Solution:

Clearly the statement is represented by

$p \rightarrow (\sim q \wedge (r \vee s))$

$\therefore$ option (D)

Q. 21 Let the coefficients of the middle terms in the expansion of $\left(\frac{1}{\sqrt{6}}+\beta x\right)^4$, $(1-3\beta x)^2$ and $\left(1-\frac{\beta}{2}x\right)^6$, $\beta > 0$, respectively from the first three terms of an A.P. If d is the common difference of this A.P., then $50-\frac{2d}{\beta^2}$ is equal to _____.

Correct Answer:

57

Solution:

Middle term for $\left(\frac{1}{\sqrt{6}}+\beta x\right)^4$ is 3rd term coefficient $= {}^4C_2 \cdot \left(\frac{1}{\sqrt{6}}\right)^2 \cdot (\beta)^2$

$=\beta^2$

For $(1-3\beta x)^2$, middle term is 2^{nd} term coefficient $= -6\beta$

For $\left(1-\frac{\beta}{2}x\right)^6$, 4 th term is middle term cofficient $= {}^6C_3 \cdot \left(-\frac{\beta}{2}\right)^3 = -\frac{5}{2}\beta^3$

As these are in AP

$\therefore -|^2\,\beta^3 = \beta^2 - \frac{5}{2}\beta^3$

$\Rightarrow -12 = \beta - \frac{5}{2}\beta^2$

$\Rightarrow \frac{5}{2}\beta^2 - \beta - 12 = 0$

$\Rightarrow 5\beta^2 - 2\beta - 24 = 0$

$\Rightarrow \beta = \frac{12}{5}$

$d = -6\beta - \beta^2.$

$\therefore 50 - \frac{2d}{\beta^2} = 50 + \frac{2(\beta^2+6\beta)}{\beta^2}$

$= 50 + 2 + \frac{12}{\beta}$

$= 57$

Q. 22 A class contains b boys and g girls. If the number of ways of selecting 3 boys and 2 girls from the class is 168, then $b+3\,g$ is equal to _____.

Correct Answer:
17

Solution:
Number of ways $= {}^bC_3 \cdot {}^gC_2 = 168$

$\Rightarrow \frac{b(b-1)(b-2)}{6} \cdot \frac{g(g-1)}{2} = 168$

$\Rightarrow b(b-1)(b-2)g(g-1) = 8 \cdot 7 \cdot 6 \cdot 3 \cdot 2$

$\Rightarrow b = 8, g = 3$

$\Rightarrow b + 3g = 8 + 9 = 17$

Q. 23 Let the tangents at the points P and Q on the ellipse $\frac{x^2}{2}+\frac{y^2}{4}=1$ meet at the point $R(\sqrt{2}, 2\sqrt{2}-2)$. If S is the focus of the ellipse on its negative major axis, then SP^2+SQ^2 is equal to _____.

Correct Answer: 13

Solution:

Equation of chard of contact is T=0

$$\Rightarrow \quad \frac{\sqrt{2}x}{2}+\frac{(2\sqrt{2}-2)y}{4}=1$$

$$\Rightarrow \quad 2\sqrt{2}x+(2\sqrt{2}-2)y=4$$

$$\Rightarrow \quad x=\frac{4-(2\sqrt{2}-2)y}{2\sqrt{2}}$$

Finding point of intersection with ellipse

$$\frac{(4-(2\sqrt{2}-2)y)^2}{8\times 2}+\frac{y^2}{4}=1$$

$$\Rightarrow \frac{(2-(\sqrt{2}-1)y)^2}{4}+\frac{y^2}{4}=1$$

$$\Rightarrow \quad 4+(\sqrt{2}-1)^2y^2-4(\sqrt{2}-1)y+y^2=4$$

$$\Rightarrow \quad (3-2\sqrt{2}+1)y^2-4(\sqrt{2}-1)y=0$$

$$\Rightarrow \quad 2\sqrt{2}(\sqrt{2}-1)y^2-4(\sqrt{2}-1)y=0$$

$$\Rightarrow 2\sqrt{2}(\sqrt{2}-1)y(y-\sqrt{2})=0$$

$$\Rightarrow y=0,\sqrt{2}$$

$$\Rightarrow x=\sqrt{2},1$$

$$\therefore P(\sqrt{2},0), Q(1,\sqrt{2})$$

S is $(0,-be)$

$$e=\sqrt{1-\frac{2}{4}}=\frac{1}{\sqrt{2}}$$

$\therefore$ S is $(0,-\sqrt{2})$

$$\therefore PS^2+SQ^2=4+9=13$$

Q. 24 If $1+\left(2+{}^{49}C_1+{}^{49}C_2+\ldots+{}^{49}C_{49}\right)\left({}^{50}C_2+{}^{50}C_4+\ldots+{}^{50}C_{50}\right)$ is equal to $2^n\cdot m$, where m is odd, then $n+m$ is equal to _____.

Correct Answer:
99

Solution:

$$1+\left(2+{}^{204}C_1+\ldots+{}^{49}C_{49}\right)\left({}^{50}C_2+{}^{50}C_4+\ldots-\cdots+{}^{50}C_{50}\right)$$
$$=1+\left(1+{}^{49}C_0+{}^{49}C_1+\ldots+{}^{49}C_{49}\right)\left({}^{50}C_0+{}^{50}C_2+\cdots+{}^{50}C_{50}-1\right)$$
$$=1+\left(1+2^{49}\right)\left(2^{49}-1\right)$$
$$=1+\left(\left(2^{49}\right)^2-1^2\right)$$
$$=2^{98}$$
$$\therefore n=98, m=1$$
$$\Rightarrow n+m=99$$

Q. 25 Two tangent lines L_1 and L_2 are drawn from the point $(2,0)$ to the parabola $2y^2=-x$. If the lines L_1 and L_2 are also tangent to the circle $(x-5)^2+y^2=r$, then $17r$ is equal to _____.

Correct Answer: 9

Solution:

$$y^2 = -\frac{1}{2}x \Rightarrow a = \frac{1}{8}$$

Let tangent be $y = mx - \frac{a}{m}$

$$\Rightarrow y = mx - \frac{1}{8m}$$

Parses through $(2, 0)$

$$\Rightarrow 0 = 2m - \frac{1}{8m}$$

$$\Rightarrow 2m = \frac{1}{8m}$$

$$\Rightarrow m^2 = \frac{1}{16}$$

$$\Rightarrow m = \frac{1}{4}, -\frac{1}{4}$$

Form $= \frac{1}{4}, \quad y = \frac{x}{4} - \frac{1}{2}$

This is tangent to $(x-5)^2 + y^2 = \gamma$

$\therefore \sqrt{\gamma} =$ distance between line and $(5, 0)$

$$\Rightarrow \sqrt{r} = \frac{\left|\frac{5}{4} - \frac{1}{2}\right|}{\sqrt{\frac{1}{16} + 1}}$$

$$\Rightarrow \sqrt{r} = \frac{\frac{3}{4}}{\frac{\sqrt{17}}{4}}$$

$$\Rightarrow \quad 17r = 9$$

Q. 26 If $\frac{6}{3^{12}} + \frac{10}{3^{11}} + \frac{20}{3^{10}} + \frac{40}{3^9} + \ldots + \frac{10240}{3} = 2^n \cdot m$, where m is odd, then $m.n$ is equal to _____.

Correct Answer:
12

Solution:

$$\frac{6}{3^{12}}+\frac{10}{3^{11}}+\frac{20}{3^{10}}+\cdots+\frac{10240}{3^{1}}$$

$$=\frac{1}{3^{12}}+\left(\frac{5}{3^{12}}+\frac{10}{3^{11}}+\cdots+\frac{10240}{3^{1}}\right)$$

G.p. with $r=2\times 3=6, n=12$

$$=\frac{1}{3^{12}}+\frac{5}{3^{12}}\left(\frac{6^{12}-1}{6-1}\right)$$

$$=\frac{1}{3^{12}}+\frac{6^{12}-1}{3^{12}}$$

$$=\frac{6^{12}}{3^{12}}$$

$$=2^{12}$$

$\therefore m=1, n=12$

$\therefore m\cdot n=12$

Q. 27 Let $S=\left[-\pi,\frac{\pi}{2}\right)-\left\{-\frac{\pi}{2},-\frac{\pi}{4},-\frac{3\pi}{4},\frac{\pi}{4}\right\}$. Then the number of elements in the set $A=\{\theta\in S:\tan\theta(1+\sqrt{5}\tan(2\theta))=\sqrt{5}-\tan(2\theta)\}$ is _____.

Correct Answer:

-

Solution:

$$\tan\theta=\frac{\sqrt{5}-\tan 2\theta}{1+\sqrt{5}\tan 2\theta}$$

Let $\tan\alpha=\sqrt{5}\Rightarrow\frac{\pi}{4}<\alpha<\frac{\pi}{2}$

$$\tan\theta=\tan(\alpha-2\theta)$$

$$\alpha-2\theta=n\pi+\theta$$

$$3\theta=\alpha-n\pi$$

$$\theta=\frac{\alpha}{3}-n\left(\frac{\pi}{3}\right)$$

For $-\pi\leqslant\theta<\frac{\pi}{2}$

$n=0,1,2,3,4$

$\therefore$ 5 solutions

Q. 28 Let $z = a + ib, b \neq 0$ be complex numbers satisfying $z^2 = \bar{z} \cdot 2^{1-|z|}$. Then the least value of $n \in N$, such that $z^n = (z+1)^n$, is equal to _____.

Correct Answer:
6

Solution:

$$z^2 = \bar{z} \cdot 2^{|-|z|}$$

Take mod of both sides

$$|z|^2 = |z| \cdot \left|2^{1-iz|}\right|$$

$$|z| = 2^{1-|z|}$$

Using hit and trial , $|z| = 1$ is solution

$$|z| = 1$$

$\therefore$ Equation becomes

$$z^2 = \bar{z}$$

$$(a^2 - b^2) + i(2ab) = a - ib$$

$$\therefore 2ab = -b \Rightarrow a = -\frac{1}{2}$$

$$(\because b \neq 0)$$

And $a^2 - b^2 = a$

$$a^2 - b^2 = a$$

$$\frac{1}{4} - b^2 = -\frac{1}{2}$$

$$\Rightarrow b^2 = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$\Rightarrow b = \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}$$

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$$\therefore z = -\frac{1}{2} + \frac{i\sqrt{3}}{2} \text{ or } -\frac{1}{2} - \frac{i\sqrt{3}}{2}$$

$$z = \omega \text{ or } \omega^2$$

For $z = \omega$

$$z^n = (1+z)^n$$

$$\Rightarrow \omega^n = (1+\omega)^n$$

$$\Rightarrow \omega^n = \left(-\omega^2\right)^n$$

$$\Rightarrow \omega^n = (-1)^n(\omega)^{2n}$$

$$\Rightarrow \omega^n = \frac{1}{(-1)^n}$$

Least n is 6

For

$$z = \omega^2$$

$$z^n = (1+z)^n$$

$$\Rightarrow \left(\omega^2\right)^n = \left(1+\omega^2\right)^n$$

$$\Rightarrow \omega^{2n} = (-\omega)^n$$

$$\Rightarrow w^{2n} = (-1)^n w^n$$

$$\Rightarrow \omega^n = (-1)^n$$

Least n=6

$$\therefore z = -\frac{1}{2} + \frac{i\sqrt{3}}{2} \text{ or } -\frac{1}{2} - \frac{i\sqrt{3}}{2}$$

$$z = \omega \text{ or } \omega^2$$

For $z = \omega$

$$z^n = (1+z)^n$$

$$\Rightarrow \omega^n = (1+\omega)^n$$

$$\Rightarrow \omega^n = \left(-\omega^2\right)^n$$

$$\Rightarrow \omega^n = (-1)^n(\omega)^{2n}$$

$$\Rightarrow \omega^n = \frac{1}{(-1)^n}$$

Least n is 6

For

$$z = \omega^2$$

$$z^n = (1+z)^n$$

$$\Rightarrow \left(\omega^2\right)^n = \left(1+\omega^2\right)^n$$

$$\Rightarrow \omega^{2n} = (-\omega)^n$$

$$\Rightarrow w^{2n} = (-1)^n w^n$$

$$\Rightarrow \omega^n = (-1)^n$$

Least n=6

Q. 29 A bag contains 4 white and 6 black balls. Three balls are drawn at random from the bag. Let X be the number of white balls, among the drawn balls. If σ^2 is the variance of X, then $100\sigma^2$ is equal to _____.

Correct Answer:

56

Solution:

$$P(x=0)=\frac{^6C_3}{^{10}C_3}=\frac{1}{6}$$

$$P(x=1)=\frac{^6C_2\cdot{}^4C_1}{^{10}C_3}=\frac{1}{2}$$

$$P(x=2)=\frac{^6C_1\cdot{}^4C_2}{^{10}C_3}=\frac{3}{10}$$

$$P(x=3)=\frac{^4C_3}{^{10}C_3}=\frac{1}{30}$$

$$\text{Mean, } \mu=\Sigma pixi$$

$$=\frac{1}{6}\times 0+\frac{1}{2}\times 1+\frac{3}{10}\times 2+\frac{1}{30}\times 3$$

$$=\frac{1}{2}+\frac{6}{10}+\frac{1}{10}$$

$$=\frac{12}{10}$$

$$\sigma^2=\sum p_i{x_i}^2-\mu^2$$

$$=\frac{1}{2}\times 1+\frac{3}{10}\times 4+\frac{1}{30}\times 9-\frac{144}{100}$$

$$=\frac{56}{100}$$

$$\Rightarrow 100\sigma^2=56$$

Q. 30 The value of the integral $\int_0^{\frac{\pi}{2}} 60\frac{\sin(6x)}{\sin x}dx$ is equal to _____.

Correct Answer: 104

Solution:

$$
\begin{aligned}
I &= 60\int_0^{\pi/2}\frac{\sin 6x}{\sin x}dx \\
&= 120\int_0^{\pi/2}\frac{\sin 3x\cos 3xdx}{\sin x} \\
&= 120\int_0^{\pi/2}\left(3-4\sin^2 x\right)\left(4\cos^3 x-3\cos x\right)dx \\
&= 120\int_0^{\pi/2}\left(3-4\sin^2 x\right)\left(4\cos^2 x-3\right)\cos xdx \\
&= \sin x = t \Rightarrow t \\
&= 120\int_0^1\left(3-4t^2\right)\left(4\left(1-t^2\right)-3\right)dt \\
&= 120\int_0^1\left(3-4t^2\right)\left(1-4t^2\right)dt \\
&= 120\int_0^1\left(3-4t^2-12t^2+16t^4\right)dt \\
&= 120\int_0^1\left(3-16t^2+16t^4\right)dt \\
&= 120\left[3t-\frac{16t^3}{3}+\frac{16t^5}{5}\right]_0^1 \\
&= 120\left(3-\frac{16}{3}+\frac{16}{5}\right) \\
&= 104
\end{aligned}
$$

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29 July - Morning Shift

Physics

Q. 1 Given below are two statements : One is labelled as **Assertion (A)** and other is labelled as **Reason (R).**

Assertion (A) : Time period of oscillation of a liquid drop depends on surface tension (S), if
density of the liquid is ρ and radius of the drop is r, then $\mathrm{T}=\mathrm{K}\sqrt{\rho \mathrm{r}^3/\mathrm{S}^{3/2}}$ is dimensionally correct, where K is dimensionless.
Reason (R): Using dimensional analysis we get R.H.S. having different dimension than that of time period.
In the light of above statements, choose the correct answer from the options given below.

Option 1:
Both (A) and (R) are true and (R) is the correct explanation of (A)

Option 2:
Both (A) and (R) are true but (R) is not the correct explanation of (A)

Option 3:
(A) is true but (R) is false

Option 4:
(A) is false but (R) is true

Correct Answer:
(A) is false but (R) is true

Solution:

$$\mathrm{T}=\mathrm{K}\sqrt{\frac{\rho \mathrm{r}^3}{\mathrm{S}^{3/2}}}$$

$$[\mathrm{S}]=\left[\frac{\mathrm{F}}{\ell}\right]=\left[\mathrm{M}^1\mathrm{L}^0\mathrm{T}^{-2}\right]$$

$$[\mathrm{S}] = \left[\mathrm{M}^{1}\mathrm{L}^{-3}\mathrm{T}^{0}\right]$$

$$[\mathrm{r}] = [\mathrm{L}']$$

$$\left[\sqrt{\frac{\rho \mathrm{r}^{3}}{\mathrm{S}^{3/2}}}\right] = \left[\rho^{1/2}\mathrm{r}^{3/2}\mathrm{S}^{-3/4}\right]$$

$$= \left[\mathrm{M}^{1/2}\mathrm{L}^{-3/2}\right]\left[\mathrm{L}^{3/2}\right] \times \left[\mathrm{M}^{-3/4}\mathrm{T}^{-3/2}\right]$$

$$= \left[\mathrm{M}^{-1/4}\mathrm{T}^{-3/2}\right]$$

$$\left[\sqrt{\frac{\rho \mathrm{r}^{3}}{\mathrm{S}^{3/2}}}\right] \neq [\mathrm{T}]$$

Using dimensional analysis we get RHS having different dimension than that of time period

$\therefore (\mathrm{A})$ is false but (R) is true

Hence 4 is correct option.

Q. 2 A ball is thrown up vertically with a certain velocity so that, it reaches a maximum height h.

Find the ratio of the times in which it is at height $\frac{\mathrm{h}}{3}$ while going up and coming down respectively.

Option 1:

$\frac{\sqrt{2}-1}{\sqrt{2}+1}$

Option 2:

$\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

Option 3:

$\frac{\sqrt{3}-1}{\sqrt{3}+1}$

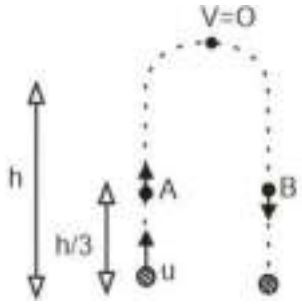
Option 4:

$\frac{1}{3}$

Correct Answer:

$$\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$$

Solution:



$$S = ut + \frac{1}{2}at^2$$

$$\frac{+h}{3} = ut + \frac{1}{2}(-g)t^2$$

$$2h = 6ut - 3gt^2$$

$$3gt^2 - 6ut + 2h = 0$$

$$t = \frac{-(-6u) \pm \sqrt{36u^2 - 4 \times 3g \times (2h)}}{2 \times 39}$$

$$t = \frac{6u \pm \sqrt{36u^2 - 24gh}}{6g} \rightarrow (1)$$

$$v^2 = u^2 + 2as$$

$$0 = u^2 + 2(-g)(+h)$$

$$u^2 = 2gh \rightarrow (2)$$

From eq (1) and (2)

$$t = \frac{6\sqrt{2gh} \pm \sqrt{48gh}}{6g}$$

$$t_1 = \frac{6\sqrt{2gh} - \sqrt{48gh}}{6g}$$

$$t_2 = \frac{6\sqrt{2gh} + \sqrt{48gh}}{6g}$$

$$\frac{t_1}{t_2} = \frac{6 - \sqrt{24}}{6 + \sqrt{24}} = \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$$

Hence 2 is correct option

Q. 3 If $t = \sqrt{x} + 4$, then $\left(\frac{dx}{dt}\right)_{t=4}$ is :

Option 1:
4

Option 2:
Zero

Option 3:
8

Option 4:
16

Correct Answer:
Zero

Solution:

$t = \sqrt{x} + 4$

$x = (t-4)^2$

$\frac{dx}{dt} = 2(t-4)$

at $t = 4s$,

$\frac{dx}{dt} = 0$

Hence 2 is correct option.

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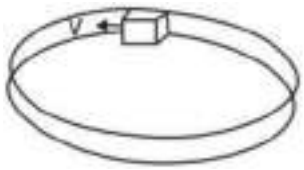
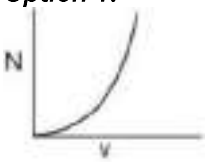
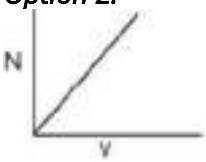
Q. 4 A smooth circular groove has a smooth vertical wall as shown in figure. A block of mass m moves against the wall with a speed v. Which of the following curve represents the correct relation between the normal reaction on the block by the wall (N) and speed of the block (v) ?



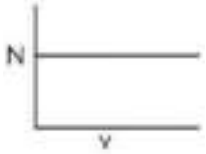
Option 1:



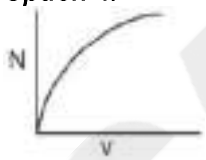
Option 2:



Option 3:



Option 4:



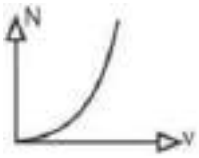
Correct Answer:

Solution:
The normal reaction on the block by the wall provides necessary centripetal force

$$\mathrm{N}=\frac{\mathrm{mv}^2}{\mathrm{R}}$$

$R \rightarrow$Radius of the circle

$N \ \alpha \ v^2$



Hence 1 is correct option

Q. 5 A ball is projected with kinetic energy E, at an angle of $60°$ to the horizontal. The kinetic energy of this ball at the highest point of its flight will become :

Option 1:
Zero

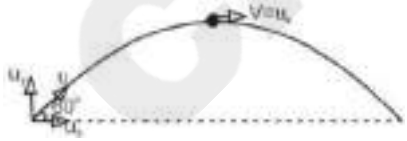
Option 2:
$\frac{E}{2}$

Option 3:
$\frac{E}{4}$

Option 4:
E

Correct Answer:
$\frac{E}{4}$

Solution:



$u_x = u\cos 60° = \frac{u}{2}$

$u_y = u\cos 60° = \frac{\sqrt{3}u}{2}$

$E = \frac{1}{2}mu^2$

At the highest point ,

$$\text{Kinetic energy} = \frac{1}{2}\text{mv}^2$$

$$= \frac{1}{2}\text{m}\left(\text{u}\cos 60^\circ\right)^2$$

$$= \frac{1}{4}\left(\frac{1}{2}\text{mu}^2\right)$$

$$= \frac{1}{4}(\text{E}) = \frac{\text{E}}{4}$$

Hence 3 is correct option

Q. 6 Two bodies of mass $1\ \text{kg}$ and $3\ \text{kg}$ have position vectors $\hat{\text{i}} + 2\hat{\text{j}} + \hat{\text{k}}$ and $-3\hat{\text{i}} - 2\hat{\text{j}} + \hat{\text{k}}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector :

Option 1:
$\hat{i} + 2\hat{j} + \hat{k}$

Option 2:
$-3\hat{\text{i}} - 2\hat{\text{j}} + \hat{\text{k}}$

Option 3:
$-2\hat{\text{i}} + 2\hat{\text{k}}$

Option 4:
$-2\hat{\text{i}} - \hat{\text{j}} + 2\hat{\text{k}}$

Correct Answer:
$\hat{i} + 2\hat{j} + \hat{k}$

Solution:

$\bar{\text{r}}_1 = \hat{\text{i}} + 2\hat{j} + \hat{\text{k}}$

$\bar{\text{r}}_2 = -3\hat{\text{i}} - 2\hat{j} + \hat{\text{k}}$

$$\bar{\text{r}}_{\text{com}} = \frac{\text{m}_1\bar{\text{r}}_1 + \text{m}_2\bar{\text{r}}_2}{\text{m}_1 + \text{m}_2}$$

$$= \frac{(\hat{i} + 2\hat{j} + \hat{k}) + 3(-3\hat{i} - 2\hat{j} + \hat{k})}{4}$$

$$= \frac{-8\hat{i} - 4\hat{j} + 4\hat{k}}{4}$$

$$\bar{r}_{com} = -2\hat{i} - \hat{j} + \hat{k}$$

$$|\bar{r}_{com}| = \sqrt{4+1+1} = \sqrt{6}$$

The magnitude of position vector of centre of mass of this system $|\bar{r}_{com}| = \sqrt{6}$ is similar to the magnitude of vector $(\hat{i} + 2\hat{j} + \hat{k})$

Hence 1 is correct option.

Q. 7 Given below are two statements : One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R): Because the angle of contact between the oil/grease and water is obtuse.

In the light of the above statements, choose the correct answer from the option given below.

Option 1:
Both (A) and (R) are true and (R) is the correct explanation of (A)

Option 2:
Both (A) and (R) are true but (R) is not the correct explanation of (A)

Option 3:
(A) is true but (R) is false

Option 4:
(A) is false but (R) is true

Correct Answer:
Both (A) and (R) are true and (R) is the correct explanation of (A)

Solution:
→ For cleaning of cloth, wetting liquid is required which acute angle of contact.
→ clothes containing oil orgrease cannot be cleaned by water because the angle of contact between the oil/grease and wates is obtuse.
→Hence, both A and R are true and R is the correct explanation of A

→Hence 1 is correct option.

Q. 8 If the length of a wire is made double and radius is halved of its respective values. Then, the Young's modulus of the material of the wire will :

Option 1:
Remain same

Option 2:
Become 8 times its initial value

Option 3:
Become $\frac{1}{4}^{\text{th}}$ of its initial value

Option 4:
Become 4 times its initial value

Correct Answer:
Remain same

Solution:
→ Young's modulus depends on the material of wire

→ It is independent of dimension of wire

Q. 9 The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α, is given by :

Option 1:
$2\pi\sqrt{L/(g\cos\alpha)}$

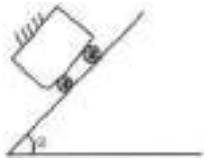
Option 2:
$2\pi\sqrt{L/(g\sin\alpha)}$

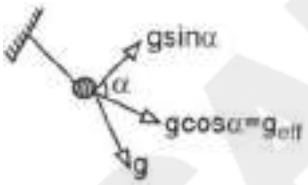
Option 3:
$2\pi\sqrt{L/g}$

Option 4:
$2\pi\sqrt{L/(g\tan\alpha)}$

Correct Answer:
$2\pi\sqrt{L/(g\cos\alpha)}$

Solution:





$$g_{eff} = \sqrt{g^2 + (g\sin\alpha)^2 + 2g \times g\sin\alpha\cos(90+\alpha)}$$
$$= g\cos(\alpha)$$

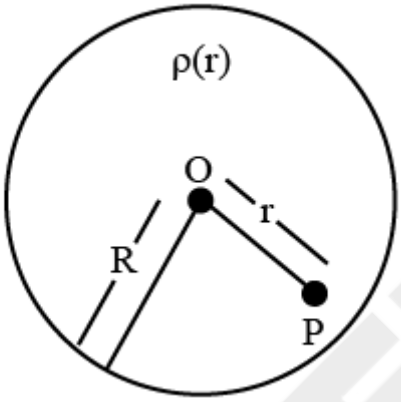
$$T = 2\pi\sqrt{\frac{\ell}{g_{eff}}} = 2\pi\sqrt{\frac{\ell}{g\cos\alpha}}$$

Hence 1 is correct option.

Q. 10 A spherically-symmetric charge distribution is considered with charge density varying as

$$\rho(r)=\begin{cases}\rho_0\left(\frac{3}{4}-\frac{r}{R}\right) & \text{for } r\le R\\ \text{zero} & \text{for } r>R\end{cases}$$

Where, $\mathrm{r}(\mathrm{r}<\mathrm{R})$ is the distance from the centre O (as shown in the figure). The electric field at point P will be :



Option 1:

$\frac{\rho_0 \mathrm{r}}{4\varepsilon_0}\left(\frac{3}{4}-\frac{\mathrm{r}}{\mathrm{R}}\right)$

Option 2:

$\frac{\rho_0 \mathrm{r}}{3\varepsilon_0}\left(\frac{3}{4}-\frac{\mathrm{r}}{\mathrm{R}}\right)$

Option 3:

$\frac{\rho_0 \mathrm{r}}{4\varepsilon_0}\left(1-\frac{\mathrm{r}}{\mathrm{R}}\right)$

Option 4:

$\frac{\rho_0 \mathrm{r}}{5\varepsilon_0}\left(1-\frac{\mathrm{r}}{\mathrm{R}}\right)$

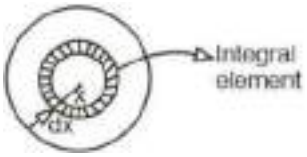
Correct Answer:

$\frac{\rho_0 \mathrm{r}}{4\varepsilon_0}\left(1-\frac{\mathrm{r}}{\mathrm{R}}\right)$

Solution:

$$\rho(r) = \rho_o\left(\frac{3}{4} - \frac{r}{R}\right) \text{ For } \quad r \leq R$$

$$= \text{Zero} \qquad\qquad r > R$$



Change in the integral element is,

$$dq = \rho 4\pi x^2 dx$$

$$dq = \rho_0\left(\frac{3}{4} - \frac{x}{R}\right) \times 4\pi x^2 dx$$

$$q = \int dq = \rho_0 \times 4\pi \int_0^r \left(\frac{3}{4}x^2 dx - \frac{x^3}{R}dx\right)$$

$$= 4\pi\rho_0\left(\frac{3}{4}\left[\frac{x^3}{3}\right]_0^r - \left[\frac{x^4}{4R}\right]_0^r\right)$$

$$q = 4\pi\rho_0 r^3\left[\frac{1}{4} - \frac{r}{4R}\right)$$

By Gauss's law,

$$\oint \bar{E}.d\bar{A} = \frac{q_{enc}}{\varepsilon_0}$$

$$E \times (4\pi r^2) = \frac{4\pi\rho_o r^3}{4}\left(1 - \frac{r}{R}\right)$$

$$E = \frac{\rho_0 r}{4\varepsilon_o}\left(1 - \frac{r}{R}\right)$$

Hence 3 is the correct option.

Q. 11 Given below are two statements.
Statement I : Electric potential is constant within and at the surface of each conductor.
Statement II : Electric field just outside a charged conductor is perpendicular to the surface of the conductor at every point.

In the light of the above statements, choose the most appropriate answer from the options given below.

Option 1:
Both statement I and statement II are correct

Option 2:
Both statement I and statement II are incorrect

Option 3:
Statement I is correct but statement II is incorrect

Option 4:
Statement I is incorrect but statement II is correct

Correct Answer:
Both statement I and statement II are correct

Solution:
$\rightarrow$ Electric field inside the conductor is zero. Therefore, electric potential is constant within and at the Surface of each conductor

$\rightarrow$ Conductor is an equipotential Surface. Electric field is always perpendicular to the equipotential surface

$\rightarrow$ Therefore, both statement I and Statement II are correct

Hence 1 is correct option.

Q. 12 Two metallic wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the these wires respectively, the effective conductivity of the combination is :

Option 1:
$\frac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

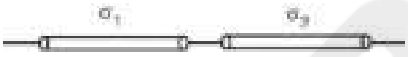
Option 2:
$\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

Option 3:
$\frac{\sigma_1+\sigma_2}{2\sigma_1\sigma_2}$

Option 4:
$\frac{\sigma_1+\sigma_2}{\sigma_1\sigma_2}$

Correct Answer:
$\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

Solution:



For wire in series combination

$$R_{eff} = R_1 + R_2$$

$$\frac{2\ell}{\sigma_{eff}\, A} = \frac{\ell}{\sigma_1 A} + \frac{\ell}{\sigma_2 A}$$

$$\sigma_{eff} = \frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$$

Hence 2 is correct option

Q. 13 An alternating emf $E = 440 \sin 100\ \pi t$ is applied to a circuit containing an inductance of $\frac{\sqrt{2}}{\pi}H$. If an a.c. ammeter is connected in the circuit, its reading will be :

Option 1:
$4.4\ A$

Option 2:
$1.55\ A$

Option 3:
$2.2\ A$

Option 4:
$3.11\ A$

Correct Answer:
$2.2\ A$

Solution:
$E = 440 \sin(100\pi t)$

$L = \frac{\sqrt{2}}{\pi}H$

$\omega = 100\pi$

$X_L = \omega L = 100\sqrt{2}$

$E_0 = 400$

AC ammeter reading is

$$I_{rms} = \frac{1\,(E_o)}{\sqrt{2}\,(x_L)} = \frac{I_o}{\sqrt{2}} = 2.2A$$

Hence 3 is correct option

Q. 14 A coil of inductance $1\mathrm{H}$ and resistance 100Ω is connected to a battery of $6\ \mathrm{V}$. Determine approximately :
(a) The time elapsed before the current acquires half of its steady - state value.
(b) The energy stored in the magnetic field associated with the coil at an instant $15\ \mathrm{ms}$ after the circuit is switched on. (Given $\ln 2 = 0.693, \mathrm{e}^{-3/2} = 0.25$)

Option 1:
$\mathrm{t} = 10\ \mathrm{ms}; \mathrm{U} = 2\ \mathrm{mJ}$

Option 2:
$\mathrm{t} = 10\ \mathrm{ms}; \mathrm{U} = 1\ \mathrm{mJ}$

Option 3:
$\mathrm{t} = 7\ \mathrm{ms}; \mathrm{U} = 1\ \mathrm{mJ}$

Option 4:
$\mathrm{t} = 7\ \mathrm{ms}; \mathrm{U} = 2\ \mathrm{mJ}$

Correct Answer:
$\mathrm{t} = 7\ \mathrm{ms}; \mathrm{U} = 1\ \mathrm{mJ}$

Solution:
$\mathrm{L} = 1\mathrm{H}$

$\mathrm{R} = 100\Omega$

$\mathrm{V} = 6\mathrm{V}$

(a) Growth of current in R-L Circuit is represented by equation

$$\mathrm{i} = \mathrm{i}_0\left(1 - \mathrm{e}^{-\mathrm{t}/\tau}\right)$$

$$\text{steady state current} = \mathrm{i}_0 = \frac{\mathrm{V}}{\mathrm{R}} = 6 \times 10^{-2}$$

$$\mathrm{i} = \frac{\mathrm{i}_0}{2} = \mathrm{i}_0\left(1 - \mathrm{e}^{-\mathrm{t}/\tau}\right)$$

$$\frac{1}{2} = \mathrm{e}^{-\mathrm{t}/\tau}$$

$$\ln(2) = \frac{\mathrm{t}}{\tau} = \frac{\mathrm{t}}{\mathrm{L}/\mathrm{R}}$$

$$\ln(2) = \frac{100t}{1}$$

$$t = \frac{0.693}{100} \cong 7 \text{ ms}$$

the time elapsed before the current acquires half of its steady state value

(b) The energy stored in the magnetic field associated with coil at $t = 15m$

$$i = i_0\left(1 - e^{-t/\tau}\right)$$

$$= 6 \times 10^{-2}\left(1 - e^{-15/10}\right)$$

$$i = 6 \times 10^{-2}\left(\frac{3}{4}\right) = 4.5 \times 10^{2} \text{ A}$$

$$\text{Energy stored } = \frac{1}{2}Li^2$$

$$= \frac{1}{2} \times 1 \times \frac{81}{4} \times 10^{-4}$$

$$= \frac{81}{8} \times 10^{-4}$$

$$= 10^{-3}J$$

Hence 3 is correct option.

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Q. 15 Match **List - I with List - II :**

List - I	**List - II**
(a) UV rays	(i) Diagnostic tool in medicine
(b) X-rays	(ii) Water purification
(c) Microwave	(iii) Communication, Radar
(d) Infrared wave	(iv) Improving visibility in foggy days

Choose the correct answer from the options given below :

Option 1:
(a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

Option 2:
(a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

Option 3:
(a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

Option 4:
(a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

Correct Answer:
(a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

Solution:
UV rays $\longrightarrow$ Water purification

x-rays $\longrightarrow$ Diagnostic tool in medicine

Microware $\longrightarrow$ Communication, Radar

Infrared wave $\longrightarrow$ Improving visibility in foggy days.

Hence 2 is correct option.

Q. 16 The kinetic energy of emitted electron is E when the light incident on the metal has wavelength λ. To double the kinetic energy, the incident light must have wavelength :

Option 1:
$\frac{\mathrm{hc}}{\mathrm{E}\lambda-\mathrm{hc}}$

Option 2:
$\frac{\mathrm{hc}\ \lambda}{\mathrm{E}\lambda+\mathrm{hc}}$

Option 3:
$\frac{\mathrm{h}\lambda}{\mathrm{E}\lambda+\mathrm{hc}}$

Option 4:
$\frac{\mathrm{hc}\lambda}{\mathrm{E}\ \lambda-\mathrm{hc}}$

Correct Answer:
$\frac{\mathrm{hc}\ \lambda}{\mathrm{E}\lambda+\mathrm{hc}}$

Solution:
From Einstein's Photoelectric effect,

$$\frac{\mathrm{hc}}{\lambda}=\phi_{\mathrm{o}}+\mathrm{E}\rightarrow(1)$$

Let the wavelength be λ',when the kinetic energy is $2\mathrm{E}$

$$\frac{\mathrm{hc}}{\lambda'}=\phi_{0}+2\mathrm{E}\rightarrow 2$$

Eq. (2) - (1)

$$\frac{hc}{\lambda'}-\frac{hc}{\lambda}=E$$

$$\frac{hc}{\lambda'}=E+\frac{hc}{\lambda}$$

$$\lambda'=\frac{hc\lambda}{E\lambda+hc}$$

Q. 17 Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level.

Option 1:
$3:4$

Option 2:
$4:3$

Option 3:
$1:4$

Option 4:
$4:1$

Correct Answer:
$3:4$

Solution:
From $n=2$ to $n=1$

$$\text{Energy of photon}=\frac{hc}{\lambda}=Rch\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$$

$$E_1=Rch\left(\frac{1}{1}-\frac{1}{4}\right)=\frac{3}{4}Rch$$

$$E_2=Rch\left(\frac{1}{1}-\frac{1}{\infty}\right)=Rch$$

$$\frac{E_1}{E_2}=\frac{3}{4}$$

Hence 1 is correct option

Q. 18 Find the modulation index of an AM wave having $8\ \mathrm{V}$ variation where maximum amplitude of the AM wave is $9\ \mathrm{V}$.

Option 1:
0.8

Option 2:
0.5

Option 3:
0.2

Option 4:
0.1

Correct Answer:
0.8

Solution:

$$\text{Varation } = \mathrm{V_{max}} - \mathrm{V_{min}} = 8\ \mathrm{V}$$

$$\mathrm{V_{max}} = 9\ \mathrm{V}$$

$$\therefore \mathrm{V_{min}} = 1\ \mathrm{V}$$

$$\mu = \frac{\mathrm{V_{max}} - \mathrm{V_{min}}}{\mathrm{V_{max}} + \mathrm{V_{min}}} = \frac{8}{10} = 0.8$$

Hence 1 is correct option

Q. 19 A travelling microscope has 20 divisions per cm on the main scale while its vernier scale has total 50 divisions and 25 vernier scale divisions are equal to 24 main scale divisions, what is the least count of the travelling microscope?

Option 1:
$0.001\ \mathrm{cm}$

Option 2:

$0.002\ \text{mm}$

Option 3:

$0.002\ \text{cm}$

Option 4:

$0.005\ \text{cm}$

Correct Answer:

$0.002\ \text{cm}$

Solution:

$$1\text{MSD} = \frac{1\ \text{cm}}{20} = 0.05\ \text{cm}$$

$$25\text{VSD} = 24\text{MSD}$$

$$\text{LC} = 1\text{MSD} - 1\text{VSD}$$

$$= \frac{1}{25}\text{MSD} = \frac{0.05}{25}\ \text{cm}$$

$$\text{LC} = 2 \times 10^{-3}\ \text{cm}$$

Hence 3 is correct option.

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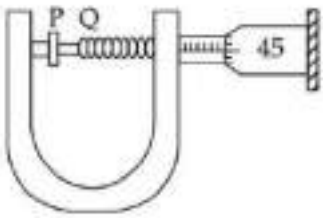
Q. 20 In an experiment to find out the diameter of wire using screw gauge, the following observations were noted :



(A) Screw moves 0.5 mm on main scale in one complete rotation
(B) Total divisions on circular scale $= 50$
(C) Main scale reading is 2.5 mm
(D) 45^{th} division of circular scale is in the pitch line
(E) Instrument has 0.03 mm negative error
Then the diameter of wire is :

Option 1:
2.92 mm

Option 2:
2.54 mm

Option 3:
2.98 mm

Option 4:
3.45 mm

Correct Answer:
2.98 mm

Solution:
$\text{pitch} = 0.5$ mm

$\text{No.of CSD} = 50$

$\text{MSR} = 2.5$ mm

$\text{CSD} = 45$

Zero error $= -0.03$ mm

$$\text{CSR} = \text{CSD} \times \text{LC} = 45 \times \frac{0.5}{50} \text{ mm} = 0.45 \text{ mm}$$

$$\text{Diameter of wire} = \text{MSR} + \text{CSR} + \text{Zero correction}$$

$$= 2.5\ \text{mm} + 0.45\ \text{mm}(0.03\ \text{mm})$$

$$\text{Diameter of wire} = 2.98\ \text{mm}$$

Hence 3 is correct option

Q. 21 An object is projected in the air with initial velocity u at an angle θ. The projectile motion is such that the horizontal range R, is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the case. The value of the angle of projection, at which the second object is projected, will be_____ degree.

Correct Answer:
15

Solution:

$$\mathrm{R} = \frac{\mathrm{u}^2 \sin 2\theta}{\mathrm{g}}$$

for maximum value of range, $\theta = 45^\circ$

$$\therefore \mathrm{R}_1 = \frac{\mathrm{u}^2}{\mathrm{g}} \rightarrow (1)$$

$$\mathrm{R}_2 = \frac{\mathrm{R}_1}{2} = \frac{\mathrm{u}^2 \sin 2\theta}{\mathrm{g}}$$

$$\frac{\mathrm{u}^2}{2\mathrm{g}} = \frac{\mathrm{u}^2 \sin 2\theta}{\mathrm{g}}$$

$$2\theta = 30^\circ \text{ or } 2\theta = 150^\circ$$

$$\theta = 15^\circ \text{ or } \theta = 75^\circ$$

The value of the angle of projection, at which the second object is projected will be 15 degree

Q. 22 If the acceleration due to gravity experienced by a point mass at a height h above the surface of earth is same as that of the acceleration due to gravity at a depth $\alpha h\,(h < R_e)$ from the earth surface. The value of α will be_____ (use $R_e = 6400\ km$)

Correct Answer:
2

Solution:
$g_h = g_d$

$\text{depth } = d = \alpha h$

$(h << R_e)$

$\therefore g\left(1 - \frac{2h}{R}\right) = g\left(1 - \frac{d}{R}\right)$

$1 - \frac{2h}{R} = 1 - \frac{\alpha h}{R}$

$\alpha = 2$

The value of α is 2

Q. 23 The pressure P_1 and density d_1 of diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes suddenly to $P_2\,(> P_1)$ and d_2 respectively during an adiabatic process. The temperature of the gas increases and becomes_____ times of its initial temperature.

(given $\frac{d_2}{d_1} = 32$)

Correct Answer:
4

Solution:
For adiabatic process

$$PV^{v} = P^{1-v}T^{v} = k$$

Also for a given mass of gas

$$v\ \alpha\ \frac{1}{d}$$

$$d \rightarrow \text{ density of gas}$$

$$p \times \frac{1}{d^{v}} = k$$

$$p \propto d^{v}$$

$$\therefore d^{v-v^2}T^{v} = K$$

$$d_1^{v-v^2}T_1^{v} = d_2^{v-v^2}T_2^{v}$$

$$\frac{d_2^{v^2-v}}{d_1^{v^2-v}} = \left(\frac{T_2}{T_1}\right)^{v}$$

$$\left(\frac{d_2}{d_1}\right)^{v-1} = \left(\frac{T_2}{T_1}\right)$$

$$(32)^{2/5} = \frac{T_2}{T_1}$$

$$\frac{T_2}{T_1} = 4$$

The temperature of the gas increases and becomes 4 times of its initial temperature

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Q. 24 One mole of a monoatomic gas is mixed with three moles of a diatomic gas. The molecular specific heat of mixture at constant volume is $\frac{\alpha^2}{4}\mathrm{RJ/mol\ K}$; then the value of α will be_____ (Assume that the given diatomic gas has no vibrational mode).

Correct Answer:
3

Solution:
For monoatomic gas ,

$$\mathrm{V_1} = \frac{5}{3}, \mathrm{n_1} = 1$$

For diatomic gas,

$$\mathrm{V_2} = \frac{7}{5}, \mathrm{n_2} = 3$$

$$\mathrm{U} = \mathrm{U_1} + \mathrm{U_2}$$

$$\mathrm{n_{mix}}\ (\mathrm{c_v})_{\mathrm{mix}}\,\mathrm{T} = \mathrm{n_1}\,(\mathrm{c_v})_1\,\mathrm{T} + \mathrm{n_2}\,(\mathrm{c_v})_2\,\mathrm{T}$$

$$\frac{(\mathrm{n_1}+\mathrm{n_2})\,\mathrm{R}}{\mathrm{v_{mix}}-1} = \frac{\mathrm{n_1 R}}{\mathrm{v_1}-1} + \frac{\mathrm{n_2 R}}{\mathrm{v_2}-1}$$

$$\frac{4}{\mathrm{V_{mix}}-1} = \frac{1}{2/3} + \frac{3}{2/5}$$

$$= \frac{3}{2} + \frac{15}{2}$$

$$\frac{4}{\mathrm{v_{mix}}-1} = 9$$

$$\frac{4}{9} + 1 = \mathrm{V_{mix}}$$

$$\mathrm{v_{mix}} = \frac{13}{9}$$

$$(\mathrm{C_v})_{\mathrm{mix}} = \frac{\mathrm{R}}{\mathrm{v_{mix}}-1} = \frac{\alpha^2}{4}\mathrm{R}$$

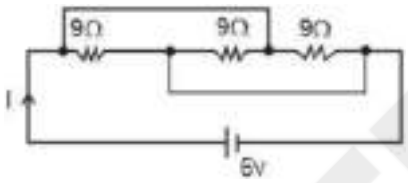
$$\frac{1}{\left(\frac{13}{9}-1\right)}=\frac{\alpha^2}{4}$$

$$\frac{9}{4}=\frac{\alpha^2}{4}$$

$$\alpha=3$$

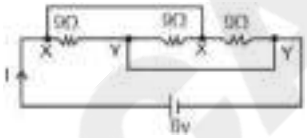
The value of α is 3

Q. 25 The current I flowing through the given circuit will be _____A.



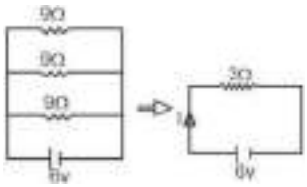
Correct Answer:
2

Solution:



$$\mathrm{X}=6\mathrm{v}$$

$$\mathrm{Y}=0$$



$$I = \frac{6V}{3\Omega} = 2A$$

The current I flowing through the given circuit will be $2A$

Q. 26 A closely wounded circular coil of radius $5\ cm$ produces a magnetic field of $37.68 \times 10^{-4}\ T$ at its center. The current through the coil is _____A. [Given, number of turns in the coil is 100 and $\pi = 3.14$]

Correct Answer:
3

Solution:

$R = 5\ cm = 5 \times 10^{-2}\ m$

$B = 37.68 \times 10^{-4}\ T$

$n = 100$

$$B = \frac{\mu_0 ni}{2R}$$

$$37.68 \times 10^{-4} = \frac{4\pi \times 10^{-7} \times 100 \times i}{2 \times 5 \times 10^{-2}}$$

$$\frac{376.8 \times 10^{-6}}{4\pi \times 10^{-5}} = i$$

$i = 3\ A$

Q. 27 Two light beams of intensities $4I$ and $9I$ interfere on a screen. The phase difference between these beams on the screen at point A is zero and at point B is π. The difference of resultant intensities, at the point A and B, will be _____ I.

Correct Answer:
24

Solution:

$$I_{resultant} = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$$

At point A,

$$I_A = 4I + 9I + 2\sqrt{4I}\sqrt{9I}\cos 0°$$

$I_A = 25I$

At point B

$I_B = 4I + 9I + 2\sqrt{4I}\sqrt{9I}\cos\pi$

$I_B = I$

The difference of resultant intensities at A & B is $I_A - I_B = 24I$

Q. 28 A wire of length $314\ cm$ carrying current of $14\ A$ is bent to form a circle. The magnetic moment of the coil is_____ $A-m^2$. [Given $\pi = 3.14$]

Correct Answer:
11

Solution:
$L = 314\ cm = 2\pi R;\ \ R = 50\ cm$

$I = 14\ A$

$M = \text{Magnetic moment of the coil} = IA$

$$= 14 \times \pi(0.5)^2$$

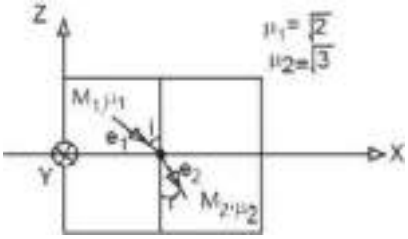
$$= \frac{14}{4} \times \frac{22}{7}$$

$M = 11A - m^2$

Q. 29 The $X - Y$ plane be taken as the boundary between two transparent media M_1 and M_2. M_1 in $Z \geqslant 0$ has a refractive index of $\sqrt{2}$ and M_2 with $Z < 0$ has a refractive index of $\sqrt{3}$. A ray of light travelling in M_1 along the direction given by the vector $\vec{P} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$, is incident on the plane of separation. The value of difference between the angle of incident in M_1 and the angle of refraction in M_2 will be_____ degree.

Correct Answer:
15

Solution:



unit vector along incident ray ,

$$\hat{e}_1 = \frac{\bar{p}}{|\bar{p}|} = \frac{4\sqrt{3}}{10}\hat{i} - \frac{3\sqrt{3}}{10}\hat{j} - \frac{\hat{k}}{2}$$

$$\hat{n} = \hat{k}(\text{unit vector along normal})$$

By law of refraction (vector form)

$$\mu_1 |(\hat{e}_1 \times \hat{n})| = \mu_2 |(\hat{e}_2 \times \hat{n})|$$

$$|\hat{e}_1 \times \hat{n}| = \sin i$$

$$\left| \frac{4}{10}\sqrt{3}(-\hat{j}) - \frac{3\sqrt{3}}{10}\hat{i} \right| = \sin i$$

$$\sin \hat{i} = \frac{3}{4} = \frac{i}{1} = 53^\circ \rightarrow (1)$$

$$|\hat{e}_2 \times \hat{n}| = \frac{\mu_1}{\mu_2} |\hat{e}_1 \times \hat{n}|$$

$$\sin r = \sqrt{\frac{2}{3}} \times \frac{3}{4} = \frac{\sqrt{3}}{2\sqrt{2}}$$

$$\sin r = 38^\circ$$

$$i - r = 15^\circ$$

Q. 30 If the potential barrier across a p-n junction is $0.6\ \text{V}$. Then the electric field intensity, in the depletion region having the width of $6 \times 10^{-6}\ \text{m}$, will be_____$\times 10^5\ \text{N/C}$

Correct Answer:
1

Solution:

$\Delta \text{V} = 0.6\ \text{V}$

$\text{d} = 6 \times 10^{-6}\ \text{m}$

$$\text{E} = \frac{\Delta \text{V}}{\text{d}} = \frac{0.6}{6 \times 10^{-6}} = \frac{6 \times 10^5}{6}$$

$\text{E} = 1 \times 10^5\ \text{N/C}$

Chemistry

Q. 1 Which of the following pair of molecules contain odd electron molecule and an expanded octet molecule?

Option 1:
BCl_3 and SF_6

Option 2:
NO and H_2SO_4

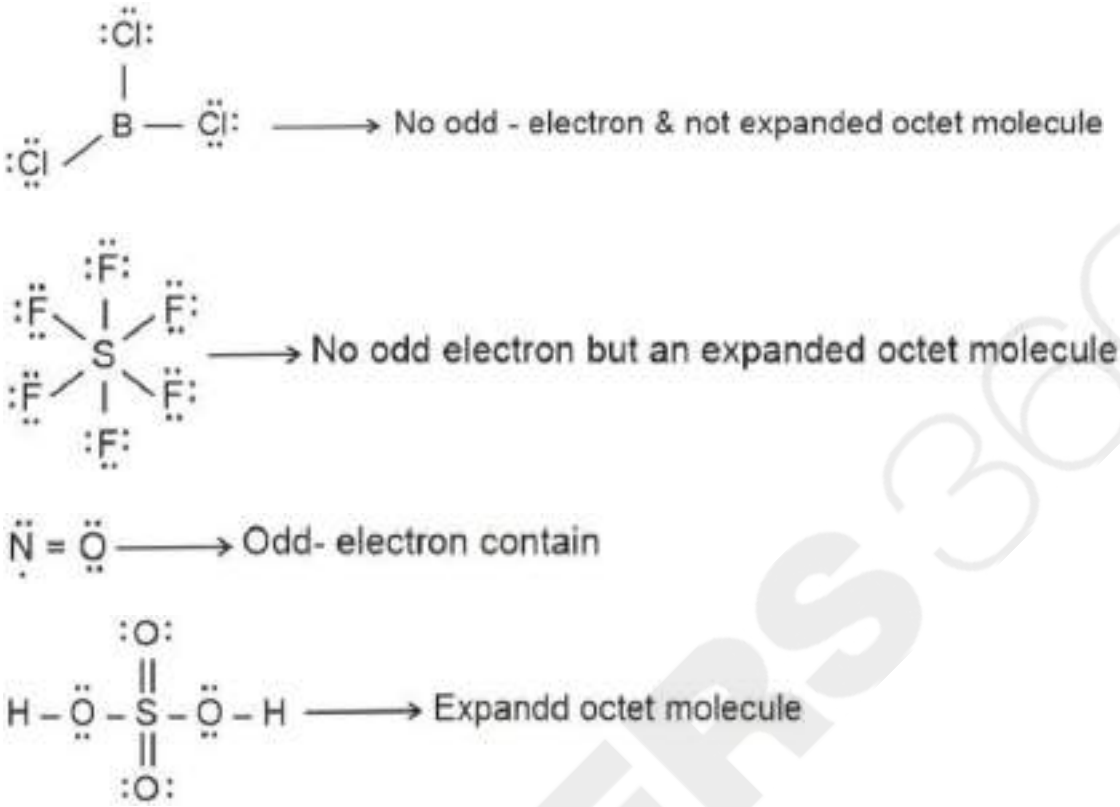
Option 3:
SF_6 and H_2SO_4

Option 4:
BCl_3 and NO

Correct Answer:

NO and H_2SO_4

Solution:



So, option (2) has a pair of molecules which contain odd electron (NO) and expanded octet molecule H_2SO_4.

Option (2) is correct.

Q. 2 $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$20\ g \quad 5\ g$

Consider the above reaction, the limiting reagent of the reaction and number of moles NH_3 formed respectivly are :

Option 1:

$H_2, 1.42$ moles

Option 2:

$H_2, 0.71$ moles

Option 3:

$N_2, 1.42$ moles

Option 4:
$N_2, 0.71$ moles

Correct Answer:
$N_2, 1.42$ moles

Solution:
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

$$\begin{array}{ll} 20\text{g} & 5\text{g} \\ \downarrow & \downarrow \\ \frac{20}{28}\text{mole} & \frac{5}{6}\text{mole} \\ \downarrow & \downarrow \\ 0.71\text{mole} & 0.83\text{mole} \end{array}$$

$0.71 < 0.83$ Hence N_2 is Limiting reagent at this reaction.

Product depend on limiting reagent.
From 1 ,ol of N_2 , 2 mole of NH_3 is forming.
So, from 0.71 mol N_2 , 1.42 mole of NH_3 will form.

So, option (3) is correct.

Q. 3 $100\ \text{mL}$ of $5\%\ (\text{w/v})$ solution of NaCl in water was prepared in $250\ \text{mL}$ beaker. Albumin from the egg was poured into NaCl solution and stirred well. This resulted in a/an :

Option 1:
Lyophilic sol

Option 2:
Lyophobic sol

Option 3:
Emulsion

Option 4:
Precipitate

Correct Answer:
Lyophilic sol

Solution:
This is method of preparation of Lyophilic solution using egg Albumin and 100 ml of 5 % (w/v) solution of NaCl in water in a 250 ml beaker.

Option (1) is correct.

Q. 4 The first ionization enthalpy of Na, Mg and Si, respectively, are : $496, 737$ and $786\ \mathrm{kJ\ mol^{-1}}$. The first ionization enthalpy $\left(\mathrm{kJ\ mol^{-1}}\right)$ of Al is :

Option 1:
487

Option 2:
768

Option 3:
577

Option 4:
856

Correct Answer:
577

Solution:
The order of first ionisation is -

$$\begin{array}{cccccc} & \mathrm{Na} & < & \mathrm{Al} & < & \mathrm{Mg} & < & \mathrm{Si} \\ & \downarrow & & & & \downarrow & & \downarrow \\ \mathrm{IE}_1 \rightarrow & 496 & & & & 737 & & 786 \end{array}$$

So, IE_1 of Al should be between 496 and 737 KJ|mol.
among the option only 577 is suitable for above criteria.

Option (3) is correct.

Q. 5 In metallurgy the term "gangue" is used for :

Option 1:
Contamination of undesired earthy materials.

Option 2:
Contamination of metals, other than desired metal.

Option 3:
Minerals which are naturally occuring in pure form

Option 4:
Magnetic impurities in an ore.

Correct Answer:
Contamination of undesired earthy materials.

Solution:
An ore rarely contains only a desired substance. it is usually contaminated with earthy or undesired material is known as gangue.

So, in metallurgy the term "gangue" is used for contamination of undesired earthy materials.

Option (1) is correct.

Q. 6 The reaction of zinc with excess of aqueous alkali, evolves hydrogen gas and gives :

Option 1:
$Zn(OH)_2$

Option 2:
ZnO

Option 3:
$[Zn(OH)_4]^{2-}$

Option 4:
$[ZnO_2]^{2-}$

Correct Answer:
$[ZnO_2]^{2-}$

Solution:
The reaction will be

$$Zn + 2NaOH \rightarrow \underset{\downarrow}{\underset{\text{sodium zincate}}{Na_2ZnO_2}} + H_2\uparrow$$
$$[ZnO_2]^{2-}$$

It is preparation method of H_2.

Option (4) is correct.

Q. 7 Lithium nitrate and sodium nitrate, when heated separately, respectively, give :

Option 1:
$LiNO_2$ and $NaNO_2$

Option 2:
Li_2O and Na_2O

Option 3:
Li_2O and $NaNO_2$

Option 4:
$LiNO_2$ and Na_2O

Correct Answer:
Li_2O and $NaNO_2$

Solution:
Lithium nitrate when heated gives lithium oxide Li_2O whereas other alkali metal nitrates decompose to give thr corresponding nitrate.

$$4LiNO_3 \longrightarrow 2H_2O + 4NO_2 + O_2$$
$$2NaNO_3 \longrightarrow 2NaNO_2 + O_2$$

Option (3) is correct.

Q. 8 Number of lone pairs of electrons in the central atom of SCl_2, O_3, ClF_3 and SF_6, respectively, are:

Option 1:
$0, 1, 2$ and 2

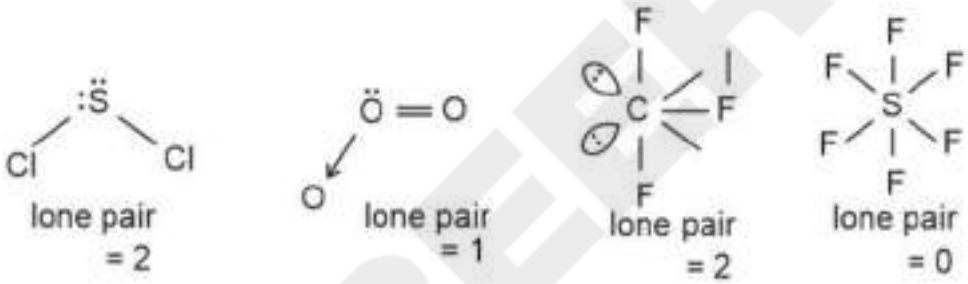
Option 2:
$2, 1, 2$ and 0

Option 3:
$1, 2, 2$ and 0

Option 4:
$2, 1, 0$ and 2

Correct Answer:
$2, 1, 2$ and 0

Solution:



So, lone pair on central atoms are 2,1,2,0

So option (2) is correct.

Q. 9 In following pairs, the one in which both transition metal ions are colourless is :

Option 1:
Sc^{3+}, Zn^{2+}

Option 2:
Ti^{4+}, Cu^{2+}

Option 3:
V^{2+}, Ti^{3+}

Option 4:
Zn^{2+}, Mn^{2+}

Correct Answer:
Sc^{3+}, Zn^{2+}

Solution:
We know the colours of some of the transition metal ions.

$$Sc^{3+} \longrightarrow 3d^0 \longrightarrow \text{colourless}$$
$$Ti^{4+} \longrightarrow 3d^0 \longrightarrow \text{colourless}$$
$$Ti^{3+} \longrightarrow 3d^1 \longrightarrow \text{purple}$$
$$V^{2+} \longrightarrow 3d^3 \longrightarrow \text{violet}$$
$$Hn^{2+} \longrightarrow 3d^5 \longrightarrow \text{pink}$$
$$Cu^{2+} \longrightarrow 3d^9 \longrightarrow \text{blue}$$
$$Zn^{2+} \longrightarrow 3d^{10} \longrightarrow \text{colourless}$$

So, pair Sc^{3+} and Zn^{2+} are colourless.
Option (1) is correct.

Q. 10 In neutral or faintly alkaline medium, $KMnO_4$ being a powerful oxidant can oxidize, thiosulphate almost quantitatively, to sulphate. In this reaction overall change in oxidation state of manganese will be :

Option 1:
5

Option 2:
1

Option 3:
0

Option 4:
3

Correct Answer:
3

Solution:
The reaction will be -

$$8MnO_4^- + 3S_2O_3^{2-} + H_2O \longrightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$$

So change in oxidation state of Mn will be -
$+7\left(\mathrm{MnO_4^-}\right) \longrightarrow +4\left(\mathrm{MnO_4}\right)$
$\mathrm{Change} = 7 - 4 = 3$
Option (4) is correct.

Q. 11 Which among the following pairs has only herbicides?

Option 1:
Aldrin and Dieldrin

Option 2:
Sodium chlorate and Aldrin

Option 3:
Sodium arsinate and Dieldrin

Option 4:
Sodium chlorate and sodium arsinite.

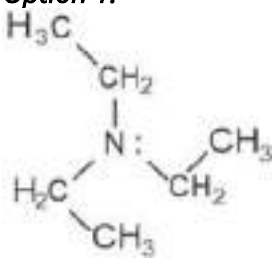
Correct Answer:
Sodium chlorate and sodium arsinite.

Solution:
Sodium chlorate $\left(\mathrm{NaClO_3}\right)$ and Sodium arsinite $\left(\mathrm{Na_3AsO_3}\right)$ are herbicides.

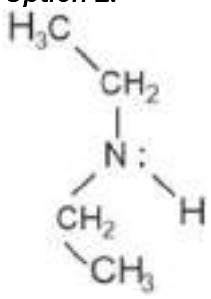
So option (4) is correct.

Q. 12 Which among the following is the strongest Bronsted base?

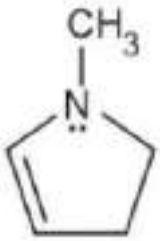
Option 1:



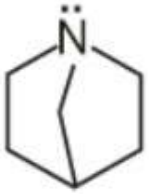
Option 2:



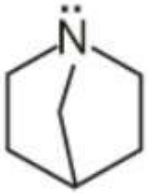
Option 3:



Option 4:

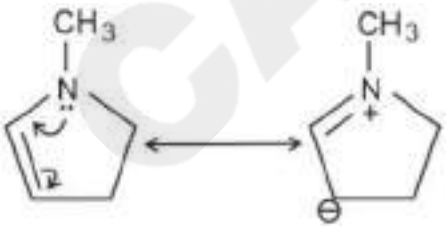


Correct Answer:



Solution:

Option (3) will be least basic due to lone pair is in resonance



Between (1) an (2), (2) will more basic due to less hinderance.
In (4) steric hinderance very less due to ring.

So free rotation is not allowed.
Hence overall (4) will be most basic.

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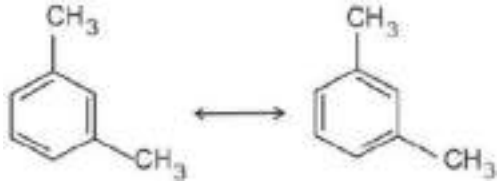
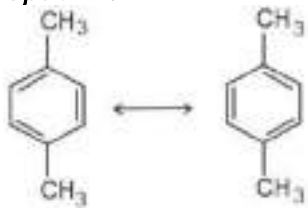
Option (4) is correct.

Q. 13 Which among the following pairs of the structures will give different products on ozonolysis ? (Consider the double bonds in the structures are rigid and not delocalized.)

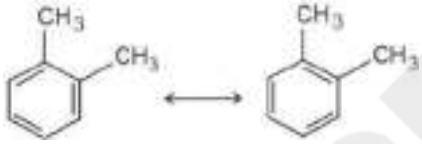
Option 1:



Option 2:

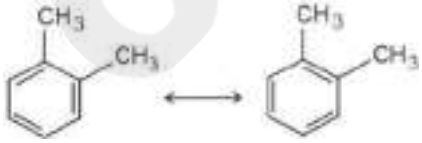


Option 3:



Option 4:

Correct Answer:



Solution:

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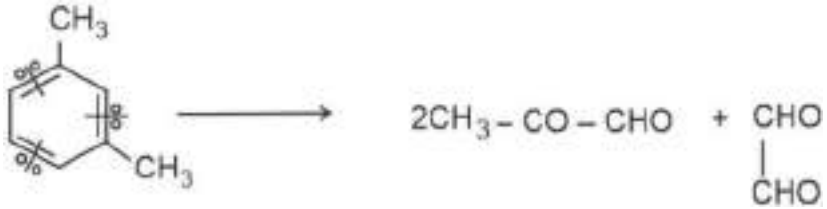
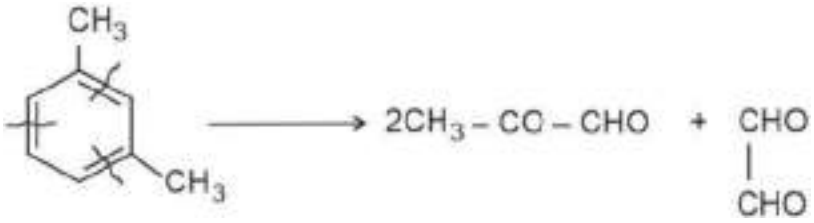
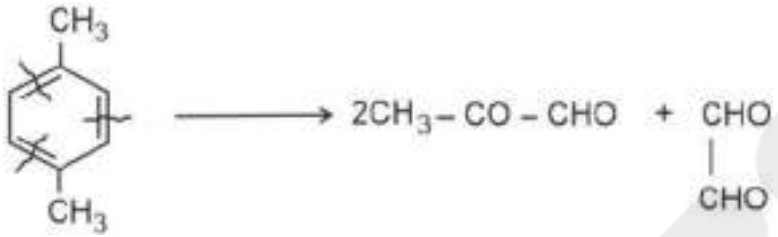
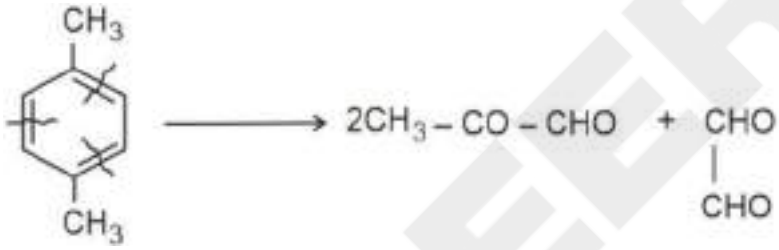
(1)



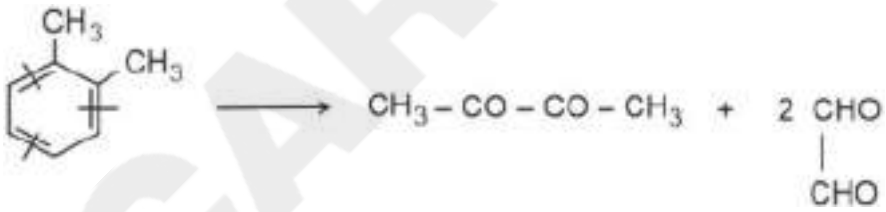


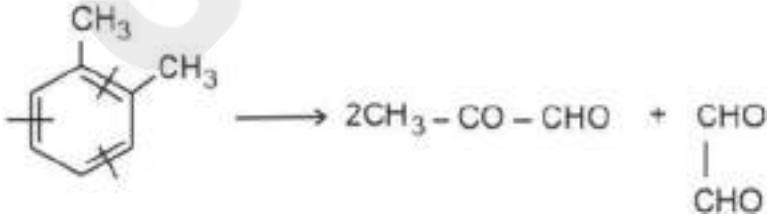
(2)





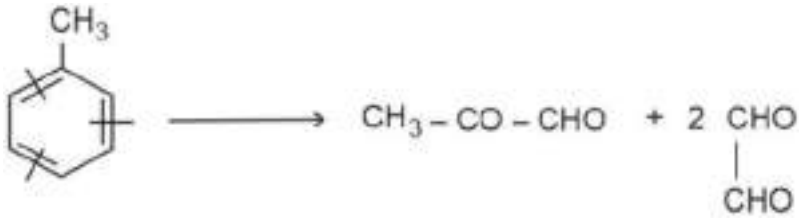
(3)

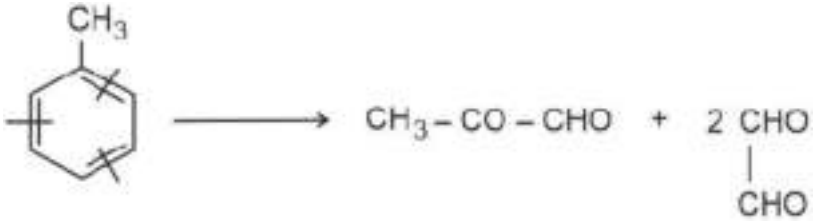




different products giving the pair.

(4)



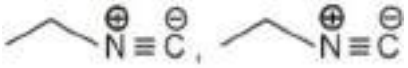


So option (3) is correct.

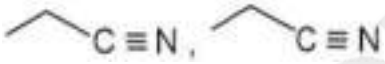
Q. 14

Considering the above reactions, the compound $'A'$ and compound $'B'$ respectively are :

Option 1:



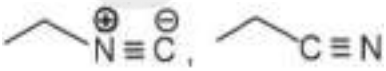
Option 2:



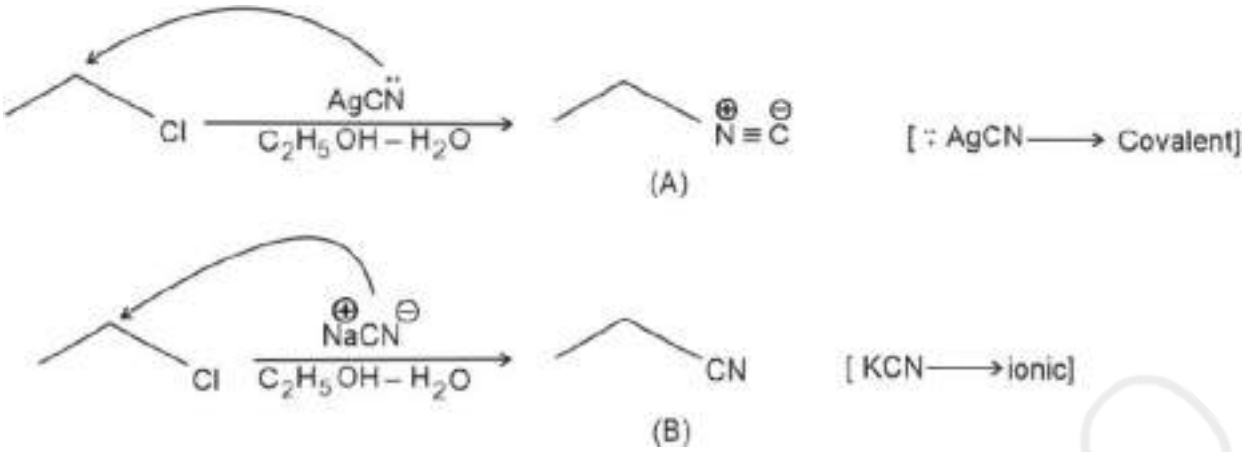
Option 3:

Option 4:

Correct Answer:

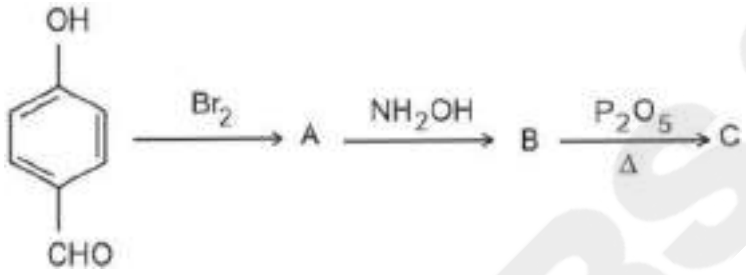


Solution:



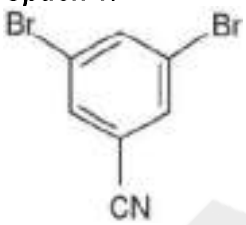
So, option (3) is correct.

Q. 15

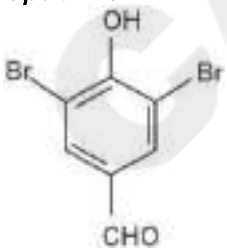


Consider the above reaction sequence, the Product $'C'$ is: :

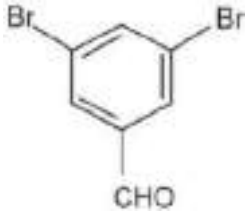
Option 1:



Option 2:



Option 3:

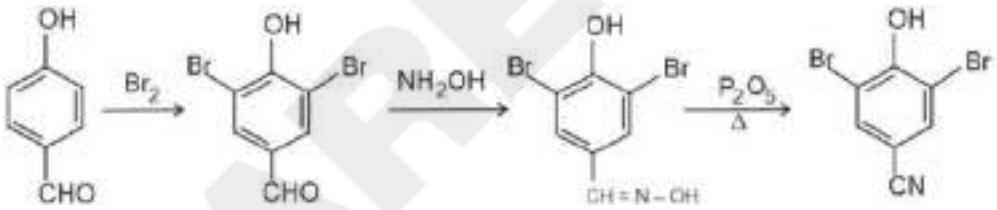


Option 4:

Correct Answer:

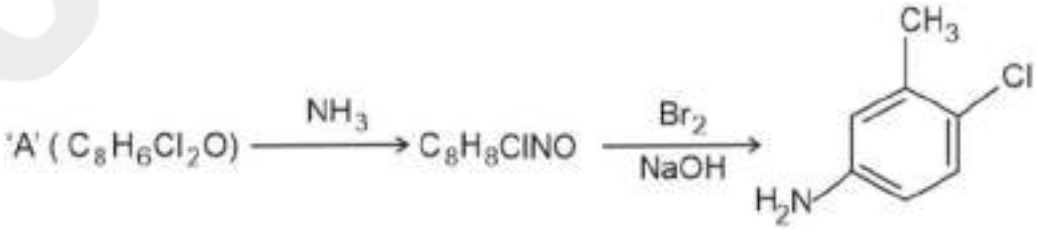
Solution:

The reaction will be -



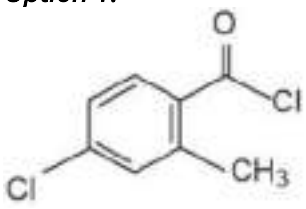
So, option (4) is correct.

Q. 16

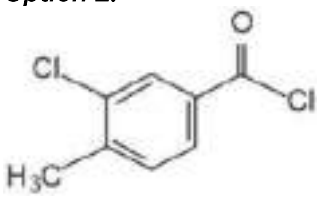


Consider the above reaction, the compound $'A'$ is :

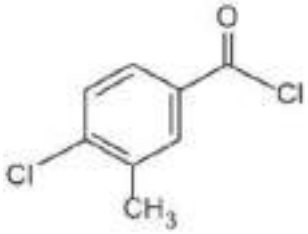
Option 1:



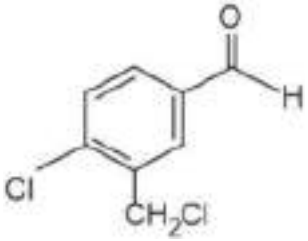
Option 2:



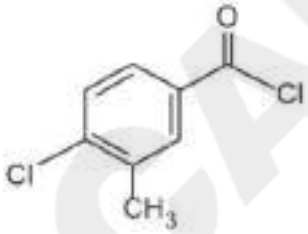
Option 3:



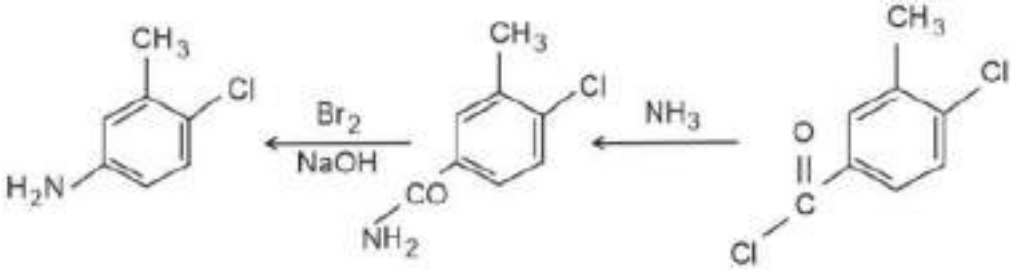
Option 4:



Correct Answer:



Solution:



Option (3) is correct.

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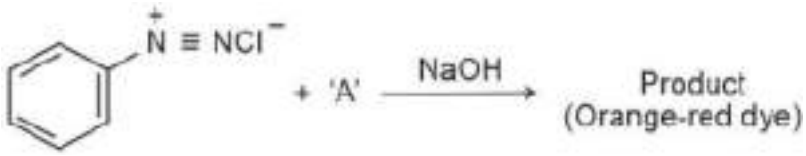
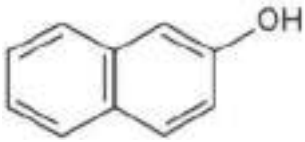
Q. 17

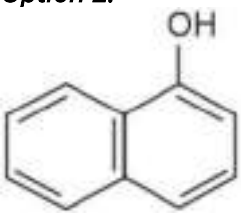


Which among the following represent reagent $'A'$?

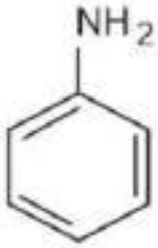
Option 1:



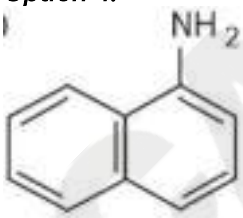
Option 2:



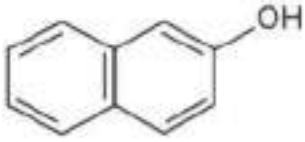
Option 3:



Option 4:

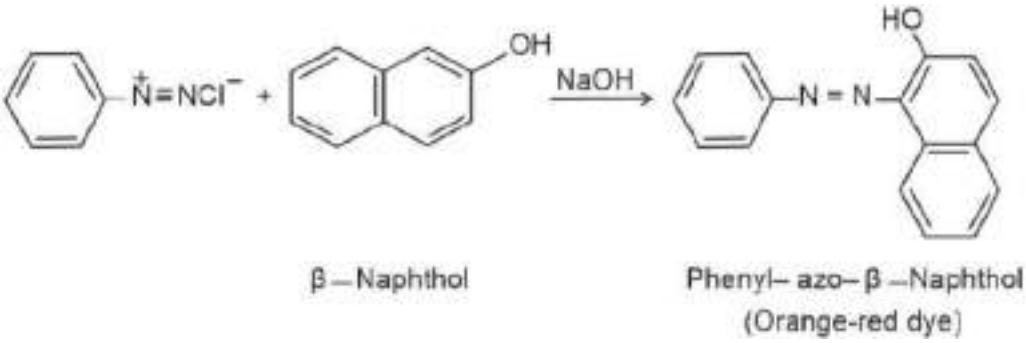


Correct Answer:



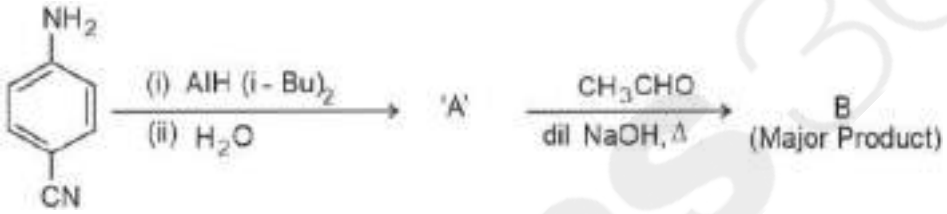
Solution:

The diazonium salt is coupled with an alkaline of β— naphthol to form an orange -red azo dye.



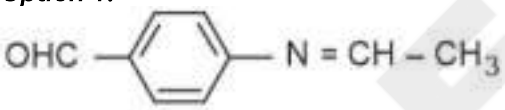
So, option (1) is correct.

Q. 18 Consider the following reaction sequence :

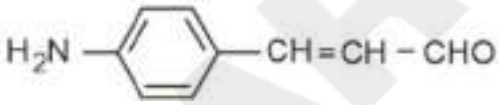


The product $'B'$ is :

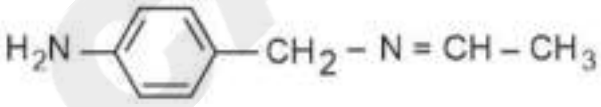
Option 1:



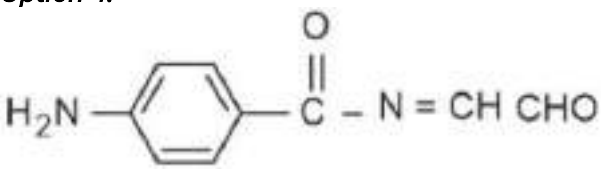
Option 2:



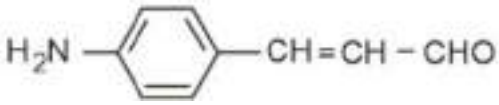
Option 3:



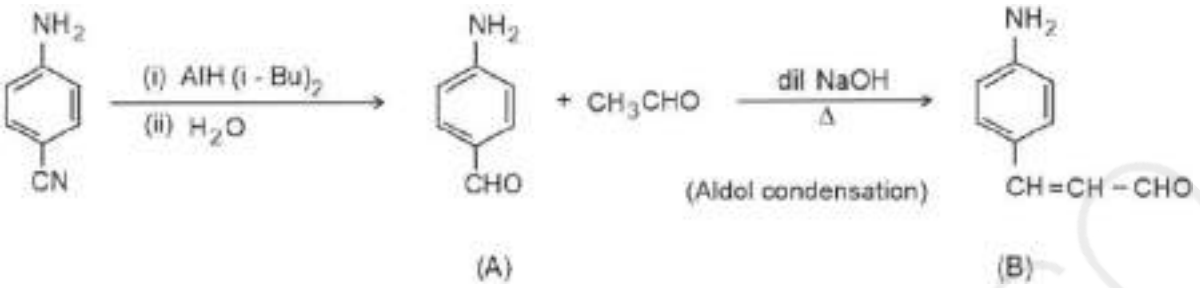
Option 4:



Correct Answer:



Solution:



Option (2) is correct.

Q. 19 Which of the following compounds is an example of hypnotic drug ?

Option 1:
Seldane

Option 2:
Amytal

Option 3:
Aspartame

Option 4:
Prontosil

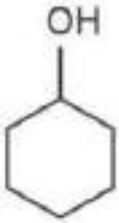
Correct Answer:
Amytal

Solution:
Veronal, amytal, nembutal, luminal are hypnotic drug.

Option (2) is correct.

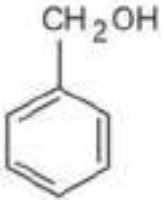
Q. 20 A compound $'X'$ is acidic and it is soluble in $NaOH$ solution, but insoluble in $NaHCO_3$ solution. Compound $'X'$ also gives violet colour with neutral $FeCl_3$ solution. The compound $'X'$ is :

Option 1:

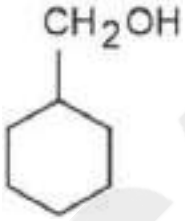


Option 2:

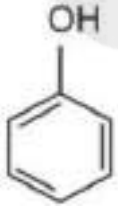
Option 3:



Option 4:



Correct Answer:



Solution:
Phenols are weakly acidic, therefore they are

Soluble in $NaOH$ solution but at the same time they are not sufficiently acidic to be soluble is

$NaHCO_3$ solution. phenols gives violet coloured complex with neutral $FeCl_3$ solution.

$$6C_6H_5OH + FeCl_3 \longrightarrow [Fe(C_6H_5O)_6]3^- + 3HCl + 3H^+$$
(violet complex)

So compound X is phenol.
So. option (2) is correct.

Q. 21 Resistance of a conductivity cell $(\text{cell constant } 129\ m^{-1})$ filled with 74.5 ppm solution of KCl is $100\ \Omega$ (labelled as solution 1). When the same cell is filled with KCl solution of 149 ppm, the resistance is $50\ \Omega$ (labelled as solution 2). The ratio of molar conductivity of solution 1 and solution 2 is i.e. $\frac{\Lambda_1}{\Lambda_2} = x \times 10^{-3}$. The value of x is _____. (Nearest integer)
Given, molar mass of KCl is $74.5\ g\ mol^{-1}$.

Correct Answer:
1000

Solution:

$$\text{Cell constant}(G) = \frac{l}{A} = RK = 129\,m^{-1}$$

$$(\text{conductivity})K = \frac{G}{R} = \frac{129}{R}$$

And we know $\Lambda_m = \frac{K \times 1000}{M}$

So, For 74.5 ppm solution of KCl

$$ppm_1 = 74.5$$

$$R_1 = 180\Omega, \Lambda_{m1} = \frac{K_1 \times 1000}{M_1}$$

and For 149 ppm solution of KCl

$$ppm_2 = 149\,ppm$$

$$R_2 = 50\Omega$$

$$\Lambda_{m2} = \frac{K_2 \times 1000}{M_2}$$

$$\text{Now, } \frac{M_1}{M_2} = \frac{ppm_1}{ppm_2} = \frac{74.5}{145.0}$$

$$\text{and } \frac{\Lambda_{m1}}{\Lambda_{m2}} = \frac{K_1}{K_2} = \frac{1}{M_1R_1} \times \frac{M_2R_2}{1}$$

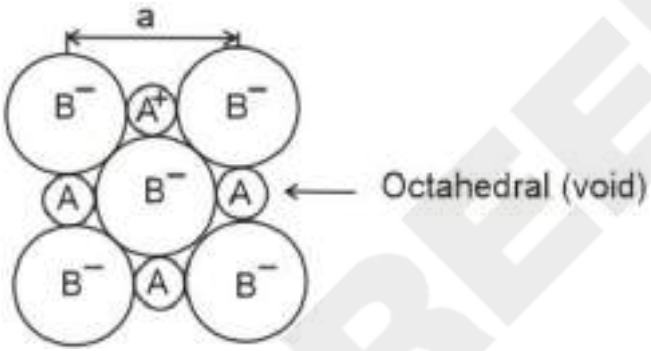
$$\Rightarrow \frac{\Lambda_{m1}}{\Lambda_{m2}} = \frac{R_2}{R_1} \times \frac{149.0}{74.5} = \frac{50}{100} \times 2 = 1 = 1000 \times 10^{-3}$$

Ans = 1000

Q. 22 Ionic radii of cation A^+ and anion B^- are 102 and $181\ pm$ respectively. These ions are allowed to crystallize into an ionic solid. This crystal has cubic close packing for B^-. A^+ is present in all octahedral voids. The edge length of the unit cell of the crystal AB is _____ pm. (Nearest Integer)

Correct Answer:
566

Solution:
This crystal has cubic close paking for B^-,
Means B^{Θ} is present all corner and face centres , and A^+ is present in all octahedral voids.



a = edge length of the unit cell of the crystal AB

$$a = 2r^- + 2r^+$$
$$a = 2 \times 181 + 2 \times 102$$
$$a = 362 + 204$$
$$a = 566\ pm$$

Ans = 566

Q. 23 The minimum uncertainty in the speed of an electron in an one dimensional region of length $2a_o$ (Where a_o = Bohr radius $52.9\ pm$) is _____ $km\ s^{-1}$.

{Given : Mass of electron $= 9.1 \times 10^{-31}$ kg, Planck's constant $h = 6.63 \times 10^{-34}$ Js }

Correct Answer:
548

Solution:
We know , Heisenberg's uncertainty Principle

$$\Delta x \times \Delta v \geq \frac{h}{4\pi m}$$

For the minimum uncertainty in the speed.

$$\Delta v = \frac{h}{4\pi m \times \Delta x}$$

given , $\Delta x = 2a_0 = 2 \times 52.9 = 105.8\,pm$
and $h = 6.63 \times 10^{-34}\,Js$

$$\Delta v = \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 105.8 \times 10^{-12}}$$
$$\Delta v = 548\,km\,s^{-1}$$

Ans = 548

Q. 24 When $600\ mL$ of $0.2\ M\ HNO_3$ is mixed with $400\ mL$ of $0.1\ M\ NaOH$ solution in a flask, the rise in temperature of the flask is _____$\times 10^{-2}{}^{\circ}C$.

(Enthalpy of neutralisation $= 57\ kJ\ mol^{-1}$ and Specific heat of water $= 4.2 JK^{-1}\ g^{-1}$)

(Neglect heat capacity of flask)

Correct Answer:
54

Solution:
The reaction will be

$$\underset{0.2M}{HNO_3} + \underset{0.1M}{NaOH} \longrightarrow H_2O$$

initial	120 m mol	40 m mol	0
final	80 m mol	0	40 m mol

So, total 40 m mol is neutralising.
So total enthalpy of neutralisation for 40 m mol will be

$\Delta H = 40\,m\,mol \times 57\,KJ\,mol$
$\Delta H = 40 \times 10^{-3} \times 57 \times 10^{3}\,J$
$\Delta H = 2280\,J$

we know the formula,
$Q = ms\Delta T = \Delta H$
m = mass of water
s = specific heat of water = $4.2\,J\,K^{-1}\,g^{-1}$
$\Delta T = ?$

$\text{Now, } m = \text{volume} \times \text{density} = 1000\,mL \times 1\,g|ml$
$m = 1000\,g$

$\text{then, } 2280 = 1000 \times 4.2 \times \Delta T$
$\Delta T = 0.54$
$\Delta T = 54 \times 10^{-2} K$
$\Delta T = 54 \times 10^{-2}\,^{\circ}C (\text{change will be same})$

Ans = 54

Q. 25 If O_2 gas is bubbled through water at $303\ K$, the number of millimoles of O_2 gas that dissolve in $1\ litre$ of water is _____
. (Nearest Integer)

(Given : Henry's Law constant for O_2 at $303\ K$ is $46.821\ k$ bar and partial pressure of $O_2 = 0.920$ bar)

(Assume solubility of O_2 in water is too small, nearly negligible)

Correct Answer:
1

Solution:
We know the formula
$P = K_H \times x$

$\text{given } K_H = 46.82 \times 10^{3} bar$
$P = 0.920\,bar$
$x = \text{mole fraction of } O_2$

$$x = \frac{n_{O_2}}{n_{O_2} + n_{H_2O}}$$

$$x = \frac{n_{O_2}}{n_{H_2O}}$$ [due to solubility of O_2 in water is too small]

$$\text{Now,} \quad 0.920 = 46.82 \times 10^3 \times \frac{n_{O_2}}{n_{H_2O}}$$

$$0.920 = 46.82 \times 10^3 \times \frac{n_{O_2}}{\frac{1000}{18}}$$

$$0.920 = 46.82 \times nO_2 \times 18$$

$$n_{O_2} = 1.09 \times 10^{-3}$$

$$n_{O_2} = 1.09\,\text{mmol} \approx 1\,\text{mmol}$$

Ans = 1

Q. 26 If the solubility product of PbS is 8×10^{-28}, then the solubility of PbS in pure water at $298\ K$ is $x \times 10^{-16}\ mol\ L^{-1}$. The value of x is _____ . (Nearest Integer)

$[\text{Given} : \sqrt{2} = 1.41]$

Correct Answer:
282

Solution:

$$PbS \longrightarrow \underset{s}{Pb^{2+}} + \underset{s}{S^{2-}}$$

$$\Rightarrow \quad K_{sp} = s \times s = s^2 = 8 \times 10^{-28}$$

$$s = 2\sqrt{2} \times 10^{-14}\,\text{mol/L}$$

$$s = 282 \times 10^{-16}\,\text{mol/L} \quad \left[\because \sqrt{2} = 1.41\right]$$

Ans = 282

Q. 27 The reaction between X and Y is first order with respect to X and zero order with respect to Y.

Experiment	$\frac{[X]}{molL^{-1}}$	$\frac{[Y]}{molL^{-1}}$	$\frac{\text{Initial rate}}{mol\ L^{-1}\ min^{-1}}$
I	0.1	0.1	2×10^{-3}
II	L	0.2	4×10^{-3}
III	0.4	0.4	$M \times 10^{-3}$
IV	0.1	0.2	2×10^{-3}

Examine the data of table and calculate ratio of numerical values of M and L . (Nearest Integer)

Correct Answer:
40

Solution:
Reaction is first order with X and Zero order with Y, So,

$$\text{rate} = K[X]^1[Y]^0$$
$$\text{rate}\ \alpha\ [X]^1$$

Now,
From Experiment I and II.

$$\frac{r_{II}}{r_I} = \frac{4 \times 10^{-3}}{2 \times 10^{-3}} = \frac{k[X]_{II}}{k[X]_I} = \frac{[X]_{II}}{[X]_I} = \frac{L}{0.1}$$
$$L = [X]_{II} = 2 \times 0.1 = 0.2$$

Now, From experiment III and IV

$$\frac{r_{III}}{r_{IV}} == \frac{k[X]_{III}}{k[X]_{IV}} = \frac{[X]_{III}}{[X]_{IV}}$$
$$\frac{M \times 10^{-3}}{2 \times 10^{-3}} = \frac{0.4}{0.1}$$
$$M = 8$$

Now, ratio of numerical values of M and L

$$\frac{M}{L} = \frac{8}{0.2} = 40$$

Ans = 40

Q. 28 In a linear tetrapeptide (Constituted with different amino acids), (number of amino acids) - (number of peptide bonds) is _____.

Correct Answer:
1

Solution:
It is tetrapeptide.
So, there will be 4 amino acids and these 4 amino acids will combine with 3 peptide bonds.

Hence, number of amino acids - number of peptide bonds
= 4-3
= 1

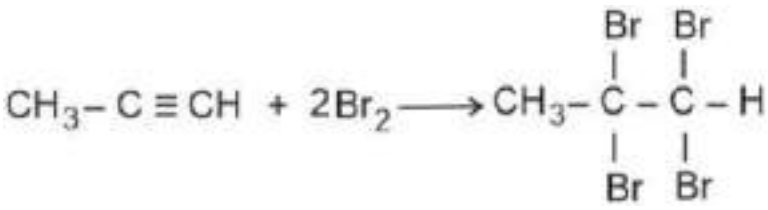
Ans = 1

Q. 29 In bromination of Propyne, with Bromine $1,1,2,2$- tetrabromopropane is obtained in 27% yield. The amount of $1,1,2,2$- tetrabromopropane obtained from $1\ g$ of Bromine in this reaction is _____$\times 10^{-1}\ g.$ (Nearest integer)

$(\text{Molar Mass : Bromine} = 80\ g/mol)$

Correct Answer:
3

Solution:
The reaction will be



So, From 2 mole of Br_2, 1 mole of $1,1,2,2-\text{tetrabromopropane}$ is obtained.
if yield is 27%,

$$\text{from 2 mole of } Br_2 \longrightarrow 1 \times 0.27 \text{ moleproduct form}$$
$$2 \times 160 \text{ gram} \longrightarrow 1 \times 0.27 \times 360 \text{ g product form}$$
$$320 \text{ gram} \longrightarrow 1 \times 0.27 \times 360 \text{ g product form}$$

$$\text{So, from 1 gram of } Br_2 \longrightarrow \frac{1 \times 0.27 \times 360}{320} \text{ g product form}$$
$$\text{from 1 gram of } Br_2 \longrightarrow 0.30 \text{ g product form}$$
$$\text{from 1 gram of } Br_2 \longrightarrow 3 \times 10^{-1} \text{ g product form.}$$

Ans = 3

Q. 30 $[Fe(CN)_6]^{3-}$ should be an inner orbital complex. Ignoring the pairing energy, the value of crystal field stabilization energy for this complex is $(-)$ _____ Δ_0. (Nearest integer)

Correct Answer:
2

Solution:
$[Fe(CN)_6]^{3-}$ is an inner orbital complex,

Due to strong field ligand CN^{Θ} the electron will be paired .

So, $\text{CFSE} = 5 \times (-0.4)\,\Delta_0 + \text{pairing energy}$

Now, for complex ingoring the pairing energy

Hence $\text{CFSE} = 5 \times (-0.4\,\Delta_0)$
$\text{CFSE} = -2\Delta_0$

Ans = 2

Maths

Q. 1 Let R be a relation from the set $\{1, 2, 3, \ldots, 60\}$to itself such that $R = \{(a, b) : b = pq$, where $p, q \geqslant 3$ are prime numbers $\}$. Then, the number of elements in R is:

Option 1:
600

Option 2:
660

Option 3:
540

Option 4:
720

Correct Answer:
660

Solution:
Number of possible values of a=60, for b=pq,
if p=3, q=3,5,7,11,13,17,19
if $p = 5 \quad q = 5, 7, 11$
if $p = 7 \quad q = 7$ Total cases$= 60 \times 11 = 660$

Q. 2 if $z = 2 + 3i$, then $z^5 + (\bar{z})^5$ is equal to :

Option 1:
244

Option 2:
224

Option 3:
245

Option 4:
265

Correct Answer:

244

Solution:

$$z^5+(\bar{z})^5 = (2+3i)^5 + (2-3i)^5$$
$$= 2\left({}^5c_0 2^5 + {}^5c_2 2^3 (3i)^2 + {}^5c_4 2^1 (3i)^{41}\right)$$
$$= 2(32 + 10 \times 8 \times (-9) + 5 \times 2 \times 81) = 244$$

Q. 3 Let A and B be two 3×3 non-zero real matrices such that AB is a zero matrix. Then

Option 1:

the system of linear equations $AX = 0$ has a unique solution

Option 2:

the system of linear equations $AX = 0$ has infinitely many solutions

Option 3:

B is an invertible matrix

Option 4:

$\text{adj}(A)$ is an invertible matrix

Correct Answer:

the system of linear equations $AX = 0$ has infinitely many solutions

Solution:

$$AB = 0 \Rightarrow |AB| = 0$$
$$|A||B| = 0$$

$$|A| = 0, \qquad |B| = 0$$

$$\text{if} |A| \neq 0, B = 0 \text{ (not possible)}$$
$$\text{if} |B| \neq 0, A = 0 \text{(not possible)}$$
$$\text{Hence} |A| = |B| = 0$$
$$\Rightarrow \text{Ax=0 has infinitely many solutions.}$$

Q. 4 If $\frac{1}{(20-a)(40-a)}+\frac{1}{(40-a)(60-a)}+\ldots+\frac{1}{(180-a)(200-a)}=\frac{1}{256}$, then the maximum value of a is:

Option 1:
198

Option 2:
202

Option 3:
212

Option 4:
218

Correct Answer:
212

Solution:

By slitting

$$\frac{1}{20}\left[\left(\frac{1}{20-\mathrm{a}}-\frac{1}{40-\mathrm{a}}\right)+\left(\frac{1}{40-\mathrm{a}}-\frac{1}{60-\mathrm{a}}\right)+\cdots\ldots+\left(\frac{1}{180-\mathrm{a}}-\frac{1}{200-\mathrm{a}}\right)\right]$$

$$\Rightarrow \frac{1}{20}\left(\frac{1}{20-\mathrm{a}}-\frac{1}{200-\mathrm{a}}\right)=\frac{1}{256}$$

$$(20-\mathrm{a})(200-\mathrm{a})=256\times 9$$

$$\mathrm{a}^2-220\mathrm{a}+1696=0$$

$$\mathrm{a}=8,212$$

Hence maximum value of a is 212 .

Q. 5 If $\lim\limits_{x\to 0}\frac{\alpha e^x+\beta e^{-x}+\gamma\sin x}{x\sin^2 x}=\frac{2}{3}$, where $\alpha,\beta,\gamma\in\mathbf{R}$, then which of the following is NOT correct ?

Option 1:
$\alpha^2+\beta^2+\gamma^2=6$

Option 2:

$\alpha\beta+\beta\gamma+\gamma\alpha+1=0$

Option 3:

$\alpha\beta^2+\beta\gamma^2+\gamma\alpha^2+3=0$

Option 4:

$\alpha^2-\beta^2+\gamma^2=4$

Correct Answer:

$\alpha\beta^2+\beta\gamma^2+\gamma\alpha^2+3=0$

Solution:

$$\lim_{x\to0}\frac{\alpha\left(1+x+\frac{x^2}{2!}+\cdots\right)+\beta\left(1-x+\frac{x^2}{2!}-\frac{x^3}{3!}+\cdots\right)+x\left(x-\frac{x^3}{3!}+\cdots\right)}{x^3}$$

Constant terms should be zoro

$\Rightarrow \alpha+\beta=0$

ceff. of x should be zero

$\Rightarrow \alpha-\beta+\gamma=0$

$$\lim_{x\to0}x^3\frac{\frac{\alpha}{3!}-\frac{\beta}{3!}+\frac{\gamma}{3!})+x^4\left(\frac{\alpha}{3!}-\frac{\beta}{3!}-\frac{\gamma}{3!}\right)}{x^3}=\frac{2}{3}$$

$$\Rightarrow \frac{\alpha}{2}+\frac{\beta}{2}=0$$

$$\frac{\alpha}{6}-\frac{\beta}{6}-\frac{\gamma}{6}=2/3$$

$\Rightarrow \alpha=1,\beta=-1,\quad \gamma=-2$

Q. 6 The integral $\int_0^{\frac{\pi}{2}}\frac{1}{3+2\sin x+\cos x}\,dx$ is equal to :

Option 1:

$\tan^{-1}(2)$

Option 2:

$\tan^{-1}(2) - \frac{\pi}{4}$

Option 3:

$\frac{1}{2}\tan^{-1}(2) - \frac{\pi}{8}$

Option 4:

$\frac{1}{2}$

Correct Answer:

$\tan^{-1}(2) - \frac{\pi}{4}$

Solution:

$$I = \int_0^{\pi/2} \frac{dx}{3 + 2\sin x + \cos x} = \int_0^{\pi/2} \frac{\sec^2 \frac{x}{2} dx}{2\tan^2 \frac{x}{2} + 4\tan \frac{x}{2} + 4}$$

$\text{Puttan } \frac{x}{2} = t, \text{ so}$

$$I = \int_0^1 \frac{dt}{(t+1)^2 + 1} = \tan^{-1}(x+1)\Big|_0^1 = \tan^{-1} 2 - \frac{\pi}{4}$$

Q. 7 Let the solution curve $y = y(x)$ of the differential equation $(1 + e^{2x})\left(\frac{dy}{dx} + y\right) = 1$ pass through the point $\left(0, \frac{\pi}{2}\right)$. Then, $\lim_{x\to\infty} e^x y(x)$ is equal to:

Option 1:

$\frac{\pi}{4}$

Option 2:

$\frac{3\pi}{4}$

Option 3:

$\frac{\pi}{2}$

Option 4:

$\frac{3\pi}{2}$

Correct Answer:

$\frac{3\pi}{4}$

Solution:

$\frac{dy}{dx}+4=\frac{1}{1+e^{2x}}$

So integrating factor is $e^{\int 1 dx}=e^{x}$

So solution is $ye^{x}=\tan^{-1}(e^{x})+c$

Now as curve is passing trrough $\left(0,\frac{\pi}{2}\right)$ So

$\Rightarrow c=\frac{\pi}{4}$

$\Rightarrow \lim_{x\to\infty}(y\cdot e^{x})=\lim_{x\to\infty}\left(\tan^{-1}(e^{x})+\frac{\pi}{4}\right)=\left(\frac{3\pi}{4}\right)$

Q. 8 Let a line L pass through the point of intersection of the lines $bx+10y-8=0$ and $2x-3y=0, b\in \mathbf{R}-\left\{\frac{4}{3}\right\}$. If the line L also passes through the point $(1,1)$ and touches the circle $17\left(x^2+y^2\right)=16$, then the eccentricity of the ellipse $\frac{x^2}{5}+\frac{y^2}{b^2}=1$ is :

Option 1:

$\frac{2}{\sqrt{5}}$

Option 2:

$\sqrt{\frac{3}{5}}$

Option 3:

$\frac{1}{\sqrt{5}}$

Option 4:

$\sqrt{\frac{2}{5}}$

Correct Answer:

$\sqrt{\frac{3}{5}}$

Solution:

Line is passing through intersection of $bx+10y-8=0$ and $2x-3y=0$ is $(bx+10y-8)+\lambda(2x-3y)=0$.

As line is Passing through $(1,1)$ so $\lambda=b+2$

Now line $(3b+4)x-(3b-4)y-8=0$ is tangent to circle $17\left(x^2+y^2\right)=16$

so $\frac{8}{\sqrt{(3b+4)^2+(3b-4)^2}}=\frac{4}{\sqrt{17}}$

$b^2=2\Rightarrow e=\sqrt{\frac{3}{5}}$

Q. 9 If the foot of the perpendicular from the point $A(-1,4,3)$ on the plane $P: 2x+my+nz=4$, is $\left(-2,\frac{7}{2},\frac{3}{2}\right)$, then the distance of the point A from the plane P, measured parallel to a line with direction ratios $3,-1,-4$, is equal to :

Option 1:

1

Option 2:

$\sqrt{26}$

Option 3:

$2\sqrt{2}$

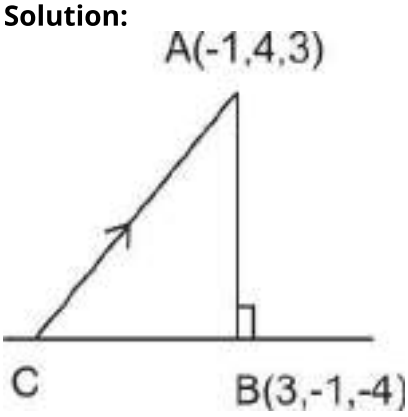
Option 4:

$\sqrt{14}$

Correct Answer:

$\sqrt{26}$

Solution:



Let B be foot of $\perp$ coordinates of B $= \left(-2, \frac{7}{2}, \frac{3}{2}\right)$

Direction ratio of line AB is $\langle 2, 1, 3\rangle$ So

$m = 1, n = 3$

So equation of AC is $\frac{x+1}{3} = \frac{y-4}{-1} = \frac{z-3}{-4} = \lambda$

So point c is $(3\lambda - 1, -1 + 9, -4, \lambda + 3)$.but c lies on the plane, so

$6\lambda - 2 - \lambda - 1 - 12\lambda + 9 = 4$

$\Rightarrow \lambda = 1 \Rightarrow C(2, 3, -1)$

$\Rightarrow \quad AC = \sqrt{26}$

Q. 10 Let $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$. If $\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is :

Option 1:

-5

Option 2:

5

Option 3:

1

Option 4:

-1

Correct Answer:

-5

Solution:

$$\tilde{a} = 3\hat{i} + \hat{j}, \quad \tilde{b} = i + 2\hat{j} + \hat{k}$$

$$\text{As } \tilde{a} \times (\tilde{b} \times \tilde{c}) = \tilde{b} + \lambda\tilde{c}$$

$$\Rightarrow \tilde{a} \cdot \tilde{c}(\tilde{b}) - (\tilde{a} \cdot \tilde{b})\tilde{c} = \tilde{b} + \lambda\tilde{c}$$

$$\Rightarrow \tilde{a} \cdot \tilde{c} = 1, \tilde{a} \cdot \tilde{b} = -\lambda$$

$$\Rightarrow (3\hat{i} + \hat{j}) \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -\lambda$$

$$\tilde{\lambda} = -5$$

Q. 11 The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is $\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then $\cot\alpha$ is equal to :

Option 1:

$\frac{6}{5}$

Option 2:

$\frac{9}{5}$

Option 3:

$\frac{4}{3}$

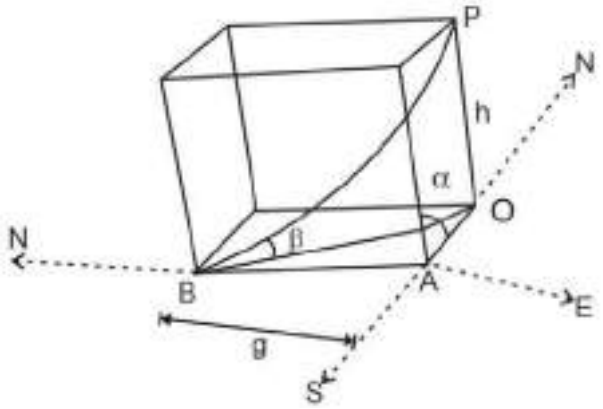
Option 4:

$\frac{7}{3}$

Correct Answer:

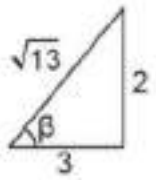
$\frac{6}{5}$

Solution:

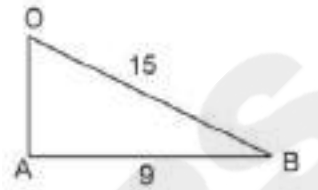


Given $OB = 15$

$\cos B = \frac{3}{\sqrt{13}}$



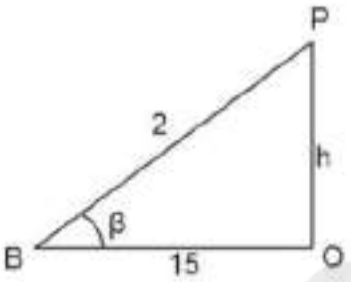
$\tan \beta = \frac{2}{3}$



$OA^2 + AB^2 = 225$

$OA^2 + 81 = 225$

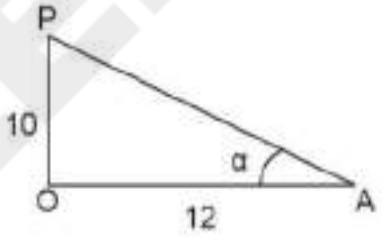
$OA = 12$



$\tan B = \frac{h}{\sqrt{15}}$

$\frac{2}{\sqrt{3}} = \frac{h}{15}$

$h = 10$



$\tan 2 = \frac{10}{12}$

$\cot \alpha = \frac{12}{10} = \frac{6}{5}$

Q. 12 The statement $(p \wedge q) \Rightarrow (p \wedge r)$ is equivalent to :

Option 1:

$q \Rightarrow (p \wedge r)$

Option 2:

$p \Rightarrow (p \wedge r)$

Option 3:
$(p \wedge r) \Rightarrow (p \wedge q)$

Option 4:
$(p \wedge q) \Rightarrow r$

Correct Answer:
$(p \wedge q) \Rightarrow r$

Solution:
$(\mathrm{p} \wedge \mathrm{q}) \Rightarrow (\mathrm{p} \wedge \mathrm{r})$
$\sim (\mathrm{p} \wedge \mathrm{q}) \vee (\mathrm{p} \wedge \mathrm{r})$
$(\sim \mathrm{p} \vee \sim \mathrm{q}) \vee (\mathrm{p} \wedge \mathrm{r})$
$(\sim \mathrm{p} \vee (\mathrm{p} \wedge \mathrm{r})) \vee \sim \mathrm{q}$
$(\sim \mathrm{p} \vee \mathrm{p}) \cap (\sim \mathrm{p} \vee \mathrm{r}) \vee \sim \mathrm{q}$
$(\sim \mathrm{p} \vee \mathrm{r}) \vee \sim \mathrm{q}$
$(\sim \mathrm{p} \vee \sim \mathrm{q}) \vee \mathrm{r}$
$\sim (\mathrm{p} \wedge \mathrm{q}) \vee \mathrm{r}$
$(\mathrm{p} \wedge \mathrm{q}) \Rightarrow \mathrm{r}$

Q. 13 Let the circumcentre of a triangle with vertices $\mathrm{A}(\mathrm{a}, 3), \mathrm{B}(\mathrm{b}, 5) \text{and } \mathrm{C}(\mathrm{a}, \mathrm{b}), \mathrm{ab} > 0 \text{ be} \mathrm{P}(1, 1)$. If the line AP intersects the line BC at the point $\mathrm{Q}\left(\mathrm{k}_1, \mathrm{k}_2\right)$, then $\mathrm{k}_1 + \mathrm{k}_2$ is equal to:

Option 1:
2

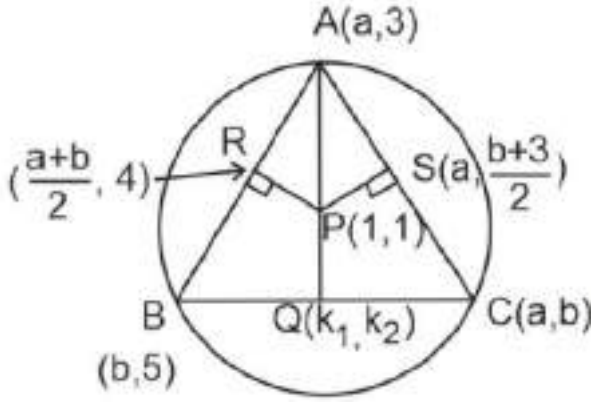
Option 2:
$\frac{4}{7}$

Option 3:
$\frac{2}{7}$

Option 4:
4

Correct Answer:
$\frac{4}{7}$

Solution:



The circumcentre of a triangle is the point of intersection the perpendicular bisector of the sides of a triangle.

$$(AC)_m \times (SP)_m = -1$$

$$\frac{1}{0} \times (sp)_m = -1$$

$$(sp)_m = 0$$

$$\frac{\frac{b+3}{2} - 1}{a - 1} = 0$$

$$b = -1$$

$$(AB)_m \times (RP)_m = -1$$

$$\left(\frac{2}{b - a}\right)\left(\frac{3}{\frac{4}{2} - 1}\right) = -1 \qquad |\because b = -1$$

$$\frac{6}{(a+1)\left(\frac{a-3}{2}\right)} = 1$$

$$a^2 - 2a - 15 = 0$$

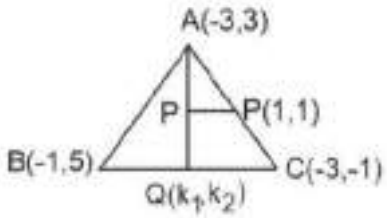
$$a = -3, \quad \text{or} \quad a = 5$$

Given $ab > 0$

$$a(-1) > 0$$

$$a < 0$$

$$a = -3 \cdot \text{accept}$$



Equation of line AP which is passes through $A(-3,3)$, and point $P(1,1)$

$$(y-1)=\left(\frac{3-1}{-3-1}\right)(x-1)$$

$$-2y+2=x-1$$

$$x+2y=3.....(1)$$

Equation of line BC

$$(y-5)=\frac{6}{2}(x+1)$$

$$y=3x+8.....(2)$$

Intersection of line AP and BC is point (K_1, K_2)

So from eq^n (1) and (2)

$\therefore$ point Q $(K_1, K_2) \Rightarrow (x, y) \Rightarrow \left(\frac{-13}{7}, \frac{17}{7}\right)$

so $K_1 + K_2 = \left(\frac{4}{7}\right)$

Q. 14 Let $\hat{a}$ *and* $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $(\hat{a}+\hat{b})$and$(\hat{a}+2\hat{b}+2(\hat{a}\times\hat{b}))$, then the value of $164\cos^2\theta$ is equal to:

Option 1:
$90+27\sqrt{2}$

Option 2:
$45+18\sqrt{2}$

Option 3:
$90+3\sqrt{2}$

Option 4:
$54+90\sqrt{2}$

Correct Answer:
$90+27\sqrt{2}$

Q. 15 If $f(\alpha)=\int_1^{\alpha}\frac{\log_{10} t}{1+t}dt, \alpha>0$, then $f\left(e^3\right)+f\left(e^{-3}\right)$ is equal to :

Option 1:
9

Option 2:
$\frac{9}{2}$

Option 3:
$\frac{9}{\log_e(10)}$

Option 4:
$\frac{9}{2\log_e(10)}$

Correct Answer:
$\frac{9}{2\log_e(10)}$

Solution:

CAREERS360

so $^{n}\Rightarrow \quad f\left(e^{3}\right)=\int_{1}^{e^{3}} \ln t dt \ldots(1)$

$f(\alpha)=\int_{1}^{\alpha} \frac{\ln t}{(\ln 10)(1+t)}$

$t=\frac{1}{x} \Rightarrow x=\frac{1}{t}$

$dt=-\frac{1}{x^{2}} dx$

$\int_{1}^{\frac{1}{\alpha}} \frac{-\ln x}{(\ln 10)\left(1+\frac{1}{x}\right)}\left(-\frac{1}{x^{2}}\right) dx$

$f(\alpha)=\frac{1}{\ln 10} \int_{1}^{\frac{1}{\alpha}} \frac{\ln x}{x(x+1)} dx$

$f\left(e^{-3}\right)=\frac{1}{\ln 10} \int_{1}^{e^{3}} \frac{\ln t dt}{t(t+1)} \ldots . (2)$

Add (1) & (2)

$f\left(e^{3}\right)+f\left(e^{-3}\right)$

$=\left(\frac{1}{\ln 10}\right) \int_{1}^{e^{3}} \frac{\tan t}{(1+t)}\left[1+\frac{1}{t}\right] dt$

$=\left(\frac{1}{\ln 10} \mid \int_{1}^{3} \frac{\ln t}{t} dt\right.$

$\ln t=\gamma$

$\frac{dt}{t}=dr$

$=\frac{1}{\ln 10} \int_{0}^{3} \gamma d\gamma$

$=\left(\frac{1}{\ln 10}\right)\left(\frac{\gamma^{2}}{2}\right)\Big|_{0}^{3}$

$=\left(\frac{1}{\log 10} \|^{\frac{9}{2}}\right)$

$=\frac{9}{2 \log e^{10}}$

$so^{n}\Rightarrow \quad f\left(e^{3}\right)=\int_{1}^{e^{3}} \ln t dt \ldots(1)$

$f(\alpha)=\int_{1}^{\alpha} \frac{\ln t}{(\ln 10)(1+t)} dt$

$t=\frac{1}{x} \Rightarrow x=\frac{1}{t}$

$dt=-\frac{1}{x^{2}} dx$

$\int_{1}^{\frac{1}{\alpha}} \frac{-\ln x}{(\ln 10)\left(1+\frac{1}{x}\right)}\left(-\frac{1}{x^{2}}\right) dx$

$f(\alpha)=\frac{1}{\ln 10} \int_{1}^{\frac{1}{\alpha}} \frac{\ln x}{x(x+1)} dx$

$f\left(e^{-3}\right)=\frac{1}{\ln 10} \int_{1}^{e^{3}} \frac{\ln t dt}{t(t+1)} \ldots . (2)$

Add (1) & (2)

$f\left(e^{3}\right)+f\left(e^{-3}\right)$

$=\left(\frac{1}{\ln 10}\right) \int_{1}^{e^{3}} \frac{\tan t}{(1+t)}\left[1+\frac{1}{t}\right] dt$

Q. 16 The area of the region $\left\{(x, y):|x-1| \leq y \leq \sqrt{5-x^{2}}\right\}$ is equal to:

Option 1:

$\frac{5}{2}\sin^{-1}\left(\frac{3}{5}\right)-\frac{1}{2}$

Option 2:

$\frac{5\pi}{4}-\frac{3}{2}$

Option 3:

$\frac{3\pi}{4}+\frac{3}{2}$

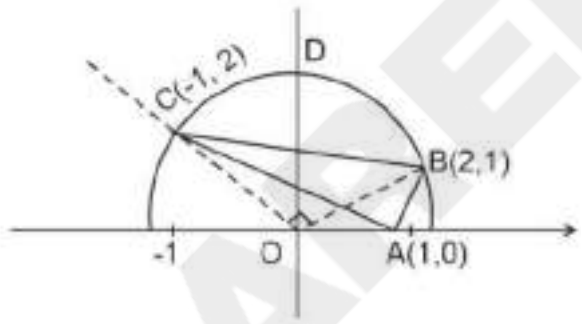
Option 4:

$\frac{5\pi}{4}-\frac{1}{2}$

Correct Answer:

$\frac{5\pi}{4}-\frac{1}{2}$

Solution:



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$|x-1| < y < \sqrt{5-x^2}$

When $|x-1| = \sqrt{5-x^2}$

$\Rightarrow (x-1)^2 = (5-x^2)$

$\Rightarrow x^2 - x - 2 = 0$

$x = 2, -1$

Required Area = Area of $\triangle ABC$ + Area of region BCD

$$= \frac{1}{2}\begin{vmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ -1 & 2 & 1 \end{vmatrix} + \frac{\pi}{4}(\sqrt{5})^2 - \frac{1}{2}(\sqrt{5})^2$$

$$= \frac{5\pi}{4} - \frac{1}{2}$$

Q. 17 Let the focal chord of the parabola $P : y^2 = 4x$ along the line $L : y = mx + c, m > 0$ meet the parabola at the points M and N. Let the line L be a tangent to the hyperbola $H : x^2 - y^2 = 4$. If O is the vertex of P and F is the focus of H on the positive x-axis, then the area of the quadrilateral OMFN is :

Option 1:
$2\sqrt{6}$

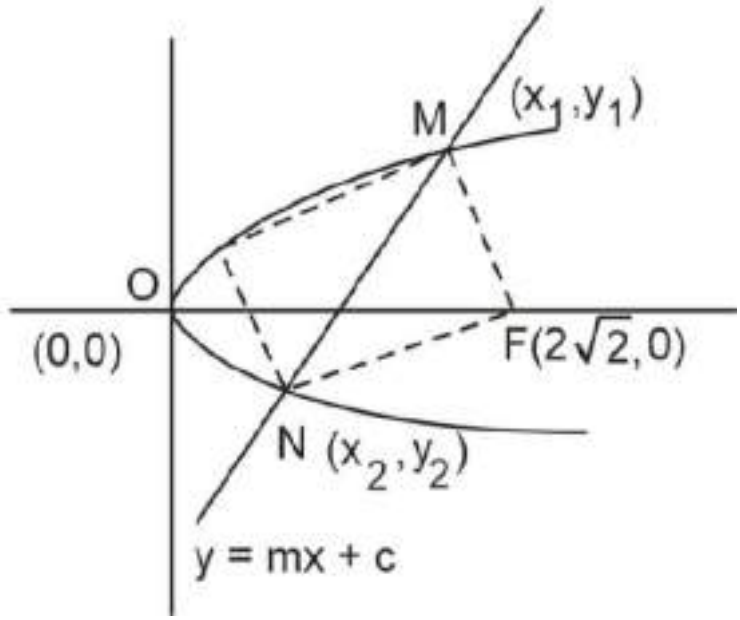
Option 2:
$2\sqrt{14}$

Option 3:
$4\sqrt{6}$

Option 4:
$4\sqrt{14}$

Correct Answer:
$2\sqrt{14}$

Solution:



$$\frac{x^2}{4} - \frac{y^2}{4} = 1$$

forcus$(a0, 0)$

f$(2\sqrt{2}, 0)$

line $L : y = mx + c$pass $(1,0)$

$0 = m + c \cdots (1)$

Line L is tangent to Hyperbola$\frac{x^2}{4} - \frac{y^2}{4} = 1$

$c = \pm\sqrt{a^2m^2 - l^2}$

$c = \pm\sqrt{4m^2 - 4}$

from (1)

$-m = \pm\sqrt{4m^2 - 4}$

$m^2 = 4m^2 - 4$

$4 = 3m^2$

$\frac{2}{\sqrt{3}} = m \quad (\text{as } m > 0)$

$c = -m$

$c = -\frac{2}{\sqrt{3}}$

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$$y = \frac{2x}{\sqrt{3}} - \frac{2}{\sqrt{3}}$$

$$y^2 = 4x$$

$$\Rightarrow \left(\frac{2x-2}{\sqrt{3}}\right)^2 = 4x$$

$$\Rightarrow x^2 + 1 - 2x = 3x$$

$$x^2 - 5x + 1 = 0$$

$$y^2 = 4\left(\frac{\sqrt{3}y + 2}{2}\right)$$

$$y^2 = 2\sqrt{3}y + 4$$

$$\Rightarrow y^2 - 2\sqrt{3}y - 4 = 0$$

$$\text{Area} = \left|\frac{1}{2}\right| \begin{matrix} 0 & x_1 & 2\sqrt{2} & x_2 & 0 \\ 0 & y_1 & 0 & y_2 & 0 \end{matrix} ||$$

$$= \left|\frac{1}{2}\left[-2\sqrt{2}y_1 + 2\sqrt{2}y_2\right]\right| = \sqrt{2}\,|y_2 - y_1| = \frac{(\sqrt{2})\sqrt{12+16}}{111}$$

$$= \sqrt{56} = 2\sqrt{14}$$

Q. 18 The number of points, where the function $f : \mathbf{R} \to \mathbf{R}, f(x) = |x-1|\cos|x-2|\sin|x-1| + (x-3)\left|x^2 - 5x + 4\right|$, is NOT differentiable, is :

Option 1:
1

Option 2:
2

Option 3:
3

Option 4:
4

Correct Answer:
1

Solution:

$$f(x) = |x-1|\cos|x-2|\sin|x-1| + (x-3)\left|x^2-5x+4\right|$$
$$= |x-1|\cos|x-2|\sin|x-1| + (x-3)|x-1|x-4|$$
$$= |x-1|\,\mathrm{Esc}\,|x-2|\sin|x-1| + (x-3)|x-4|]$$
Non differentiable at x=1 and x=4

Q. 19 Let $S = \{1, 2, 3, \ldots, 2022\}$. Then the probability, that a randomly chosen number n from the set S such that HCF $(n, 2022) = 1$, is :

Option 1:
$\frac{128}{1011}$

Option 2:
$\frac{166}{1011}$

Option 3:
$\frac{127}{337}$

Option 4:
$\frac{112}{337}$

Correct Answer:
$\frac{112}{337}$

Solution:
Total number of elements =2022

$$2022 = 2 \times 3 \times 337$$

$$\mathrm{HCF}\,(n_1 2022) = 1$$

is feasible when the value of n and 2022 has no common factor

A=Number which are divisible by 2 from {1,2,3,.....,2022}

$$n(A) = 1011$$

B=Number which are divisible by 3 from {1,2,3,.....,2022}

$$n(B) = 674$$

$A \cap B =$ Number which are tivisible by 6

from $\{1, 2, 3, \ldots 2022\}$

$6, 12, 18, \ldots .2022$

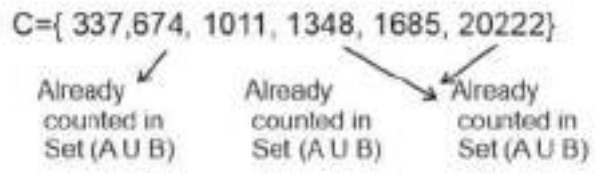
$337 = n(A \cap B)$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$= 1011 + 674 - 337$

$= 1348$

C=Number which divisible by 337 from {1,.....,1022}



Total elements which are divisible by 2 or 3 or 337\

$= 1348 + 2 = 1350$

Favourable cases = Element which are neither dinisible by 2,3 or 337

$= 2022 - 1350 = 672$

$$\text{Required probability} = \frac{672}{2022} = \frac{112}{337}$$

Q. 20 Let $f(x) = 3^{(x^2-2)^3+4}, x \in \mathbf{R}$. Then which of the following statements are true?

$P : x = 0$ is a point of local minima of f

$Q : x = \sqrt{2}$ is a point of inflection of f

$R : f'$ is increasing for $x > \sqrt{2}$

Option 1:

Only P and Q

Option 2:

Only P and R

Option 3:
Only Q and R

Option 4:
All P,Q, and R

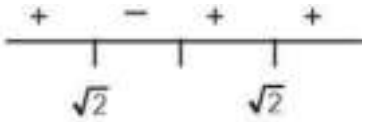
Correct Answer:
All P,Q, and R

Solution:

$$f(x) = 81 \cdot 3^{(x^2-2)^3}$$

$$f(x) = 81 \cdot 3^{(x^2-2)^3} \ln 3 \cdot 3(x^2-2)^2 \cdot 2x$$

$$= (81 \times 6)3^{(x^2-2)^3} \times (x^2-2)^2 \ln 3$$



$x = 6$ is point of local min

$$f'(x) = \underbrace{(4-6\cdot\ln 3)}_{k} \underbrace{3^{(x^2-2)^3} x (x^2-2)^2}_{g(x)}$$

$$g'(x) = 3^{(x^2-2)^3}(x^2-2)^2 + x \cdot 3^{(x^2-2)^3} \cdot 4x(x^2-2) + x \cdot (x^2-2)^2 \cdot 3^{(x^2-2)^3} \ln 3 \cdot 3(x^2-2)^2 \cdot 2x$$

$$= 3^{(x^2-2)^3}(x^2-2)\left[x^2-2+4x^2+6x^2\ln 3(x^2-2)^3\right].$$

$$g'(x) = 3^{(x^2-2)^3}(x^2-2)\left[5x^2-2+6x^2\ln 3(x^2-2)^3\right]$$

$$f''(x) = k \cdot g'(x)$$

$$f''(\sqrt{2}) = 0, f''(\sqrt{2}+) > 0, f''\left(\sqrt{2}^-\right) < 0$$

$x = \sqrt{2}$is point of infiection

$f''(x) > 0$ for $x > \sqrt{2}$ so $f'(x)$ is increasing

$$f''(\sqrt{2}) = 0, f''(\sqrt{2}+) > 0, f''\left(\sqrt{2}^-\right) < 0$$

$x = \sqrt{2}$is point of infiection

$f''(x) > 0$ for $x > \sqrt{2}$ so $f'(x)$ is increasing

Q. 21 Let $S=\left\{\theta \in(0,2 \pi): 7 \cos ^{2} \theta-3 \sin ^{2} \theta-2 \cos ^{2} 2 \theta=2\right\}$. Then, the sum of roots of all the equations $x^{2}-2\left(\tan ^{2} \theta+\cot ^{2} \theta\right) x+6 \sin ^{2} \theta=0, \theta \in S$, is_____

Correct Answer: 16

Solution:

$$7 \cos ^{2} \theta-3 \sin ^{2} \theta-2 \cos ^{2} 2 \theta=2$$

$$4 \cos ^{2} \theta+3 \cos 2 \theta-2 \cos ^{2} 2 \theta=2$$

$$2(1+\cos 2 \theta)+3 \cos 2 \theta-2 \cos ^{2} 2 \theta=2$$

$$2 \cos ^{2} 2 \theta-5 \cos 2 \theta=0$$

$$\cos 2 \theta(2 \cos 2 \theta-5)=0$$

$$2 \theta=(2 n+1) \frac{\pi}{2}$$

$$\theta=(2 n+1) \frac{\pi}{4}$$

$$S=\left\{\frac{\pi}{4}, \frac{3 \pi}{4}, \frac{5 \pi}{4}, \frac{7 \pi}{4}\right\}$$

$$x^{2}-2\left(\tan ^{2} \theta+\cot ^{2} \theta\right) x+6 \sin ^{2} \theta=0$$

$$\Rightarrow x^{2}-4 x+3=0$$

$$\text{sum of roots of all Por equations } =4 \times 4=16$$

Q. 22 Let the mean and the variance of 20 observations $x_{1}, x_{2}, \ldots, x_{20}$ be 15 and 9, respectively. For $\alpha \in \mathbf{R}$, if the mean of $\left(x_{1}+\alpha\right)^{2},\left(x_{2}+\alpha\right)^{2}, \ldots,\left(x_{20}+\alpha\right)^{2}$ is 178, then the square of the maximum value of α is equal to_____.

Correct Answer:
4

Solution:

$$\sum x_1 = 15 \times 20 = 300 \cdots (i)$$

$$\frac{\sum x_1^2}{20} - (15)^2 = 9 \cdots\cdot$$

$$\sum x_1^2 = 234 \times 20 = 4680$$

$$\frac{\sum (x_1 + \alpha)^2}{20} = 178 \Rightarrow \sum (x_1 + \alpha)^2 = 3560$$

$$\Rightarrow \sum x_1^2 + 2\alpha \sum x_1 + \sum \alpha^2 = 3560$$

$$4680 + 600\alpha + 20\alpha^2 = 3660$$

$$\Rightarrow \alpha^2 + 30\alpha + 56 = 0$$

$$\Rightarrow (\alpha + 28)(\alpha + 2) = 0$$

$$\alpha = -2, -28$$

square of maximum valu of α is 4

Q. 23 Let a line with direction ratios $a, -4\ a, -7$ be perpendicular to the lines with direction ratios $3, -1, 2\ b$ and $b, a, -2$. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane $x - y + z = 0$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to_____.

Correct Answer:
10

Solution:

$(a, -4a, -7) \perp$ to $(3, -1, 2b)$

$a = 2b$

$(a, -4a, -7) \perp$ to $(b, a, -2)$

$3a + 4a - 14b = 0$

$ab - 4a^2 + 14 = 0$

from equations (i) anal (ii)

$2b^2 - 16b^2 + 14 = 0$

$b^2 = 1$

$a^2 = 4b^2 = 4$

$\frac{x+1}{5} = \frac{y-2}{3} = \frac{z}{1} = k$

$\alpha = 5k - 1, \beta = 3k + 2, \gamma = k$

as (α, β, γ) satisfies $x - y + z = 0$

$5k - 1 - (3k + 2) + k = 0$

$k = 1$

$\therefore \alpha + \beta + \gamma = 9k + 1 = 10$

Q. 24

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\sum_{r=1}^{\infty} \frac{a_r}{2^r} = 4$, then $4a_2$ is equal to _____.

Correct Answer: 16

Solution:

$b = s = \frac{a_1}{2} + \frac{a_2}{2^2} + \frac{a_3}{2^3} + \cdots$

$\frac{s}{2} = \frac{a_4}{2^2} + \frac{a_2}{2^3} + \cdots$

$\frac{s}{2} = \frac{a_1}{2} + d\left(\frac{1}{2^2} + \frac{1}{2^3} + \cdots\right.$

$\frac{s}{2} = \frac{a_1}{2} + d\left(\frac{\frac{1}{4}}{1 - \frac{1}{2}}\right)$

$\therefore s = a_1 + d_1 = a_2 = 4$

or $4a_2 = 16$

Q. 25 Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of $\left(\sqrt[4]{2}+\frac{1}{\sqrt[4]{3}}\right)^{n}$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6}:1$. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to_____.

Correct Answer:
84

Solution:

$$\frac{T_5}{T_{n-3}}=\frac{n_{44}\left(2^{1/4}\right)^{n-4}\left(3^{-1/4}\right)^{4}}{n_{c_{n-4}}\left(2^{1/4}\right)^{4}\left(3^{-1/4}\right)^{n-4}}=\frac{\sqrt[4]{6}}{1}$$

$$\Rightarrow 2^{\frac{n-8}{4}}3^{\frac{n-8}{4}}=6^{1/4}$$

$$\Rightarrow 6^{n-8}=6$$

$$\Rightarrow n-8=1\Rightarrow n=9$$

$$T_6={}^{a}c_5\left(2^{1/4}\right)^{4}\left(3^{-1/4}\right)^{5}=\frac{84}{\sqrt[4]{3}}$$

$$\therefore \quad \alpha=84$$

Q. 26 The number of matrices of order 3×3, whose entries are either 0 or 1 and the sum of all the entries is a prime number, is_____.

Correct Answer:
282

Solution:

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{bmatrix}\ a_{1j}E\{0,1\}$$

$$\sum_{a_{ij}}=2,3,5,7$$

$$\text{Total matrix}={}^{9}C_2+{}^{9}C_3+{}^{9}C_5+{}^{9}C_7$$

$$=282$$

Q. 27 Let p and $p+2$ be prime numbers and let

$$\Delta = \begin{vmatrix} p! & (p+1)! & (p+2)! \\ (p+1)! & (p+2)! & (p+3)! \\ (p+2)! & (p+3)! & (p+4)! \end{vmatrix}$$

Then the sum of the maximum values of α and β, such that p^{α} and $(p+2)^{\beta}$ divide Δ, is_____

Correct Answer:
4

Solution:

$$\Delta = \begin{vmatrix} P! & (P+1)! & (P+2)! \\ (P+1)! & (P+2)! & (P+3)! \\ (P+2)! & (P+3)! & (P+4)! \end{vmatrix}$$

$$\Delta = P)(P+1)!(P+2)! \begin{vmatrix} 1 & 1 & 1 \\ P+1 & P+2 & P+3 \\ (P+2)(P+1) & (P+3)(P+2) & (P+4)(P+3) \end{vmatrix}$$

$$\Delta = 2P!(P+1)!(P+2)!$$

which is divisible by $p^{\alpha}\&(p+2)^{\beta}$

$\therefore \alpha = 3, \beta = 1$

Q. 28 If $\frac{1}{2\times3\times4} + \frac{1}{3\times4\times5} + \frac{1}{4\times5\times6} + \ldots + \frac{1}{100\times101\times102} = \frac{k}{101}$, then $34k$ is equal to_____

Correct Answer: 286

Solution:

$$\frac{1}{2\cdot3\cdot4} + \frac{1}{3\cdot4\cdot5} + \cdots + \frac{1}{100\cdot101\cdot102} = \frac{k}{101}$$

$$\frac{4-2}{2\cdot3\cdot4} + \frac{5-3}{3\cdot4\cdot5} + \cdots + \frac{102-100}{100\cdot101\cdot102} = \frac{2k}{101}$$

$$\frac{1}{2.3} - \frac{1}{3\cdot4} + \frac{1}{3\cdot4} - \frac{1}{4\cdot5} + \cdots \frac{1}{100\cdot101} - \frac{1}{101\cdot102} = \frac{2k}{101}$$

$$\frac{1}{2\cdot3} - \frac{1}{101\cdot102} = \frac{2k}{101}$$

$$\therefore 2k = \frac{101}{6} - \frac{1}{102}$$

$$\therefore 34k = 286$$

Q. 29 Let $S = \{4, 6, 9\}$ and $T = \{9, 10, 11, \ldots, 1000\}$. If $A = \{a_1 + a_2 + \ldots + a_k : k \in \mathbf{N}, a_1, a_2, a_3, \ldots, a_k \in S\}$, then the sum of all the elements in the set $T - A$ is equal to_____

Correct Answer:
11

Solution:
$S = \{4, 6, 9\} \quad T = \{9, 10, 11, \cdots 1000\}$
$A\{a_1 + a_2 + \cdots + a_k : K \in N\} \ \& \ a_i E \cdot S$
Here by the difinition of Set ' A'
$A = \{a : a = 4x + 64 + 9z\}$
Except the clement 11 , every element of set T is of the form $4x + 6y + 9z$ for some $x, y, z \in W$
$\therefore T - A = \{11\}$

Q. 30 Let the mirror image of a circle $c_1 : x^2 + y^2 - 2x - 6y + \alpha = 0$ in line $y = x + 1$ be $c_2 : 5x^2 + 5y^2 + 10gx + 10fy + 38 = 0$. If r is the radius of circle C_2, then $\alpha + 6r^2$ is equal to_____.

Correct Answer: 12

Solution:
Image of centre $G \equiv (1, 3)$ in $x - y + 1 = 0$ is given by

$$\frac{x_1 - 1}{1} = \frac{y_1 - 3}{-1} = \frac{-2(1 - 3 + 1)}{1^2 + 1^2}$$

$$\Rightarrow x_1 = 2, y_1 = 2$$

$$\therefore \text{ Centre of circle } c_2 \equiv (2, 2)$$

$$\therefore \text{ Equation of } c_2 \text{ be } x^2 + y^2 - 4x + 4y + \frac{38}{5} = 0$$

$$\text{Now radius of } c_2 \text{ is } \sqrt{4 + 4 - \frac{38}{5}} = \sqrt{\frac{2}{5}} = r$$

$$(\text{ radius of } 4_1)^2 = (\text{ radius of } c_2)^2$$

$$\Rightarrow 10 - \alpha = \frac{2}{5} \Rightarrow \alpha = \frac{48}{5}$$

$$\therefore \alpha + 6\gamma^2 = \frac{48}{5} + \frac{12}{5} = 12$$

29 July - Evening Shift

Physics

Q. 1 Two identical metallic spheres A and B when placed at certain distance in air repel each other with a force of F. Another identical uncharged sphere C is first placed in contact with A and then in contact with B and finally placed at midpoint between spheres A and B. The force experienced by sphere C will be:

Option 1:
$3\ \mathrm{F}/2$

Option 2:
$3\ \mathrm{F}/4$

Option 3:
F

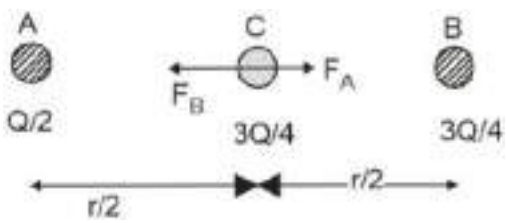
Option 4:
$2\ \mathrm{F}$

Correct Answer:
$3\ \mathrm{F}/4$

Solution:
Condition 1:

$$\mathrm{F} = \frac{\mathrm{KQ}^2}{\mathrm{r}^2}$$

Condition 2:



Force experienced by c is

$$F' = F_B - F_A$$

$$= \frac{K(3Q/4)^2}{(r/2)^2} - \frac{K\left(\frac{3Q}{4}\right)(Q/2)}{(r/2)^2}$$

$$= \frac{9KQ^2}{4r^2} - \frac{3KQ^2}{2r^2}$$

$$F' = \frac{3KQ^2}{4r^2} = \frac{3}{4}F$$

Hence 2 is correct option

Q. 2 Match List I with List II.

List-I	List-II
A. Torque	I. Nms^{-1}
B. Stress	II. Jkg^{-1}
C. Latent Heat	III. Nm
D. Power	IV. Nm^{-2}

Choose the correct answer from the options given below:

Option 1:
A-III, B-II, C-I, D-IV

Option 2:
A-III, B-IV, C-II, D-I

Option 3:
A-IV, B-I, C-III, D-II

Option 4:
A-II, B-III, C-I, D-IV

Correct Answer:
A-III, B-IV, C-II, D-I

Solution:

$\text{Torque} \rightarrow \text{Nm}$

$\text{Stress} \rightarrow \text{N/m}^2$

$\text{Latent heat} \rightarrow \text{J/kg}$

$\text{Power} \longrightarrow \text{Nms}^{-1}$

Hence 2 is correct option

Q. 3 Two identical thin metal plates has charge q_1 and q_2 respectively such that $q_1 > q_2$. The plates were brought close to each other to form a parallel plate capacitor of capacitance C. The potential difference between them is :

Option 1:
$\frac{(q_1 + q_2)}{C}$

Option 2:
$\frac{(q_1 - q_2)}{C}$

Option 3:
$\frac{(q_1 - q_2)}{2C}$

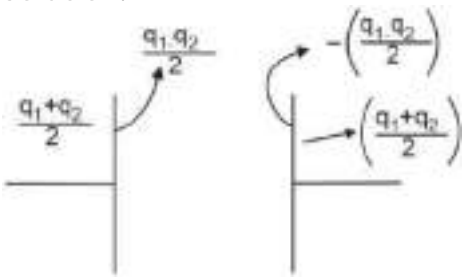
Option 4:

$\frac{2\left(q_1-q_2\right)}{C}$

Correct Answer:

$\frac{\left(q_1-q_2\right)}{2C}$

Solution:



$q = C(\Delta v)$

$\Delta v = \frac{q}{c} = \frac{(q_1-q_2)}{2c}$

The inner surface charge is taken into consideration as outer surface charges has no contribution in the potential difference between the plates of capacitor

Hence 3 is correct option

Q. 4 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Alloys such as constantan and manganin are used in making standard resistance coils.
Reason R: Constantan and manganin have very small value of temperature coefficient of resistance.
In the light of the above statements, choose the correct answer from the options given below.

Option 1:
Both A and R are true and R is the correct explanation of A.

Option 2:
Both A and R are true but R is NOT the correct explanation of A.

Option 3:
A is true but R is false.

Option 4:
A is false but R is true.

Correct Answer:
Both A and R are true and R is the correct explanation of A.

Solution:
$\rightarrow$ As constantan and manganin has very small value of temperature coefficient of resistance they are used in making standard resistance coils.

$\rightarrow$ Both $\mathrm{A\ and\ R}$ are true and R is the correct explanation of A

Hence 1 is correct option

Q. 5 A $1\ \mathrm{m}$ long wire is broken into two unequal parts $\mathrm{X\ and\ Y}$. The X part of the wire is streched into another wire W. Length of W is twice the length of X and the resistance of W is twice that of Y. Find the ratio of length of $\mathrm{X\ and\ Y}$.

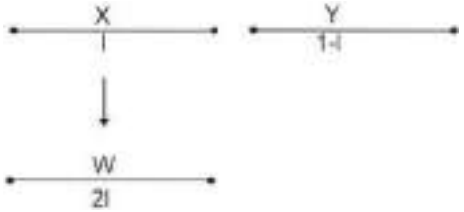
Option 1:
$1:4$

Option 2:
$1:2$

Option 3:
$4:1$

Option 4:
$2:1$

Correct Answer:
$1:2$

Solution:


$R_w = 2R_Y$ (Given)

$$\frac{R_x}{R_Y} = \frac{\ell}{1-\ell} \rightarrow (1)$$

$$\frac{R_W}{R_X} = \frac{(2\ell)^2}{(\ell)^2} = 4 \rightarrow (2)$$

From the above equations

$R_W = 4R_X = 2R_Y$

$$\frac{R_X}{R_Y} = \frac{1}{2} = \frac{\ell}{1-\ell}$$

$1 - \ell = 2\ell$

$3\ell = 1$

$$\ell = \frac{1}{3}$$

$$\frac{\ell_x}{\ell_Y} = \frac{\ell}{1-\ell} = \frac{1/3}{2/3} = \frac{1}{2}$$

Hence 2 is correct option

Q. 6 A wire X of length $50\ \text{cm}$ carrying a current of $2\ \text{A}$ is placed parallel to a long wire Y of length $5\ \text{m}$. The wire Y carries a current of $3\ \text{A}$. The distance between two wires is $5\ \text{cm}$ and currents flow in the same direction. The force acting on the wire Y is

Option 1:

$1.2 \times 10^{-5}\ \text{N}$ directed towards wire X

Option 2:

1.2×10^{-4} N directed away from wire X.

Option 3:

1.2×10^{-4} N directed towards wire X.

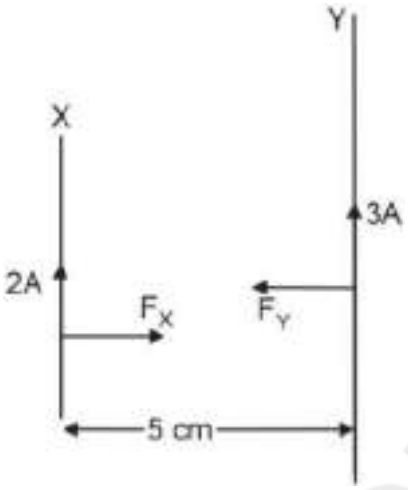
Option 4:

2.4×10^{-5} N directed towards wire X

Correct Answer:

1.2×10^{-5} N directed towards wire X

Solution:



$\ell_x = 50\text{cm}$

$\ell_Y = 500\text{cm} = 5\text{m}$

The force acting on the wire Y is equal and opposite to force acting on wire x

$\text{i.e}, F_X = F_Y$

$F_X = B\, I\, \ell_x$

$F_X = \left(\frac{\mu_0 i_y}{2\pi r}\right) I_x\, \ell_x$

$= \frac{2 \times 10^{-7} \times 3 \times 2 \times 0.5}{5 \times 10^{-2}}$

$F_Y = F_X = 1.2 \times 10^{-5}$ N

Like current carrying conductor attract each other

Hence 1 is correct option

Q. 7 A juggler throws balls vertically upwards with same initial velocity in air. When the first ball reaches its highest position, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is

Option 1:
$g/2n$

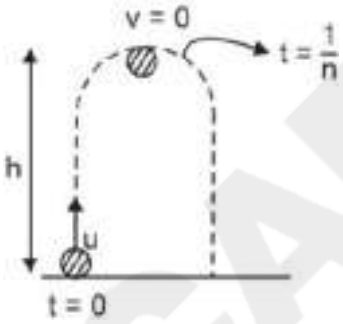
Option 2:
g/n

Option 3:
$2\,gn$

Option 4:
$g/2n^2$

Correct Answer:
$g/2n^2$

Solution:



$$V = u + at$$

If n balls are thrown in 1 second then time between two consecutive ball is $\frac{1}{n}$ i.e time taken by the first ball when it reaches the highest position is $\frac{1}{n}s$

$$0 = u + (-g) + \left(\frac{1}{n}\right)$$

$$u = \frac{9}{n} \rightarrow (1)$$

$$v^2 = u^2 + 2as$$

$$0 = \frac{g^2}{n^2} + 2(-g)h; h = \frac{9}{2n^2}$$

Hence 4 is correct option.

Q. 8 A circuit element X when connected to an a.c. supply of peak voltage $100\ V$ gives a peak current of $5\ A$ which is in phase with the voltage. A second element Y when connected to the same a.c. supply also gives the same value of peak current which lags behind the voltage by $\frac{\pi}{2}$.If X and Y are connected in series to the same supply, what will be the rms value of the current in ampere?

Option 1:
$\frac{10}{\sqrt{2}}$

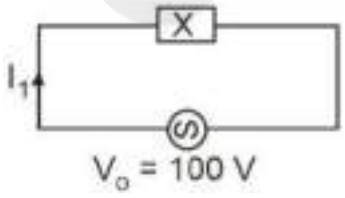
Option 2:
$\frac{5}{\sqrt{2}}$

Option 3:
$5\sqrt{2}$

Option 4:
$\frac{5}{2}$

Correct Answer:
$\frac{5}{2}$

Solution:



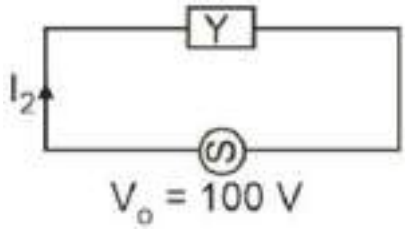
$$I_1 = 5A = \frac{100v}{R}$$

$$R = 20\Omega$$

$I_1 \rightarrow$ Peak current

Current and voltage are in same phase

$\therefore$ Element X is resistor



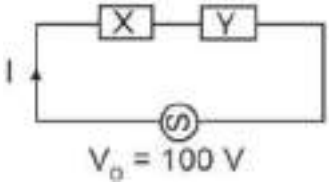
Current lags behind the voltage by $\pi/2$

Element Y$\rightarrow$ Inductor

$I_2 = 5A$

$I_2 \rightarrow$peak current

$X_L = 20\Omega$



$I \rightarrow$ Peak current

$$I = \frac{V_0}{\sqrt{R^2 + x_L^2}} = \frac{100}{20\sqrt{2}} = \frac{5}{\sqrt{2}}$$

$$I_{rms} = \frac{I}{\sqrt{2}} = \frac{5}{2}\ A$$

Hence 4 is correct option.

Q. 9 An unpolarised light beam of intensity $2I_0$ is passed through a polaroid P and then through another polaroid Q which is oriented in such a way that its passing axis makes an angle of 30° relative to that of P. The intensity of the emergent light is

Option 1:

$\frac{I_0}{4}$

Option 2:

$\frac{I_0}{2}$

Option 3:

$\frac{3I_0}{4}$

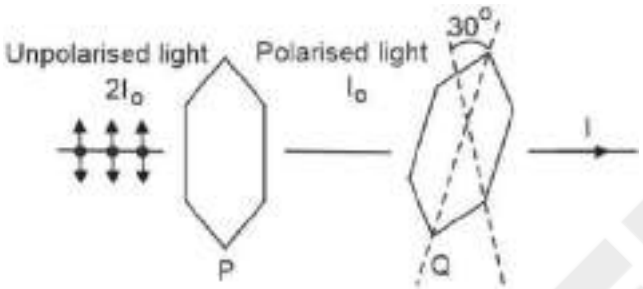
Option 4:

$\frac{3I_0}{2}$

Correct Answer:

$\frac{3I_0}{4}$

Solution:



By Malus Law,

$$I = I_o \cos^2 \Theta$$

$$= I_o (\cos 30^\circ)^2$$

$$I = \frac{3I_o}{4}$$

Hence 3 is correct option

Q. 10 An α particle and a proton are accelerated from rest through the same potential difference. The ratio of linear momenta acquired by above two particals will be:

Option 1:

$\sqrt{2} : 1$

Option 2:

$2\sqrt{2} : 1$

Option 3:

$4\sqrt{2}:1$

Option 4:

$8:1$

Correct Answer:

$2\sqrt{2}:1$

Solution:

$q_\alpha = 2e, m_\alpha = 4m_p$

$q_p = e$

$P = \sqrt{2m\,(kE)} = \sqrt{2mqv}$

$$\frac{P_\alpha}{P_p} = \sqrt{\frac{2m_\alpha q_\alpha V}{2m_p q_p V}}$$

$$\frac{p_\alpha}{p_p} = \sqrt{4 \times 2} = 2\sqrt{2}$$

Hence 2 is correct option

Q. 11 Read the following statements:
(A) Volume of the nucleus is directly proportional to the mass number.
(B) Volume of the nucleus is independent of mass number.
(C) Density of the nucleus is directly proportional to the mass number.
(D) Density of the nucleus is directly proportional to the cube root of the mass number.
(E) Density of the nucleus is independent of the mass number.
Choose the correct option from the following options.

Option 1:

(A) and (D) only.

Option 2:
(A) and (E) only.

Option 3:
(B) and (E) only.

Option 4:
(A) and (C) only

Correct Answer:
(A) and (E) only.

Solution:
$\rightarrow$Radius of nucleus is,

$$\mathrm{R} = \mathrm{R}_0 \mathrm{A}^{1/3}$$

$\rightarrow$Volume of nucleus $\propto \mathrm{R}^3 \propto \mathrm{A}$

$\rightarrow$ Density of nucleus is constant for all elements.

$\rightarrow$ A & E as correct statements

Hence 2 is correct option

Q. 12 An object of mass $1\ \mathrm{kg}$ is taken to a height from the surface of earth which is equal to three times the radius of earth. The gain in potential energy of the object will be [If, $\mathrm{g} = 10\ \mathrm{ms}^{-2}$ and radius of earth $= 6400\ \mathrm{km}$]

Option 1:
$48\mathrm{MJ}$

Option 2:
$24\mathrm{MJ}$

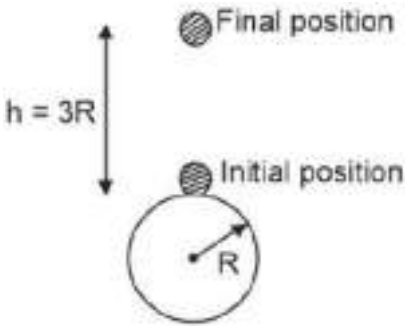
Option 3:
$36\mathrm{MJ}$

Option 4:
$12\mathrm{MJ}$

Correct Answer:

$48\mathrm{MJ}$

Solution:



$$\text{Initial P.E.} = \mathrm{PE_i} = -\frac{\mathrm{GMm}}{\mathrm{R}}$$

$$\text{Final PE.} = \mathrm{PE_f} = \frac{-\mathrm{GMm}}{(\mathrm{R}+\mathrm{h})} = \frac{-\mathrm{GMm}}{4\mathrm{R}}$$

$$\Delta\mathrm{PE} = \mathrm{PE_f} - \mathrm{PE_i}$$

$$= \frac{3}{4}\frac{\mathrm{GMm}}{\mathrm{R}}$$

$$= \frac{3\mathrm{mgR}}{4} = \frac{3}{4} \times 1 \times 10 \times 6400 \times 10^3$$

$$\triangle\mathrm{PE} = 48\mathrm{MJ}$$

Q. 13 A ball is released from a height h. If $\mathrm{t_1}$ and $\mathrm{t_2}$ be the time required to complete the first half and second half of the distance respectively. Then, choose the correct relation between $\mathrm{t_1}$ and $\mathrm{t_2}$.

Option 1:

$\mathrm{t_1} = (\sqrt{2})\mathrm{t_2}$

Option 2:

$\mathrm{t_1} = (\sqrt{2}-1)\mathrm{t_2}$

Option 3:

$t_2 = (\sqrt{2}+1)t_1$

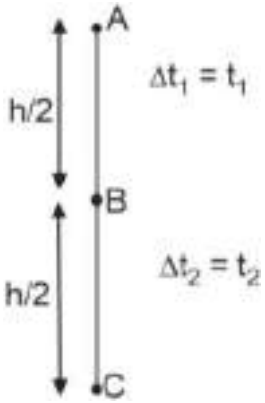
Option 4:

$t_2 = (\sqrt{2}-1)t_1$

Correct Answer:

$t_2 = (\sqrt{2}-1)t_1$

Solution:



From A to B

$$S = ut + \frac{1}{2}at^2$$

$$-\frac{h}{2} = 0 + \frac{1}{2}(-g)t_1^2$$

$$t_1 = \sqrt{\frac{h}{g}} \rightarrow (1)$$

From A to C

$$s = ut + \frac{1}{2}at^2$$

$$-h = 0 + \frac{1}{2}(-g)\,(t_1+t_2)^2$$

$$t_1 + t_2 = \sqrt{\frac{2h}{g}} \rightarrow (2)$$

From eq (1) and (2)

$$t_2 = \sqrt{\frac{2h}{g}} - \sqrt{\frac{h}{g}} \rightarrow (3)$$

$$\frac{t_1}{t_2} = \frac{1}{\sqrt{2}-1}$$

$$t_2 = (\sqrt{2}-1)t_1$$

Q. 14 Two bodies of masses $m_1 = 5\ kg$ and $m_2 = 3\ kg$ are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be : [Take $g = 10\ ms^{-2}$]

Option 1:
30 N

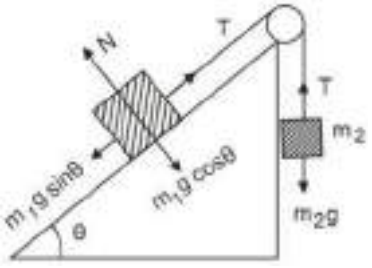
Option 2:
40 N

Option 3:
50 N

Option 4:
60 N

Correct Answer:
40 N

Solution:



For a system at rest

$m_1 g \cos\theta = N \rightarrow (1)$

($N \rightarrow$ Force by inclined plane on the block)

$T = m_2 g = 30N \rightarrow (2)$

$m_1 g \sin\theta = T$

$50 \sin\theta = 30$

$\sin\theta = \frac{3}{5}$

$\theta = 37^\circ$

$N = m_1 g \cos\theta$

$= 50 \times \cos 37^\circ$

$N = 40N$

The force exerted by the inclined plane on the body of mass m_1 will be $40\ N$

Q. 15 If momentum of a body is increased by 20%, then its kinetic energy increases by

Option 1:
36%

Option 2:
40%

Option 3:
44%

Option 4:
48%

Correct Answer:
44%

Solution:
$\frac{\Delta p}{p} \times 100 = 20\%$

$p_i = p$

$p_f = 1.2p$

$KE_i = \frac{p_i^2}{2m}$

$KE_f = \frac{p_f^2}{2m}$

$\frac{KE_F}{KE_i} = \frac{p_f^2}{p_i^2} = \frac{1.44p^2}{p^2}$

$\Delta KE = KE_f - KE_i$

$= 0.44\, KE_i$

$\frac{\Delta KE}{KE_i} \times 100 = 44\%$

The kinetic energy increases by 44%

Q. 16 The torque of a force $5i + 3\hat{j} - 7\hat{k}$ about the origin is τ. If the force acts on a particle whose position vector is $2\hat{i} + 2\hat{j} + \hat{k}$, then the value of τ will be

Option 1:
$11\hat{i} + 19\hat{j} - 4\hat{k}$

Option 2:
$-11\hat{i} + 9\hat{j} - 16\hat{k}$

Option 3:
$-17\hat{i} + 19\hat{j} - 4\hat{k}$

Option 4:

$17\hat{i}+9\hat{j}+16\hat{k}$

Correct Answer:

$-17\hat{i}+19\hat{j}-4\hat{k}$

Solution:

$\vec{F}=5\hat{i}+3\hat{j}-7\hat{k}$

$\vec{r}=2\hat{i}+2\hat{j}+\hat{k}$

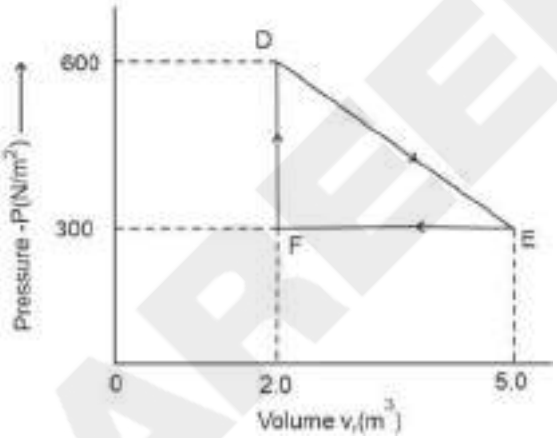
$\vec{\tau}=\vec{r}\times\vec{F}$

$=(2\hat{i}+2\hat{j}+\hat{k})\times(5\hat{i}+3\hat{j}-7\hat{k})$

$=(6\hat{k}+14\hat{j}-10\hat{k}-14\hat{i}+5\hat{j}-3\hat{i})$

$\vec{\tau}=(-17\hat{i}+19\hat{j}-4\hat{k})$

Q. 17 A thermodynamic system is taken from an original state D to an intermediate state E by the linear process shown in the figure. Its volume is then reduced to the original volume from E to F by an isobaric process. The total work done by the gas from D to E to F will be



Option 1:

$-450\ \text{J}$

Option 2:

$450\ \text{J}$

Option 3:

$900\ \text{J}$

Option 4:
$1350\ \text{J}$

Correct Answer:
$450\ \text{J}$

Solution:
In process DEF

$$\text{W} = (\text{Area of loop})$$

$$= \frac{1}{2} \times 3 \times 300$$

$$\text{W} = 450\ \text{J}$$

Hence 2 is correct option

Q. 18 The vertical component of the earth's magnetic field is $6 \times 10^{-5}\ \text{T}$ at any place where the angle of dip is 37°. The earth's resultant magnetic field at that place will be (Given $\tan 37^{\circ} = \frac{3}{4}$)

Option 1:
$8 \times 10^{-5}\ \text{T}$

Option 2:
$6 \times 10^{-5}\ \text{T}$

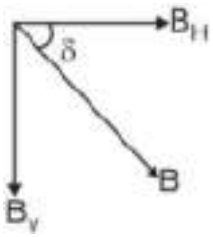
Option 3:
$5 \times 10^{-4}\ \text{T}$

Option 4:
$1 \times 10^{-4}\ \text{T}$

Correct Answer:
$1 \times 10^{-4}\ \text{T}$

Solution:
$\delta = 37^{\circ}$

$$\text{B}_{\text{V}} = 6 \times 10^{-5}\ \text{T}$$



$$\sin\delta = \frac{B_V}{B}$$

$$B = \frac{B_V}{\sin\delta} = \frac{6\times10^{-5}}{3/5}$$

$$B = 1\times10^{-4}T$$

Hence 4 is correct option.

Q. 19 The root mean square speed of smoke particles of mass 5×10^{-17} kg in their Brownian motion in air at NTP is approximately. [Given $k = 1.38\times10^{-23}JK^{-1}$]

Option 1:
$60\ mm\ s^{-1}$

Option 2:
$12\ mm\ s^{-1}$

Option 3:
$15\ mm\ s^{-1}$

Option 4:
$36\ mm\ s^{-1}$

Correct Answer:
$15\ mm\ s^{-1}$

Solution:
$m = 5\times10^{-17}$ kg

$k_1 = 1.38\times10^{-23}$ J/k

$$\frac{1}{2}mv_{rms}^2 = \frac{3}{2}kT$$

$$v_{rms} = \sqrt{\frac{3kT}{m}}$$

$$= \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 293}{5 \times 10^{-17}}}$$

$$\cong 15 \text{ mm/s}$$

Hence 3 is correct option

Q. 20 Light enters from air into a given medium at an angle of $45°$ with interface of the air-medium surface. After refraction, the light ray is deviated through an angle of $15°$ from its original direction. The refractive index of the medium is :

Option 1:
1.732

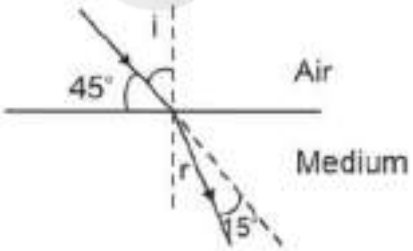
Option 2:
1.333

Option 3:
1.414

Option 4:
2.732

Correct Answer:
1.414

Solution:



$i = 45°$

$r = 30^\circ$

$\mu_1 \sin i = \mu_2 \sin r$

$1\,(\sin 45^\circ) = \mu \times \sin 30^\circ$

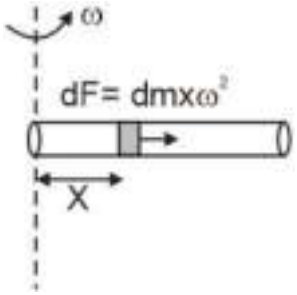
$\mu = \sqrt{2} = 1.414$

Hence 3 is correct option

Q. 21 A tube of length $50\ \text{cm}$ is filled completely with an incompressible liquid of mass $250\ \text{g}$ and closed at both ends. The tube is then rotated in horizontal plane about one of its ends with a uniform angular velocity $x\sqrt{F}\ \text{rad}\ s^{-1}$. If F be the force exerted by the liquid at the other end then the value of x will be_____.

Correct Answer:
4

Solution:



$$F = \int dF = \int dnx\omega^2$$

$$= \int \left(\frac{M}{L}dx\right) x\omega^2$$

$$= \frac{M\omega^2}{L}\int_0^L xdx$$

$$F = \frac{ML\omega^2}{2}$$

$$F = \frac{(0.25) \times (0.5) \times (x^2F)}{2}$$

$$16 = x^2$$

$$x = 4$$

The value of x is 4

Q. 22 Nearly 10% of the power of a $110\ \mathrm{W}$ light bulb is converted to visible radiation. The change in average intensities of visible radiation, at a distance of $1\ \mathrm{m}$ from the bulb to a distance of $5\ \mathrm{m}$ is $a \times 10^{-2}\ \mathrm{W/m^2}$. The value of 'a' will be_____.

Correct Answer:
84

Solution:

$$\mathrm{P_{delivered}} = \eta\, \mathrm{P_{soura}}$$

$$= \frac{1}{10} \times 110 = 11\mathrm{W}$$

$$\mathrm{Intensity} = \frac{\mathrm{P_{delivered}}}{4\pi \mathrm{r}^2}$$

$$\mathrm{At\ r = 1m, I_1} = \frac{11}{4\pi\,(1)^2}$$

$$\mathrm{I_1} = \frac{11}{4 \times \frac{22}{7}} = \frac{7}{8}\ \mathrm{W/m^2}$$

$$\mathrm{At\ r = 5\ m,}$$

$$\mathrm{I_2} = \frac{11}{4\pi(5)^2} = \frac{11}{4 \times \frac{22}{7} \times 25}$$

$$\mathrm{I_2} = \frac{7}{2} \times 10^{-2}\ \mathrm{W/m^2}$$

$$\mathrm{I_1 - I_2} = \frac{7}{8} - \frac{7}{2} \times 10^{-2}$$

$$= \left(\frac{700}{8} - \frac{28}{8}\right) \times 10^{-2}$$

$$= \frac{672}{8} \times 10^{-2}$$

$$\mathrm{I_1 - I_2} = 84 \times 10^{-2}\ \mathrm{W/m^2}$$

The value of a is 84

Q. 23 A metal wire of length $0.5\ \mathrm{m}$ and cross-sectional area $10^{-4}\ \mathrm{m}^2$ has breaking stress $5 \times 10^8 \mathrm{Nm}^{-2}$. A block of is attached at one end of the string and is rotating in a horizontal circle. The maximum linear velocity of block will be_____ ms^{-1}

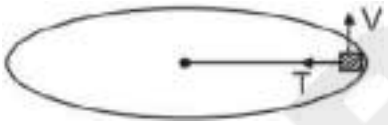
Correct Answer:
50

Solution:

$\ell = 0.5\mathrm{m}$

$\mathrm{A} = 10^{-4}\mathrm{m}^2$

$\mathrm{Stress} = 5 \times 10^8$



$$\mathrm{T} = \frac{\mathrm{mv}^2}{\ell}$$

$$\mathrm{Stress} = \frac{\mathrm{I}}{\mathrm{A}} = \frac{\mathrm{mv}^2}{\ell \mathrm{A}}$$

$$5 \times 10^8 = \frac{10\mathrm{v}^2}{(0.5) \times 10^{-4}}$$

$$2.5 \times 10^3 = \mathrm{V}^2$$

$$\mathrm{V}^2 = 2500$$

$$\mathrm{V} = 50\ \mathrm{m/s}$$

The maximum linear velocity will be $50\ \mathrm{m/s}$

Q. 24 The velocity of a small ball of mass $0.3\ \mathrm{g}$ and density $8\ \mathrm{g/cc}$ when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $1.3\ \mathrm{g/cc}$, then the value of viscous force acting on the ball will be $\mathrm{x} \times 10^{-4}\ \mathrm{N}$, The value of x is_____. [use $\mathrm{g} = 10\ \mathrm{m/s^2}$]

Correct Answer:
25

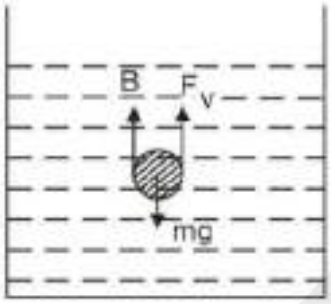
Solution:

$$\mathrm{m} = 0.3\mathrm{g} = 3 \times 10^{-4}\mathrm{kg}$$

$$\mathrm{d} = 8\frac{\mathrm{g}}{\mathrm{cc}} = 8000\frac{\mathrm{kg}}{\mathrm{m}^3}$$

$$\text{Volume of ball} = \frac{\mathrm{m}}{\mathrm{d}} = \frac{3 \times 10^{-4}}{8 \times 10^3} = \frac{3}{8} \times 10^{-7}$$

$$\mathrm{V} = \frac{4}{3}\pi \mathrm{r}^3 = \frac{3}{8} \times 10^{-7}$$



At terminal velocity,

$$\mathrm{B} + \mathrm{F_v} = \mathrm{mg}$$

$$\mathrm{F_v} = \mathrm{mg} - \rho \mathrm{vg}$$

$$= 3 \times 10^{-3} - 1300 \times \frac{3}{8} \times 10^{-7} \times 10$$

$$= 3 \times 10^{-3} - \frac{39}{8} \times 10^{-4}$$

$$= (3 - 0.4875) \times 10^{-3}$$

$$= 2.5125 \times 10^{-3}$$

$$\mathrm{F_v} = 25.125 \times 10^{-4}$$

$x \simeq 25$

Q. 25 A modulating signal $2\sin\left(6.28 \times 10^{\circ}\right)t$ is added to the carrier signal $4\sin\left(12.56 \times 10^{9}\right)t$ for amplitude modulation. The combined signal is passed through a non-linear square law device. The output is then passed through a band pass filter. The bandwidth of the output signal of band pass filter will be_____MHz.

Correct Answer:
2

Solution:

$\omega_m = 6.28 \times 10^6 = 2\pi f_m$

$f_m = 10^6 \text{ Hz}$

$\omega_c = 12.56 \times 10^9 = 2\pi f_c$

$f_c = 2 \times 10^9 \text{ Hz}$

$= 2000 \times 10^6 \text{ Hz}$

Bandwidth of the output signal

$= (f_c + f_m) - (f_c - f_m) = 2f_m$

$= 2 \times 10^6 H_2$

$= 2MH_z$

Q. 26 The speed of a transverse wave passing through a string of length 50 cm and mass 10 g is 60 ms^{-1}. The area of cross-section of the wire is 2.0 mm^2 and its Young's modulus is $1.2 \times 10^{11} \text{Nm}^{-2}$. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5} \text{ m}$. The value of x is_____.

Correct Answer:
15

Solution:

$\ell = 0.5 \text{ m}$

$\text{m} = 10\text{g} = 10^{-2} \text{ kg}$

$\text{V} = 60\frac{\text{m}}{\text{s}} = \sqrt{\frac{\text{T}}{\mu}}$

$\text{A} = 2 \times 10^{-6} \text{ m}^2$

$\text{Y} = 1.2 \times 10^{11} \text{ N/m}^2$

$\Delta\ell = \text{x} \times 10^{-5} \text{ m}$

$\mu\text{V}^2 = \text{T}$

$\left(\frac{10^{-2}}{0.5}\right)(60)^2 = \text{T}$

$\frac{36}{0.5} = \text{T}$

$\text{T} = 72\text{H}$

$\text{Y} = \frac{\text{F}}{\text{A}} \times \frac{\ell}{\Delta\ell}$

$1.2 \times 10^{11} = \frac{72}{2 \times 10^{-6}} \times \frac{0.5}{\text{x} \times 10^{-5}}$

$\text{x} = \frac{18}{1.2}$

$\text{x} = 15$

CAREERS360

Q. 27 The metallic bob of simple pendulum has the relative density 5 . The time period of this pendulum is $10\ \mathrm{s}$. If the metallic bob is immersed in water, then the new time period becomes $5\sqrt{\mathrm{x}}\ \mathrm{s}$. The value of x will be_____.

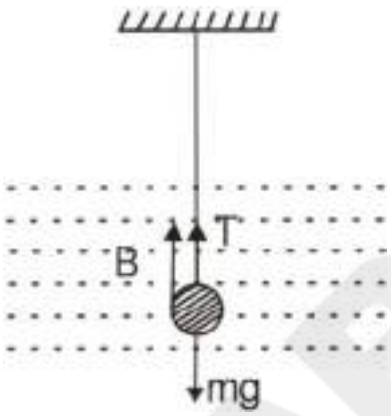
Correct Answer:
5

Solution:

$$\mathrm{T} = 10\mathrm{s}$$

$$\rho_{\mathrm{bob}} = 5$$

$$\mathrm{T} = 2\pi\sqrt{\frac{\ell}{\mathrm{g}}} \rightarrow (1)$$



$$\mathrm{T_{eff}} = \mathrm{mg} - \mathrm{B}$$

$$= \rho_{\mathrm{b0b}}\mathrm{vg} - \rho_{\mathrm{l}}\mathrm{vg}$$

$$\mathrm{T_{eff}} = (\rho_{\mathrm{bob}} - \rho_{\ell})\,\mathrm{vg}$$

$$\mathrm{T'} = 2\pi\sqrt{\frac{\mathrm{m}\ell}{\mathrm{T_{eff}}}}$$

$$\mathrm{T'} = 2\pi\sqrt{\frac{\rho_{\mathrm{b0b}}\mathrm{v}\ell}{(\rho_{\mathrm{bob}} - \rho_{\ell})\,\mathrm{vg}}}$$

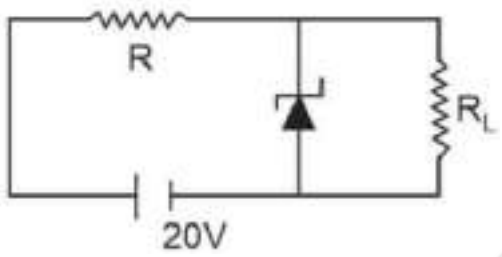
$$T' = 2\pi\sqrt{\frac{5}{(5-1)}} \times \sqrt{\frac{\ell}{g}}$$

$$= \frac{10}{2} \times \sqrt{5}$$

$$T' = 5\sqrt{5}\ s$$

$$x = 5$$

Q. 28 A $8\ V$ Zener diode along with a series resistance R is connected across a $20\ V$ supply (as shown in the figure). If the maximum Zener current is $25\ mA$, then the minimum value of R will be_____ Ω.



Correct Answer:
480

Solution:
$I_2 = 25 \times 10^{-3}\ A$

$I_R = 12\ V$

For $(I_2)_{max}$, I_c will be minimum and I is maximum

$$IR = 25 \times 10^{-3}R = 12$$

$$R = \frac{12000}{25} = 480\Omega$$

The minimum value of R will be will be $480\ \Omega$

Q. 29 Two radioactive materials A and B have decay constants 25λ and 16λ respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of B to that of A will be "e" after a time $\frac{1}{a\lambda}$. The value of a is_____.

Correct Answer:
9

Solution:

$\lambda_A = 25\lambda$

$\lambda_B = 16\lambda$

$\frac{N_B}{N_A} = \frac{N_0 e^{-\lambda_B t}}{N_0 e^{-\lambda_A t}} = e$

$e^{(\lambda_A - \lambda_B)t} = e$

but $t = \frac{1}{a\lambda}$(Given)

$(\lambda_A - \lambda_B)\, t = 1$

$t = \frac{1}{(\lambda_A - \lambda_B)} = \frac{1}{a\lambda}$

$\lambda_A - \lambda_B = a\lambda$

$a\lambda = a\lambda$

$a = 9$

Q. 30 A capacitor of capacitance $500\mu F$ is charged completely using a dc supply of $100V$. It is now connected to an inductor of inductance $50mH$ to form an LC circuit. The maximum current in LC circuit will be_____A.

Correct Answer:
10

Solution:

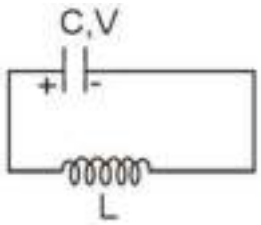
$C = 500\mu F$

$v = 100v$

$L = 50mH$

The potential drop capacitor at steady state is $100v$

$t = 0$



$(U_{max})_{capacitor} = (U_{max})_{Inductor}$

$\frac{1}{2}CV^2 = \frac{1}{2}LI_0^2$

$I_0 = \sqrt{\frac{C}{L}}V$

$\text{Maximum current} = I_0 = \sqrt{\frac{500 \times 10^{-6}}{50 \times 10^{-3}}} \times 100$

$I_0 = 10\ A$

Chemistry

Q. 1 Consider the reaction $4HNO_3(l) + 3KCl(s) \rightarrow Cl_2(g) + NOCl(g) + 2H_2O(g) + 3KNO_3(s)$ The amount of HNO_3 required to produce $110.0\ g$ of KNO_3 is
(Given: Atomic masses of H, O, N and K are $1, 16, 14$ and 39, respectively.)

Option 1:
$32.2\ g$

Option 2:
$69.4\ g$

Option 3:
$91.5\ g$

Option 4:
$162.5\ \text{g}$

Correct Answer:
$91.5\ \text{g}$

Solution:
Given Reaction is

$$\underset{\text{xg}}{4HNO_3(l)} + 3KCl(s) \longrightarrow Cl_2(g) + NOCl(g) + 2H_2O(g) + \underset{\text{110g}}{3KNO_3(s)}$$

According to Stoichiometry

$$\frac{m_{HNO_3}}{4} = \frac{m_{KNO_3}}{3}$$

$$m_{HNO_3} = 63\text{g/mol} \quad m_{KNO_3} = 101\text{g/mol}$$

$$\frac{w_{HNO_3}}{63 \times 4} = \frac{110\text{g}}{101 \times 3}$$

$$\frac{w_{HNO_3}}{63 \times 4} = \frac{110\text{g}}{101 \times 3}$$

$$w_{HNO_3} = 91.48\text{g}$$
$$w_{HNO_3} = 91.5\text{g}$$

The correct answer is option (3).

Q. 2 Given below are the quantum numbers for 4 electrons.

A. $n = 3, 1 = 2, \ m_1 = 1, \ m_s = +1/2$
B. $n = 4, 1 = 1, \ m_1 = 0, \ m_s = +1/2$
C. $n = 4, 1 = 1, \ m_1 = 0, \ m_s = +1/2$
D. $n = 3, 1 = 1, \ m_1 = -1, \ m_s = +1/2$

The correct order of increasing energy is

Option 1:
$D < B < A < C$

Option 2:
$D < A < B < C$

Option 3:
$B < D < A < C$

Option 4:
$B < D < C < A$

Correct Answer:
$D < A < B < C$

Solution:
According to $(m + l)$ role , lower the value of n,
lower is energy of orbital.

If $(m + l)$ value are same , then lower is value of n
lower is energy of orbital.

A	$n = 3$	$l = 2$	$n + l = 5$
B	$n = 4$	$l = 1$	$n + l = 5$
C	$n = 4$	$l = 2$	$n + l = 6$
D	$n = 3$	$l = 1$	$n + l = 4$

Energy of electrons is
$D < A < B < C$

Correct option is 2.

Q. 3 $C(s) + O_2(g) \rightarrow CO_2(g) + 400\ kJ$

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g) + 100\ kJ$

When coal of purity 60% is allowed to burn in presence of insufficient oxygen, 60% of carbon is converted into $'CO'$ and the remaining is converted into $'CO_2'$. The heat generated when $0.6\ kg$ of coal is burnt is _____.

Option 1:
$1600\ kJ$

Option 2:
$3200\ kJ$

Option 3:
$4400\ kJ$

Option 4:
$6600\ \text{kJ}$

Correct Answer:
$6600\ \text{kJ}$

Solution:
Given sample of coal weight 600 g of purity 60 % .

$$\text{Actual carbon present} = 360\,\text{g C}$$

of

360 g of C , 60 % burns to give CO while rest to form completely to give CO_2.

$$C + \frac{1}{2}O_2 \longrightarrow CO \quad 100\,\text{KJ/mol}$$

210 g

18 mole

$$\Delta H = 18 \times 100$$
$$= 1800\,\text{KJ}$$

$$C + O_2 \longrightarrow CO_2 \quad 100\,\text{KJ/mol}$$

144g

12 mole

$$\Delta H = 12 \times 400$$
$$= 4800\,\text{KJ}$$

$$\text{Total heat generated} = (4800 + 1800)\text{kJ}$$
$$= 6600\,\text{KJ}$$

Correct option is (D).

Q. 4 $200\ \text{mL}$ of $0.01\ \text{M}HCl$ is mixed with $400\ \text{mL}$ of $0.01 M H_2SO_4$. The pH of the mixture is ____.

Given: $\log 2 = 0.30, \log 3 = 0.48, \log 5 = 0.70, \log 7 = 0.84, \log 11 = 1.04$

Option 1:
1.14

Option 2:
1.78

Option 3:
2.34

Option 4:
3.02

Correct Answer:
1.78

Solution:

$$\text{molarity of mixing} = \frac{m_1v_1 + m_2v_2}{v_1 + v_2}$$

$$\frac{0.01 \times 200 + 2 \times 0.01 \times 400}{600} = \frac{10}{600} = \frac{1}{60}\text{m}$$

$$\begin{aligned}\text{Now, pH} &= -\log[\mu t] \\ &= -\log\frac{1}{60} = \log 60 = \log 10 + \log 2 + \log 3 \\ &= 1 + 0.3 + 0.48 \\ \text{pH} &= 1.78\end{aligned}$$

Option (B) is correct.

Q. 5 Given below are the critical temperatures of some of the gases:

Gas	Critical temperature (K)
He	5.2
CH_4	190.0
CO_2	304.2
NH_3	405.5

The gas showing least adsorption on a definite amount of charcoal is

Option 1:
He

Option 2:
CH_4

Option 3:
CO_2

Option 4:
NH_3

Correct Answer:
He

Solution:
Extent of Adsorption of gas α Critical temperature of gas.
More is critical temeperature of gas , easily it can be equified.
Critical Temperature (Tc) is the maximum temperature at which gas can be equified on applying suitable pressure.

Ease of Adsorption is $NH_3 > CO_2 > CH_4 > He$

Hence option (A) is correct.

Q. 6 In liquation process used for tin (Sn), the metal

Option 1:
is reacted with acid.

Option 2:
is dissolved in water.

Option 3:
is brought to molten form which is made to flow on a slope.

Option 4:
is fused with $NaOH$

Correct Answer:
is brought to molten form which is made to flow on a slope.

Solution:
Liquation or swoating applicable for metals such as $S_n, P6 \text{ and } \beta$, which have low melting points as compared to their impurities
In this impure metal block is kept of slaping and heated above melting point, metal melts and pour down the slope.

Correct option is (C).

Q. 7 Given below are two statements.
Statement I: Stannane is an example of a molecular hydride.
Statement II: Stannane is a planar molecule.
In the light of the above statement, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both Statement I and Statement II are true.

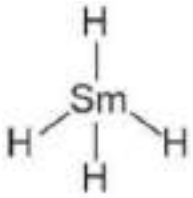
Option 2:
Both Statement I and Statement II are false.

Option 3:
Statement I is true but Statement II is false.

Option 4:
Statement I is false but Statement II is true.

Correct Answer:
Statement I is true but Statement II is false.

Solution:
Stannane is S_nH_4, which is a molecular hydride.
but stannane is not planar but has a tetrahedral geometry.



Hence, option (C) is correct.

Q. 8 Portland cement contains 'X' to enhance the setting time. What is 'X'?

Option 1:
$CaSO_4 \cdot \frac{1}{2}H_2O$

Option 2:
$CaSO_4 \cdot 2H_2O$

Option 3:
$CaSO_4$

Option 4:
$CaCO_3$

Correct Answer:
$CaSO_4 \cdot 2H_2O$

Solution:
Gypsum $(COSO_4 \cdot 2H_2O)$ is added portland cement to increase seting time of cement.

Hence option (B) is correct.

Q. 9 When borax is heated with CoO on a platinum loop, blue coloured bead formed is largely due to

Option 1:
B_2O_3

Option 2:
$Co\,(BO_2)_2$

Option 3:
CoB_4O_7

Option 4:
$Co\,[B_4O_5(OH)_4]$

Correct Answer:
$Co\,(BO_2)_2$

Solution:
Borax bead test

$$Na_2B_4O_7 \xrightarrow{\Delta} 2\,NaBO_2 + B_2O_3$$

Sodium metaborate

$$CoO + B_2O_3 \xrightarrow{\Delta} Co\,(BO_2)_2$$

Cobalt metoborate

(blue)

Hence option (2) is correct.

Q. 10 Which of the following $3\,\mathrm{d}$-metal ion will give the lowest enthalpy of hydration $(\Delta_{\mathrm{hyd}}\mathrm{H})$ when dissolved in water ?

Option 1:
Cr^{2+}

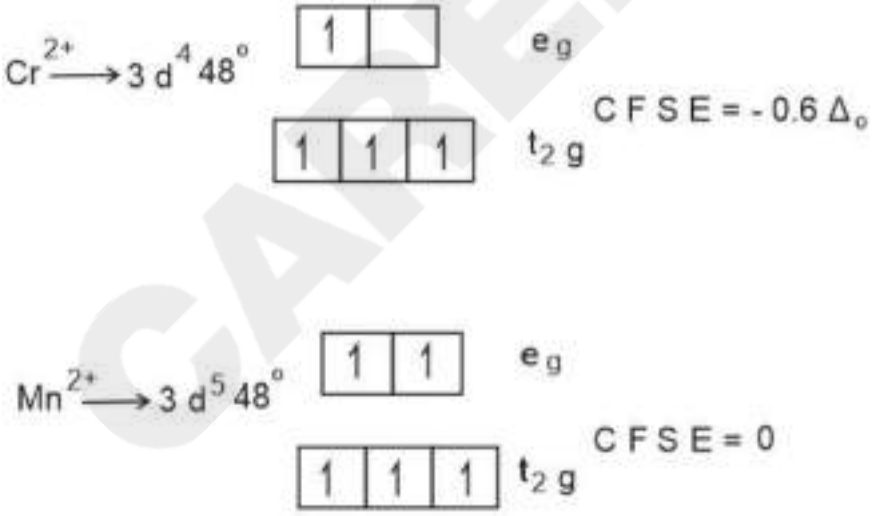
Option 2:
Mn^{2+}

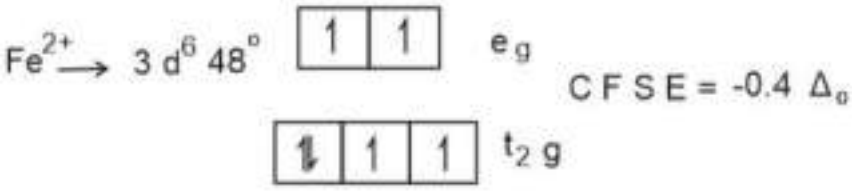
Option 3:
Fe^{2+}

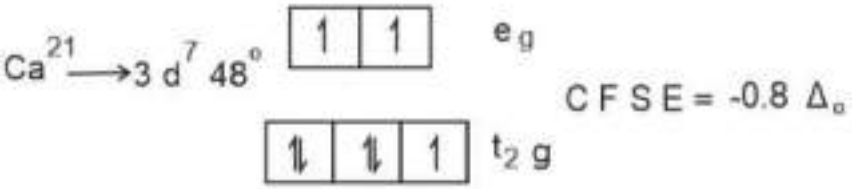
Option 4:
Co^{2+}

Correct Answer:
Mn^{2+}

Solution:
Water act as a weak ligand.CFSE of metal ions depend on strength of ligand.







Since the CFSE of $\mathrm{Mn^{2+}}$ is the least , $\Delta \mathrm{H_{Hyd}}$ of it is also lowest.

Hence option (B) is correct.

Q. 11 Octahedral complexes of $\mathrm{copper(II)}$ undergo structural distortion $\mathrm{(Jahn-Teller)}$.
Which one of the given $\mathrm{copper(II)}$ complexes will show the maximum structural distortion? $\mathrm{(en-ethylenediamine;\ H_2\ N-CH_2-CH_2-NH_2)}$

Option 1:
$\mathrm{[Cu\,(H_2O)_6]\,SO_4}$

Option 2:
$\mathrm{[Cu(en)\,(H_2O)_4]\,SO_4}$

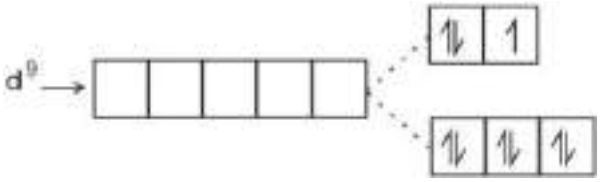
Option 3:
$\mathrm{cis-[Cu(en)_2Cl_2]}$

Option 4:
$\mathrm{trans-\ [Cu(en)_2Cl_2]}$

Correct Answer:
$\mathrm{[Cu\,(H_2O)_6]\,SO_4}$

Solution:

Electronic configuration of $\mathrm{Cu\ (II)}$ is $\mathrm{d^9}$



The maximum John-Teller distortion will be shown maximum by $\mathrm{[Cu\ (H_2O)_6]\ SO_4}$, as $\mathrm{H_2O}$ is ,monodentate ligand. While in other complexes ethylenediamine (en) is present which forms chelate and due to chelotion, distortion is lessened because ring Strain can happen.

Hence option (A) is correct.

Q. 12 Dinitrogen is a robust compound, but reacts at high altitudes to form oxides. The oxide of nitrogen that can damage plant leaves and retard photosynthesis is

Option 1:
NO

Option 2:
$\mathrm{NO_3^-}$

Option 3:
$\mathrm{NO_2}$

Option 4:
$\mathrm{NO_2^-}$

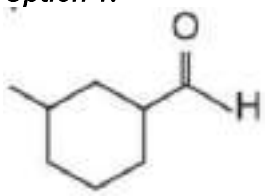
Correct Answer:
$\mathrm{NO_2}$

Solution:
$\mathrm{NO_2}$ formed of high altitudes damages leaves and reload photosynthesis.

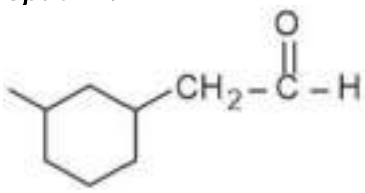
Hence option (C) is correct.

Q. 13 Correct structure of $\gamma - \mathrm{methylcyclohexane}$ carbaldehyde is

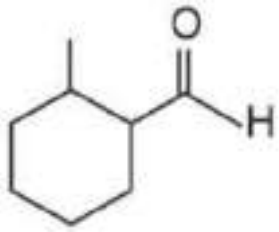
Option 1:



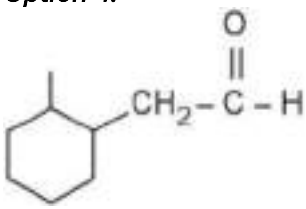
Option 2:



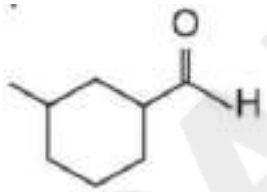
Option 3:



Option 4:

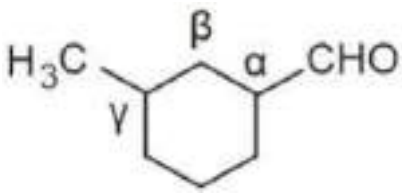


Correct Answer:



Solution:

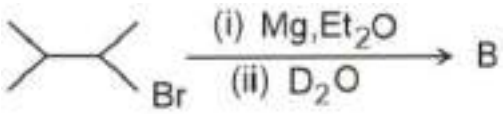
γ – methylcyclohexane carbaldehyde is



Hence option (1) is correct.

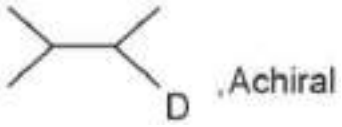
Q. 14 Compound $'A'$ undergoes following sequence of reactions to give compound $'B'$ The correct structure and chirality of compound $'B'$ is

$[$ where Et is $-C_2H_5$ $]$

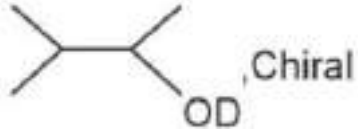


Compound 'A'

Option 1:

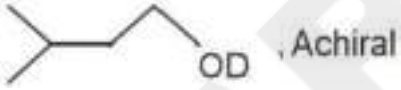


Option 2:



Option 3:

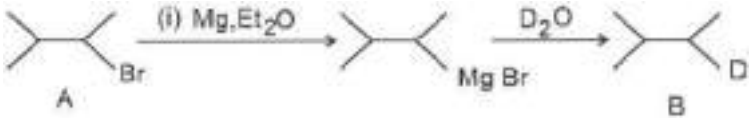
Option 4:

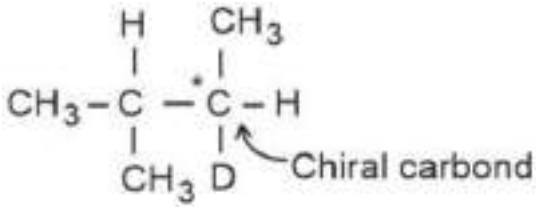


Correct Answer:

Solution:

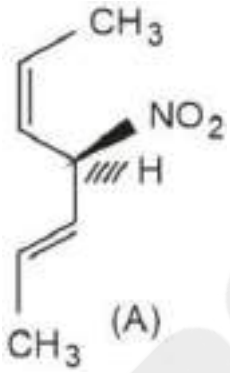
On Reaction of haloalkane with Mg in presence of tay ether followed by reaction on with part on Janor generates alkane.



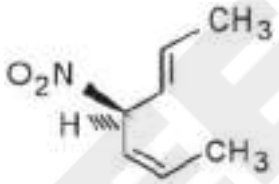
Compound B is 

Hence option (C) is correct.

Q. 15 Given below are two statements.

Statement I: The compound 

is optically active.

Statement II 

is mirror image of above compound A.

In the light of the above statement, choose the ***most appropriate*** answer from the options given below.

Option 1:
Both Statement I and Statement II are correct.

Option 2:
Both Statement I and Statement II are incorrect.

Option 3:
Statement I is correct but Statement II is incorrect.

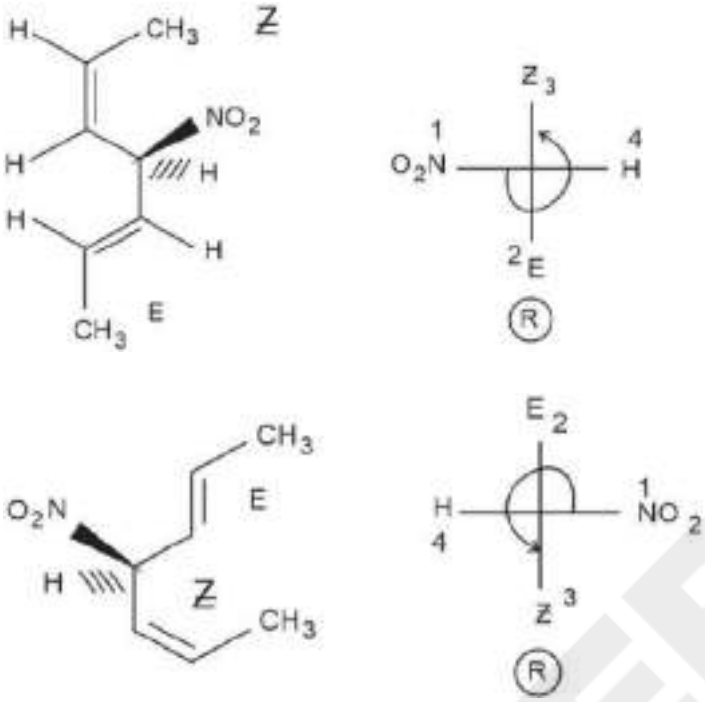
Option 4:
Statement I is incorrect but Statement II is correct.

Correct Answer:

Statement I is correct but Statement II is incorrect.

Solution:

Compound A is chirol and optically active because the alkanes are E nad Z respectively but the comp own in Statement $-\mathrm{II}$ is, identical to A.



Since they are identical and not enantiomers , hence are not mirror images.

So option (C) is correct.

Q. 16 When enthanol is heated with conc. $\mathrm{H_2SO_4}$, a gas is produced. The compound formed, when this gas is treated with cold dilute aqueous solution of Baeyer's reagent, is

Option 1:

formaldehyde

Option 2:

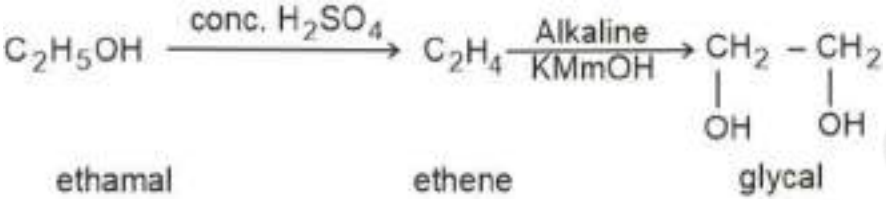
formic acid

Option 3:
glycol

Option 4:
ethanoic acid

Correct Answer:
glycol

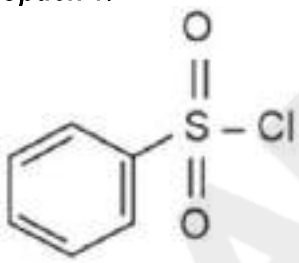
Solution:



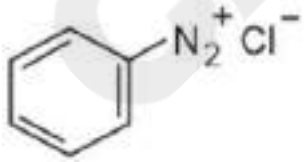
Hence option (C) is correct.

Q. 17 The Hinsberg reagent is

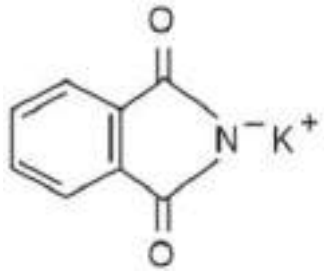
Option 1:



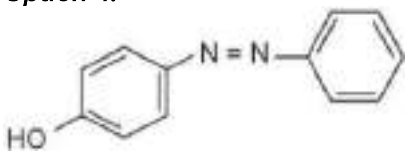
Option 2:



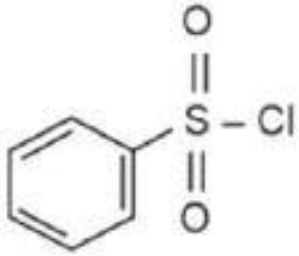
Option 3:



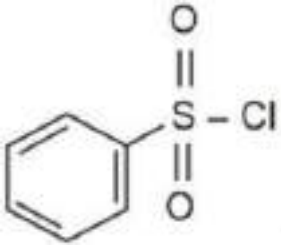
Option 4:



Correct Answer:



Solution:
Hinsberg reagent is



Which is used to differentiate between $1^{\circ}, 2^{\circ} \text{and } 3^{\circ}$ Amines.

Hence option (A) is correct.

Q. 18 Which of the following is **NOT** a natural polymer?

Option 1:
Protein

CAREERS360

Option 2:
Starch

Option 3:
Rubber

Option 4:
Rayon

Correct Answer:
Rayon

Solution:
Rayon is a man-made Polymer. Monomer of Rayon is Cellulose acetate.
Hence option (D) is correct.

Q. 19 Given below are two statements. One is labelled as **Assertion** A and the other is labelled as **Reason R.**
Assertion A: Amylose is insoluble in water.
Reason R: Amylose is a long linear molecule with more than 200 glucose units. In the light of the above statements, choose the ***correct*** answer from the options given below.

Option 1:
Both **A** and **R** are correct and **R** is the correct explanation of **A.**

Option 2:
Both **A** and **R** are correct but **R** is NOT the correct explanation of **A.**

Option 3:
A is correct but **R** is not correct.

Option 4:
A is not correct but **R** is correct.

Correct Answer:
A is not correct but **R** is correct.

Solution:
Starch consist of two types of Poly socch oxide molecules Amylose and Amylopectin.

Amylose is soluble in water but not Amylopectin.

Amylose is linear polymer consisting of 06 out 200 units.

Hence option (D) is correct.

Q. 20 A compound $'X'$ is a weak acid and it exhibits colour change at pH close to the equivalence point during neutralization of $NaOH$ with CH_3COOH. Compound $'X'$ exists in ionized form in basic medium. The compound $'X'$ is

Option 1:
methyl orange

Option 2:
methyl red

Option 3:
phenolphthalein

Option 4:
erichrome Black T

Correct Answer:
phenolphthalein

Solution:
For titration weak acid and strong base, phenolphthalein is used. Also phenolphthalein orange Color in Basic medium.

Indicator	PH Range
Methyl Orange	3.2 - 4.4
Methyl Red	4.8 - 6.0
phenolphthalein	8.2-10
erichrome Black T	7-11

Hence option (C) is correct.

Q. 21 'x' g of molecular oxygen (O_2) is mixed with 200 g of neon (Ne). The total pressure of the non-reactive mixture of O_2 and Ne in the cylinder is 25 bar. The partial pressure of Ne is 20 bar at the same temperature and volume. The value of 'x' is _____.

[Given : Molar mass of $O_2 = 32\ g\ mol^{-1}$. Molar mass of $Ne = 20\ g\ mol^{-1}$]

Correct Answer:
80

Solution:
According to Question
Total pressure is 25 bar and $P_{NC} = 20\,bar$ hence
$P_{O_2} = 5\,bar$

From Dalton law of partial pressure
$P_{gas} \propto x_g$

Hence
$$\frac{P_{NC}}{P_{O2}} = \frac{M_{NC}}{M_{O2}}$$
$$\frac{20\,bar}{5\,bar} = \frac{200\ g \times 32\ g/mol}{20\ g/mol \times xg}$$
$$x = 80g$$

Q. 22 Consider, $PF_5, BrF_5, PCl_3, SF_6, [ICl_4]^-$, ClF_3 and IF_5.
Amongst the above molecule $(s)/ion(s)$, the number of molecule $(s)/ion(s)$ having $sp^3\ d^2$ hybridisation is _____.

Correct Answer:
6

Solution:

Molecule	Steric Numer	Hybridization
PF_5	$= \frac{5+5}{2} = 5$	sp^3d
BrF_5	$= \frac{7+5}{2} = 6$	sp^3d^2

Pu_4	$= \frac{5+3}{2} = 4$	sp^3
SF_6	$= \frac{6+6}{2} = 6$	sp^3d^2
Iu_4^-	$= \frac{7+4+1}{2} = 6$	sp^3d^2
CF_3	$= \frac{7+3}{2} = 5$	sp^3d
IF_5	$= \frac{7+5}{2} = 6$	sp^3d^2

BrF_5, SF_6, Fu_4^- and IF_5 have sp^3d^2 hybridization.

Q. 23 1.80 g of solute A was dissolved in 62.5 cm^3 of ethanol and freezing point of the solution was found to be 155.1 K. The molar mass of solute A is ___ $g\ mol^{-1}$.
[Given: Freezing point of ethanol is 156.0 K.
Density of ethanol is 0.80 $g\ cm^{-3}$.
Freezing point depression constant of ethanol is 2.00 $K\ kg\ mol^{-1}$]

Correct Answer:
80

Solution:
$\text{wt.of solute} = 1.80\,\text{g}$
$\text{val.of solvent} = 62.5\,\text{ml}$
$\text{density of solvent} = 0.8\ \text{g/ml}$
$\text{weight of solvent} = \text{d} \times \text{v} = 50\ \text{g}$
$\text{Depression constant} = (\text{k}_\text{f}) = 2\,\text{k}\,\text{kg mol}^{-1}$
$\text{Depression in freezing point} = 0.9\,\text{k}$

$$\Delta t_f = k_f \times \frac{\text{wt.of solute} \times 1000}{\text{molar mass of solute} \times \text{wt.of solvent (g)}}$$

$$0.9 = \frac{2\,\text{kg/mol} \times 1.8\ \text{g} \times 1000}{\text{m} \times 50\ \text{g}}$$

$$\text{m} = \frac{2 \times 1.8 \times 1000}{0.8 \times 50}\,\text{g/mol}$$

$$= 80\,\text{g/mol}$$

Hence answer is (80).

Q. 24 For a cell, $Cu(s)\left|Cu^{2+}(0.001M)\right|\left|Ag^{+}(0.01M\right|Ag(s))$ the cell potential is found to be $0.43\ V$ at $298\ K$. The magnitude of standard electrode potential for Cu^{2+}/Cu is ____ $\times 10^{-2}\ V$.

$$\left[\text{Given : } E_{Ag^{\ominus}/Ag} = 0.80\ V \text{ and } \frac{2.303RT}{F} = 0.06\ V\right]$$

Correct Answer:
34

Solution:
For the cell

$$Cu(s)\left|Cu^{2+}(0.001M)\right|\left|Ag^{+}(0.01M)\right|Ag(s)$$

$$\begin{aligned}\text{Anode}: & Cu \longrightarrow Cu^{2+} + 2e^{-}\\ \text{Cathode}: & \underline{2Ag^{+} + 2e^{-} \longrightarrow 2Ag}\\ & Cu + 2Ag^{+} \longrightarrow Cu^{2+} + 2Ag\end{aligned}$$

$$E_{cell} = E^{0}_{cell} - \frac{0.06}{M}\log\frac{[Cu^{2+}]}{[Ag^{+}]^{2}}$$

$$0.43 = E^{0}_{cell} - \frac{0.06}{2}\log\frac{10^{-3}}{(10^{-2})^{2}}$$

$$E^{0}_{cell} = 0.43 + 0.03\log 10 = 0.46\ V$$

$$E^{0}_{cell} = E^{0}_{Ag^{+}|Ag} - E^{0}_{Cu^{2+}|Cu}$$

$$0.46 = 0.8 - E^{0}_{Cu^{2+}|Cu}$$

$$E^{0}_{Cu^{2+}|Cu} = 0.34\ V = 34 \times 10^{-2}\ V$$

Hence answer is 34.

Q. 25 Assuming $1\mu g$ of trace radioactive element X with a half life of 30 years is absorbed by a growing tree. The amount of X remaining in the tree after 100 years is _____ $\times 10^{-1}\mu g$.

$$[\text{Given: } \ln 10 = 2.303; \log 2 = 0.30]$$

Correct Answer:
1

Solution:

$$t_{1/2} = 30 \text{ years } k = \frac{\ln 2}{30 \text{ years}}$$

$$A_0 = 14 \text{ g} \qquad t = 100 \text{ year}$$

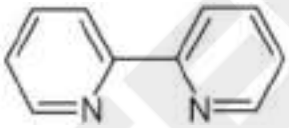
$$\frac{2.303 \log 2}{30} = \frac{2.303}{100} \quad \frac{1\mu g}{A_t}$$

$$\frac{10}{3} \log 2 = \log \frac{1}{A_t} \qquad 0.01 = \log \frac{1}{A_t}$$

$$A_t = 0.1\,\mu g = 1 \times 10^{-1} \mu g$$

Hence answer is 1.

Q. 26 Sum of oxidation state (magnitude) and coordination number of cobalt in $Na\,[Co(bpy)Cl_4]$ is _____.

(Given: bpy = )

Correct Answer:
9

Solution:
For $Na[Co\,(bpy)\,Cl_4]$ oxidation state of Co is x

$$+1 \times 1 + x \times 1 + 0 \times 1 + (-1 \times 4) = 0$$
$$x = +3$$

Coordination number of Co is 6
Sum of $C.\ N$ and $O.S$ is $6 + 3 = 9$

Hence, Correct answer is 9.

Q. 27 Consider the following sulphur based oxoacids.

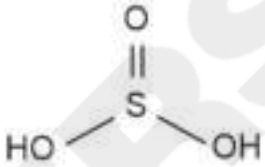
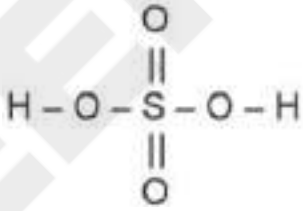
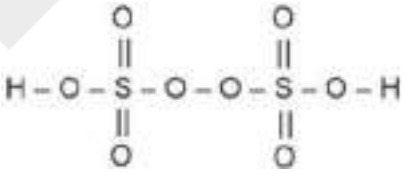
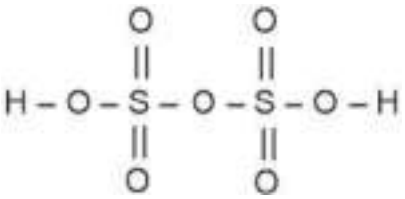
$H_2SO_3, H_2SO_4, H_2\ S_2O_8$ and $H_2\ S_2O_7$.

Amongst these oxoacids, the number of those with peroxo $(O-O)$ bond is

_____

Correct Answer:
1

Solution:
If calculated oxidation state of element is greater than maximum oxidation number , then peroxy linkages are present.

Aciol	oxidation state of S	Structure
H_2SO_3	+4	
H_2SO_4	+6	
$H_2S_2O_8$	+7	
$H_2S_2O_7$	+7	

Only $H_2S_2O_8$ has peroxy linkage.

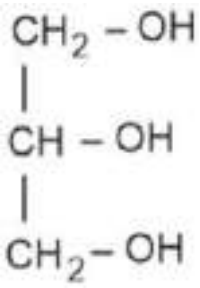
Q. 28 A $1.84\ \text{mg}$ sample of polyhydric alcoholic compound 'X' of molar mass $92.0\ \text{g/mol}$ gave $1.344\ \text{mL}$ of H_2 gas at STP. The number of alcoholic hydrogens present in compound 'X' is _____.

Correct Answer:
3

Solution:
$\text{polyhydric alcoholic} \longrightarrow H_2$

$$m \times m_{\text{Alcohol}} = 2 \times mH_2$$
$$m \times \frac{1.84 \times 10^{-3}}{92} = 2 \times \frac{1.344\,\text{ml}}{22400\,\text{ml}}$$
$$m = 6$$

No. of Alcoholic Hydrogen present are 6.
But if molecular mass is $92\ \text{g/mol}$ then expected polyhydric alcohol must be glyceral.

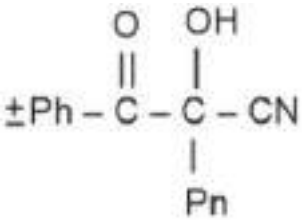


Q. 29 The number of stereoisomers formed in a reaction of $(\pm)\text{Ph}(C=O)C(OH)(CN)\text{Ph}$ with HCN is _____.

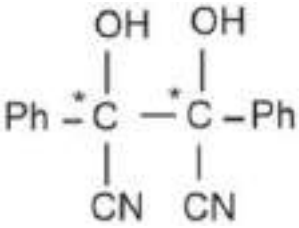
[where Ph is $-C_6H_5$]

Correct Answer:
3

Solution:

On reaction of  with HCN will form meso compound as well along with

stereoisomers

 There are 2 chiral centres in product.

If n is even

No. of enantiomers $= 2^{n-1} = 2^1 = 2$
No.of meso compounds $= 2^{\frac{n}{2}-1} = 2^0 = 1$

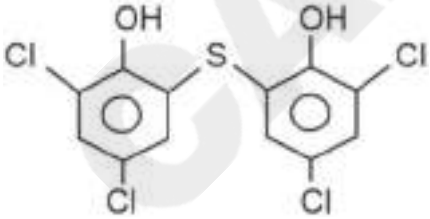
Total number of optical isomers $= 2 + 1 = 3$

Hence correct answer is 3.

Q. 30 The number of chlorine atoms in bithionol is _____.

Correct Answer:
4

Solution:
Structure of Bithionol is



There are 4 Chlorine atoms.
Hence answer is 4.

Maths

Q. 1 If $z \neq 0$ be a complex number such that $\left|z-\frac{1}{z}\right|=2$, then the maximum value of $|z|$ is :

Option 1:
$\sqrt{2}$

Option 2:
1

Option 3:
$\sqrt{2}-1$

Option 4:
$\sqrt{2}+1$

Correct Answer:
$\sqrt{2}+1$

Solution:

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$$||z\left|-\frac{1}{|z|}\right| \leq \left|z-\frac{1}{z}\right| \quad \text{(Triangle Law)}$$

$$|z|-\frac{1}{|z|}| \leq 2$$

Let $|z| = r$

$$\left|r-\frac{1}{r}\right| \leq 2$$

$$\Rightarrow -2 \leq r-\frac{1}{r} \leq 2 \leq 2$$

$$\Rightarrow -2 \leq r-\frac{1}{r} \text{ and } r-\frac{1}{r} \leq 2$$

$$\Rightarrow -2r \leq r^2-1 \text{ and } r^2-1 \leq 2r$$

$$\Rightarrow r^2+2r-1 \geqslant 0 \text{ and } r^2-2r-1 \leq 0$$

$$\Rightarrow (r \leq -\sqrt{2}-1 \text{ or } r \geqslant \sqrt{2}-1) \text{ and } -\sqrt{2}+1 \leq r \leq \sqrt{2}+1$$

$$\Rightarrow \sqrt{2}-1 \leq r \leq \sqrt{2}+1$$

$$\therefore \text{ Maximum} = \sqrt{2}+1$$

$$\therefore \text{ option (D)}$$

Q. 2 Which of the following matrices can **NOT** be obtained from the matrix $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$ by a single elementary row operation?

Option 1:

$$\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$$

Option 2:

$$\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Option 3:

$$\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$$

Option 4:

$$\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$$

Correct Answer:

$$\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$$

Solution:

(A) can be obtained using $R_1 \to R_1 + R_3$

(B) can be obtained by $R_1 \leftrightarrow R_2$

(D) can be obtained by $R_2 \to R_2 + 2R_1$

$\therefore$ option (C)

Q. 3 If the system of equations

$x + y + z = 6$

$2x + 5y + \alpha z = \beta$

$x + 2y + 3z = 14$

has infinitely many solutions, then $\alpha + \beta$ is equal to

Option 1:

8

Option 2:

36

Option 3:

44

Option 4:

48

Correct Answer:

44

Solution:

For infinite solutions

$\Delta = 0$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 2 & 5 & \alpha \\ 1 & 2 & 3 \end{vmatrix} = 0$$

$$\Rightarrow (15 - 2\alpha) - 1(6 - \alpha) + (4 - 5) = 0$$

$$\Rightarrow 15 - 2^{\alpha} - 6 + \alpha - 1 = 0$$

$$\Rightarrow \alpha = 8$$

Also $\Delta z = 0$

$$\Rightarrow \begin{vmatrix} 1 & 1 & 6 \\ 2 & 5 & \beta \\ 1 & 2 & 14 \end{vmatrix} = 0$$

$$\Rightarrow (70 - 2\beta) - 1(28 - \beta) + 6(4 - 5) = 0$$

$$\Rightarrow 36 - \beta = 0$$

$$\Rightarrow \beta = 36$$

$$= \alpha + \beta = 44$$

$\therefore$ option (c)

Q. 4 Let the function $f(x) = \begin{cases} \frac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & \text{if } x \neq 0 \\ 10 & \text{if } x = 0 \end{cases}$ be continuous at $x = 0$. Then α is equal to

Option 1:

10

Option 2:

-10

Option 3:

5

Option 4:

-5

Correct Answer:

-5

Solution:

For continuous at $x = 0$

$$\lim_{x\to 0}\left(\frac{\log(1+5x)}{x}-\frac{\log(1+\alpha x)}{\alpha}\right)=10$$

$$\Rightarrow 5-\alpha=10$$

$$\Rightarrow \alpha=-5$$

Q. 5 If $[t]$ denotes the greatest integer $\leq t$, then the value of $\int_0^1 \left[2x-\left|3x^2-5x+2\right|+1\right]dx$ is:

Option 1:
$\frac{\sqrt{37}+\sqrt{13}-4}{6}$

Option 2:
$\frac{\sqrt{37}-\sqrt{13}-4}{6}$

Option 3:
$\frac{-\sqrt{37}-\sqrt{13}+4}{6}$

Option 4:
$\frac{-\sqrt{37}+\sqrt{13}+4}{6}$

Correct Answer:
$\frac{\sqrt{37}+\sqrt{13}-4}{6}$

Solution:

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$$3x^2 - 5x + 2$$
$$= (3x - 2)(x - 1)$$
$$\therefore \left|3x^2 - 5x + 2\right| = \begin{cases} 3x^2 - 5x + 2, & 0 \leqslant x < \frac{2}{3} \\ -\left(3x^2 - 5x + 2\right) & \frac{2}{3} < x < 1 \end{cases}$$
$$I = \int_0^{\frac{2}{3}} \left[2x - \left(3x^2 - 5x + 2\right) + 1\right] dx + \int_{\frac{2}{3}}^{1} \left[2x + 3x^2 - 5x + 2 + 1\right] dx$$
$$= \int_0^{\frac{2}{3}} \left[-3x^2 + 7x - 1\right] dx + \int_{\frac{2}{3}}^{1} \left[3x^2 - 3x + 3\right] dx$$
$$= (-1)\int_0^{\frac{7-\sqrt{37}}{6}} dx + \int_{1/3}^{\frac{7-\sqrt{13}}{6}} dx + 2\int_{\frac{7-\sqrt{13}}{6}}^{2/3} dx + 2\int_{2/3}^{1} dx$$
$$= \frac{\sqrt{37} - 7}{6} + \frac{7 - \sqrt{13}}{6} - \frac{1}{3} + \frac{4}{3} - \frac{14}{6} + \frac{2\sqrt{13}}{6} + \frac{2}{3}$$
$$= \frac{\sqrt{37}}{6} - \frac{7}{6} + \frac{7}{6} - \frac{\sqrt{13}}{6} + \frac{2\sqrt{13}}{6} - \frac{2}{3}$$
$$= \frac{\sqrt{37}}{6} + \frac{\sqrt{13}}{6} - \frac{4}{6}$$
$$\therefore \text{option}(A)$$

Q. 6 Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0 = a_1 = 0$ and $a_{n+2} = 3a_{n+1} - 2a_n + 1, \forall n \geq 0$ Then $a_{25}a_{23} - 2a_{25}a_{22} - 2a_{23}a_{24} + 4a_{22}a_{24}$ is equal to

Option 1:
483

Option 2:
528

Option 3:
575

Option 4:
624

Correct Answer:
528

Solution:

$$a_{n+2} = 3a_{n+1} - 2a_n + 1$$

$$\Rightarrow \quad a_{n+2} - a_{n+1} = 2(a_{n+1} - a_n) + 1$$

Put $n = 0, 1, 2, \ldots (n-2)$

$$a_2 - a_1 = 2(a_1 - a_0) + 1$$

$$a_3 - a_2 = 2(a_2 - a_1) + 1$$

$$a_4 - a_3 = 2(a_3 - a_2) + 1$$

$$a_n - a_{n-1} = 2(a_{n-1} - a_{n-2}) + 1$$

Adding these

$$a_n - a_1 = 2(a_{n-1} - a_0) + (n-1)$$

$$\Rightarrow a_n - 2a_{n-1} = n - 1 \quad \ldots \text{(i)}$$

Now

$$a_{25}a_{23} - 2a_{25}a_{22} - 2a_{23}a_{24} + 4a_{22}a_{24}$$

$$= (a_{25} - 2a_{24})(a_{23} - 2a_{22}) \ldots \text{(ii)}$$

Put $n = 25$ in (i)

$$a_{25} - 2a_{24} = 24$$

Put $n = 23$ in (i)

$$a_{23} - 2 \cdot a_{22} = 22$$

$\therefore$ Required value $= 24.22 = 528$

Ans: (B)

Q. 7 $\sum_{r=1}^{20} (r^2 + 1)(r!)$ is equal to

Option 1:

$22! - 21!$

Option 2:

$22! - 2(21!)$

Option 3:

$21! - 2(20!)$

Option 4:

$21! - 20!$

Correct Answer:

$22! - 2(21!)$

Solution:

$$\begin{aligned}
&\left(r^2+1\right) r!_0 \\
=&\left(\gamma^2+2r+1-2r\right) r!_0 \\
=&\left((r+1)^2-2r\right) r! \\
=&(\gamma+1)^2 \cdot r! - 2r \cdot r! \\
=&(r+1)\cdot(r+1)!_0 - \gamma\cdot\gamma!_0 - \gamma\cdot r! \\
=&[(r+i)(\gamma+1)! - \gamma\cdot\gamma!_0] - [((r+1)-1)\cdot\gamma!] \\
=&[(r+1)(\gamma+1)! - \gamma\cdot\gamma!] - [(\gamma+1)! - \gamma!]
\end{aligned}$$

Applying summation from 1 to 20

$$\begin{aligned}
&= (21.21! - 1) - (21! - 1) \\
&= 20\cdot 21! \\
&= (22-2)21! \\
&= 22! - 2\cdot 21!
\end{aligned}$$

$\therefore$ option (B)

Q. 8 For $I(x) = \int \frac{\sec^2 x - 2022}{\sin^{2022} x} dx$, if $I\left(\frac{\pi}{4}\right) = 2^{1011}$, then

Option 1:

$3^{1010} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$

Option 2:

$3^{1010} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

Option 3:

$3^{1011} I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$

Option 4:

$3^{1011} I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

Correct Answer:

$3^{1010}\mathrm{I}\left(\frac{\pi}{3}\right)-\mathrm{I}\left(\frac{\pi}{6}\right)=0$

Solution:

$$\mathrm{I}(\mathrm{x})=\int\frac{\sec^2\mathrm{x}}{\sin^{2022}\mathrm{x}}\mathrm{dx}-\int\frac{202^2}{\sin^{2022}\mathrm{x}}\mathrm{dx}$$

Integrating by paris in first integral

$$=\frac{\tan\mathrm{x}}{\sin^{2022}\mathrm{x}}+\int\frac{2022}{\sin^{2023}\mathrm{x}}\cdot\cos\mathrm{x}\cdot\tan\mathrm{x}\,\mathrm{dx}-\int\frac{2022}{\sin^{2022}\mathrm{x}}\mathrm{dx}+\mathrm{c}$$

$$\mathrm{I}(\mathrm{x})=\frac{\tan\mathrm{x}}{\sin^{2022}\mathrm{x}}+\mathrm{c}$$

$$\text{Using } \mathrm{I}\left(\frac{\pi}{4}\right)=2^{1011},\mathrm{c}=0$$

$$\therefore \mathrm{I}\left(\mathrm{x}^3\right)=\frac{\tan\mathrm{x}}{\sin^{2022}\mathrm{x}}$$

cheaking, options, option(A) is correct

$\therefore$ option (A)

Q. 9 If the solution curve of the differential equation $\frac{\mathrm{dy}}{\mathrm{dx}}=\frac{\mathrm{x}+\mathrm{y}-2}{\mathrm{x}-\mathrm{y}}$ passes through the points $(2,1)$ and $(\mathrm{k}+1,2),\mathrm{k}>0$, then

Option 1:

$2\tan^{-1}\left(\frac{1}{\mathrm{k}}\right)=\log_{\mathrm{e}}\left(\mathrm{k}^2+1\right)$

Option 2:

$\tan^{-1}\left(\frac{1}{\mathrm{k}}\right)=\log_{\mathrm{e}}\left(\mathrm{k}^2+1\right)$

Option 3:

$$2\tan^{-1}\left(\frac{1}{k+1}\right)=\log_e\left(k^2+2k+2\right)$$

Option 4:

$$2\tan^{-1}\left(\frac{1}{k}\right)=\log_e\left(\frac{k^2+1}{k^2}\right)$$

Correct Answer:

$$2\tan^{-1}\left(\frac{1}{k}\right)=\log_e\left(k^2+1\right)$$

Solution:

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Let $u = x + h, v = y + k.$

$\Rightarrow du = dx$ and $dr = dy$

$\therefore \frac{dv}{du} = \frac{u - h + v - k - 2}{u - h - v + k}$

$\frac{dv}{du} = \frac{u + v + (-k - k - 2)}{(u - v) + (-h + k)}$

$\Rightarrow -h - k - 2 = 0$ and $-h + k = 0$

$\Rightarrow h = k = -1$

$\frac{dv}{du} = \frac{u + v}{u - v}$

Let $\frac{v}{u} = w$

$v = uw$

$\frac{dv}{du} = w + u\frac{dw}{du}$

$\therefore \quad w + \frac{udw}{du} = \frac{1 + w}{1 - w}$

$u\frac{dw}{du} = \frac{1 + w^2}{1 - w}$

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$$\frac{(1-w)dw}{1+u^2}=\frac{du}{u}$$

$$\int\frac{du}{1+w^2}-\frac{1}{2}\int\frac{2u}{1+w^2}du^2=\int\frac{du}{u}$$

$$\tan^{-1}(w)-\frac{1}{2}\log\left(1+w^2\right)=\log u+c$$

$$\tan^{-1}\left(\frac{u}{u}\right)-\frac{1}{2}\log\left(1+\frac{v^2}{u^2}\right)=\log(u)+c$$

$$\tan^{-1}\left(\frac{y-1}{x-4}\right)-\frac{1}{2}\log\left(1+\frac{(y-1)^2}{(x-i)^2}\right)=\log(x-1)$$

Using $(2,1)$

$$\tan^{-1}(\theta)-\frac{1}{2}\log(1)=c$$

$$\Rightarrow c=0$$

Using $(k+1,2)$

$$\tan^{-1}\left(\frac{1}{k}\right)-\frac{1}{2}\log\left(1+\frac{1}{k^2}\right)=\log(k)$$

$$2\tan^{-1}\left(\frac{1}{k}\right)-\log\left(\frac{k^2+1}{k^2}\right)-2\log k=0$$

$$2\tan^{-1}\left(\frac{1}{k}\right)=\log\left(k^2+1\right)$$

$\therefore$ option(1)

Q. 10 Let $y=y(x)$ be the solution curve of the differential equation $\frac{dy}{dx}+\left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right)y=\frac{(x+3)}{x+1}, x>-1$, which passes through the point $(0,1)$. Then $y(1)$ is equal to :

Option 1:

$\frac{1}{2}$

Option 2:

$\frac{3}{2}$

Option 3:

$\frac{5}{2}$

Option 4:

$\frac{7}{2}$

Correct Answer:

$\frac{3}{2}$

Solution:

Using partial fractions,

$$\frac{2x^2+11x+13}{x^3+6x^2+11x+6}=\frac{2}{x+1}+\frac{1}{x+2}-\frac{1}{x+3}$$

$$\text{I.F.}=e^{2\log(x+1)+\log(x+2)-\log(x+3)}$$

$$=\frac{(x+1)^2(x+2)}{(x+3)}$$

$\therefore$ Solution is

$$\frac{y(x+1)^2(x+2)}{(x+3)}=\int\frac{(x+3)}{(x+1)}\cdot\frac{(x+1)^2(x+2)}{(x+3)}dx$$

$$y\frac{(x+1)^2(x+2)}{(x+3)}=\int(x+1)(x+2)dx$$

$$y\frac{(x+1)^2(x+2)}{(x+3)}=\frac{x^3}{3}+\frac{3x^2}{2}+2x+c$$

Using $(0,1)$

$$1\cdot\frac{2}{3}=c\Rightarrow c=\frac{2}{3}$$

Put $x=1$

$$y=\frac{4\cdot3}{4}=\frac{1}{3}+\frac{3}{2}+2+\frac{2}{3}$$

$$y=\frac{1}{3}\left(\frac{27}{6}\right)=\frac{3}{2}$$

$\therefore \text{option}(B)$

Q. 11 Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2+11a+3\left(m_1^2+m_2^2\right)=220$.
If one vertex of the square is $(10(\cos\alpha-\sin\alpha), 10(\sin\alpha+\cos\alpha))$, where $\alpha\in\left(0,\frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha-\sin\alpha)x+(\sin\alpha+\cos\alpha)y=10$, then $72\left(\sin^4\alpha+\cos^4\alpha\right)+a^2-3a+13$ is equal to :

Option 1:
119

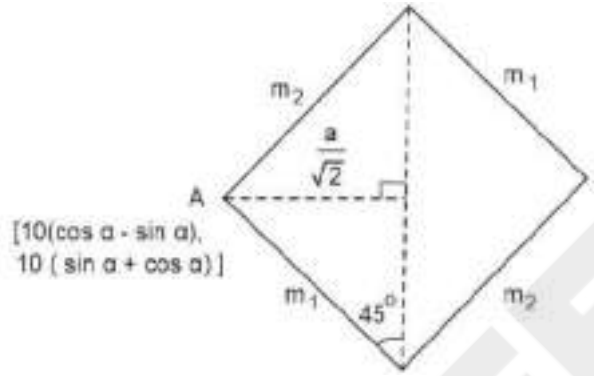
Option 2:
128

Option 3:
145

Option 4:
155

Correct Answer:
128

Solution:



$$\frac{a}{\sqrt{2}} = \frac{\left|10\left(\cos\alpha - \sin\alpha\right)^2 + 10\left(\cos\alpha + \sin\alpha\right)^2 - 10\right|}{\sqrt{\left(\cos\alpha - \sin\alpha\right)^2 + \left(\cos\alpha + \sin\alpha\right)^2}}$$

$$\Rightarrow \frac{a}{\sqrt{2}} = \frac{10}{\sqrt{2}}$$

$$\Rightarrow a = 10$$

$$\therefore \text{Equation } a^2 + 11a + (m_1^2 + m_2^2) = 220$$

$$\Rightarrow 100 + 110 + 3(m_1^2 + m_2^2) = 220$$

$$3\left(m_1^2 + m_2^2\right) = 10$$

$$\Rightarrow m_1^2 + m_2^2 = \frac{10}{3}$$

$$\Rightarrow m_1^2 + \frac{1}{m_1^2} = \frac{10}{3} \quad (\text{As } m_1 m_2 = -1\ )$$

$$\Rightarrow m_1 = \frac{1}{\sqrt{3}} \text{ or } -\frac{1}{\sqrt{3}} \text{ or } \sqrt{3} \text{ or } -\sqrt{3}$$

$\therefore (m_1, m_2)$ is $\left(\frac{1}{\sqrt{3}}, -\sqrt{3}\right)$ or $\left(-\frac{1}{\sqrt{3}}, \sqrt{3}\right)$

For $\left(-\frac{1}{\sqrt{3}}, \sqrt{3}\right)$, sides make -30° and 60° with $+x$ axis , therefore diagonal makes 15° with $+x$ axis

Also given slope of diagonal $= \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha}$

$$\Rightarrow \frac{\sin\alpha - \cos\alpha}{\sin\alpha + \cos\alpha} = \tan 15^\circ$$

$$\Rightarrow \frac{\tan\alpha - 1}{1 + \tan\alpha} = \tan 15^\circ$$

$$\Rightarrow \tan(\alpha - 45^\circ) = \tan_1 15^\circ$$

$$\Rightarrow \alpha = 60^\circ$$

$$\therefore \quad 72\left(\sin^4\alpha + \cos^4\alpha\right) + a^2 - 3a + 13$$

$$= 72\left(\frac{9}{16} + \frac{1}{16}\right) + 100 - 30 + 13$$

$$= 128$$

$\therefore$ Option (B)

Q. 12 The number of elements in the set $S = \left\{x \in \mathbb{R} : 2\cos\left(\frac{x^2 + x}{6}\right) = 4^x + 4^{-x}\right\}$ is :

Option 1:
1

Option 2:
3

Option 3:
0

Option 4:
infinite

Correct Answer:
1

Solution:

$$\cos\left(\frac{x^2+x}{6}\right)=\frac{4^x+4^{-x}}{2}$$

Minimum value of RHS (Using AM-GM) is 1 when x=0

$\therefore$ LHS = RHS can be equal only when both equal I.

Both sides equal I when x=0

x=0 is the only solution Ans:(A)

Q. 13 Let $A(\alpha,-2), B(\alpha,6)$ and $C\left(\frac{\alpha}{4},-2\right)$ be vertices of a $\triangle ABC$.If $\left(5,\frac{\alpha}{4}\right)$ is the circumcentre of $\triangle ABC$, then which of the following is NOT correct about $\triangle ABC$?

Option 1:
area is 24

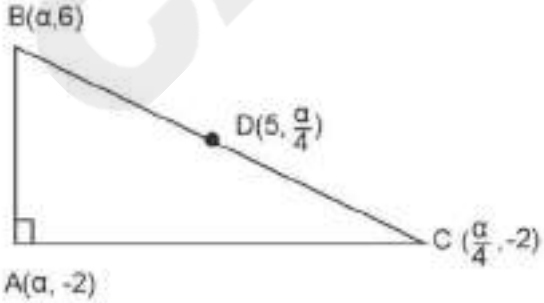
Option 2:
perimeter is 25

Option 3:
circumradius is 5

Option 4:
inradius is 2

Correct Answer:
perimeter is 25

Solution:



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clearly A is right angle and this D $\left(5, \frac{\alpha}{4}\right)$ is mid-point of BC.

$\therefore \quad \frac{6-2}{2} = \frac{\alpha}{4} \Rightarrow \alpha = 8$

$\therefore \mathrm{AB} = 8, \quad \mathrm{AC} = 6$

$\text{Area} = \frac{1}{2} \times 8 \times 6 = 24$

$\text{Perimeter} = 8 + 6 + \sqrt{8^2 + 6^2} = 24$

$\therefore$ option (B)

Q. 14 Let Q be the foot of perpendicular drawn from the point $\mathrm{P}(1, 2, 3)$ to the plane $\mathrm{x} + 2\mathrm{y} + \mathrm{z} = 14$. If R is a point on the plane such that $\angle \mathrm{PRQ} = 60°$, then the area of $\triangle \mathrm{PQR}$ is equal to :

Option 1:
$\frac{\sqrt{3}}{2}$

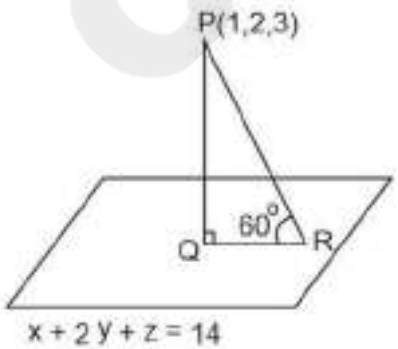
Option 2:
$\sqrt{3}$

Option 3:
$2\sqrt{3}$

Option 4:
3

Correct Answer:
$\sqrt{3}$

Solution:



$$PQ = \frac{|11+4+3-14|}{\sqrt{1+4+1}} = \frac{6}{\sqrt{6}} = \sqrt{6}$$

$$\text{Area} = \frac{1}{2} \cdot PQ \cdot QR = \frac{1}{2} \cdot \sqrt{6} \times \frac{PQ}{\tan 60^\circ}$$

$$= \frac{1}{2} \times \frac{6}{\sqrt{3}} = \sqrt{3}$$

Q. 15 If $(2,3,9), (5,2,1), (1,\lambda,8)$ and $(\lambda,2,3)$ are coplanar, then the product of all possible values of λ is :

Option 1:
$\frac{21}{2}$

Option 2:
$\frac{59}{8}$

Option 3:
$\frac{57}{8}$

Option 4:
$\frac{95}{8}$

Correct Answer:
$\frac{95}{8}$

Solution:

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A, B, C, D are coplanar implies $\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD}$ are coplanar

$$\Rightarrow \left[\begin{array}{ccc} \overrightarrow{AB} & \overrightarrow{AC} & \overrightarrow{AD} \end{array}\right] = 0$$

$$\Rightarrow \begin{vmatrix} 3 & -1 & -8 \\ -1 & \lambda - 3 & -1 \\ 2-2 & -1 & -6 \end{vmatrix} = 0$$

$$\Rightarrow 3[-6(\lambda-3)-1] + 1[6+\lambda-2]$$

$$-8[1-(2-2)(\lambda-3)] = 0$$

$$\Rightarrow 3[-6h+17] + 4 + h - 8\left[1-\left(h^2-5h+6\right)\right]$$

$$\Rightarrow -18\lambda + 51 + 4 + \lambda - 8\left[-\lambda^2+5\lambda-5\right] = 0$$

$$\Rightarrow -17\lambda + 55 + 8\lambda^2 - 40\lambda + 40 = 0$$

$$\Rightarrow 8\lambda^2 - 57\lambda + 95 = 0$$

$$\text{Product } R_1R_2 = \frac{95}{8}$$

$\therefore$ option (D)

Q. 16 Bag I contains 3 red, 4 black and 3 white balls and Bag II contains 2 red, 5 black and 2 white balls. One ball is transferred from Bag I to Bag II and then a ball is drawn from Bag II. The ball so drawn is found to be black in colour. Then the probability, that the transferred ball is red, is :

Option 1:

$\frac{4}{9}$

Option 2:

$\frac{5}{18}$

Option 3:

$\frac{1}{6}$

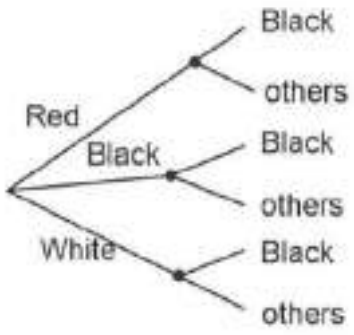
Option 4:

$\frac{3}{10}$

Correct Answer:

$\frac{5}{18}$

Solution:



$$\text{Using Baye's Thearem, required probability} = \frac{\frac{3}{10}\times\frac{5}{10}}{\frac{3}{10}\times\frac{5}{10}+\frac{4}{10}\times\frac{6}{10}+\frac{3}{10}\times\frac{5}{10}}$$

$$= \frac{15}{15+24+15} = \frac{15}{54} = \frac{5}{18}$$

$\therefore$ option (B)

Q. 17 Let $S=\{z=x+iy:|z-1+i|\geq|z|,|z|<2,|z+i|=|z-1|\}$. Then the set of all values of x, for which $w=2x+iy\in S$ for some $y\in\mathbb{R}$, is

Option 1:
$\left(-\sqrt{2},\frac{1}{2\sqrt{2}}\right]$

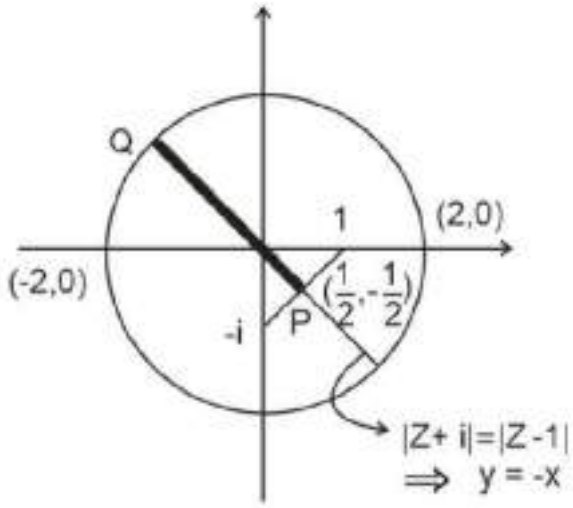
Option 2:
$\left(-\frac{1}{\sqrt{2}},\frac{1}{4}\right]$

Option 3:
$\left(-\sqrt{2},\frac{1}{2}\right]$

Option 4:
$\left(-\frac{1}{\sqrt{2}},\frac{1}{2\sqrt{2}}\right]$

Correct Answer:
$\left(-\frac{1}{\sqrt{2}},\frac{1}{4}\right]$

Solution:



The solid line denotes the required region

As $\omega = 2x + iy \in S$

Maximum value of 2 x= maximum value of real part of ω on solid line

$(2x)_{max} = \frac{1}{2}$

$x_{max} = \frac{1}{4}$

$\therefore$ option (B)

Q. 18 Let $\vec{a}, \vec{b}, \vec{c}$ be three coplanar concurrent vectors such that angles between any two of them is same. If the product of their magnitudes is 14 and $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) = 168$, then $|\vec{a}| + |\vec{b}| + |\vec{c}|$ is equal to :

Option 1:
10

Option 2:
14

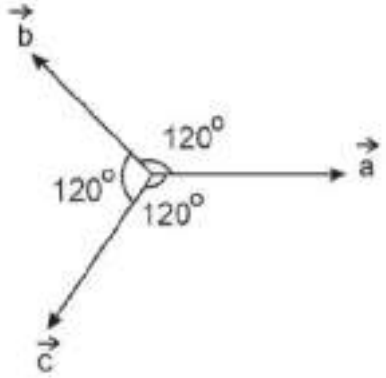
Option 3:
16

Option 4:
18

Correct Answer:

16

Solution:



$\therefore All\ of\ \vec{a}\times\vec{b}, \vec{b}\times\vec{c}\ and\ \vec{c}\times\vec{a}\ are$

$\therefore (\vec{a}\times\vec{b})\cdot(\vec{b}\times\vec{c}) = |a||b|\sin 120^\circ \cdot |b||c|\sin 120^\circ$

$= \frac{3}{4}|a||b|^2|c|$

$\therefore$ Required expression

$\Rightarrow \frac{3}{4}\left(|a||b|^2|c| + |a||b||c|^2 + |a|^2|b||c|\right) = 168$

$\Rightarrow |a||b||c|(|a| + |b| + |c|) = 224$

$\Rightarrow |a| + |b| + |c| = 16$

$\therefore$ option (c)

Q. 19 The domain of the function $f(x) = \sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ is:

Option 1:

$[1, \infty)$

Option 2:

$[-1, 2]$

Option 3:

$[-1, \infty)$

Option 4:
$(-\infty, 2]$

Correct Answer:
$[-1, \infty)$

Solution:

For domain

$$-1 \leq \frac{x^2 - 3x + 2}{x^2 + 2x + 7} \leq 1$$

As denonienatar is pasitive (Dj0)

$$-\left(x^2 + 2x + 7\right) \leq x^2 - 3x + 2 \leq x^2 + 2x + 7$$

Solving first

$$-x^2 - 2x - 7 \leqslant x^2 - 3x + 2$$

$$\Rightarrow 2x^2 - x + 9 \geqslant 0$$

$$\Rightarrow x \in R \quad (\text{as } D < 0)$$

Solving second

$$x^2 - 3x + 2 \leqslant x^2 + 2x + 7$$

$$\Rightarrow 5x + 5 \geqslant 0$$

$$\Rightarrow x \geqslant -1$$

$$\therefore \quad x \geqslant -1$$

option (C)

Q. 20 The statement $(p \Rightarrow q) \vee (p \Rightarrow r)$ is **NOT** equivalent to

Option 1:
$(p \wedge (\sim r)) \Rightarrow q$

Option 2:
$(\sim q) \Rightarrow ((\sim r) \vee p)$

Option 3:
$p \Rightarrow (q \vee r)$

Option 4:
$(p \wedge (\sim q)) \Rightarrow r$

Correct Answer:

$(\sim q) \Rightarrow ((\sim r) \vee p)$

Solution:

$(p \rightarrow q) \vee (p \rightarrow r)$

$\equiv (\sim p \vee q) \vee (\sim p \vee r)$

$\equiv \sim p \vee q \vee r$

(A) $(p \wedge \sim r) \rightarrow q$

$\equiv \sim (p \wedge \sim r) \vee q$

$\equiv \sim p \vee r \vee q$

(B) $\sim q \rightarrow (\sim r \vee p)$

$\equiv q \vee (\sim r \vee p)$

$\equiv p \vee q \vee r$

$\therefore$ option (B)

Q. 21 The sum and product of the mean and variance of a binomial distribution are $82 \cdot 5$ and 1350 respectively. Then the number of trials in the binomial distribution is_____.

Correct Answer:

96

Solution:

$$\text{Mean} = np$$

$$\text{Varianel } npq = np(1-p)$$

$$\text{Sum } = np(1+1-p) = 82.5$$

$$\Rightarrow np(2-p) = 82.5 \ldots \text{(i)}$$

$$\text{Product} = np \cdot np(1-p) = 1350$$

$$\Rightarrow \frac{(82.5)^2}{(2-p)^2}(1-p) = 1350$$

$$\Rightarrow p = \frac{5}{8}$$

$$\text{Using (i)}, n = 96$$

Q. 22 Let $\alpha, \beta(\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in N$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to_____

Correct Answer:
16

Solution:

$P_n = \alpha^n - \beta^n$

Also $\alpha^2 - \alpha - 4 = 0$

$\alpha^2 - \alpha = 4$

$\alpha^{n+2} - \alpha^{n+1} = 4\alpha^n$

similarly $\beta^{n+2} - \beta^{n+1} = 4\beta^n$

subtrating these

$(\alpha^{n+2} - \beta^{n+2}) - (\alpha^{n+1} - \beta^{n+1}) = 4(\alpha^n - \beta^n)$

$p_{n+2} - P_{n+1} = 4p_{n_1}$

Put $n = 14 \Rightarrow P_{16} - P_{15} = 4P_{14}$

Put $n = 13 \Rightarrow P_{15} - P_{14} = 4P_{13}$

Now

$$\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$$

$$= \frac{(P_{16} - P_{15})(P_{15} - P_{14})}{P_{13}P_{14}}$$

$$= \frac{4P_{14} \cdot 4P_{13}}{P_{13} \cdot P_{14}}$$

$= 16$

Ans: 16

Q. 23 Let $X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$. For $k \in \mathbb{N}$, if $X'A^kX = 33$, then k is equal to _____.

Correct Answer:
10

Solution:

$$A^2 = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I + \begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Let $\begin{bmatrix} 0 & 0 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = B$

$B^2 = 0,$

$\therefore B^3 = 0 = B^4 = B^5 = \cdots$

Now $(A^2)^P = (I + B)^P.$

$= I + pB$

$$= \begin{bmatrix} 1 & 0 & 6p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now $x'A^kx$

Let k be even and $k = 2p$

$$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 6p \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 6p+1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$= [6p + 3]$

Now $6p + 3 = 33$

$\Rightarrow p = 5$

$k = 2p = 10$

Aus: 10

$\therefore k = 2p = 10$

Q. 24 The number of natural numbers lying between 1012 and 23421 that can be formed using the digits 2,3,4,5,6 (repetition of digits is not allowed) and divisible by 55 is_____.

Correct Answer:

6

Solution:

For divisibility by 55, it should be divisible by 5 and 11

4 digit number: $\underline{a}\ \underline{b}\ \underline{c}\ \underline{5}$

If $= 1, (a, c)$ can be $(2, 4), (4, 2)$

For $b = 2, (a, c) = (3, 4), (4, 3)$

For $b = 3, (a, c) = (2, 6), (6, 2)$

5 digit numbers less than 23421 are not passible.

Ans: 6

Q. 25 If $\sum_{k=1}^{10} K^2 \left({}^{10}C_K\right)^2 = 22000L$, then L is equal to _____.

Correct Answer:
221

Solution:

$\sum k^2 \left({}^{10}C_k\right)^2$

$= \Sigma k^2 \left(\frac{10}{k} \cdot {}^{9}C_{k-1}\right)^2$

$= 100\Sigma {}^{9}C_{k-1} \cdot {}^{9}C_{k-1}$

Using product of binomial certfficiut formula

$= 100 \cdot {}^{18}C_9$

$= 100 \cdot \frac{18 \cdot 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10}{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}$

$= 221 \cdot (22000)$

$\therefore L = 221$

Ans:221

Q. 26 If $[t]$ denotes the greatest integer $\leq t$, then the number of points, at which the function $f(x) = 4|2x + 3| + 9\left[x + \frac{1}{2}\right] - 12[x + 20]$ is not differentiable in the open interval $(-20, 20)$, is_____.

Correct Answer:
79

Solution:

$4|2x+3|$ is non-differentiable at $x=-\frac{3}{2}$

$9\left[x+\frac{1}{2}\right]$ is non-differentiable at $x+\frac{1}{2}]$ is

$x=-19.5,-18.5,\ldots,-0.5,0.5,\ldots \quad \ldots,19.5$

$-12[x+20]=-12[x]-240$ is nondifferentiable at $x=-19,-18,\ldots,0,1,2.$

$\therefore$ Total $=40+39+1-1($ for $x=-\frac{-3}{2}$ is common $)$

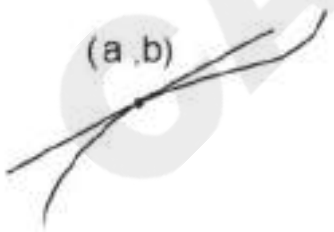
$=79$

Ans: 79

Q. 27 If the tangent to the curve $y=x^3-x^2+x$ at the point (a,b) is also tangent to the curve $y=5x^2+2x-25$ at the point $(2,-1)$, then $|2a+9b|$ is equal to_____

Correct Answer:
195

Solution:
Equation of tangent to $y=5x^2+2x-25$
at (2,-1)
$y'=10x+2=20+2=22$
$y+1=22(x-2)$
$y+1=22x-44$
$y=22x-45$
This is also tangent to $y=x^3-x^2+x$ at (a,b)



Finding point of intersection

$22x - 45 = x^3 - x^2 + x$

$\Rightarrow x^3 - x^2 - 21x + 45 = 0$

$\Rightarrow (x-3)^2(x+5) = 0$

Tangent at $x = 3 \Rightarrow a = 3$

Put $x = 3$ in $y = 22x - 45$

$\Rightarrow b = 66 - 45 = 21$

$\therefore |2a + 9b| = 6 + 189 = 195$

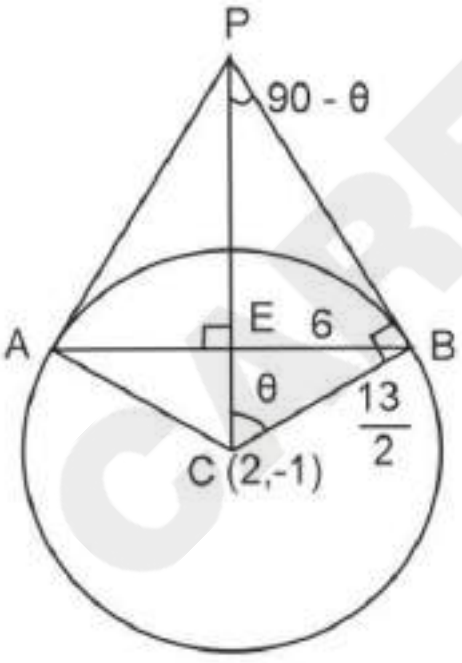
Ans:195

Q. 28 Let AB be a chord of length 12 of the circle $(x-2)^2 + (y+1)^2 = \frac{169}{4}$.

If tangents drawn to the circle at points A and B intersect at the point P, then five times the distance of point P from chord AB is equal to_____.

Correct Answer:
72

Solution:



$$\sin\theta = \frac{BE}{BC} = \frac{6}{\frac{13}{2}} = \frac{12}{13}$$

$$\text{Now } \frac{BE}{\rho E} = \tan(90-\theta)$$

$$\Rightarrow \frac{6}{PE} = \cot\theta$$

$$\Rightarrow PE = 6\tan\theta = 6\cdot\frac{12}{5} = \frac{72}{5}$$

$$\Rightarrow 5PE = 72$$

$$\text{Ans} : 72$$

Q. 29 Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + 2|\vec{b}|^2, \vec{a}\cdot\vec{b} = 3$ and $|\vec{a}\times\vec{b}|^2 = 75$. Then $|\vec{a}|^2$ is equal to_____.

Correct Answer:
14

Solution:

$$|\tilde{a}+\tilde{b}|^2 = |\tilde{a}|^2 + 2|\tilde{b}|^2$$

$$\Rightarrow |\tilde{a}|^2 + |\tilde{b}|^2 + 2\tilde{a}.\tilde{b}^2 = |\vec{a}|^2 + 2|\vec{b}|^2$$

$$\Rightarrow |\tilde{b}|^2 = +2|\tilde{a}|.|\tilde{b}|^2 = 6$$

$$\Rightarrow |\tilde{b}| = \sqrt{6}$$

$$|\tilde{a}.\tilde{b}|^2 + |\tilde{a}\times\tilde{b}|^2 = |\tilde{a}|^2|b|^2$$

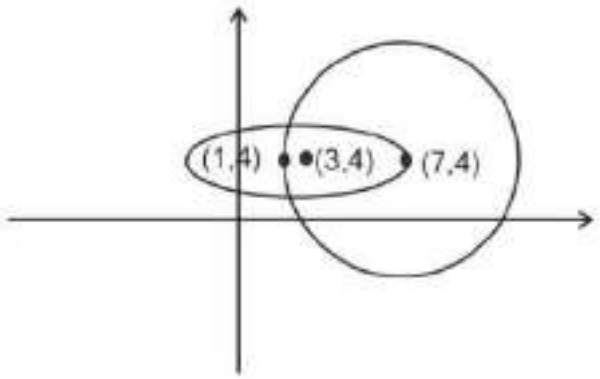
$$\Rightarrow 9 + 75 = |\tilde{a}|^2.6$$

$$\Rightarrow |\tilde{a}|^2 = \frac{84}{6} = 14$$

Q. 30 Let $S = \{(x,y) \in \mathbb{N}\times\mathbb{N} : 9(x-3)^2 + 16(y-4)^2 \le 144\}$ and $T = \{(x,y) \in \mathbb{R}\times\mathbb{R} : (x-7)^2 + (y-4)^2 \le 36\}$. Then $n(S\cap T)$ is equal to_____.

Correct Answer:
27

Solution:



For $x = 1 \Rightarrow (1,4)$

For $x = 2 \Rightarrow (2,4),(2,5),(2,3),(2,6),(2,2)$

For $x = 3 \Rightarrow (3,4),(3,5),(3,3),(3,6),(3,2)(3,7),(3,1)$

For $x = 4 \Rightarrow (4,4),(4,5),(4,3),(4,6),(4,2)$

For $x = 5 \Rightarrow (5,4),(5,5),(5,3),(5,2),(5,6)$

For $x = 6 \Rightarrow (6,4),(6,3),(6,5)$

For $x = 7 \Rightarrow (7,4)$

$\therefore$ Total 27 common points

Ans:27

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